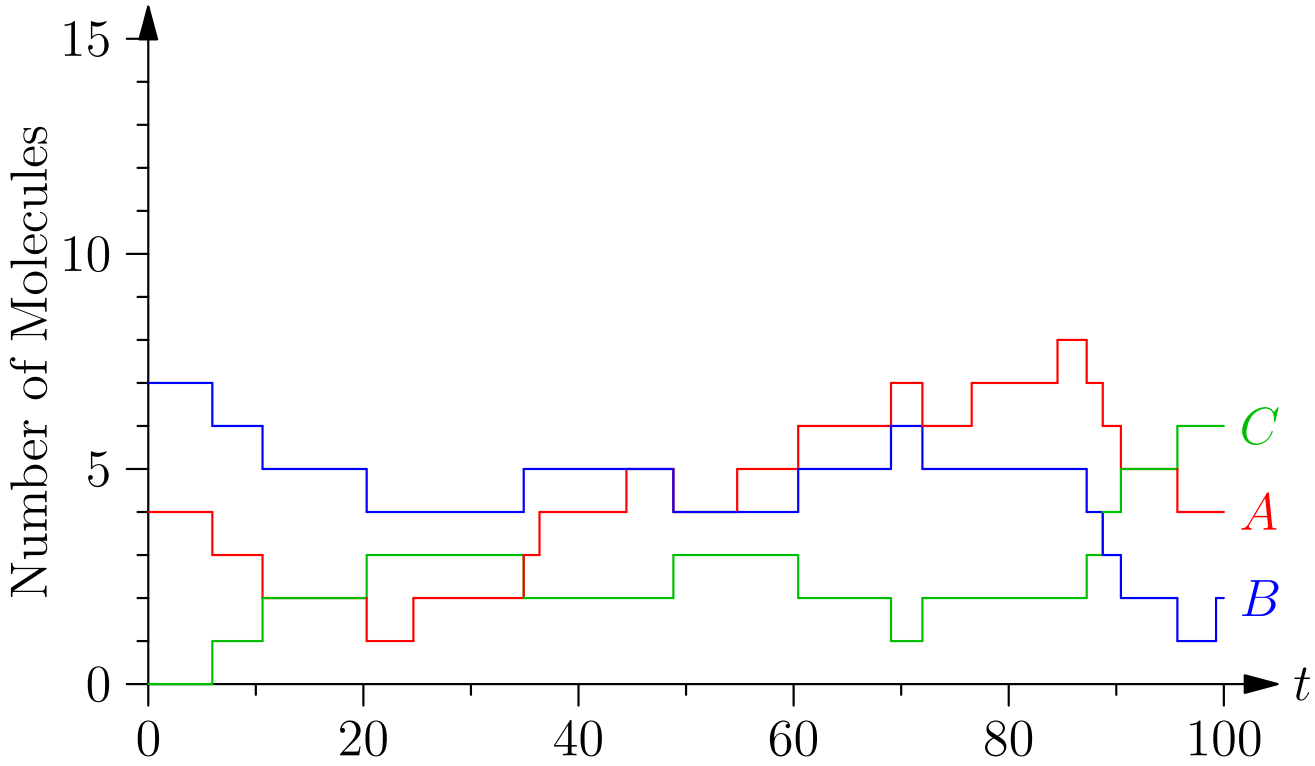
Computational Biology

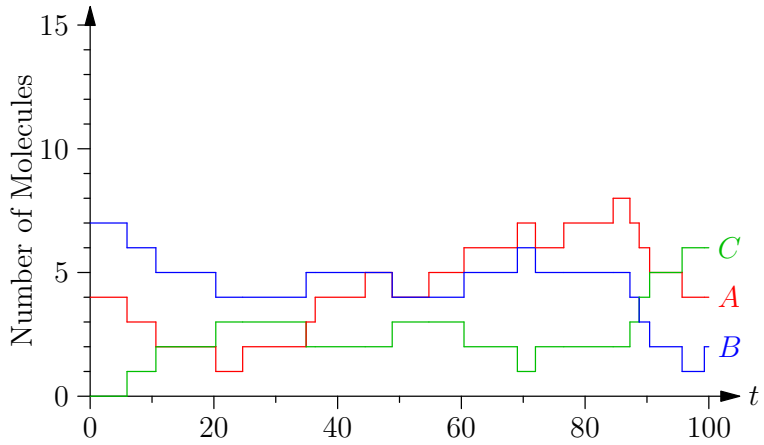
Lecture 19: Simulating Chemical Equations



Chemical Equations, Stochasticity, Gillespie Algorithm

Outline

1. **Stochastic Chemical Equations**
2. Gillespie Algorithm



Stochasticity

- A chemical equation

$$A + B \xrightarrow{k_1} C$$

  happens when molecule A by chance hits molecule B with the right momentum

- If the number of molecules are large then we can model this quite accurately by considering an ODE of what happens on average

- But for proteins and other molecules in the cell we often have a rather small number of a particular species of molecule in the cell

- In this case the ODE can be misleading

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Stochasticity in Nature

- Randomness can be problematic in making a system function
- Where this is critical then the cell relies on large molecule number
- However, there are times when randomness can serve a purpose
- In a large population of cells there can be an advantage to cells being in slightly different states
- This can give more robustness when responding to external stimuli

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Sources of Noise

- There are some variations that are common to most reactions coming from e.g.
  - ⋆ number of RNA-polymerase
  - ⋆ number of ribosomes
  - ⋆ size of cell
- This is known as **extrinsic noise**
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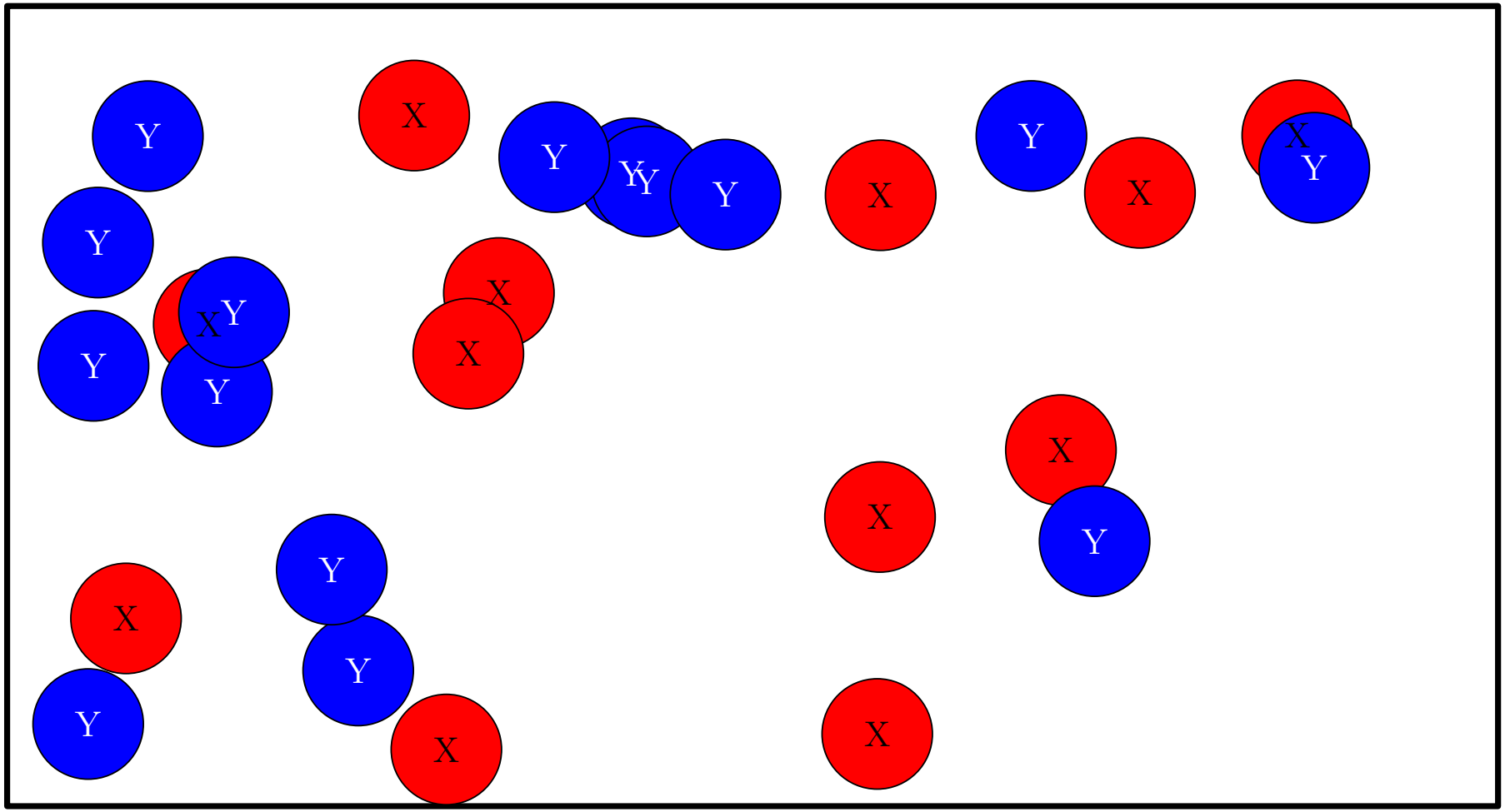

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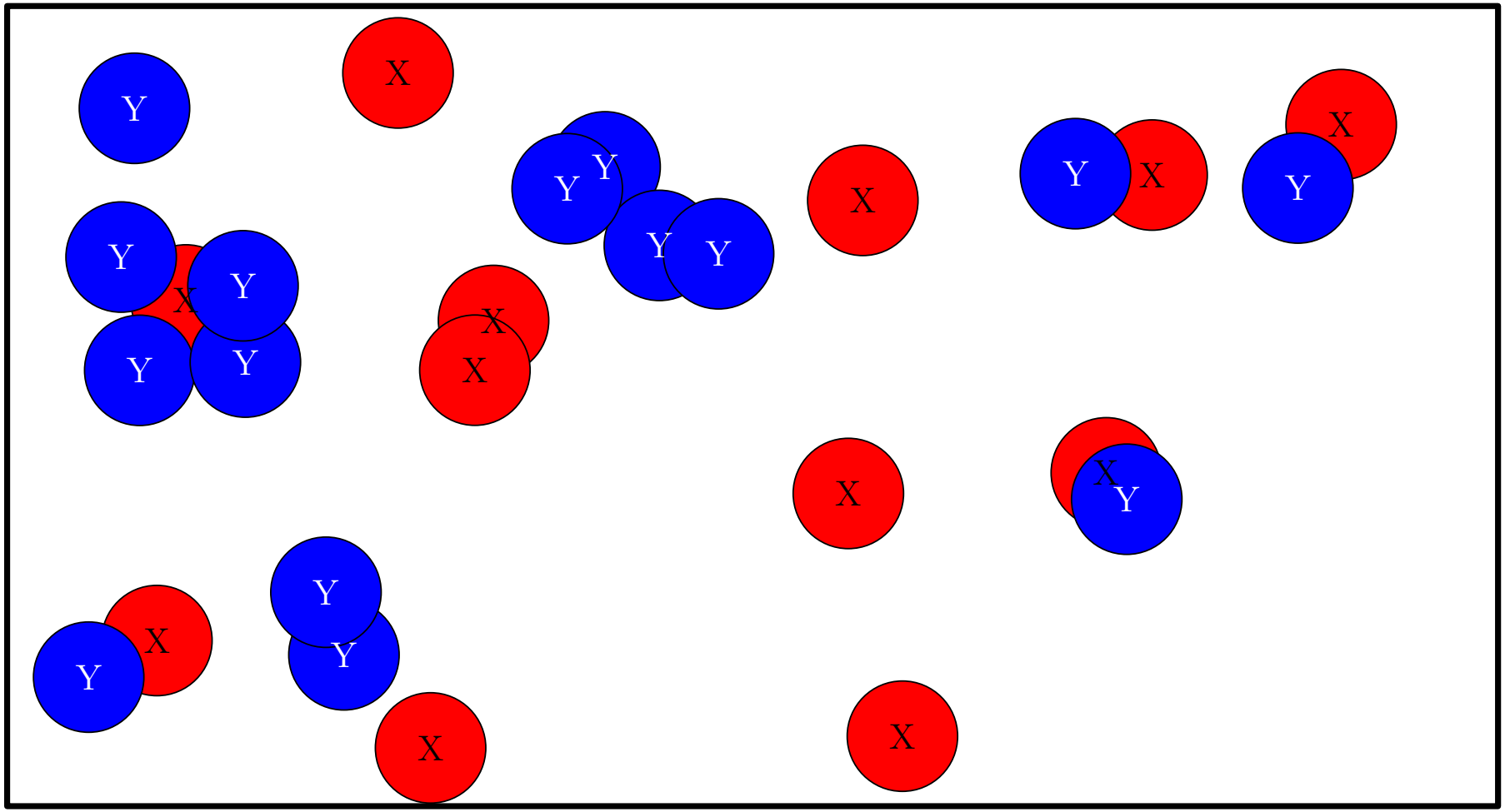

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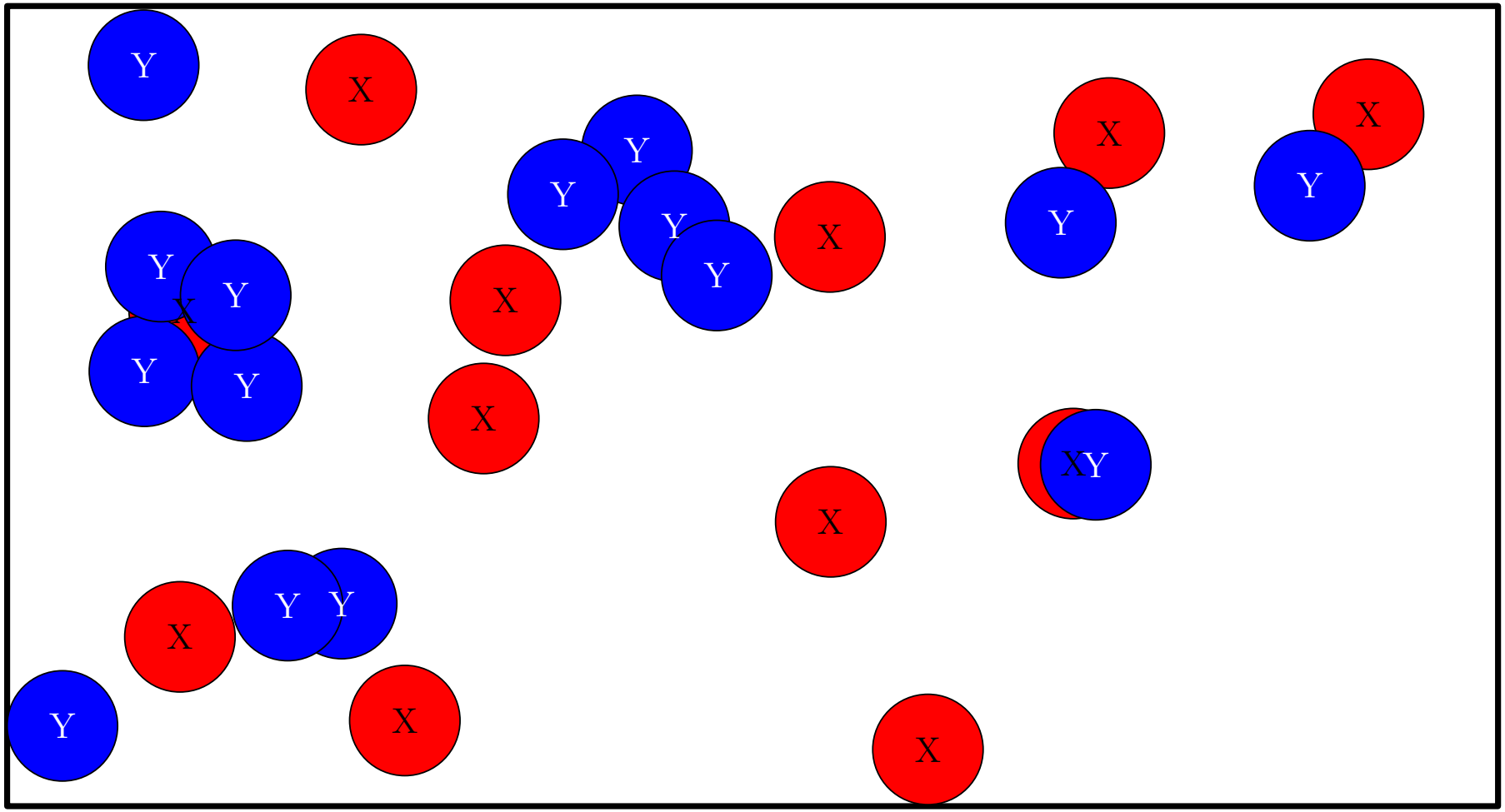

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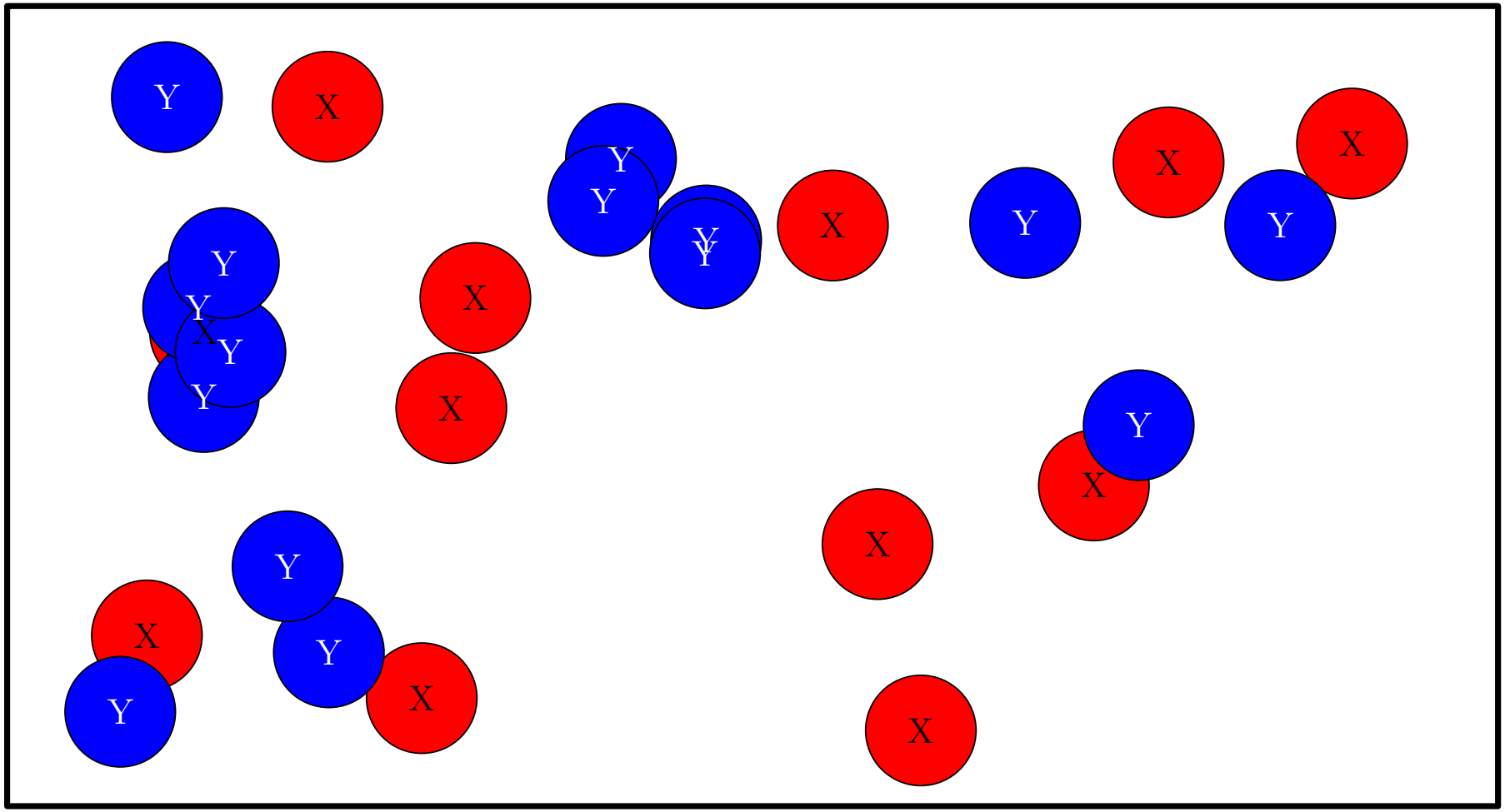

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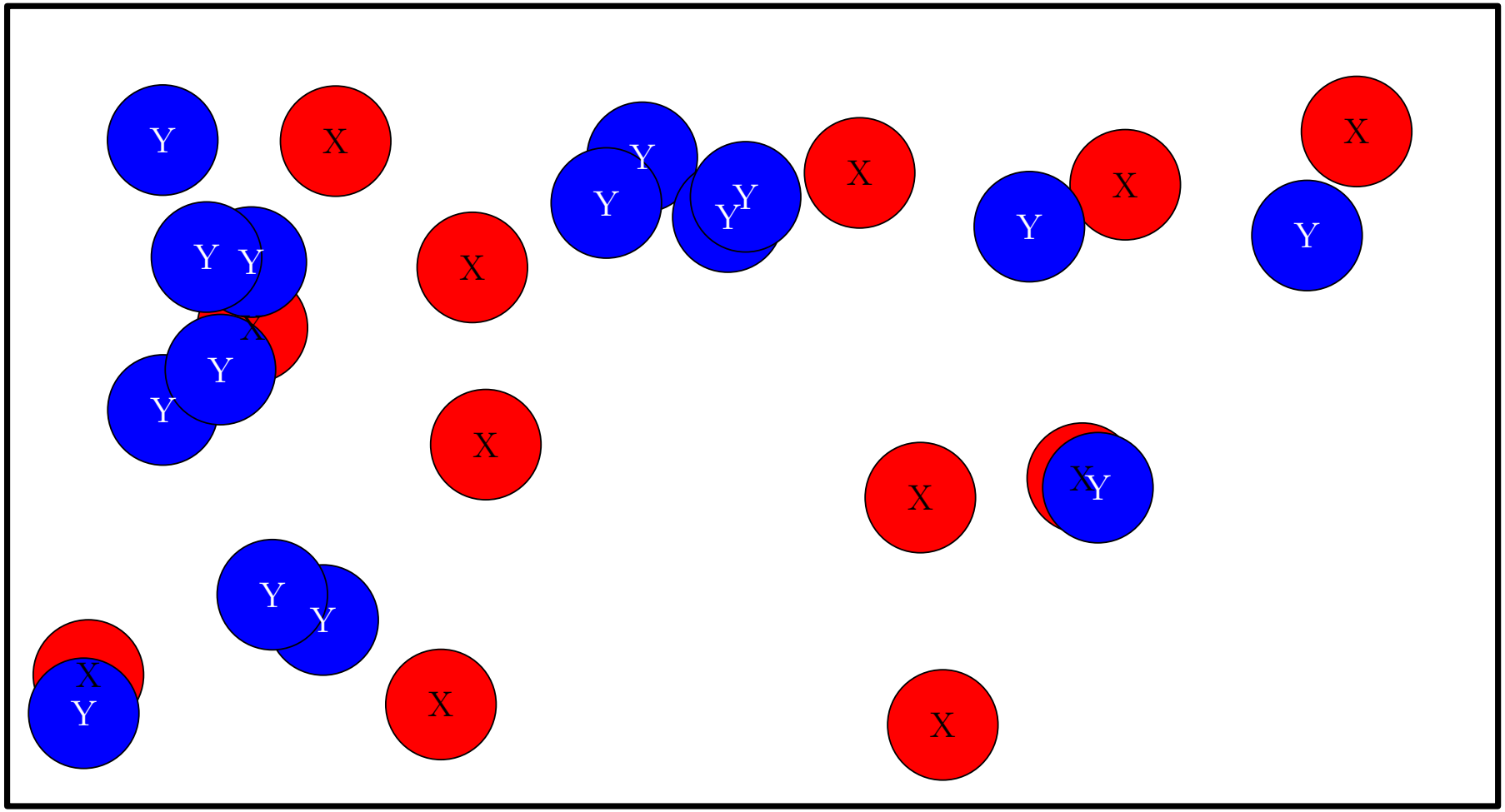

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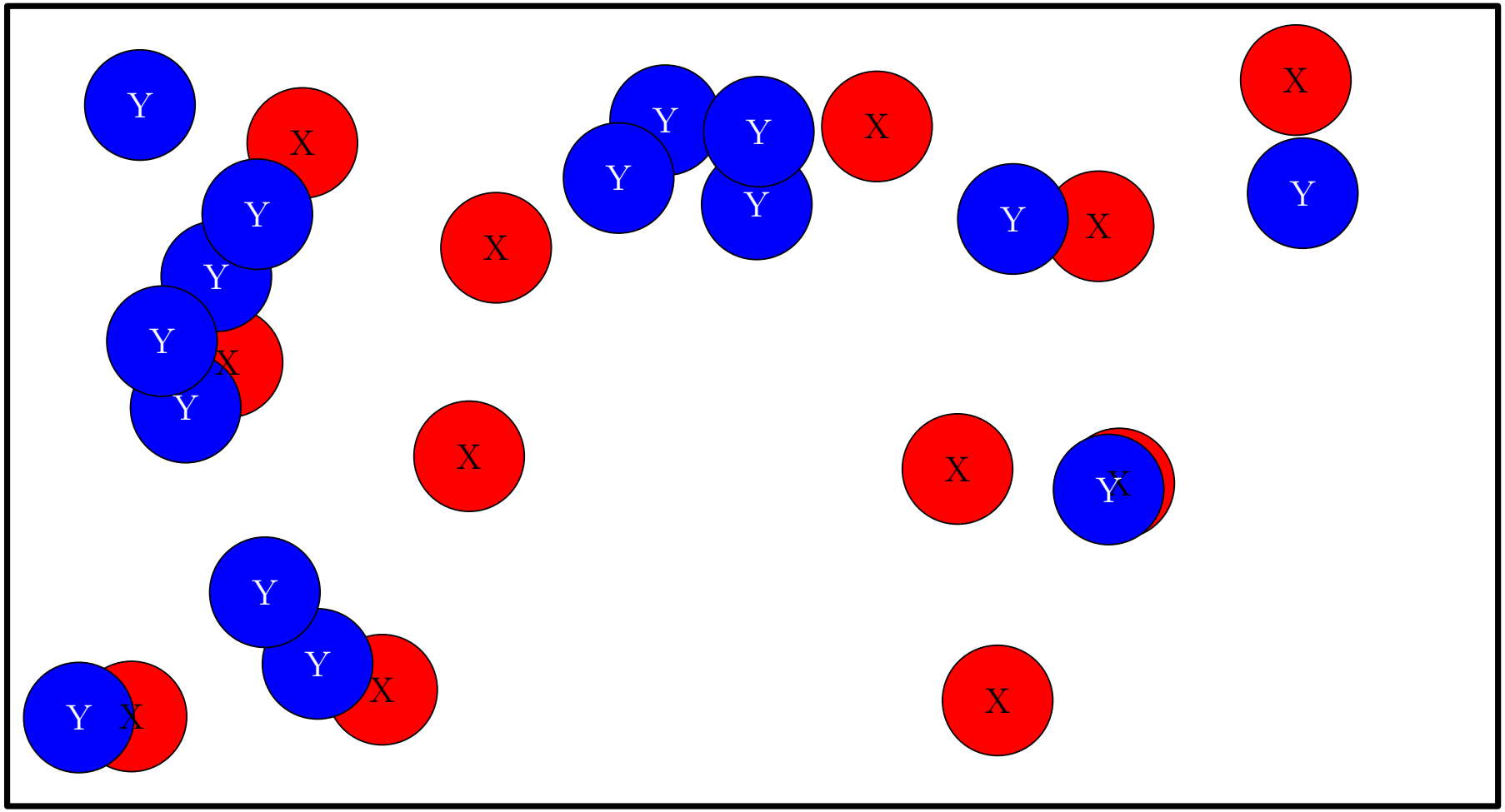

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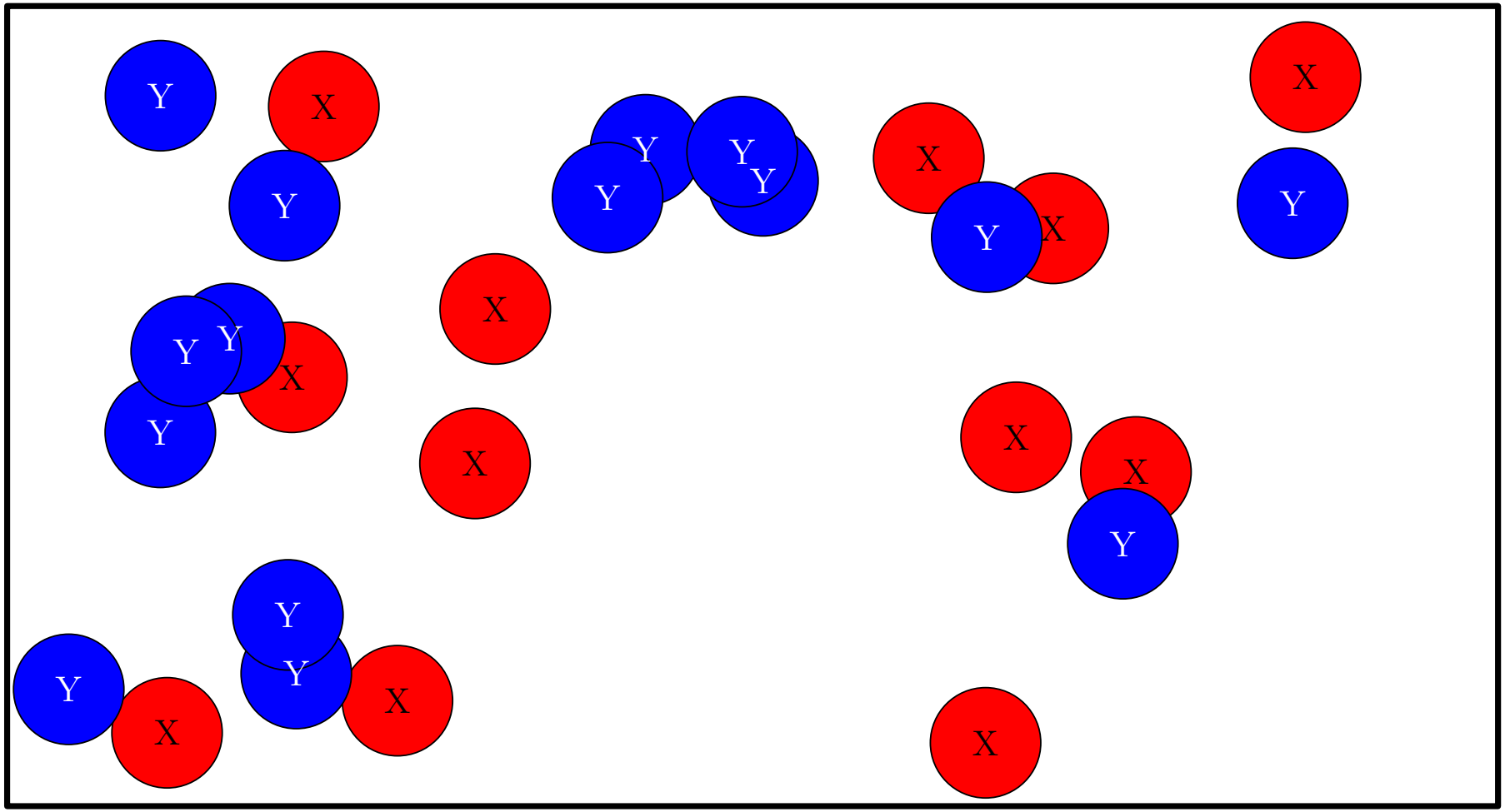

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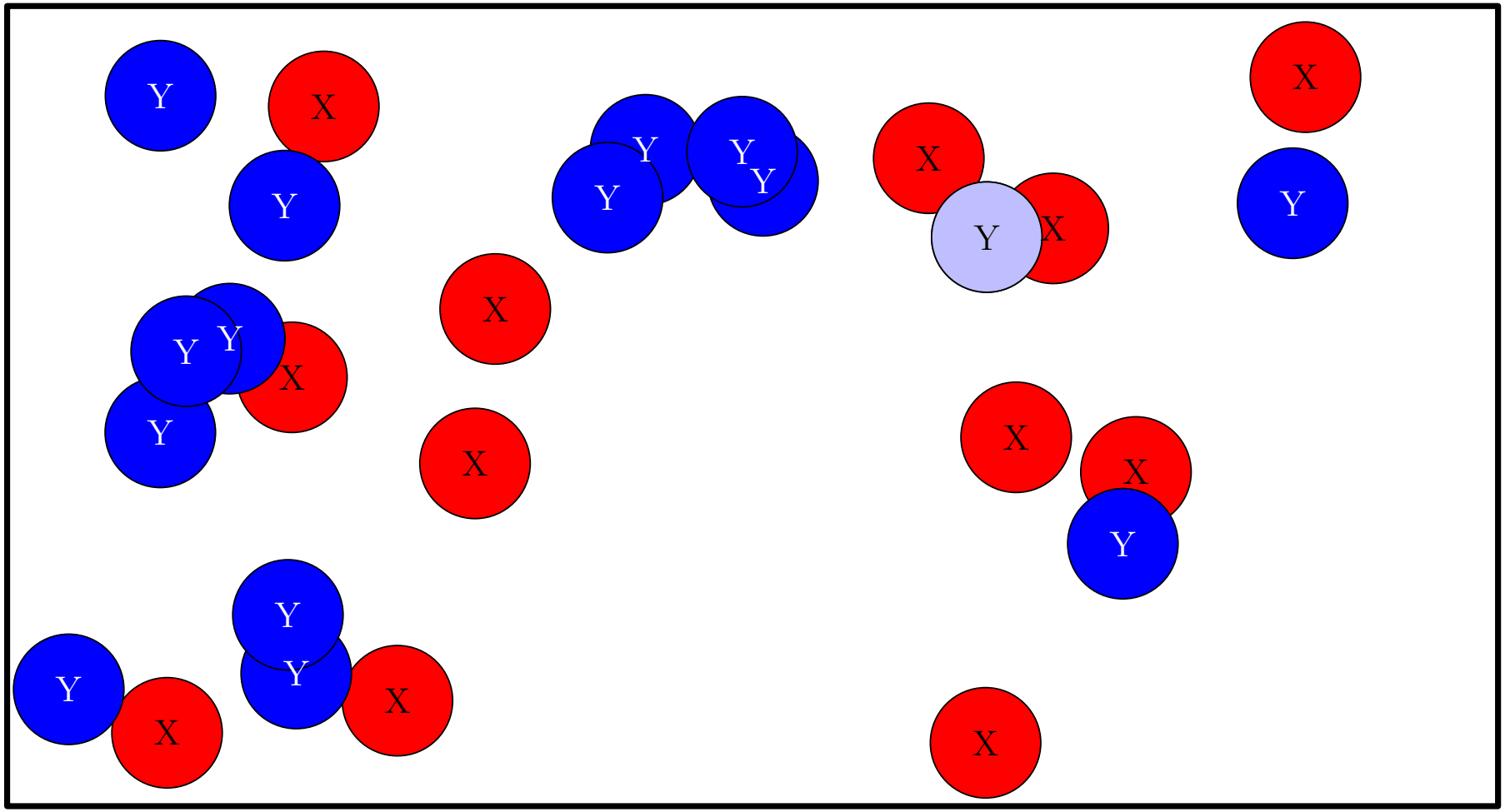

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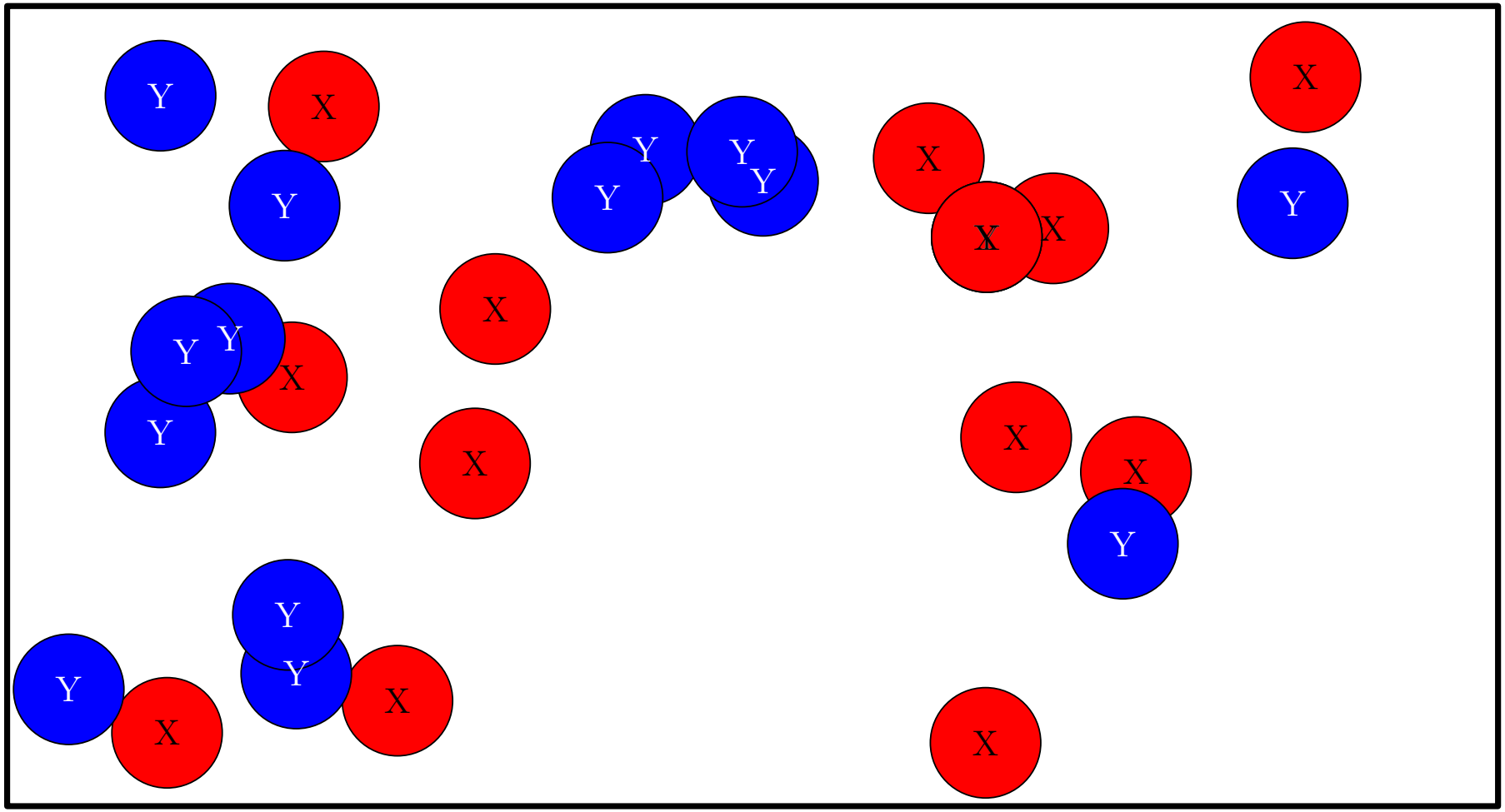

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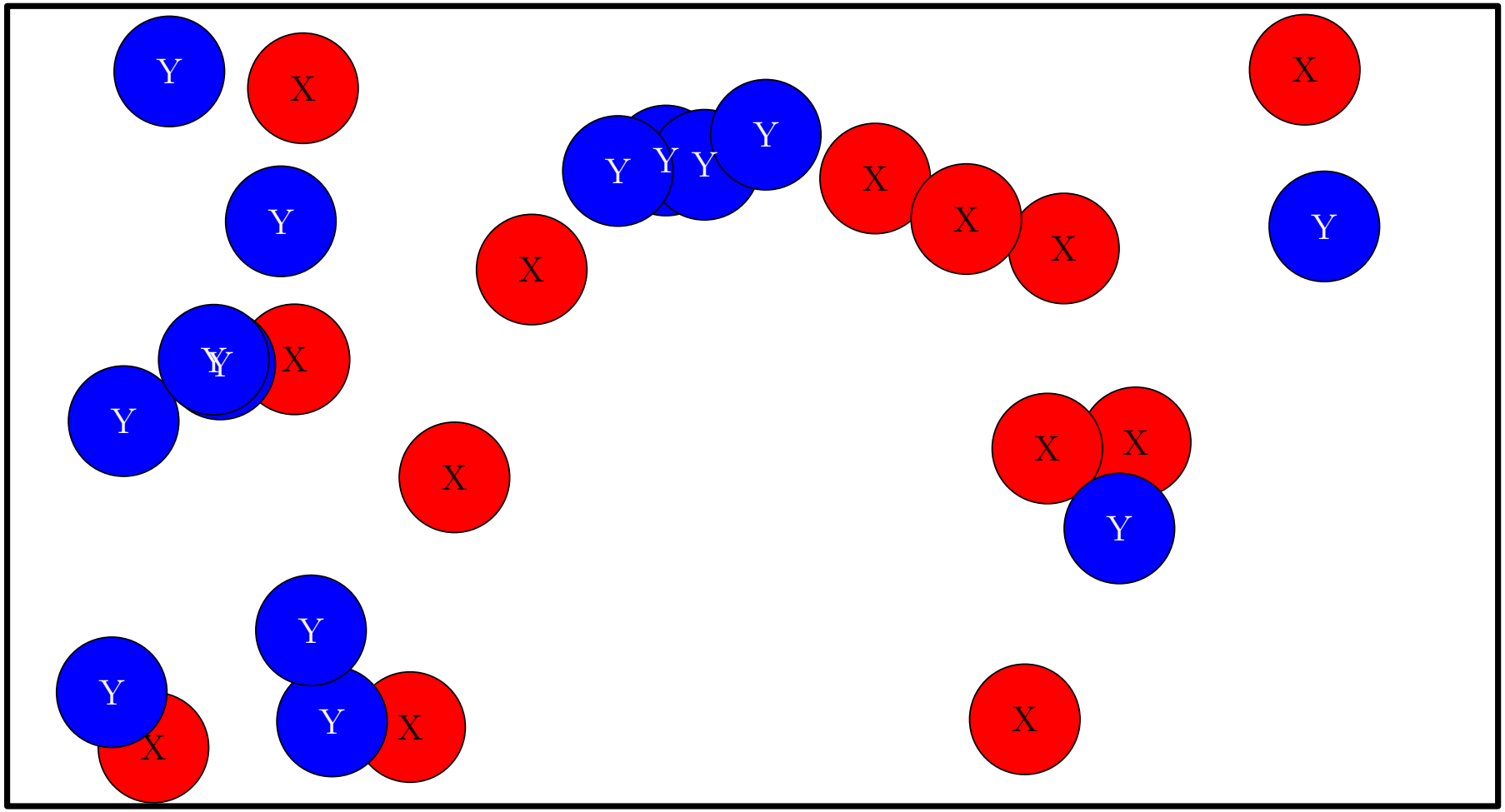

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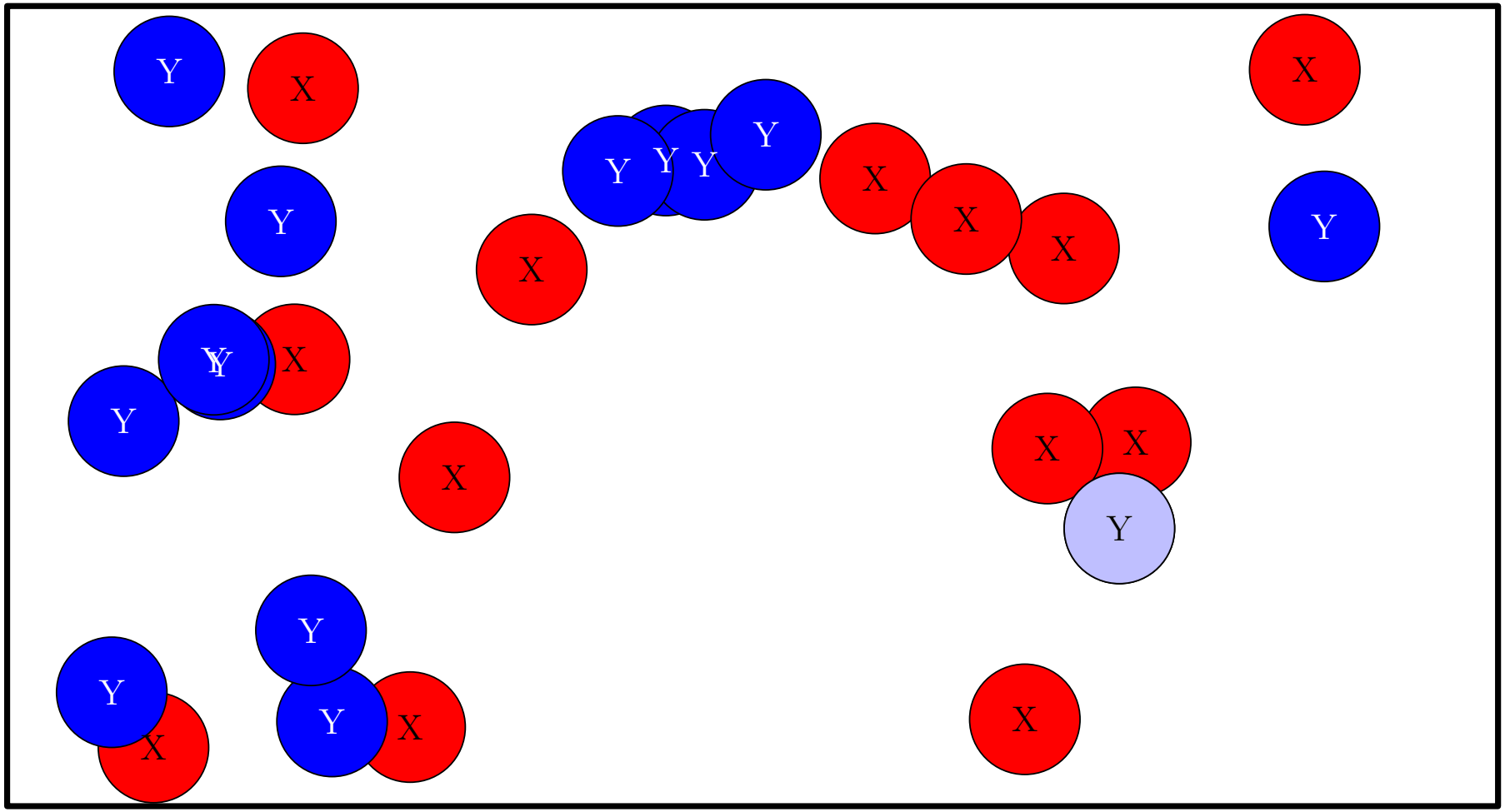

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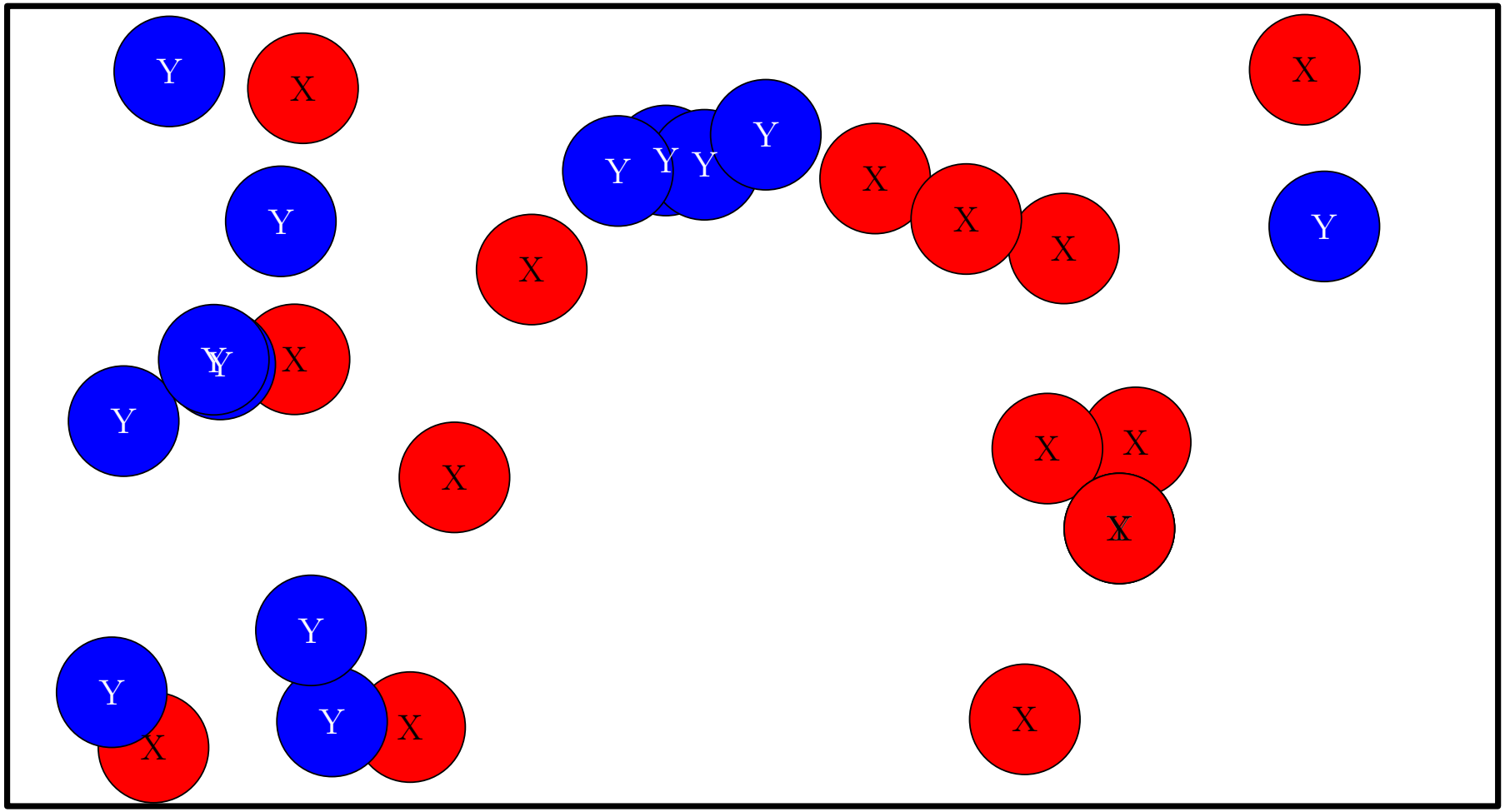

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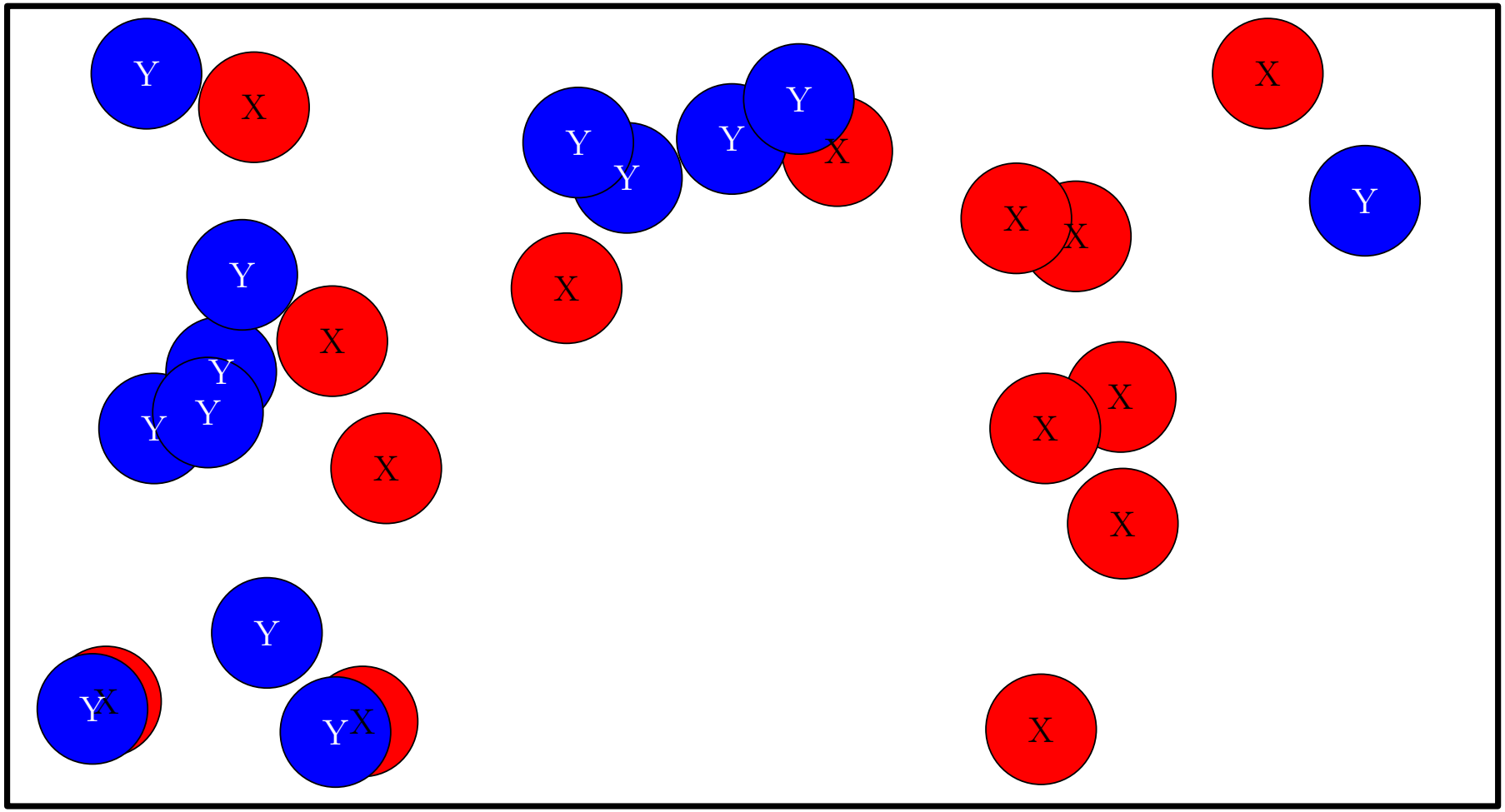

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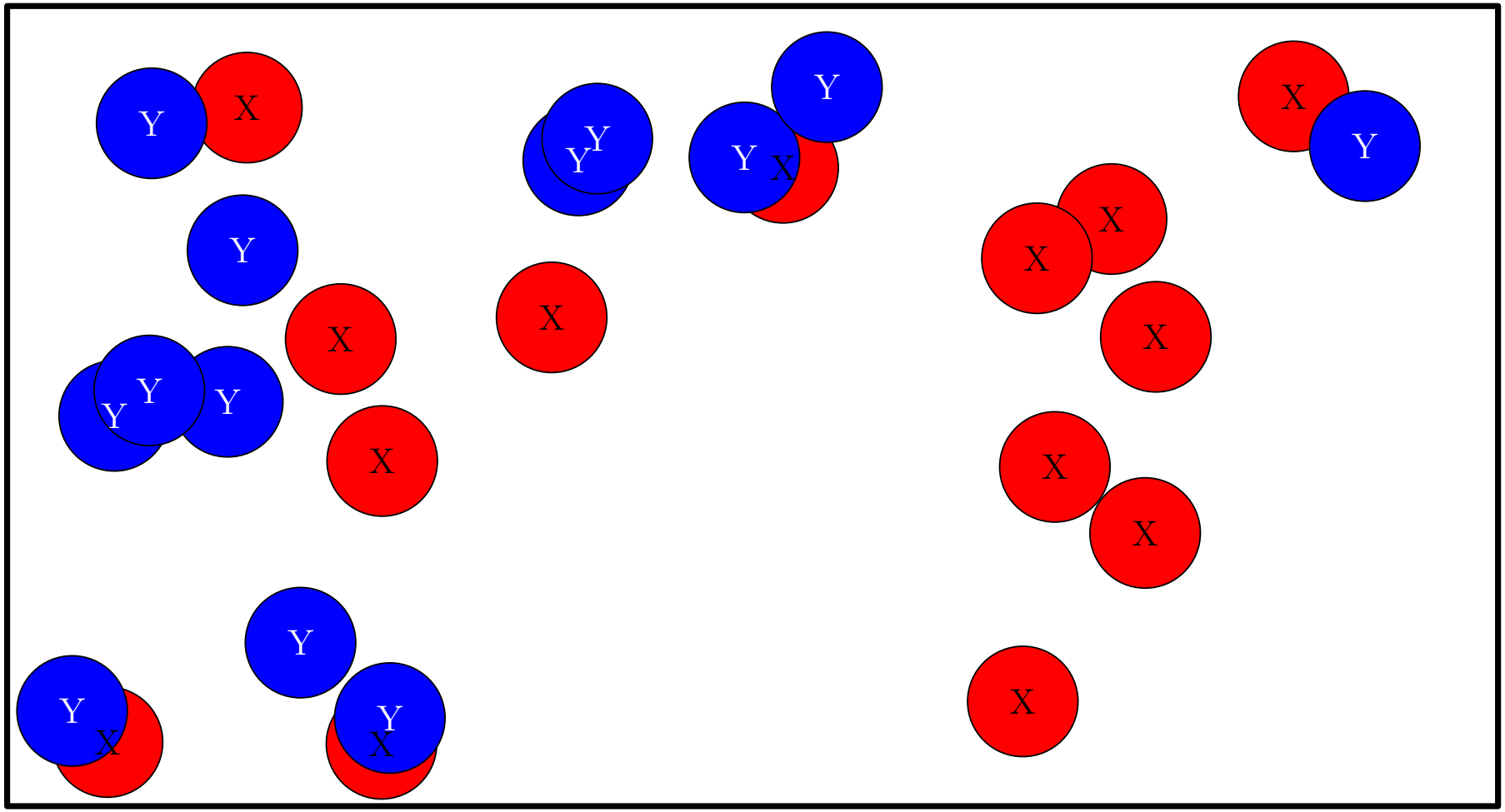

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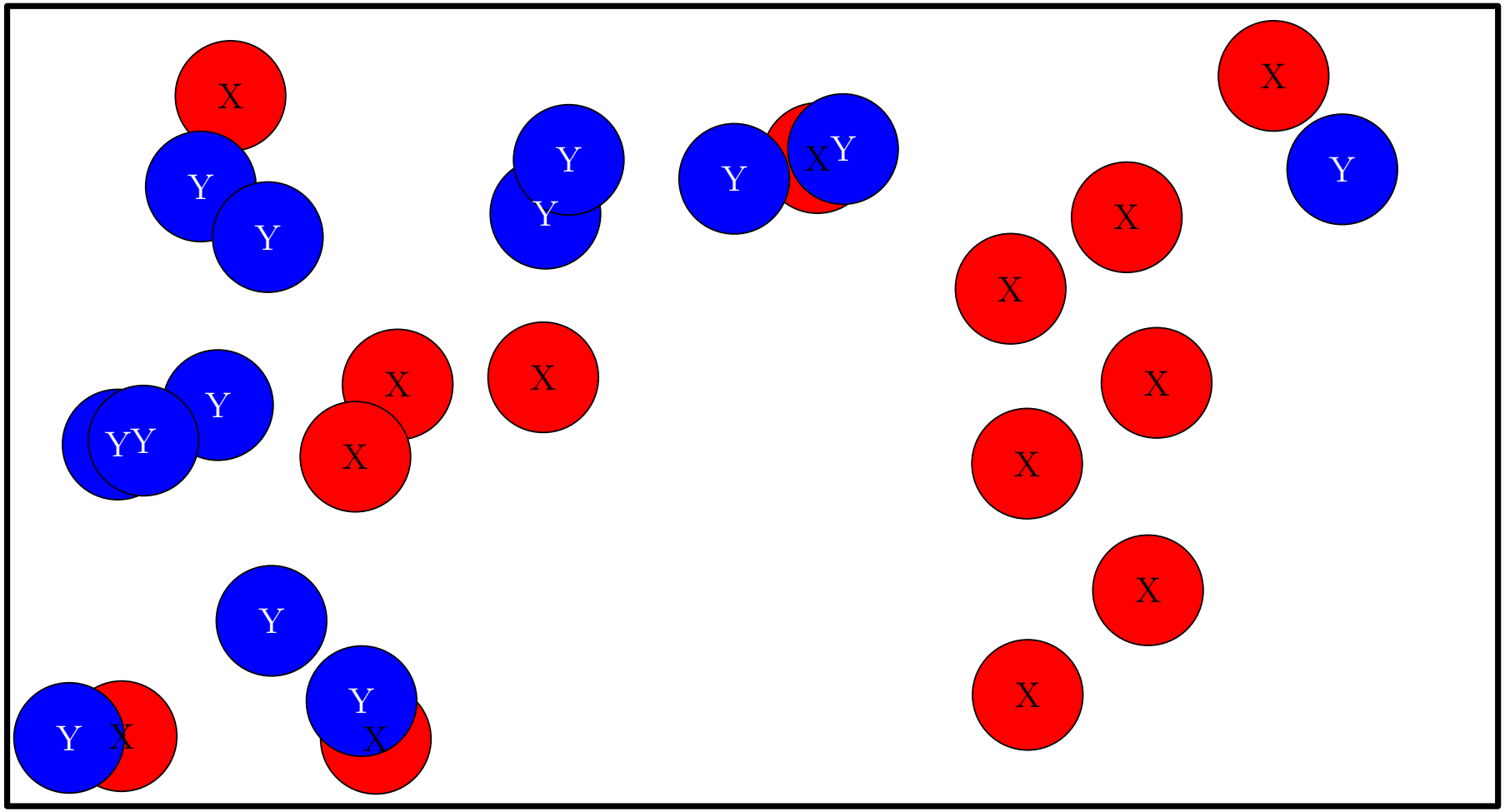

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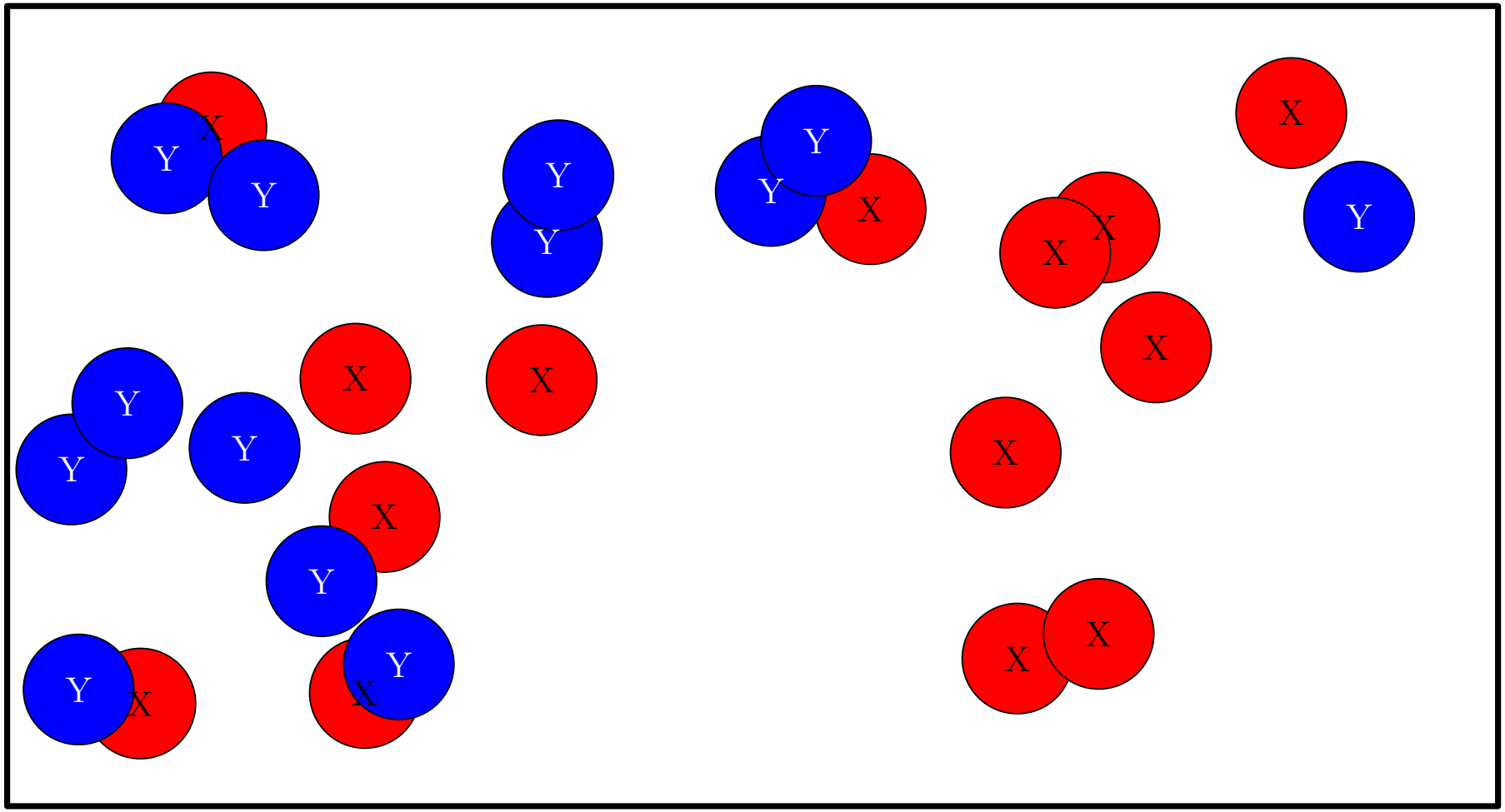

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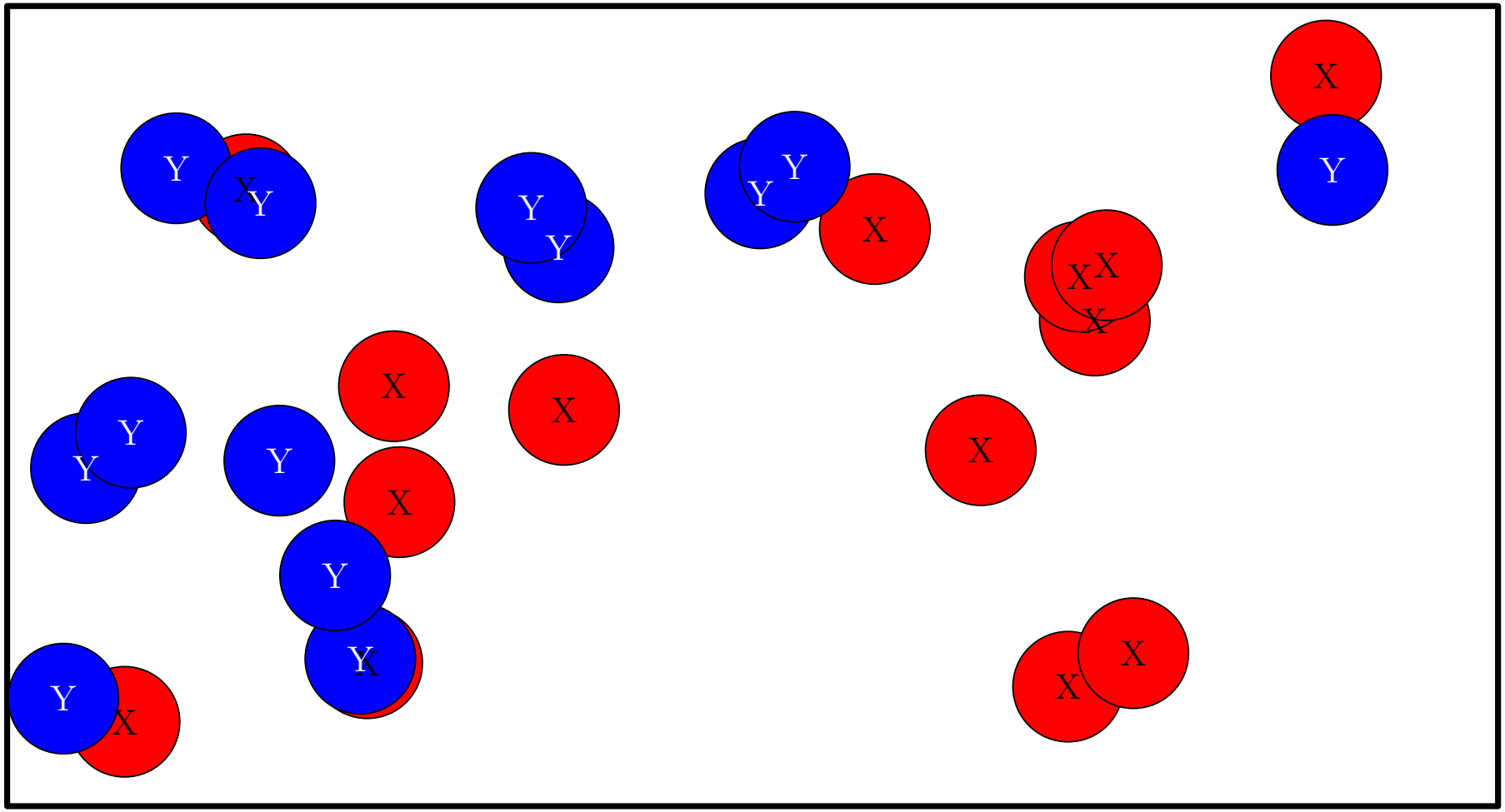

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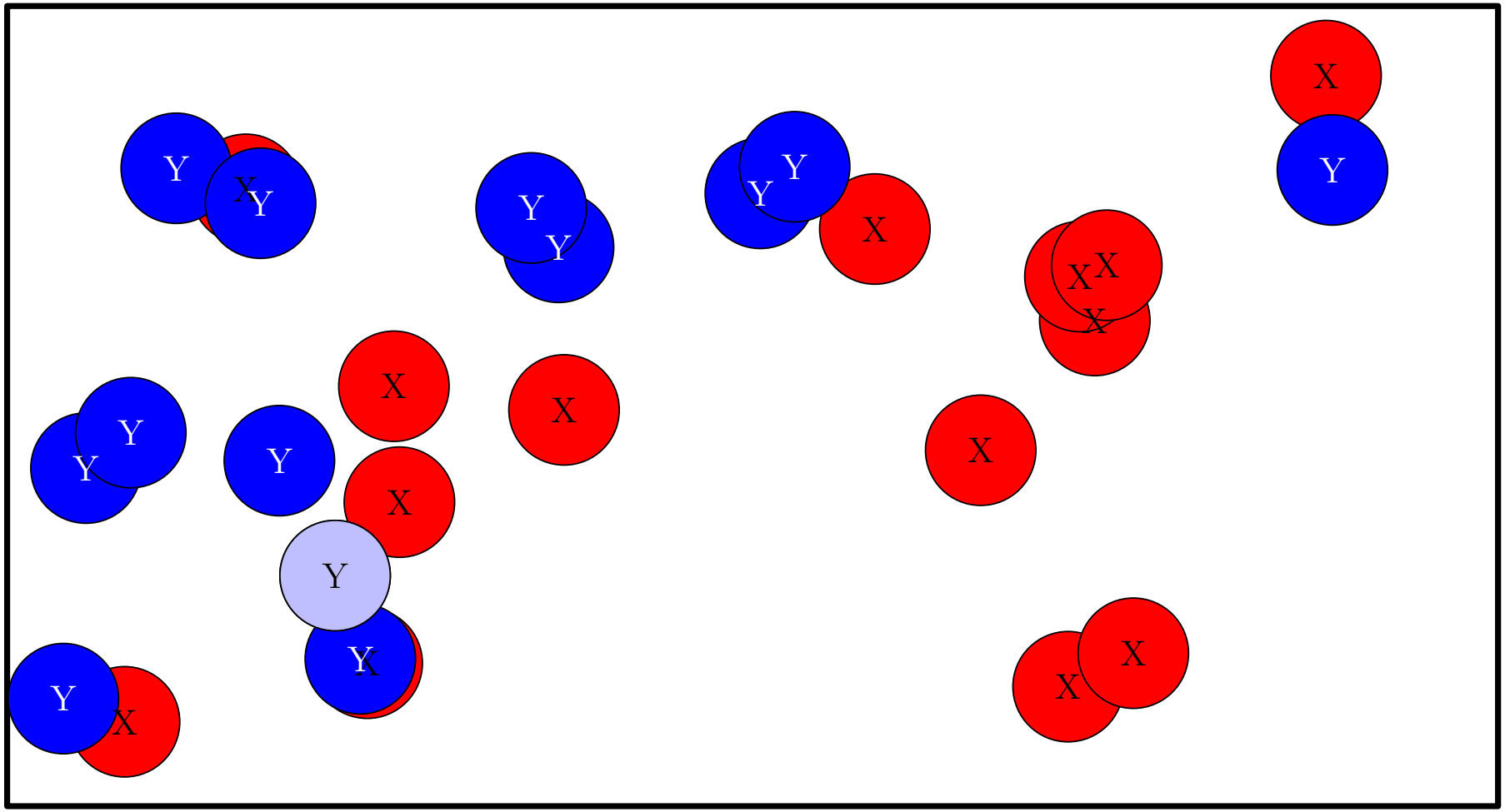

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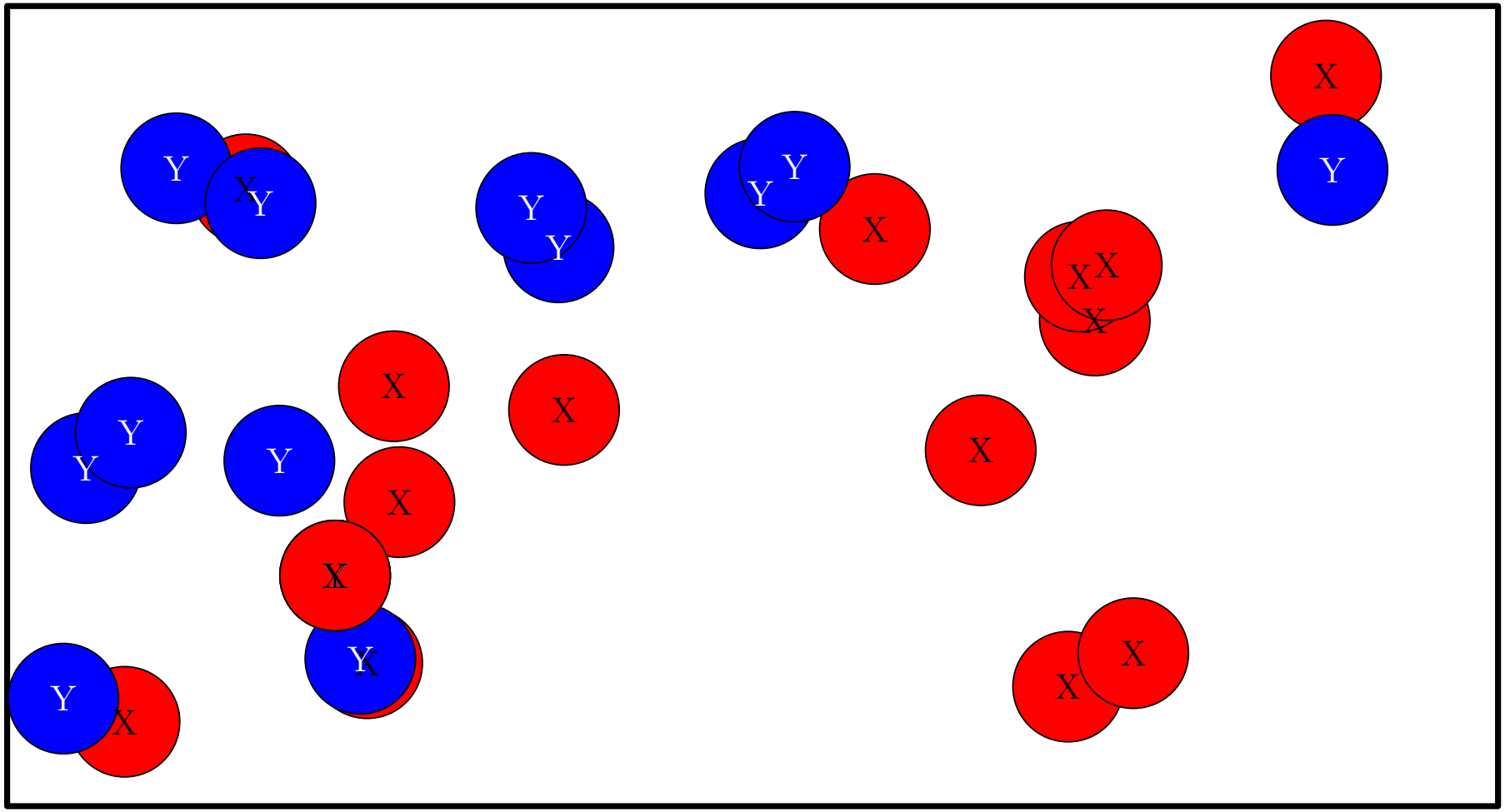

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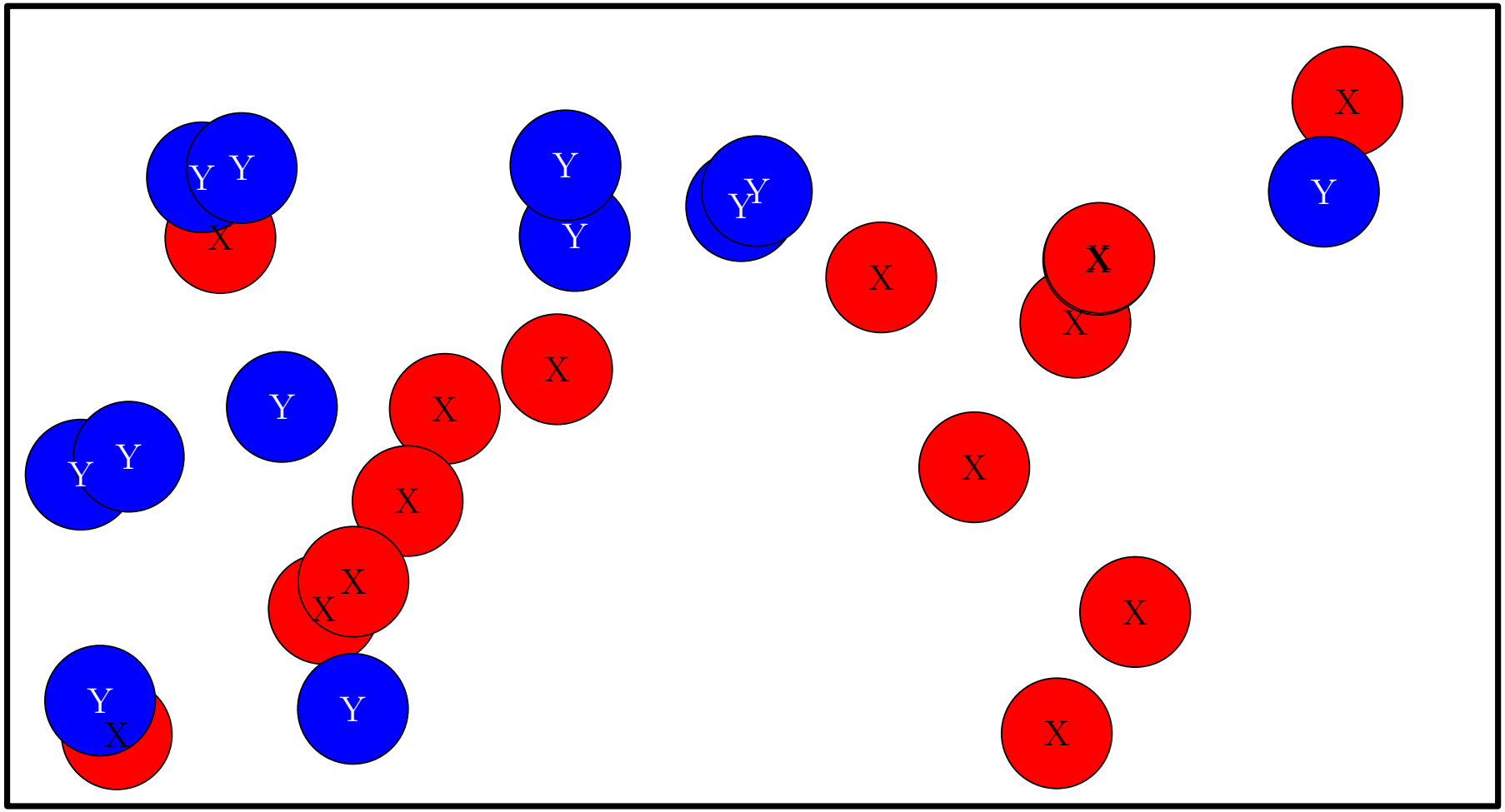

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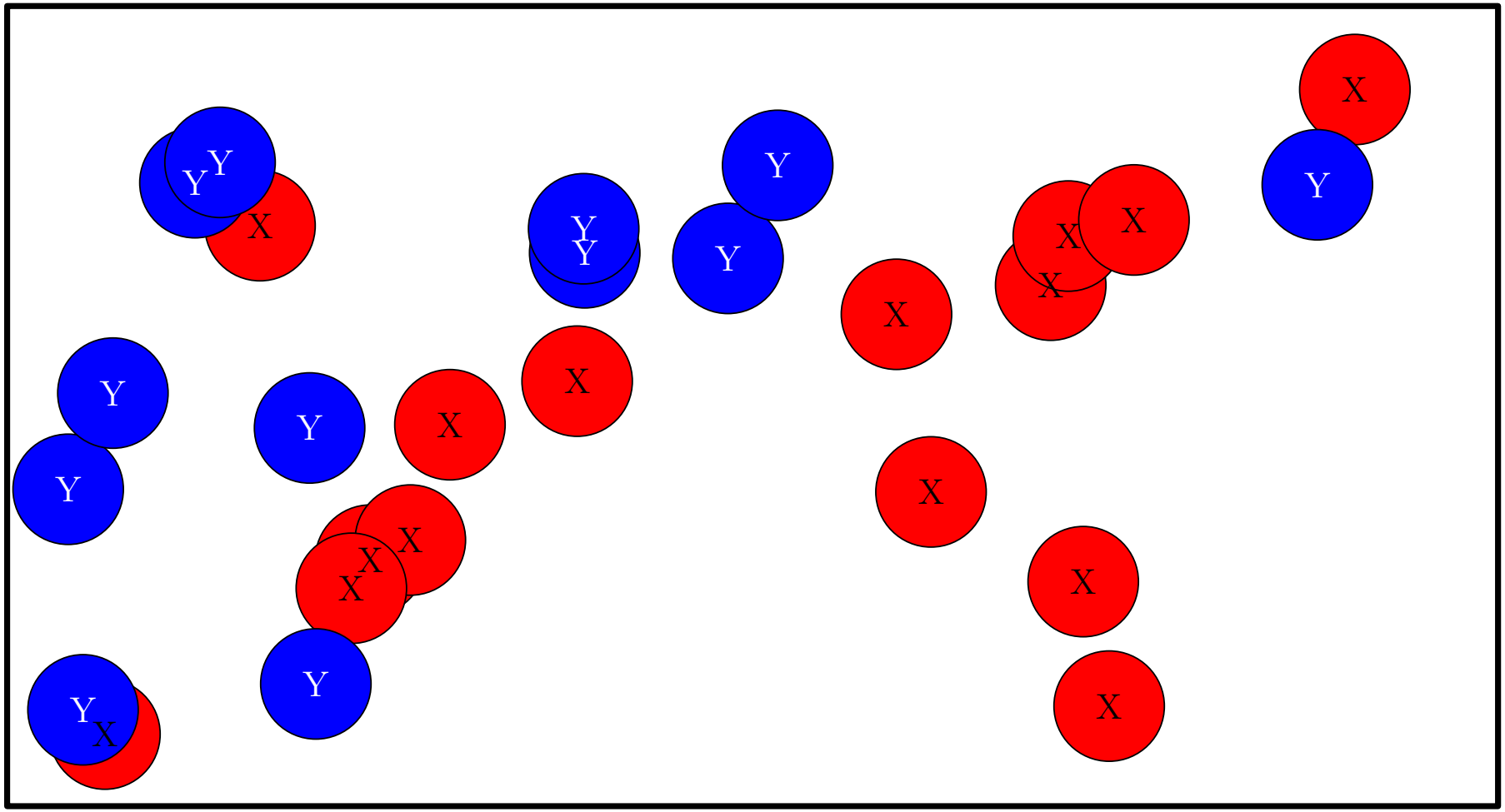

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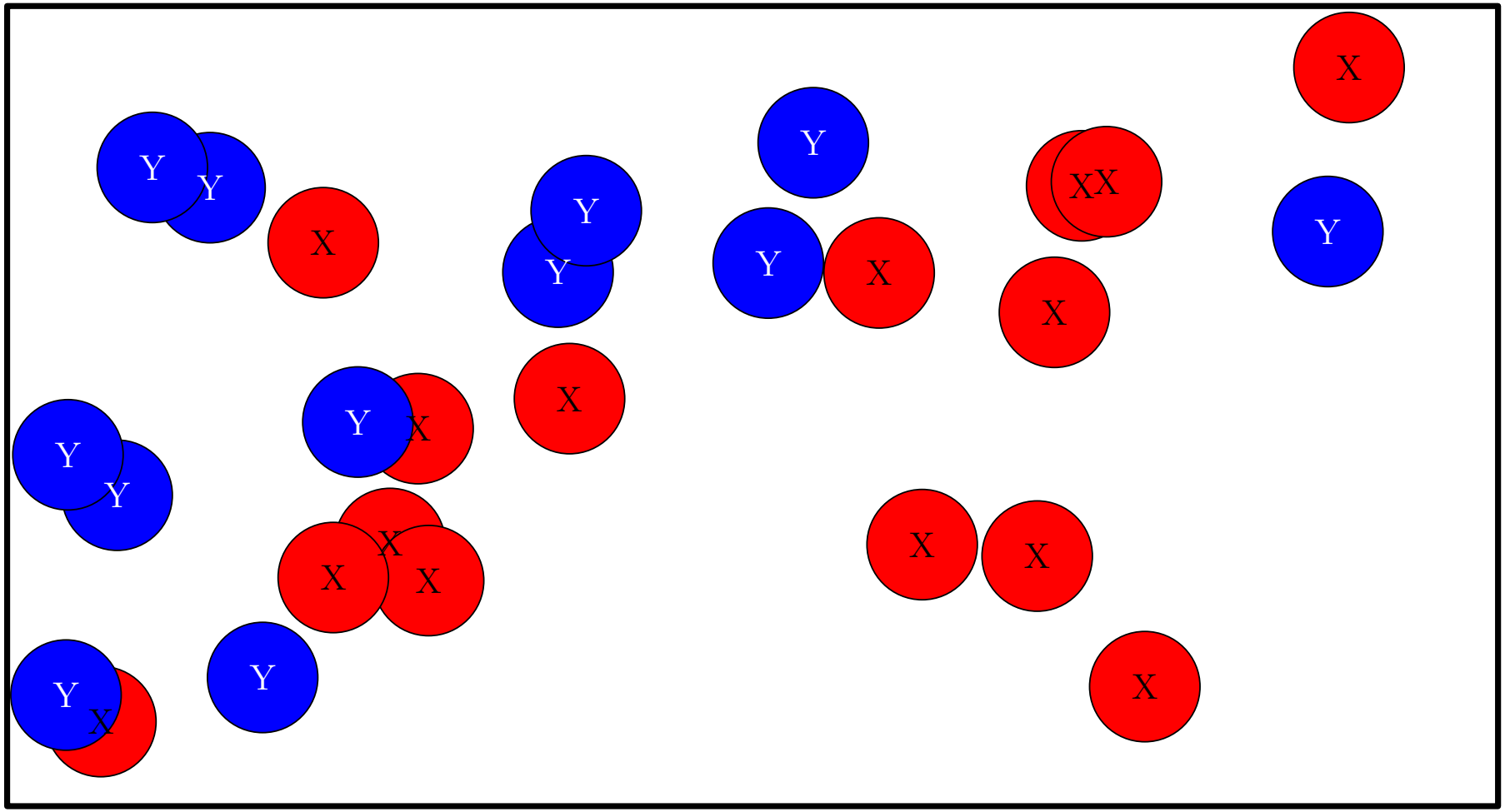

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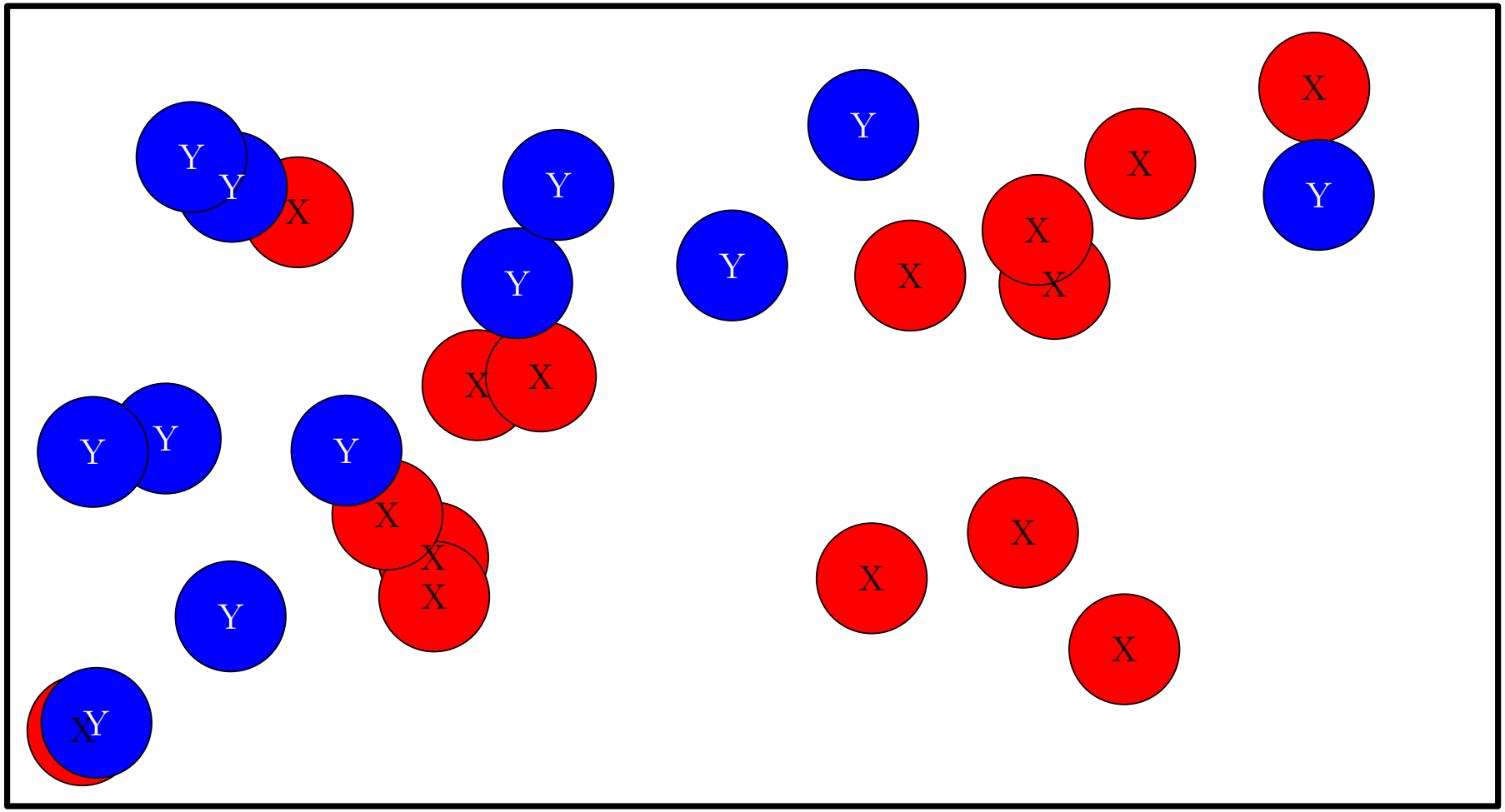

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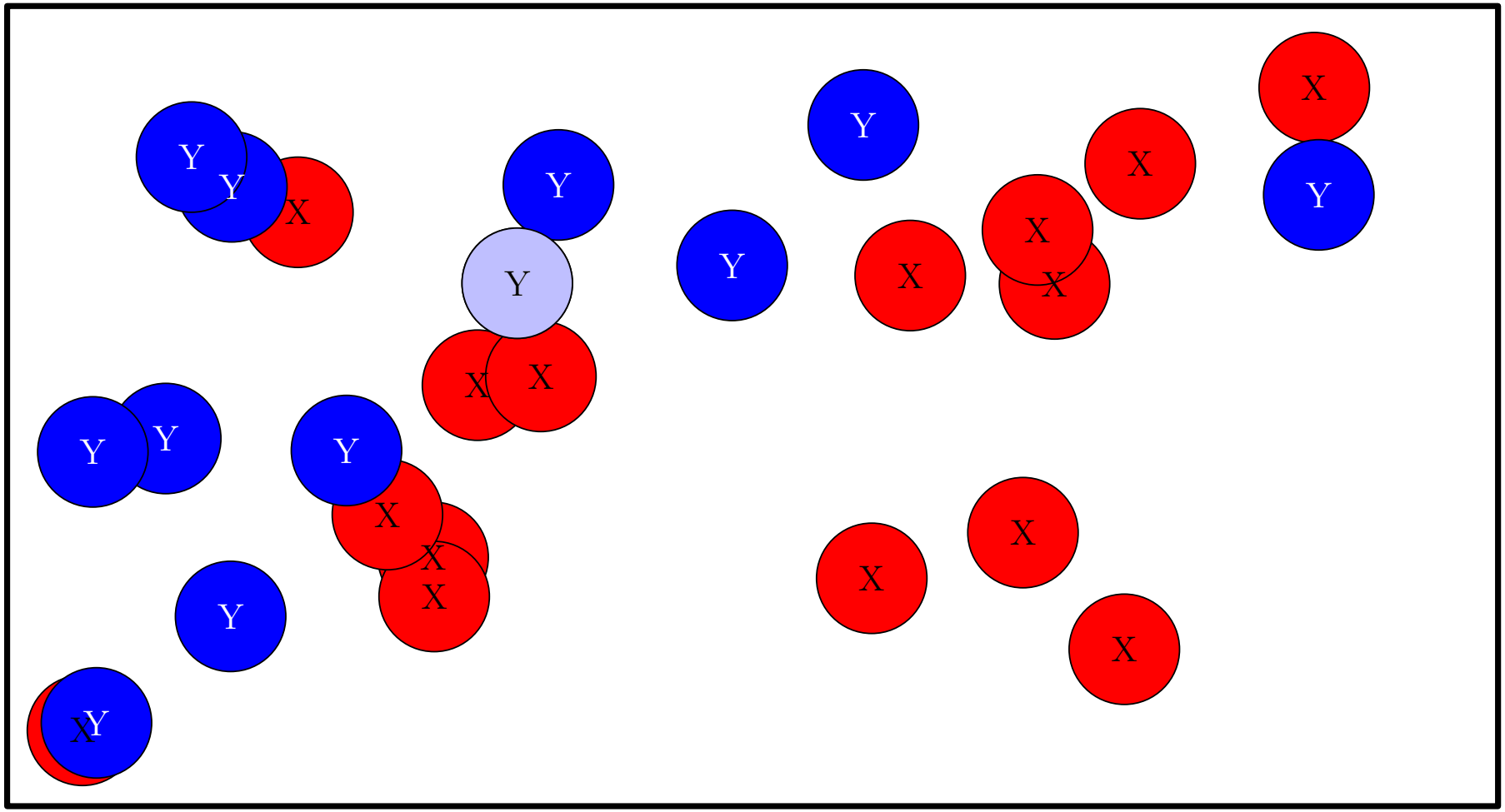

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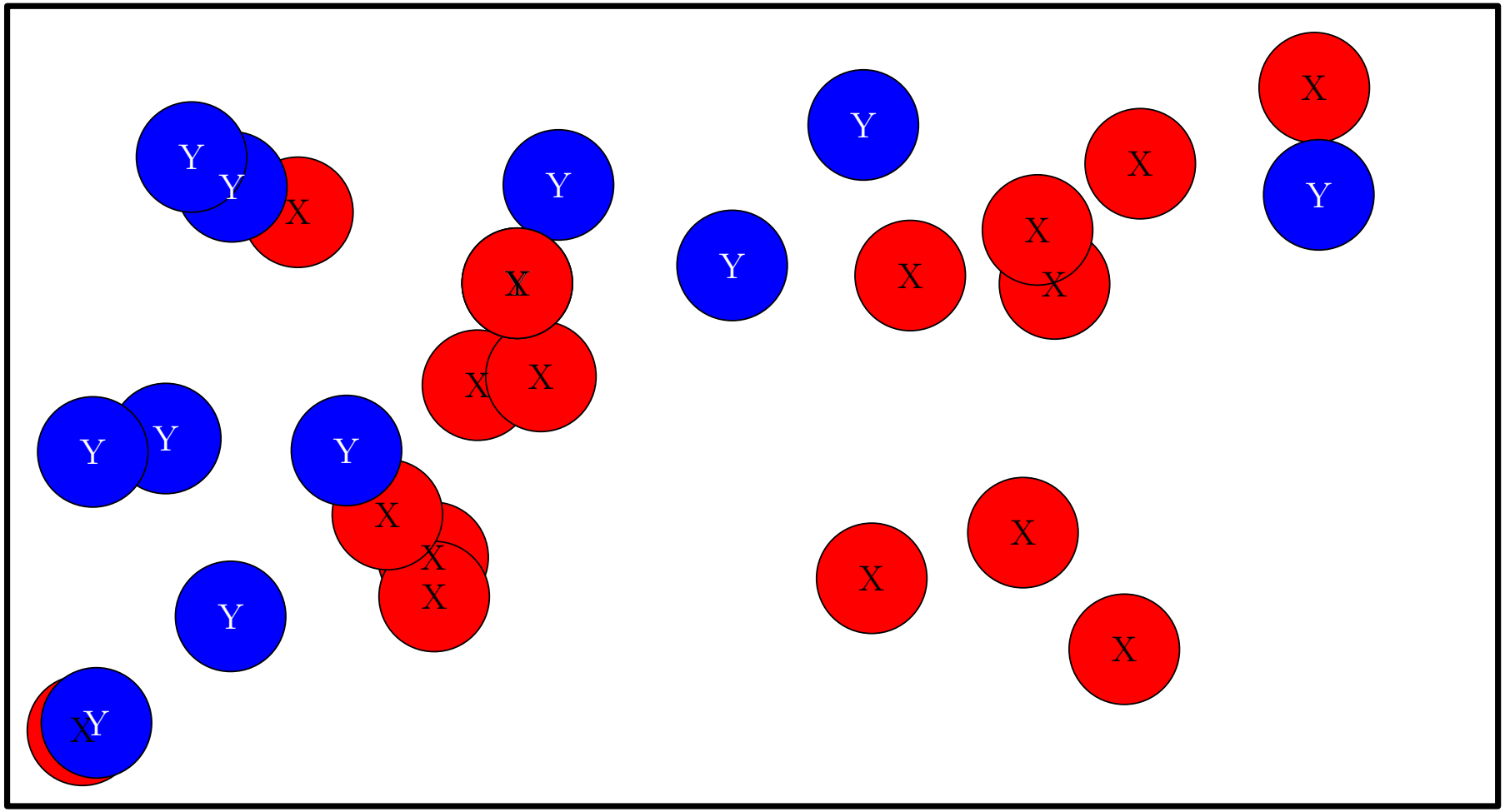

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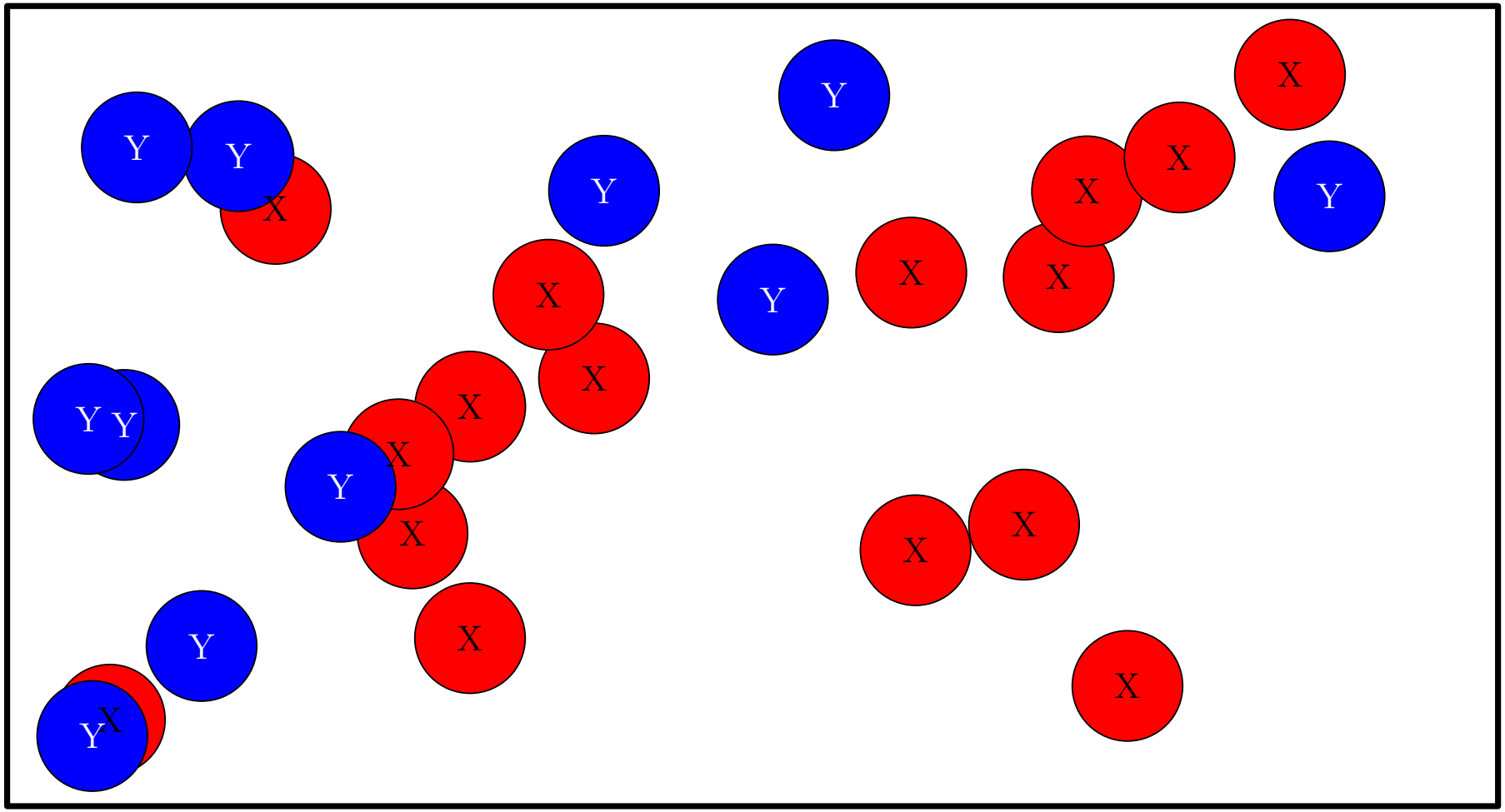

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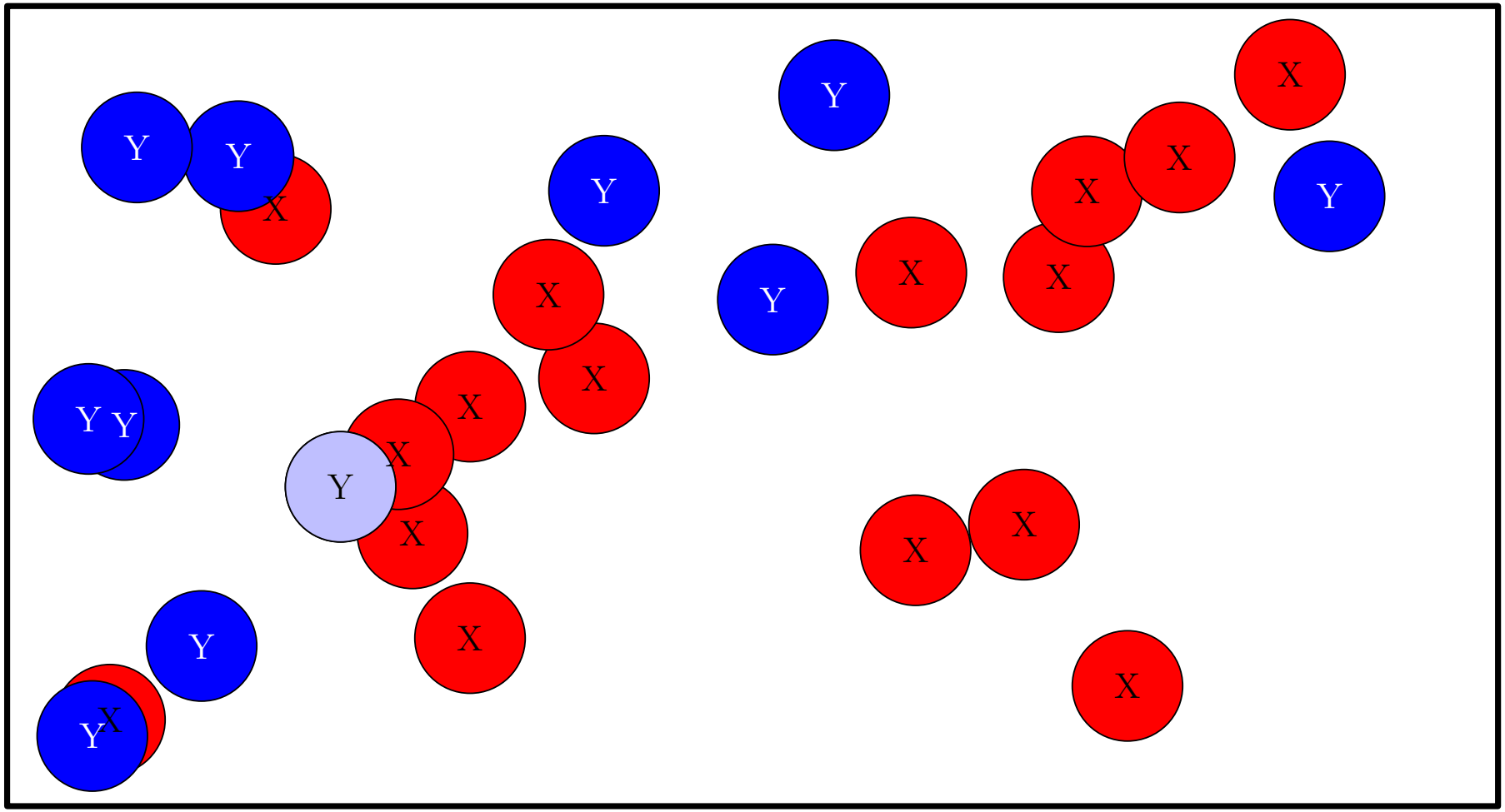

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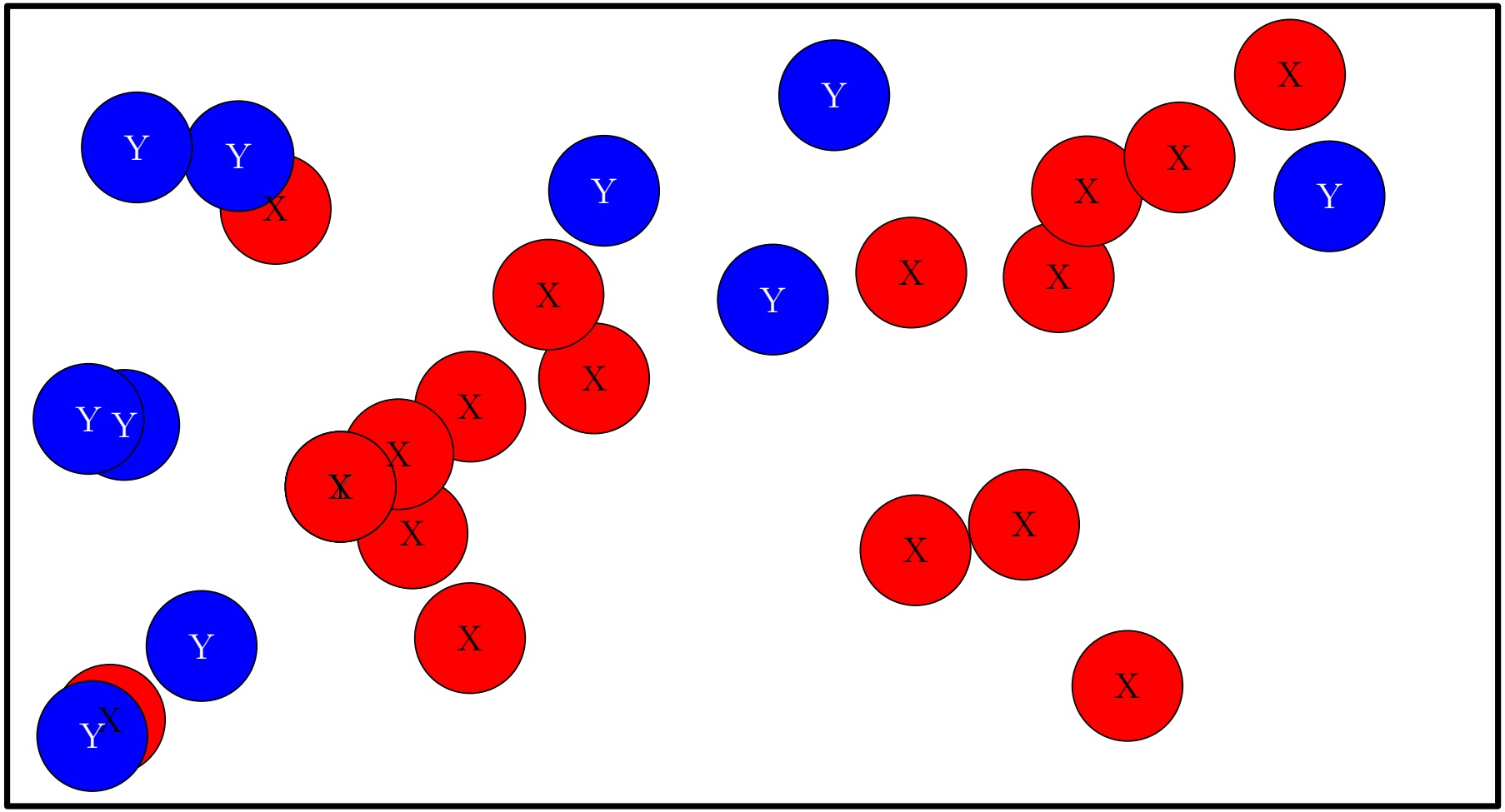

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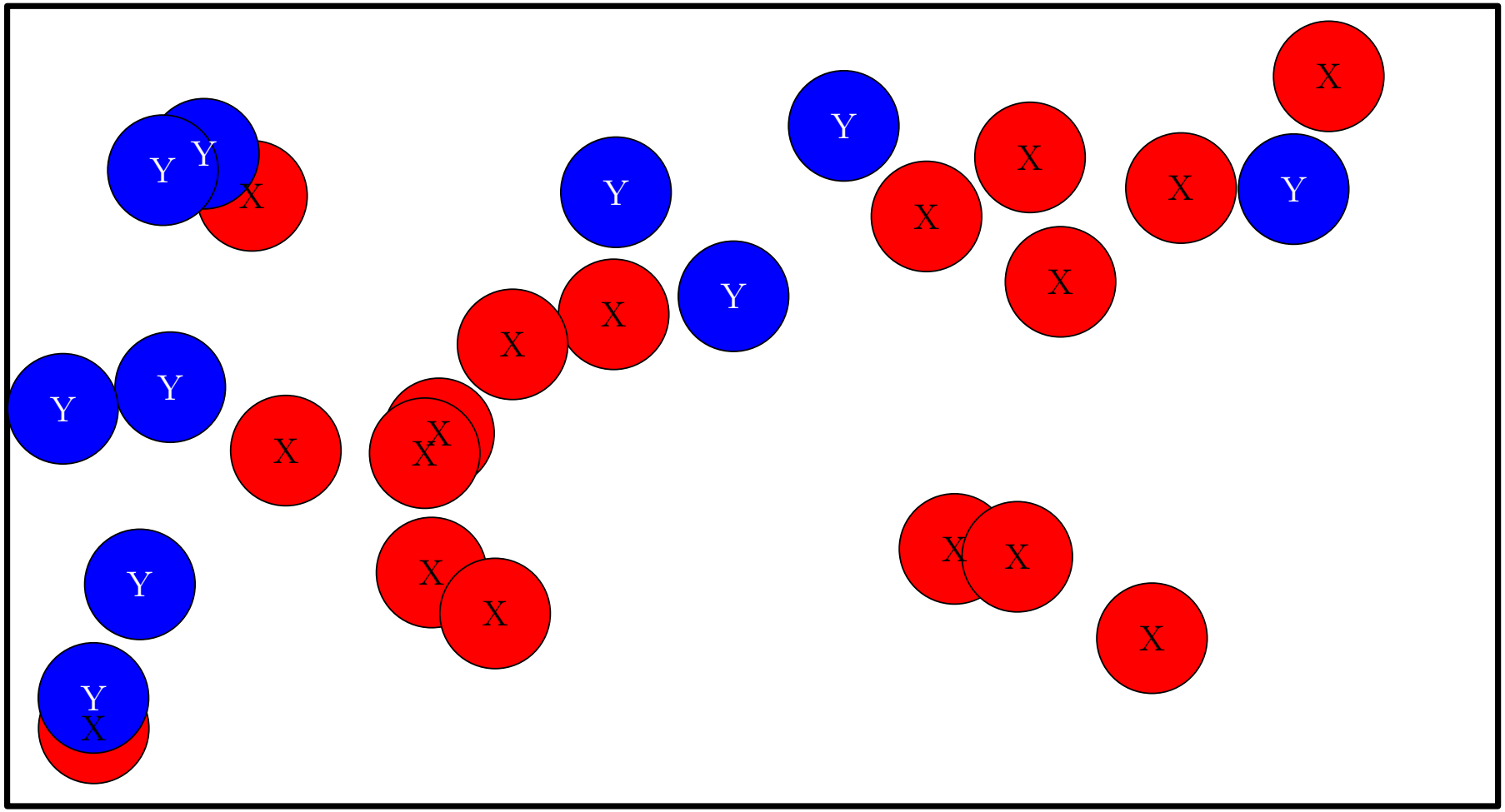

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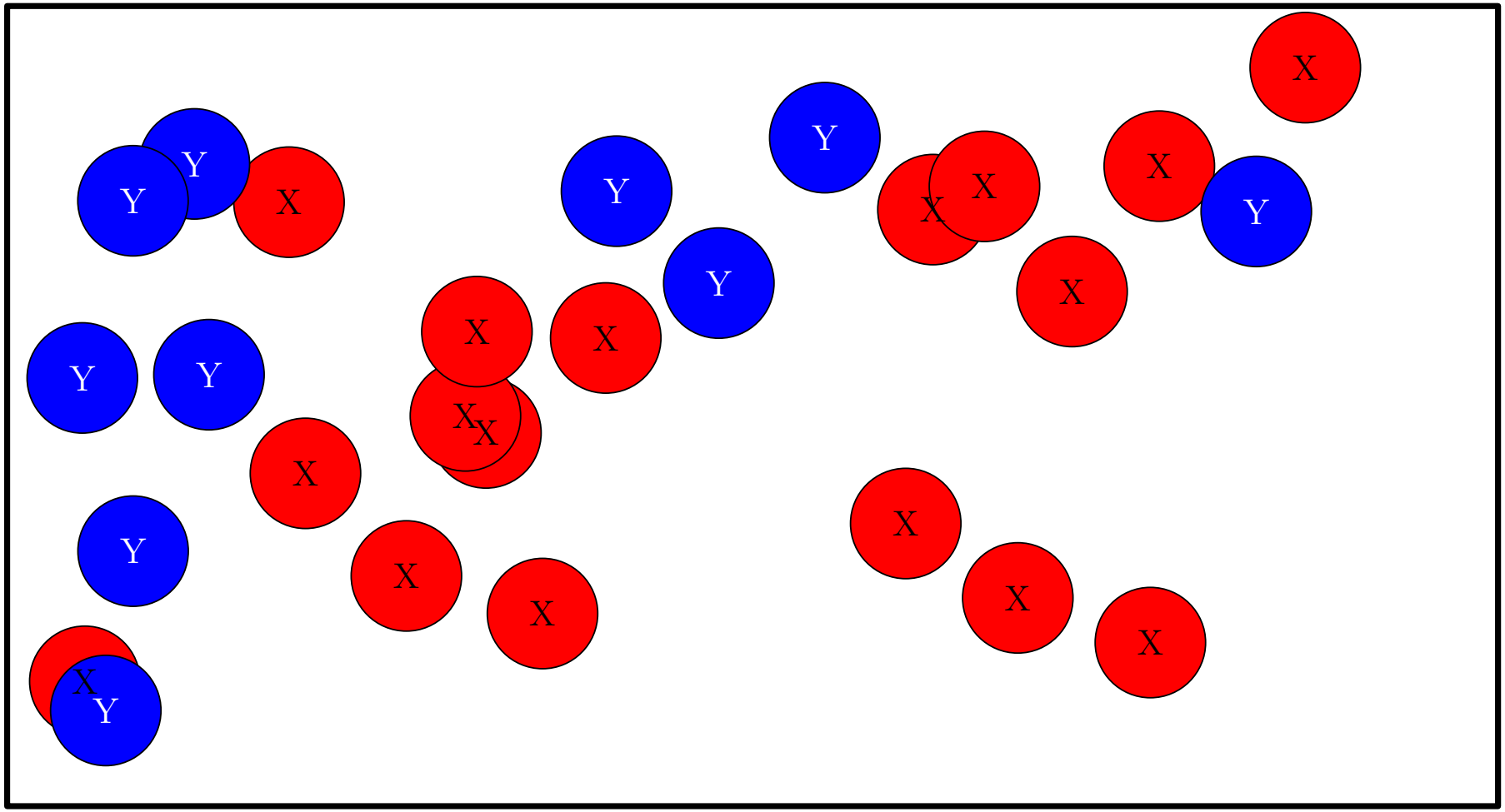
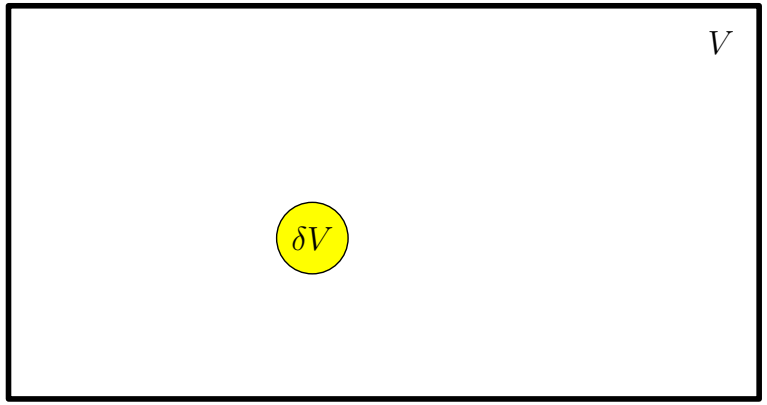


Probability of Reaction $2\,X + Y \xrightarrow{k} 3\,X$



$$\mathbb{P}\left(Y_i \in \delta V\right) = \frac{\delta V}{V}$$

- If we have n particles of type Y the probability of at least one being in δV is $n\,\delta V/V = [Y]\,\delta V$

- The probability of two particles X_1 and X_2 both being in δV is $\left(\frac{\delta V}{V}\right)^2$

- If we have m particles of type X the probability of at least two being in δV is $\frac{m\,(m-1)}{2}\left(\frac{\delta V}{V}\right)^2 \approx \frac{(\delta V)^2}{2}\,[X]^2$

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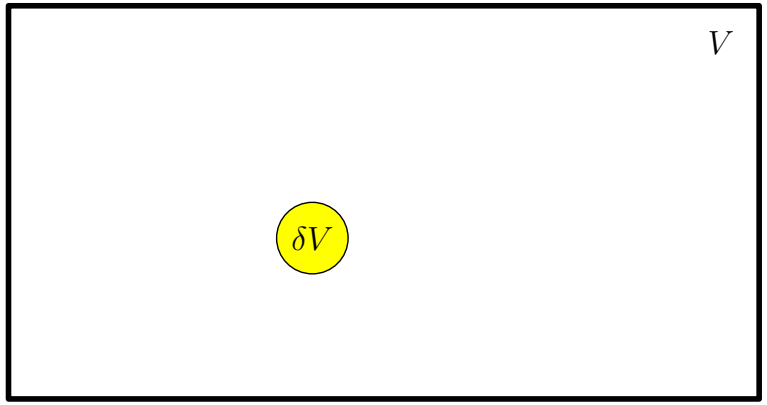

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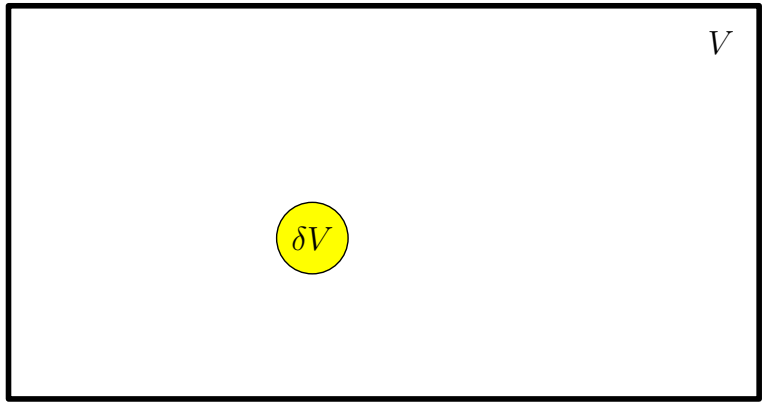

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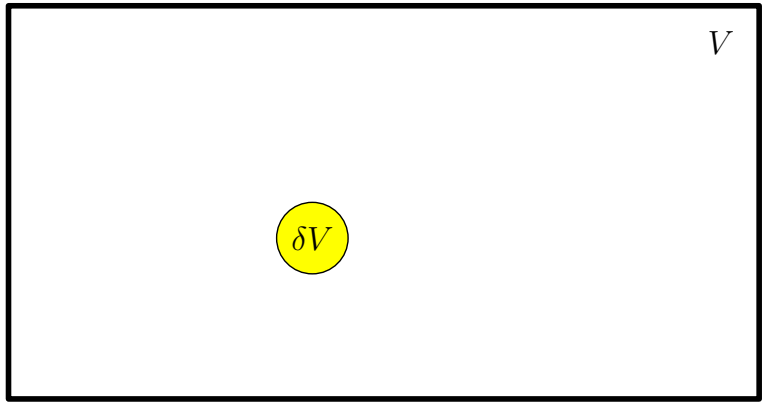

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Probability of Reaction $2\,X + Y \xrightarrow{k} 3X$

- Probability of one particle of type Y and two particles of type X being in volume δV is proportionals to

$$[Y]\,[X]^2(\delta V)^3$$

- The number of δV's in V is $N = V/\delta V$

- The probability of a reaction in V is proportional to

$$N\,[Y]\,[X]^2(\delta V)^3 = V\,[Y]\,[X]^2\,(\delta V)^2$$

- The expected number of reactions per unit volume is proportional to

$$[Y]\,[X]^2$$

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- The number of δV's in V is $N = V/\delta V$

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Reaction Rate

- A chemist would measure the rate of change of concentration $[X]$ for the reaction and divide by $[Y]\,[X]^2$ to come up with a reaction rate k

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- To work in numbers of molecules, X, we use $[X] = X/\Omega$

- $\Omega = V \times A$, where V is volume and A Avogadro's number

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Outline

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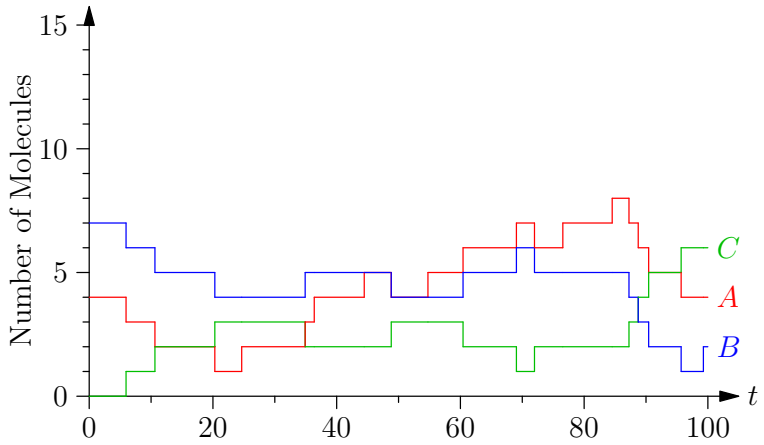
2. **Gillespie Algorithm**



Discrete Dynamic Models

- We would like to simulate the set of chemical reactions

$$\varnothing \xrightarrow{k_1} A \qquad \varnothing \xrightarrow{k_2} B \qquad A + B \xrightarrow{k_3} C \qquad C \xrightarrow{k_4} A + B$$

- At each time we want to keep note of the number of molecules $A(t)$, $B(t)$, $C(t)$
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- Events that happen at particular points in space or time are known as **point processes**

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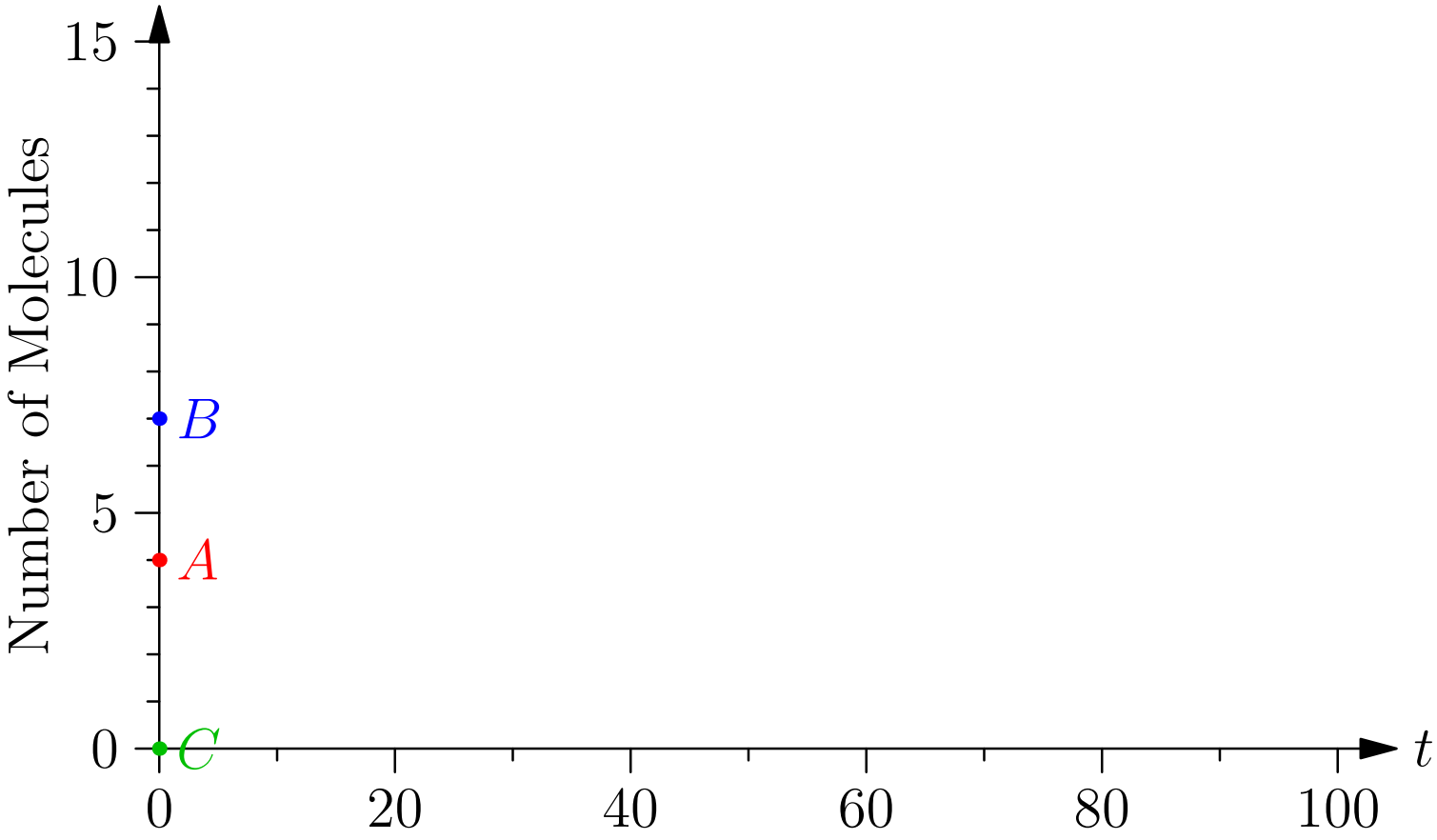

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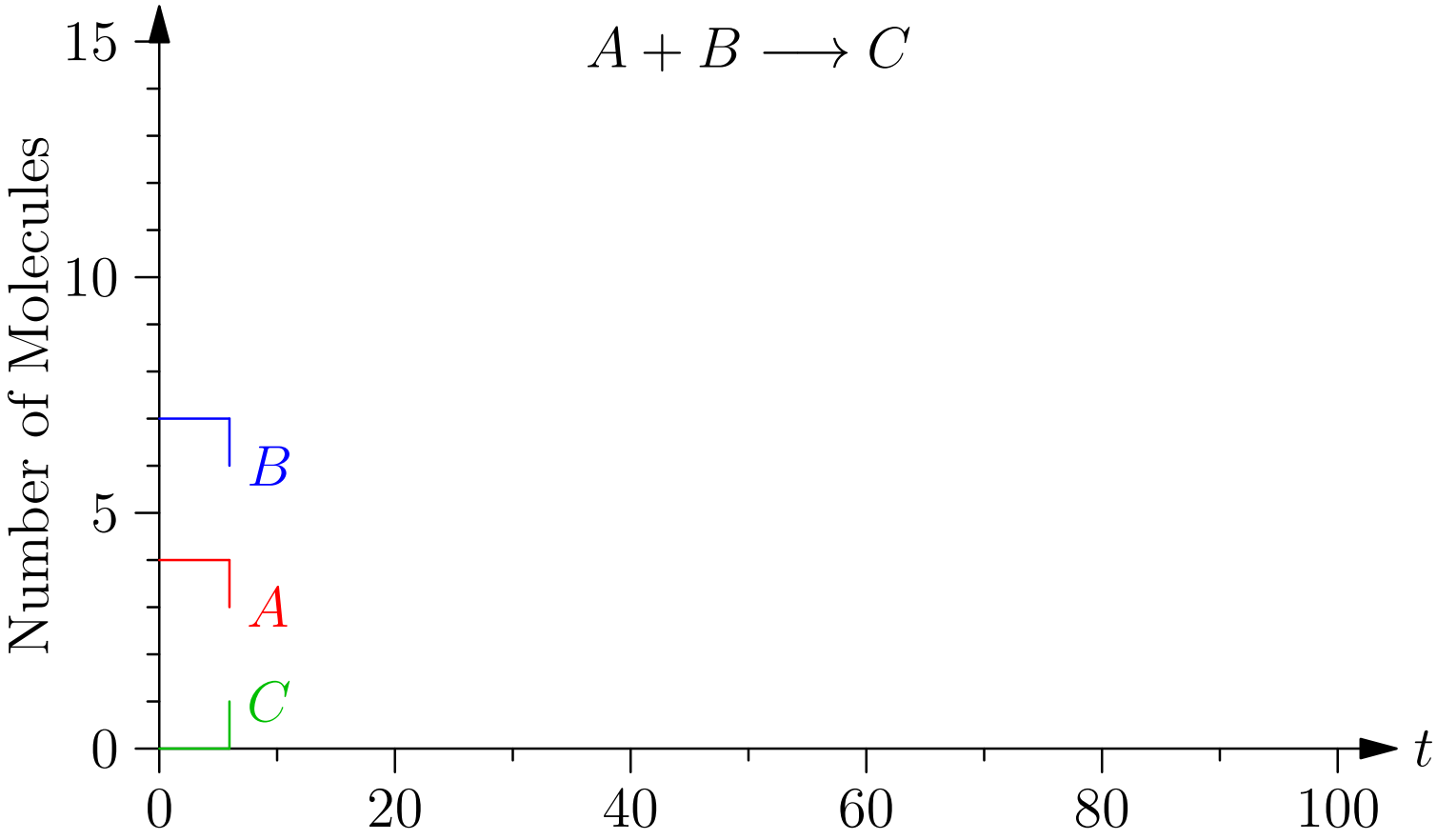

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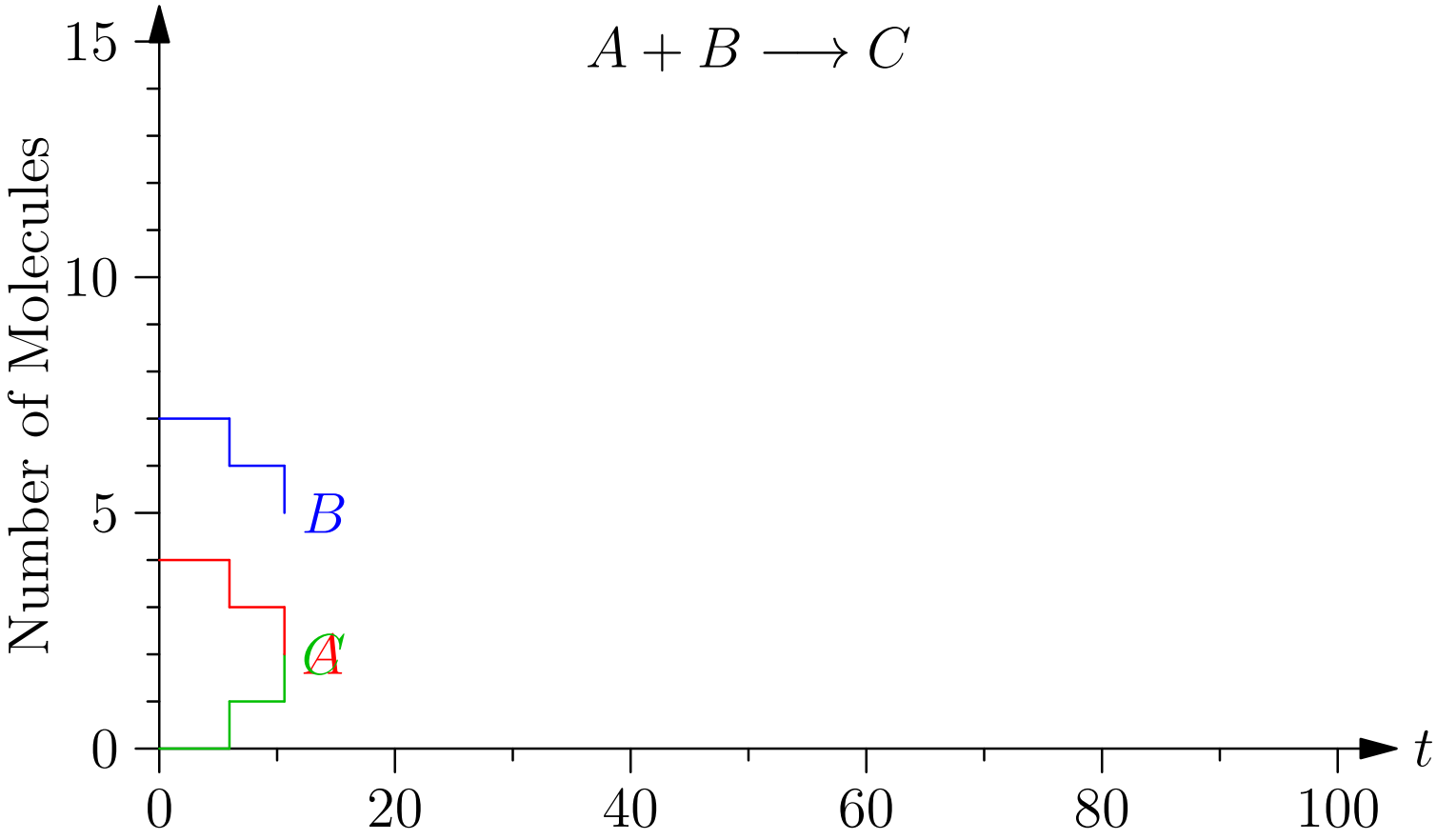

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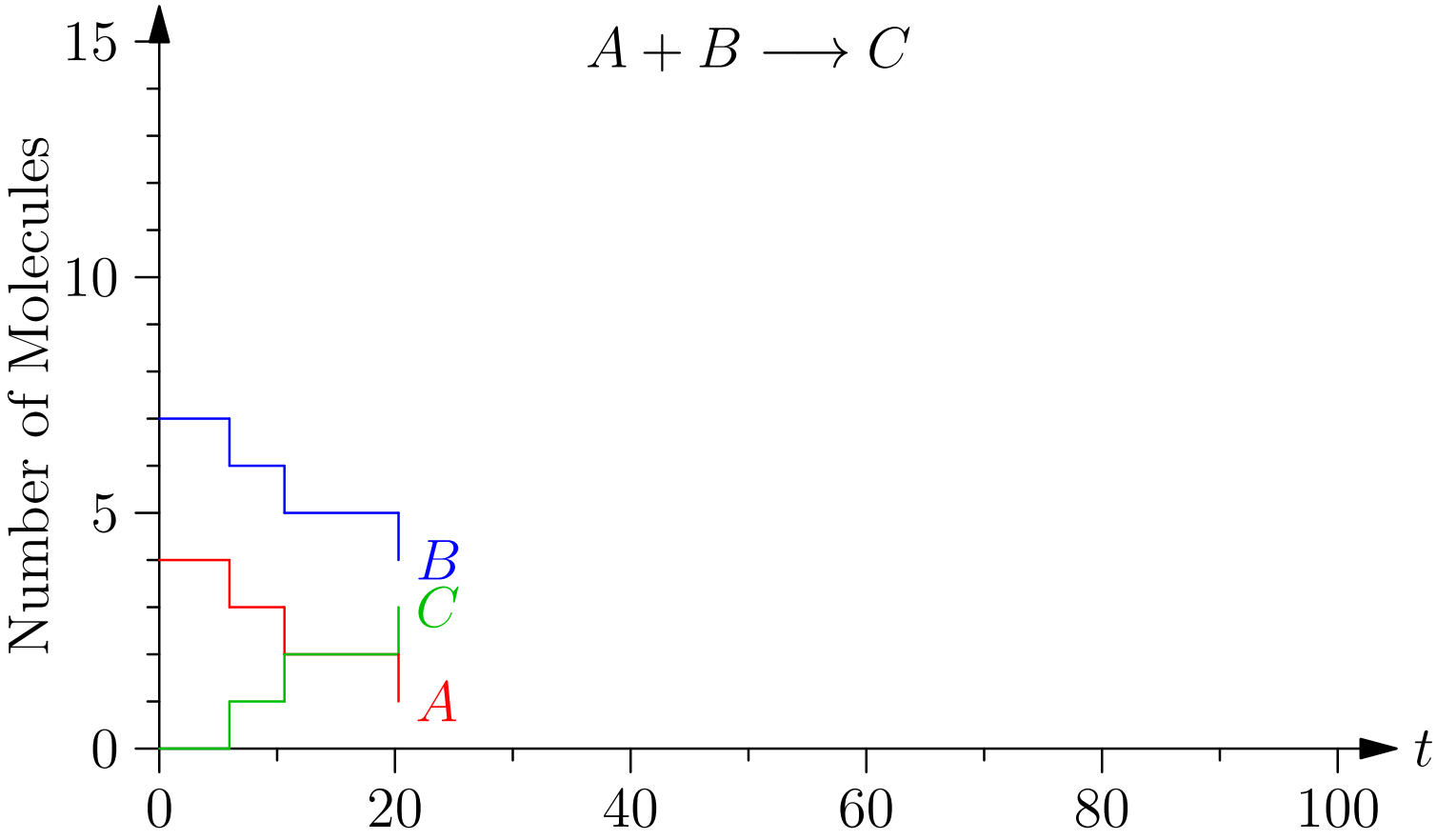

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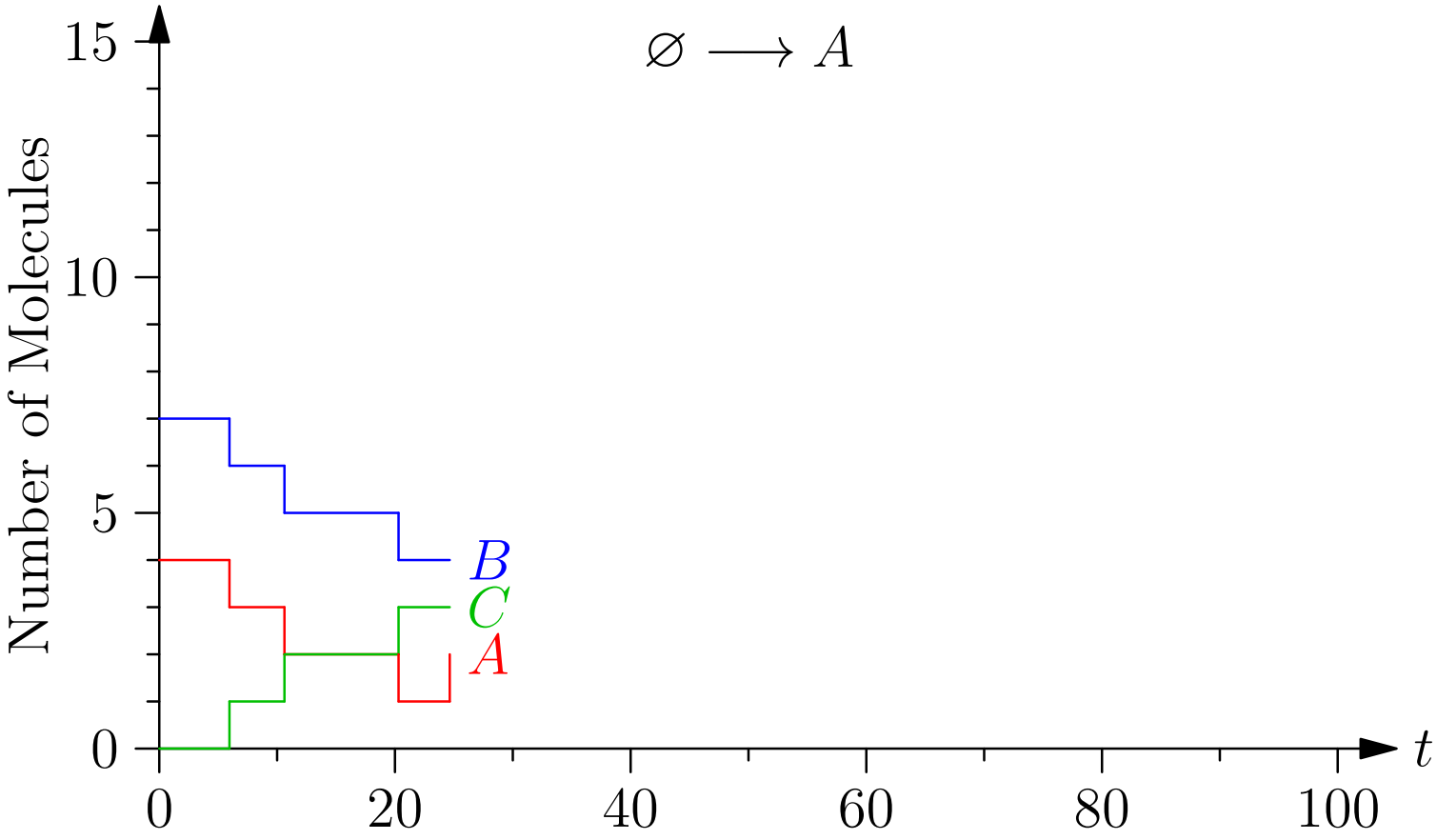

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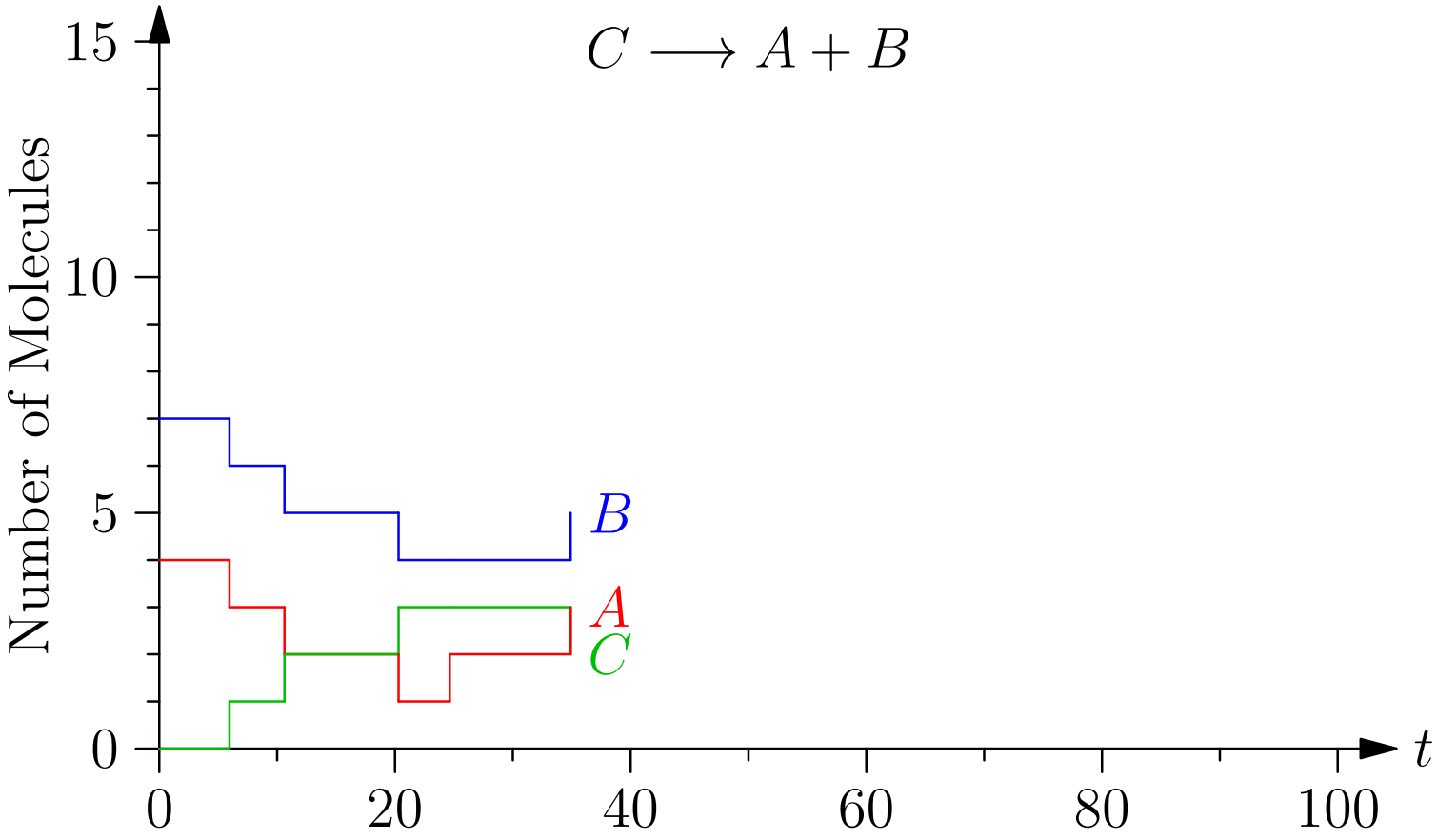

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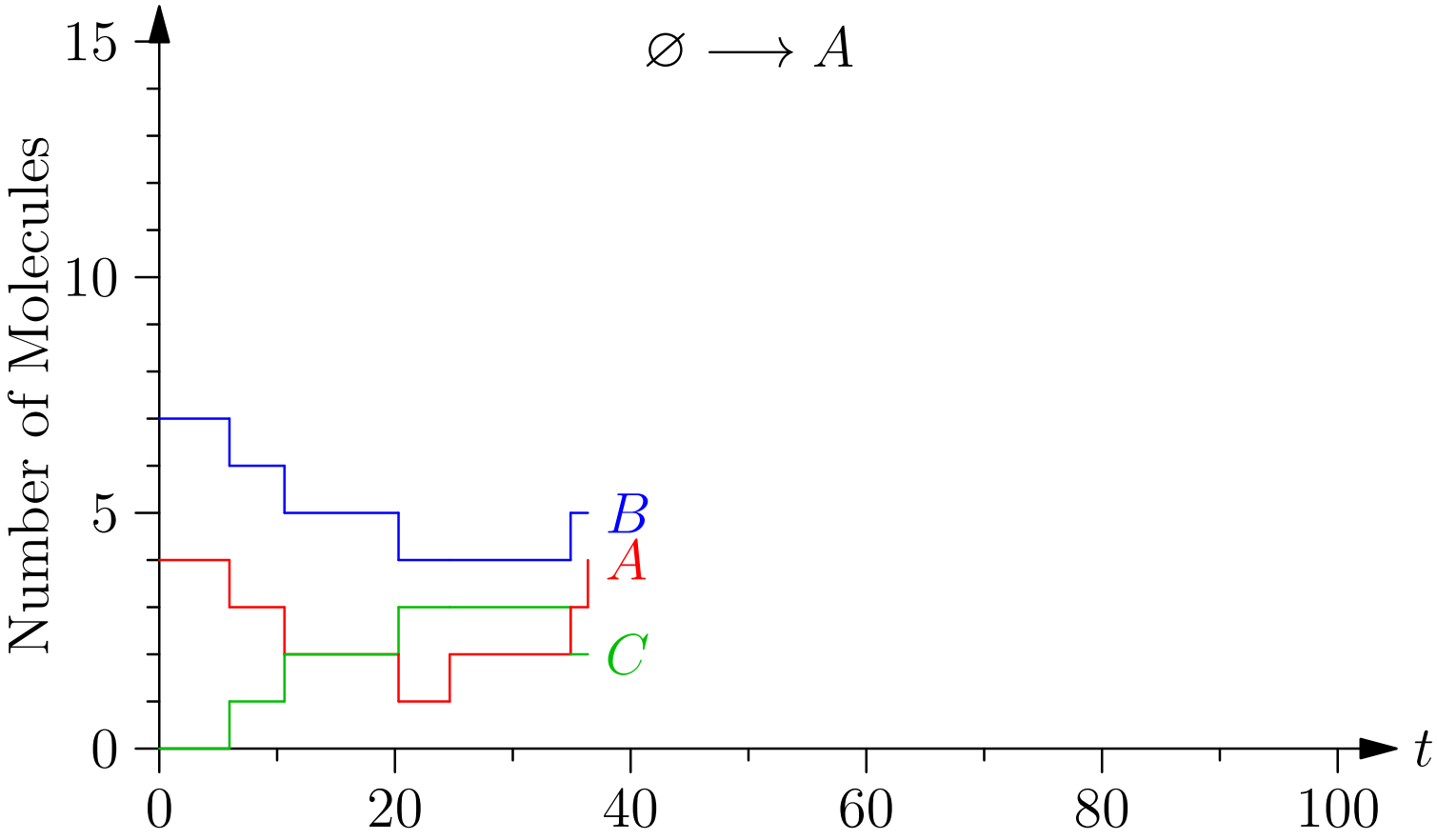

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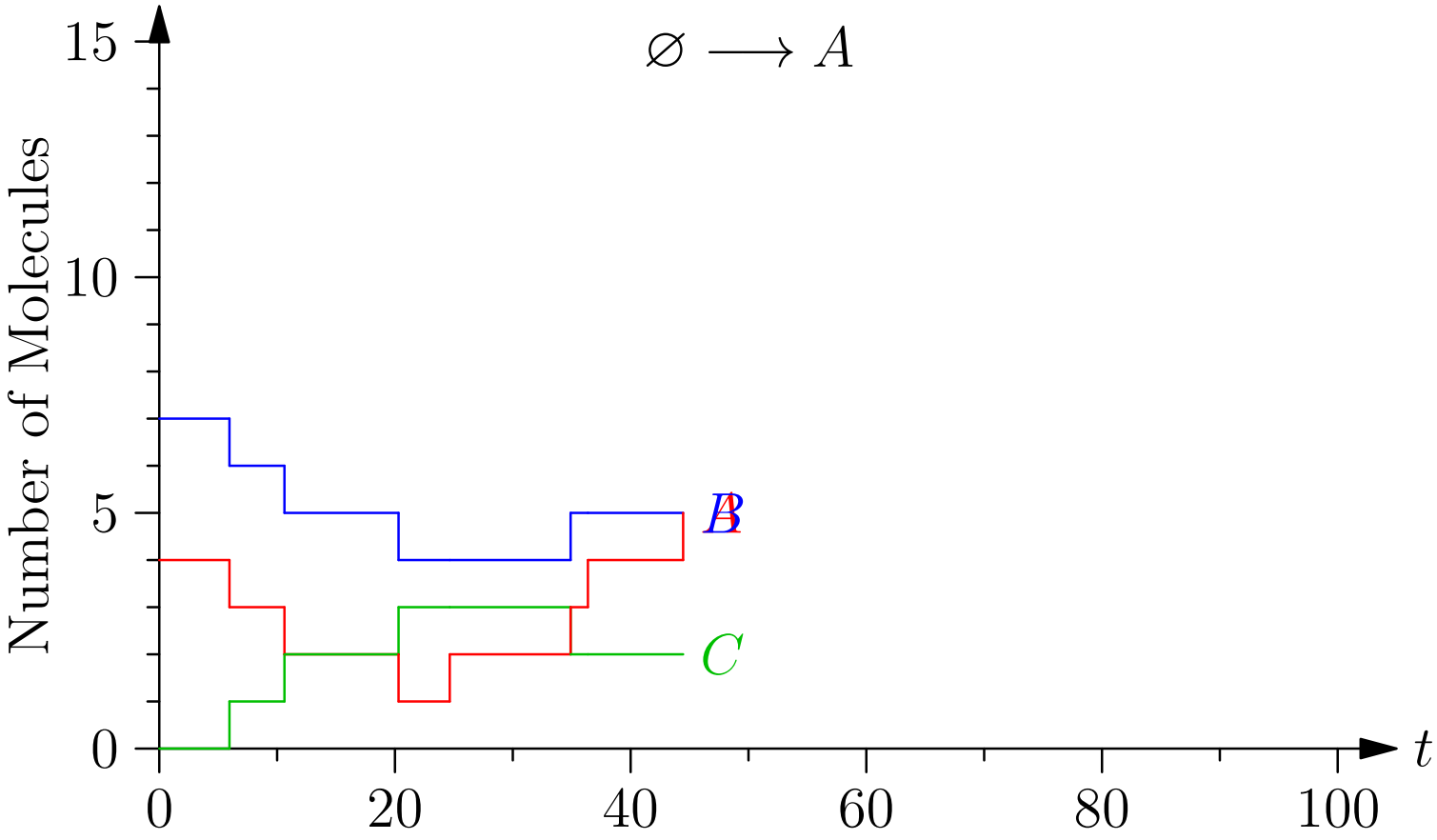

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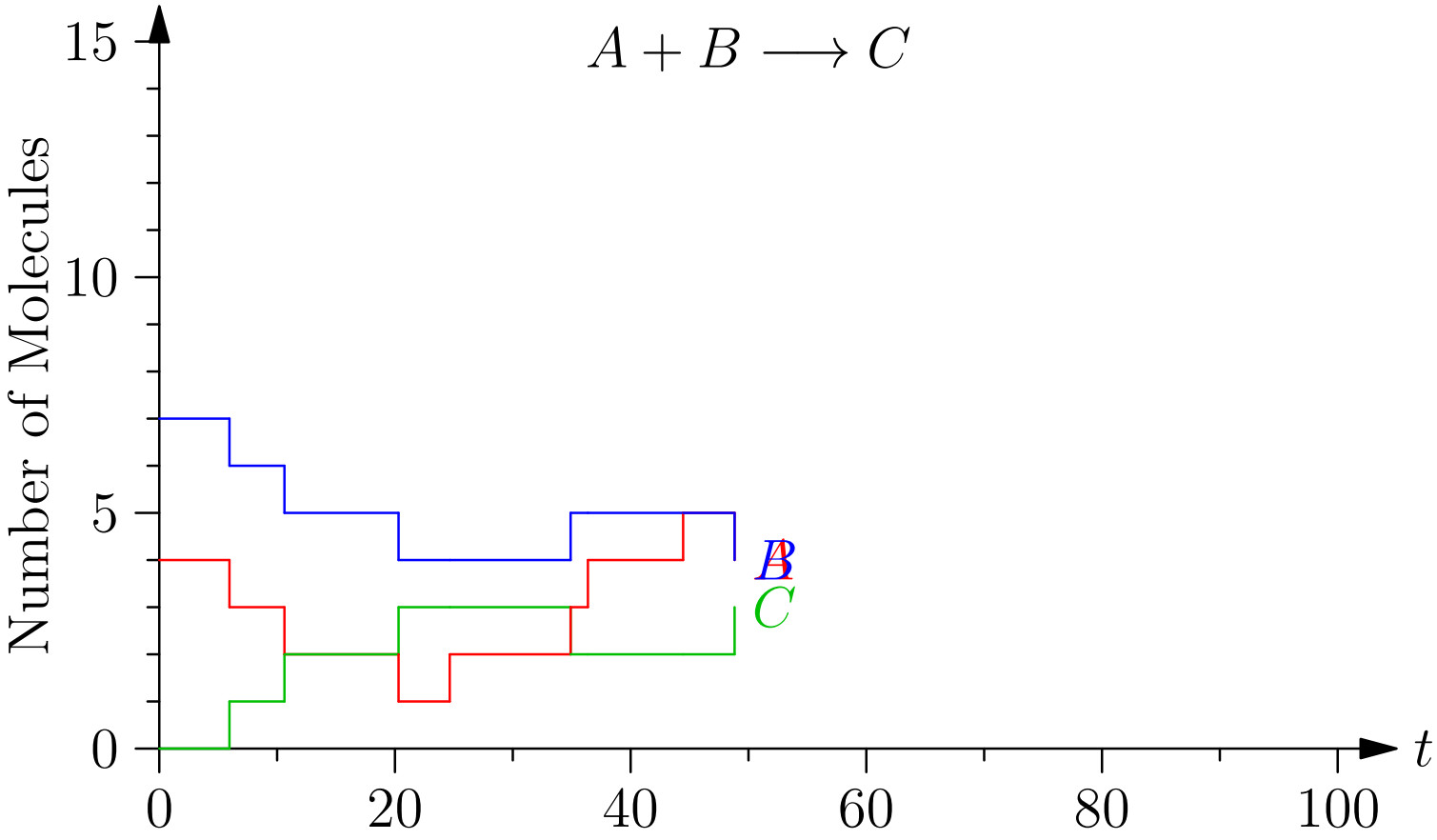

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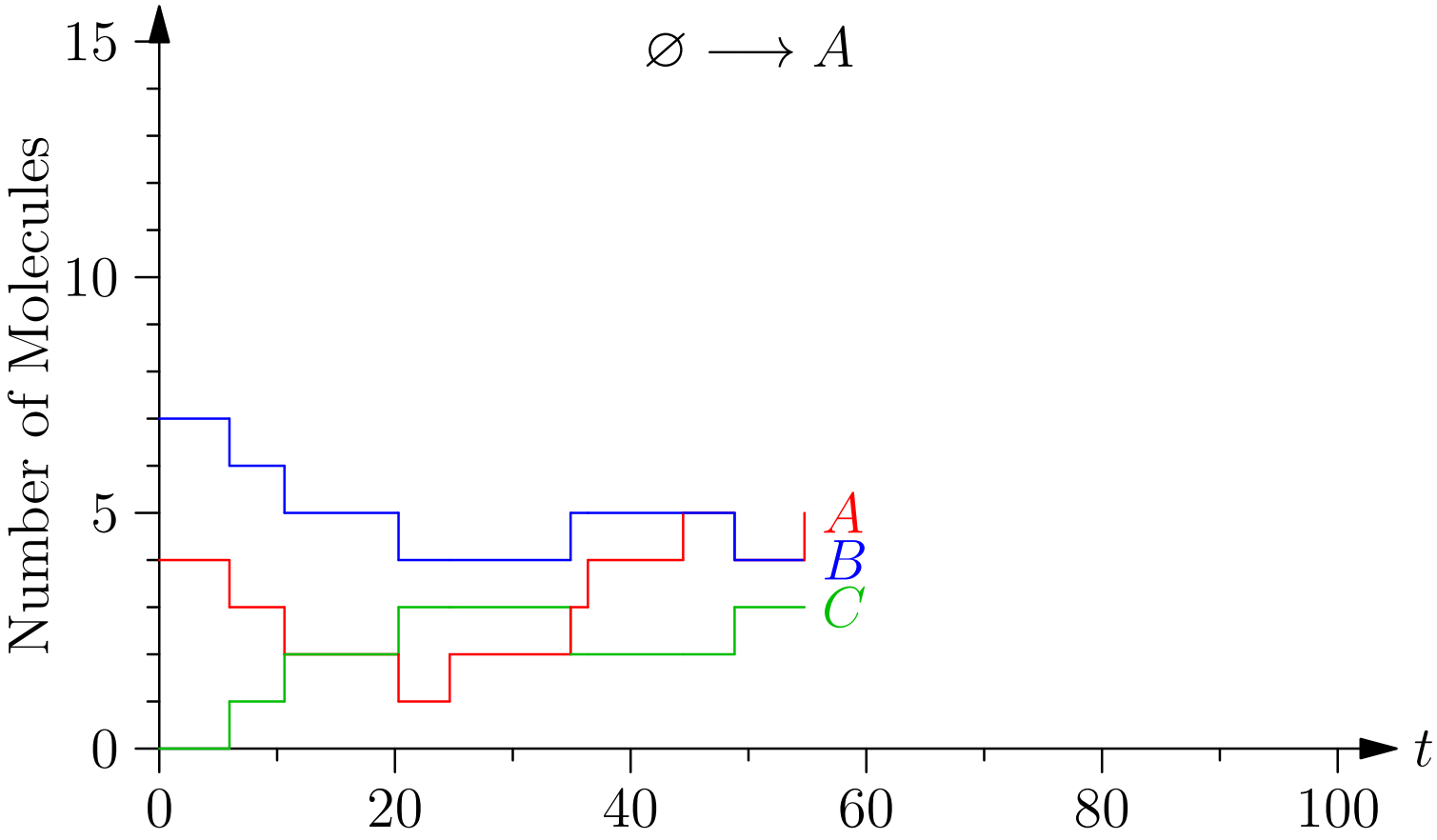

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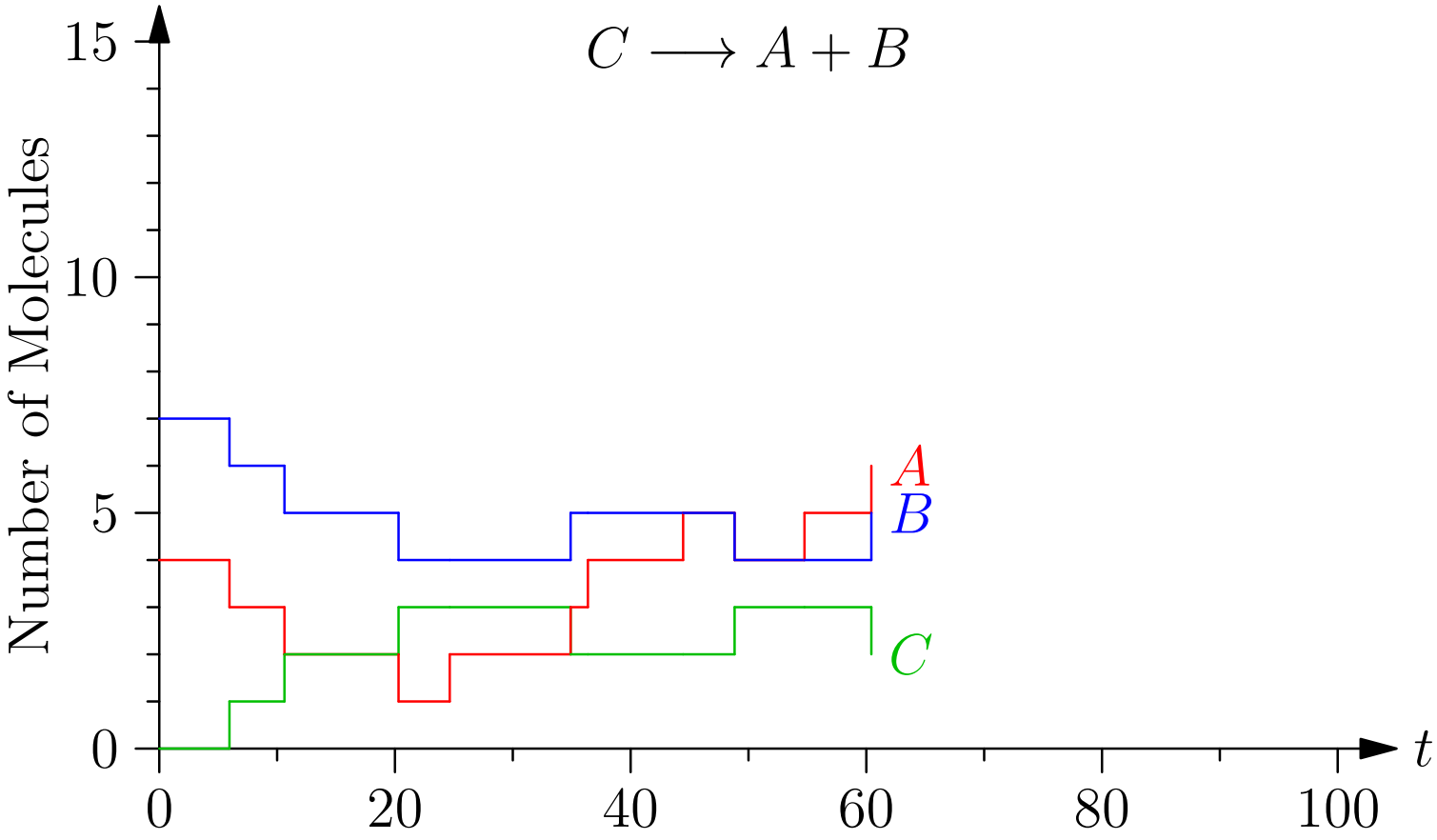

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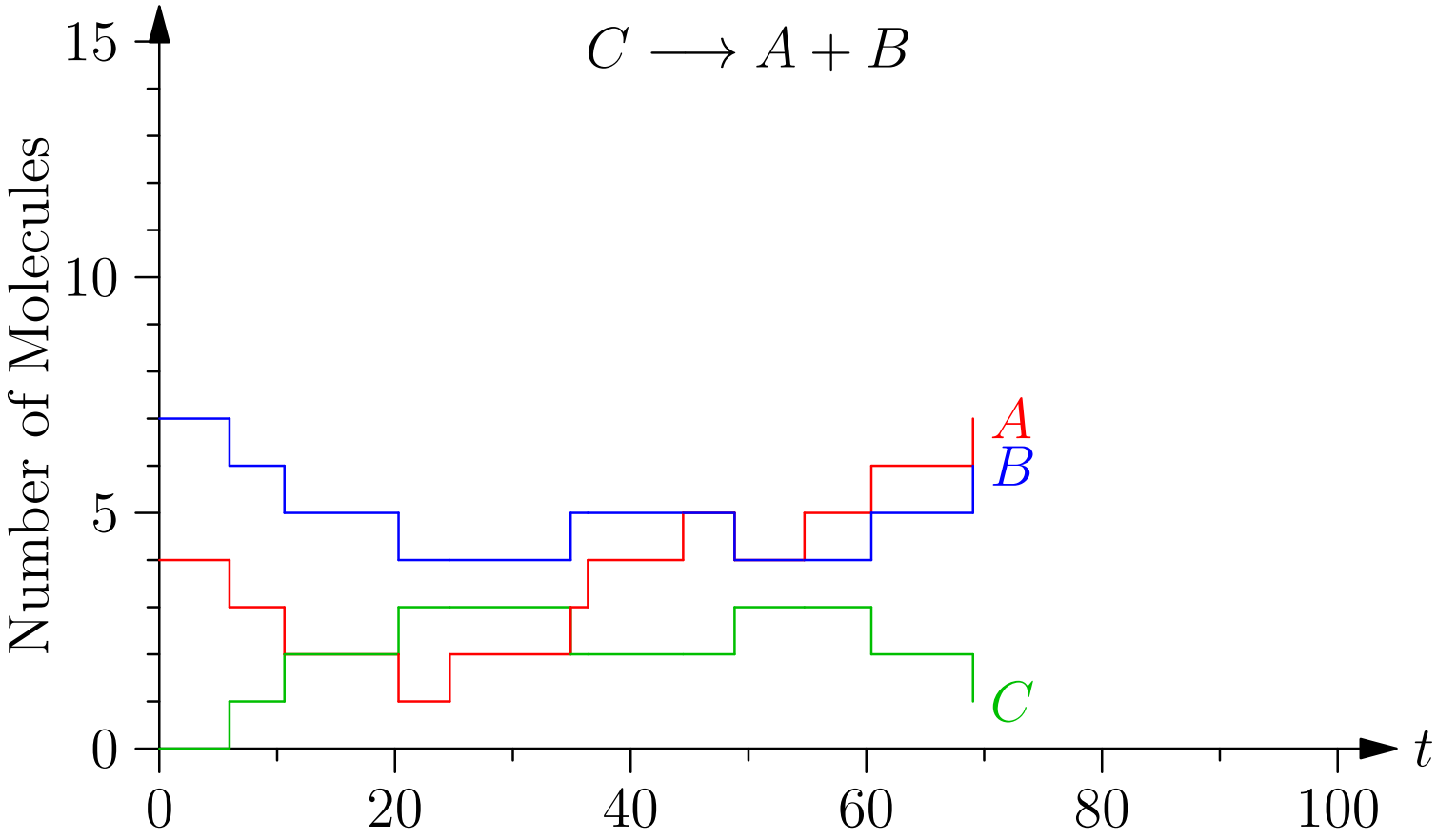

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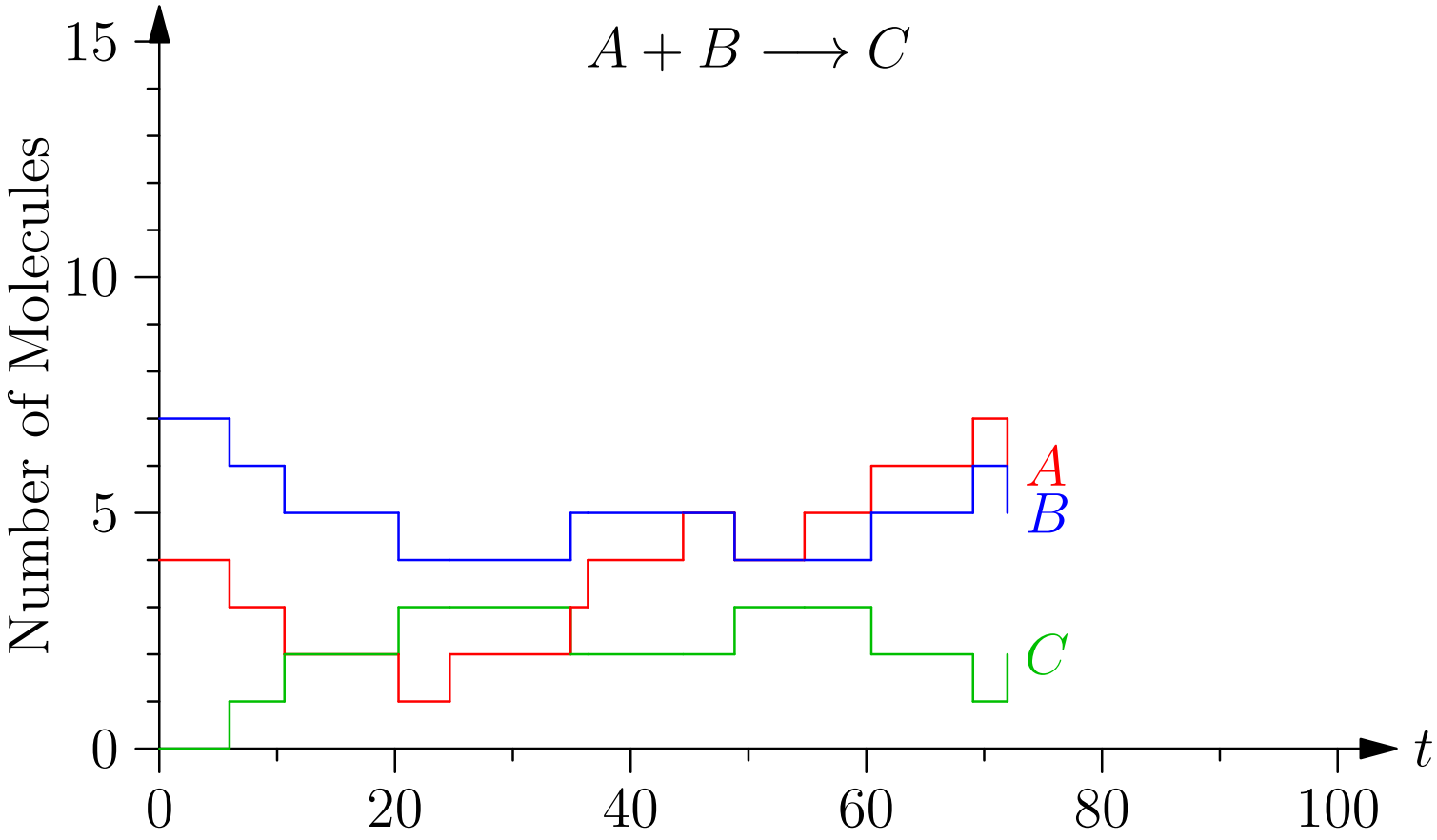

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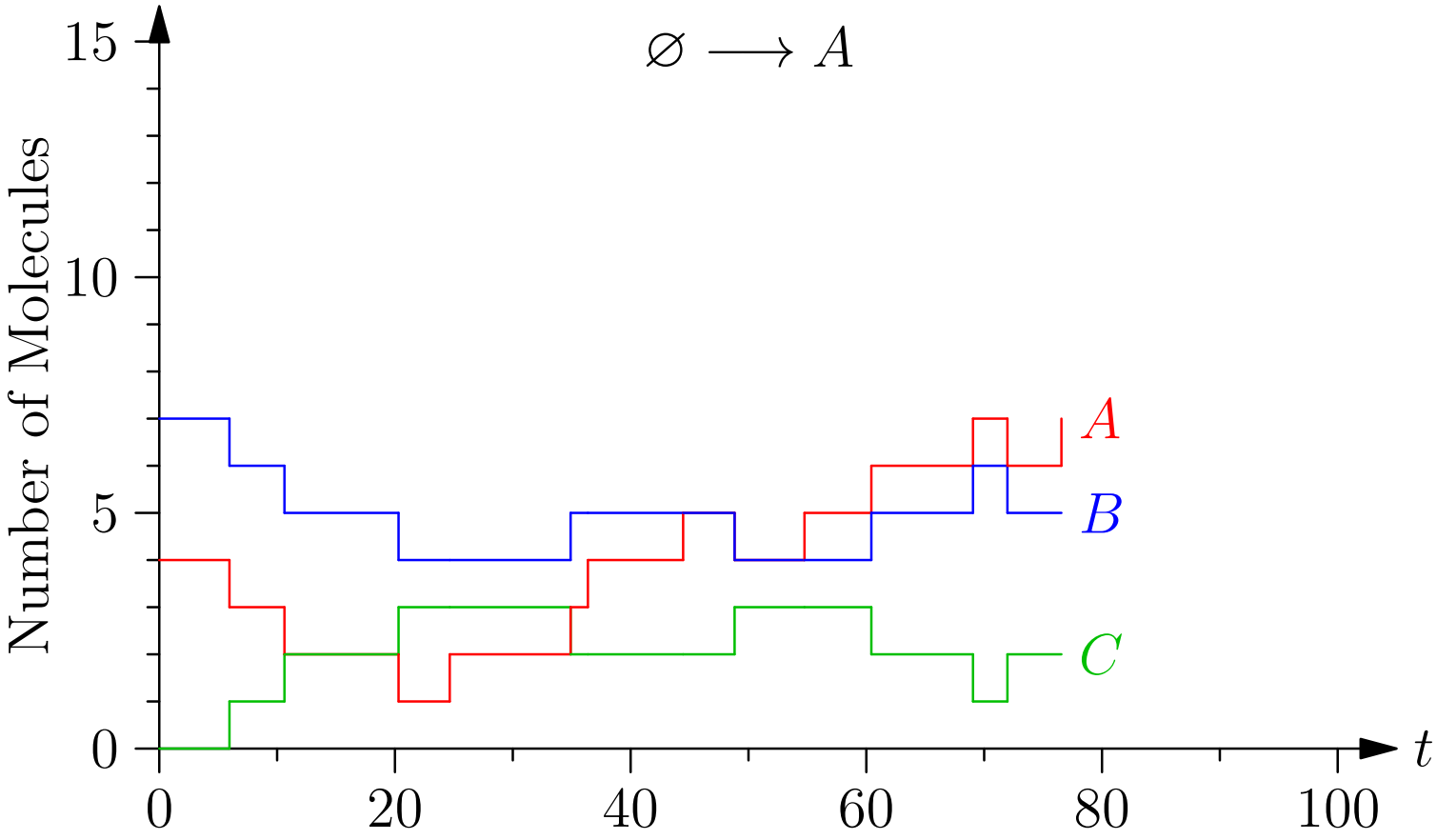

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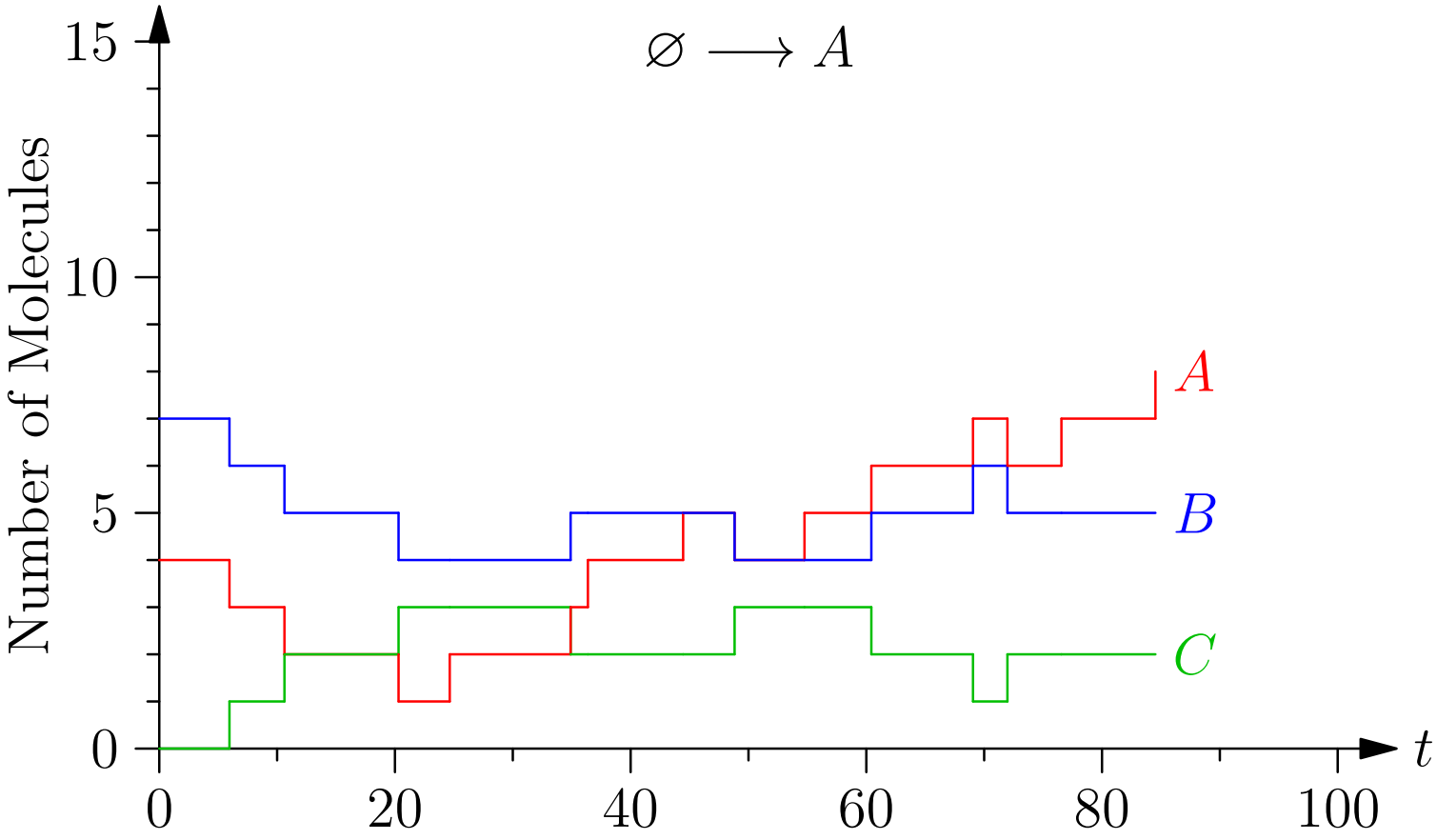

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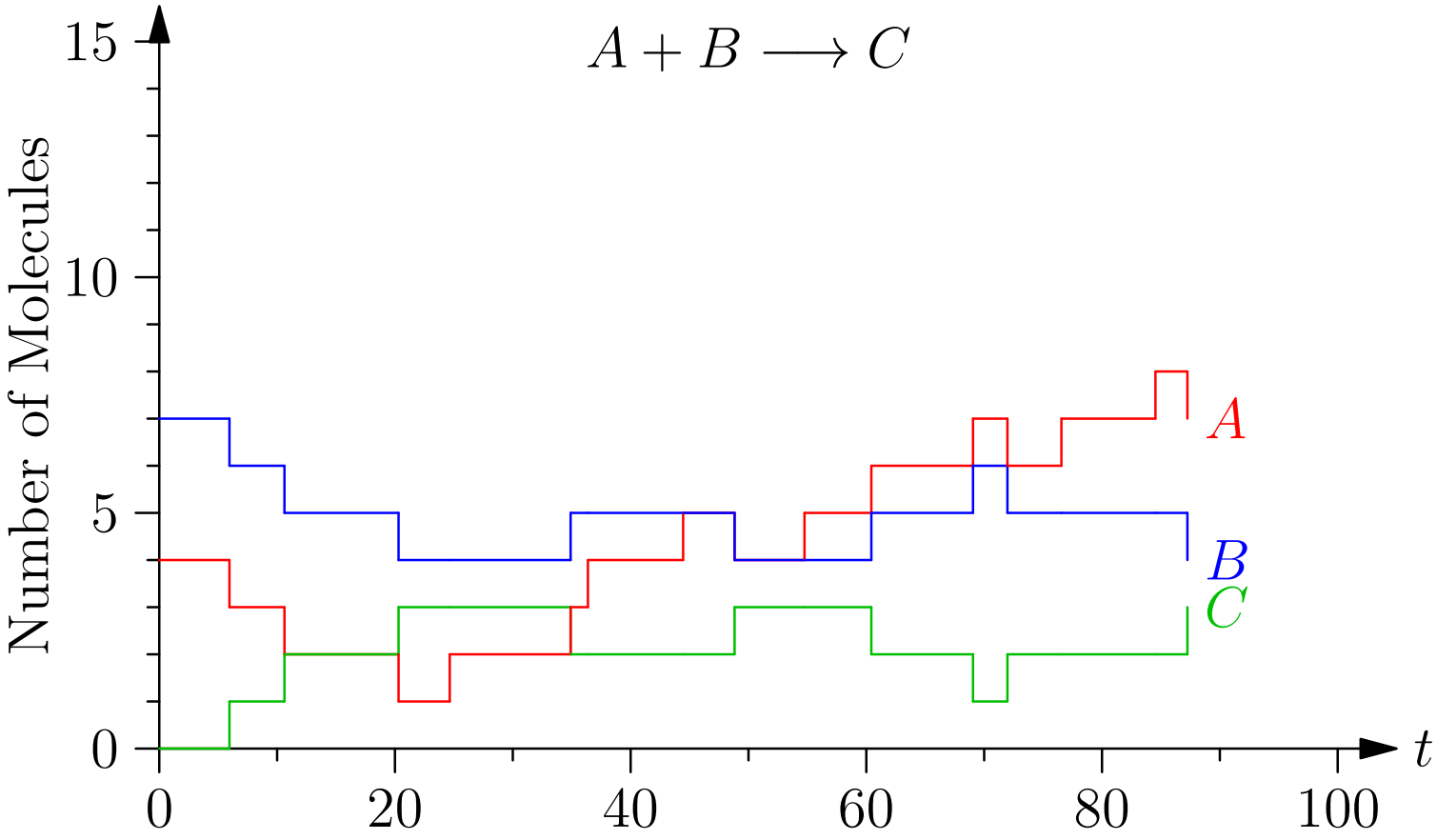

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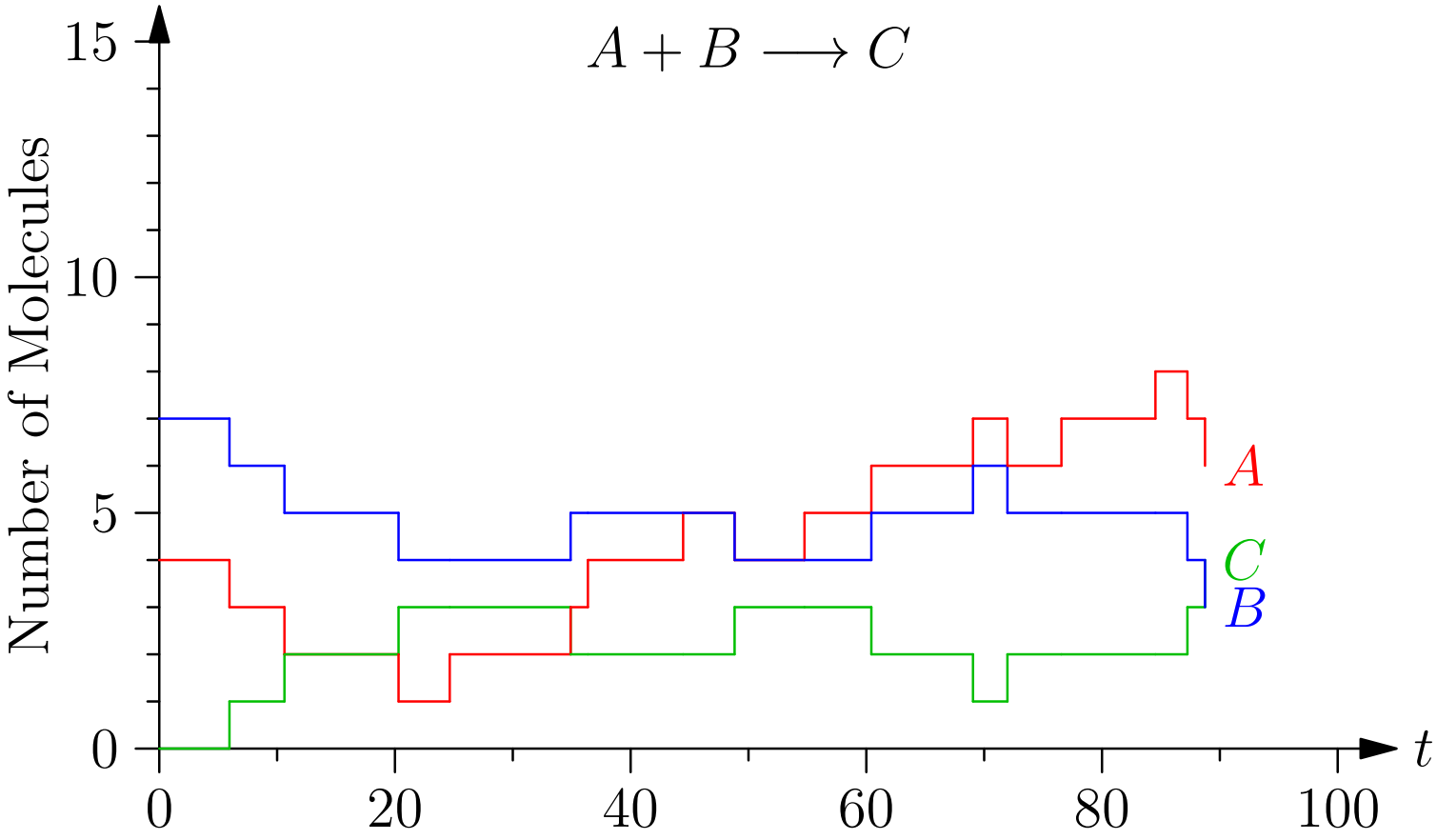

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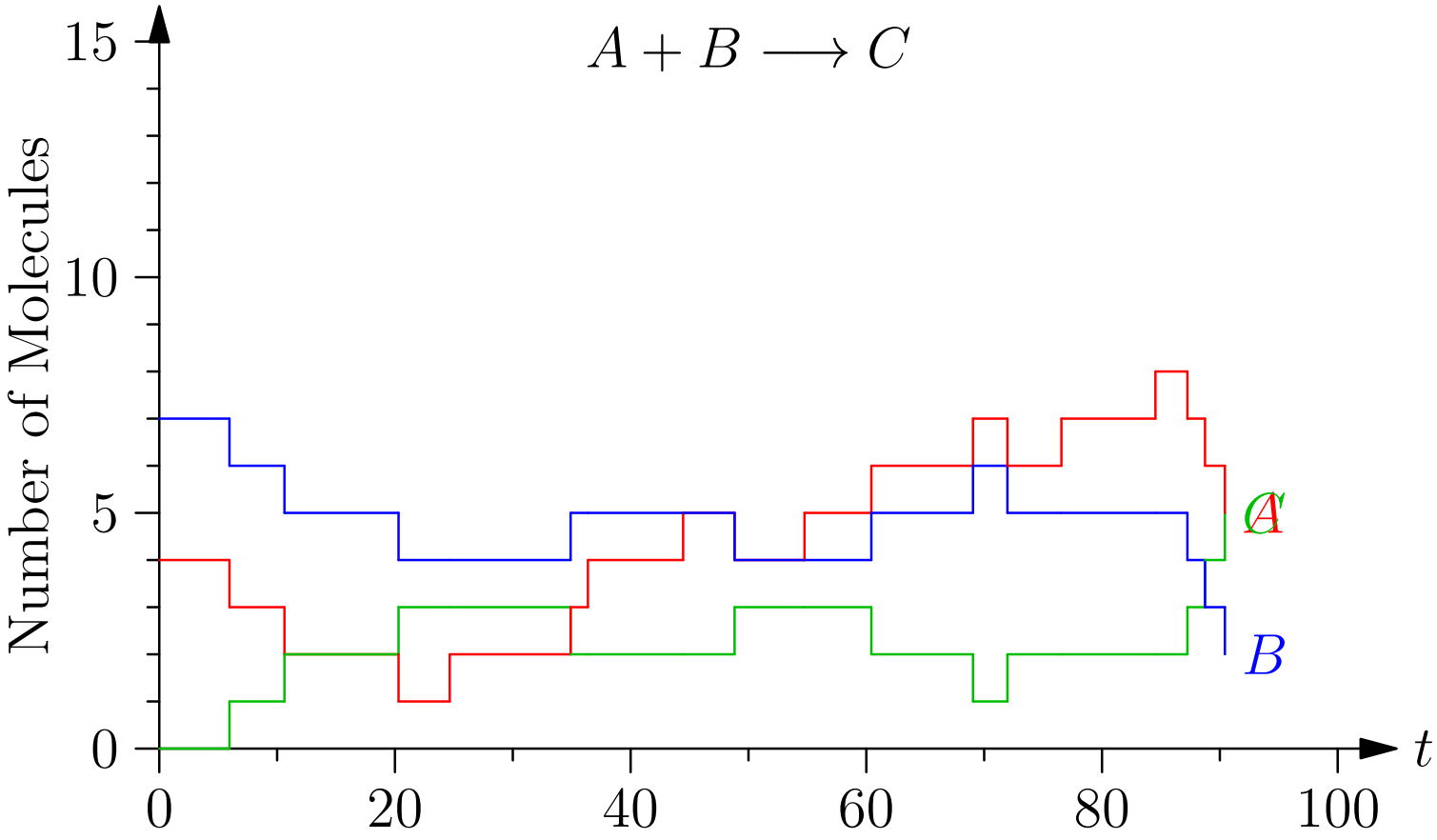

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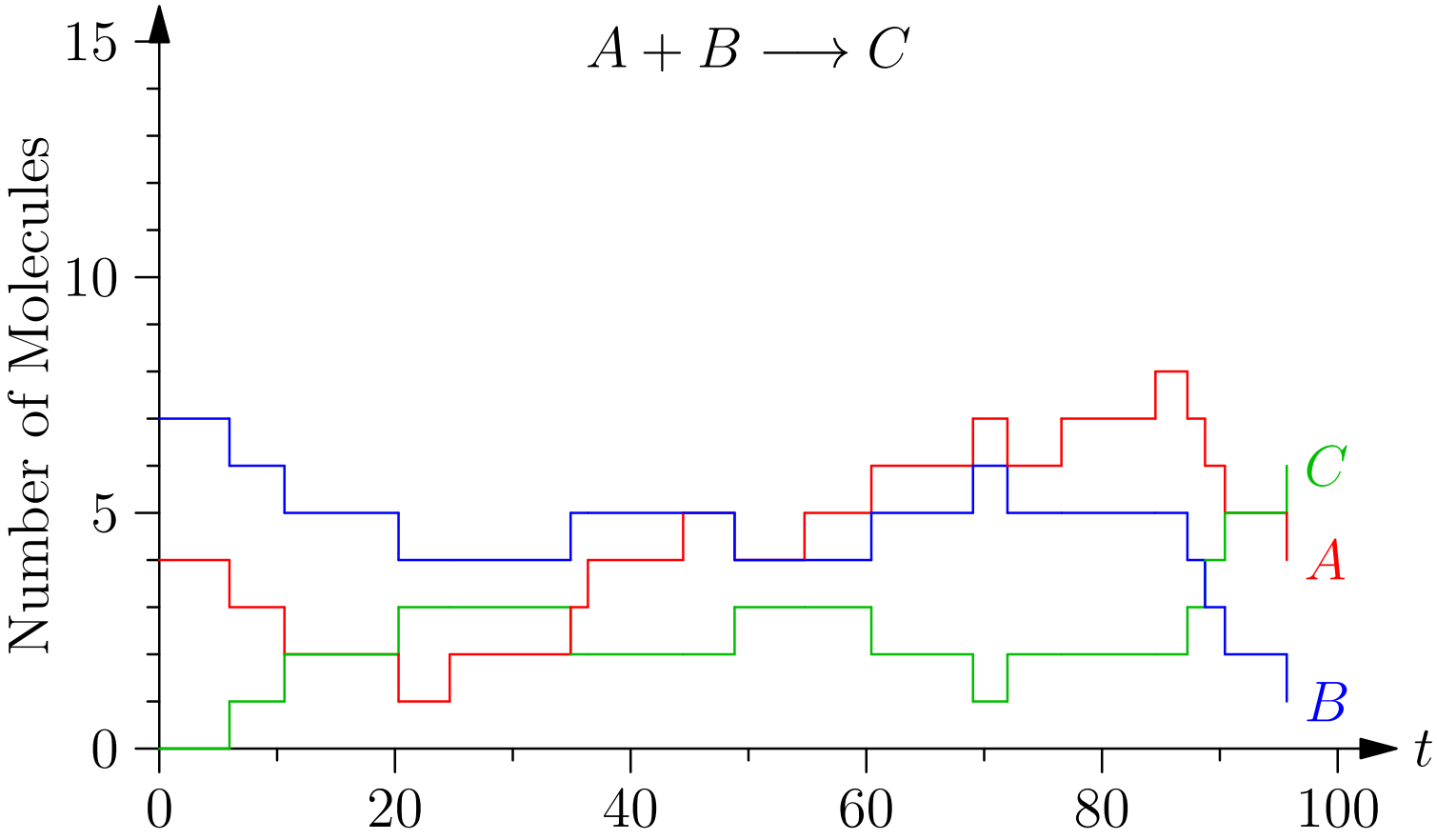

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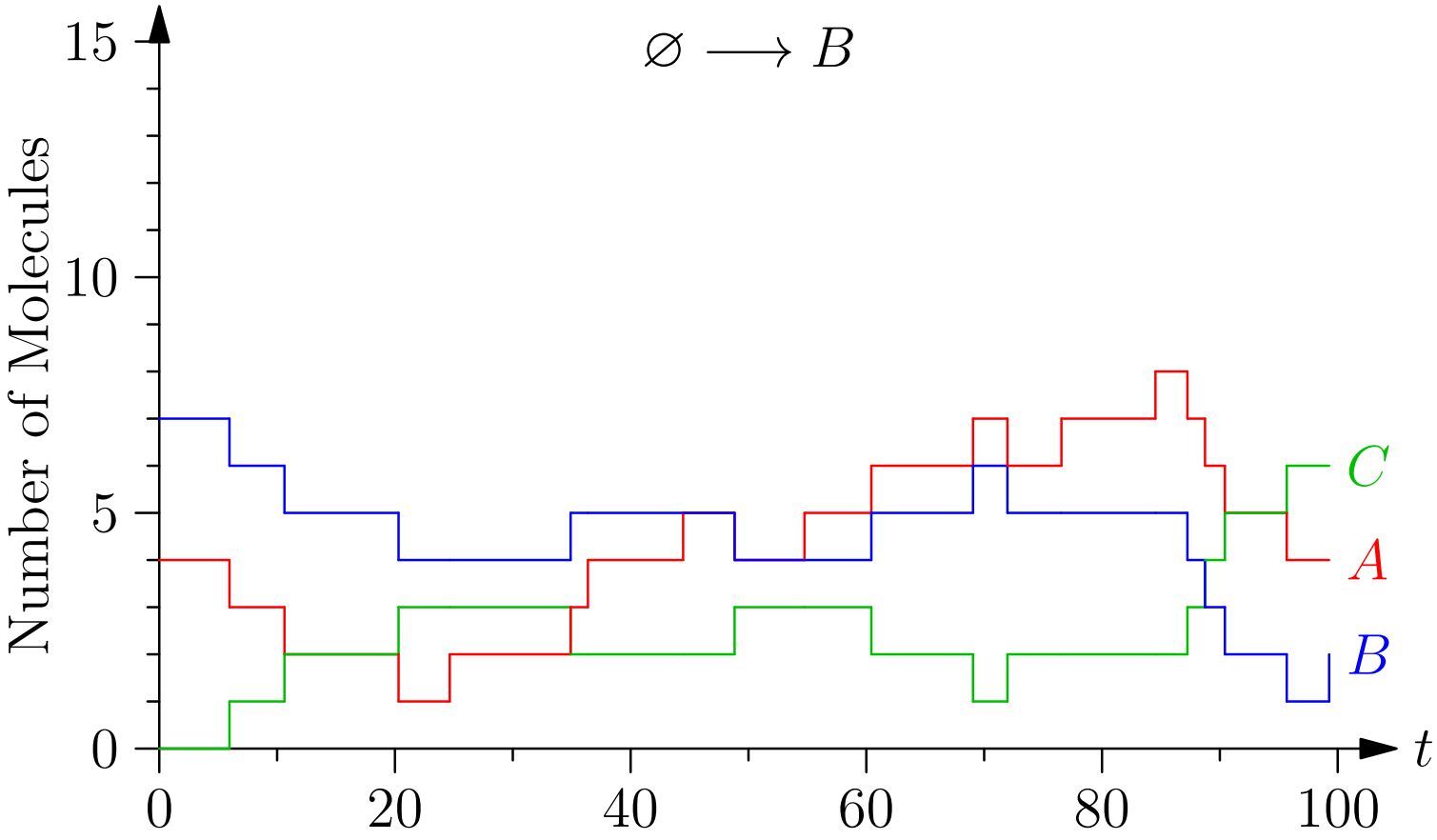

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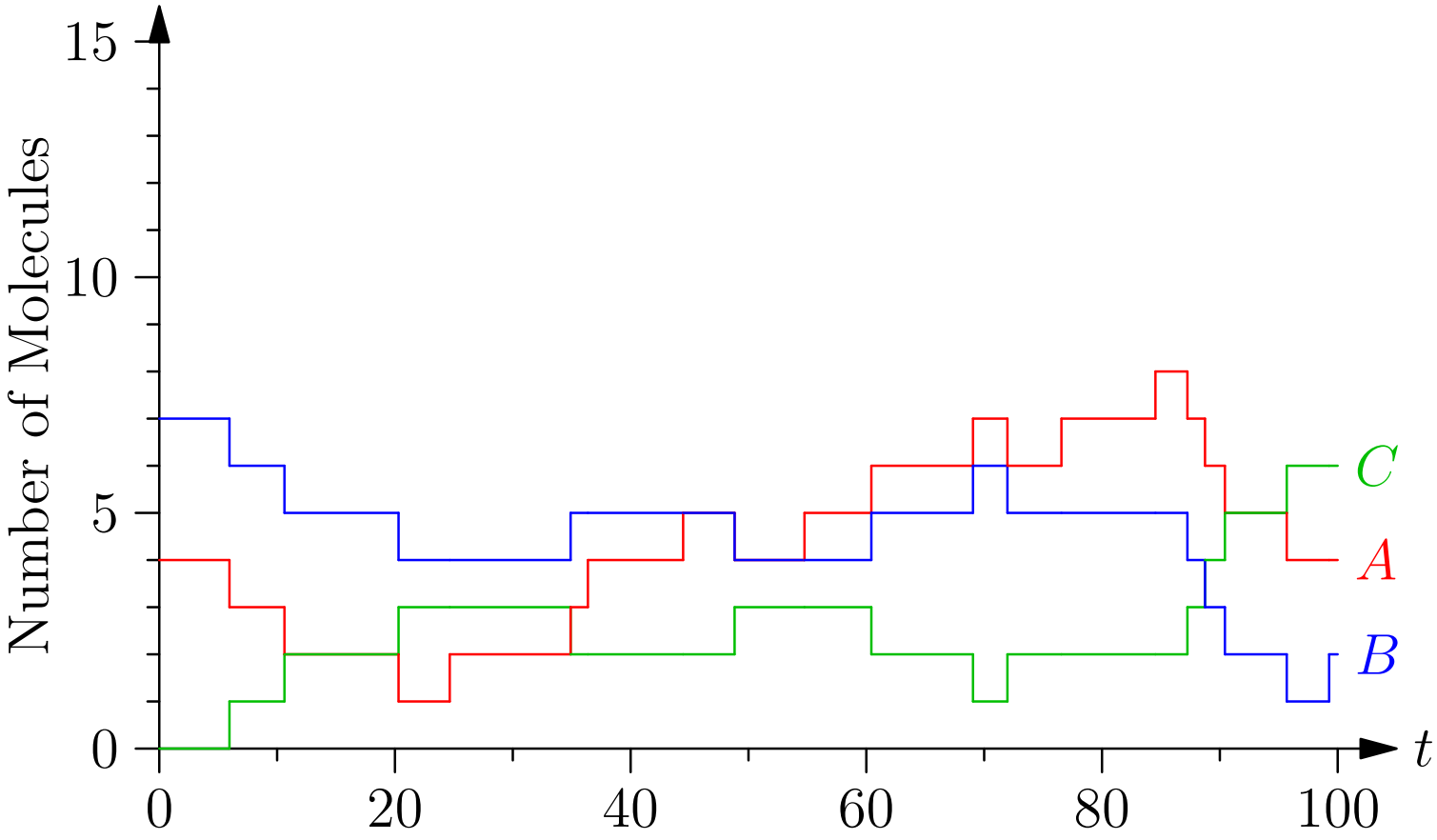

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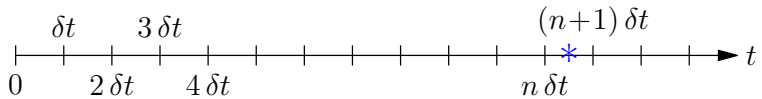
Poisson Processes

- Suppose we have a point process with a probability of occurring in time δt of $a\,\delta t$

- The probability of the first occurrence happening in time interval $n\,\delta t \leq t \leq (n+1)\,\delta t$ is



$$\mathbb{P}\left(n\,\delta t \leq t \leq (n+1)\,\delta t\right) = (1 - a\,\delta t)^n\, a\,\delta\, t = a\,\mathrm{e}^{n\log(1-a\,\delta t)}\,\delta t$$
$$\approx a\,\mathrm{e}^{-a\,n\,\delta t}\,\delta t \approx a\,\mathrm{e}^{-a\,t}\,\delta t$$

since $\log(1+\Delta) \approx \Delta + O(\Delta^2)$

- If we take the limit $\delta t \to 0$ then

$$f_T(t) = \lim_{\delta t \to 0} \frac{\mathbb{P}\left(n\,\delta t \leq t \leq (n+1)\,\delta t\right)}{\delta t}$$

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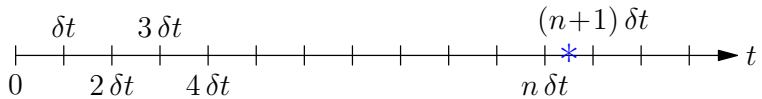

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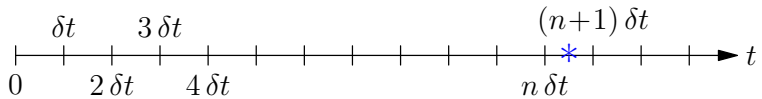

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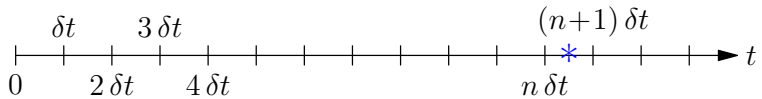

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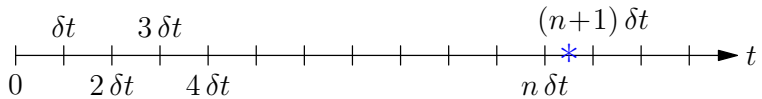

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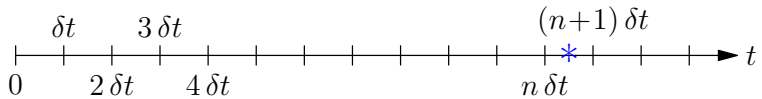

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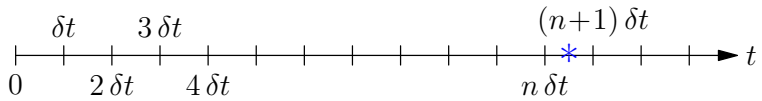

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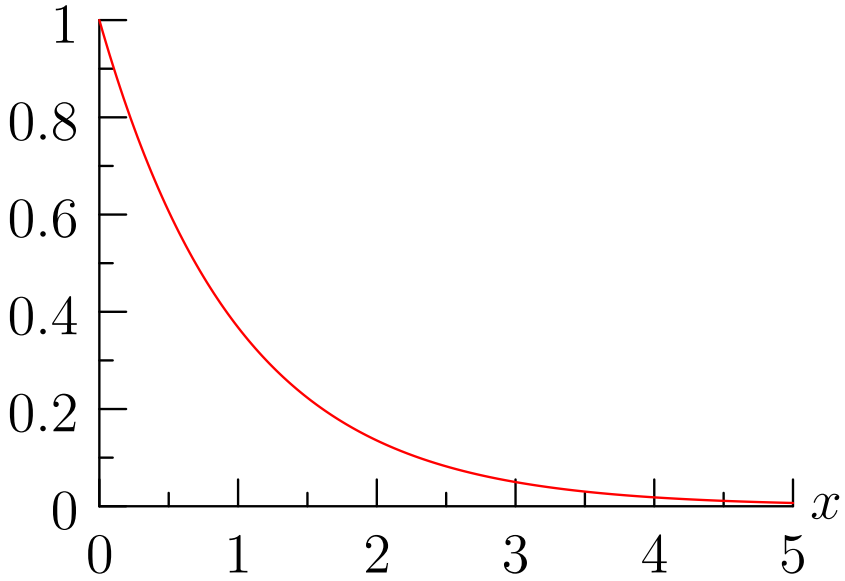

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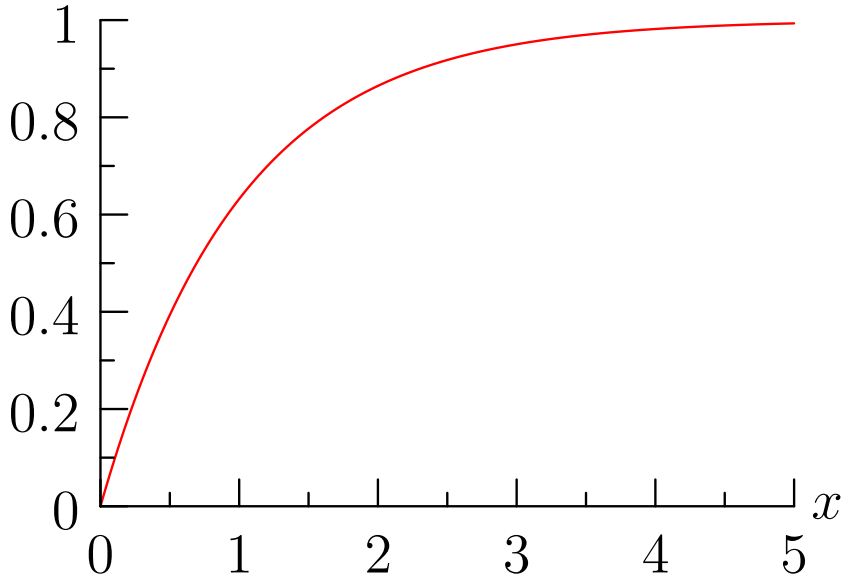

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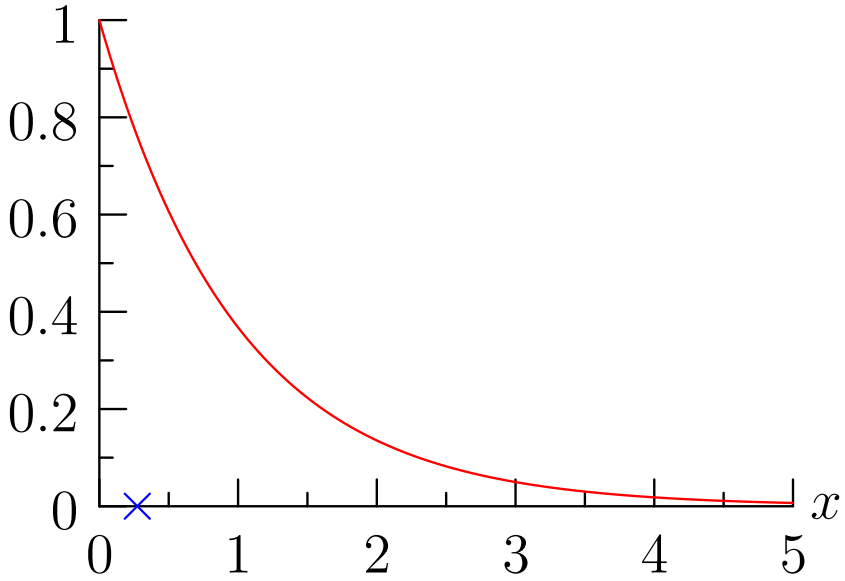

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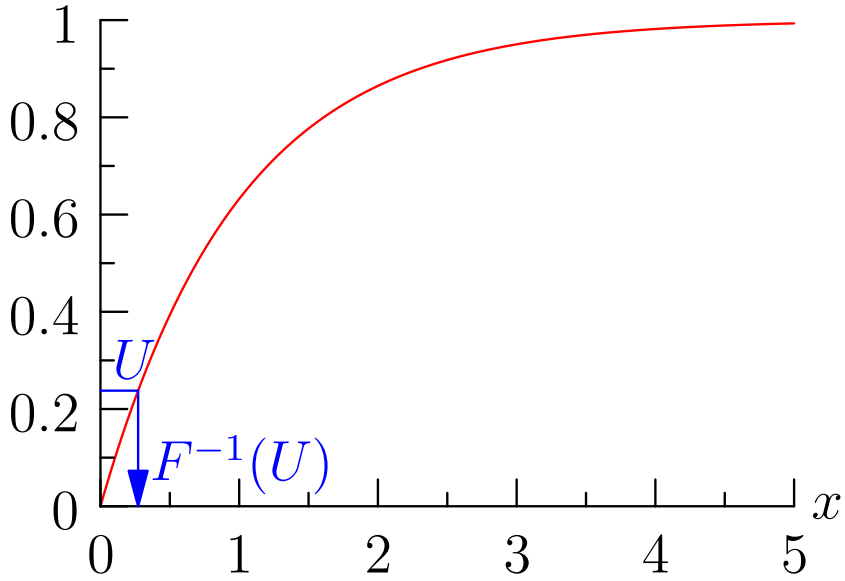

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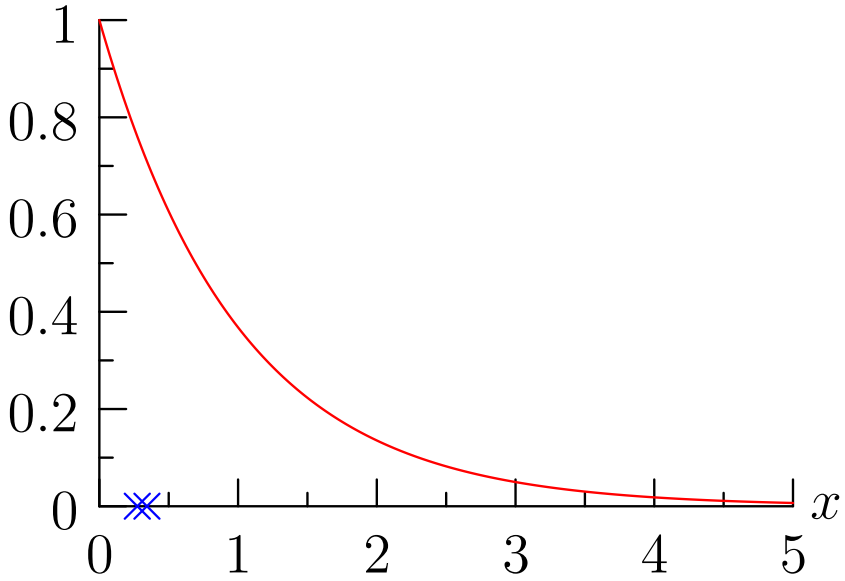

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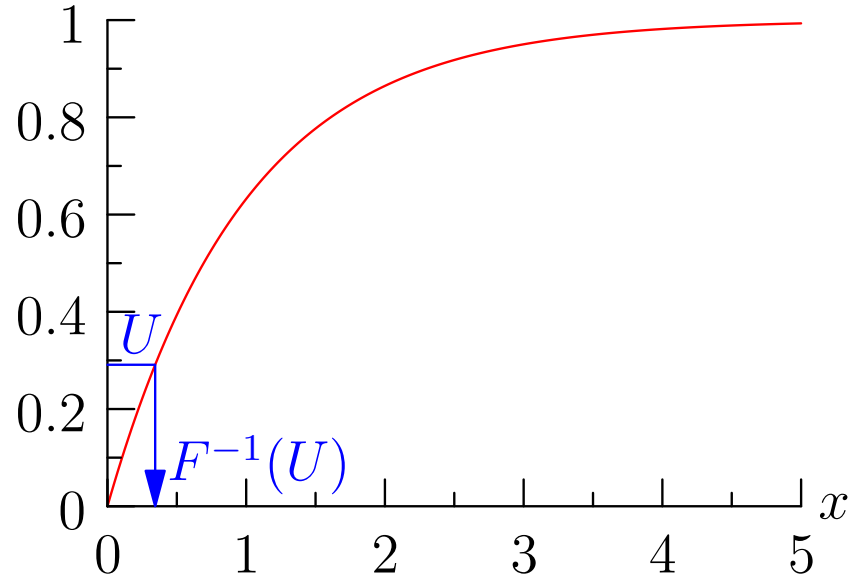

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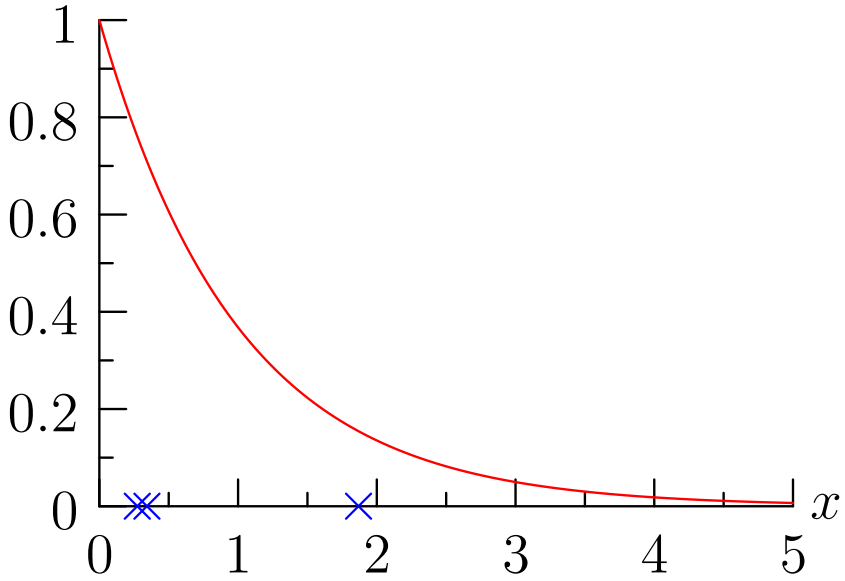

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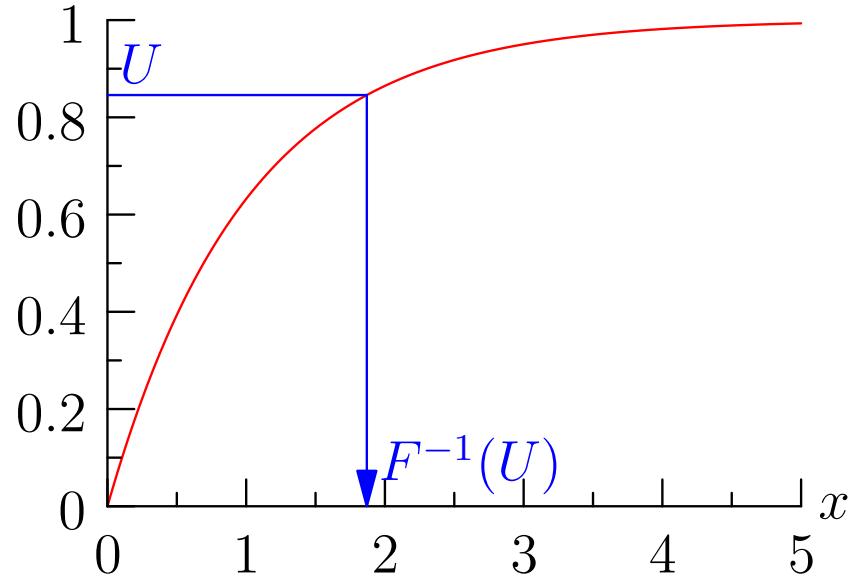

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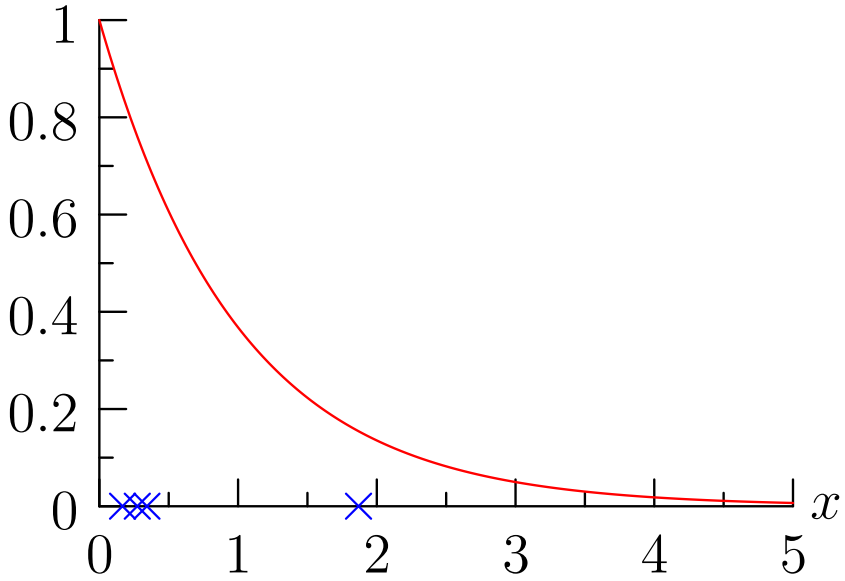

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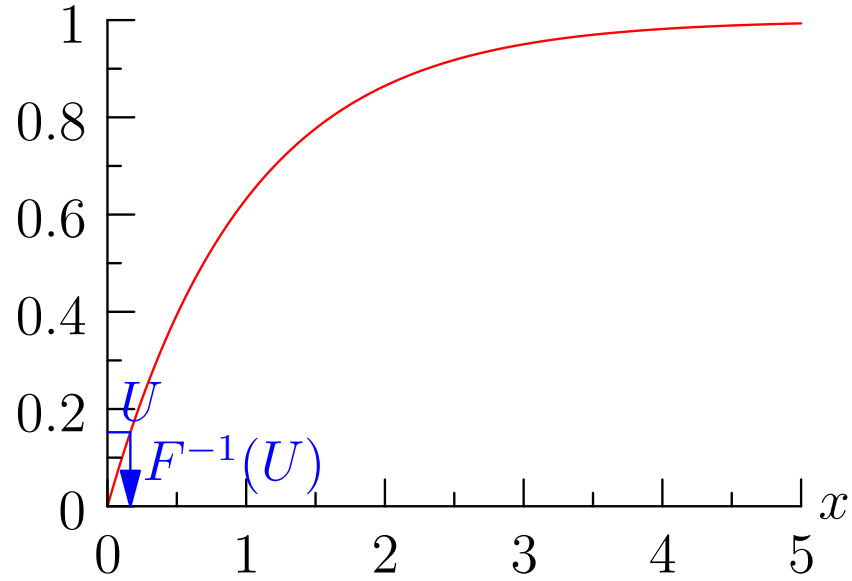

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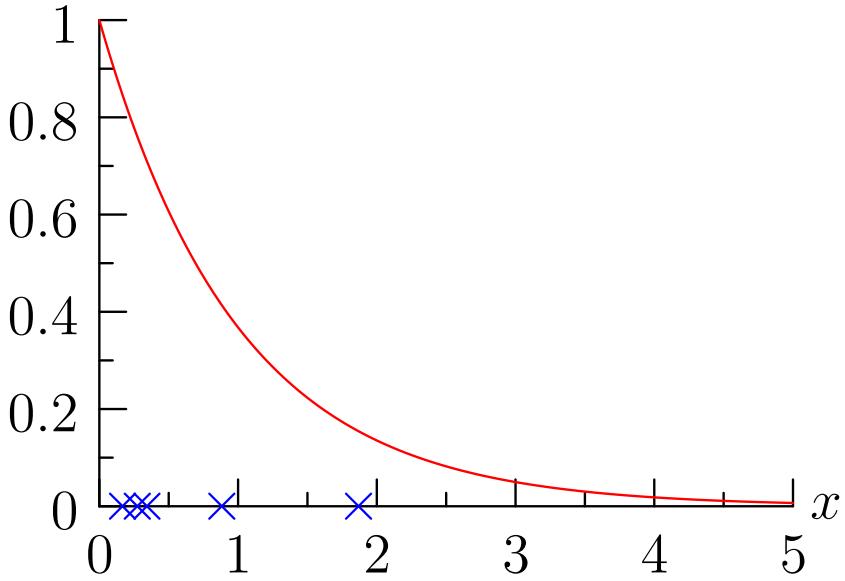

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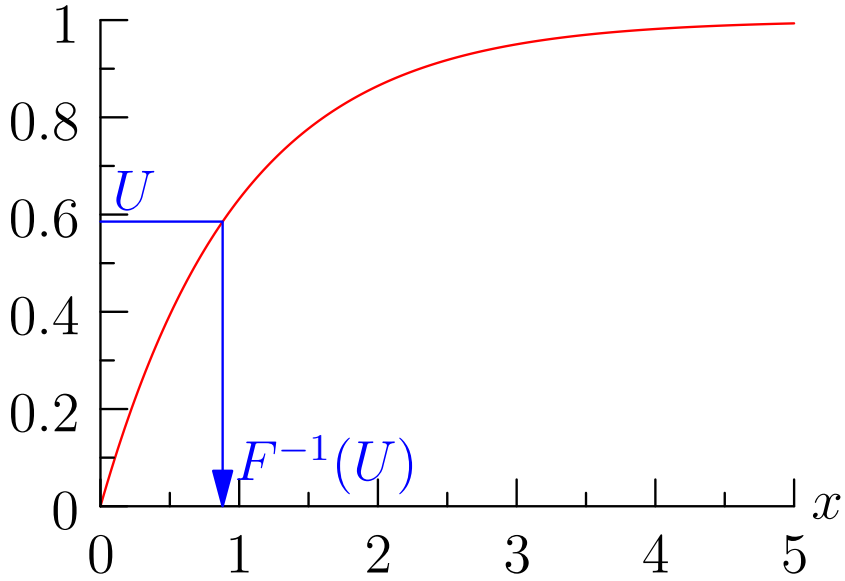

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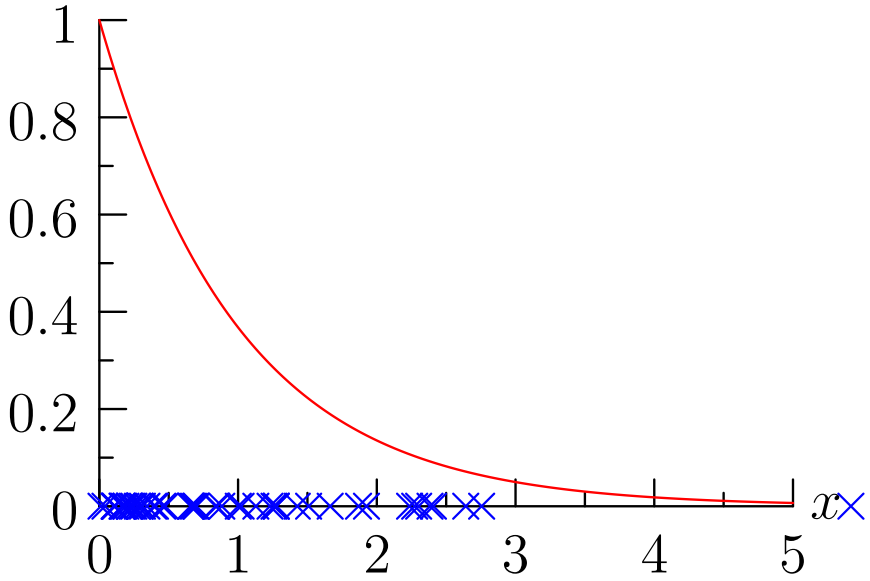

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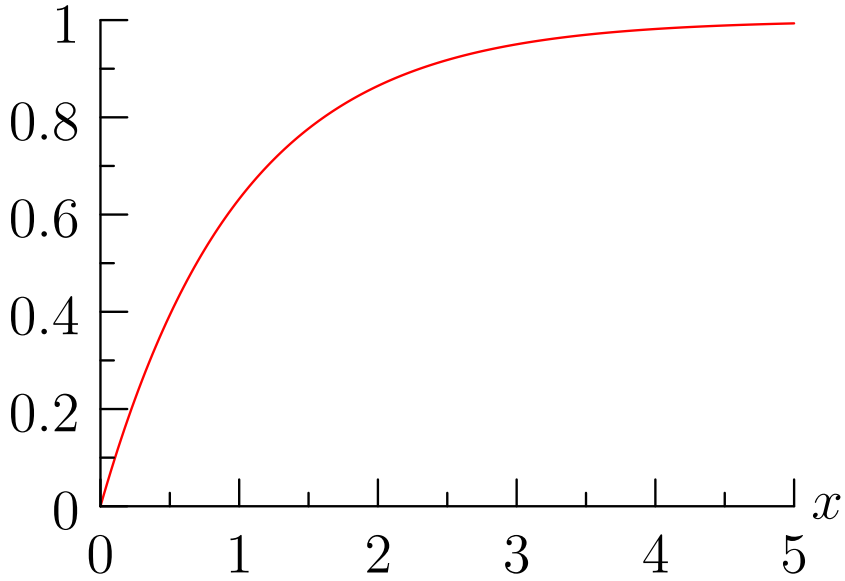

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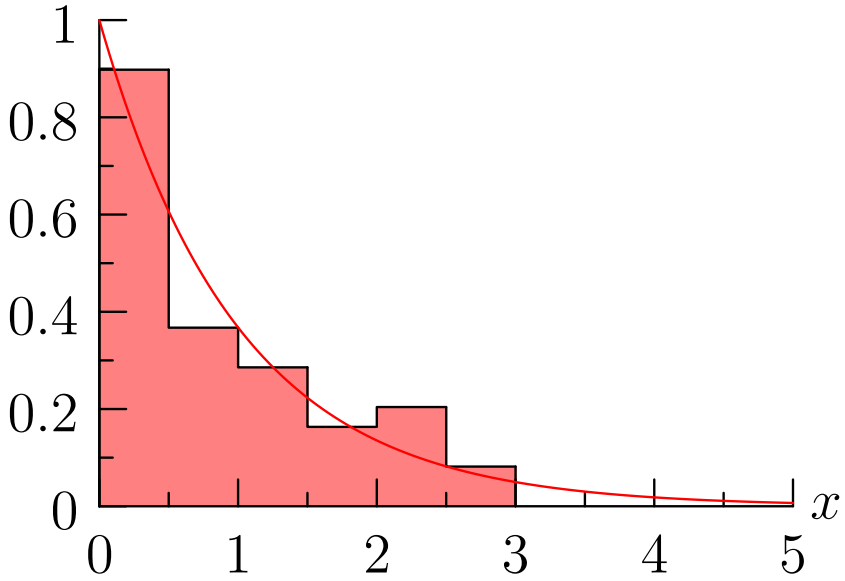

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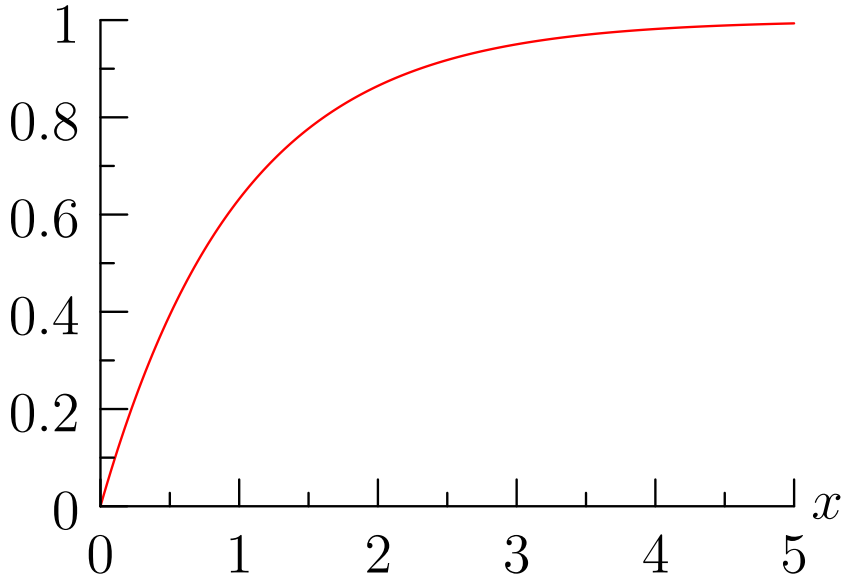

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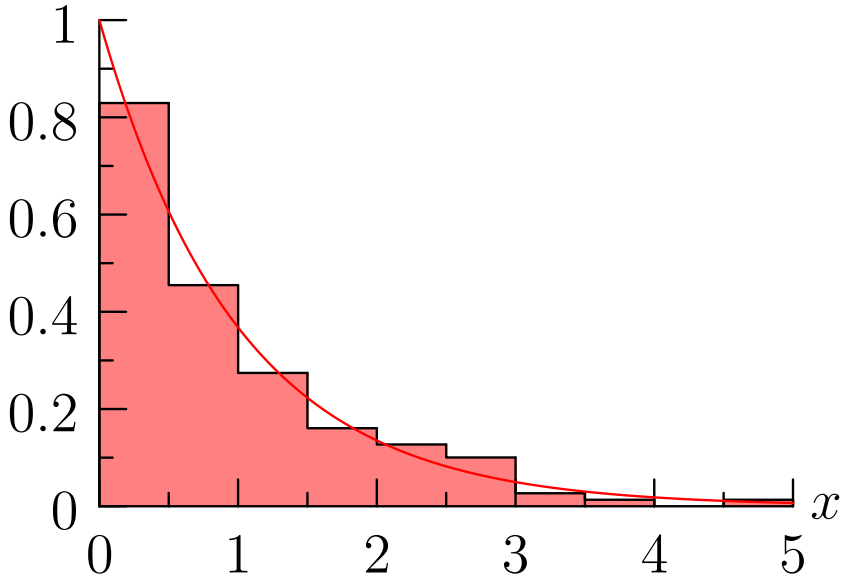

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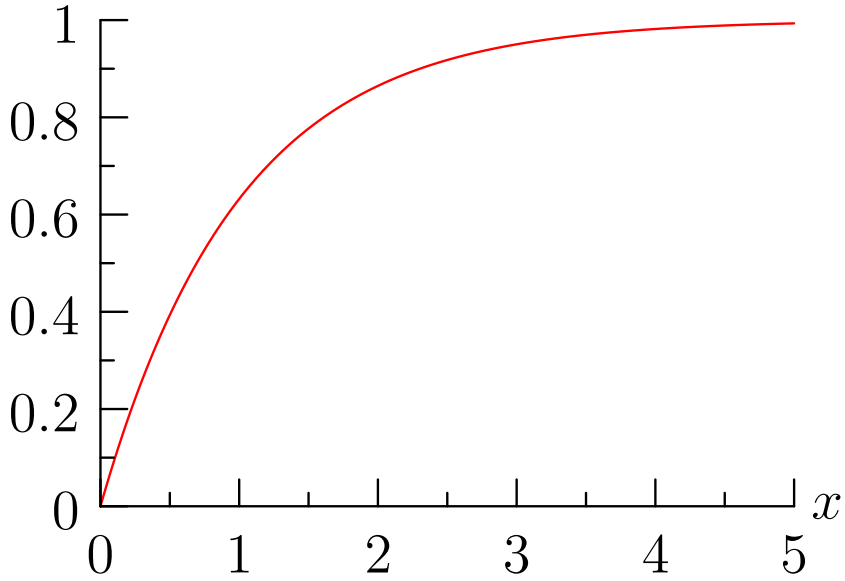

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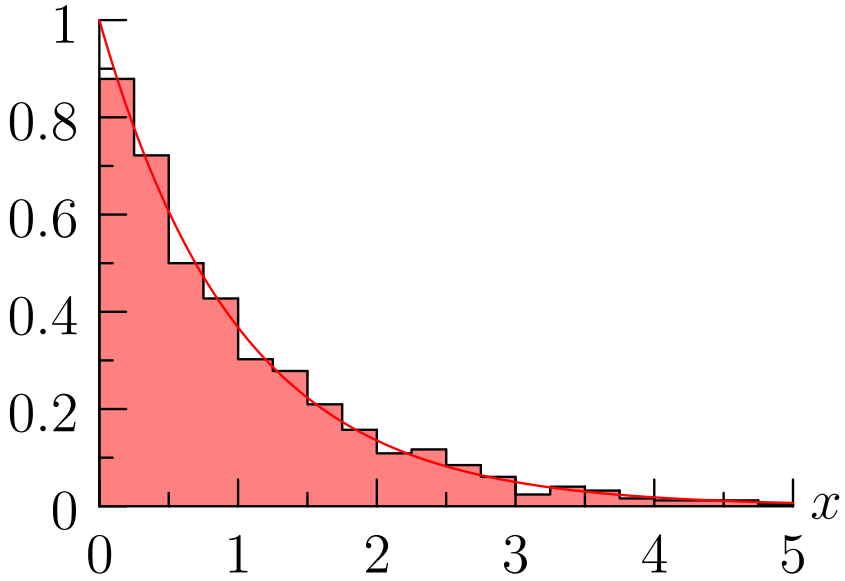

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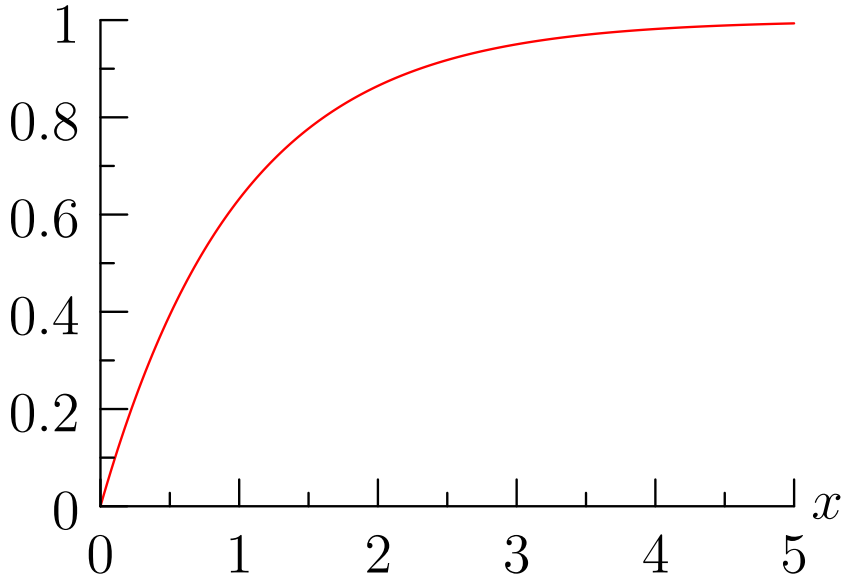

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- For exponential deviates we generate a random deviate $U \sim \mathrm{Uni}(0,1)$
- Then to generate $X \sim \mathrm{Exp}(x|1) = \mathrm{e}^{-x}$ (with $F_X(x|1) = 1 - \mathrm{e}^{-x}$) we use

$$X = F_X^{-1}(U) = -\log(1-U)$$

- To draw a random deviate $T \sim \mathrm{Exp}(t|a) = a\,\exp(-a\,t)$ we just rescale $T = X/a$
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Gillespie's Algorithm

- To simulate a reaction equation we take $\boldsymbol{X}(t) = (X_1(t), X_2(t), \ldots, X_N(t))$ to be the count of molecules at time t
- We assume we have K reactions which we label R_j
- The probability of a reaction occurring in a volume V in time interval $[t, t + \mathrm{d}t]$ we denote by $a_j(\boldsymbol{X}(t))\,\mathrm{d}t$—this is known as the propensity
- The propensity is equal to the reaction rate times the product of concentrations of chemical species that react together (i.e. on the left-hand side) times the volume

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Gillespie's Algorithm

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Gillespie's Algorithm Continued

- We compute the total propensity $a_0(\boldsymbol{X}(t)) = \sum_{j=1}^{K} a_i(\boldsymbol{X}(t))$
- We draw a random time $T \sim \mathrm{Exp(t|a_0)}$ using

$$T = -\frac{1}{a_0}\log(1-U) \qquad\qquad U \sim \mathrm{Uni}(0,1)$$

- We select the reaction that happened with probabilities

$$\mathbb{P}\left(j\right) = \frac{a_j}{a_0}$$

- We update the molecule number according to what reaction has happened

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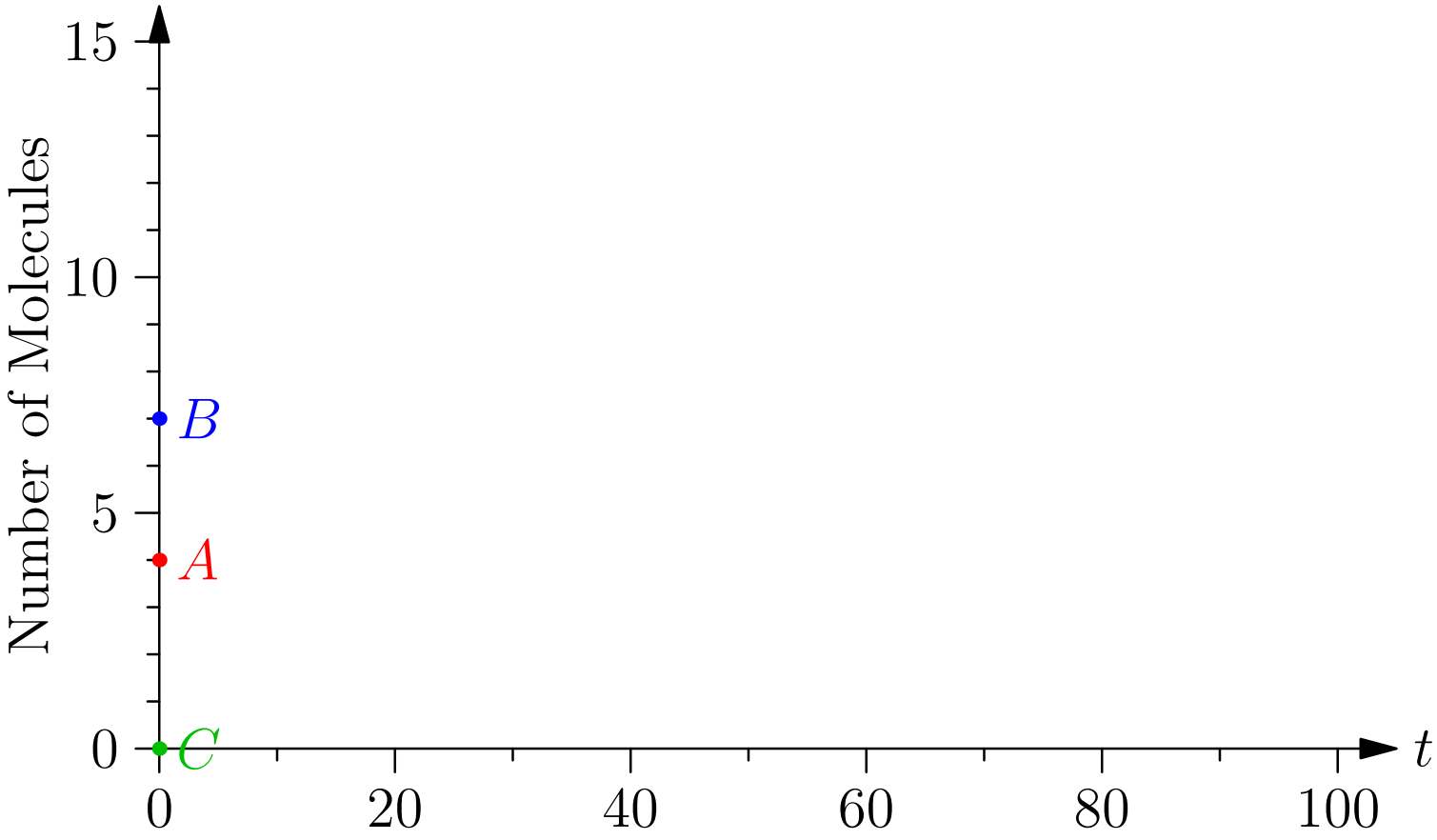

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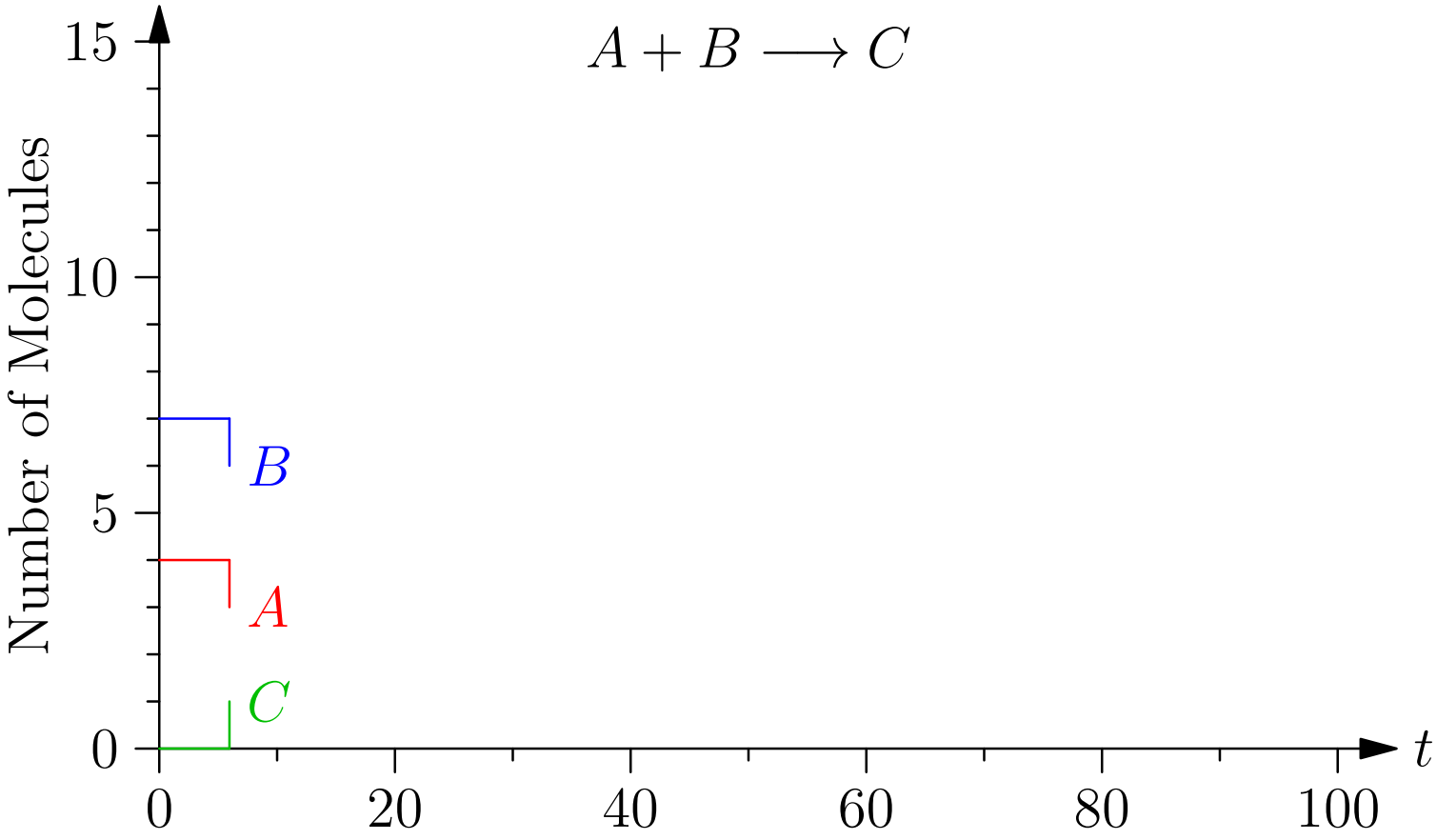

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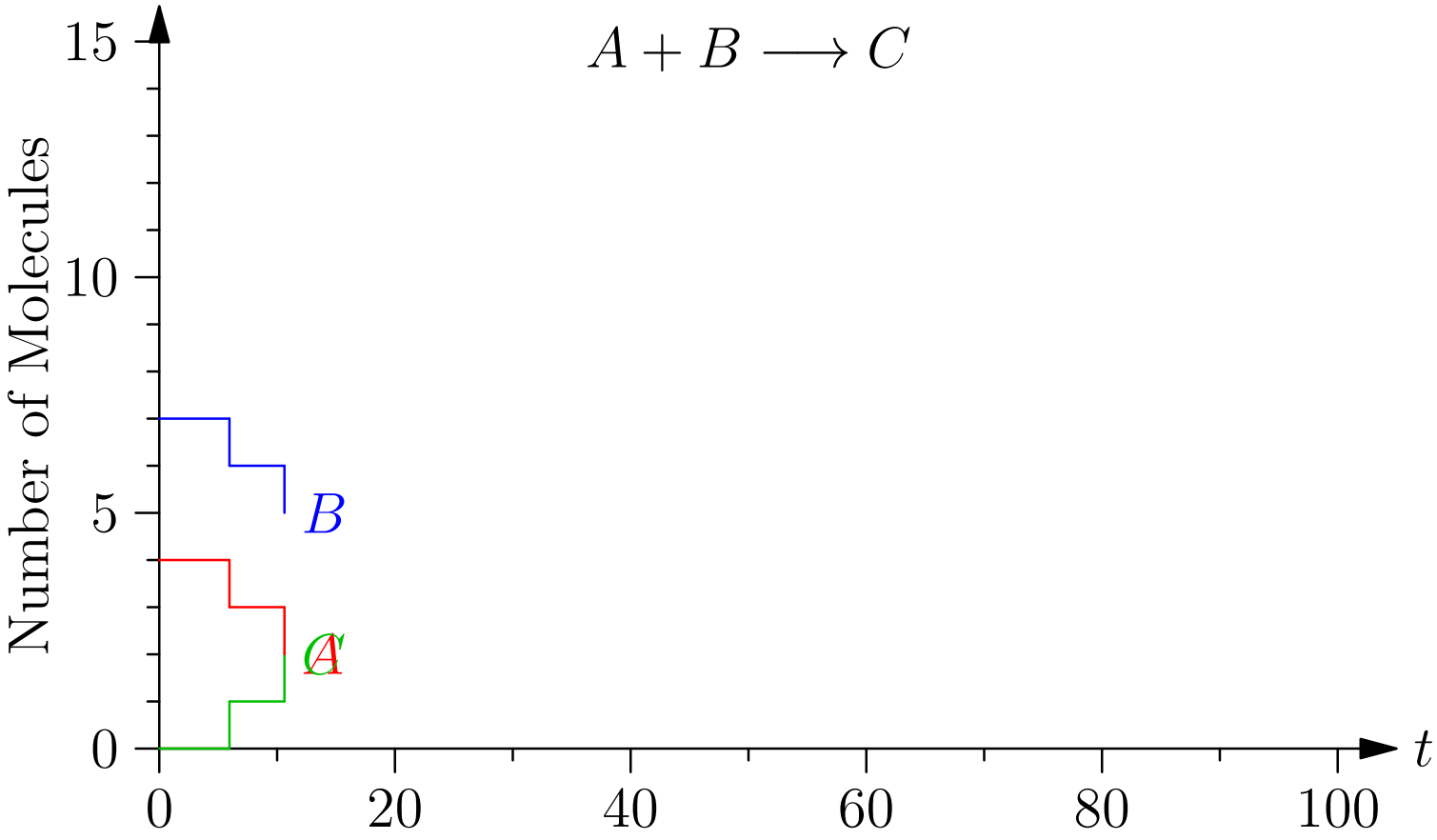

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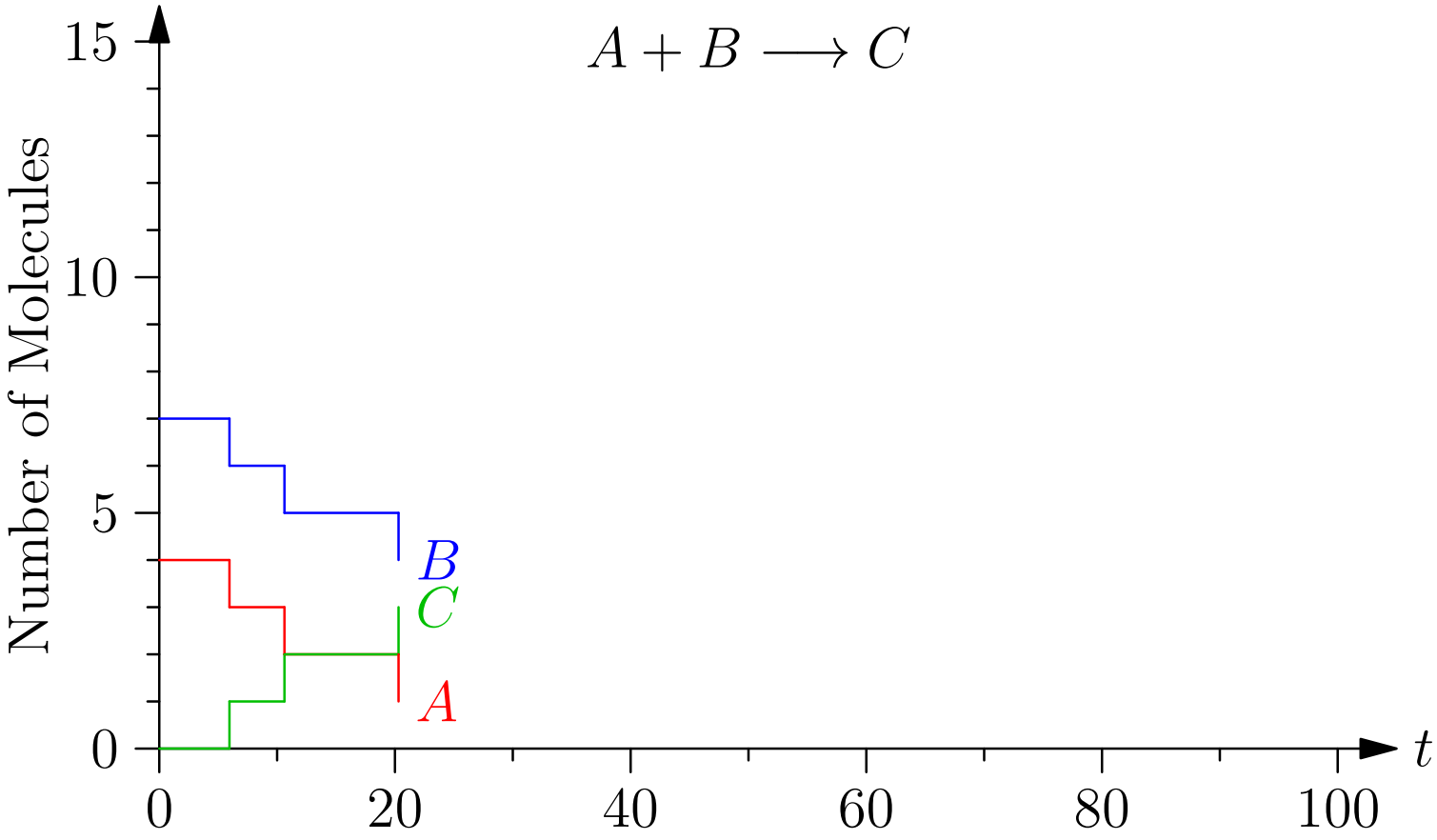

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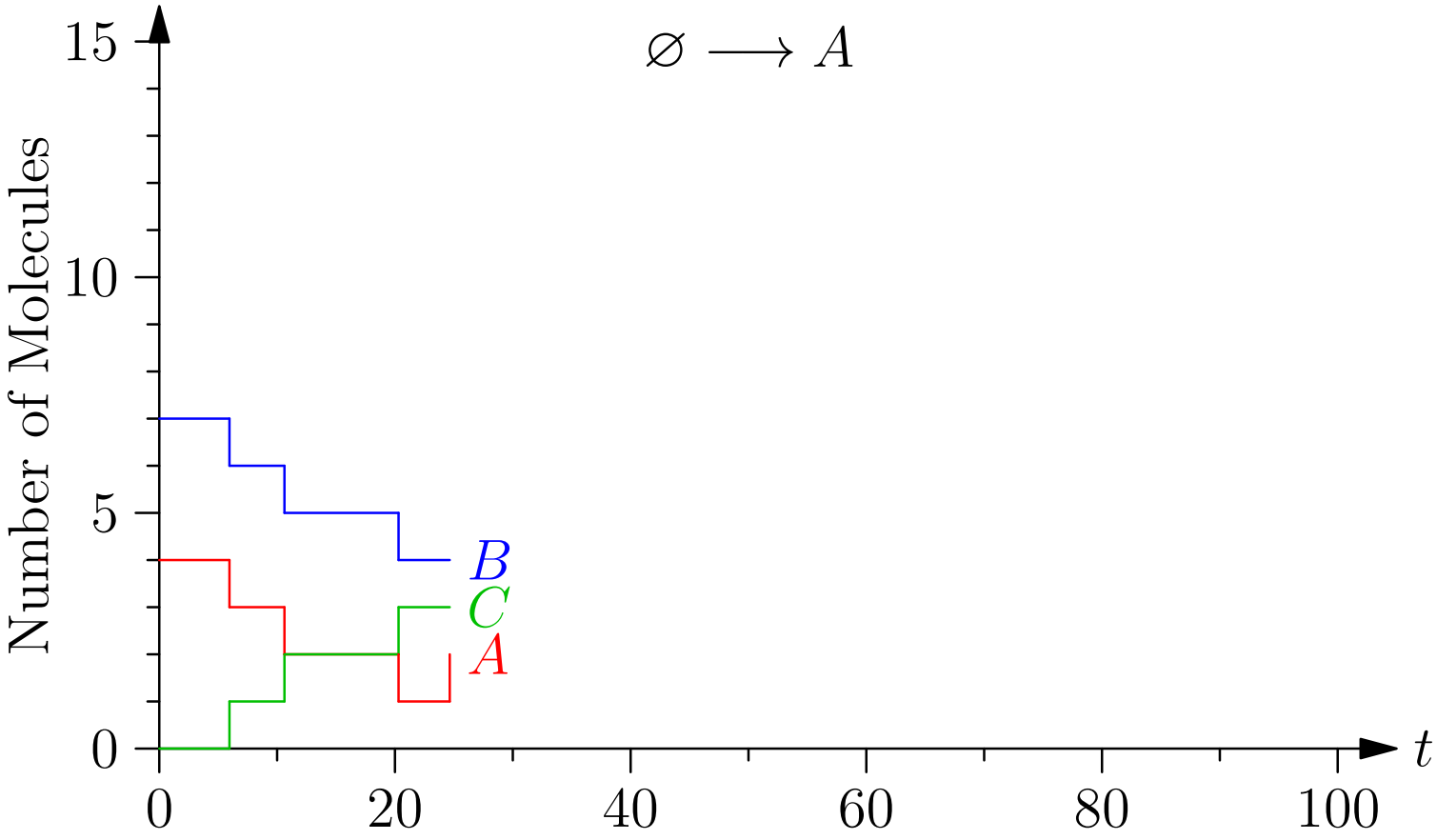

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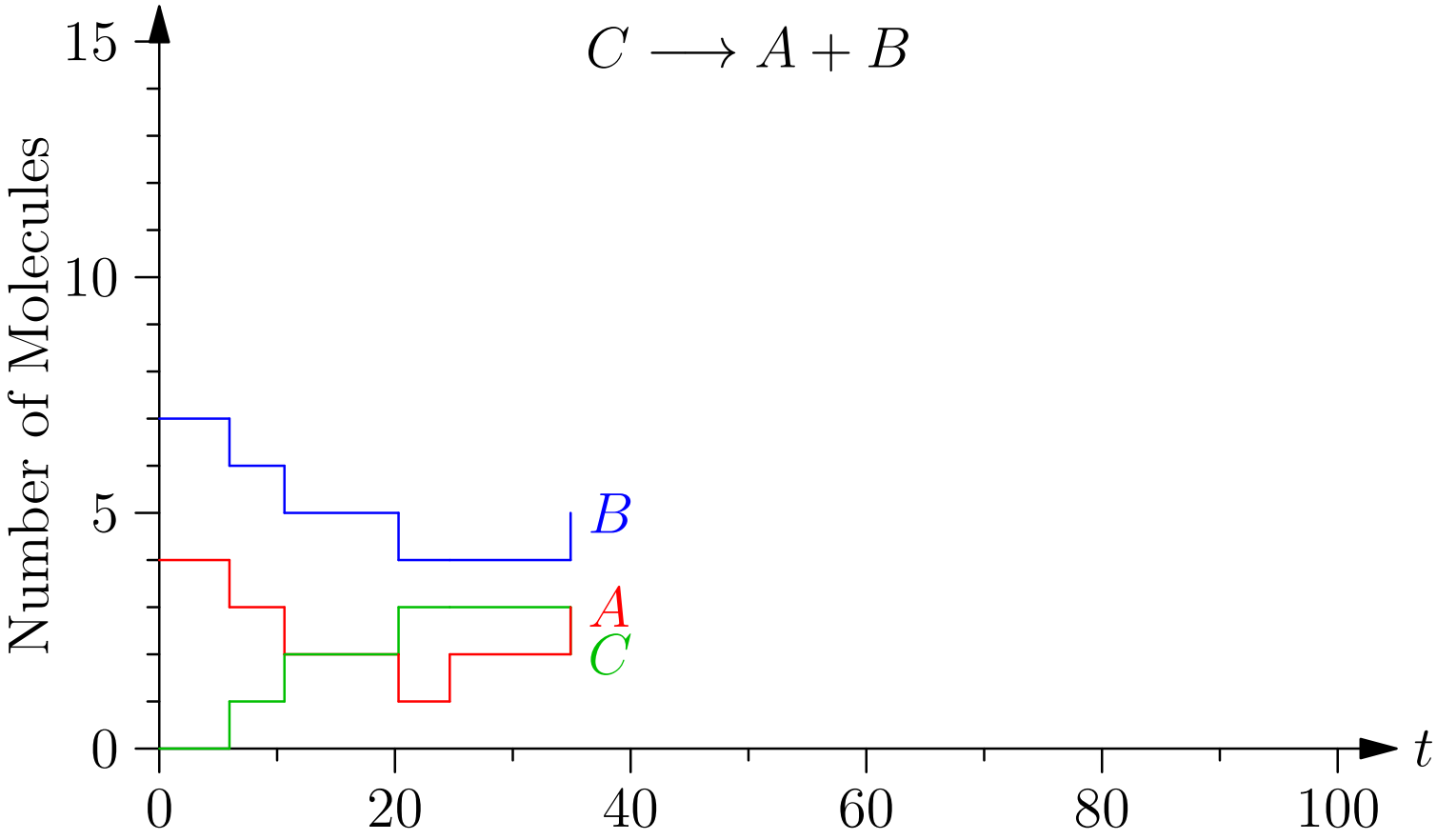

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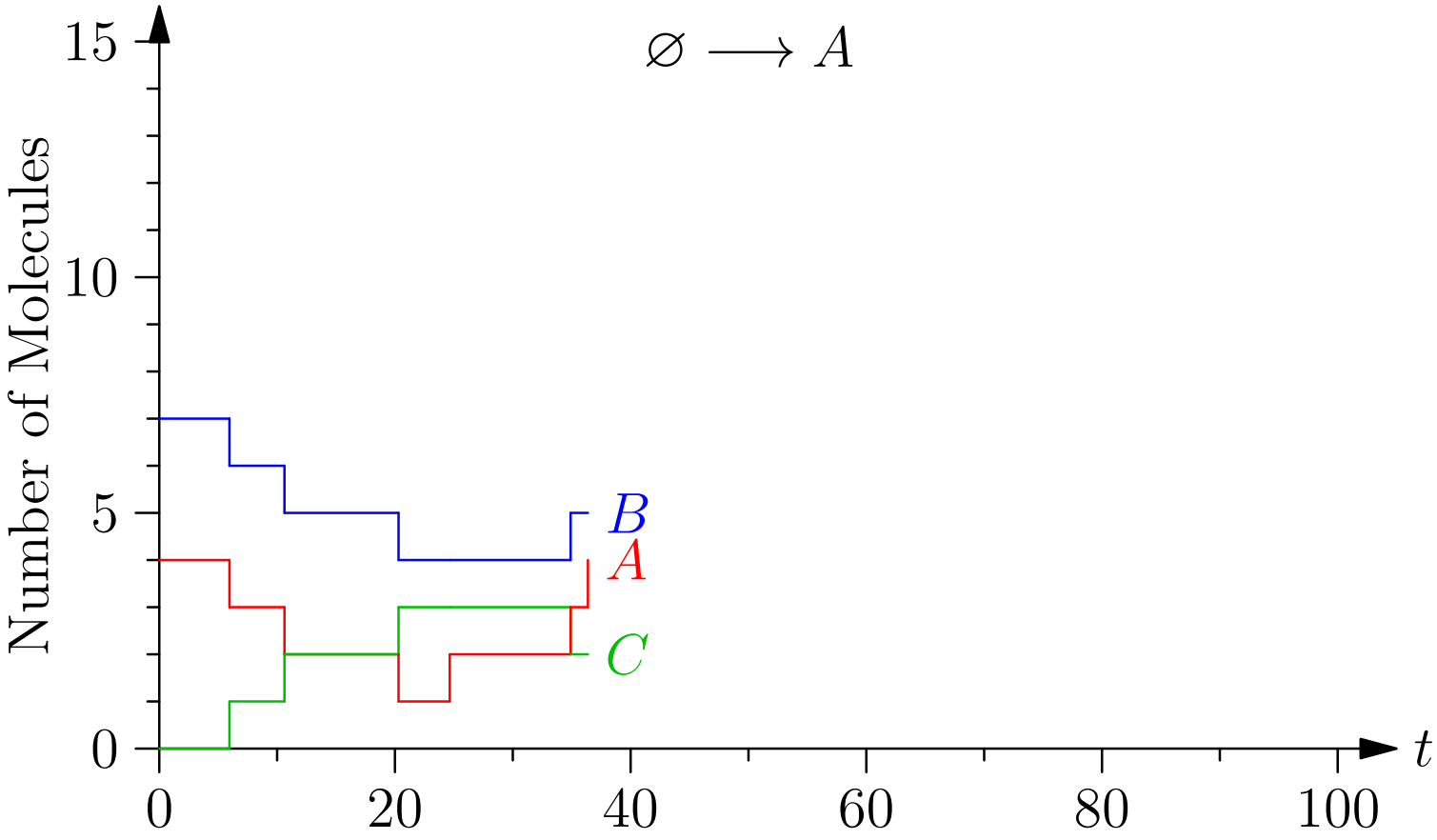

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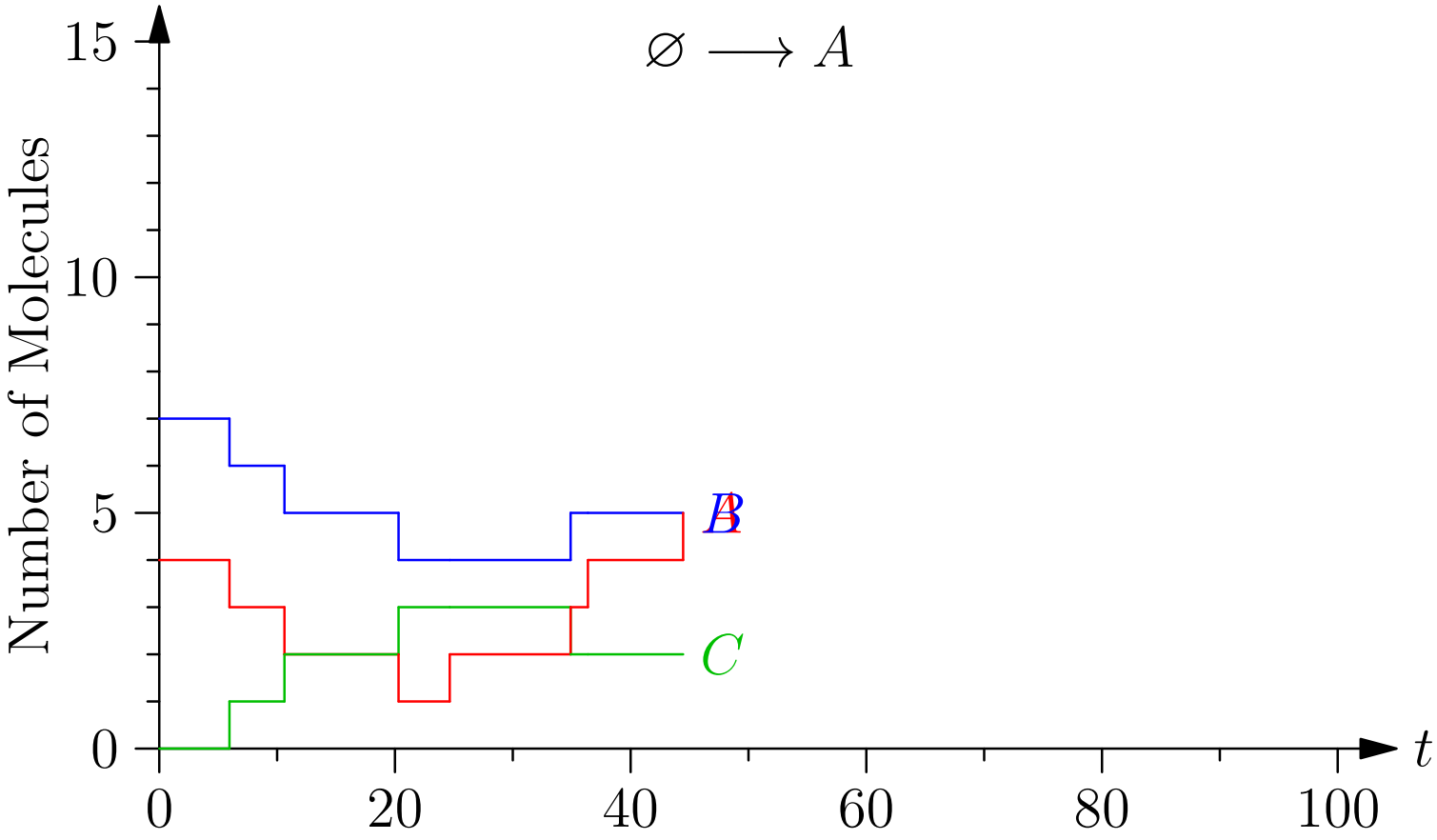

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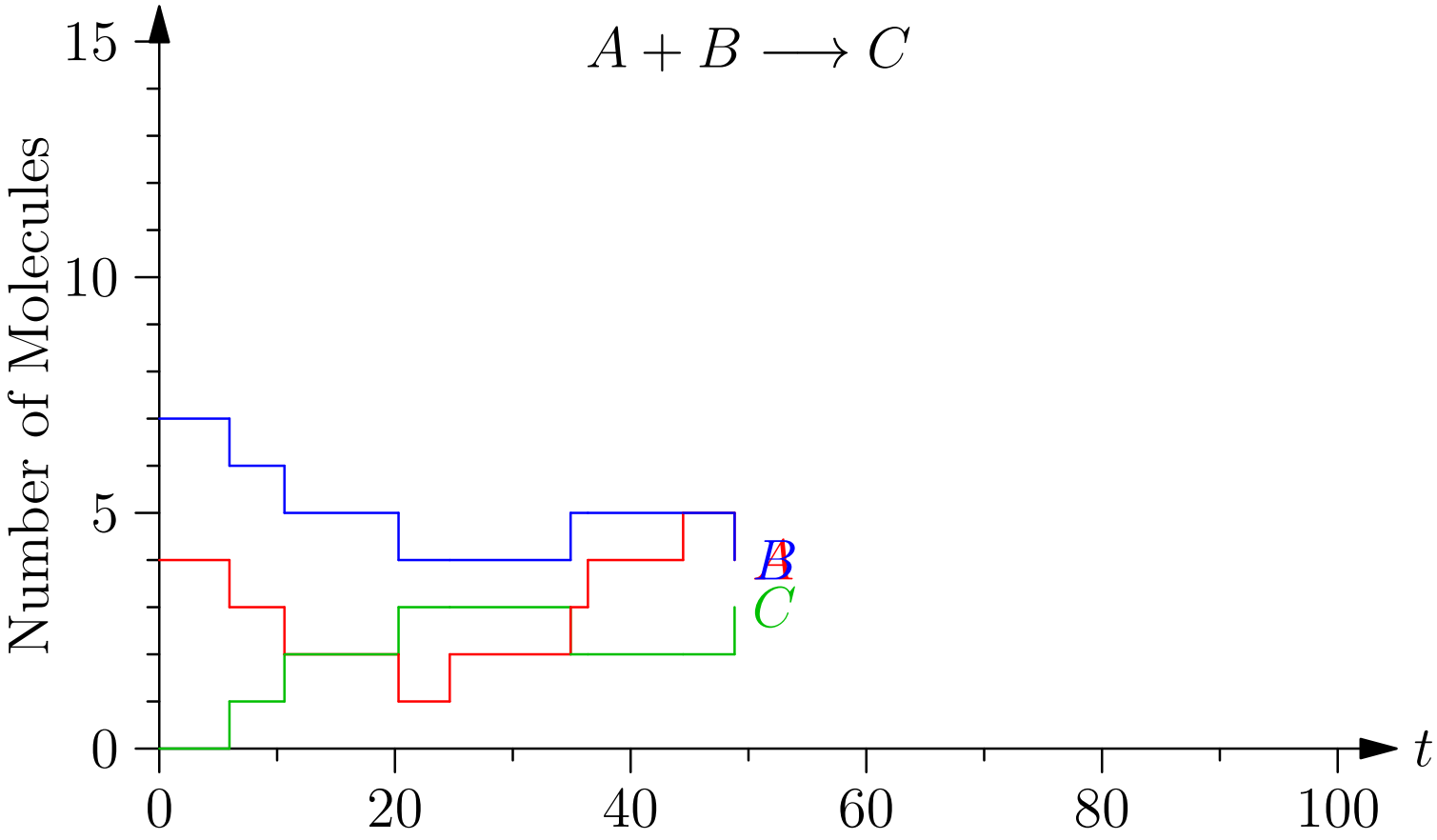

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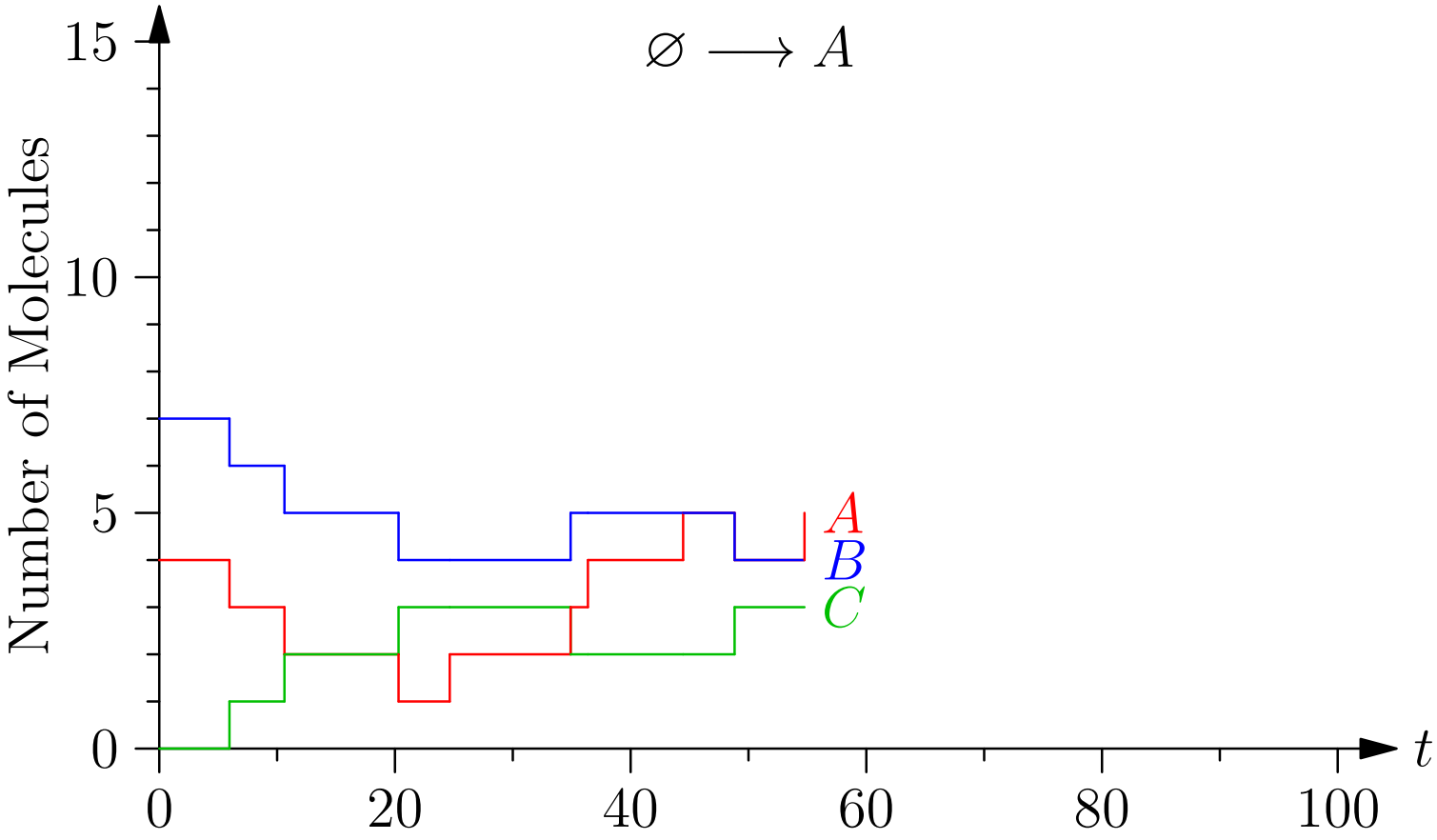

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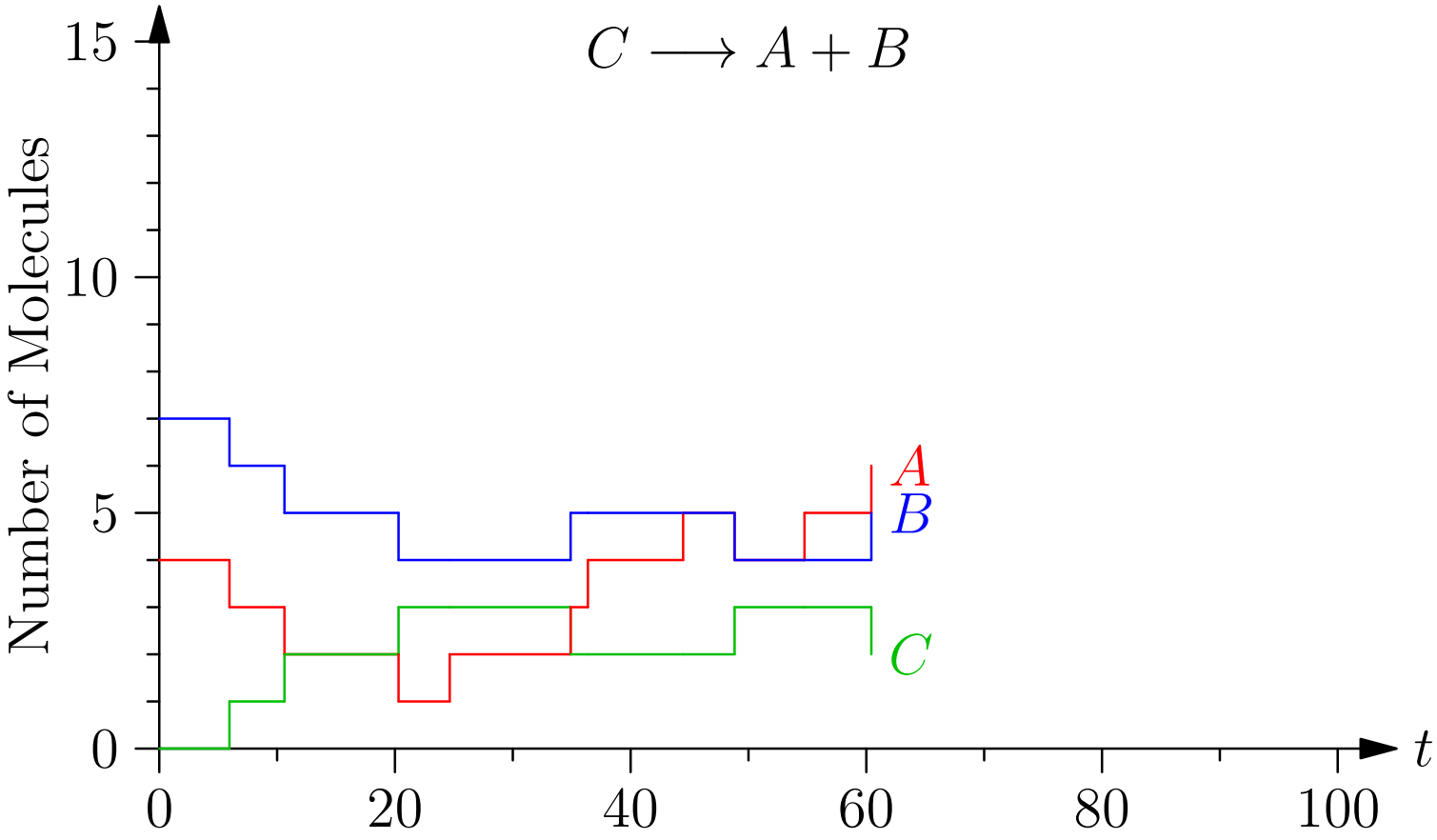

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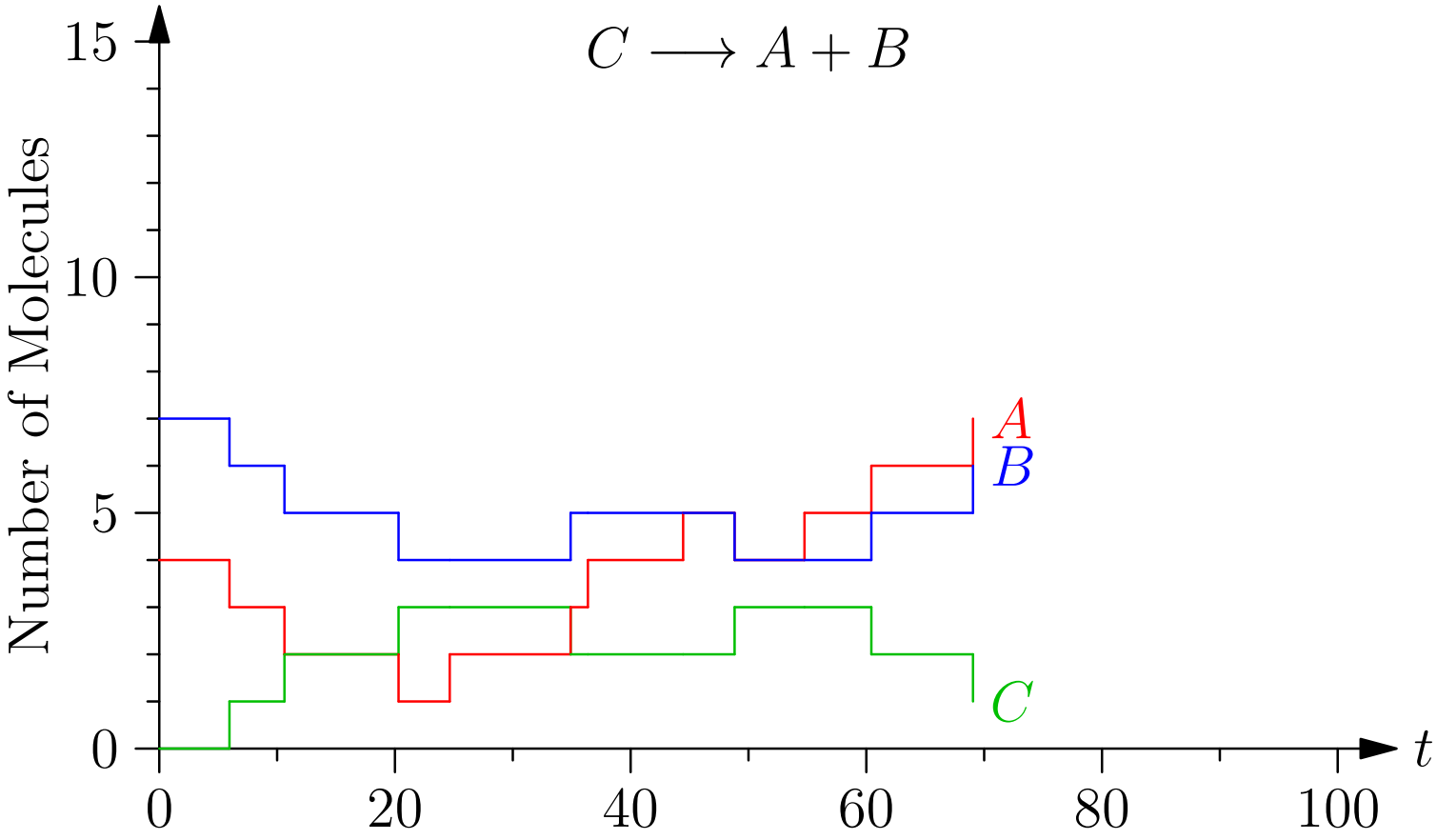

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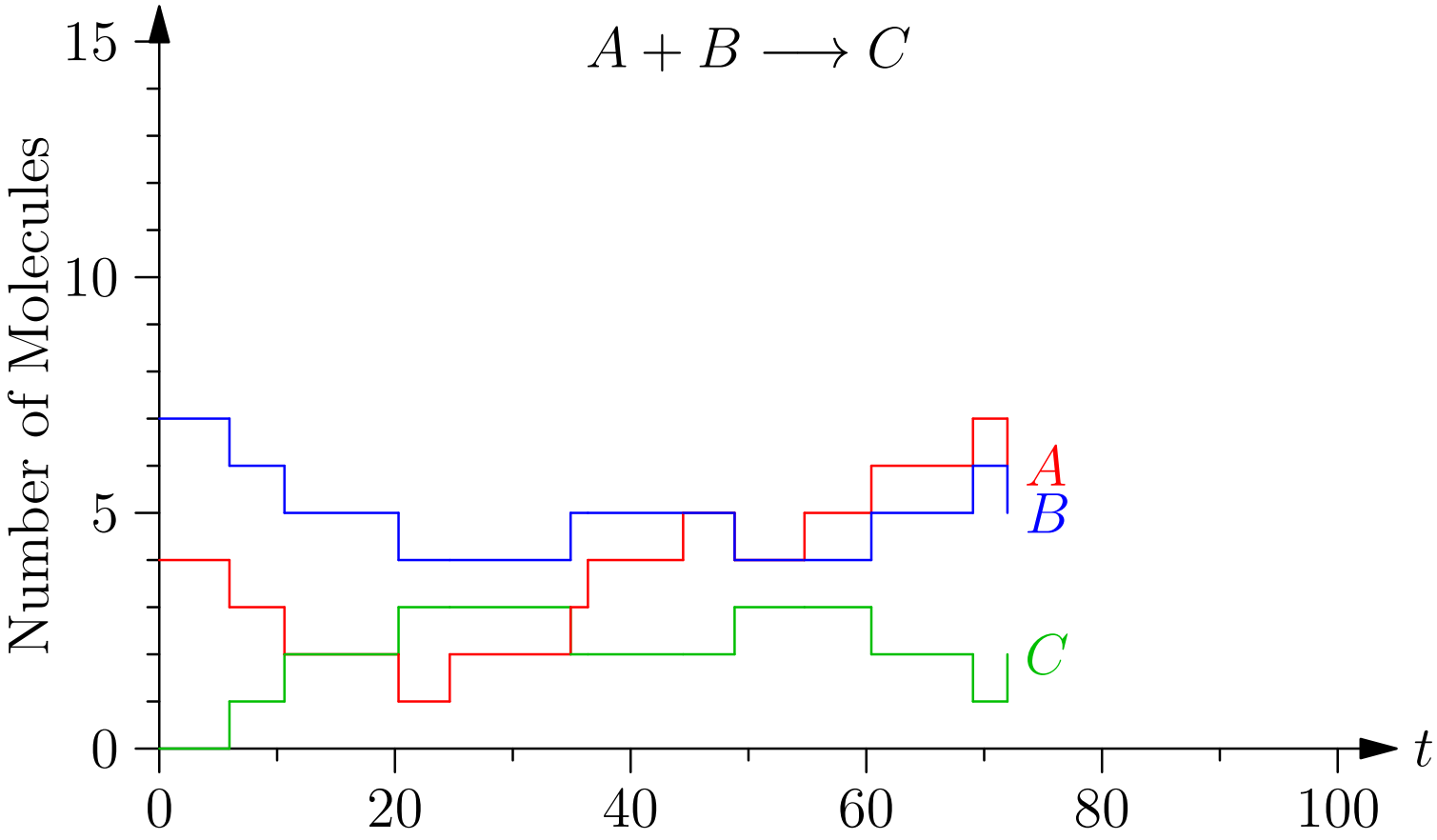

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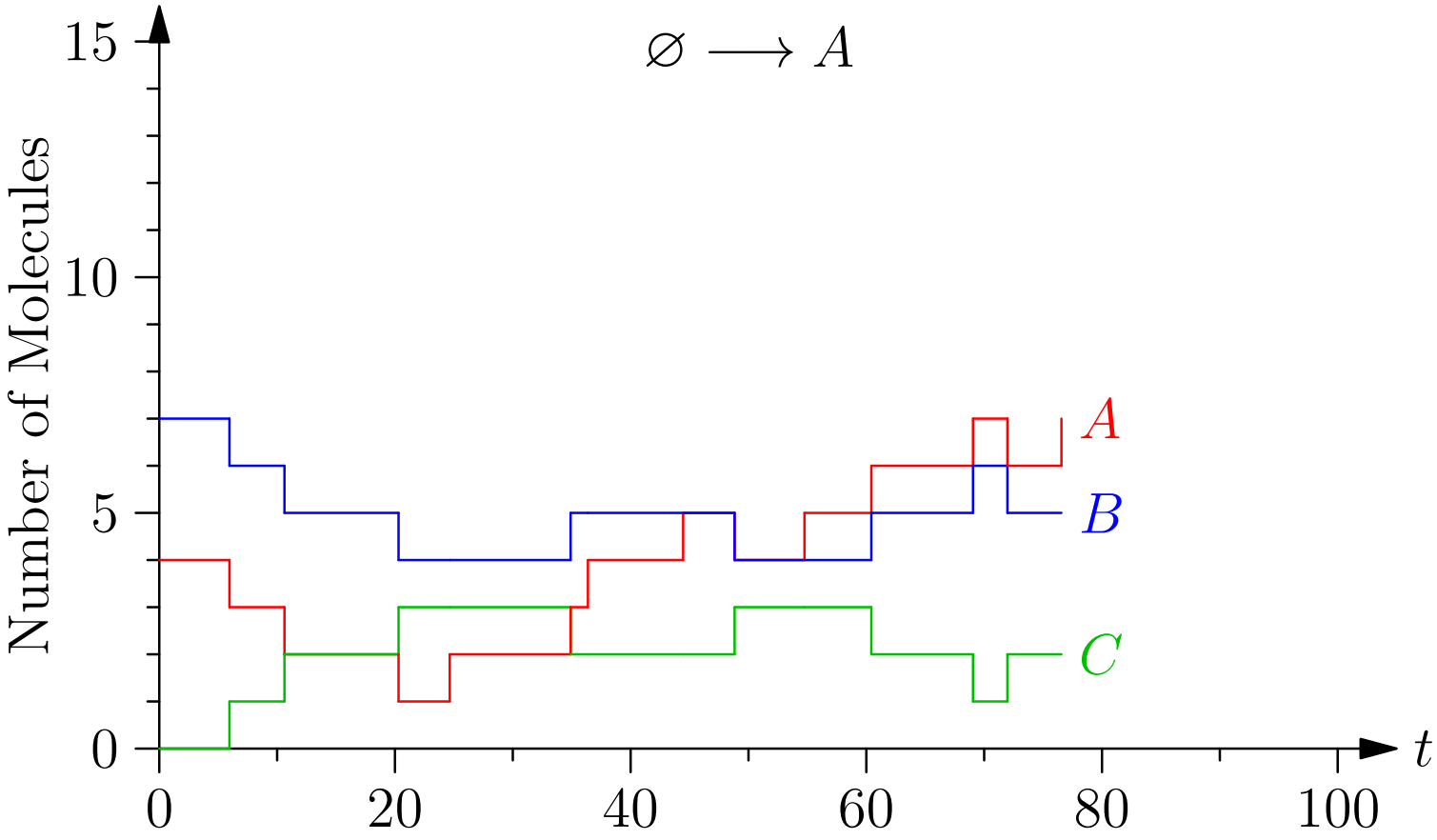

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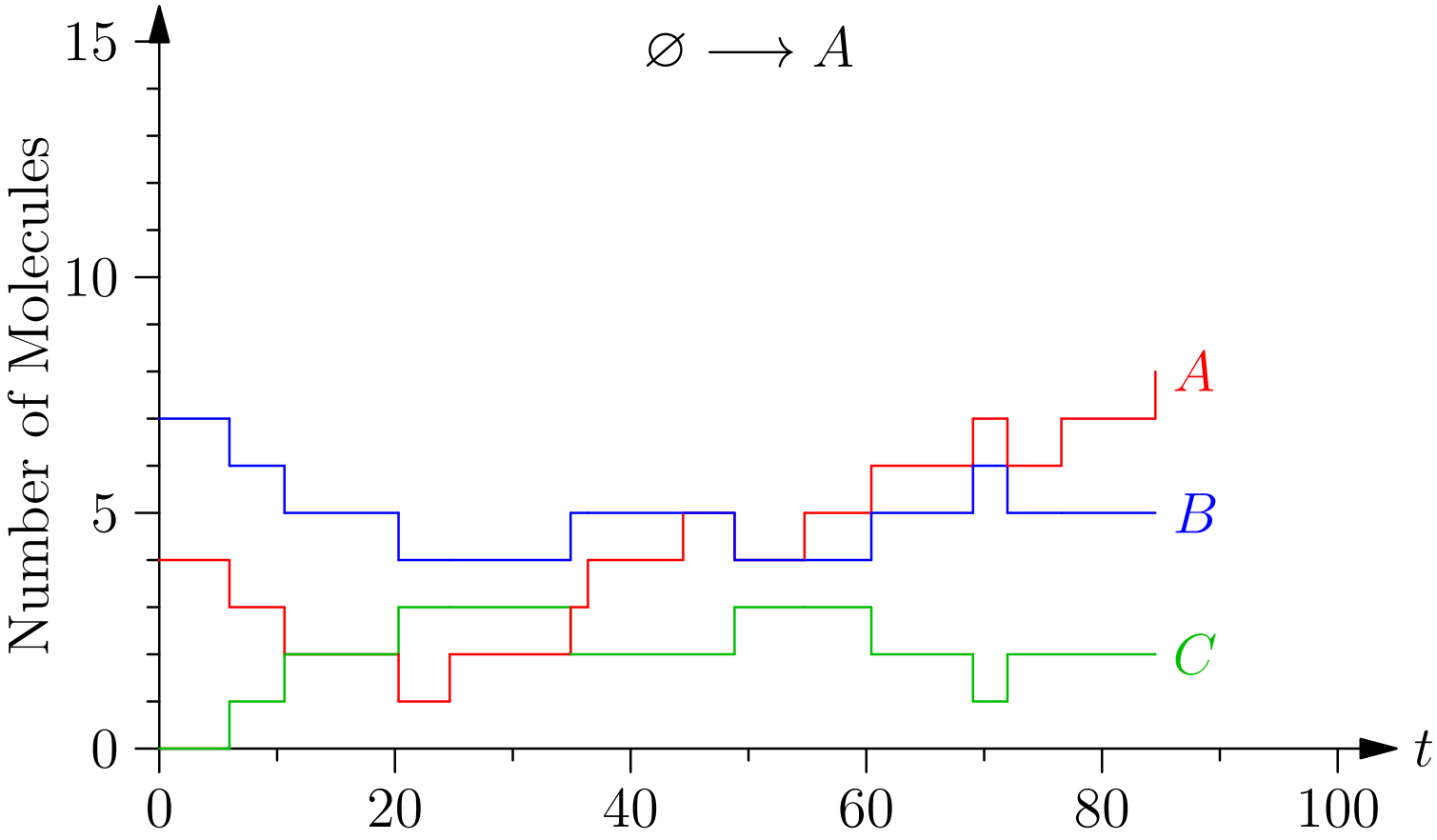

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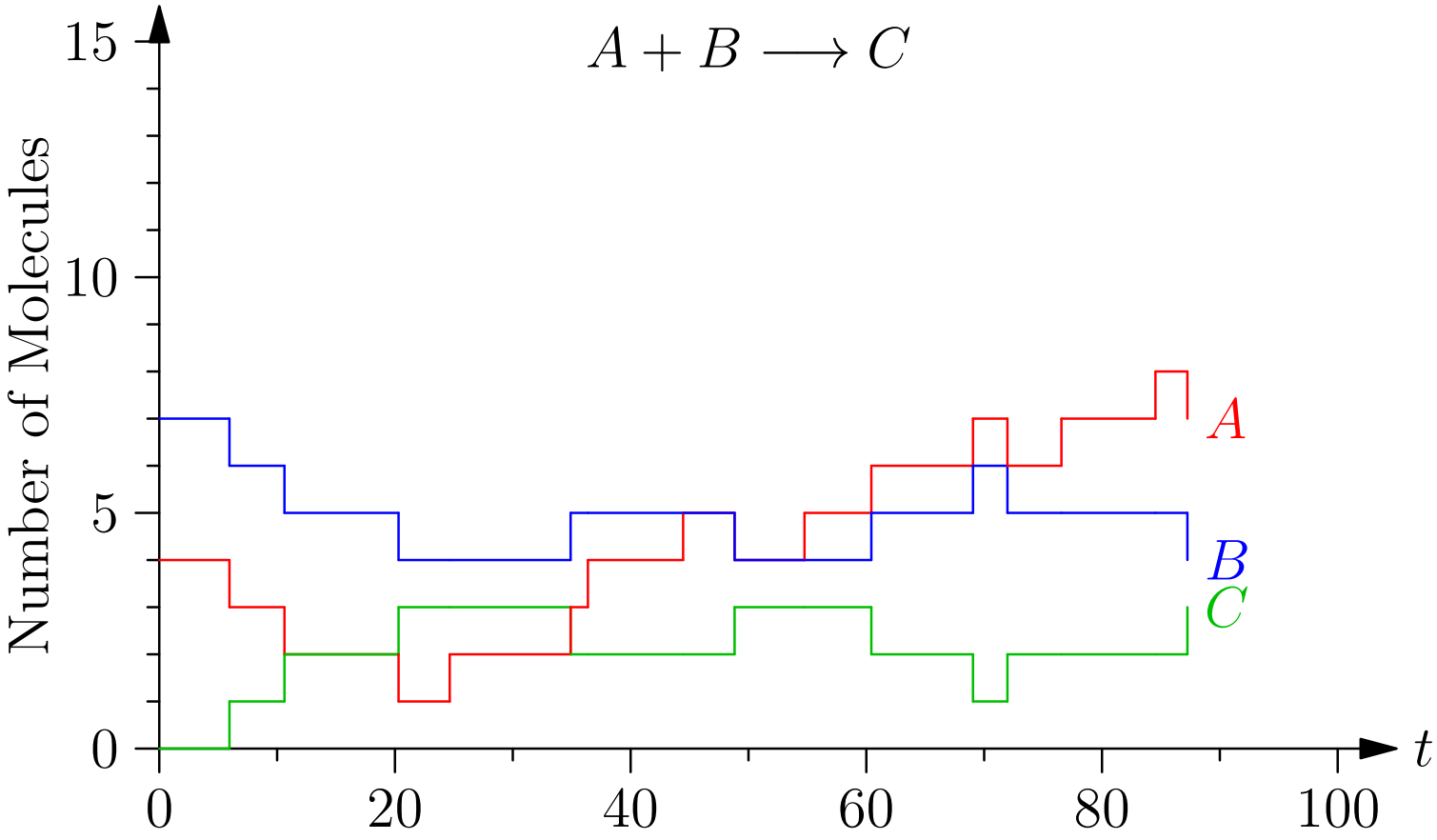

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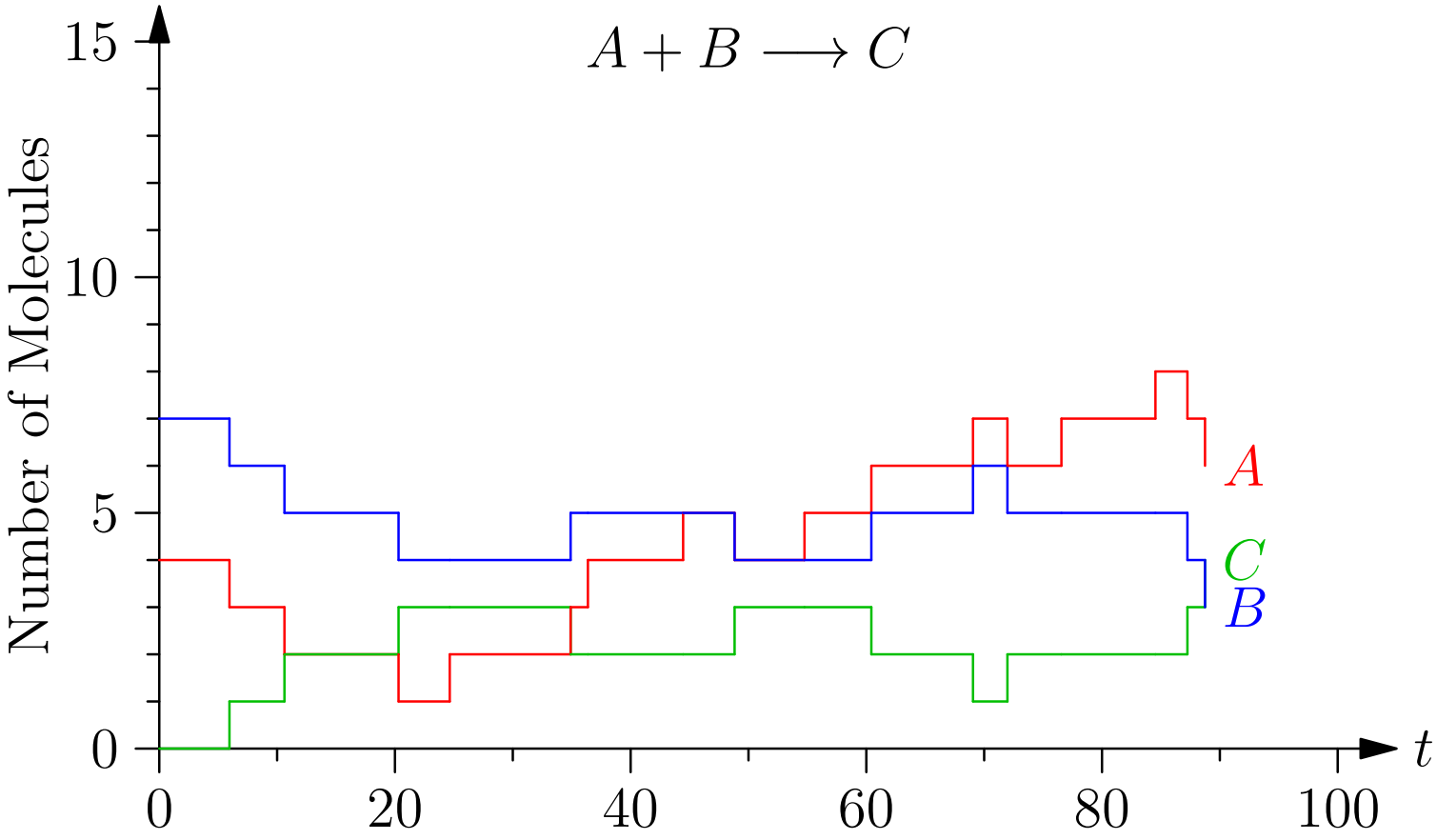

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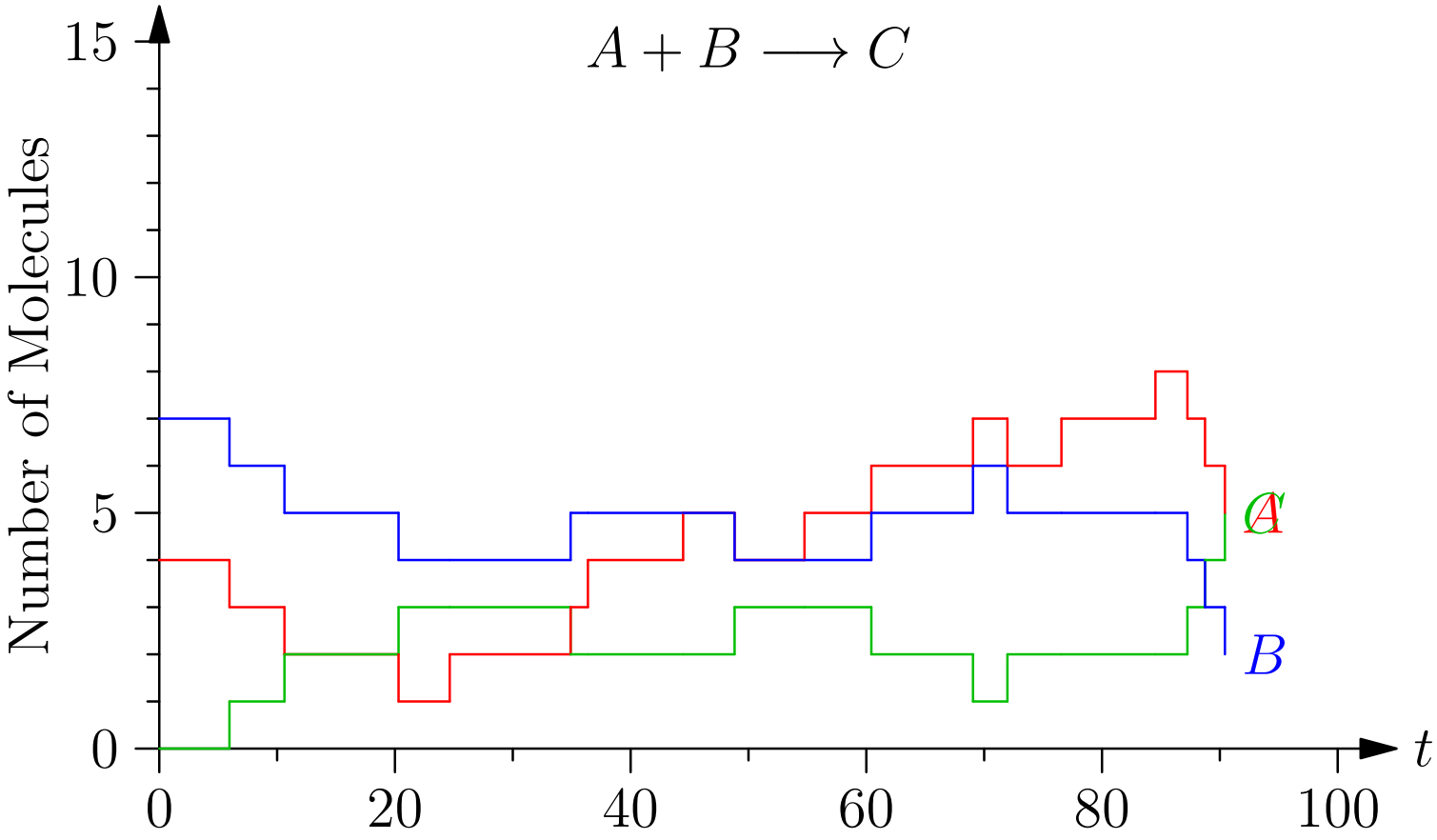

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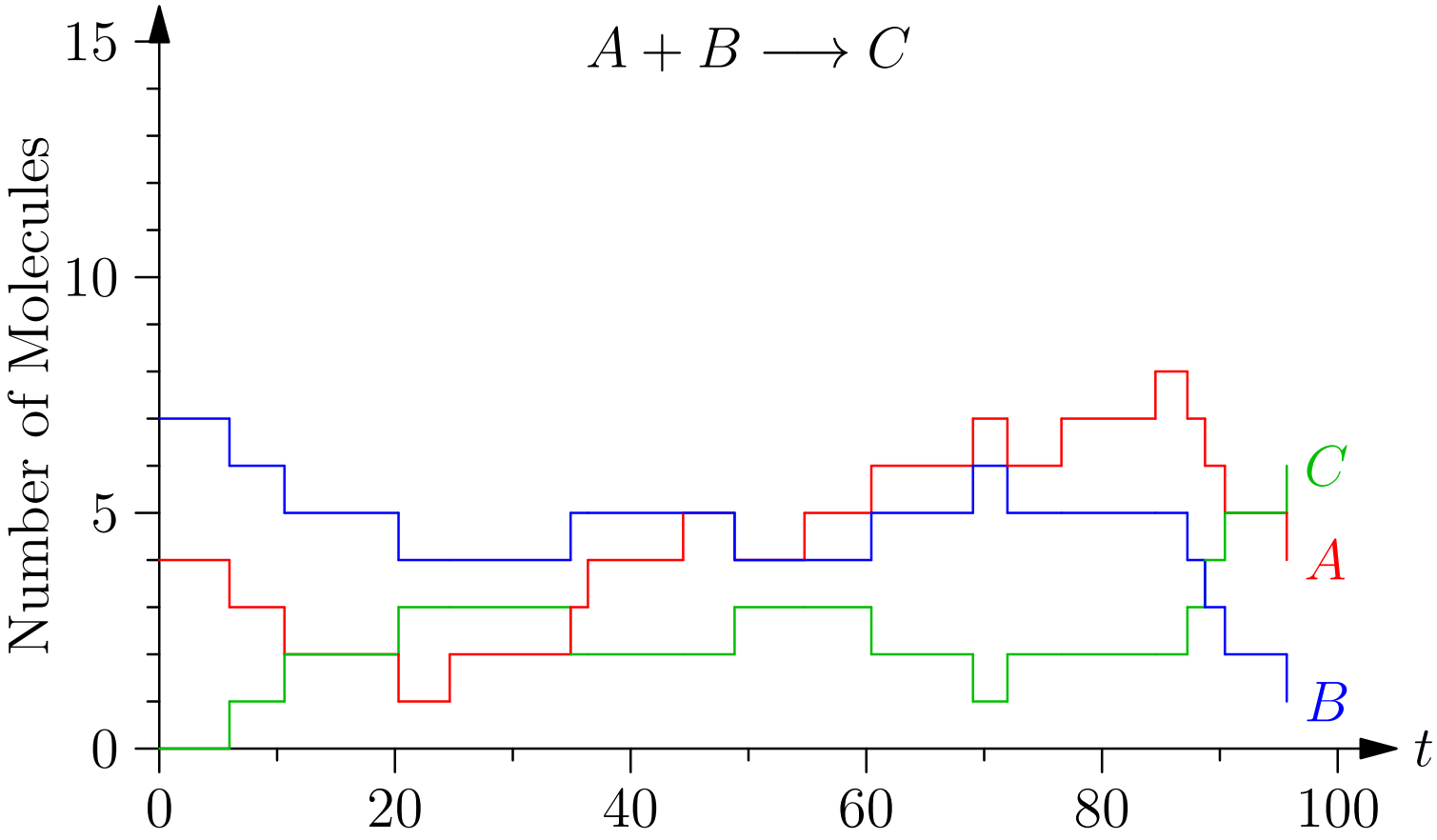

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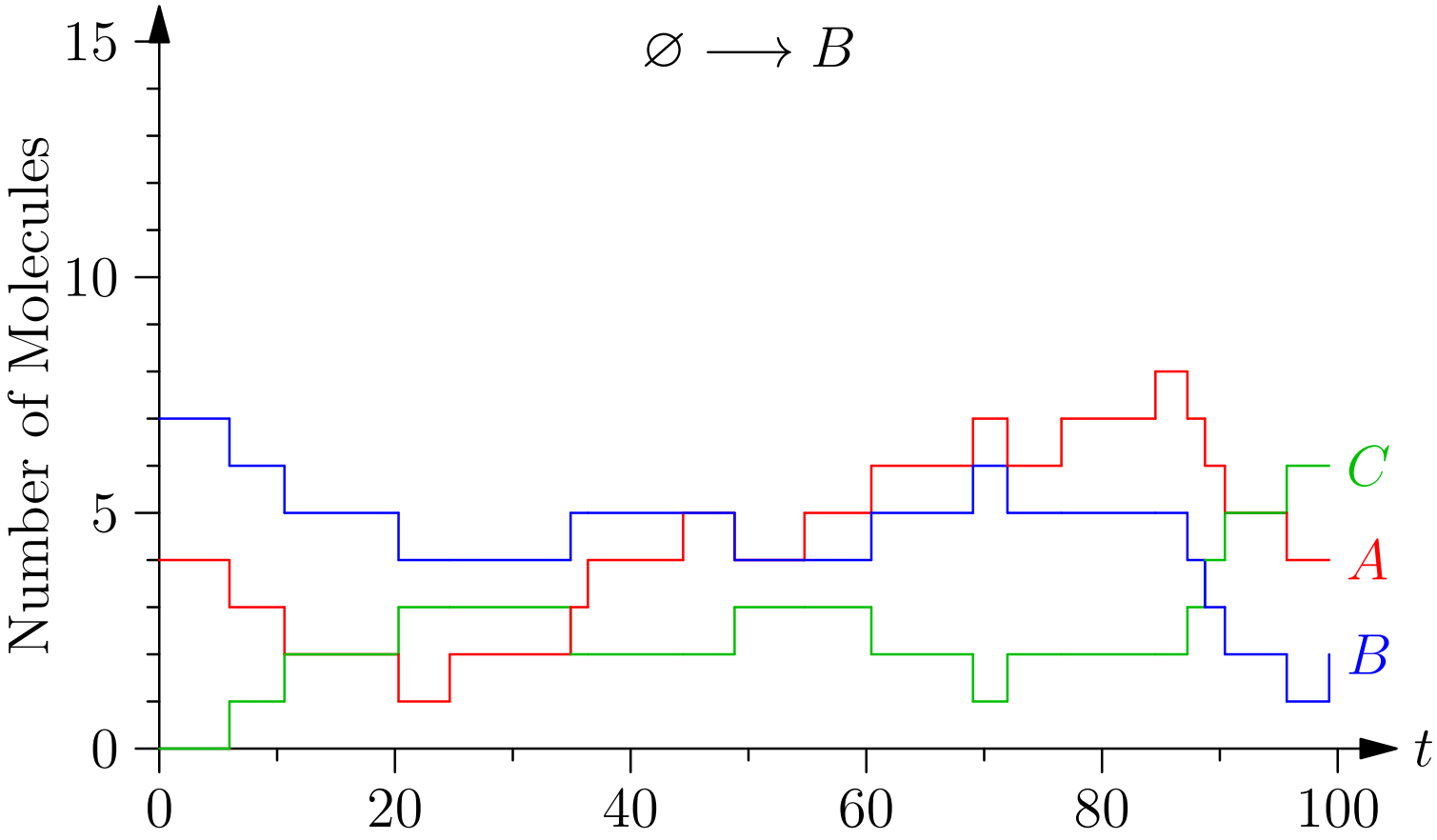

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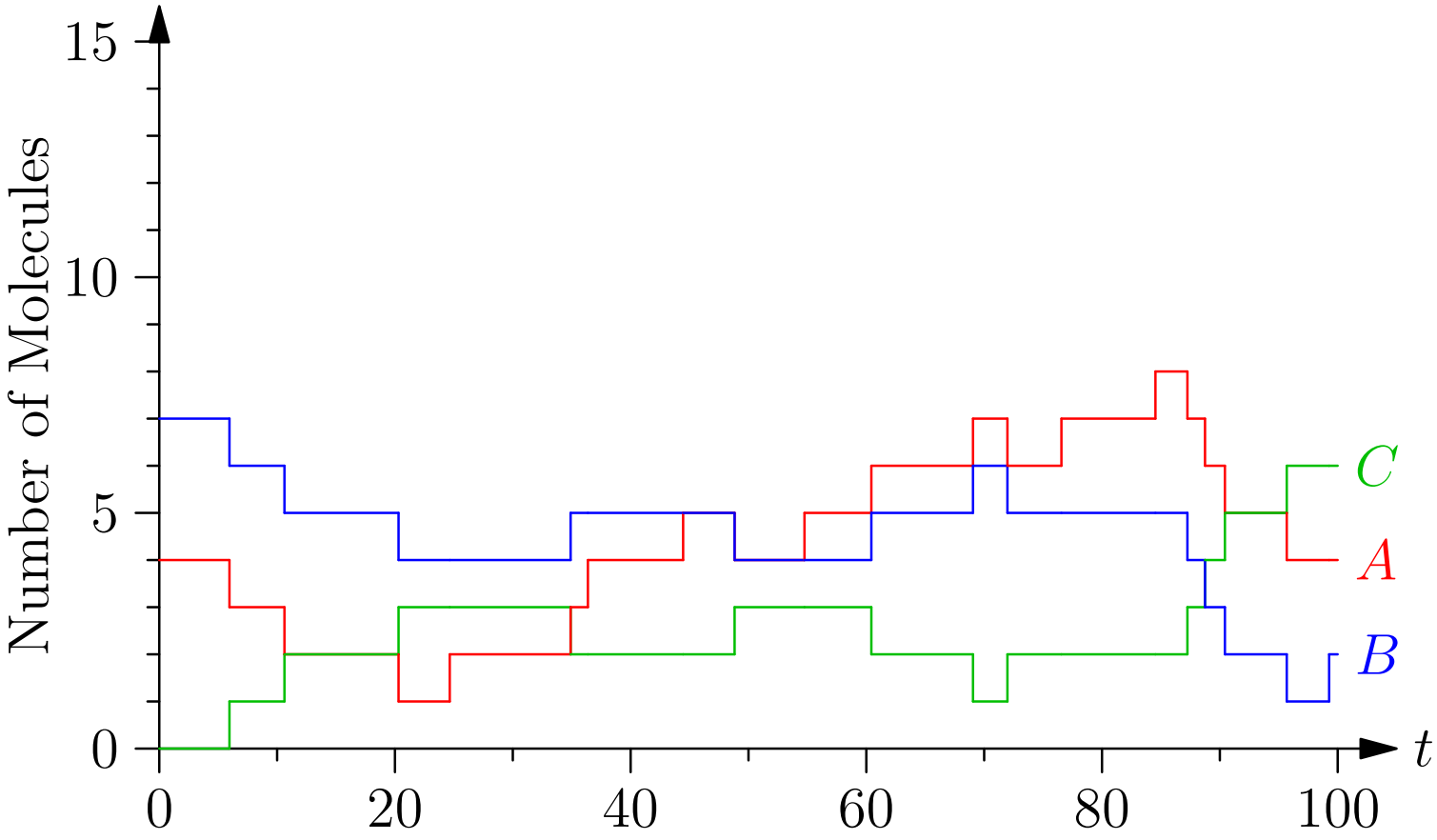

Two Species

- Recall the simple system

$$\varnothing \xrightarrow{f(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$f(X_2) = \frac{k_1}{1 + ([X_2]/K)^n}$$

- Leading to the coupled ODEs

$$\frac{\mathrm{d}[X_1(t)]}{\mathrm{d}t} = \frac{k_1}{1 + ([X_2(t)]/K)^n} - k_3\,[X_1(t)] - k_5\,[X_1(t)]$$

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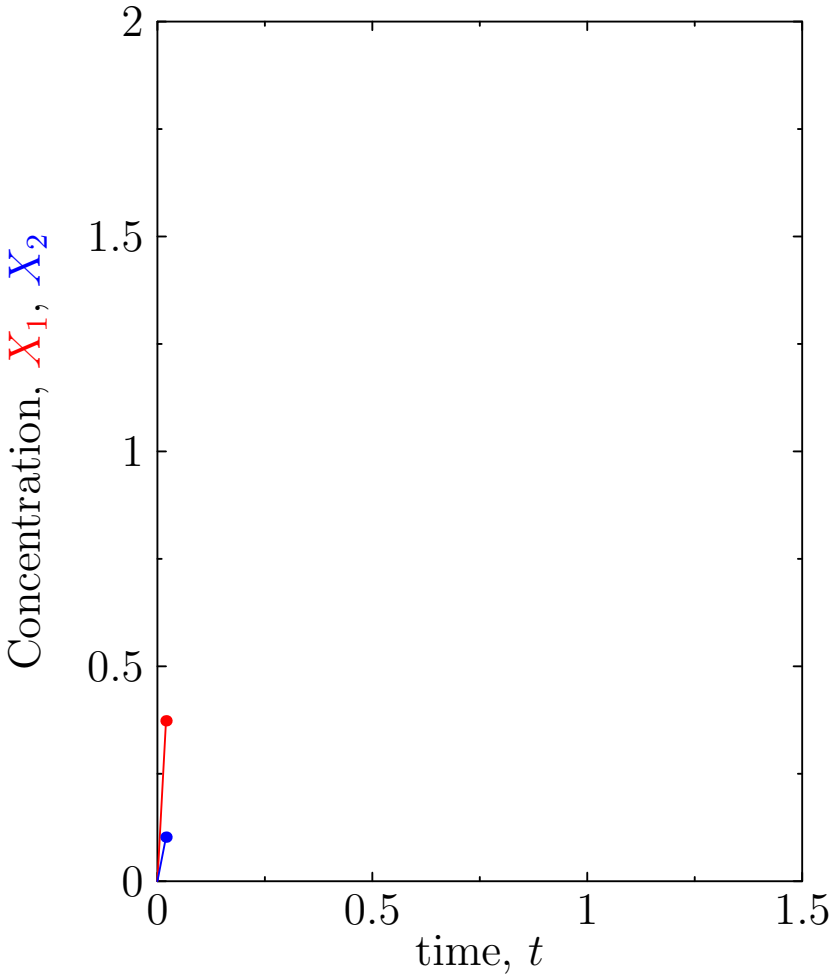
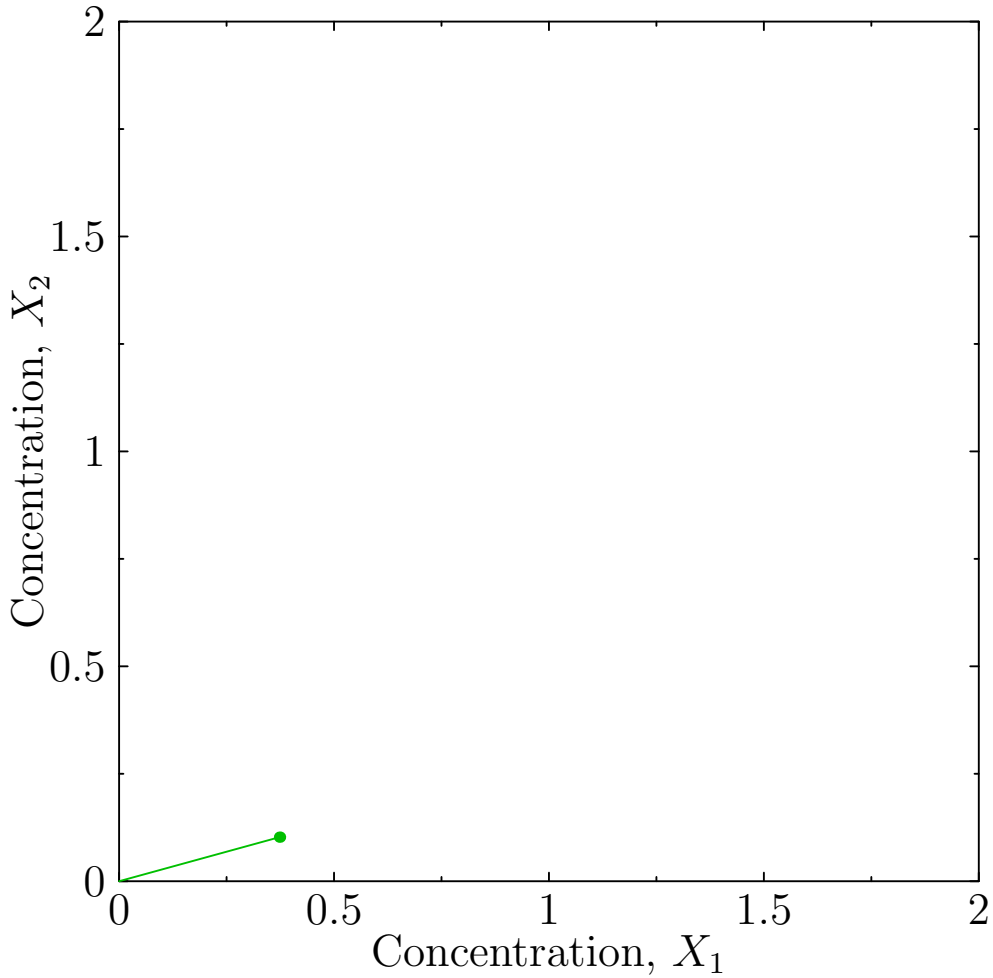
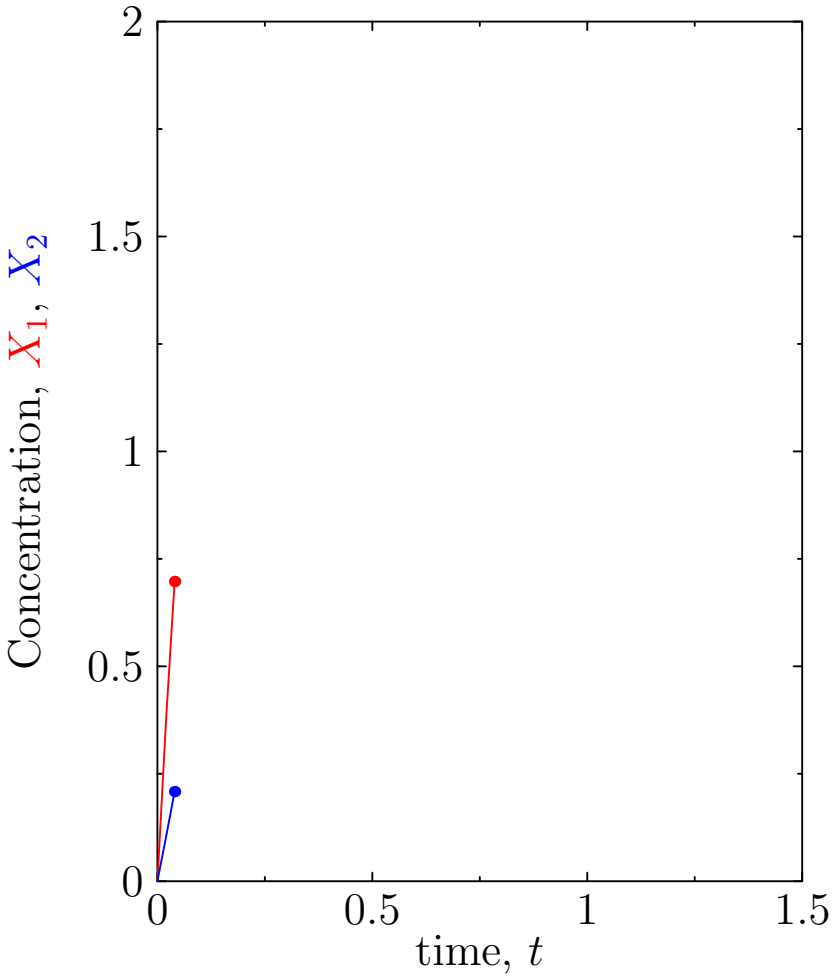
Solving the Dynamics

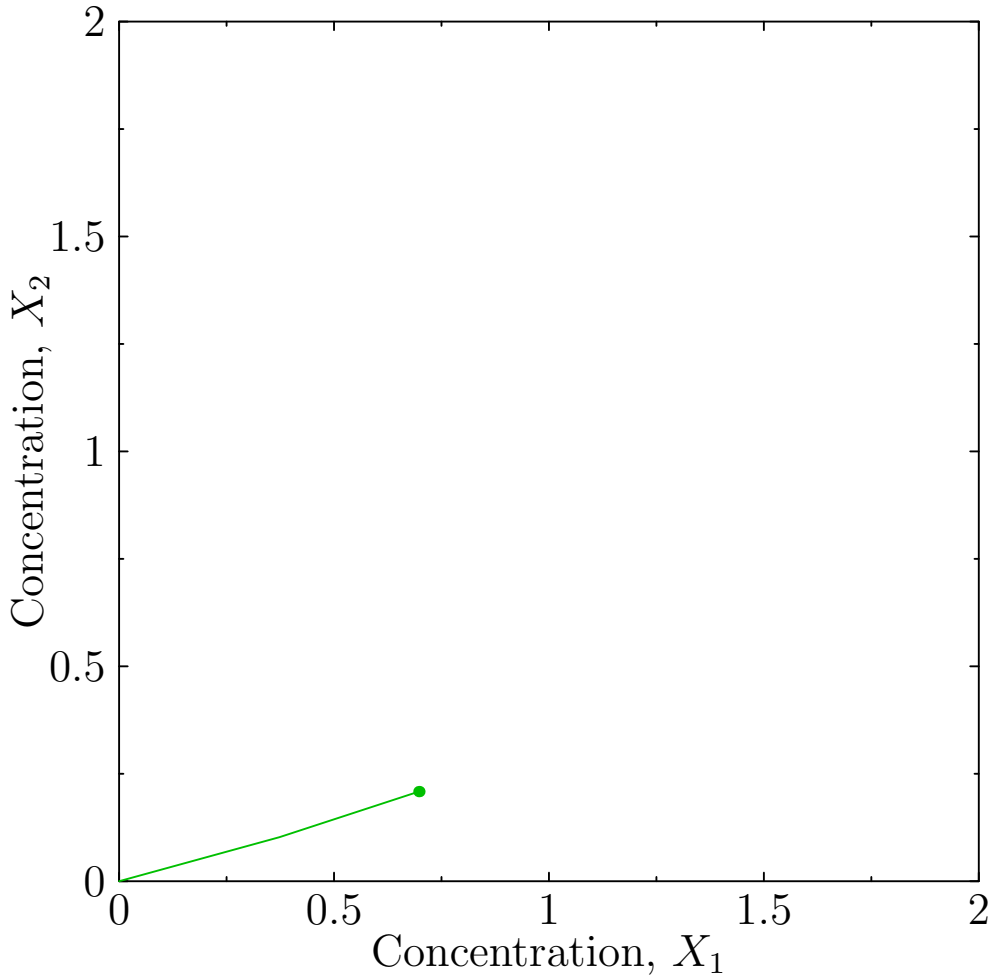




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics





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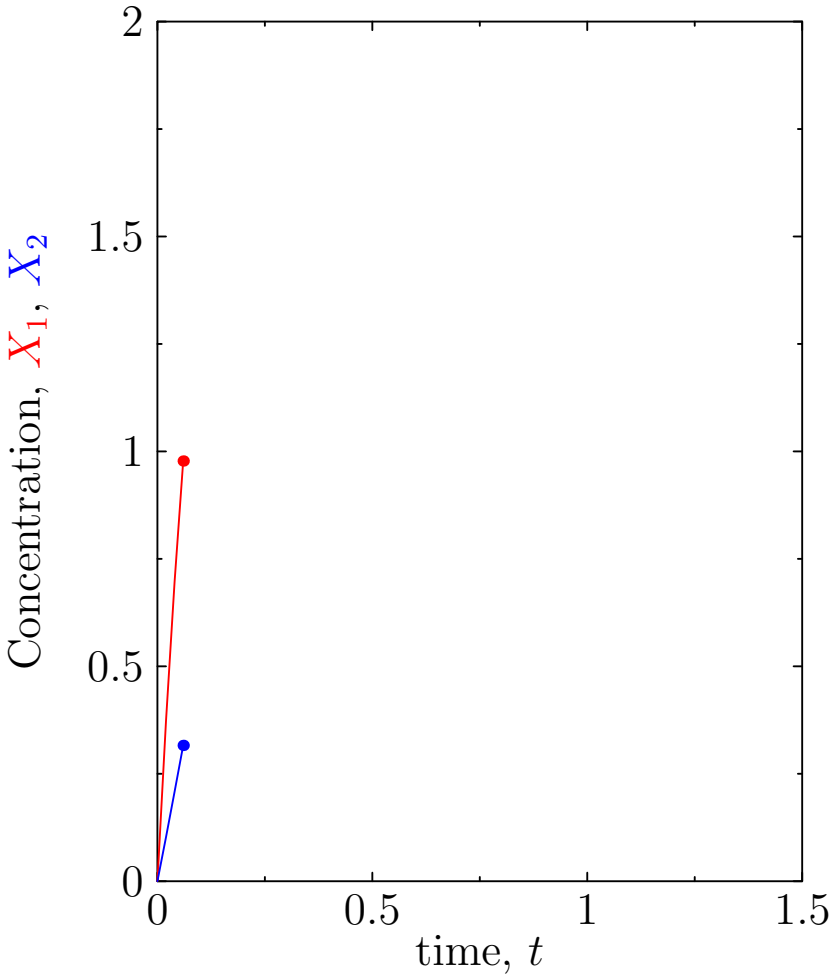

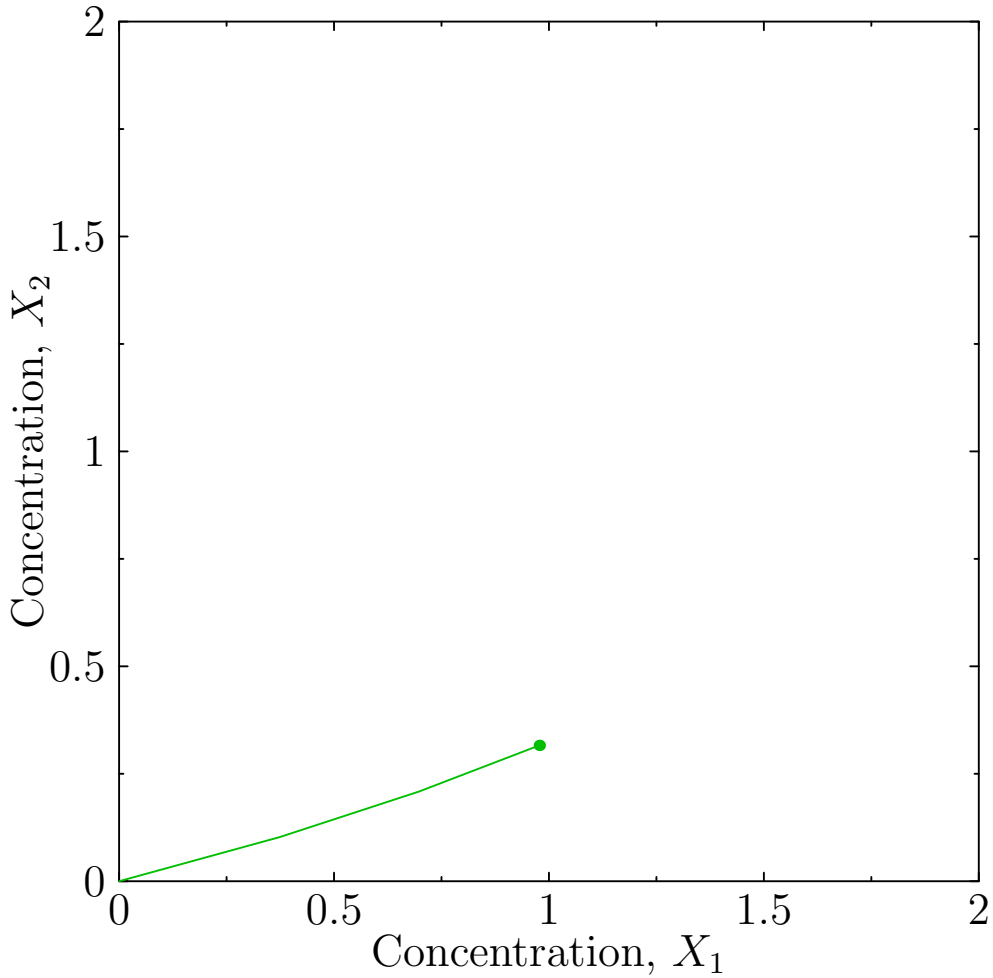

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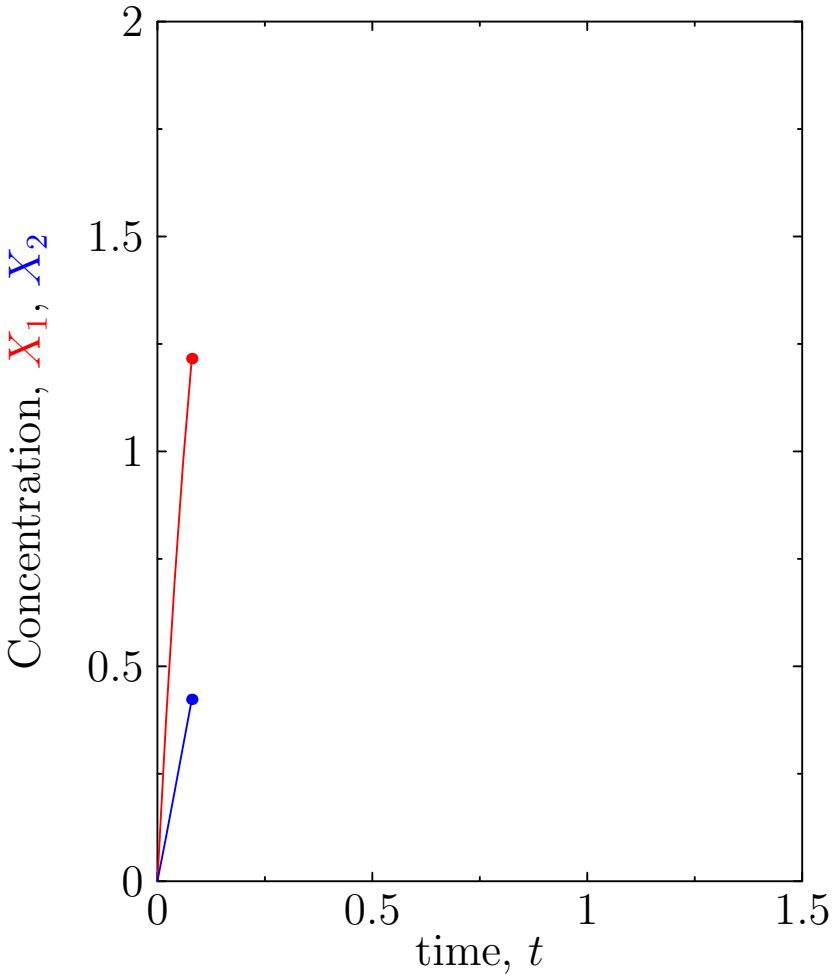

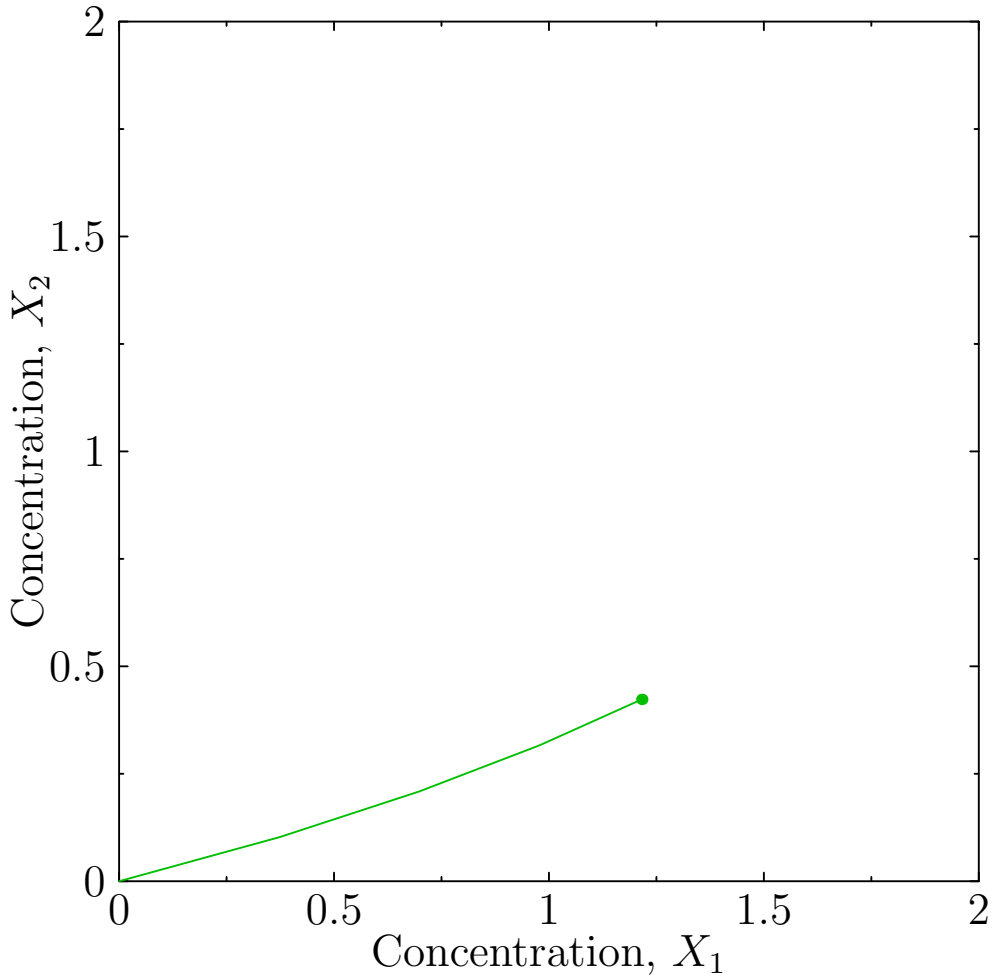

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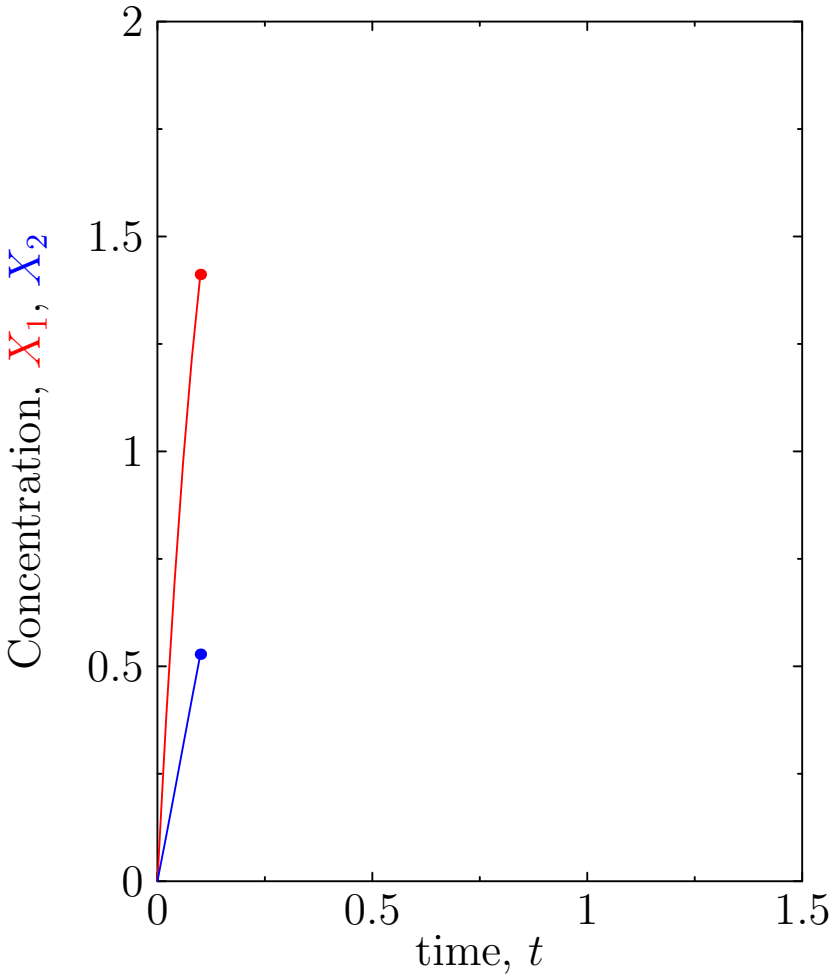

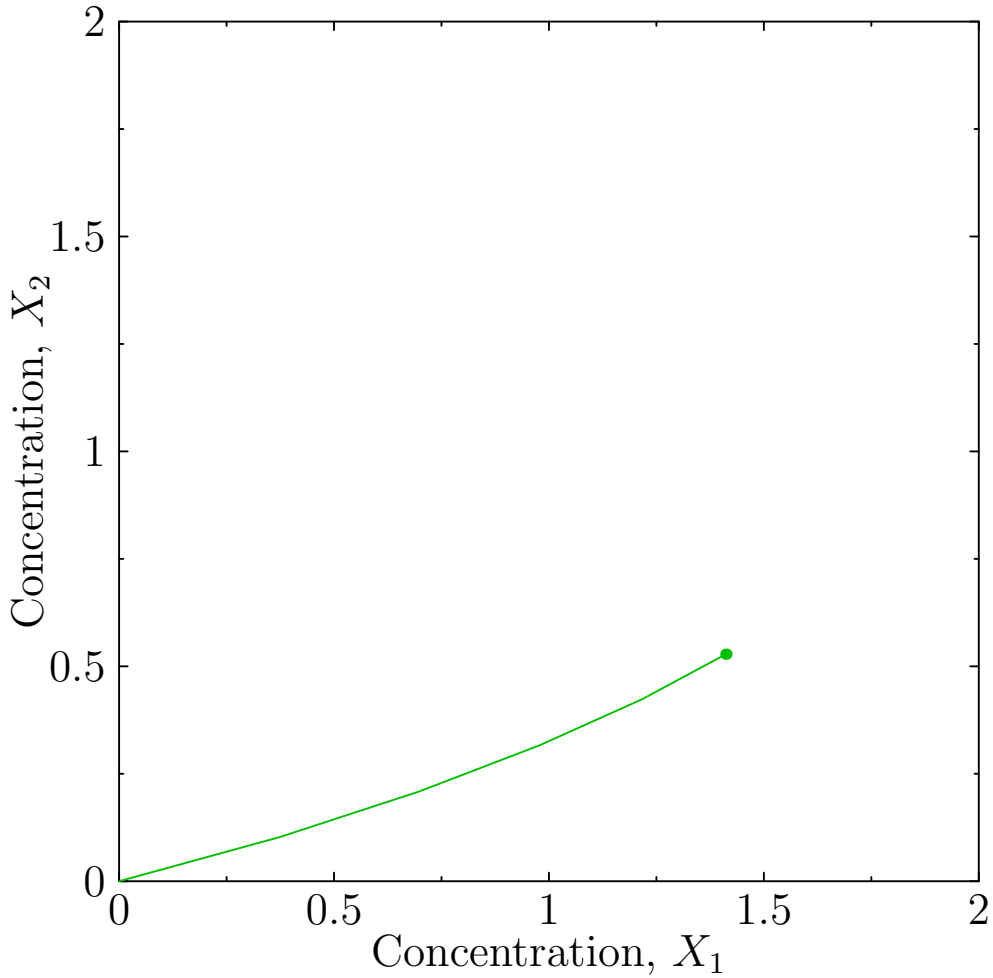

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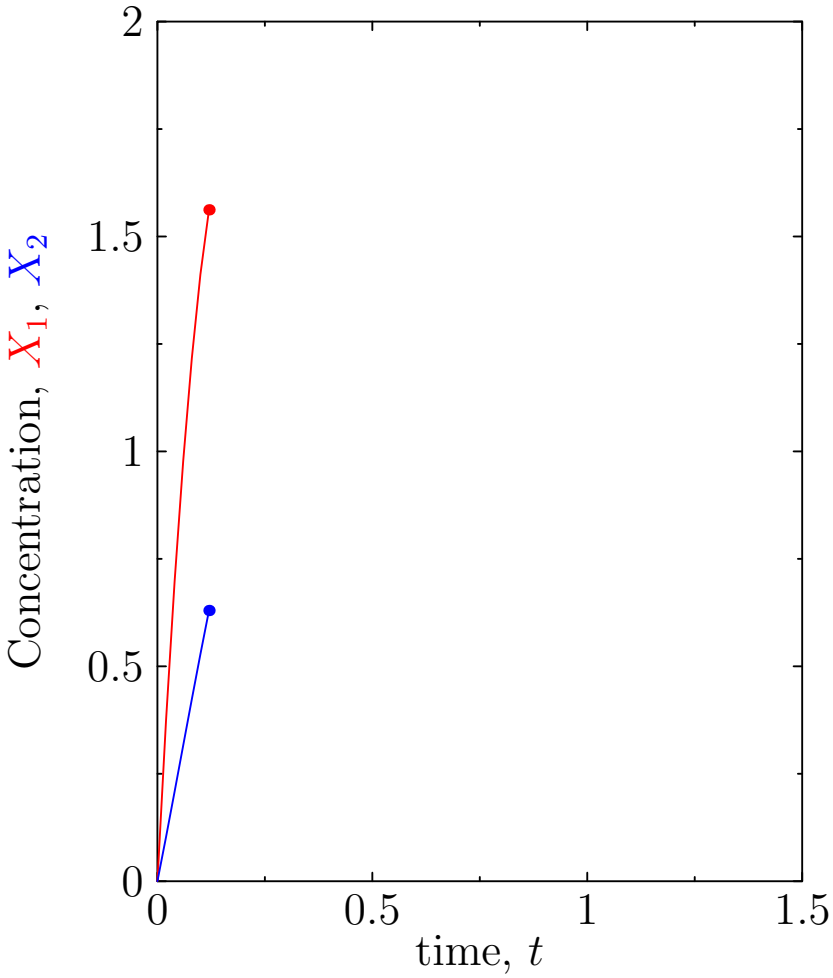

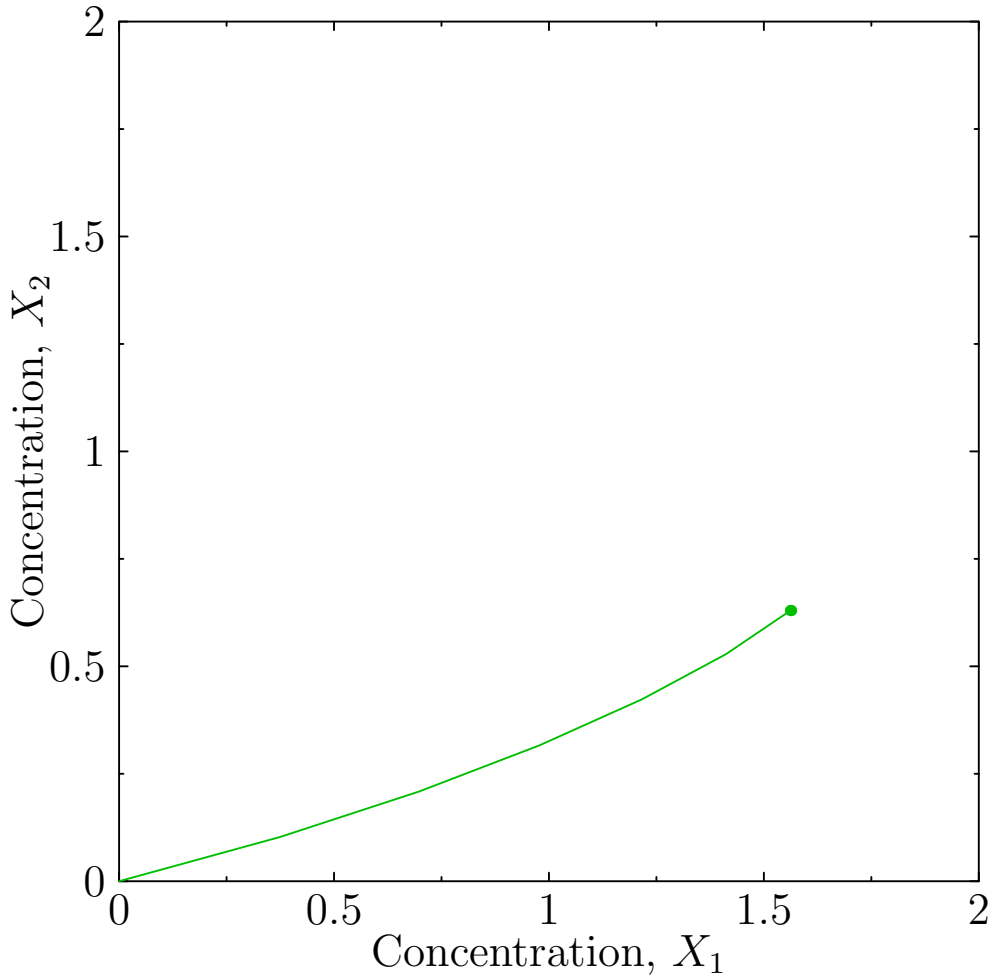

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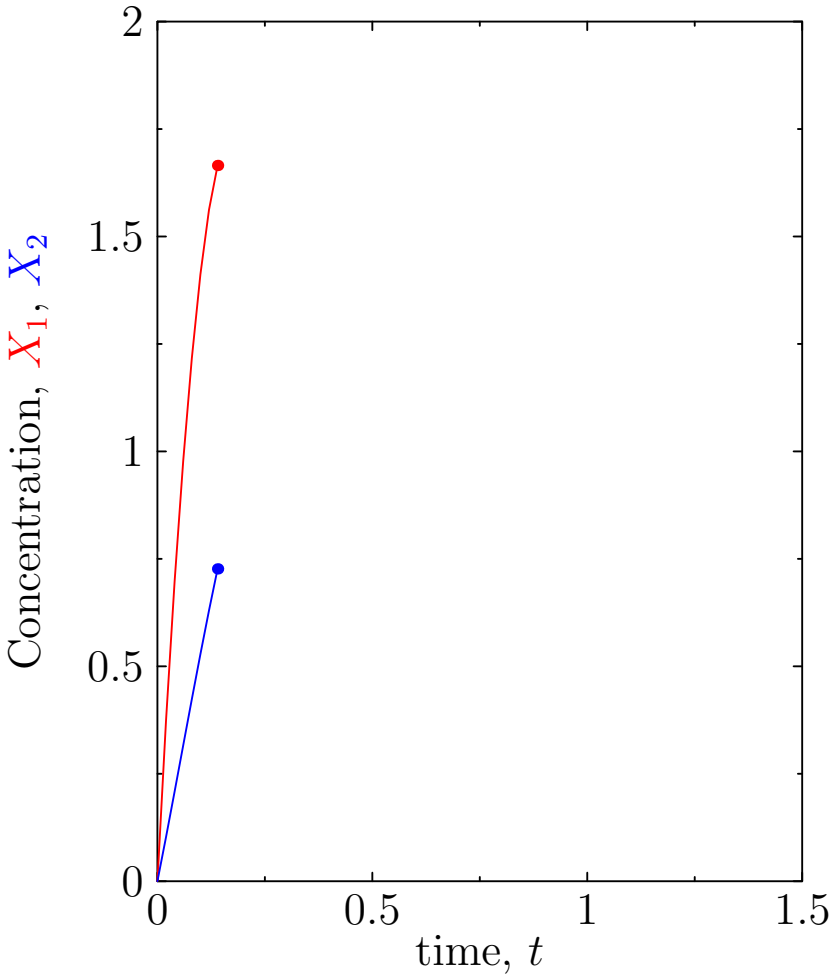

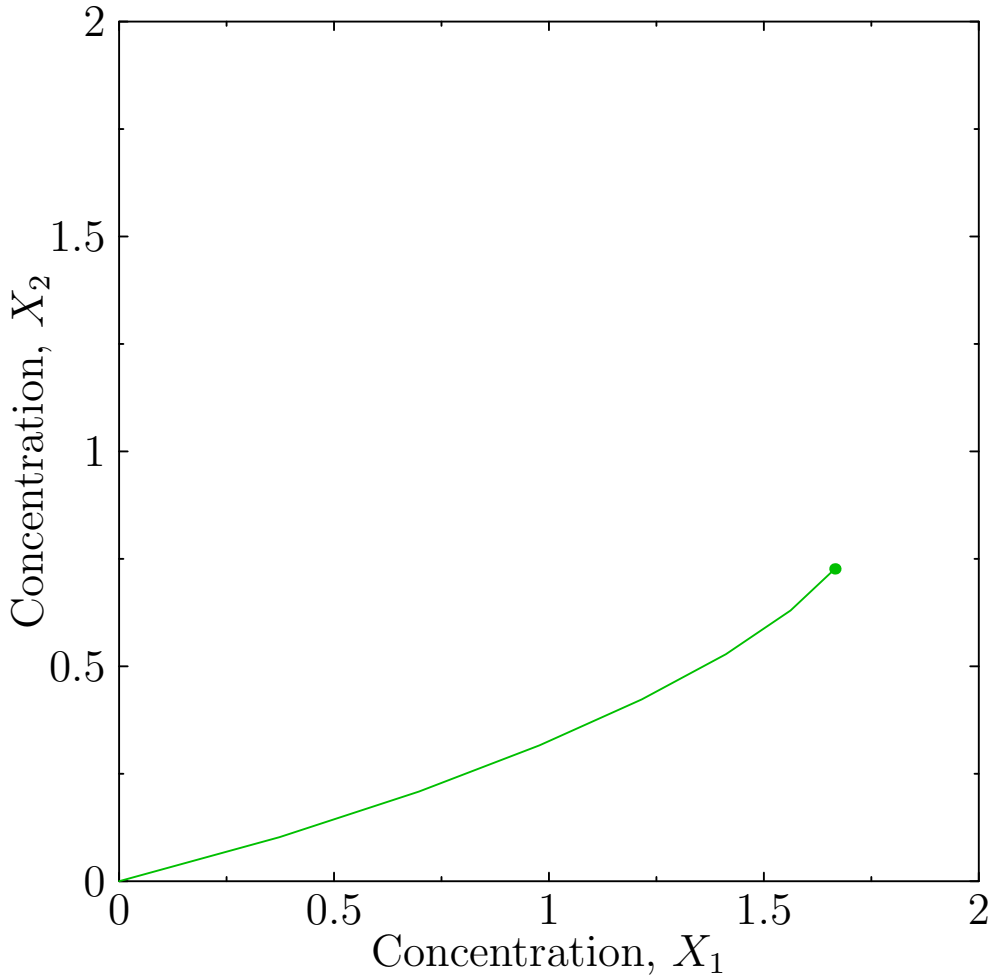

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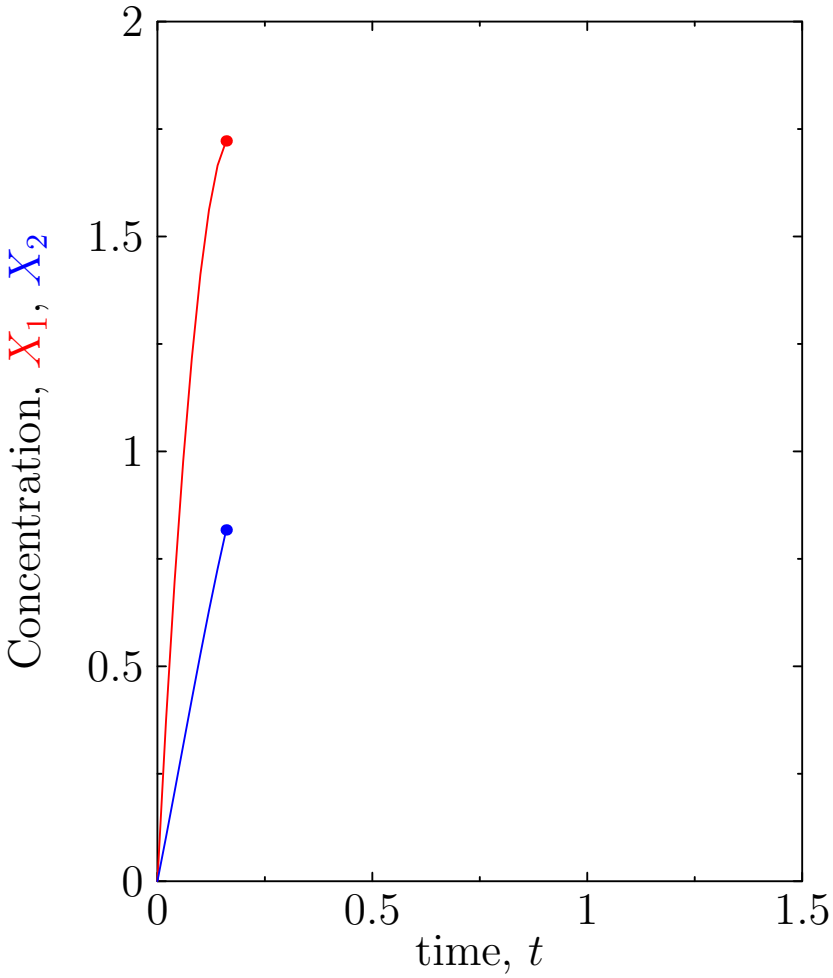

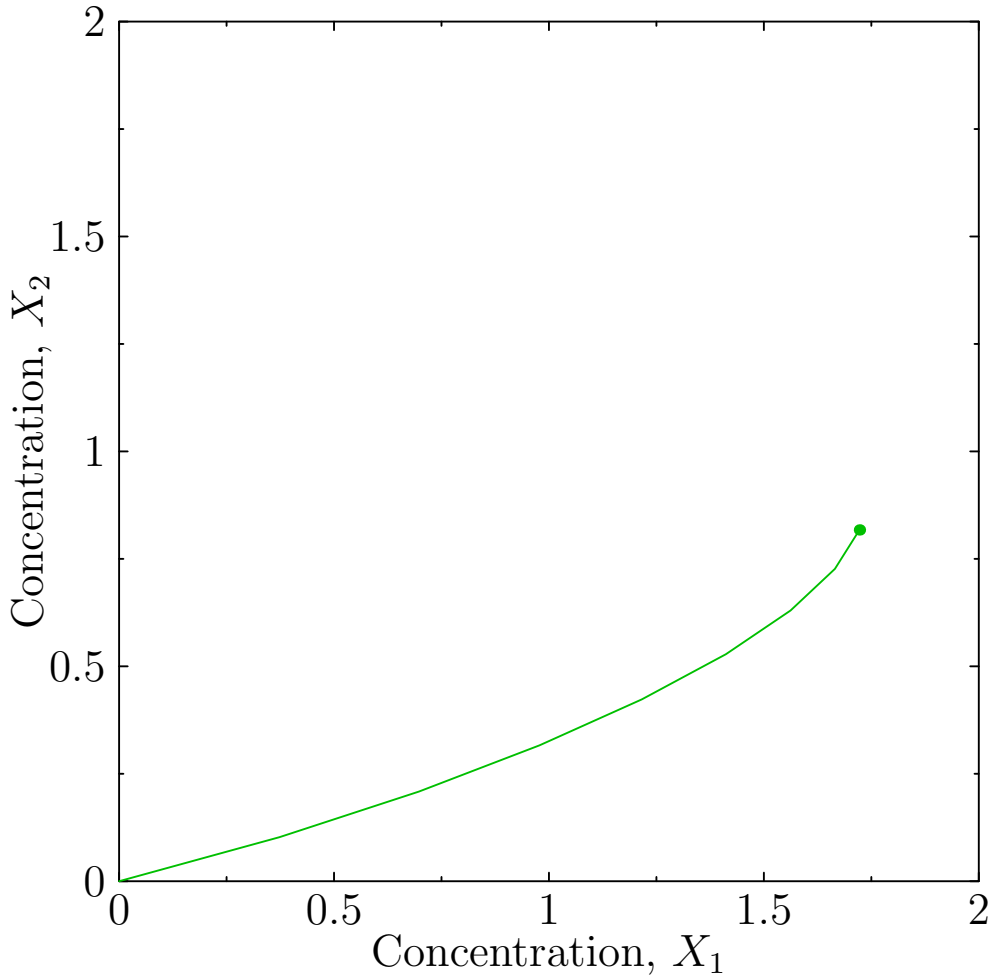

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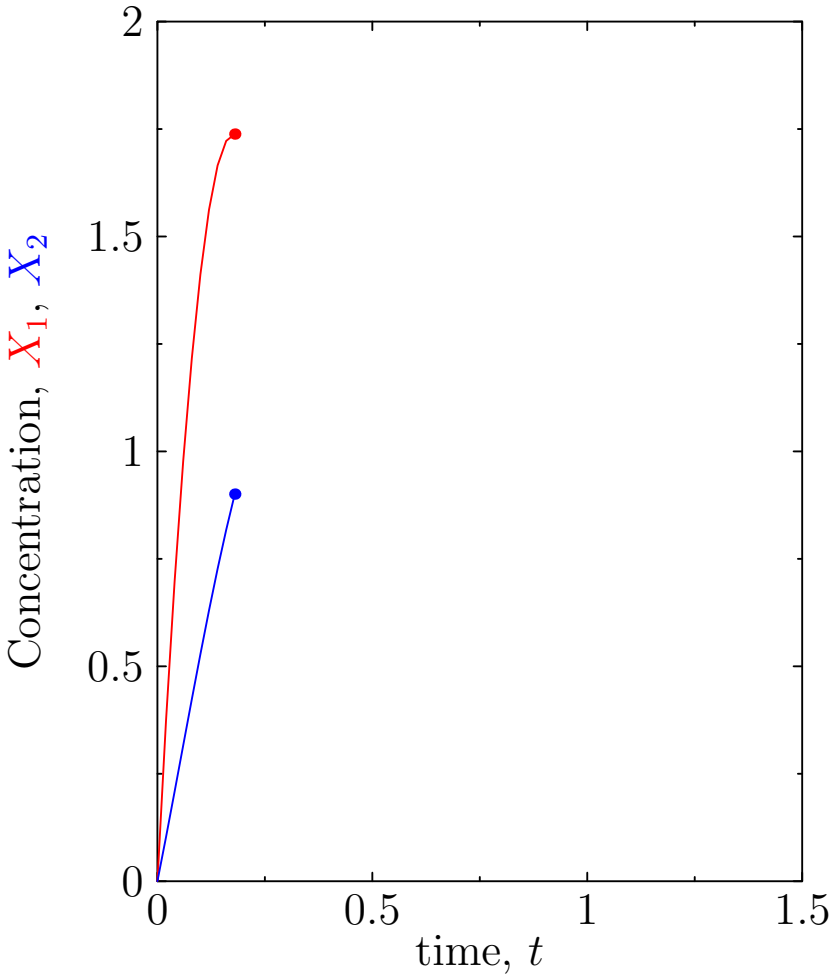

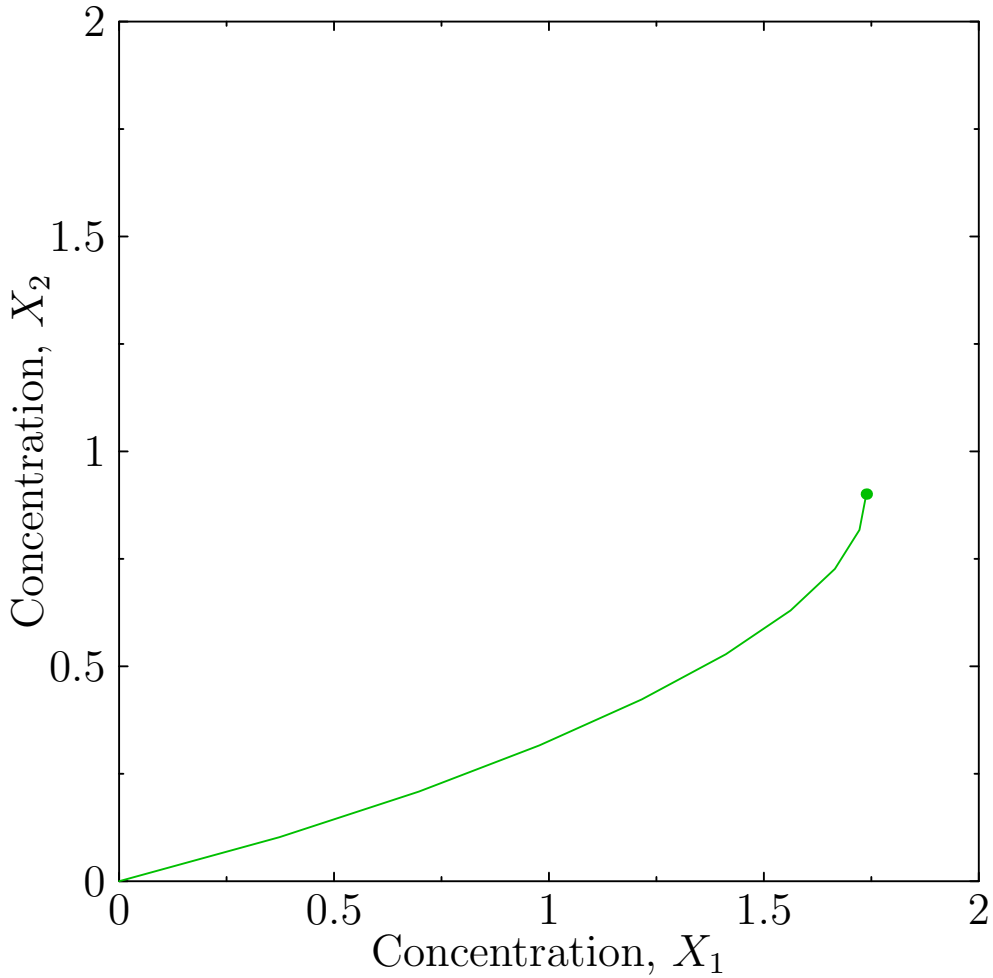

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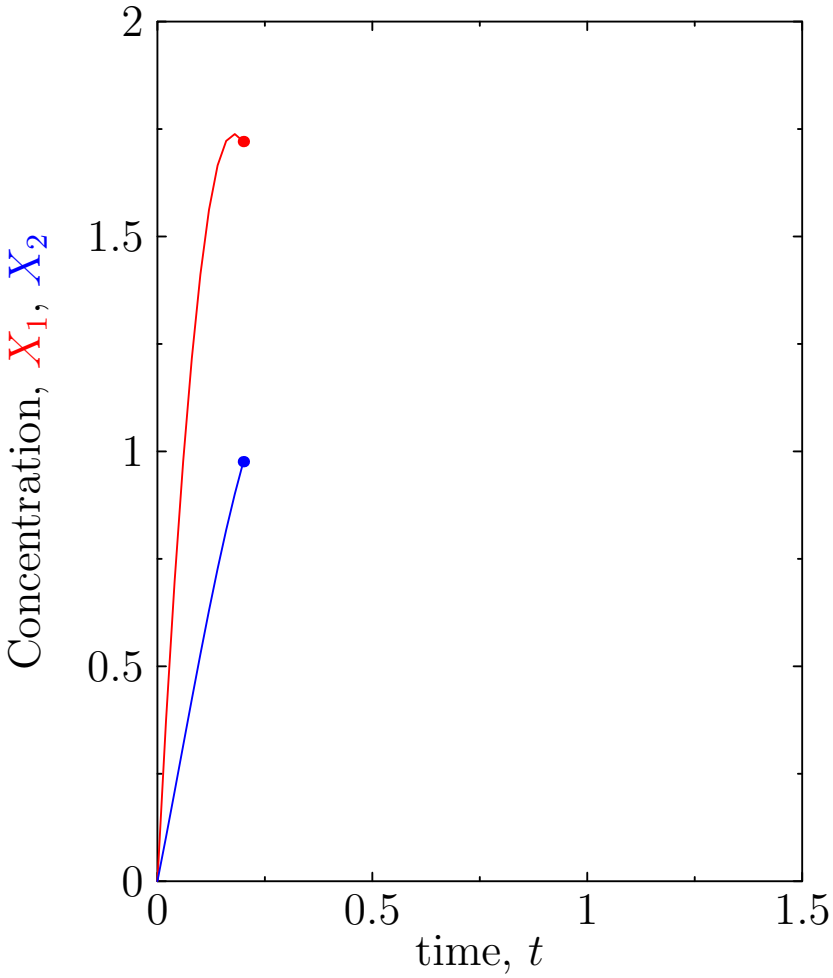

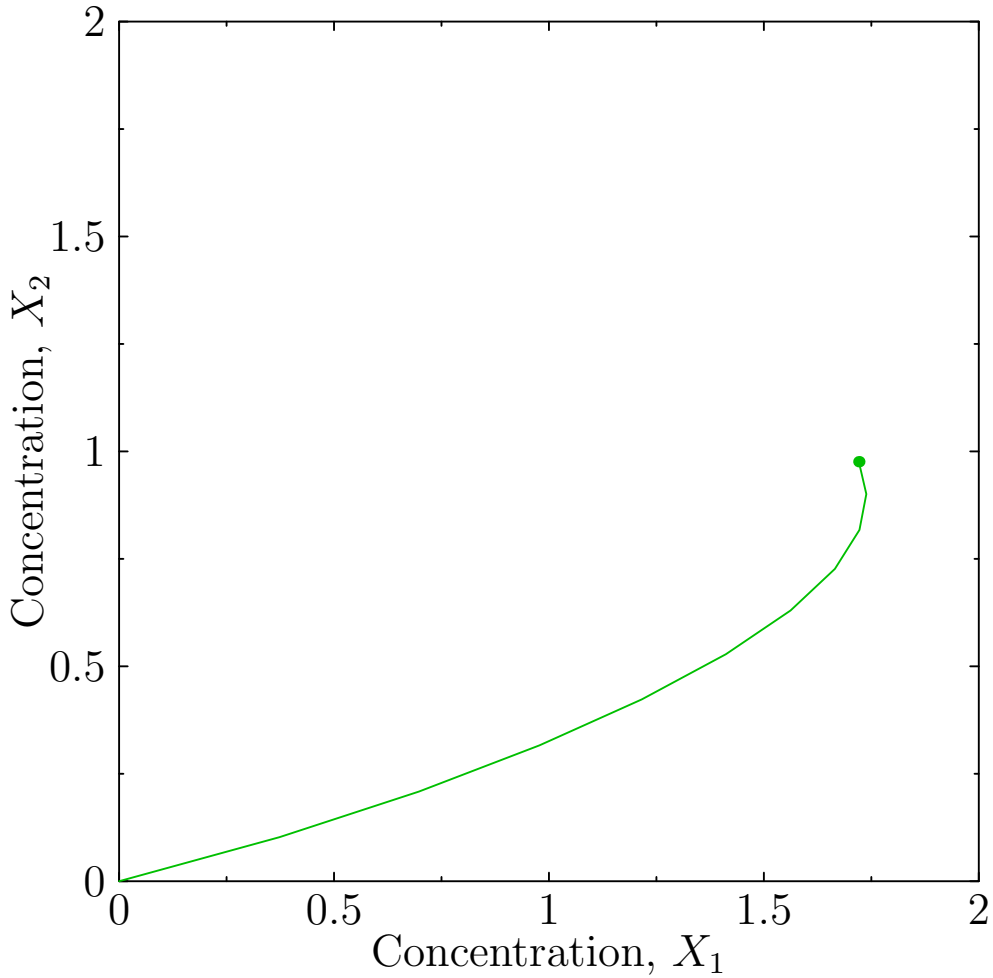

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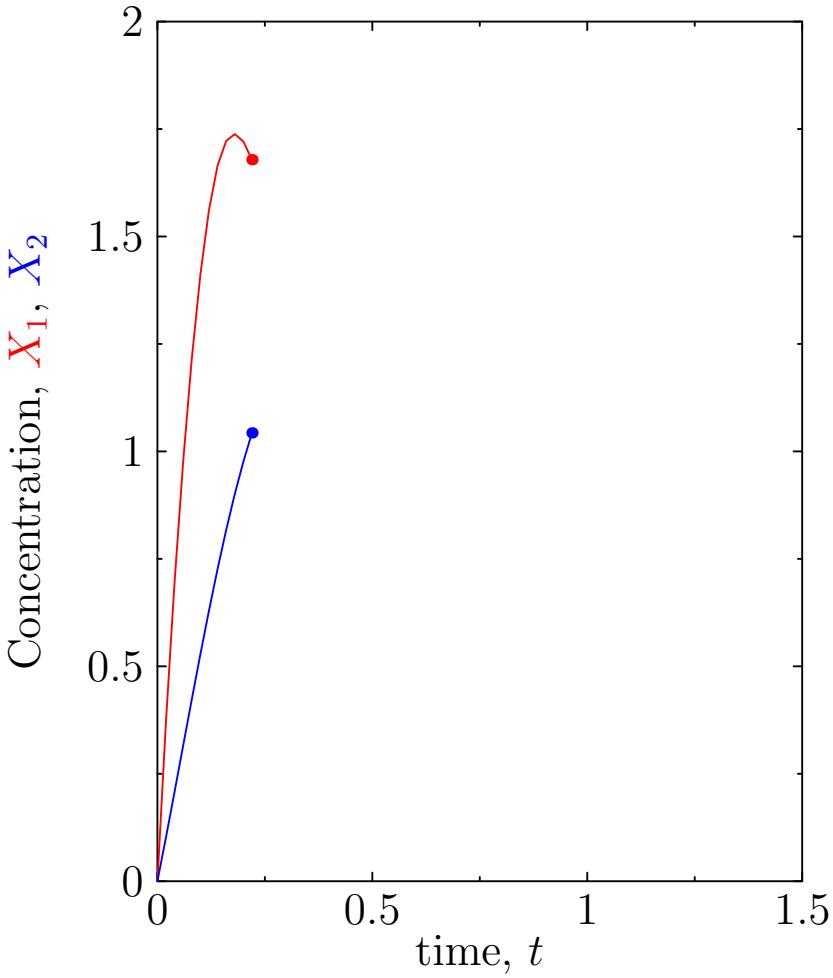

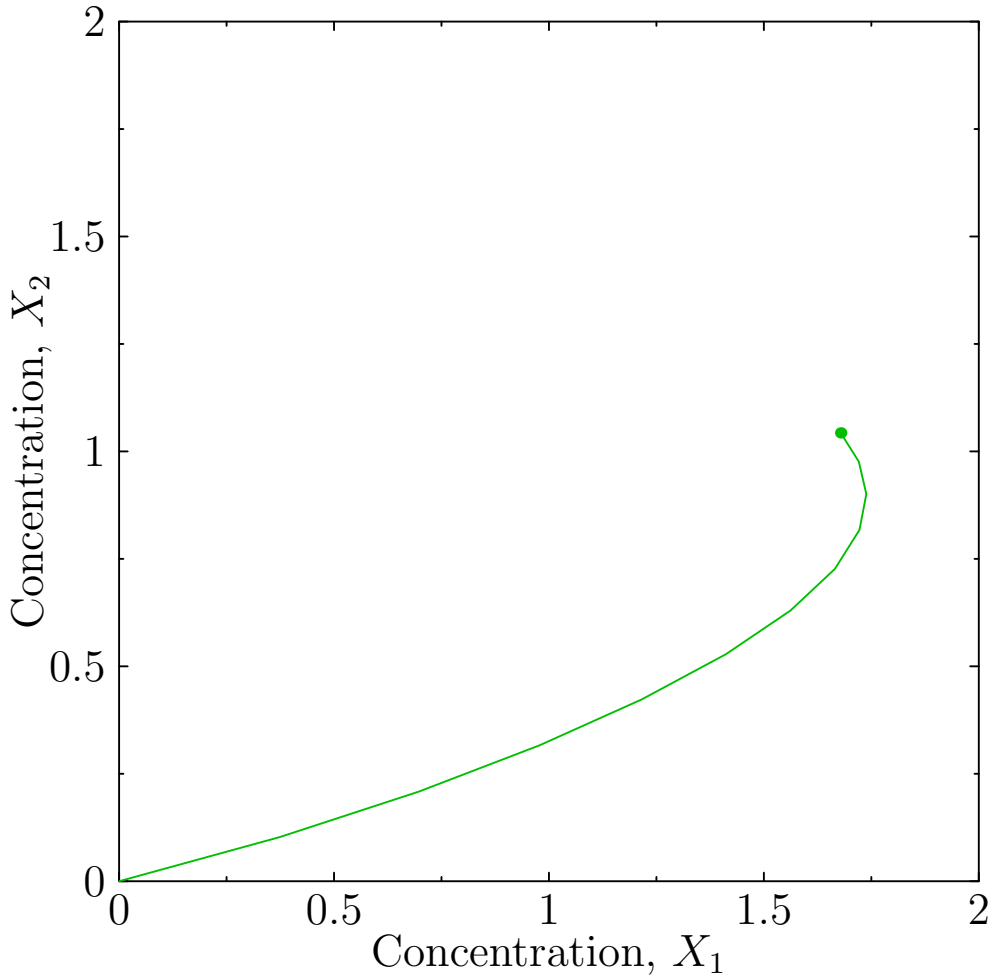

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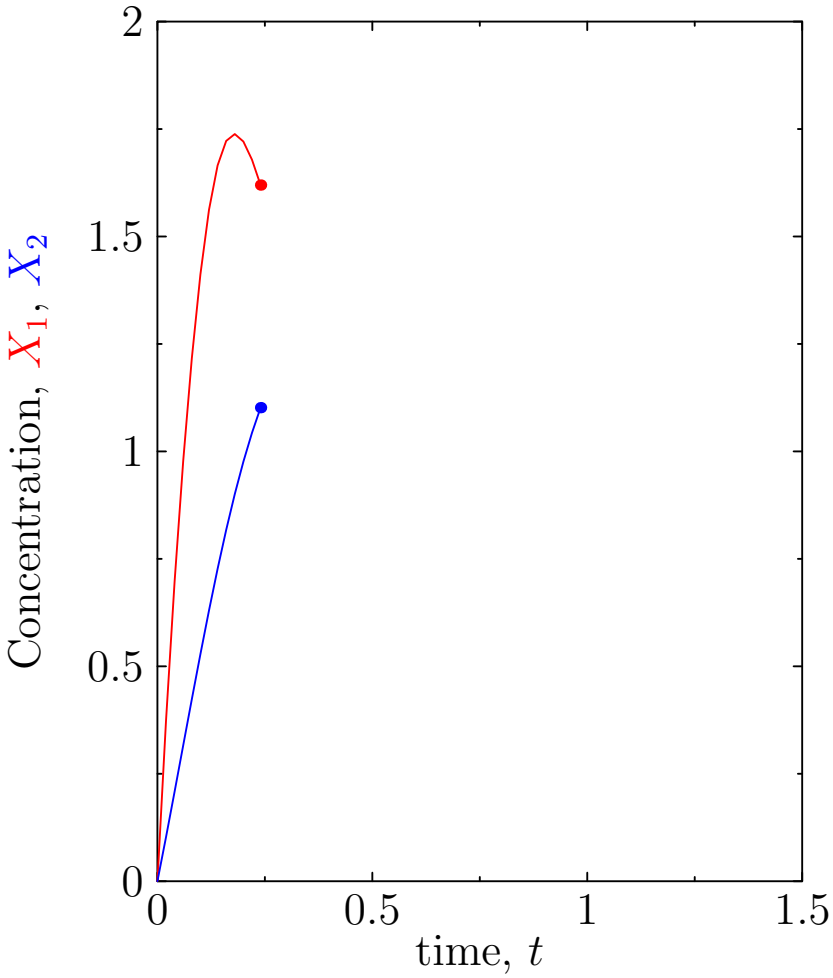

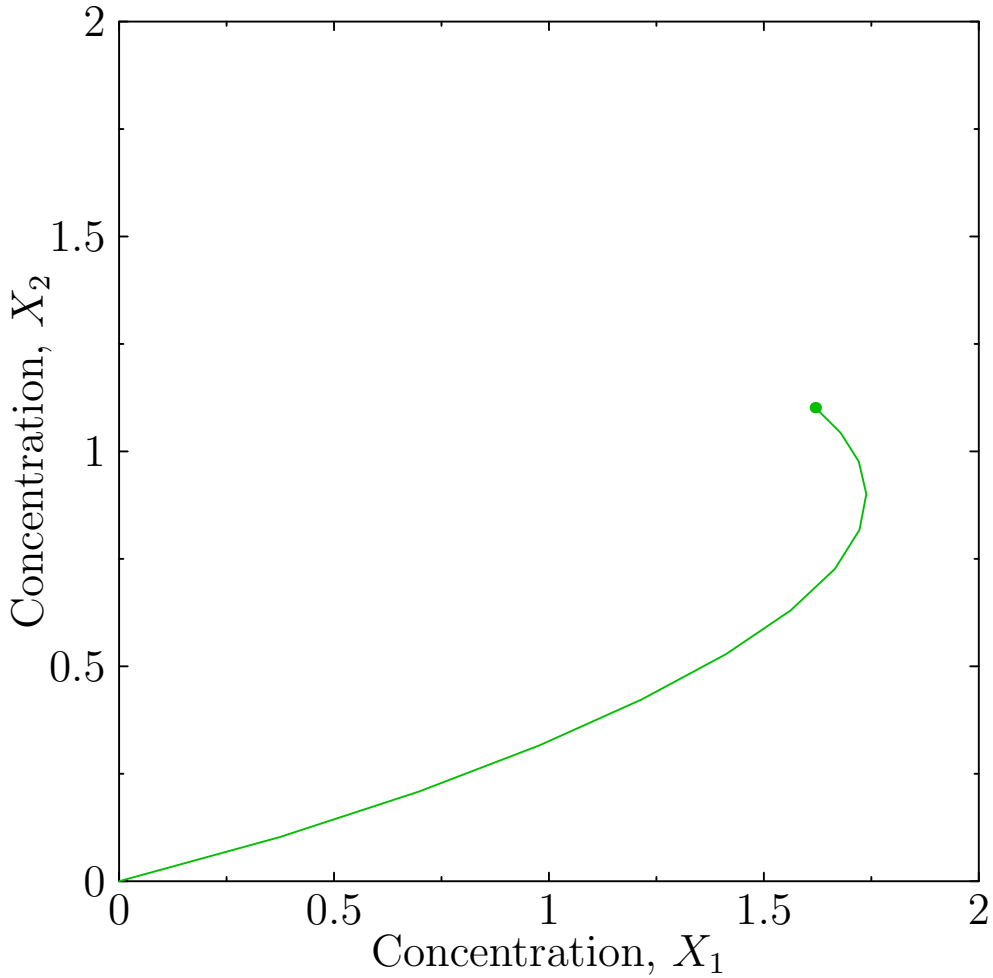

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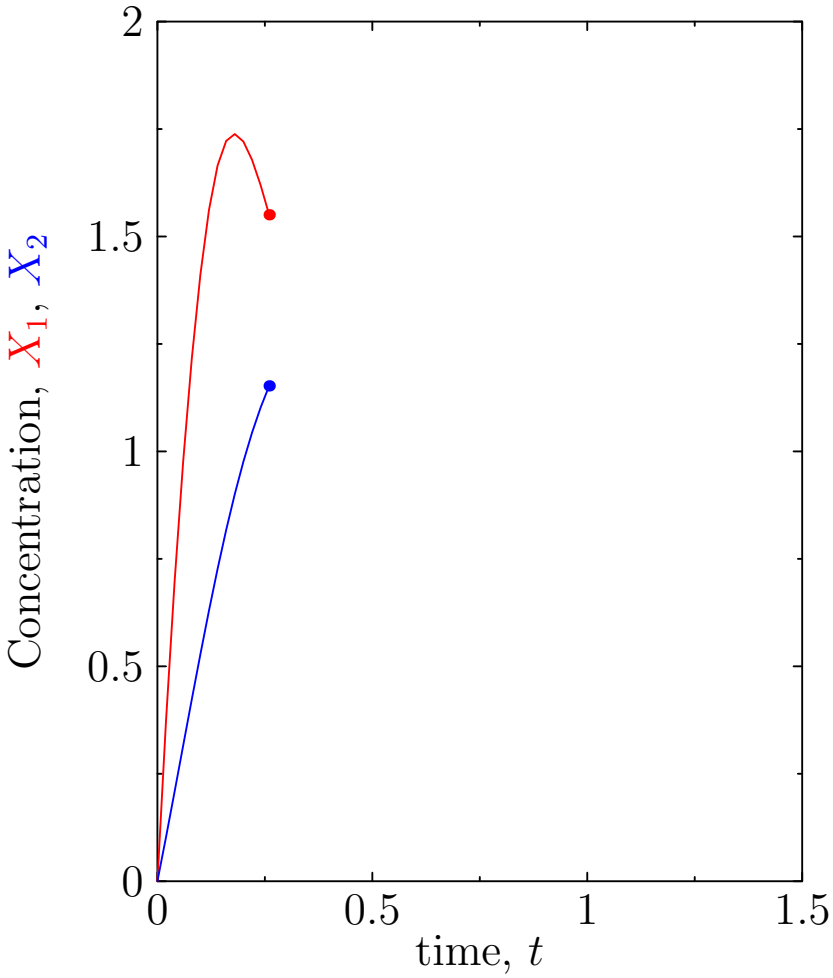

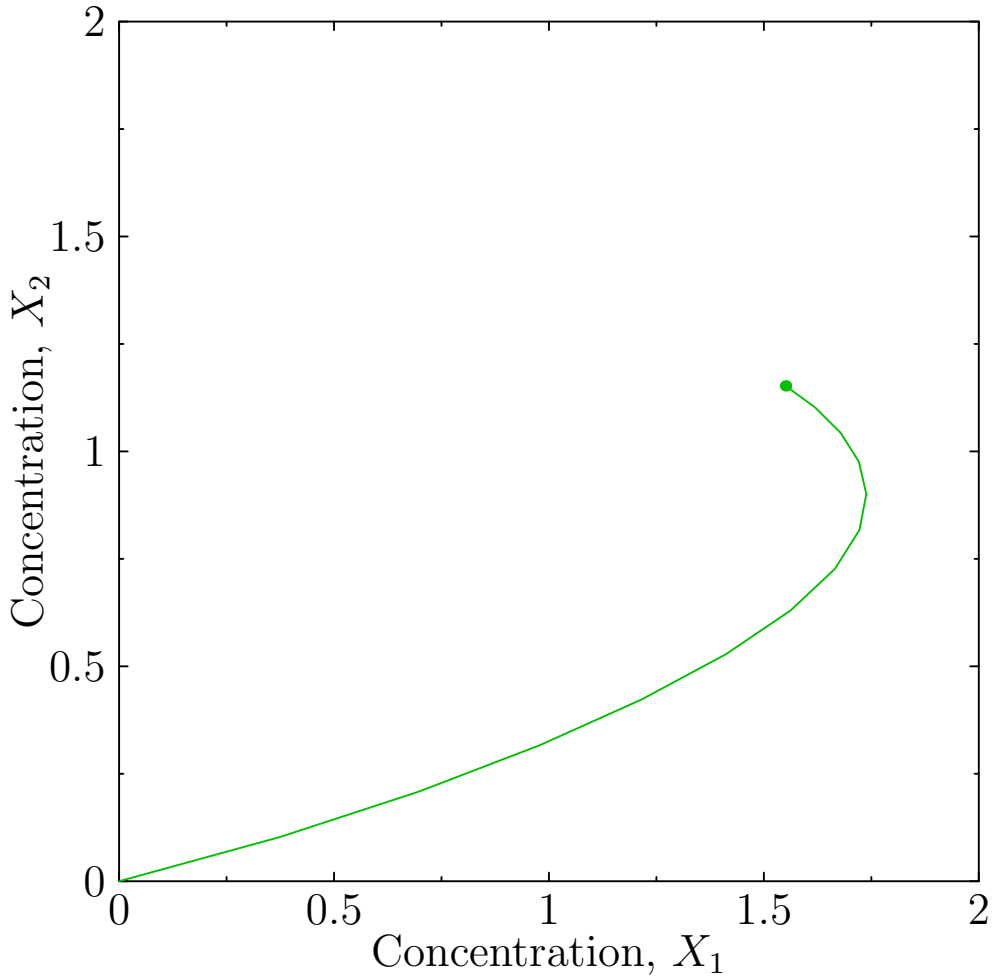

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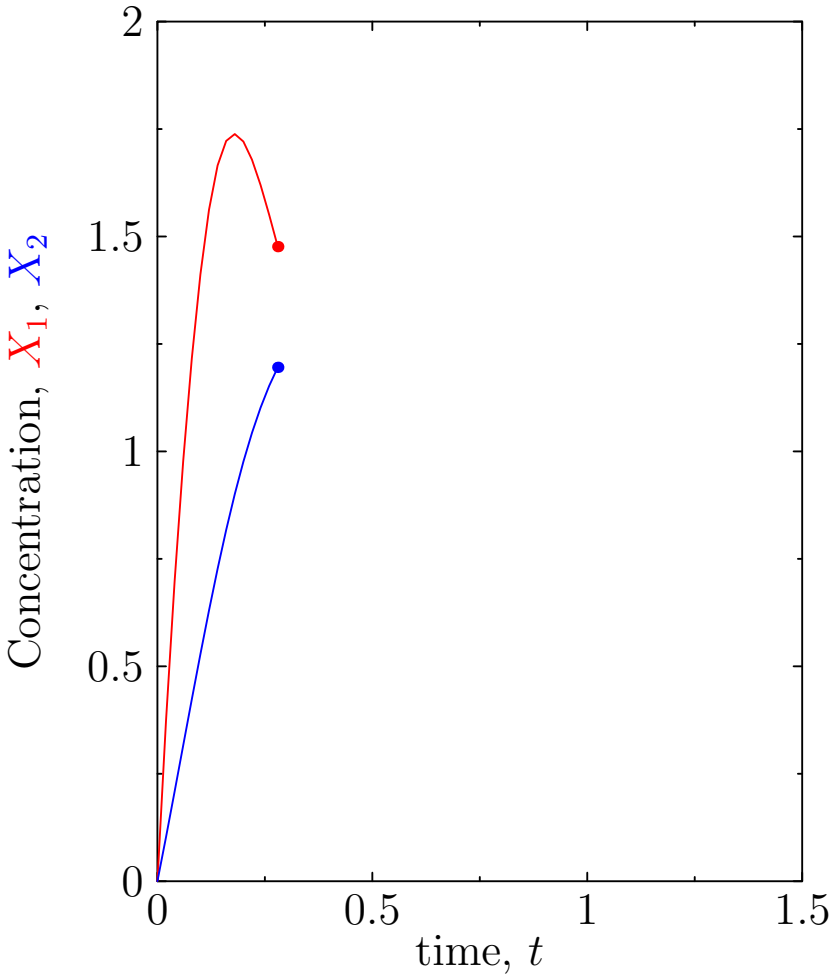

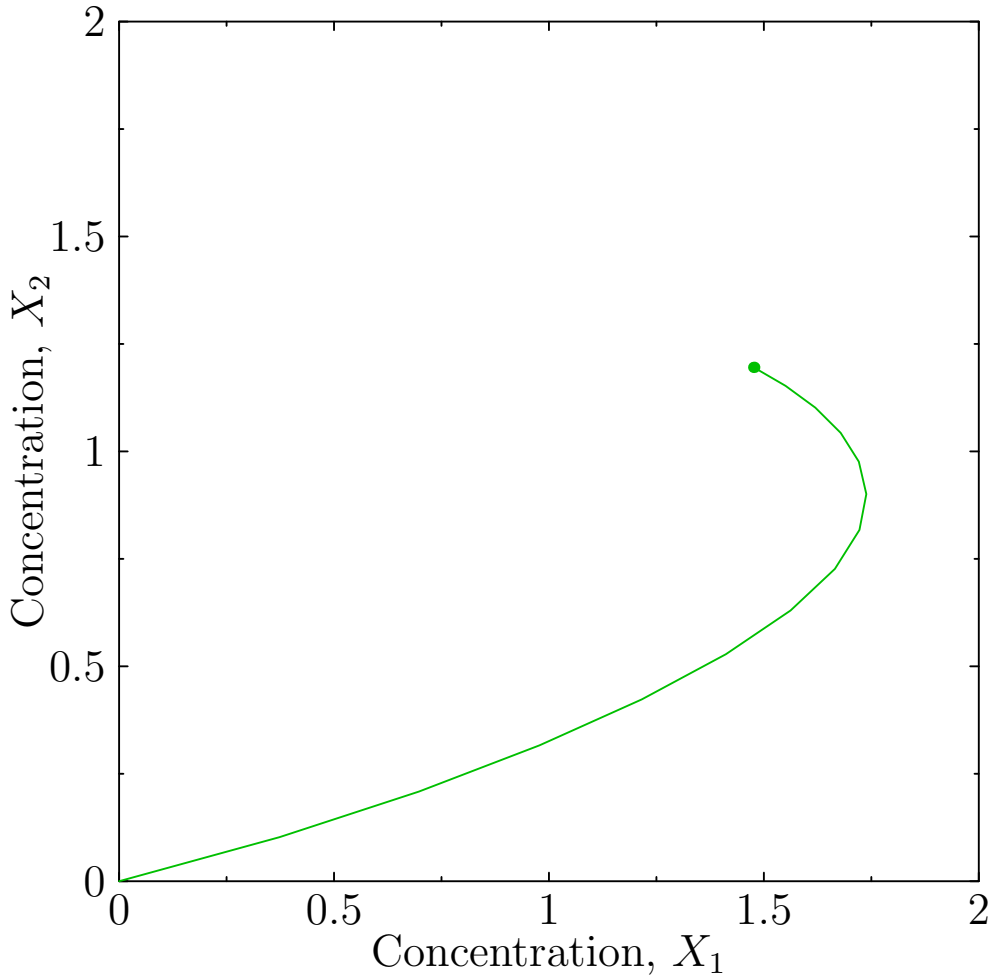

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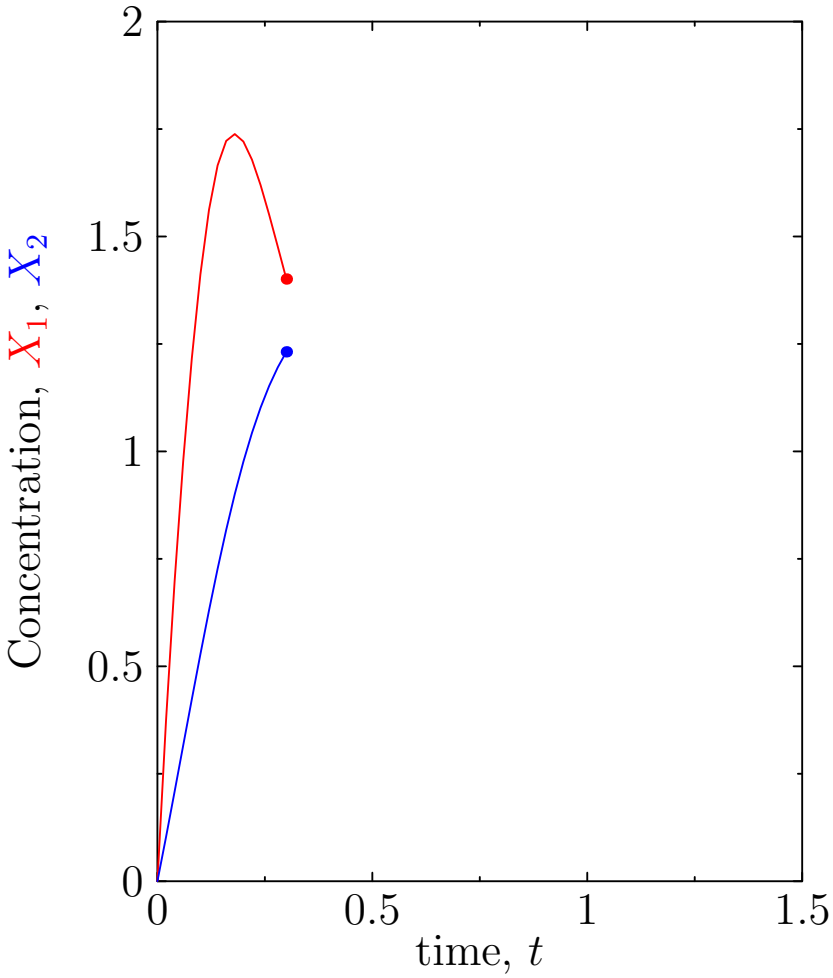

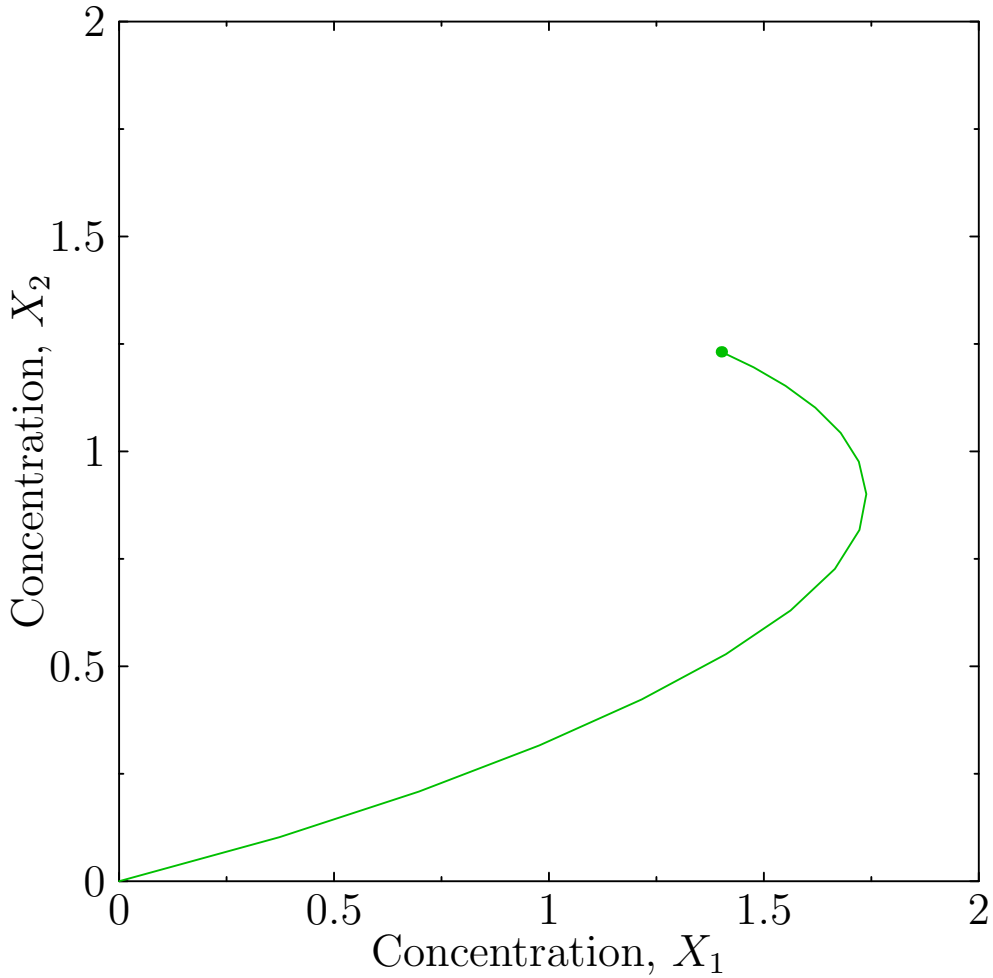

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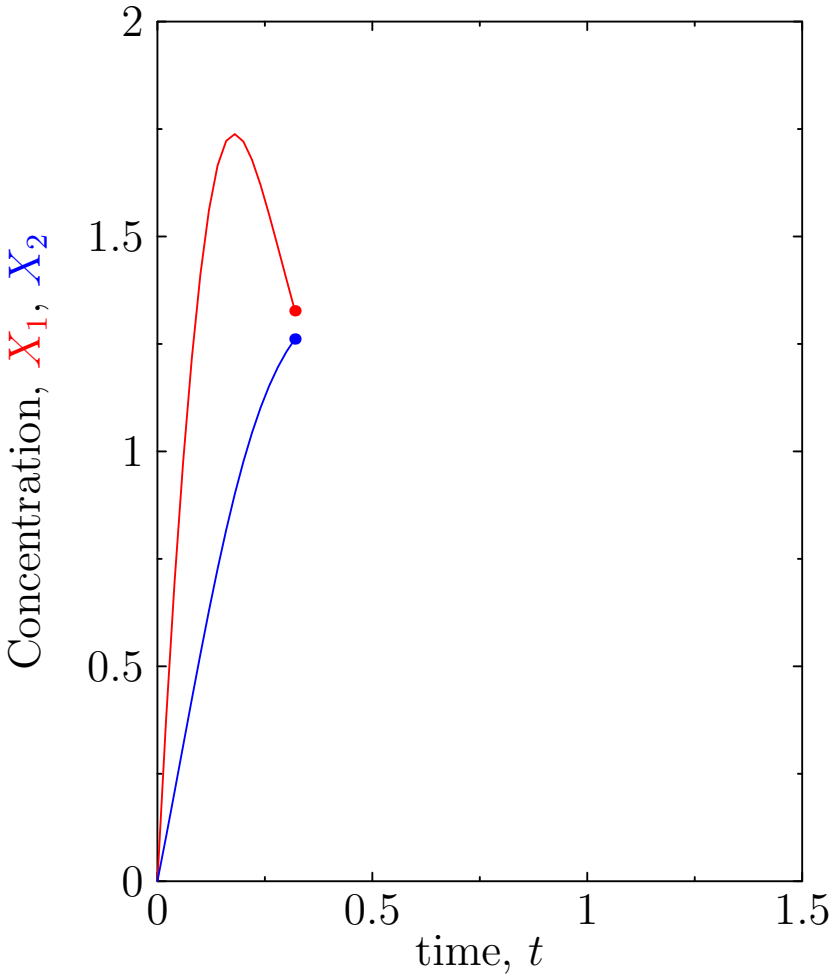

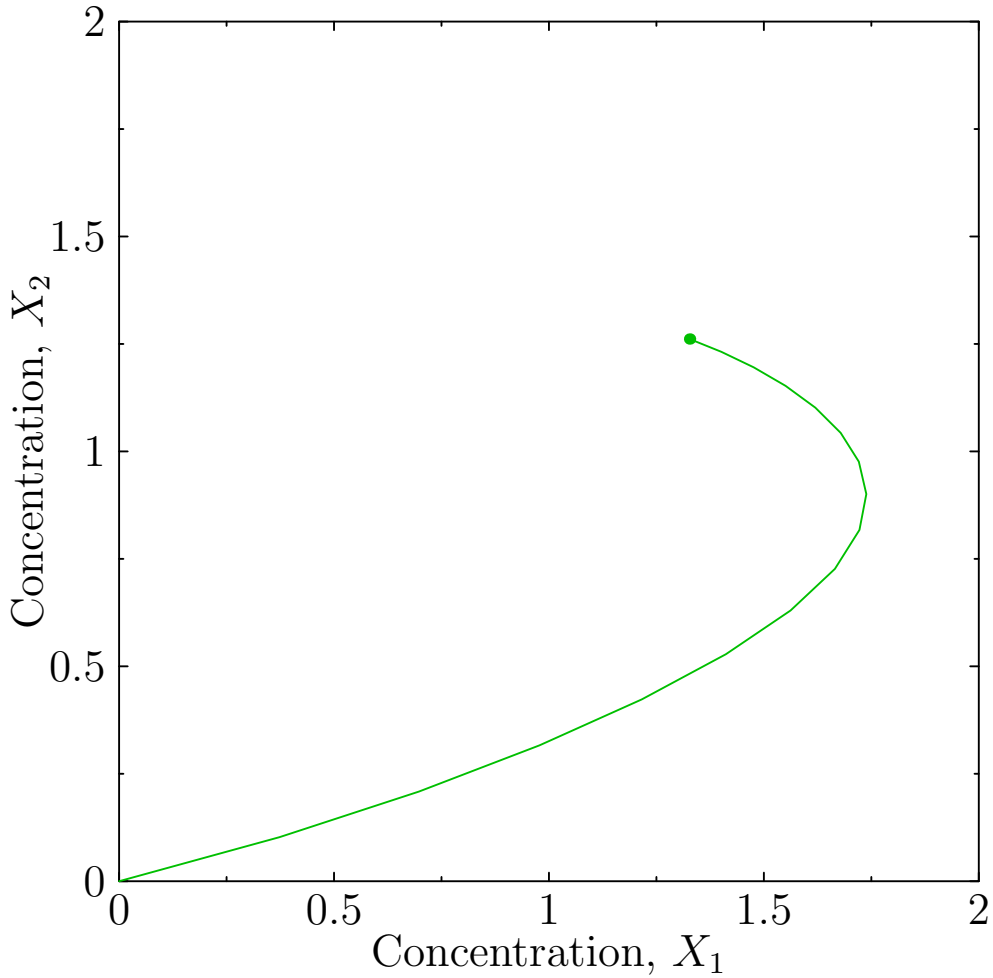

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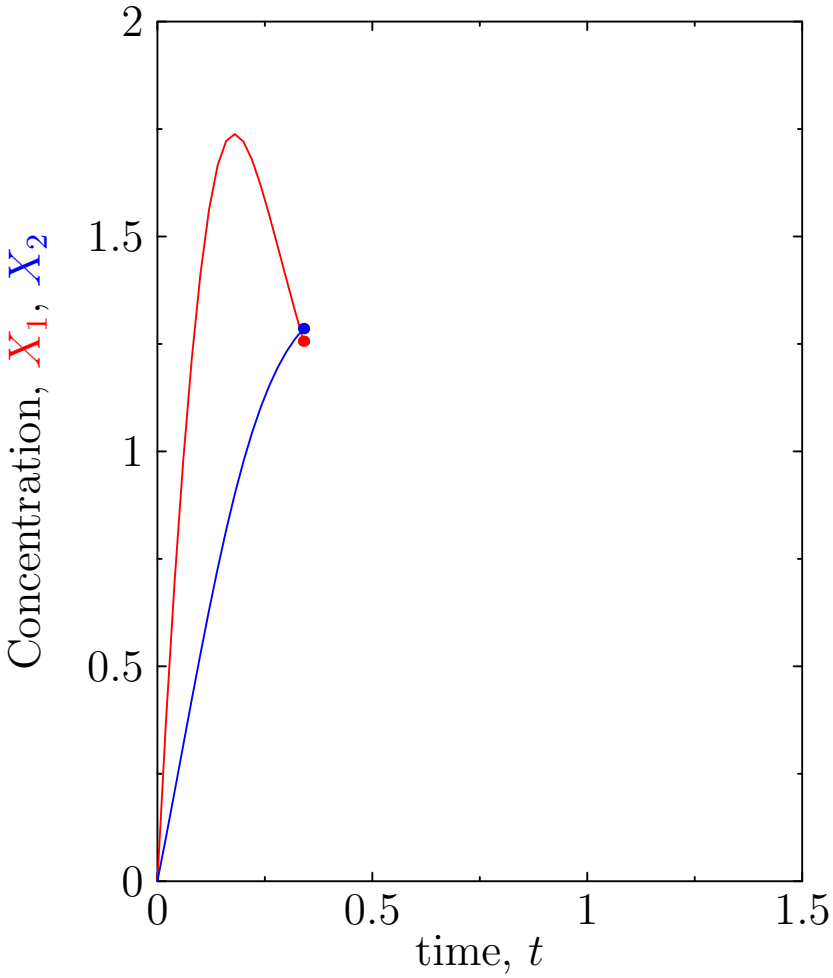

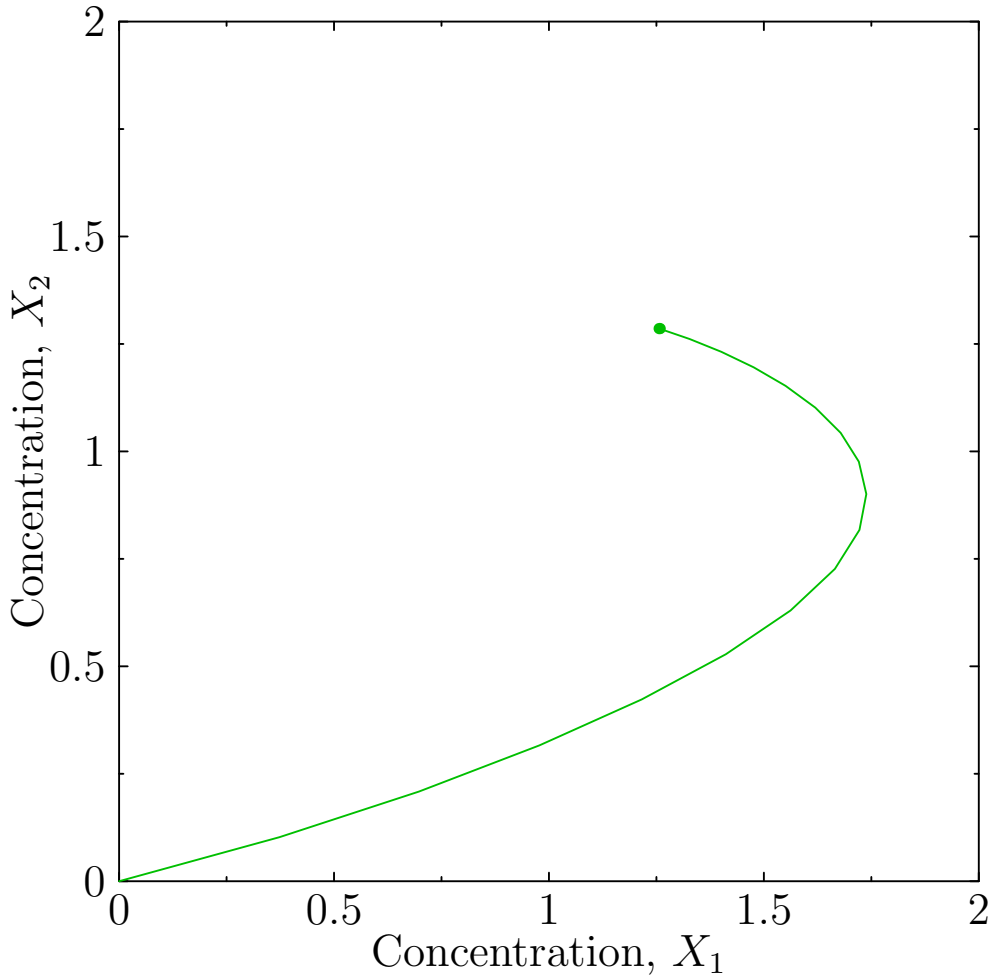

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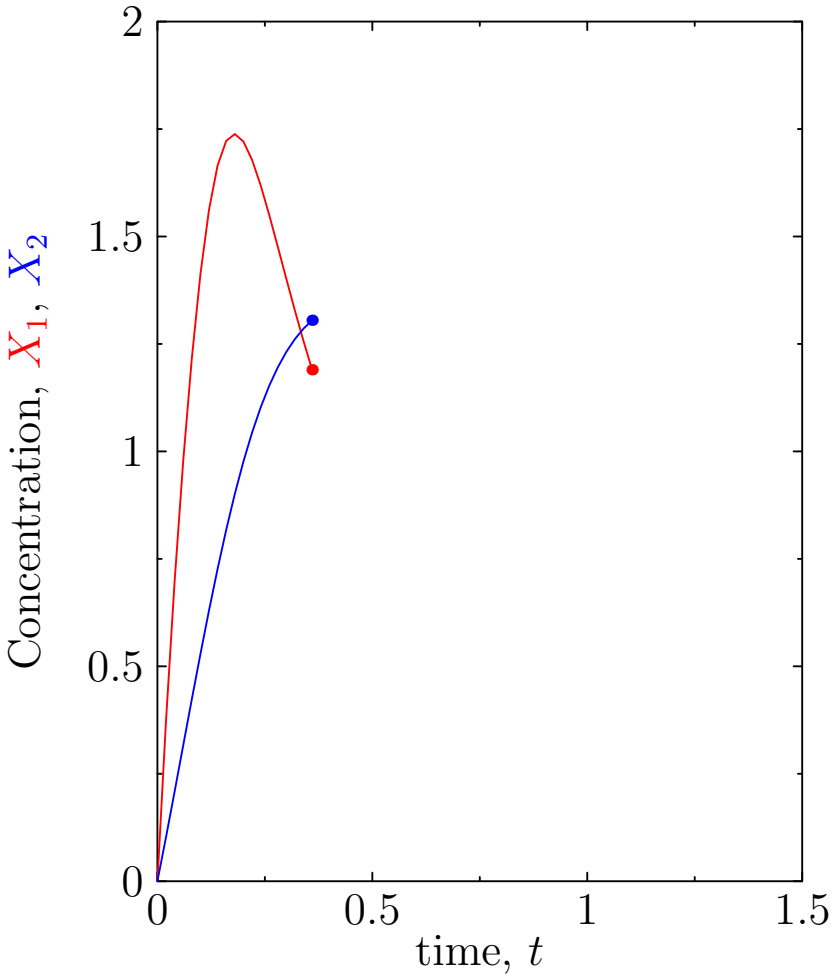

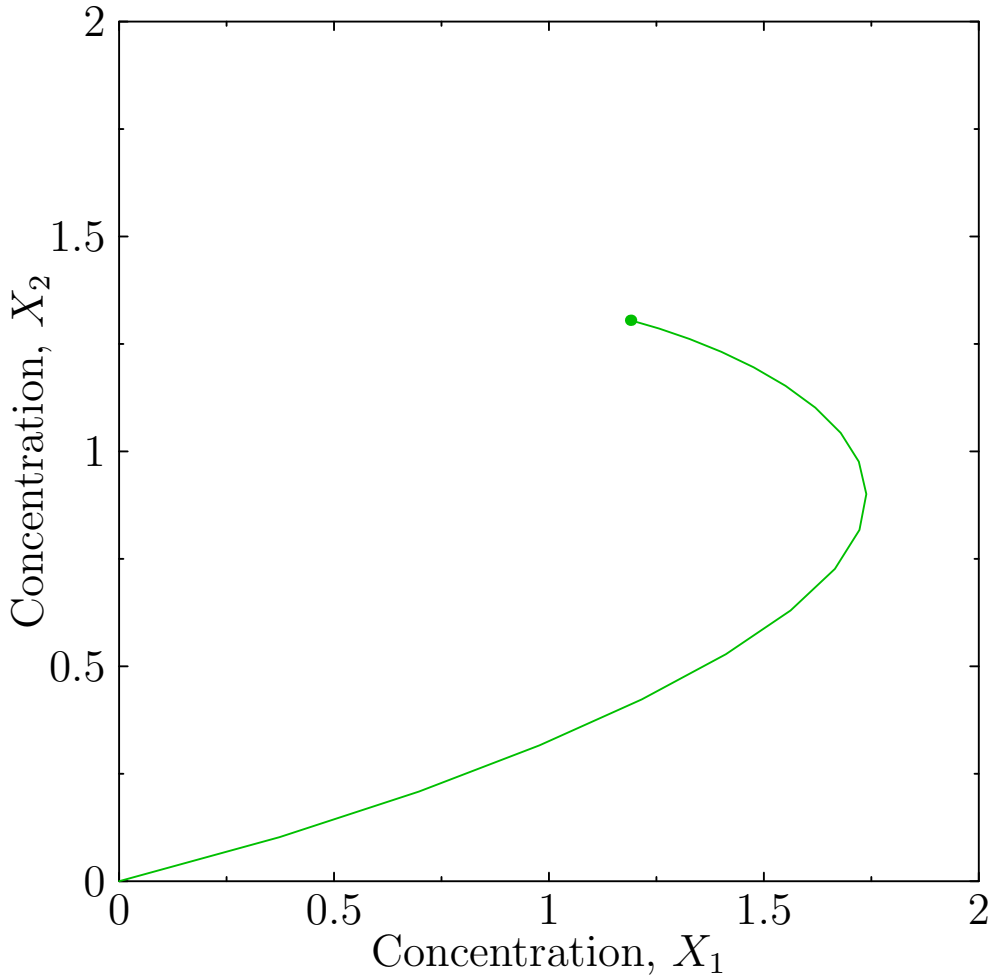

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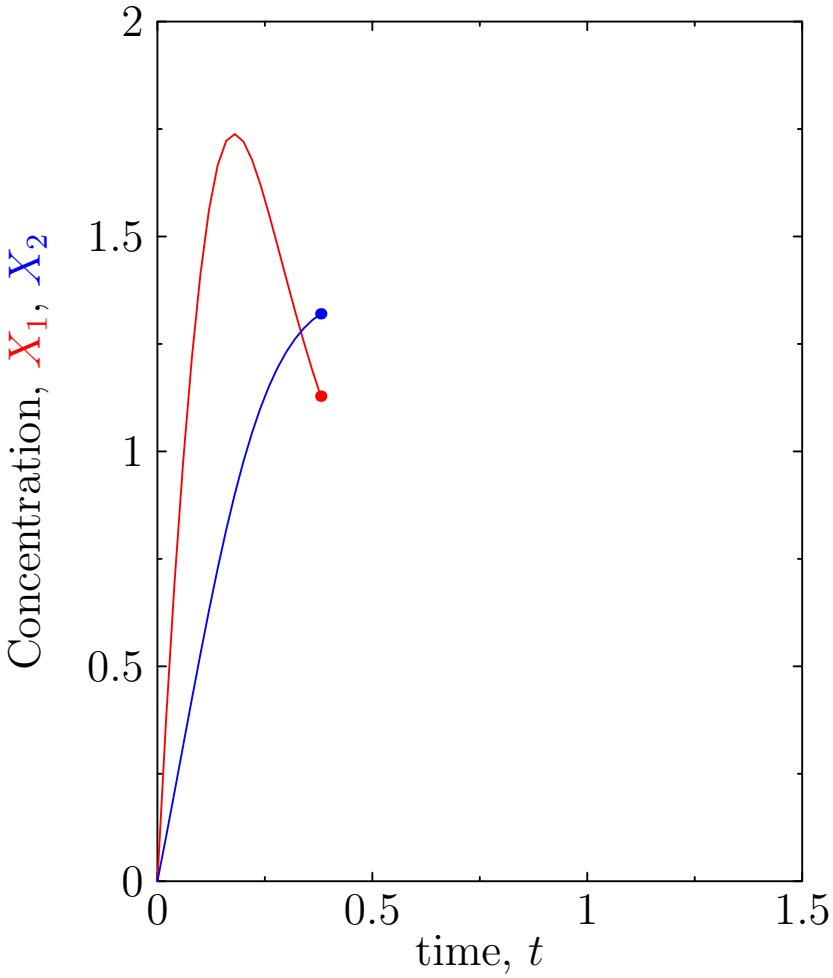

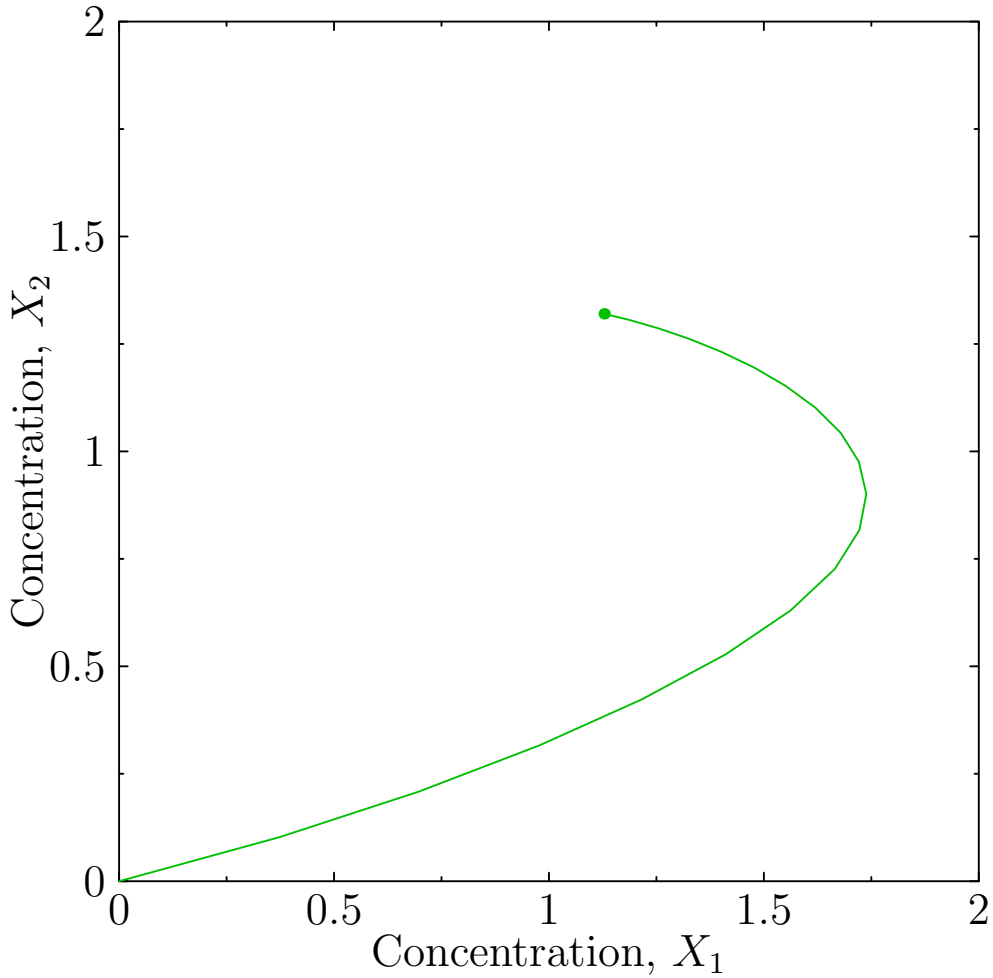

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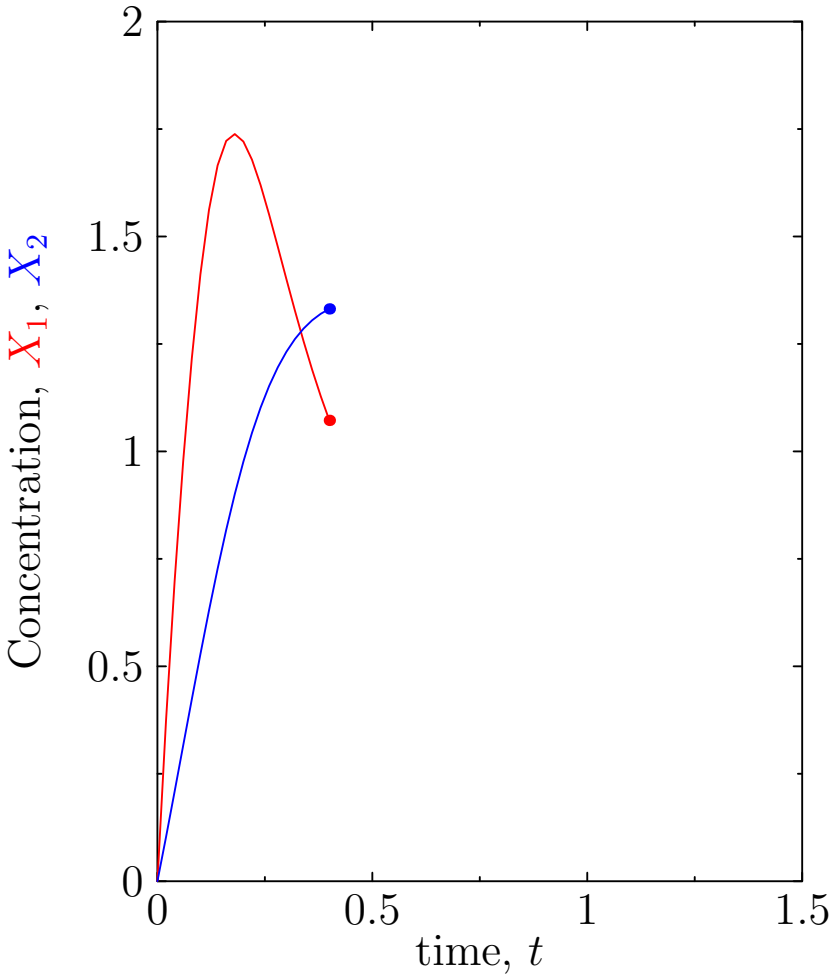

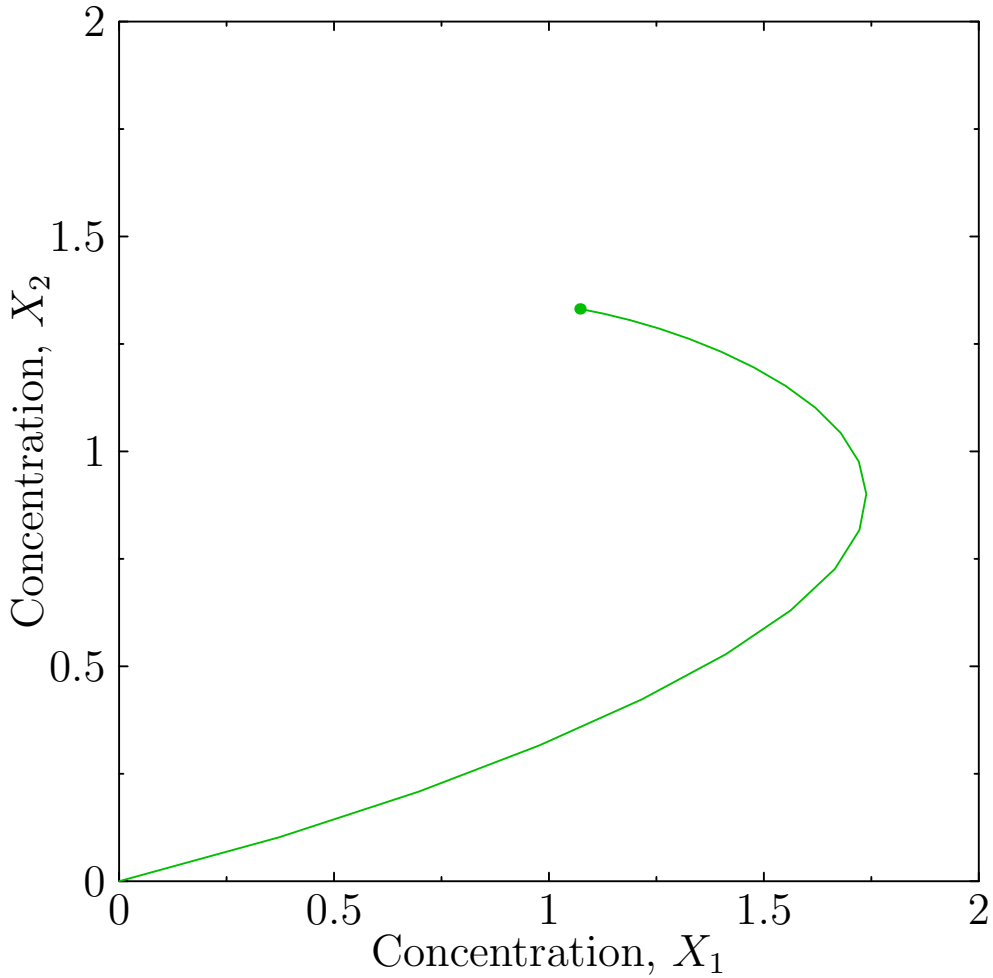

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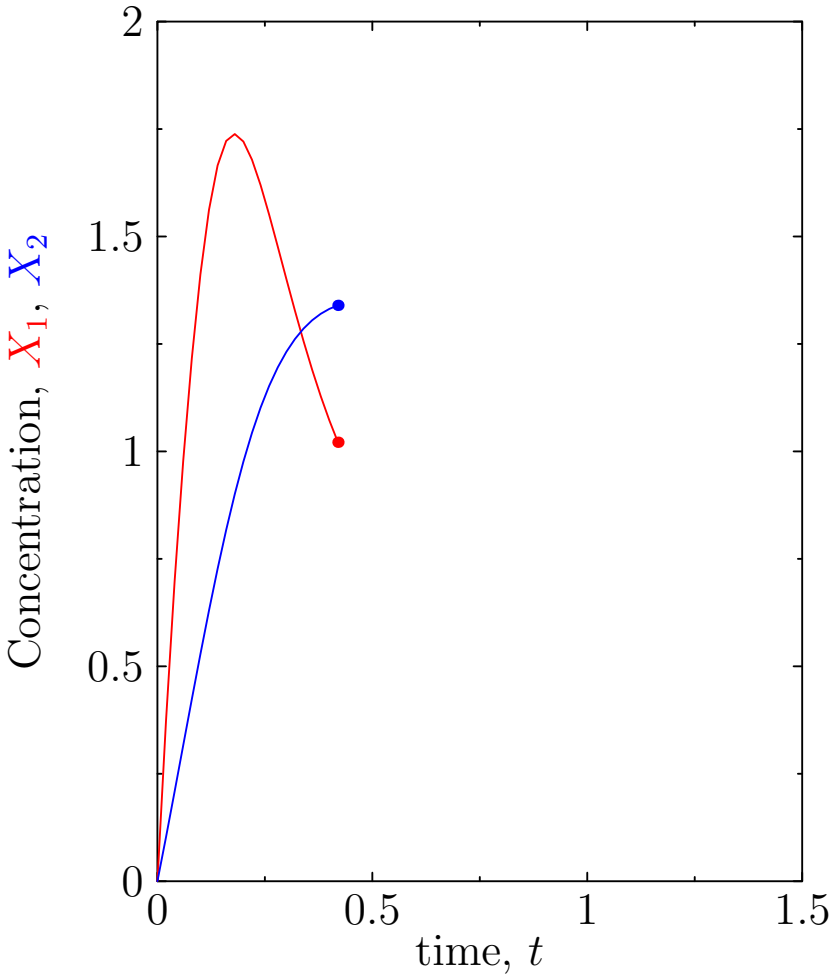

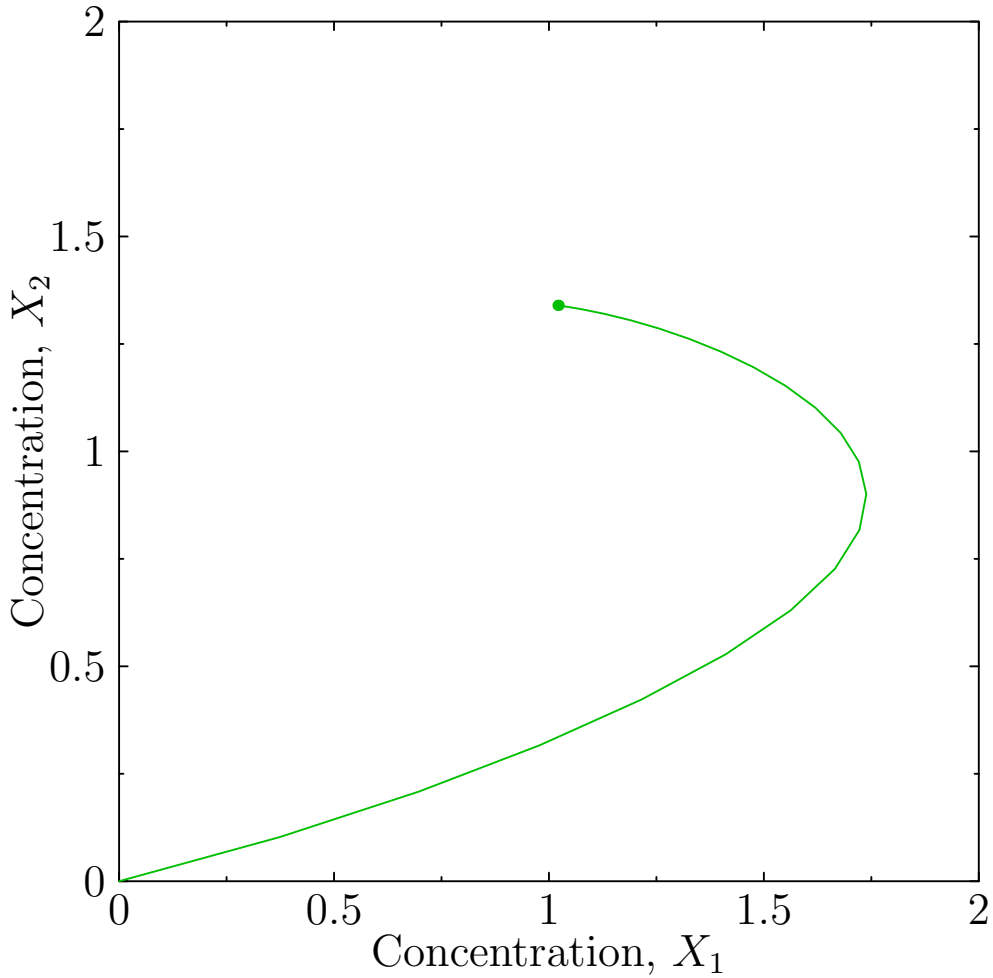

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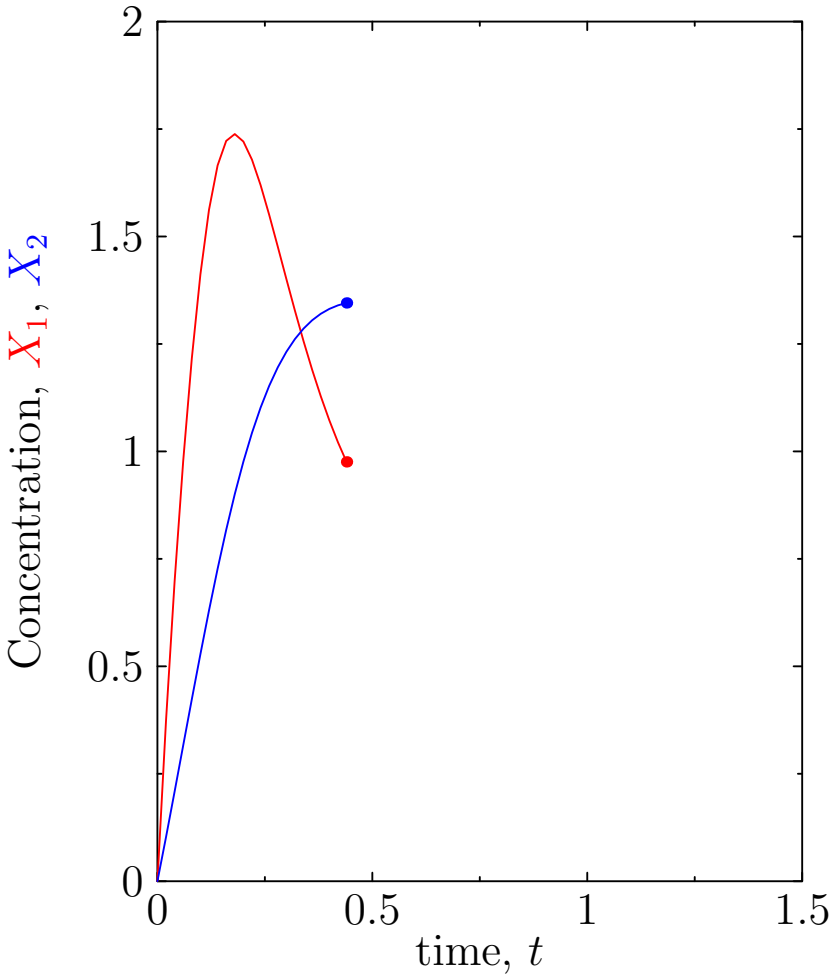

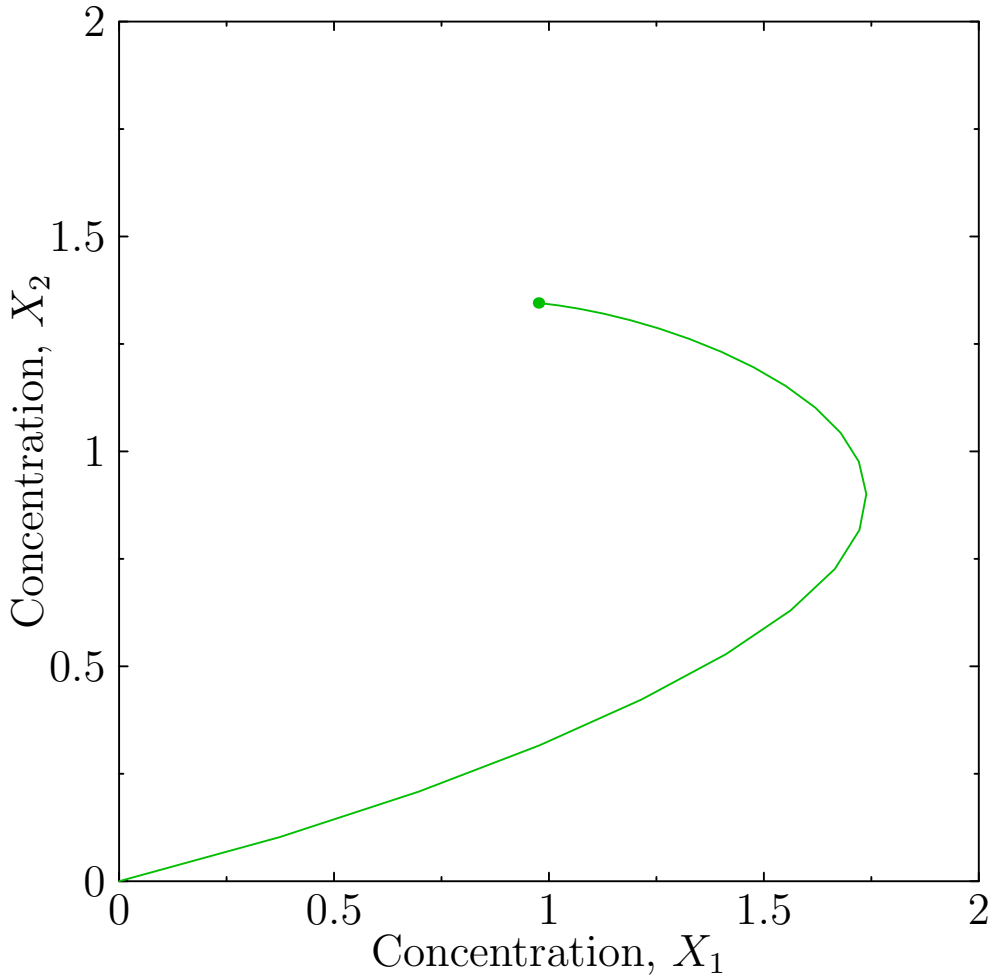

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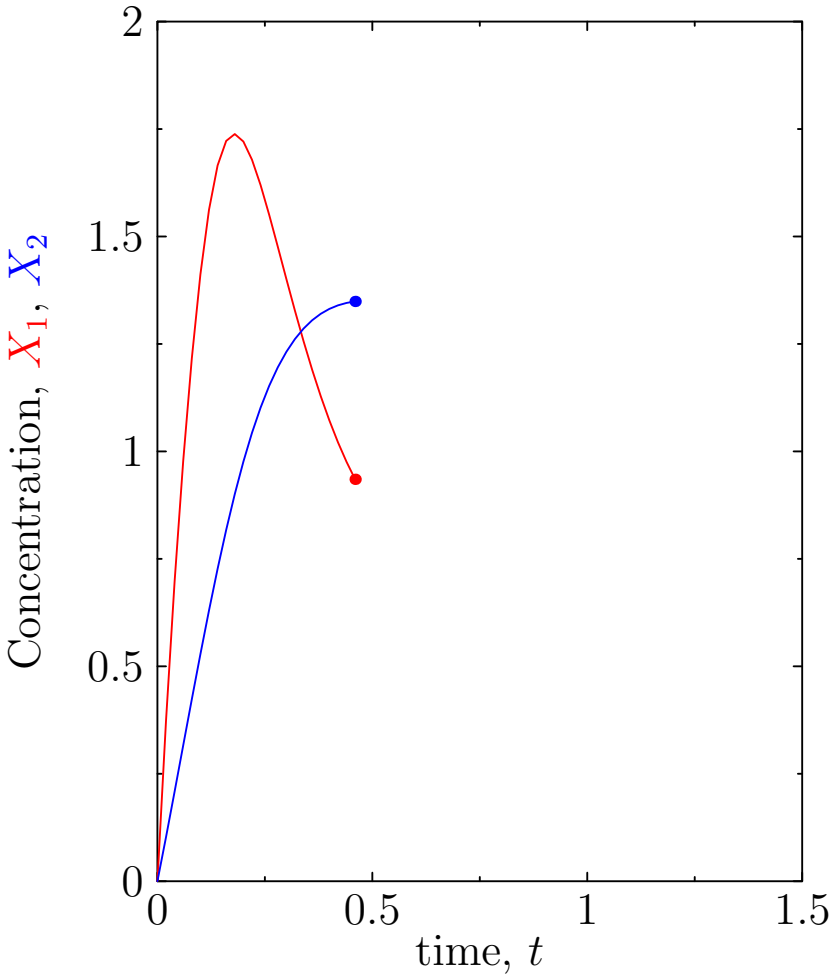

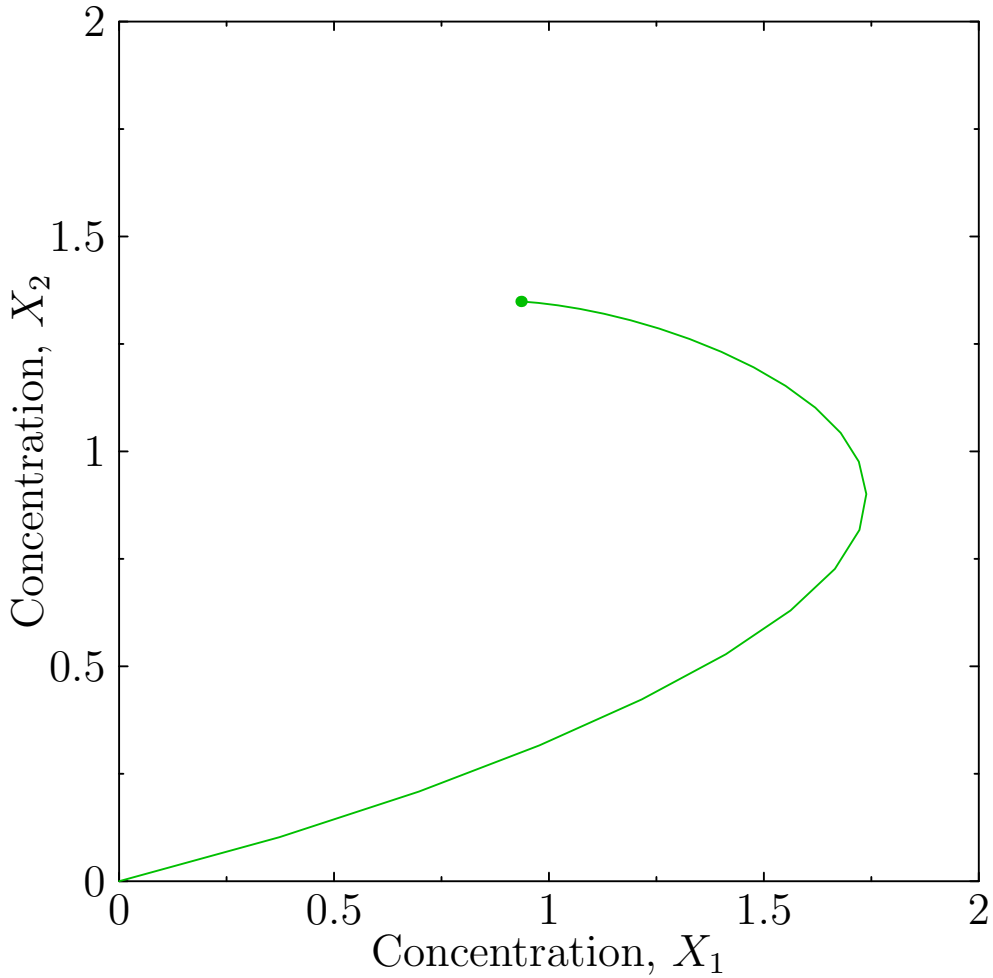

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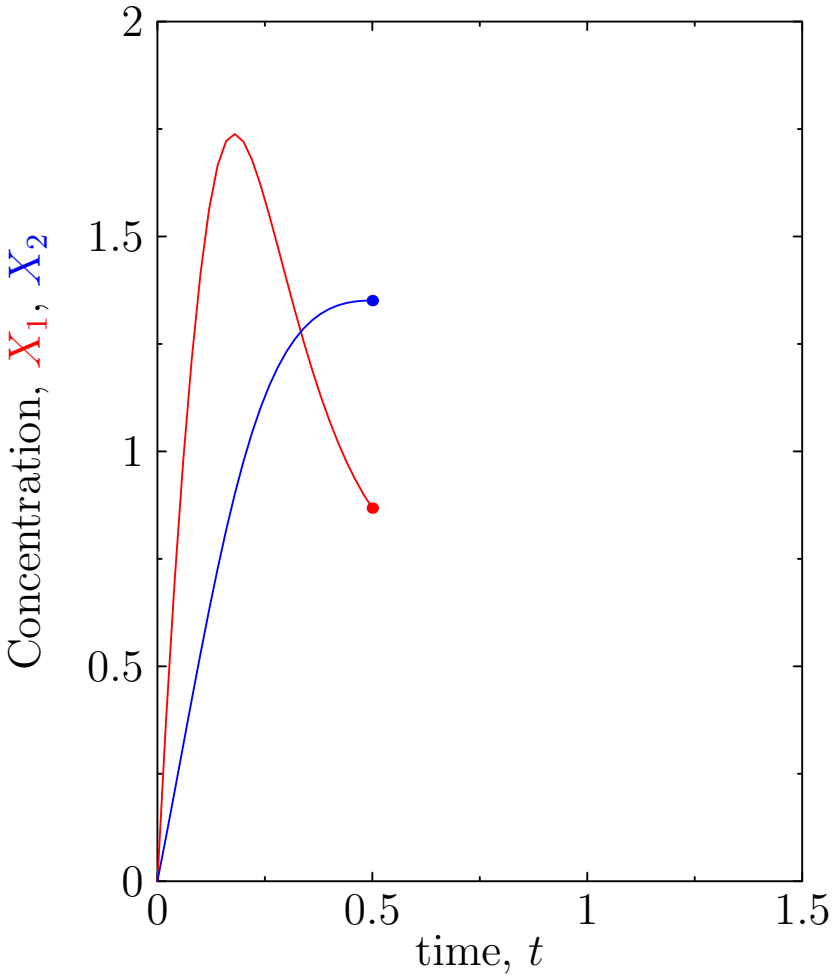

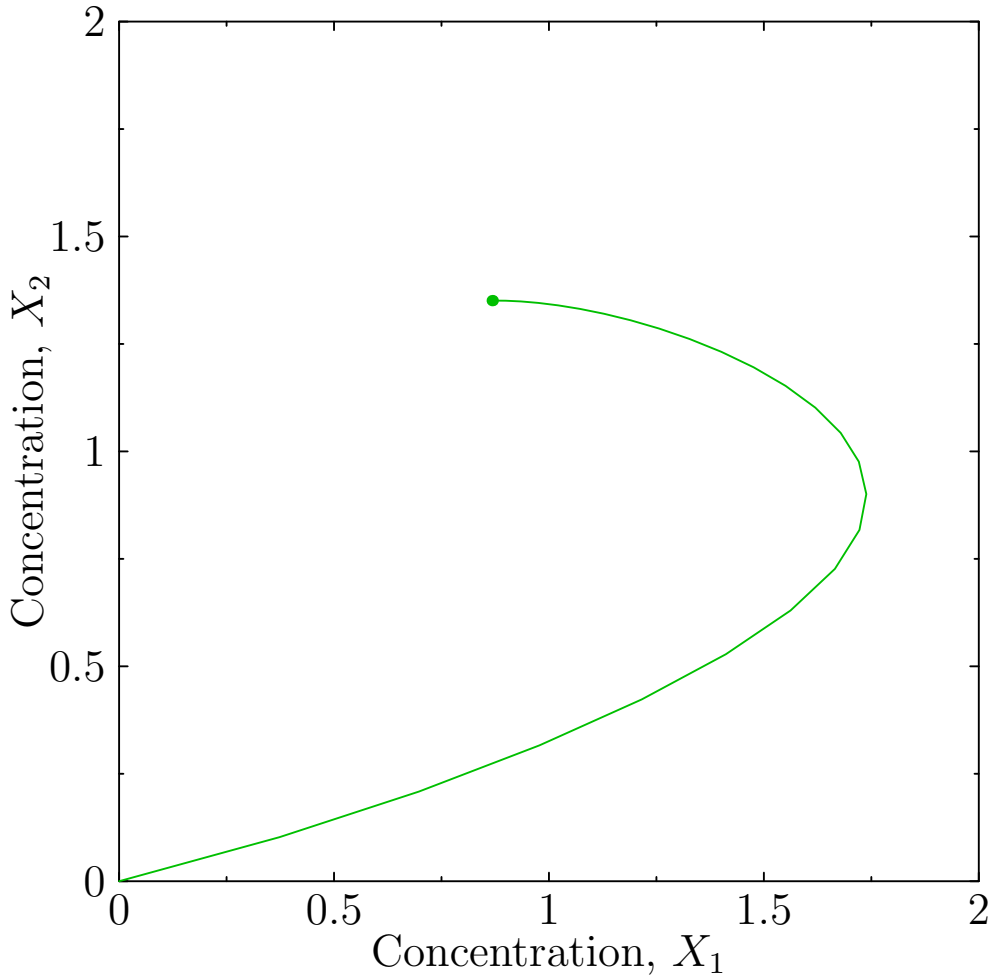

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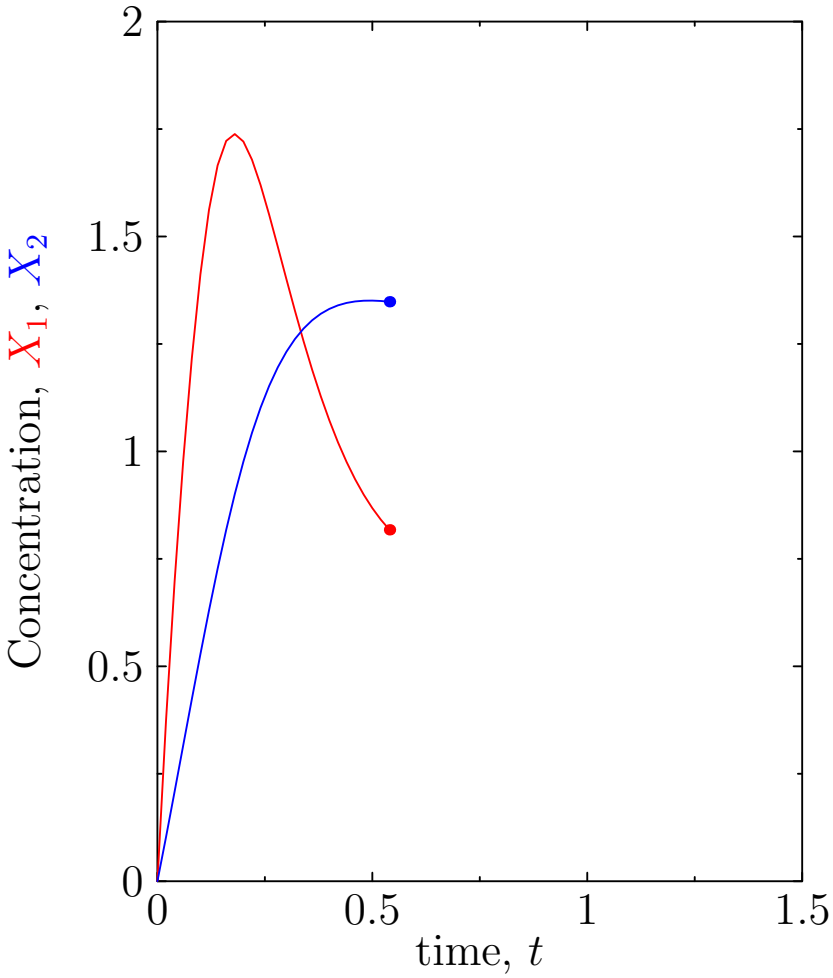

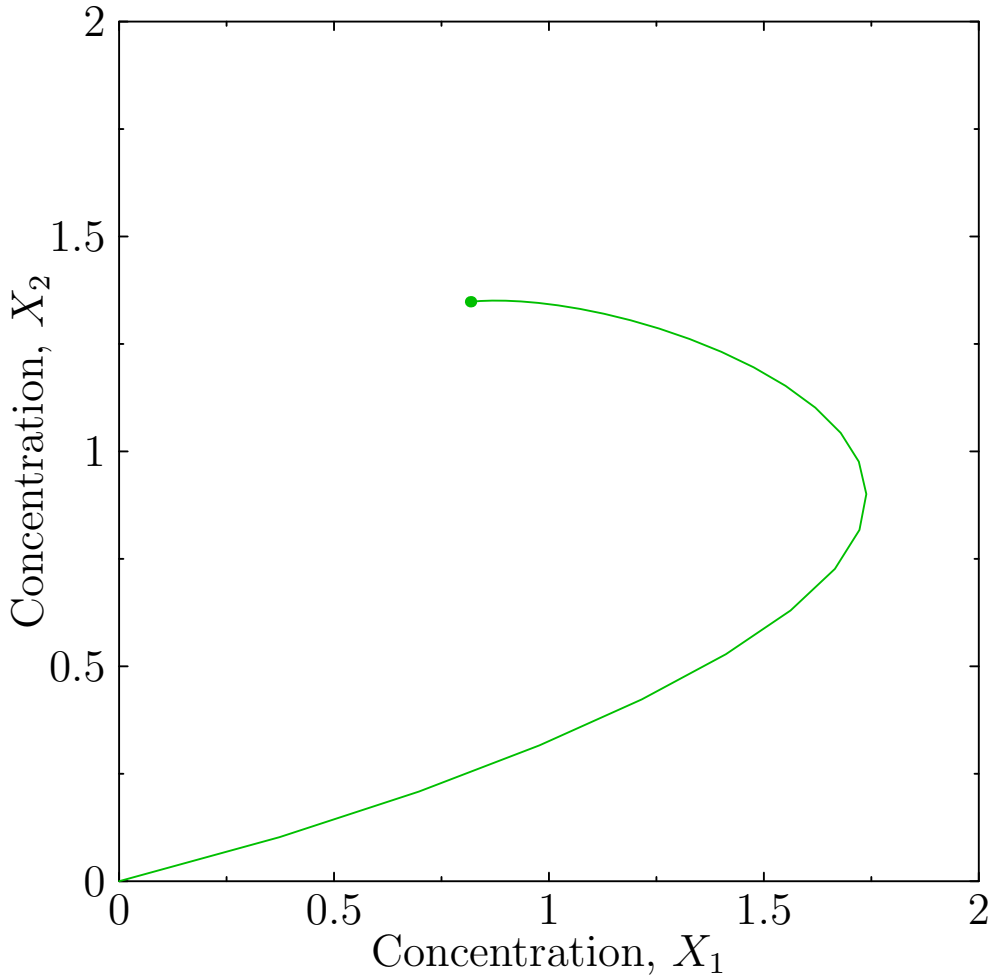

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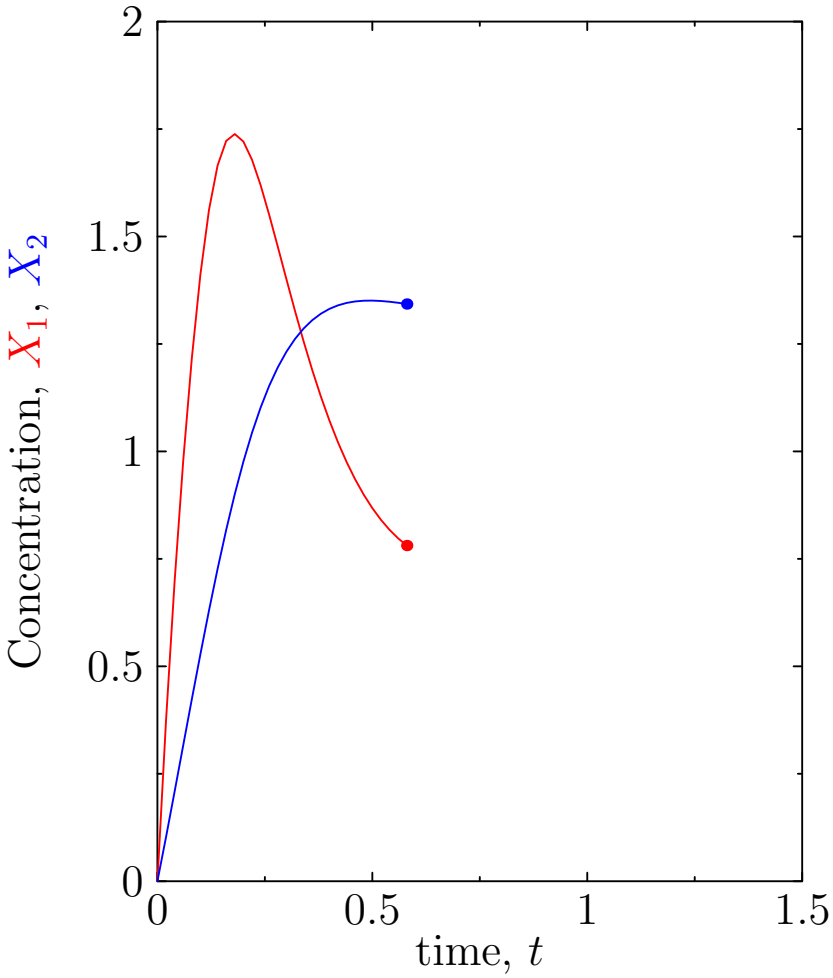

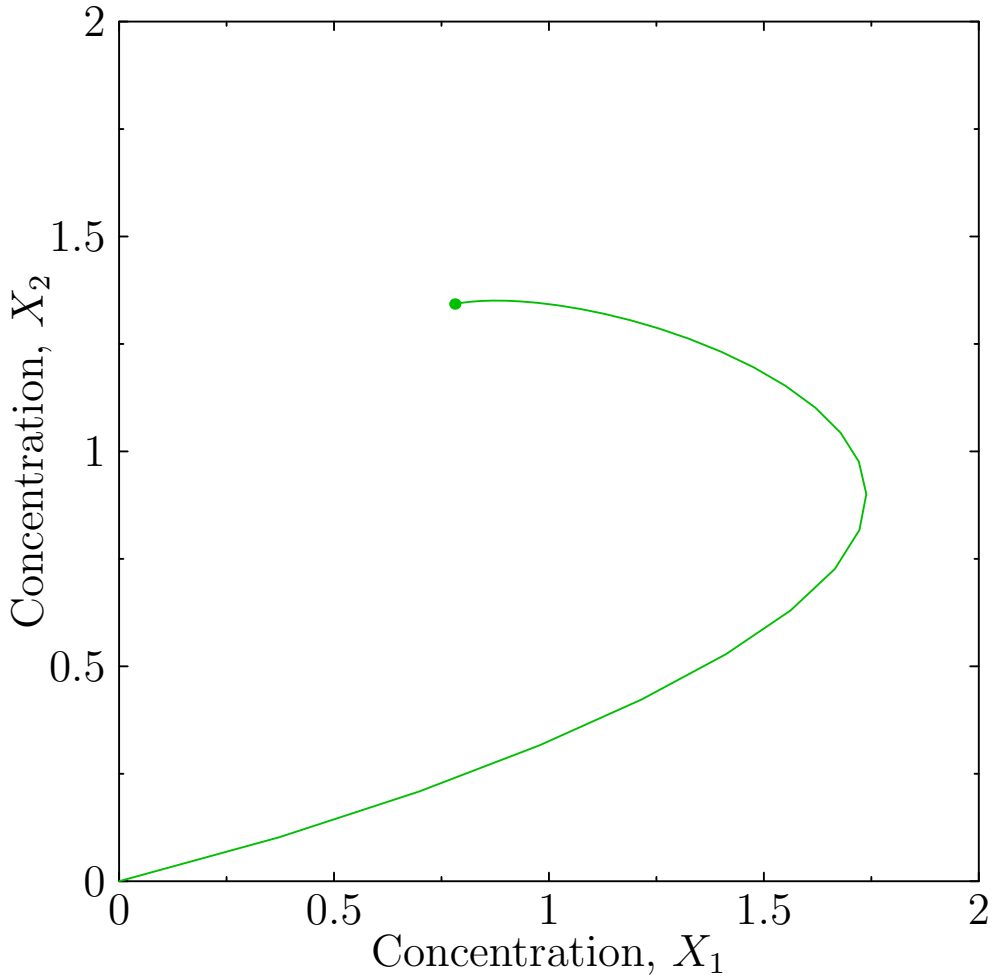

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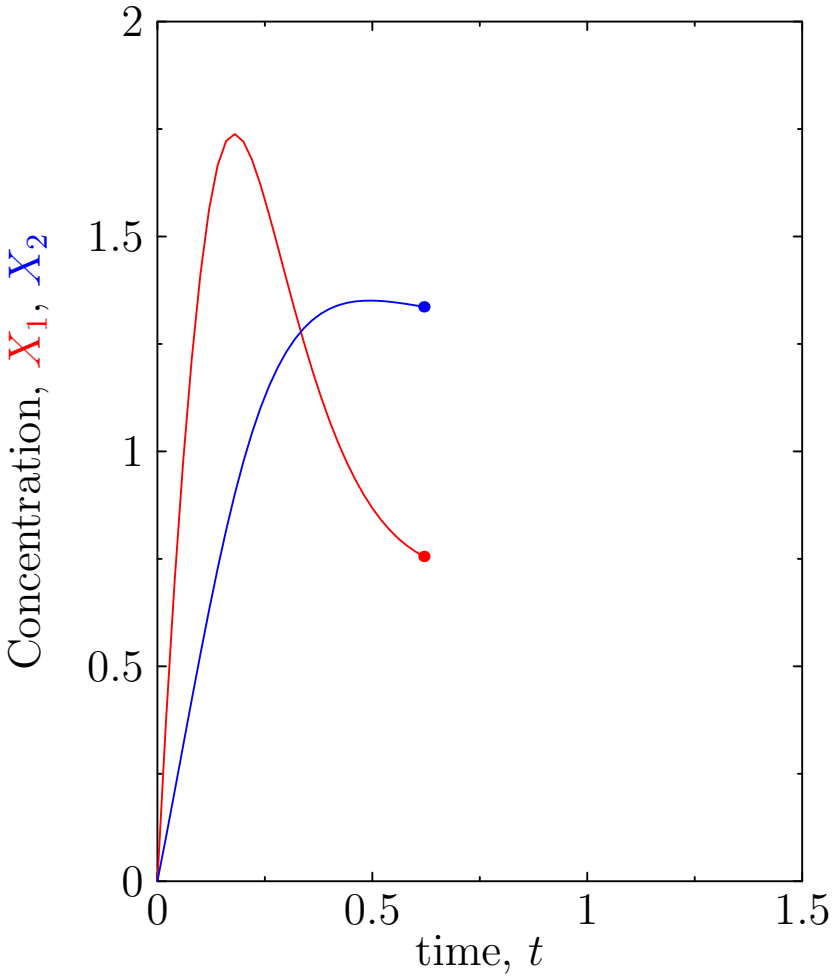

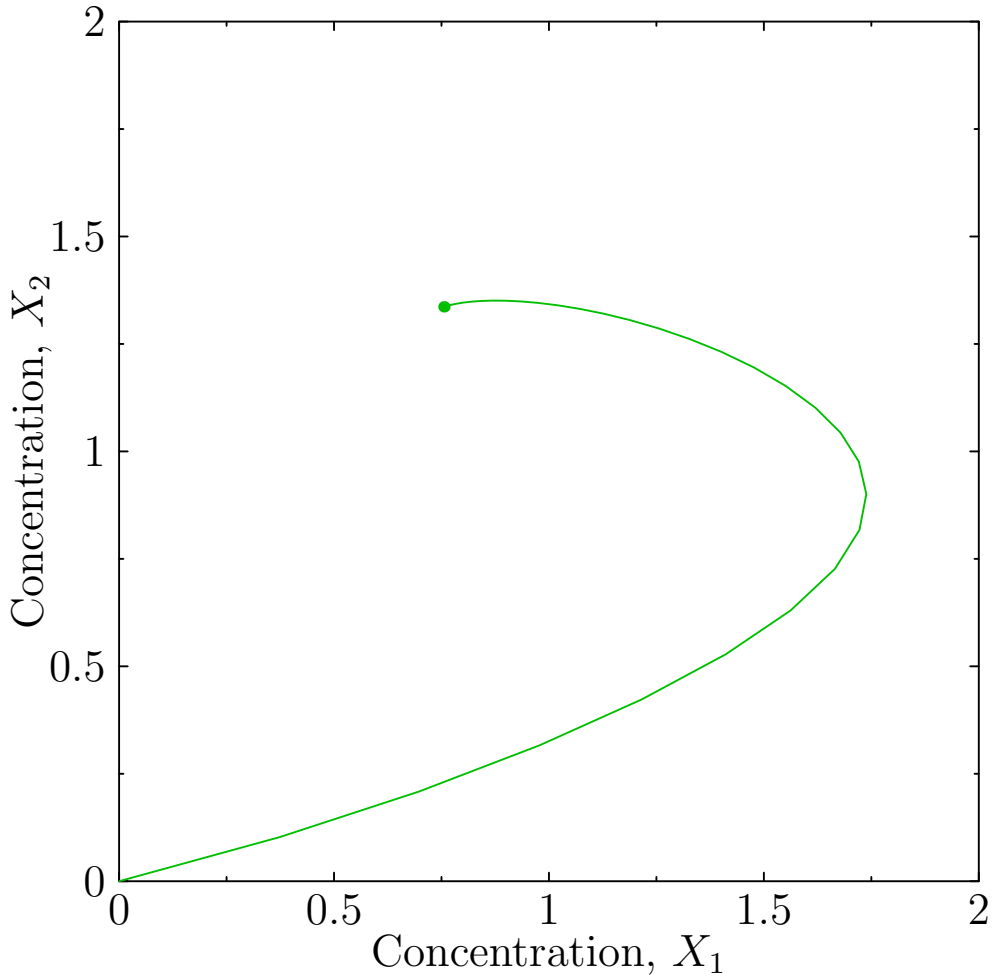

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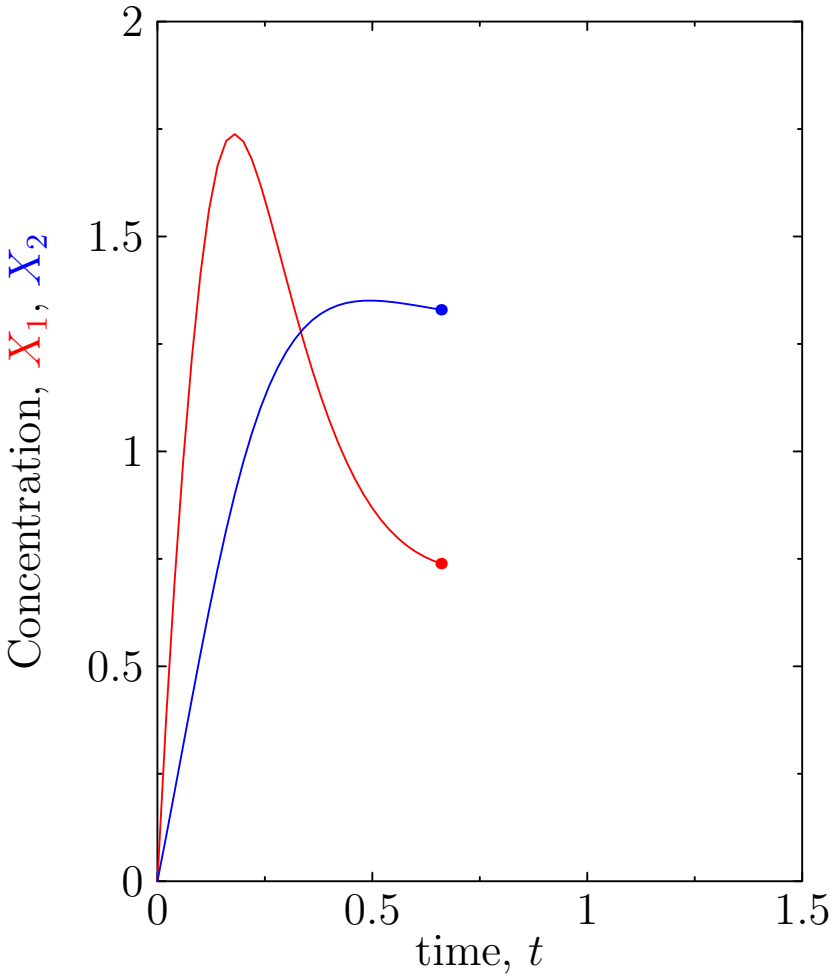

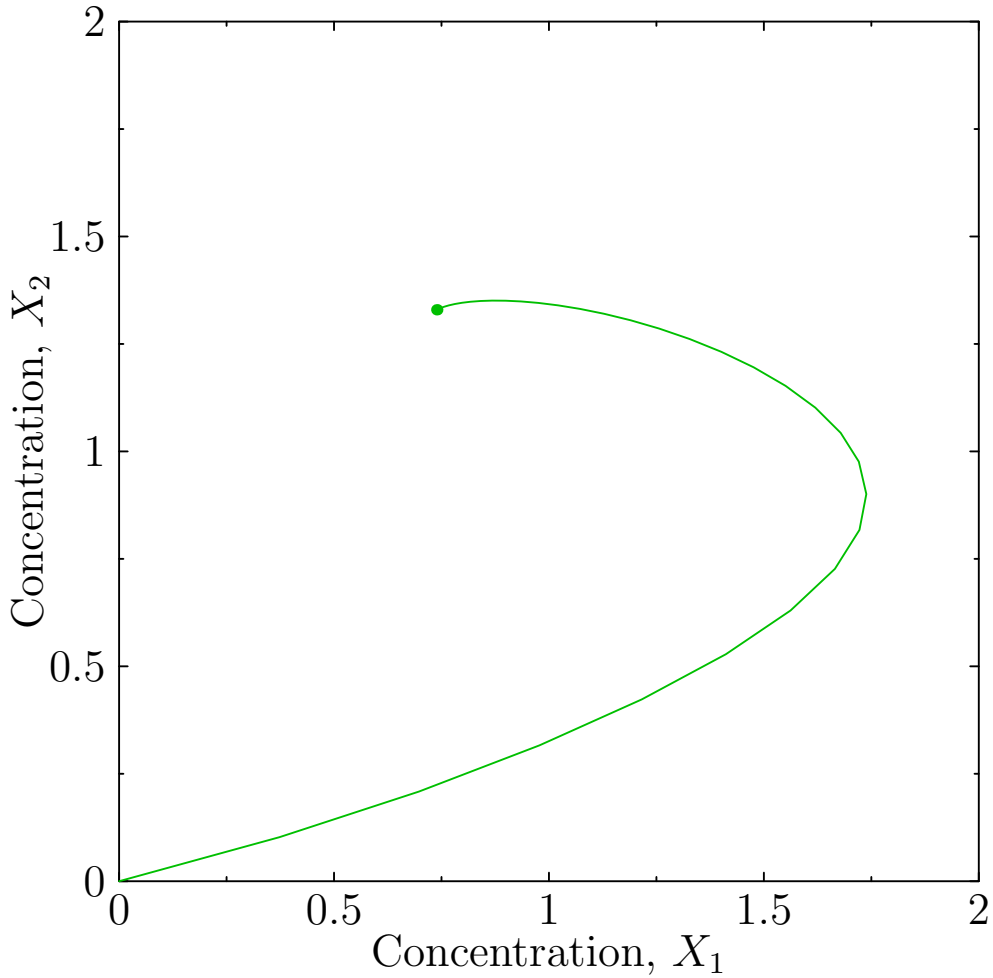

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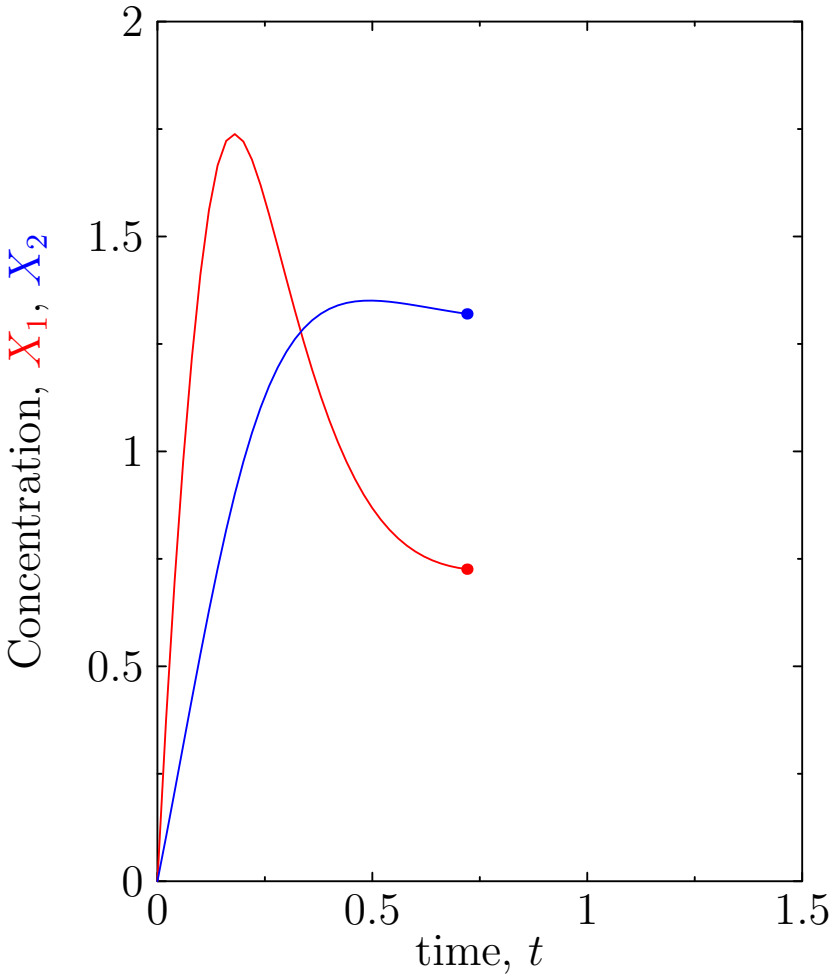

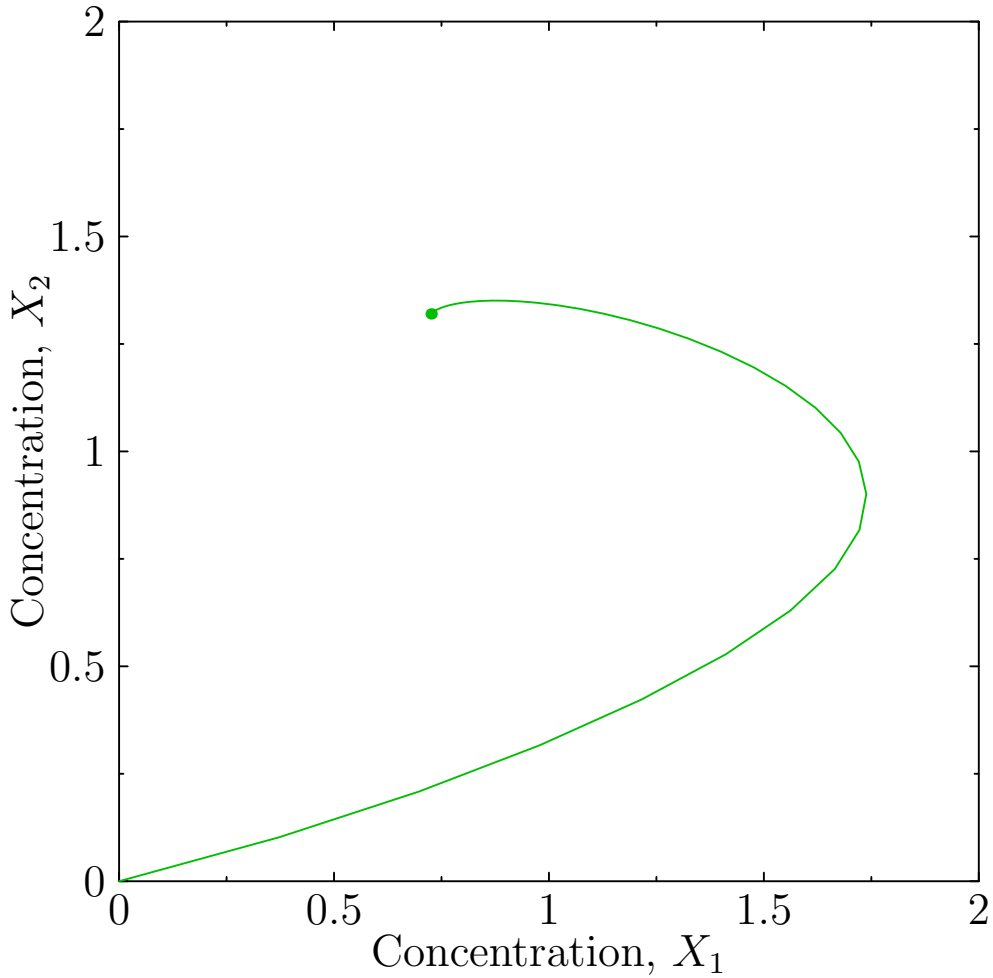

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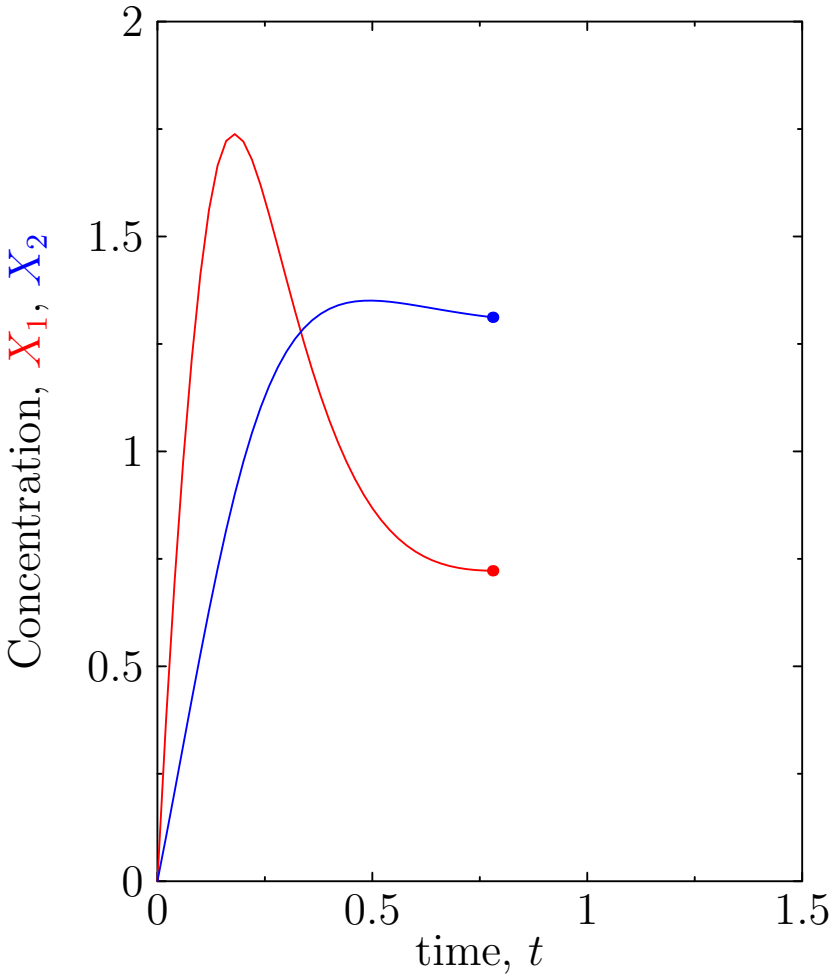

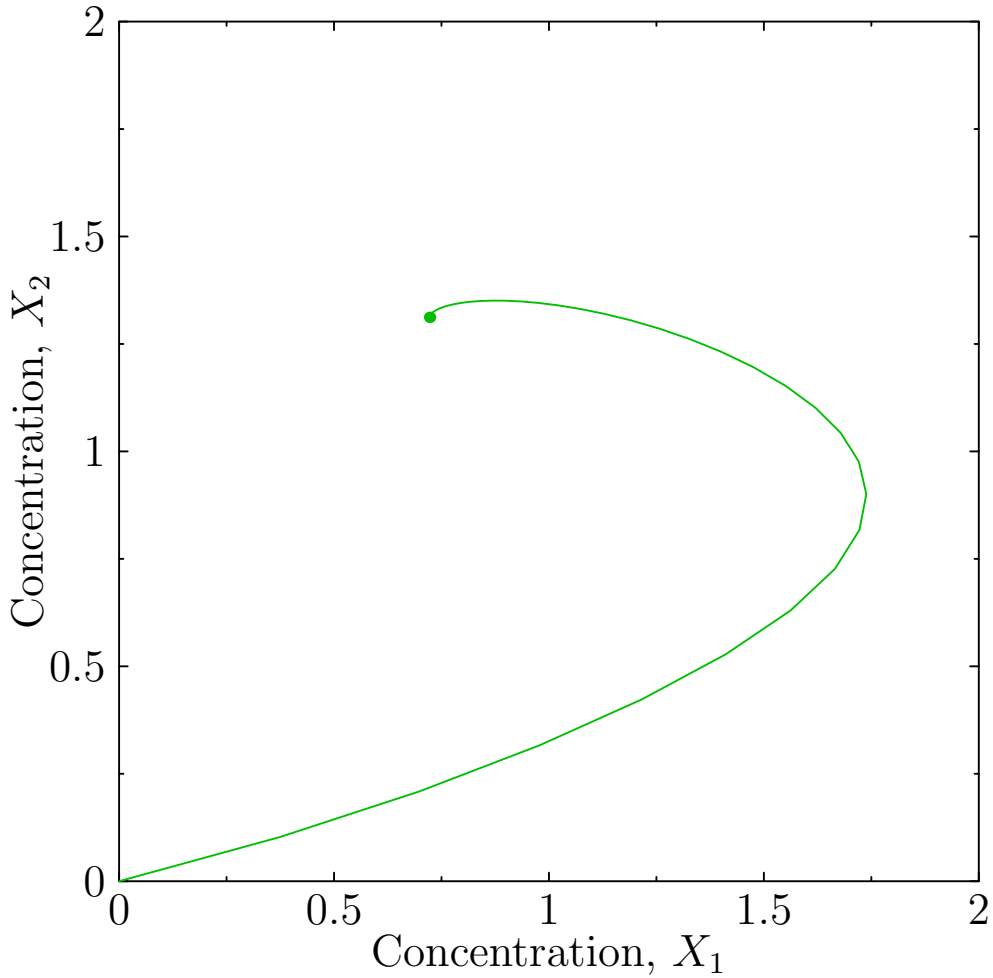

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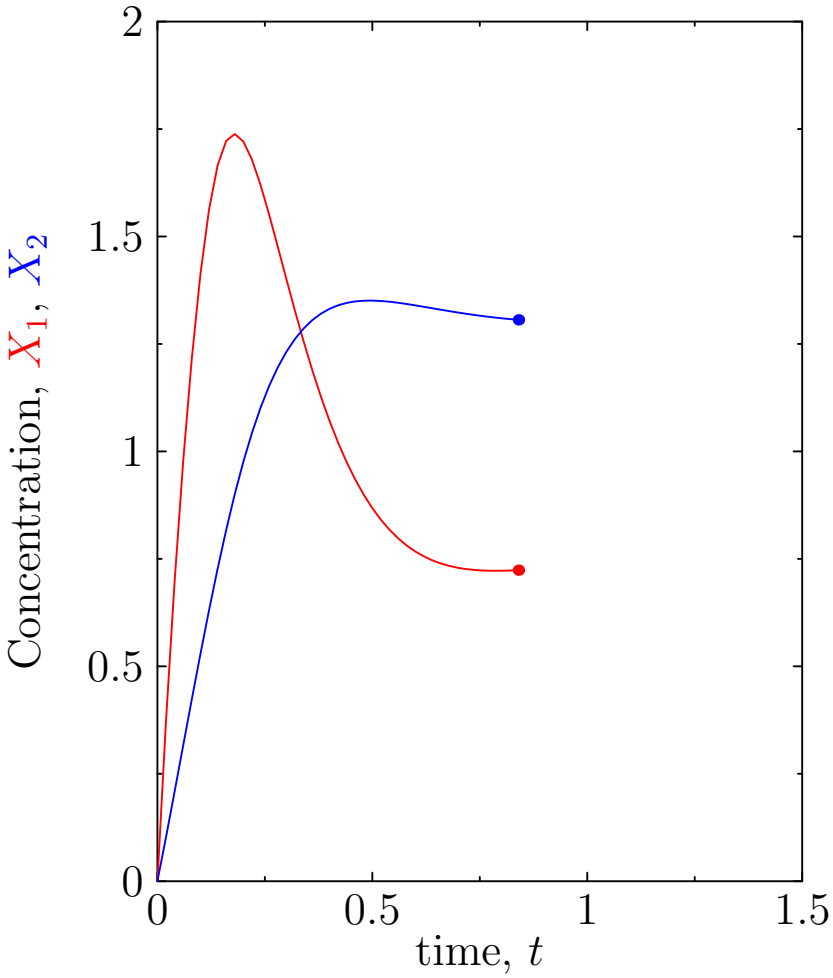

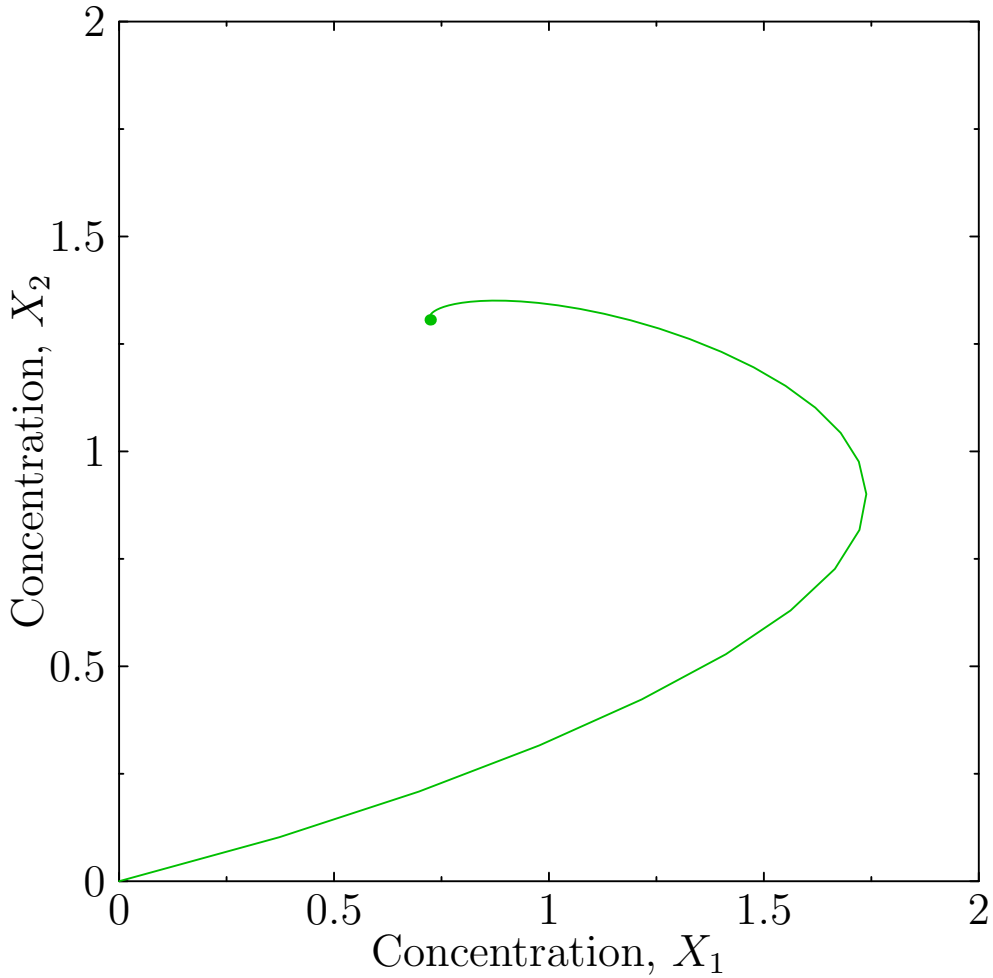

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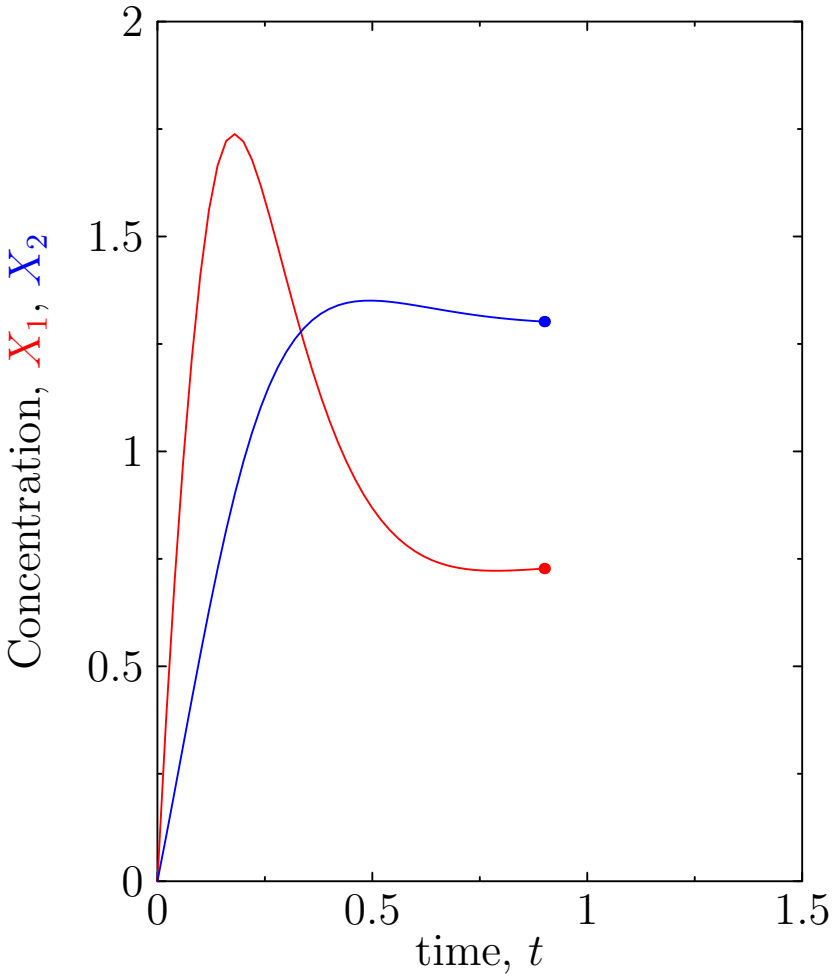

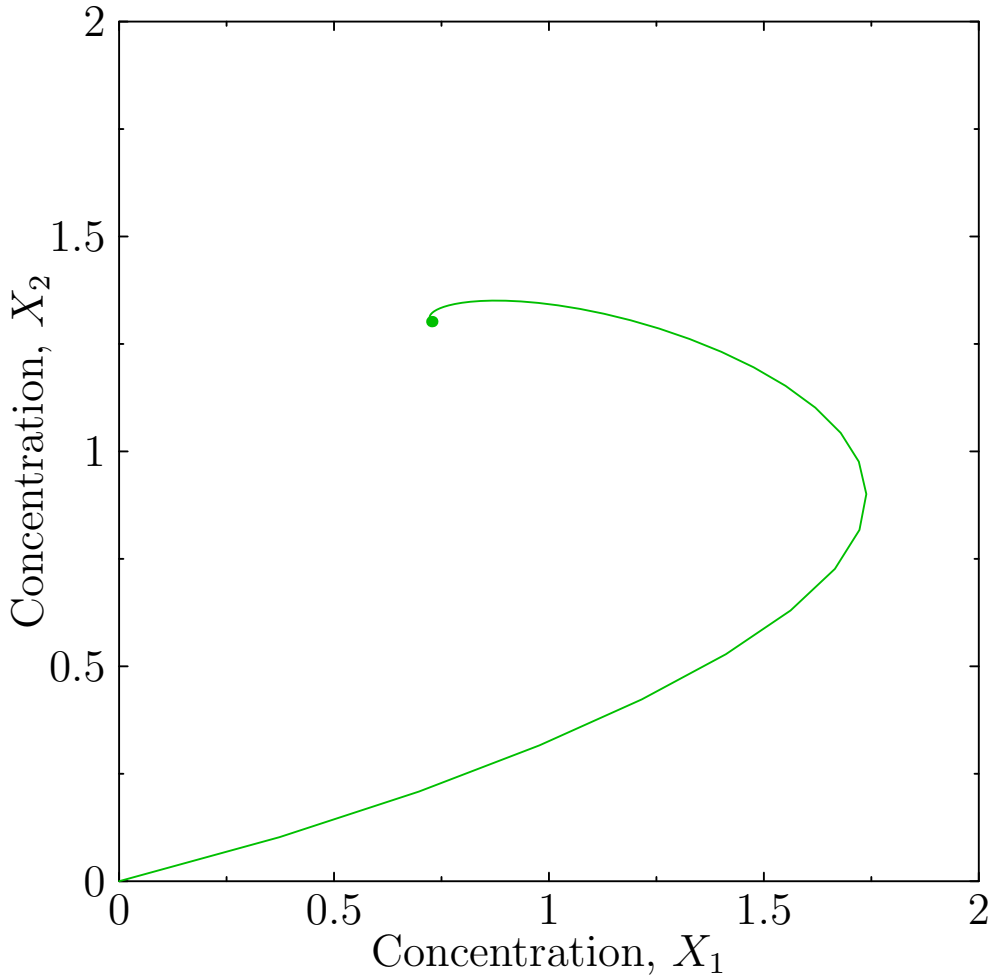

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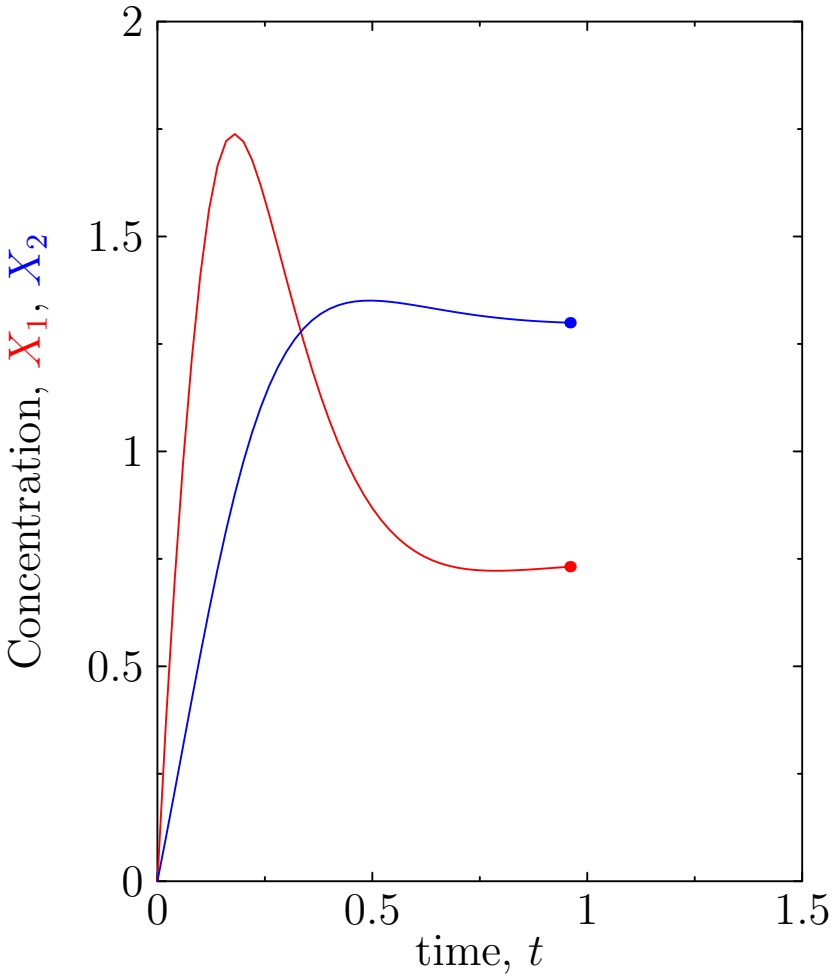

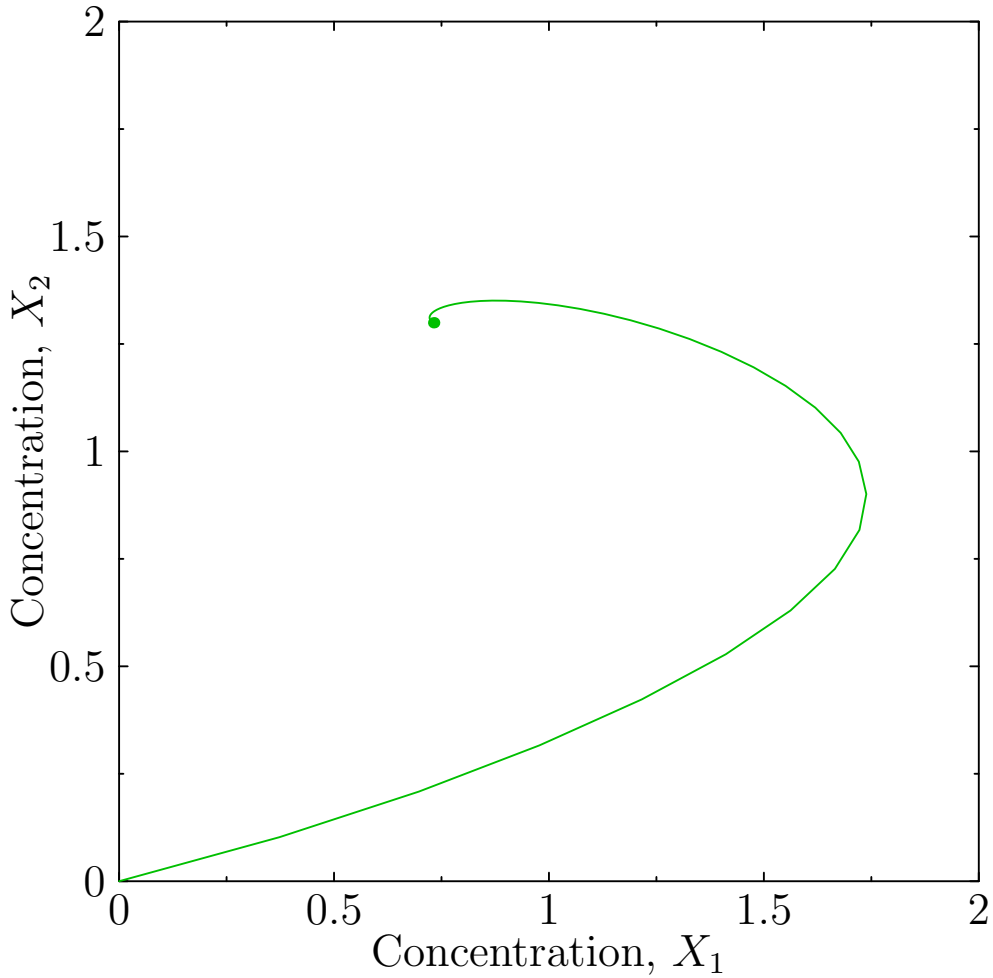

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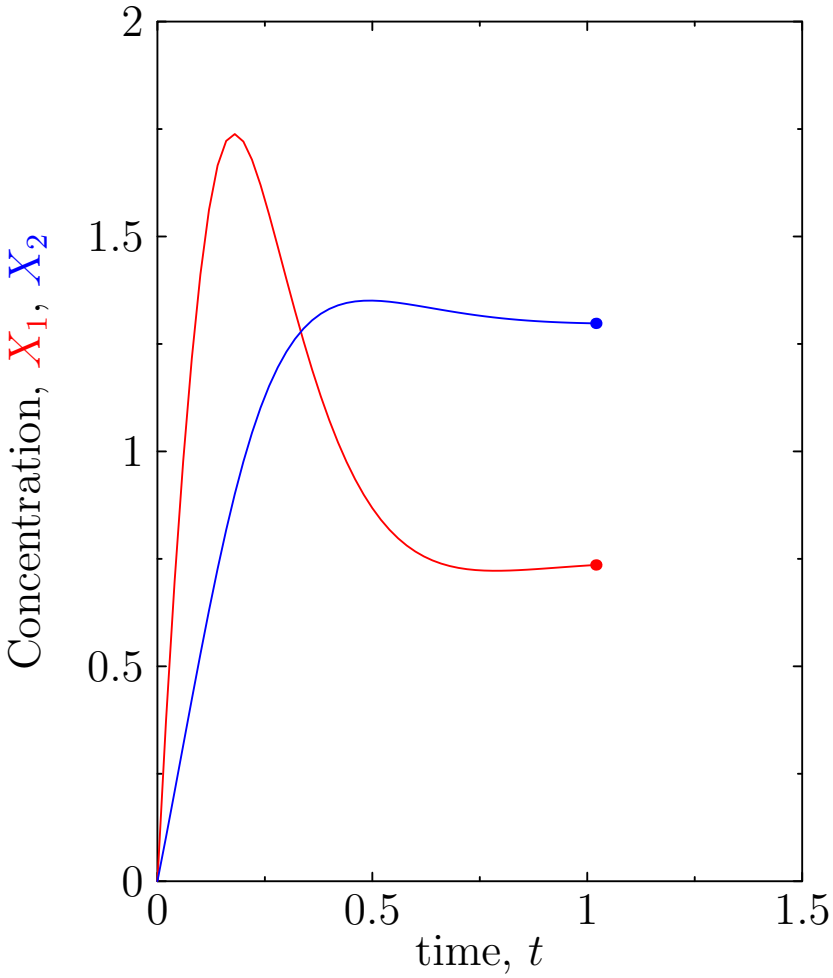

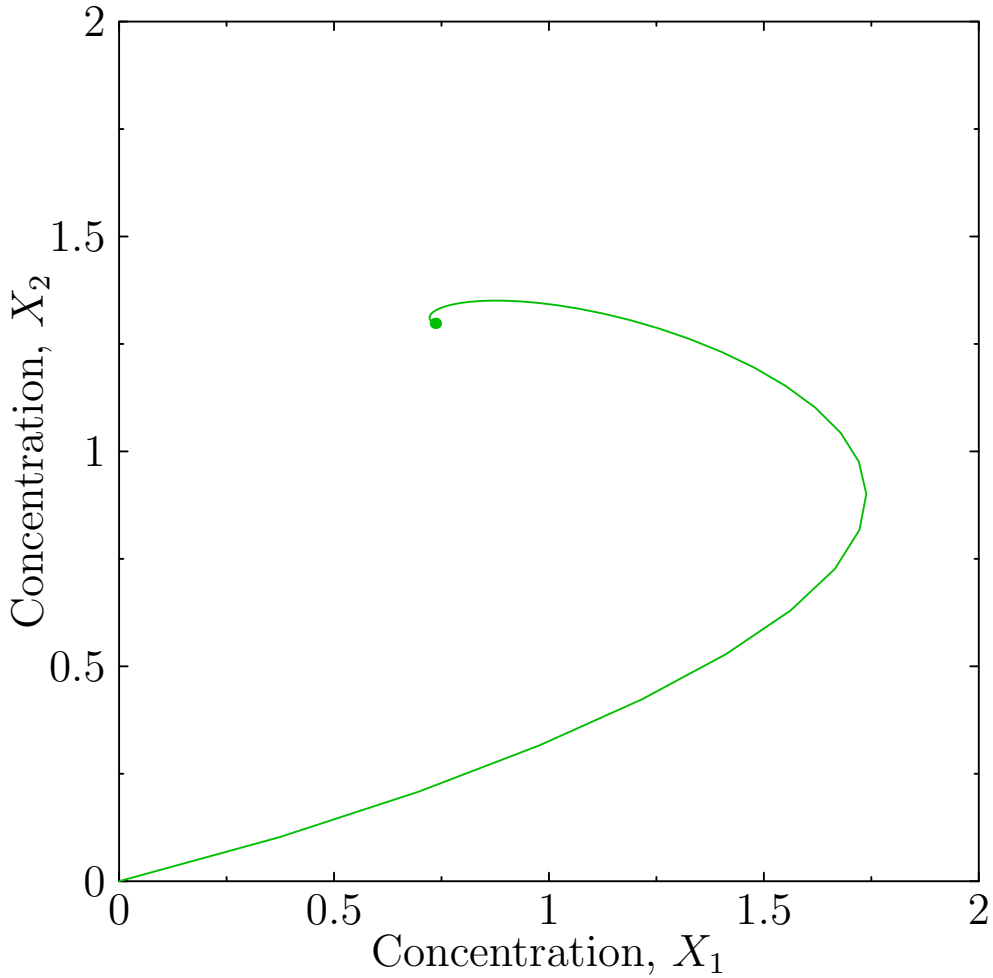

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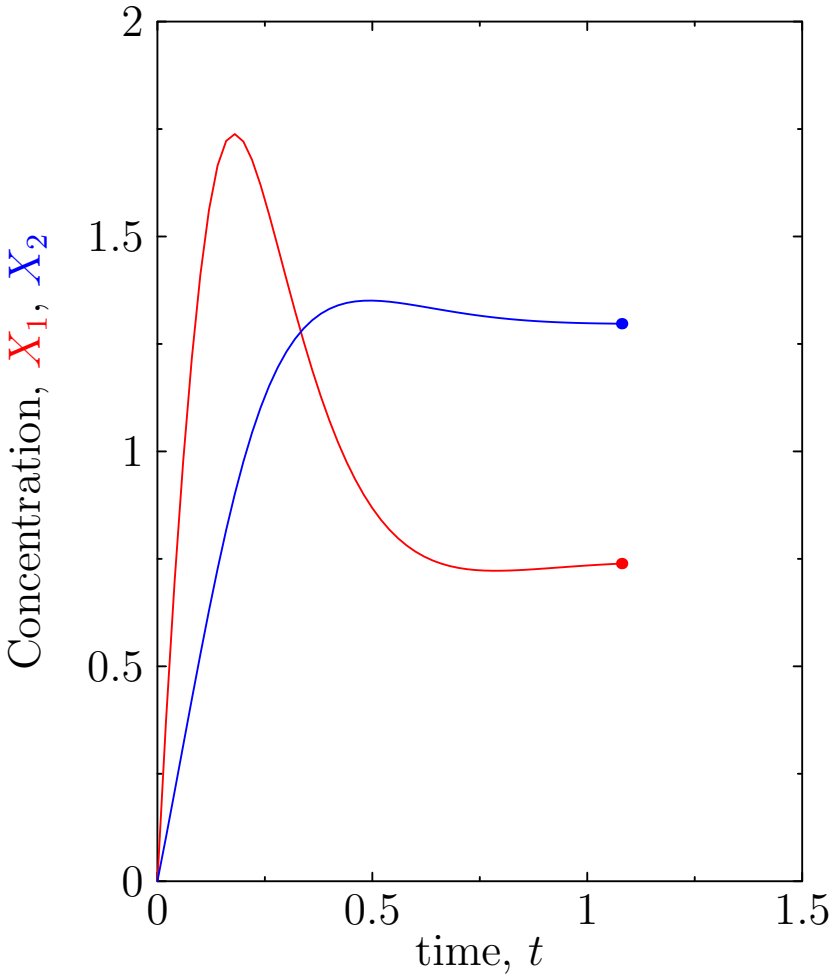

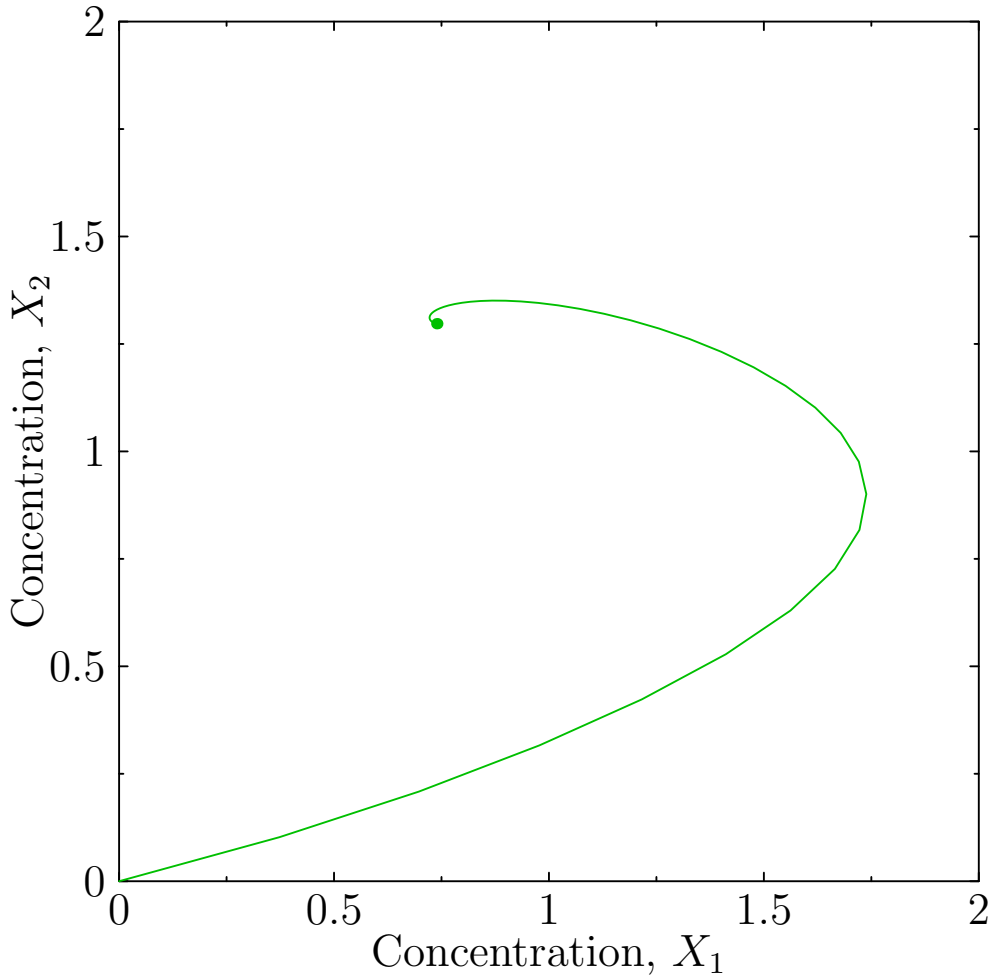

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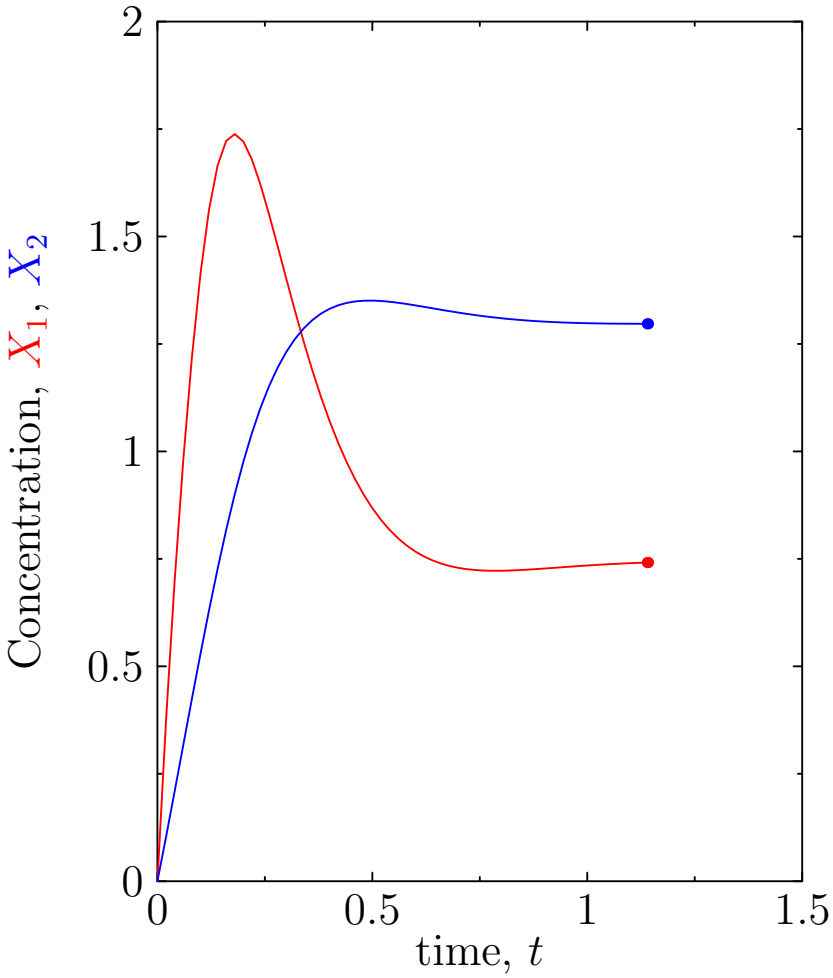

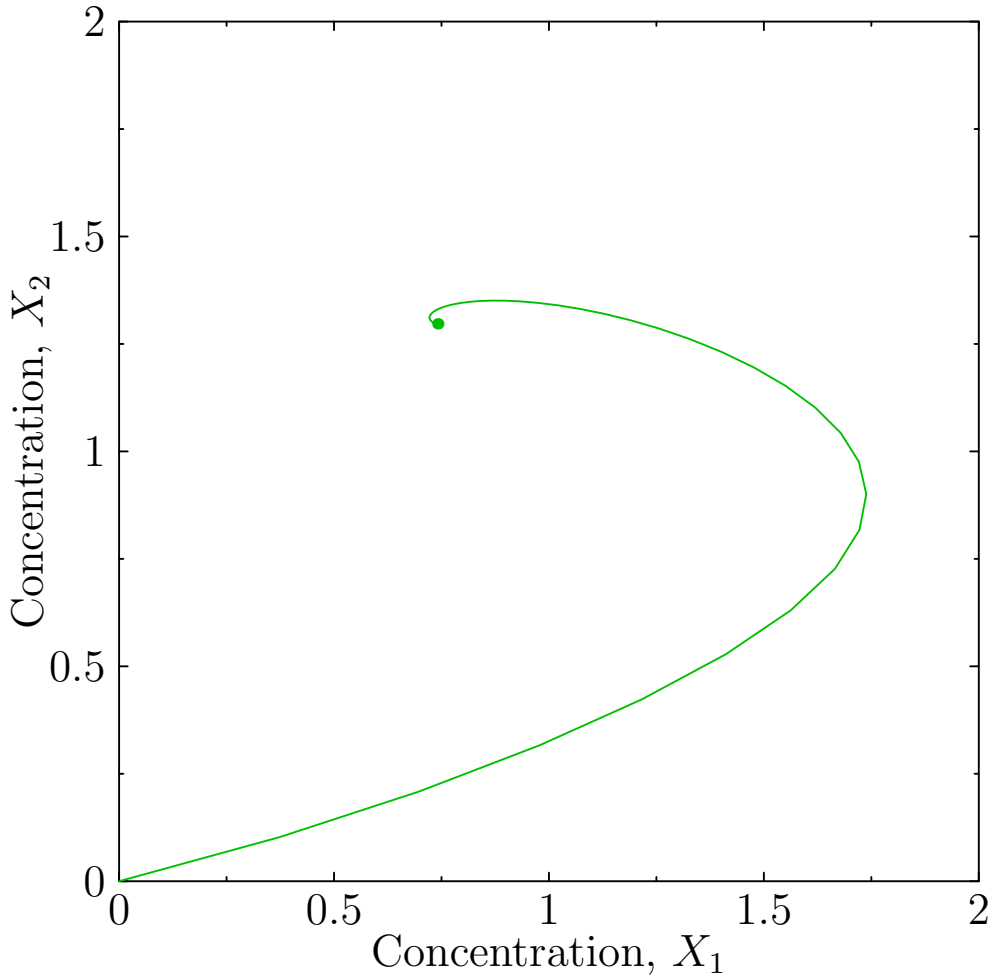

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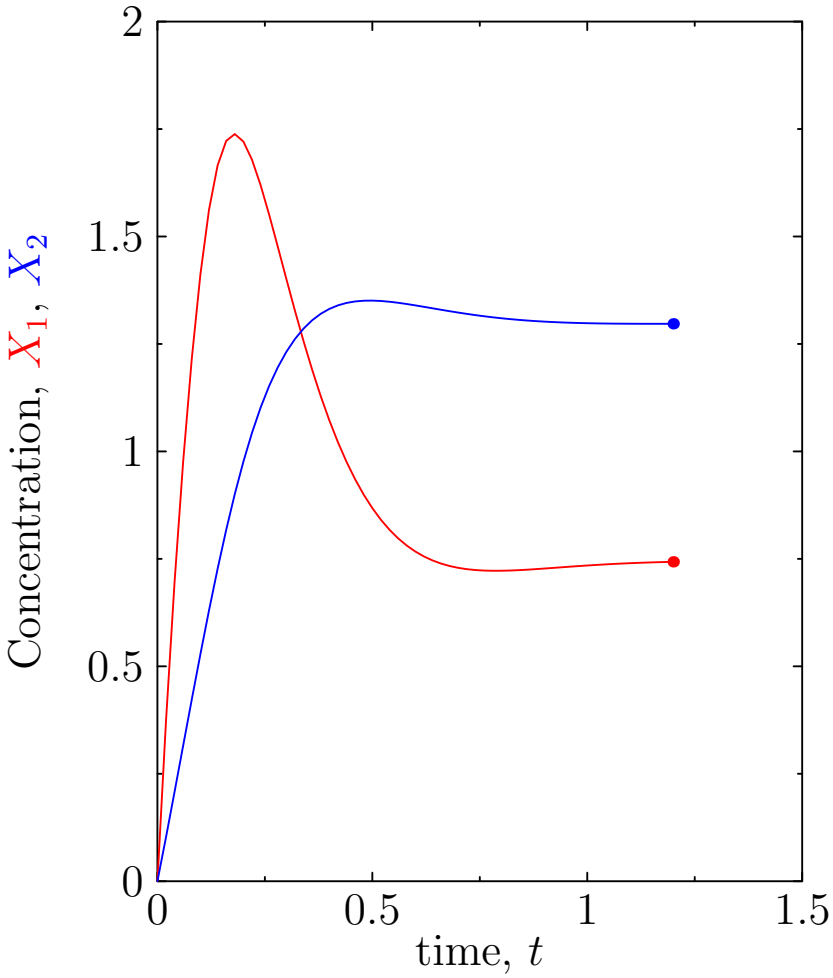

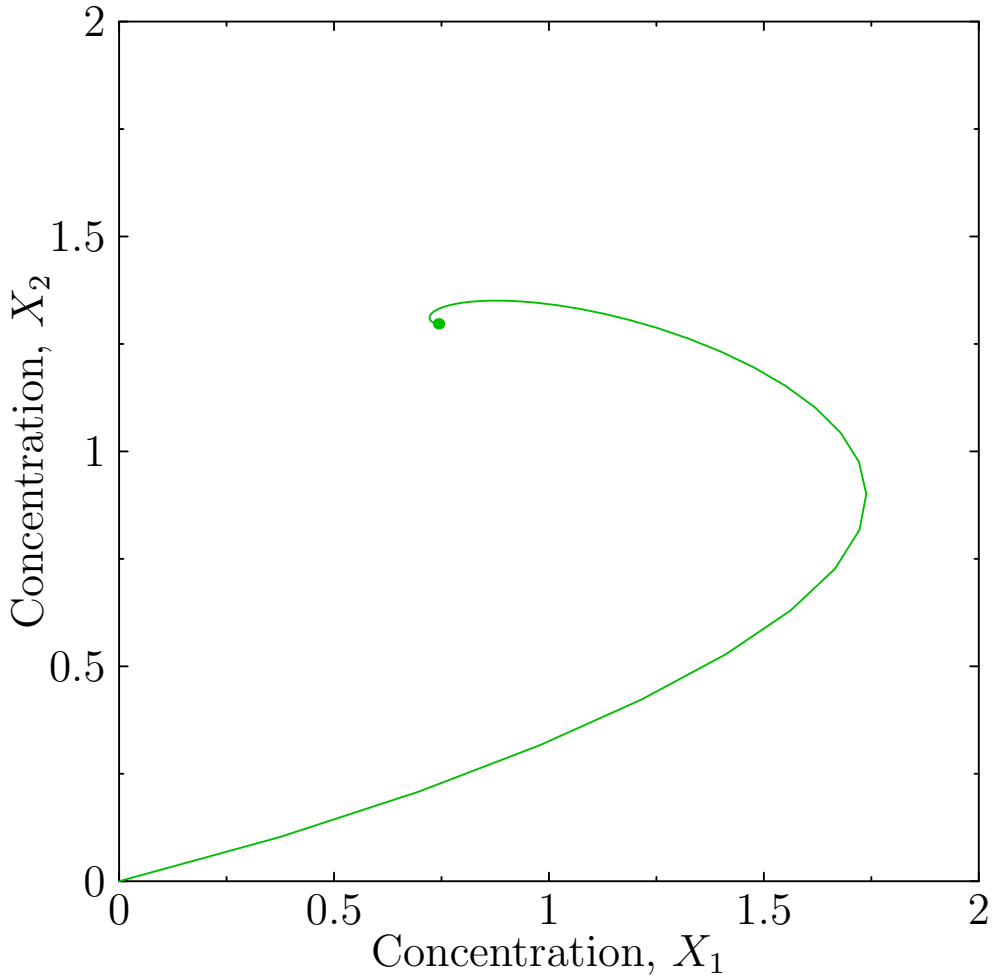

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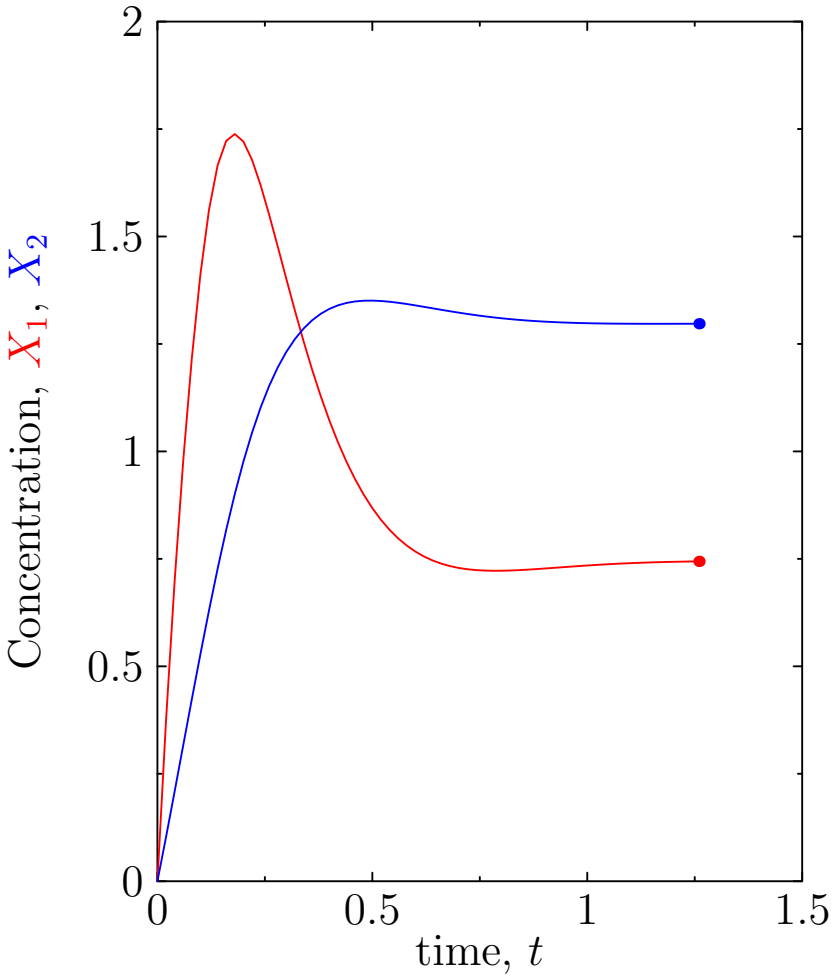

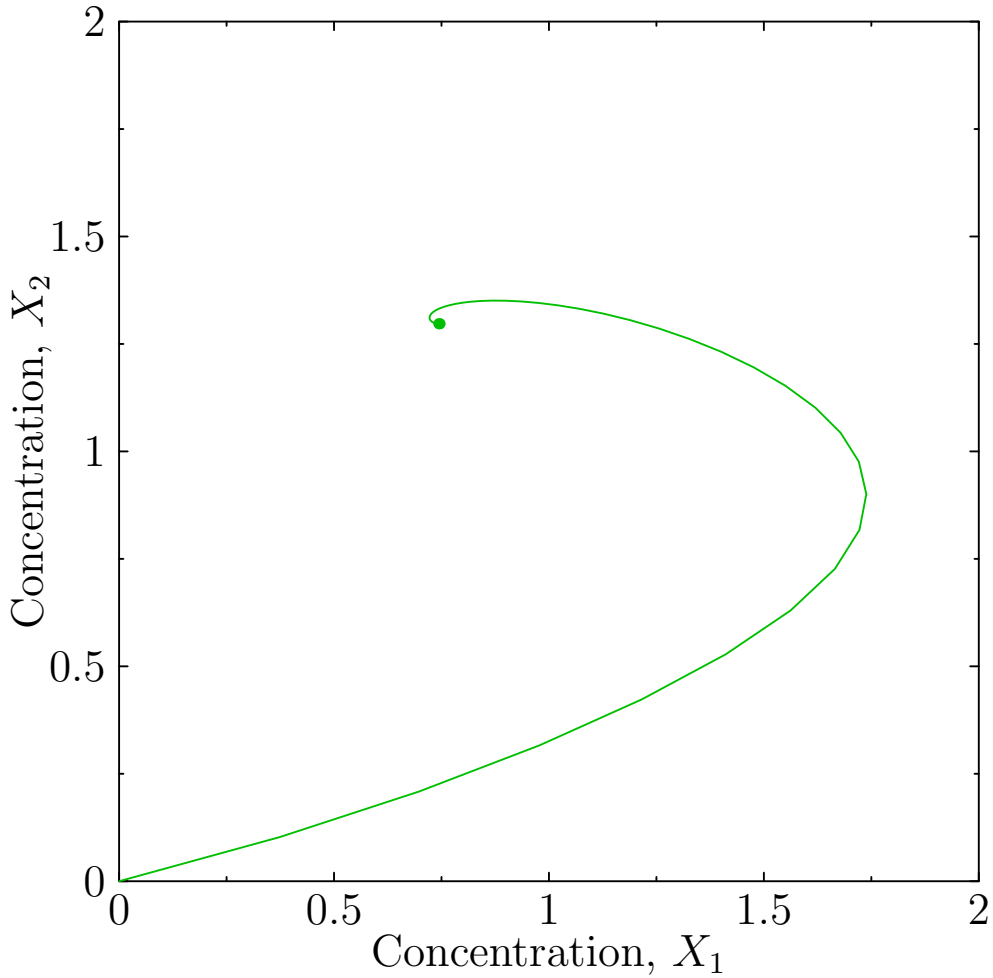

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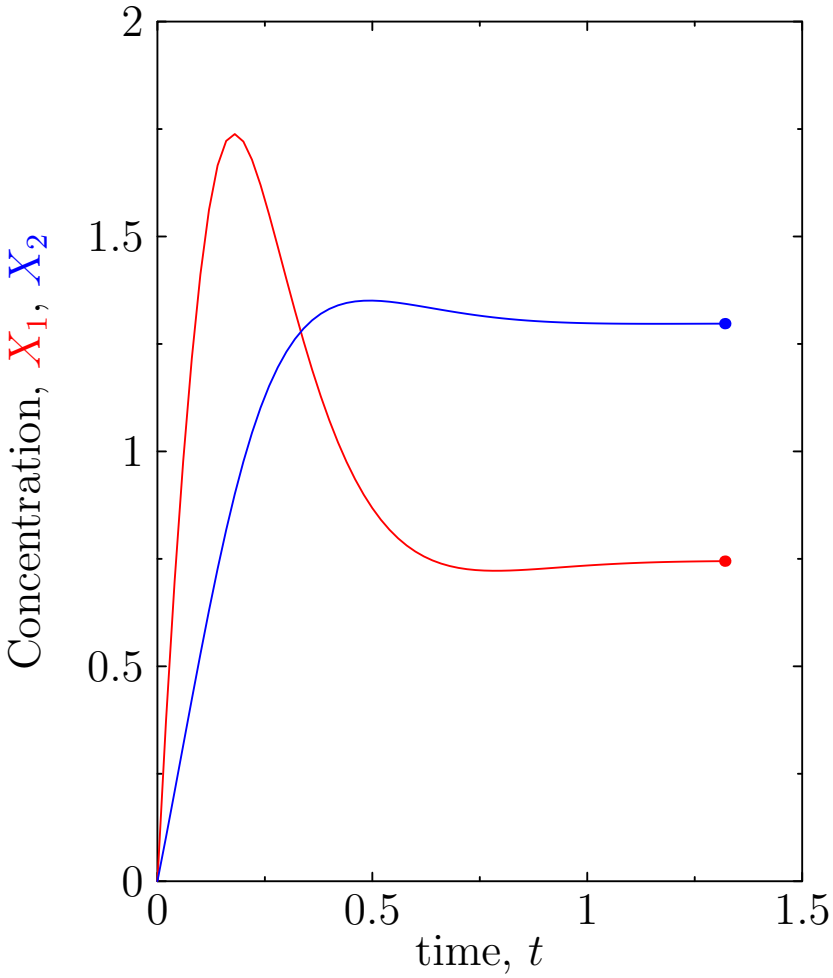

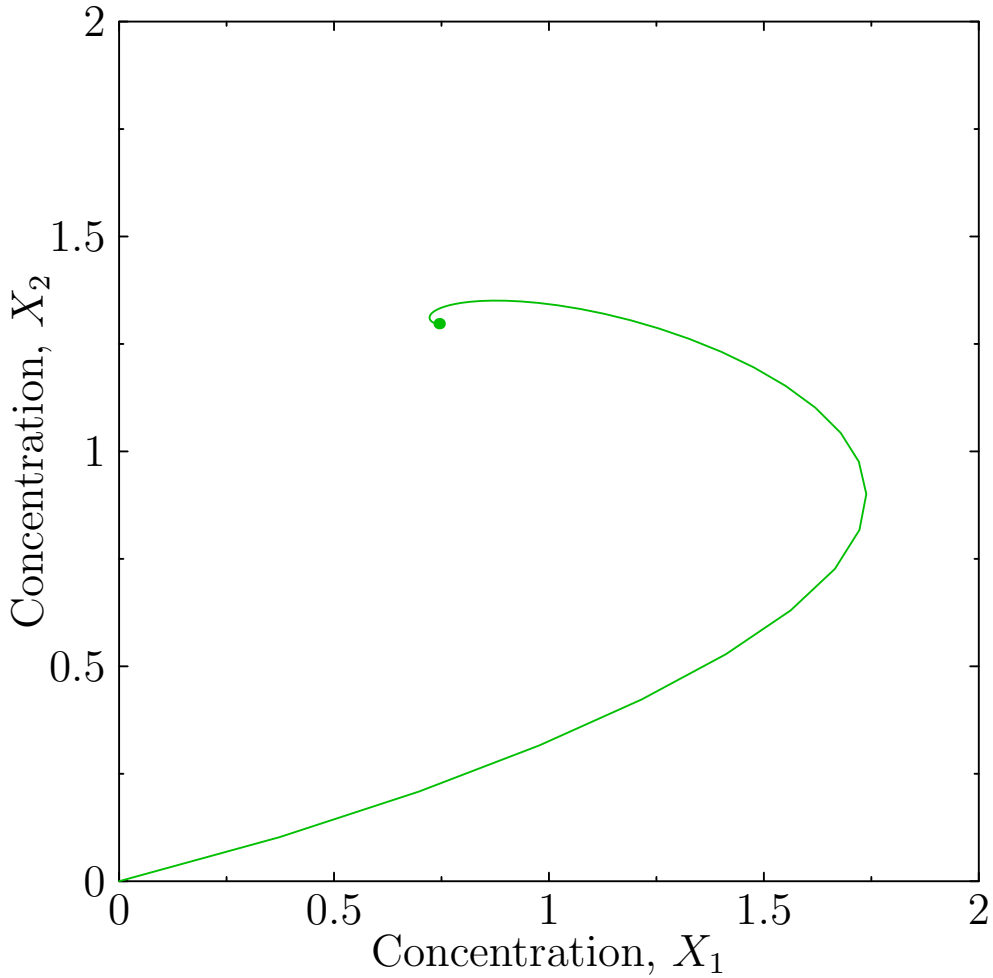

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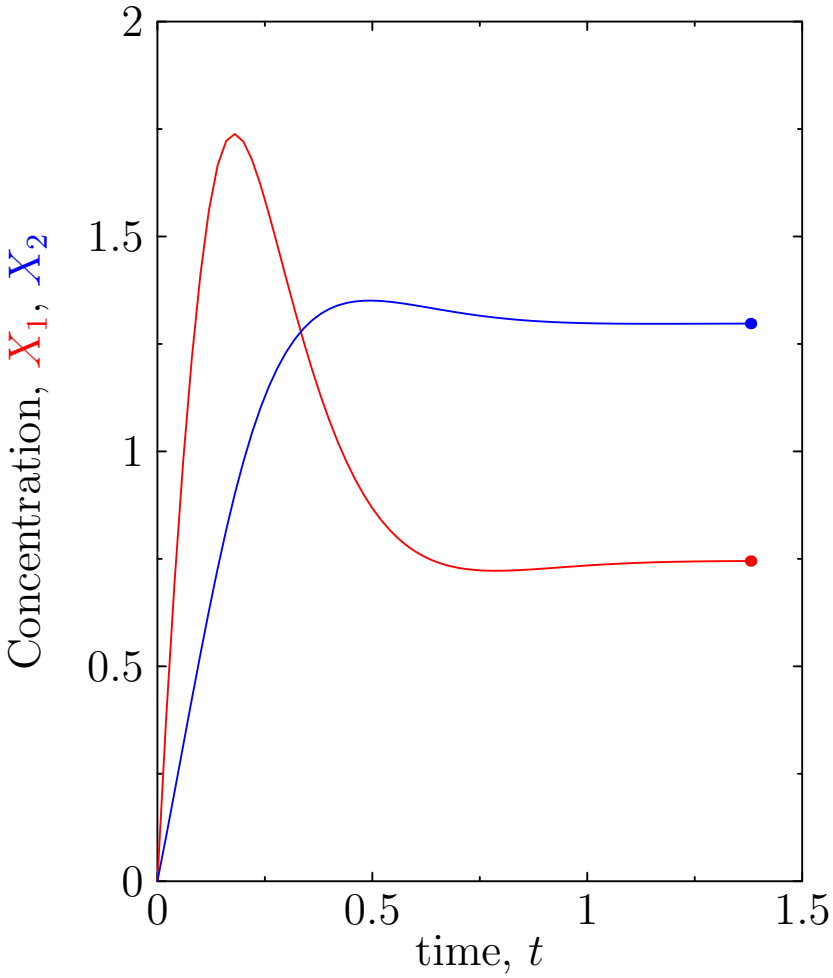

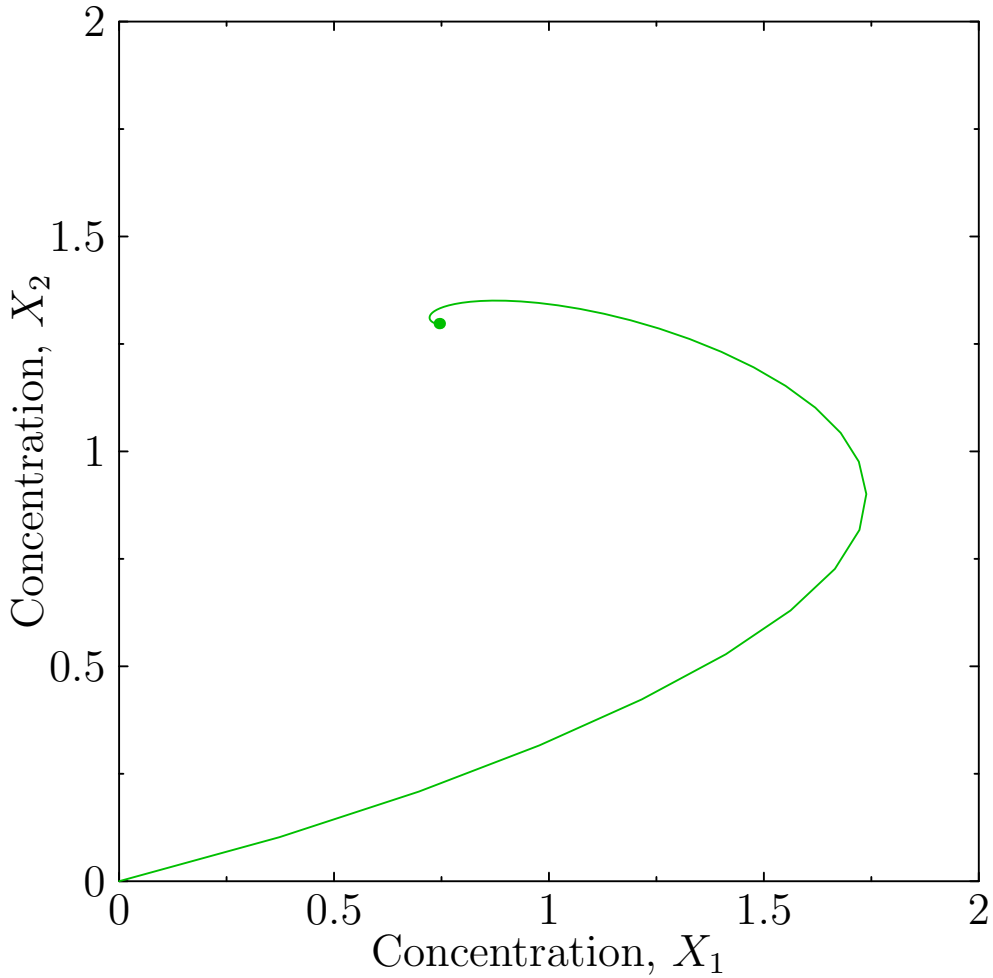

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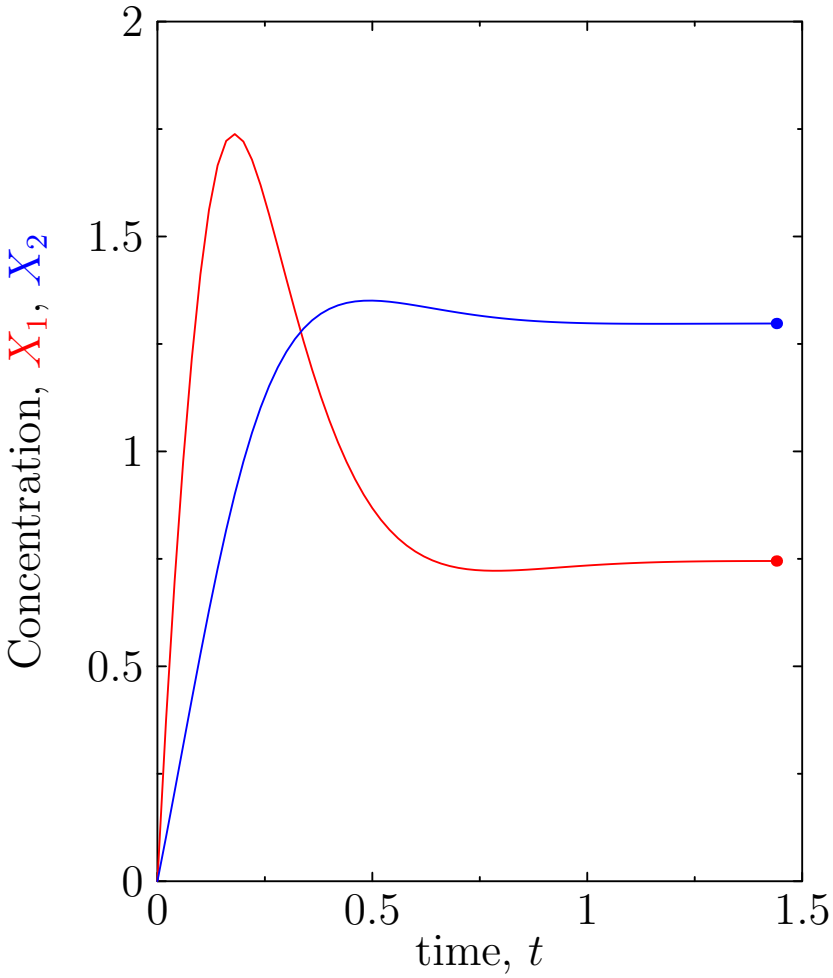

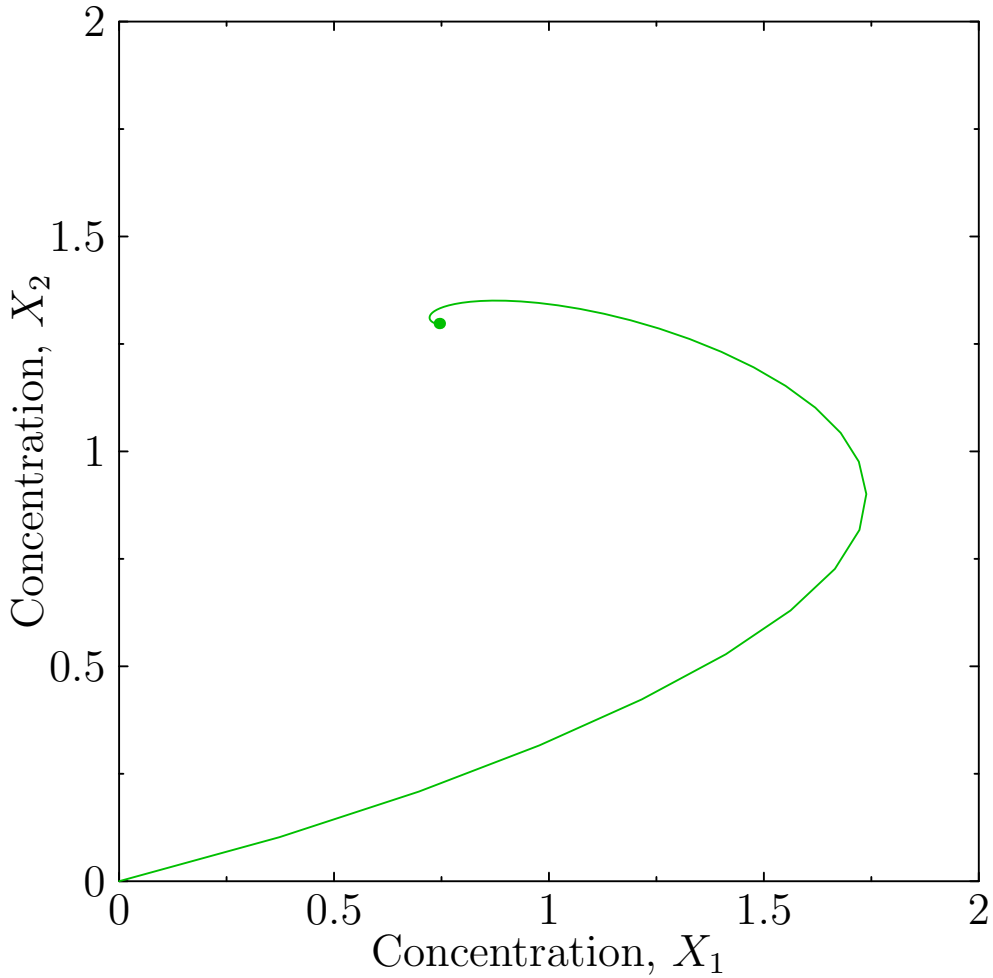

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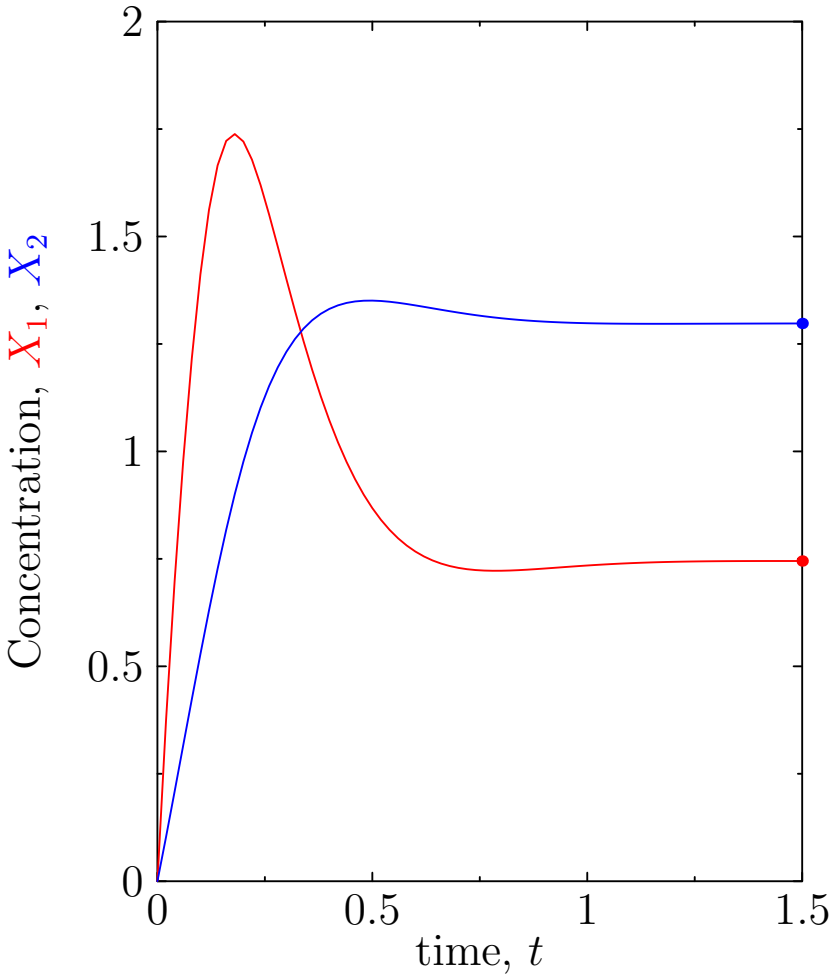

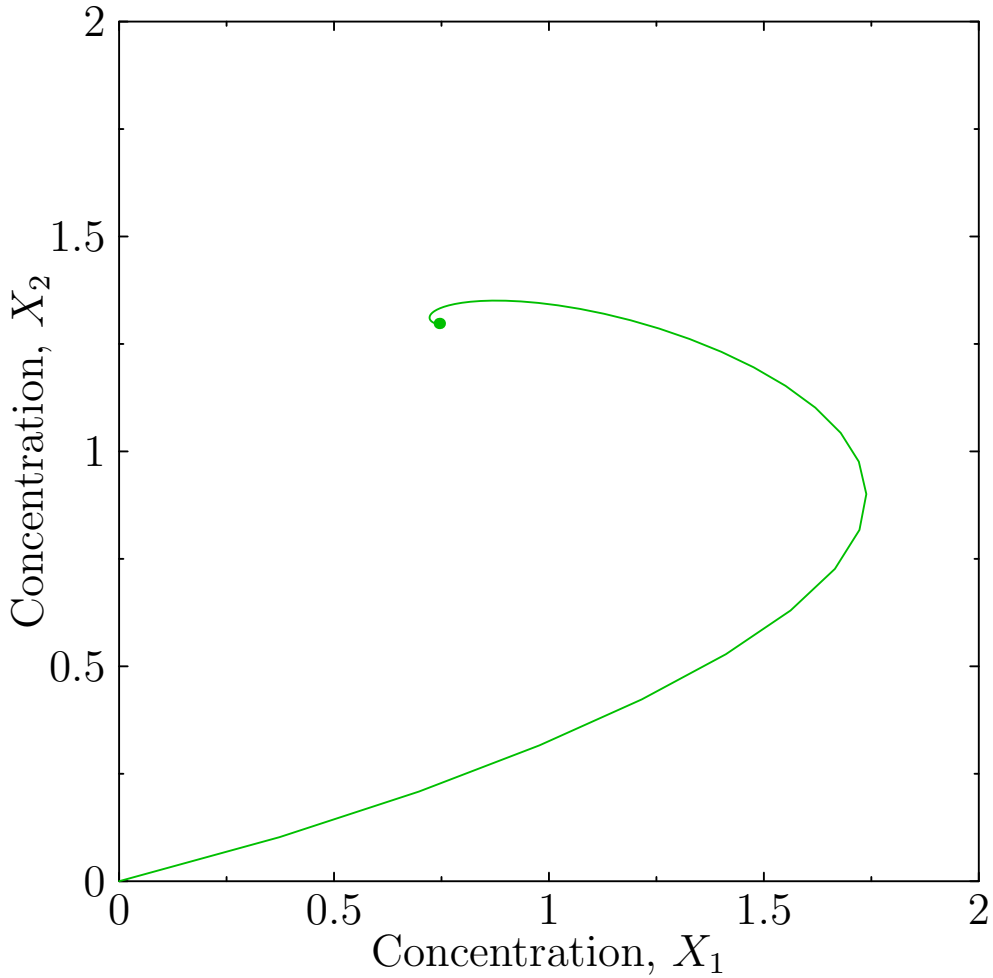

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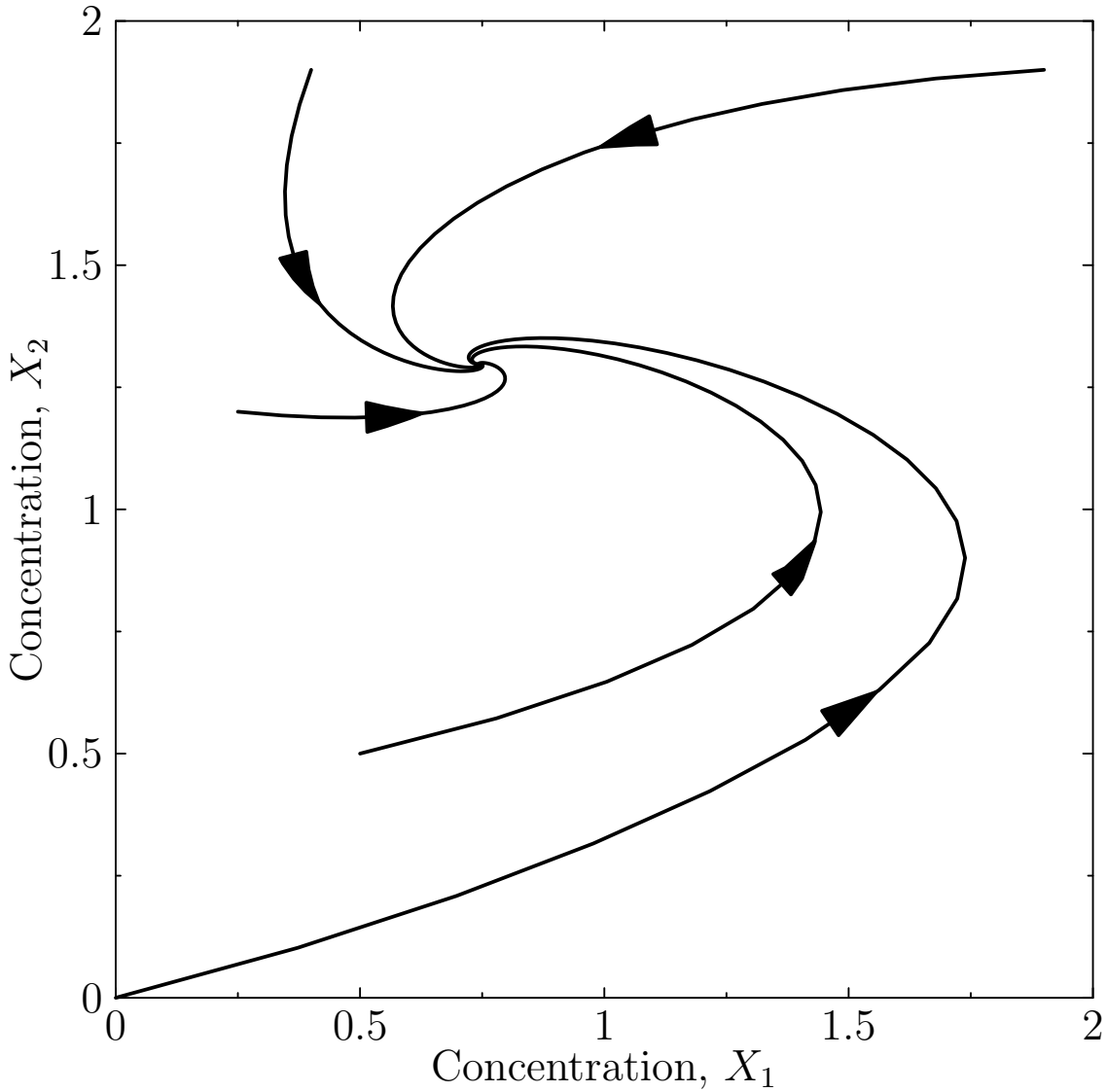
Trajectories



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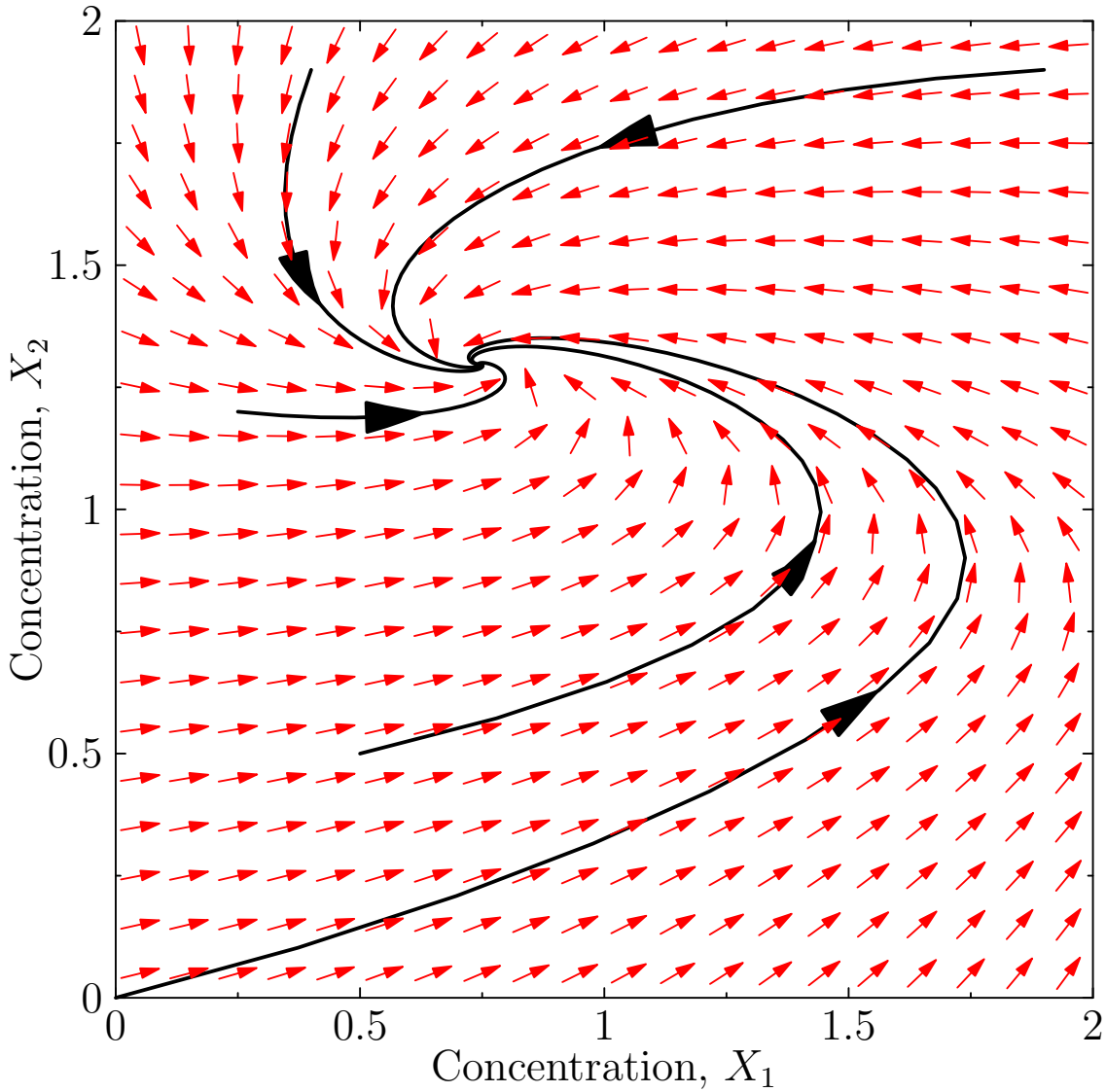
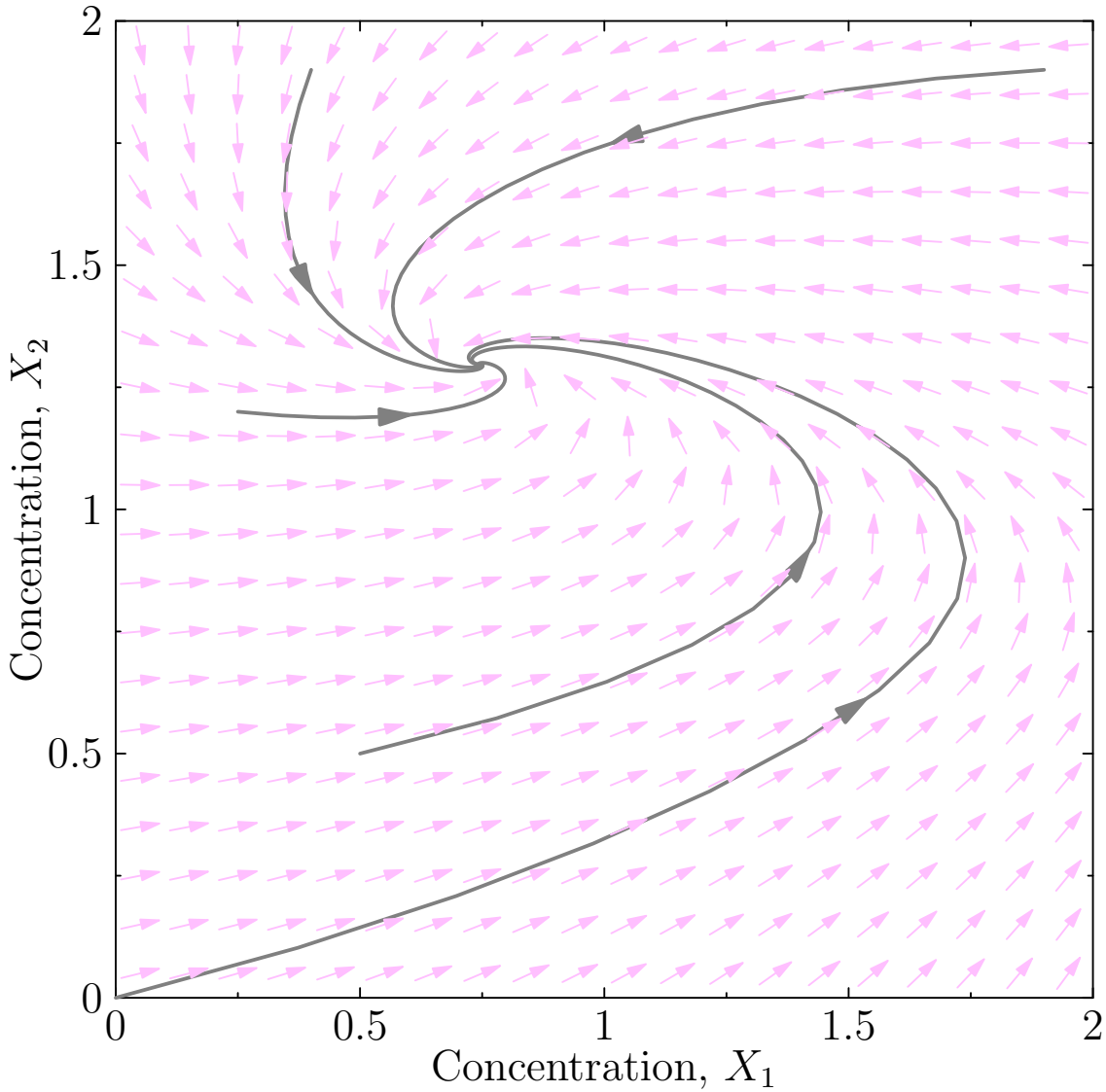


Nullclines



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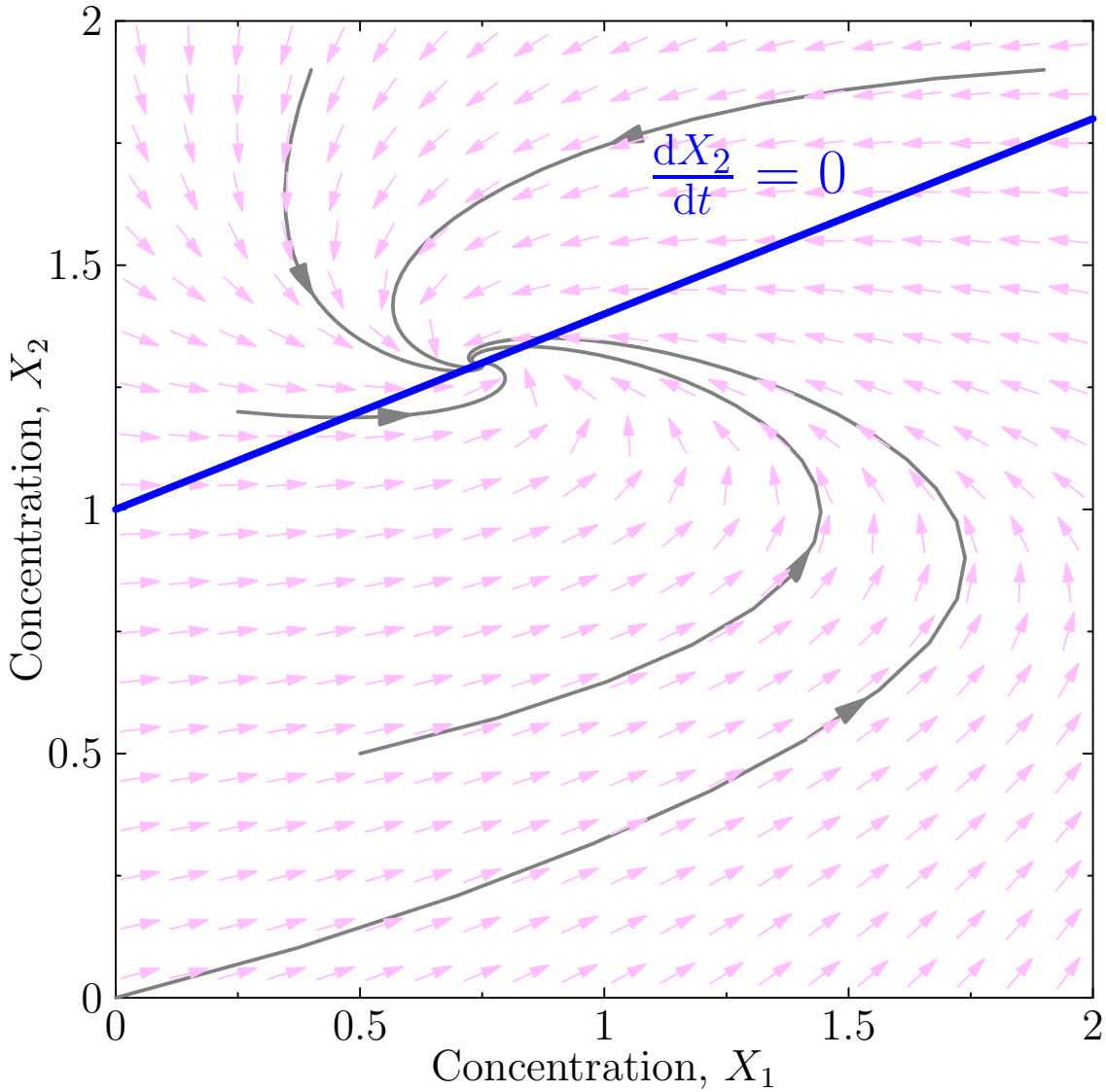
$\frac{\mathrm{d}X_2}{\mathrm{d}t} = 0$
Concentration, X_2
Concentration, X_1
0
0.5
1
1.5
2

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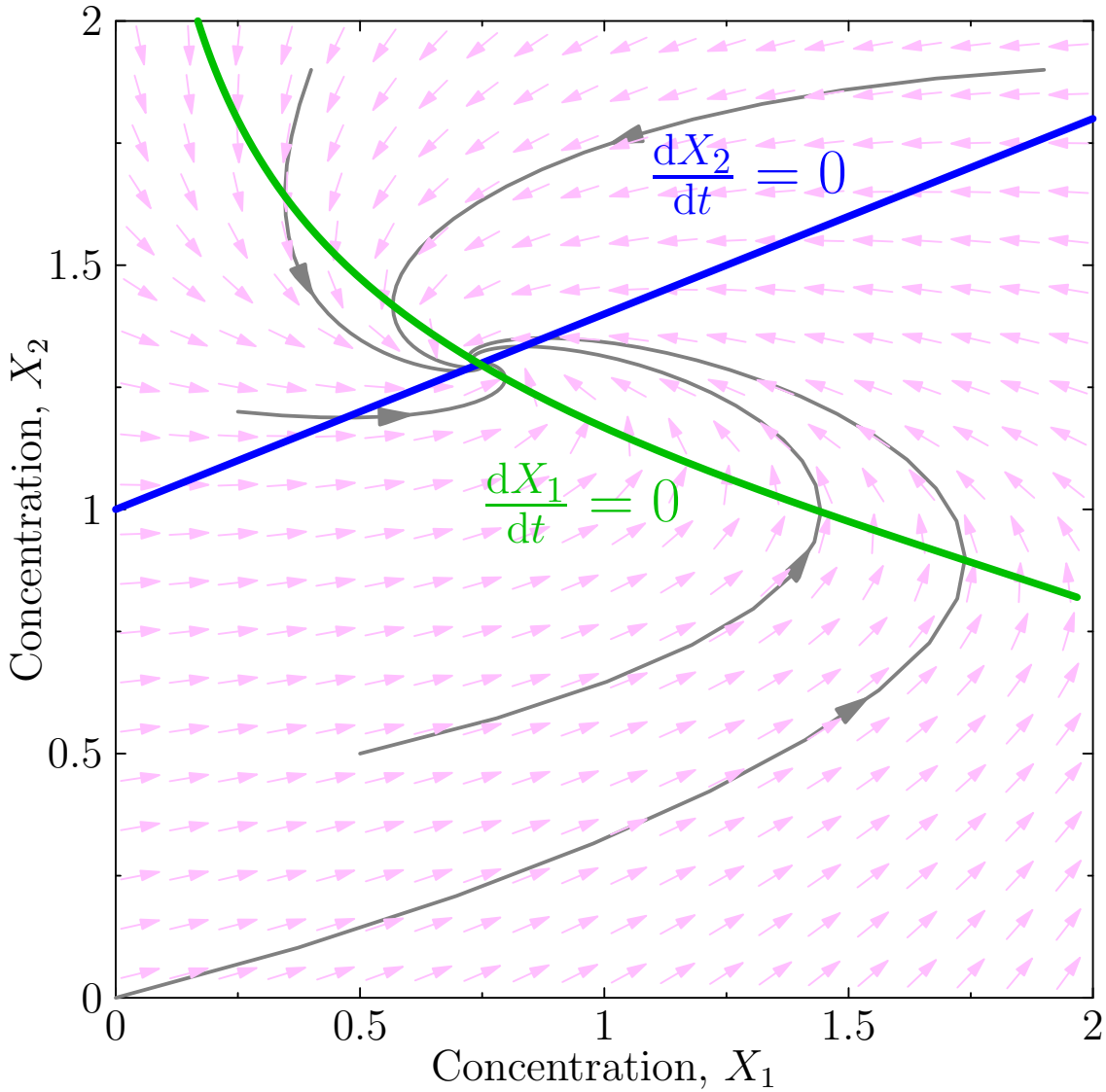
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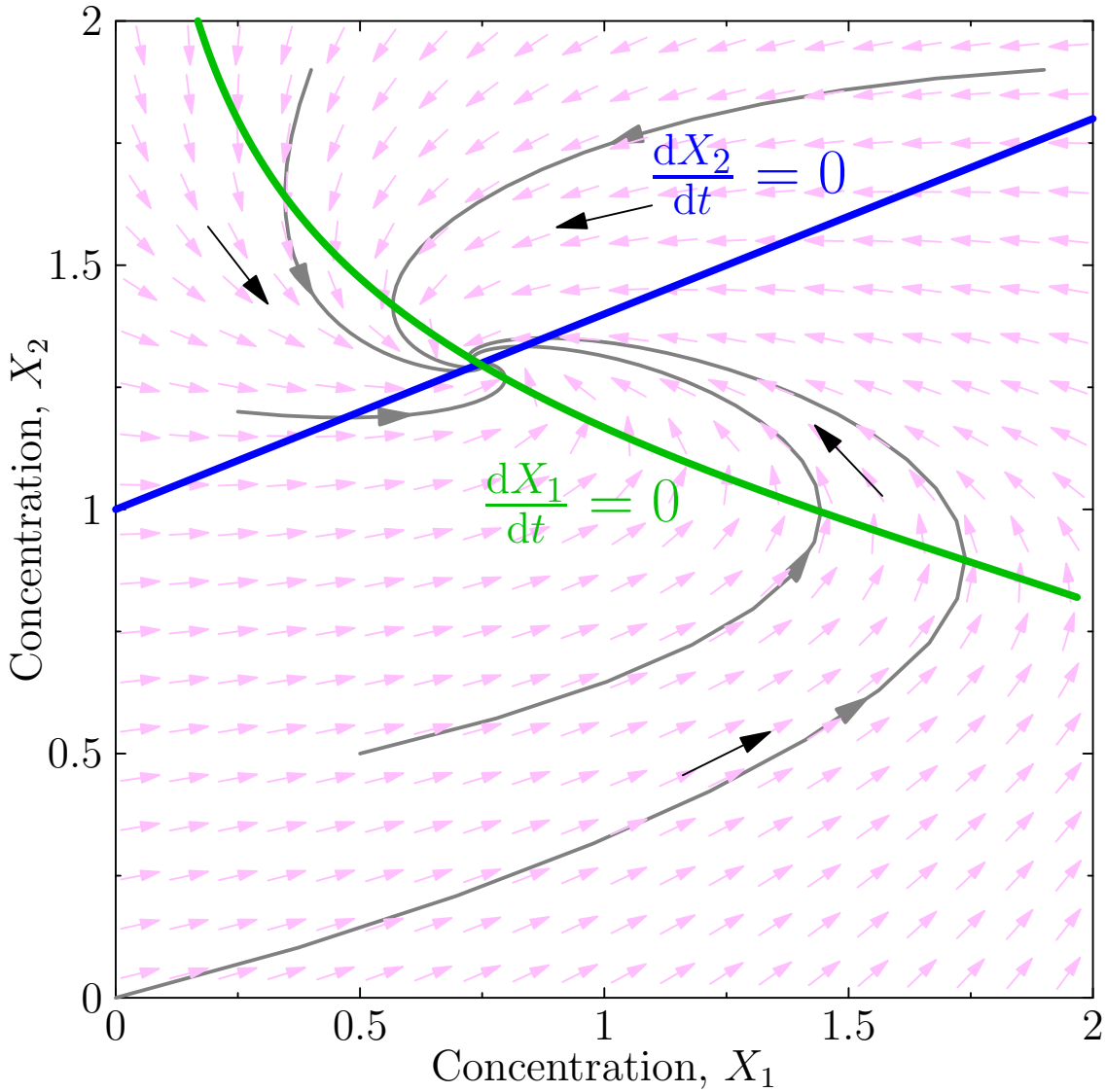
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Rescaling

- When we are working with ODEs we can work in any units of concentration and time

- Note that we can change our units of concentrations by a rescaling $[X_i] \to r\,[X_i]$

- If we rescale $K \to r\,K$, $t \to r\,t$, $k_i \to k_i/r$ for $i \in \{3, 4, 5\}$ this leaves the ODEs invariant

$$\frac{\mathrm{d}[X_1(t)]}{\mathrm{d}t} = \frac{k_1}{1 + ([X_2(t)]/K)^n} - k_3\,[X_1(t)] - k_5\,[X_1(t)]$$

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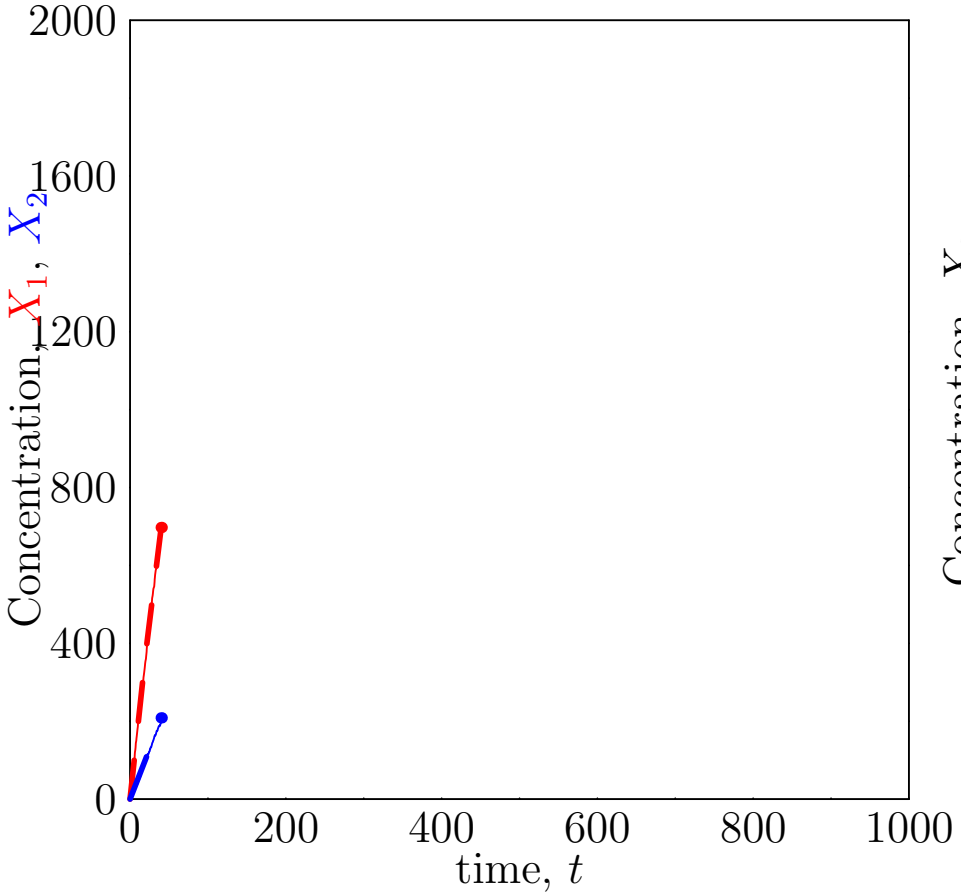
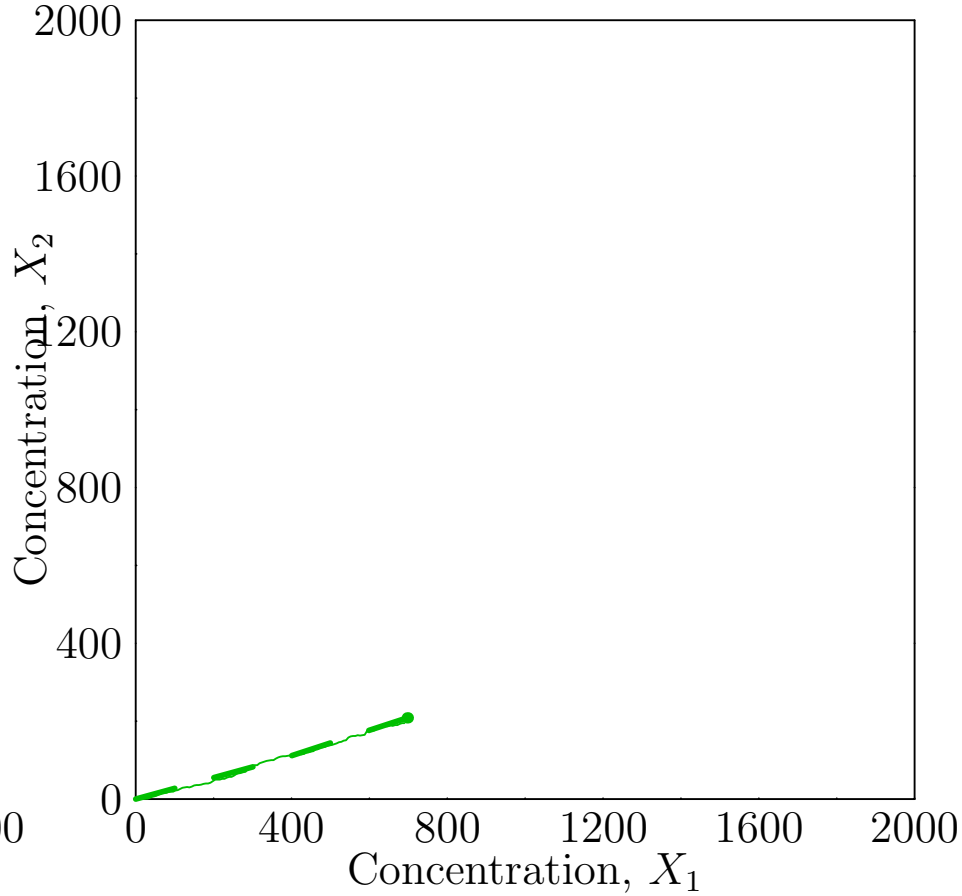
Large Numbers ($r = 1000$)





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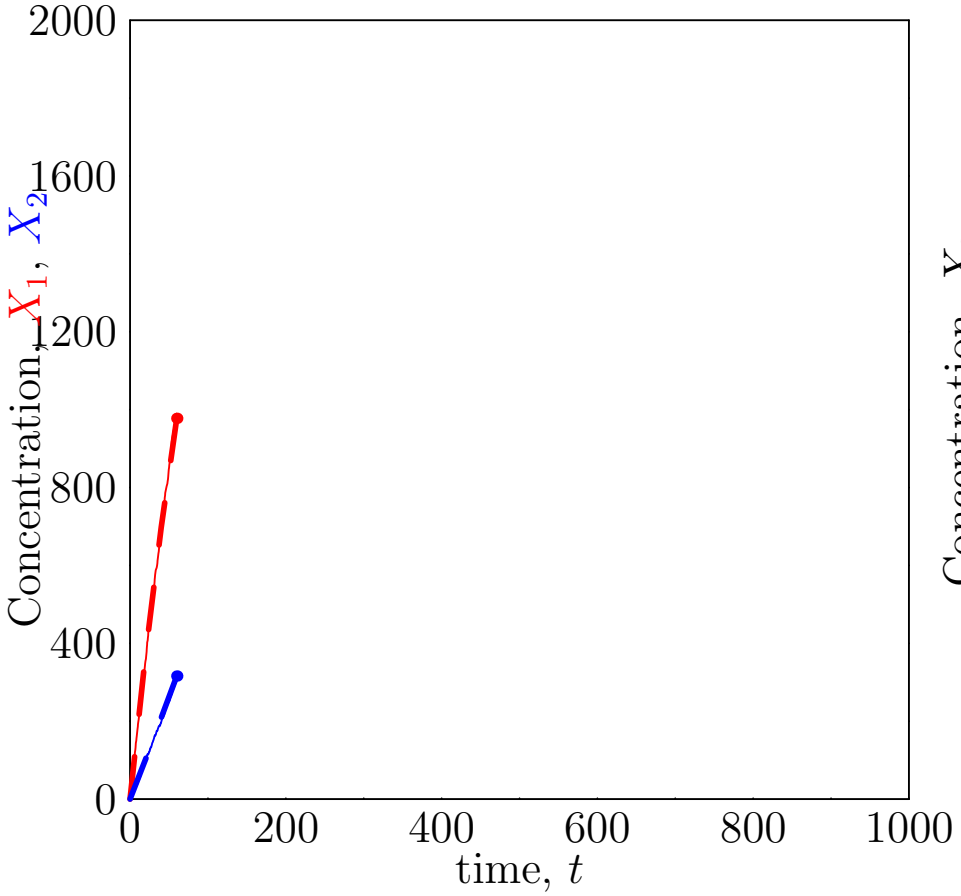

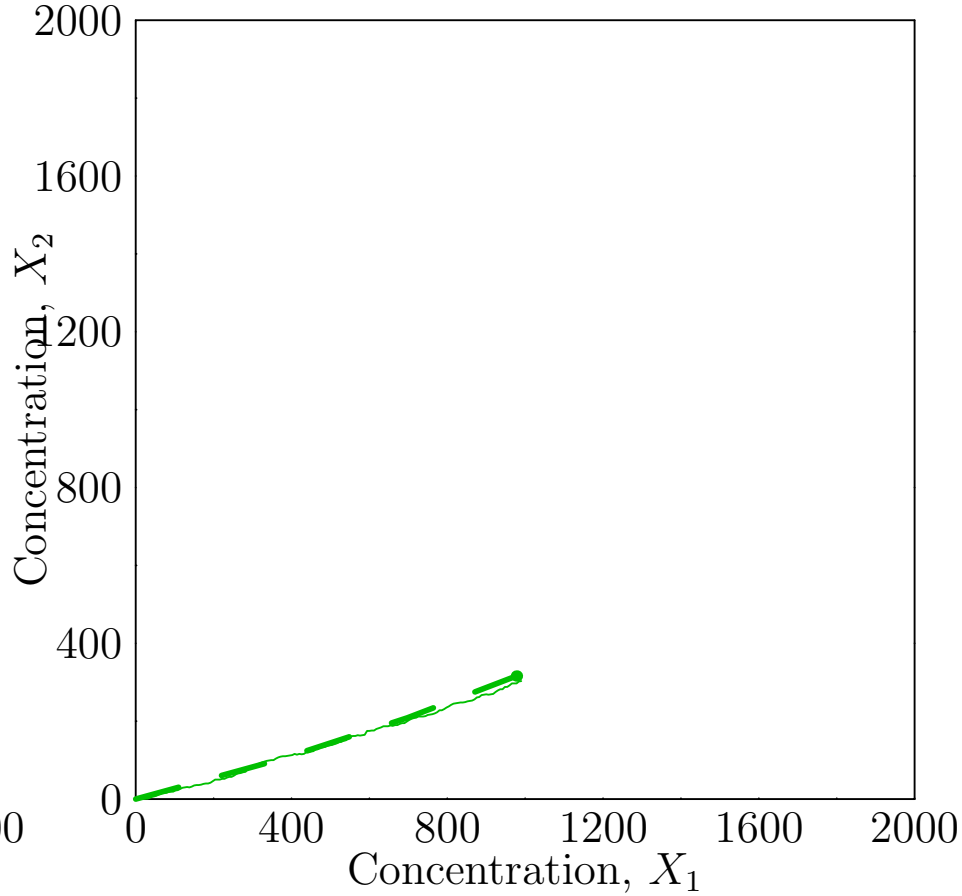

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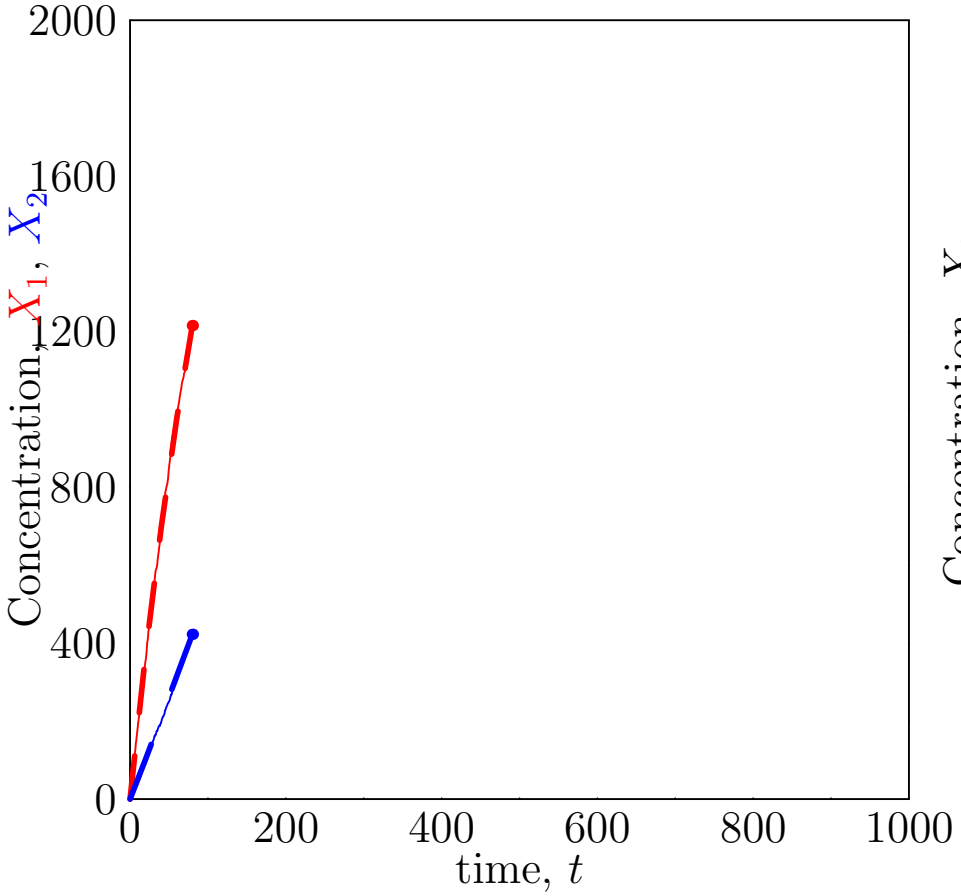

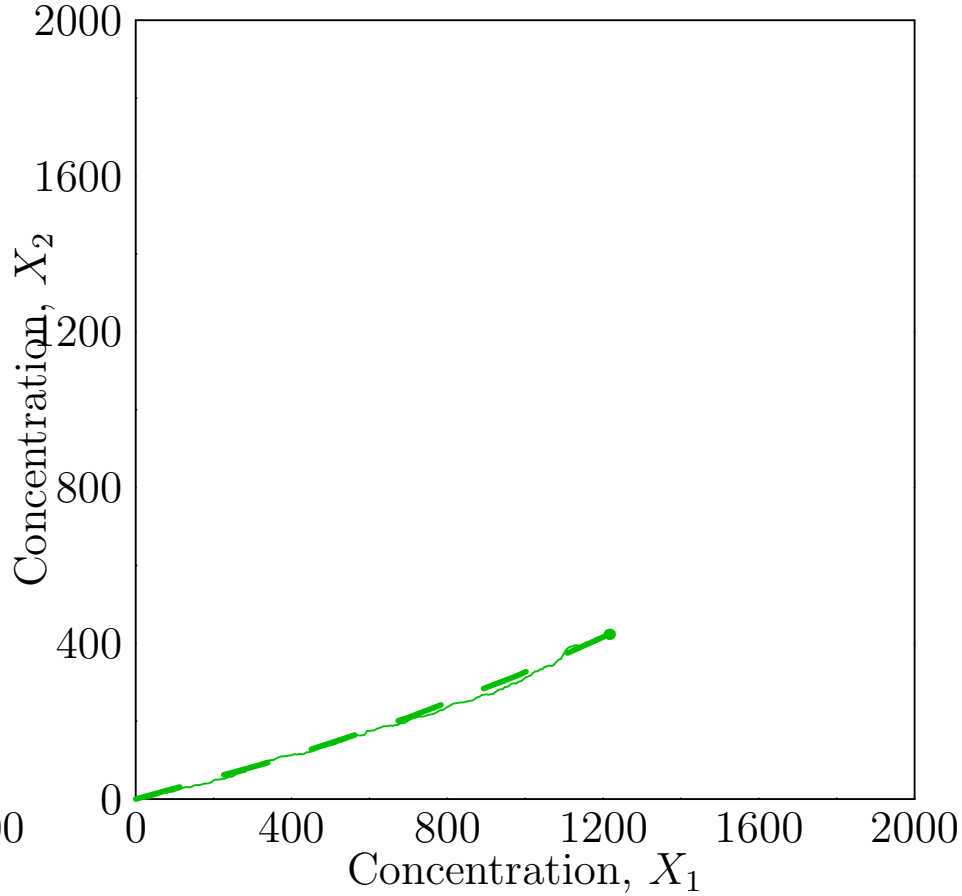

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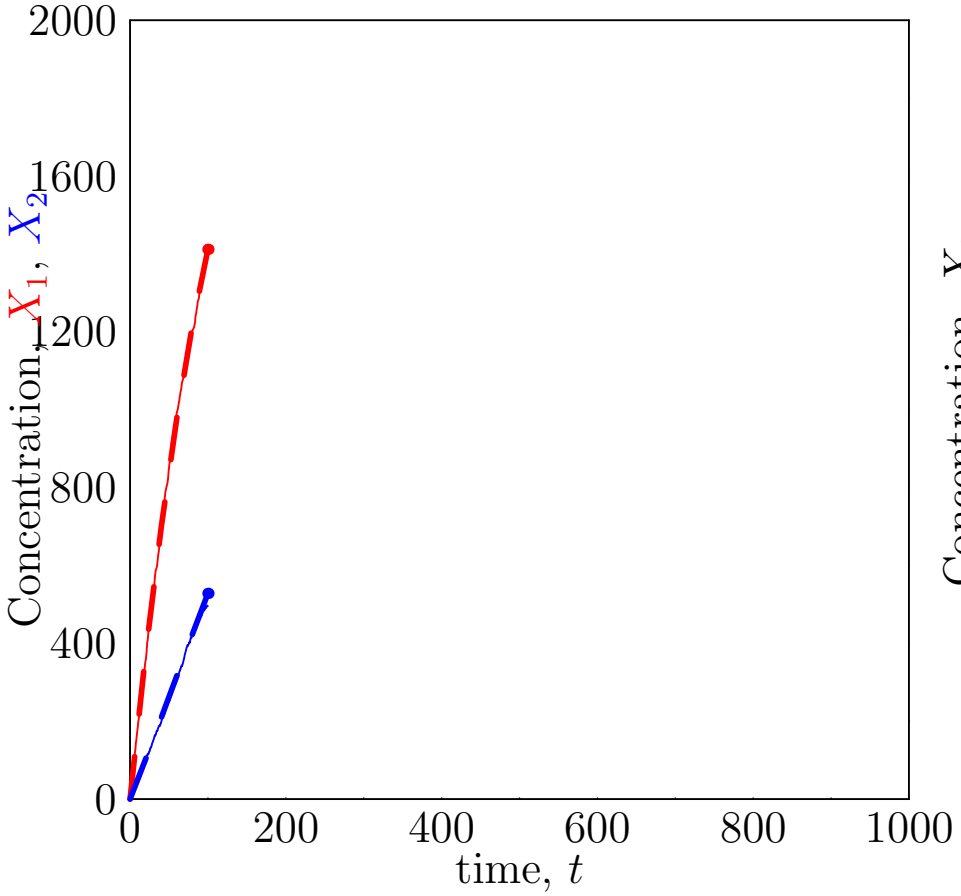

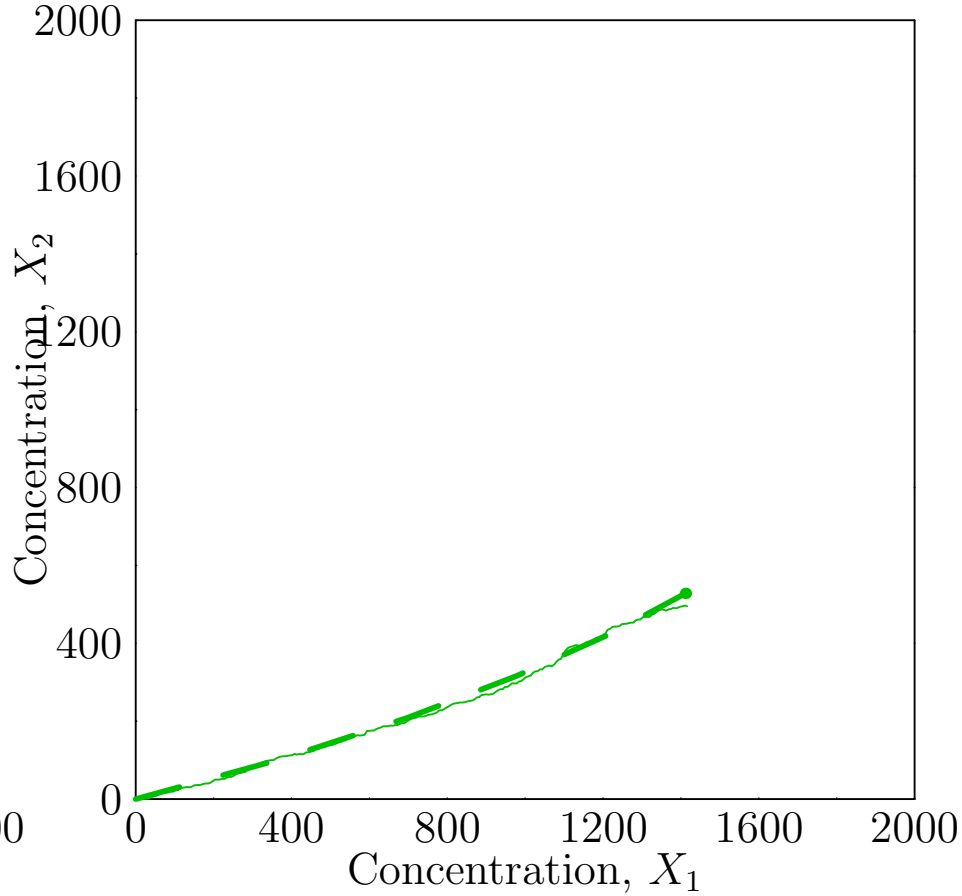

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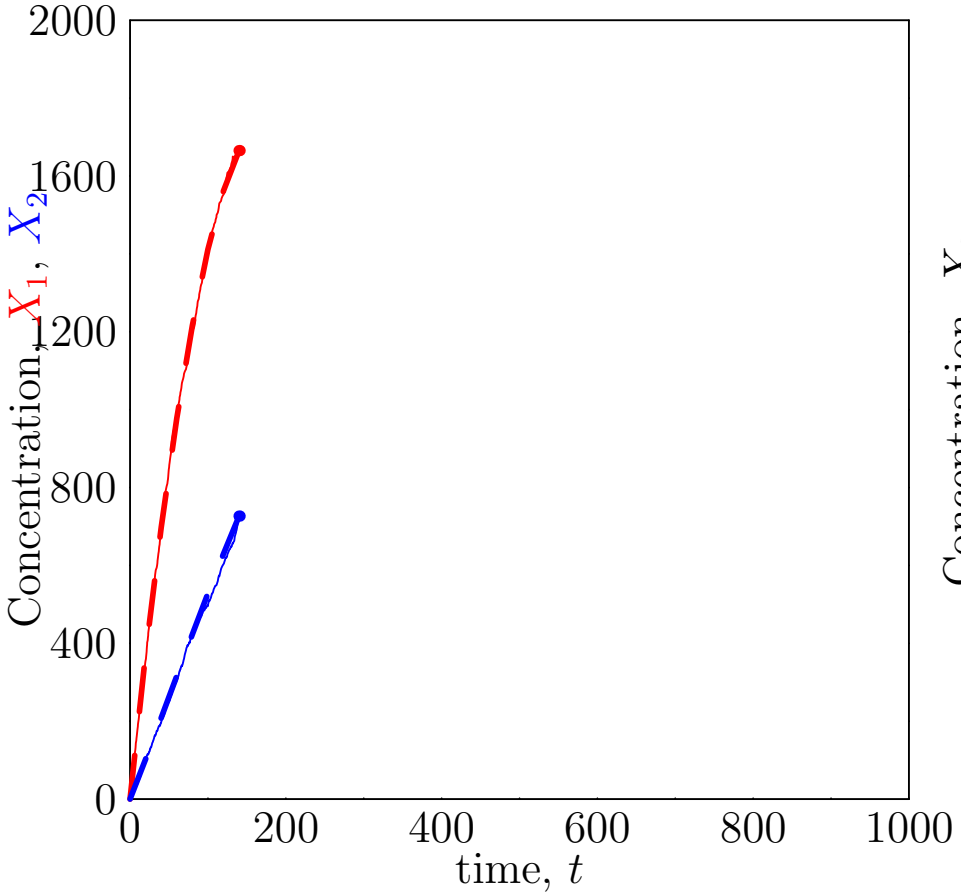

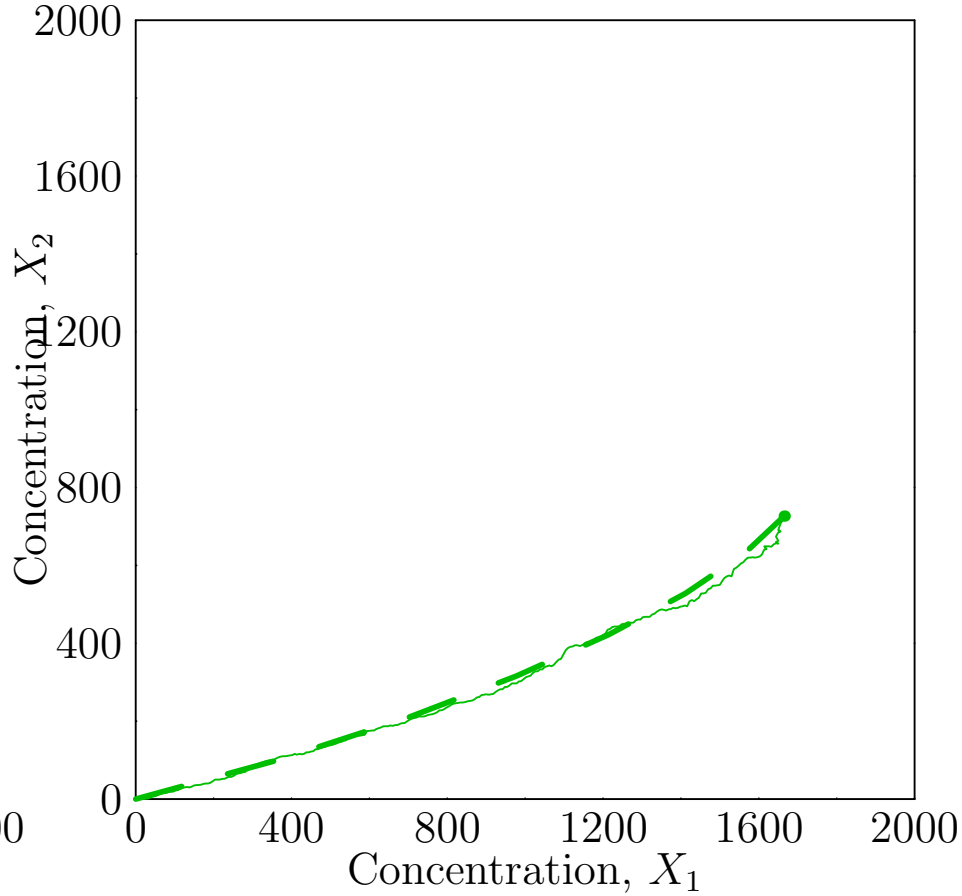

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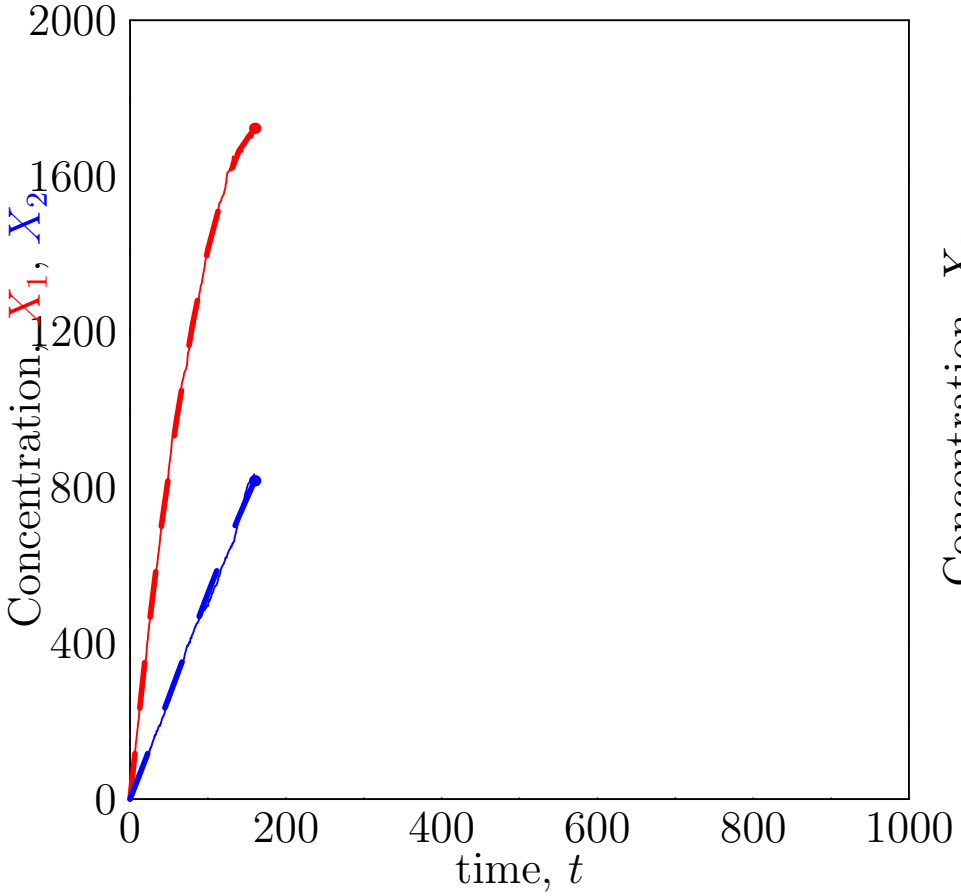

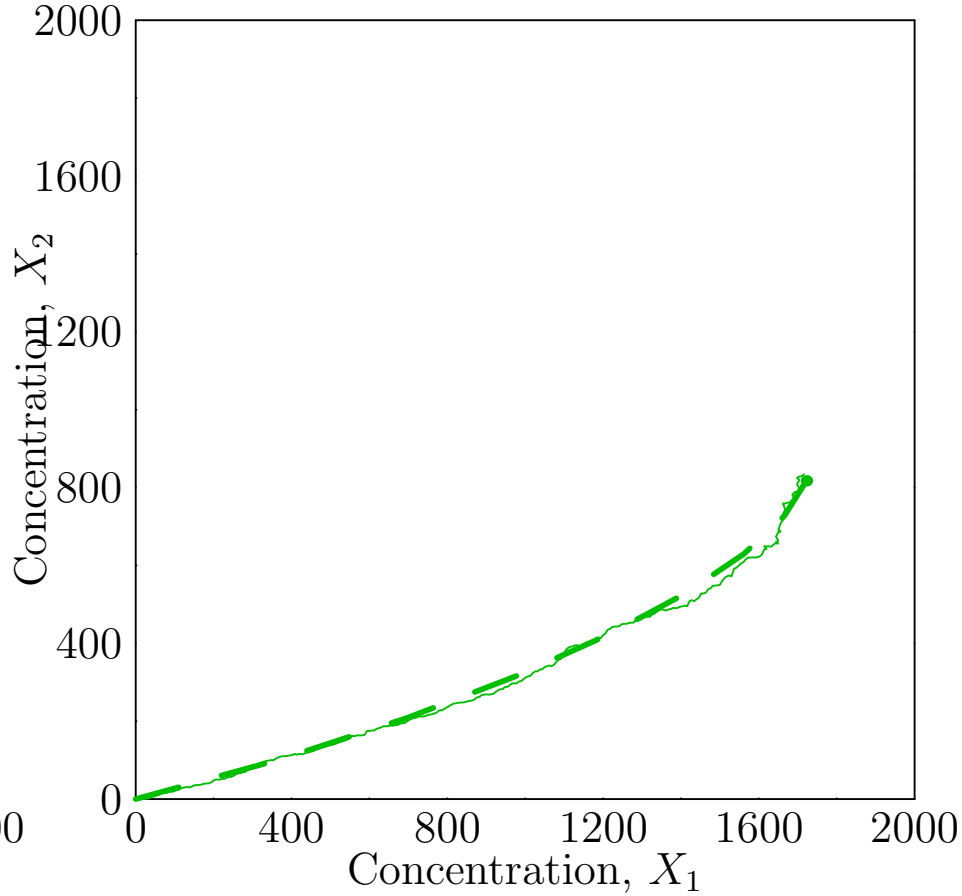

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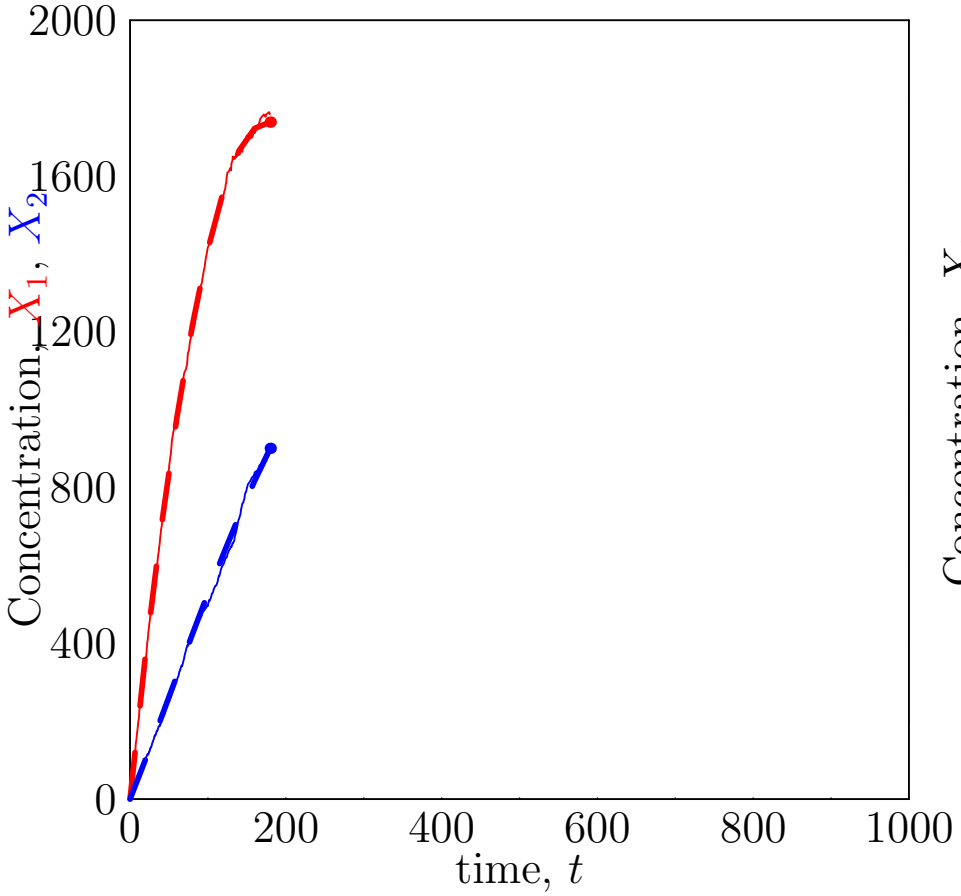

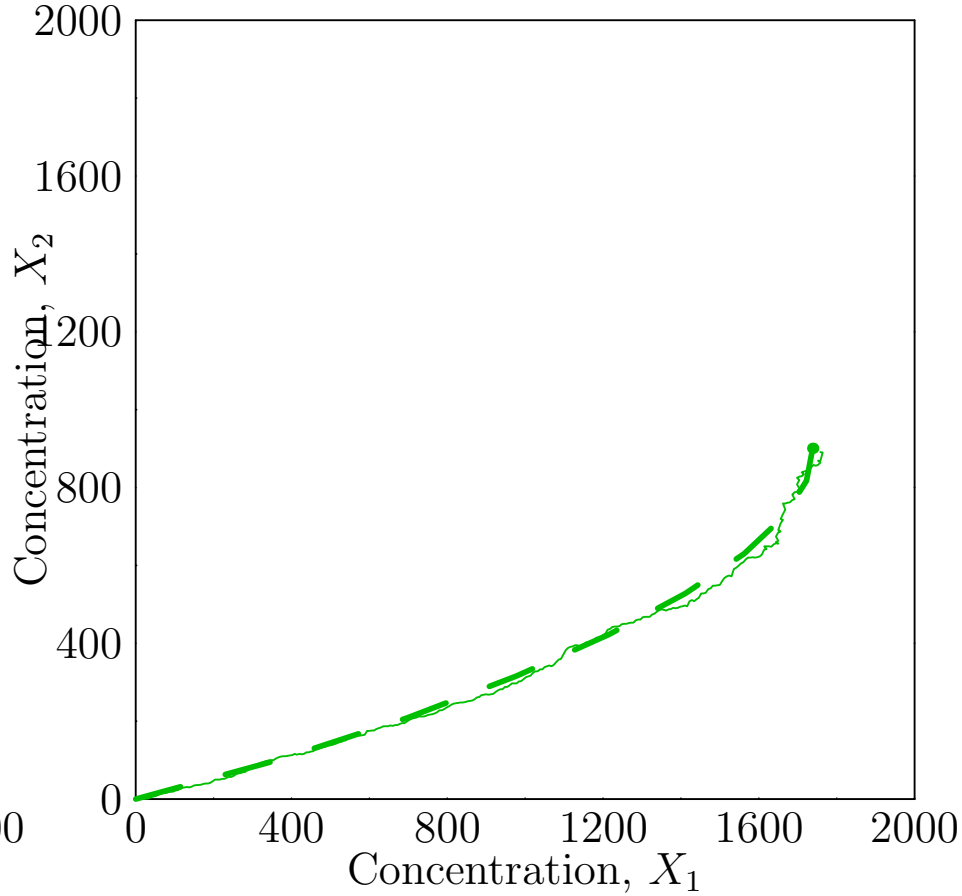

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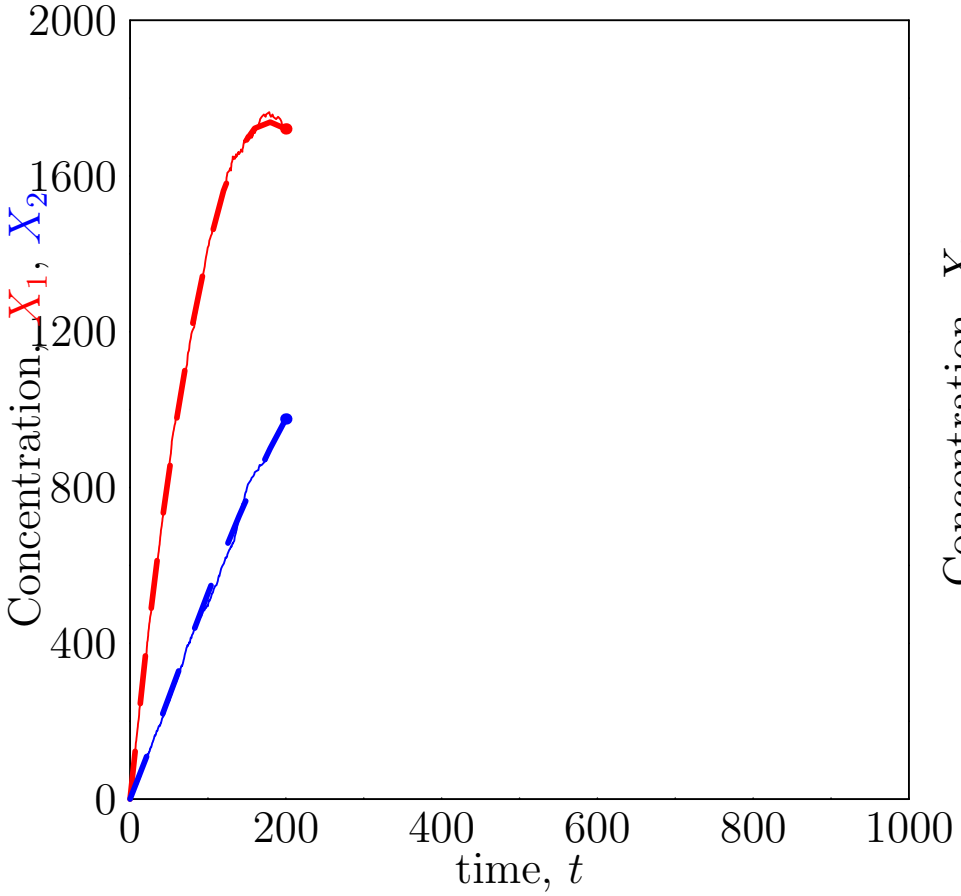

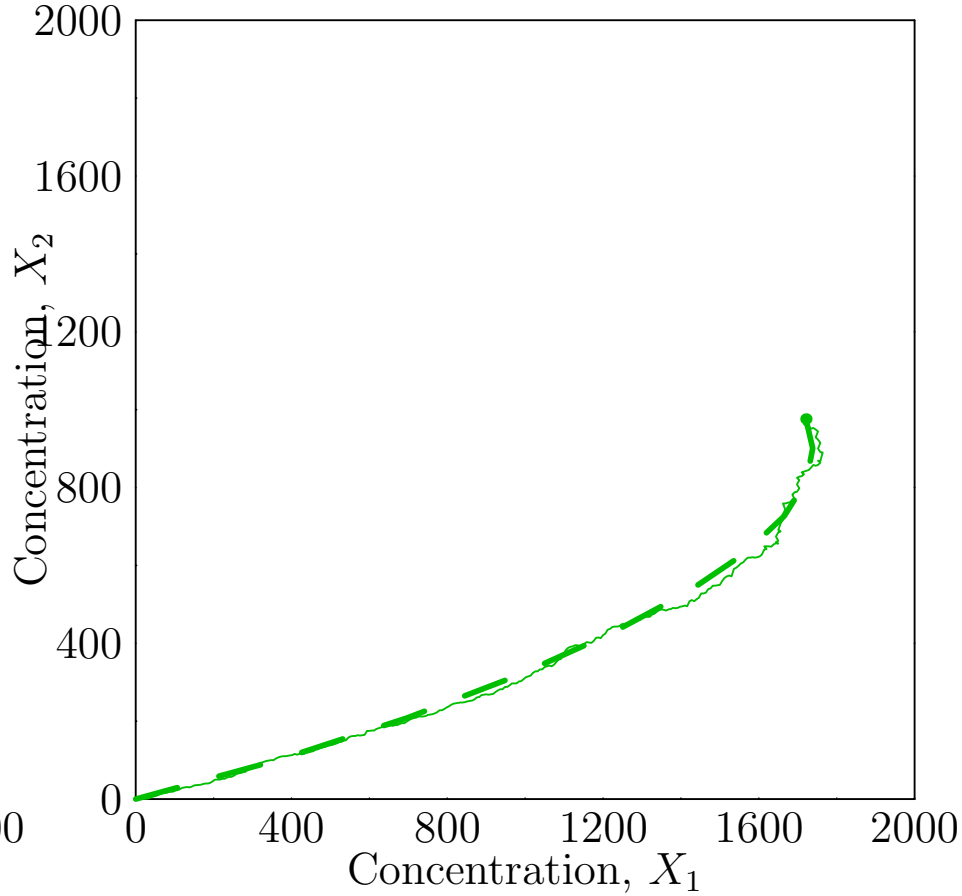

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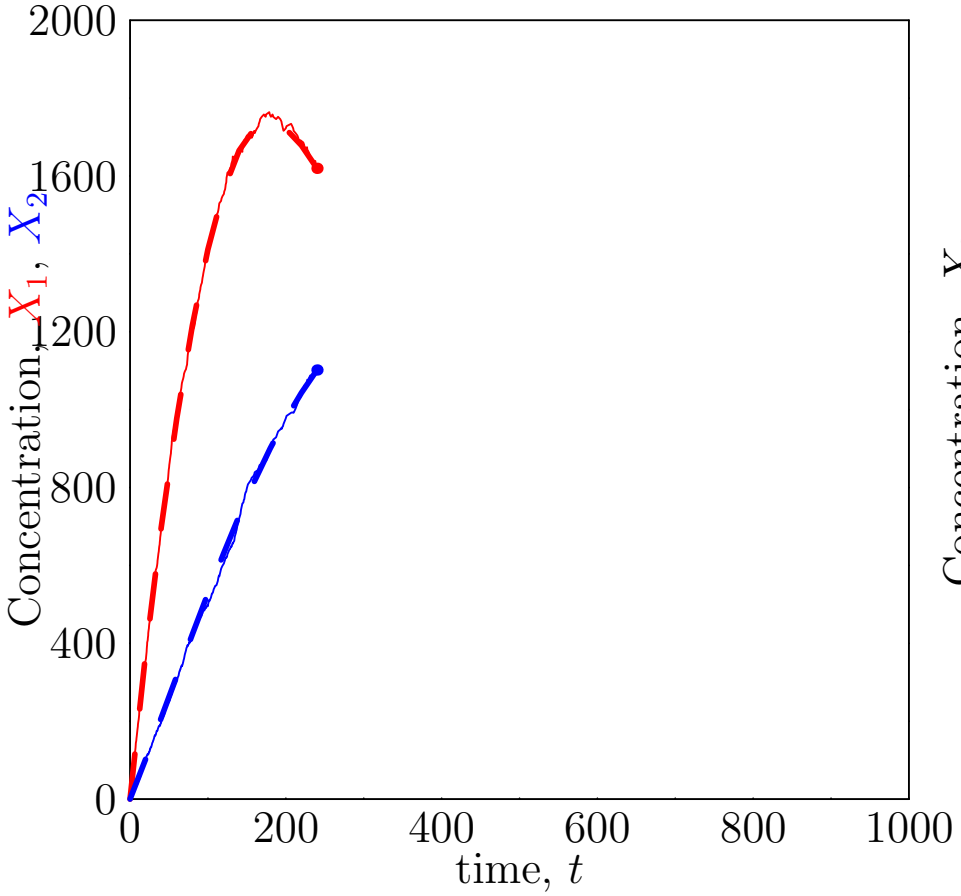

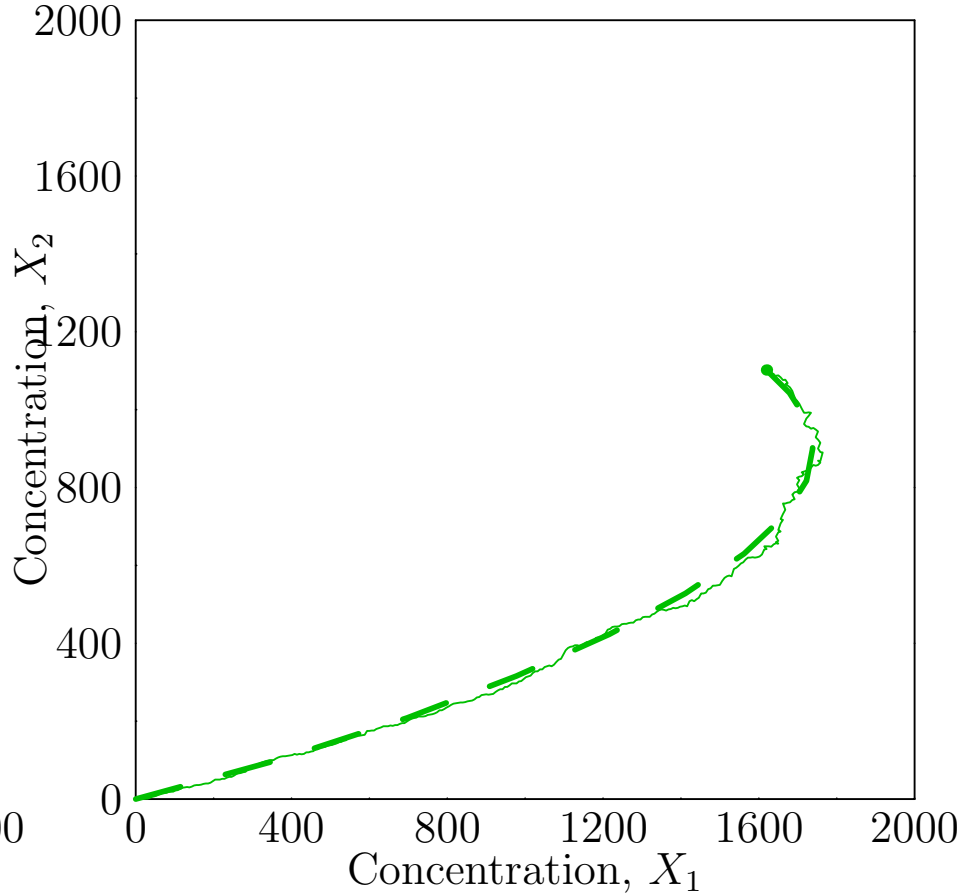

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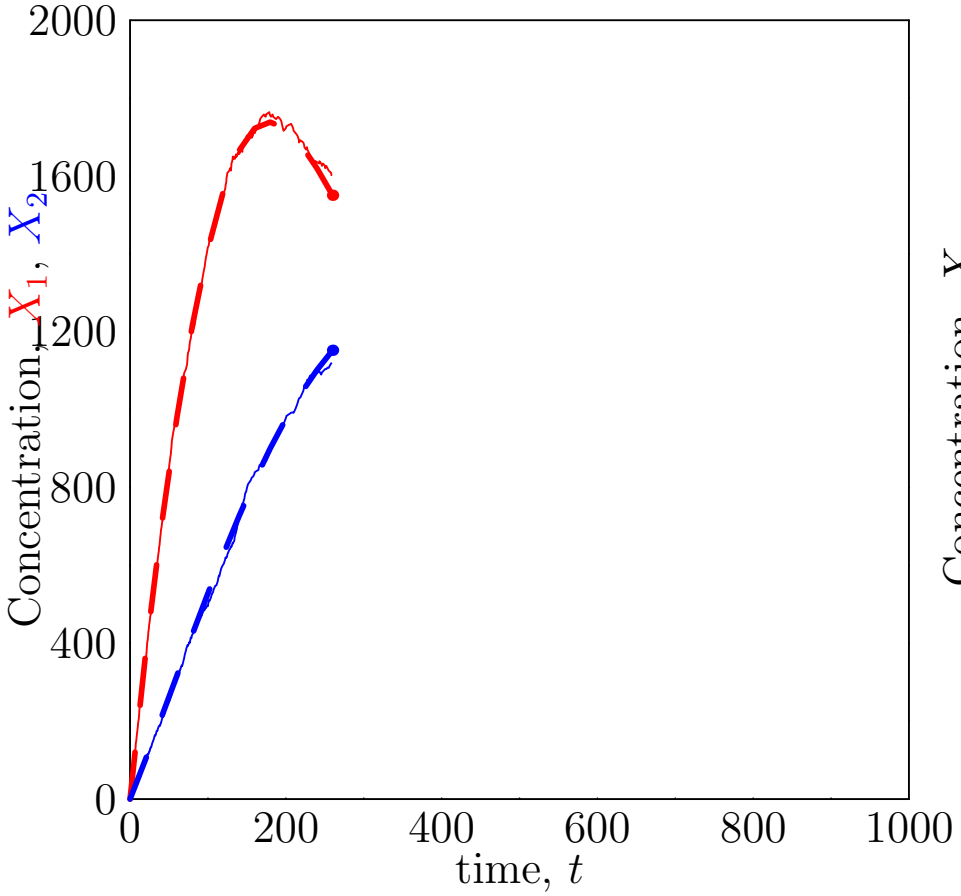

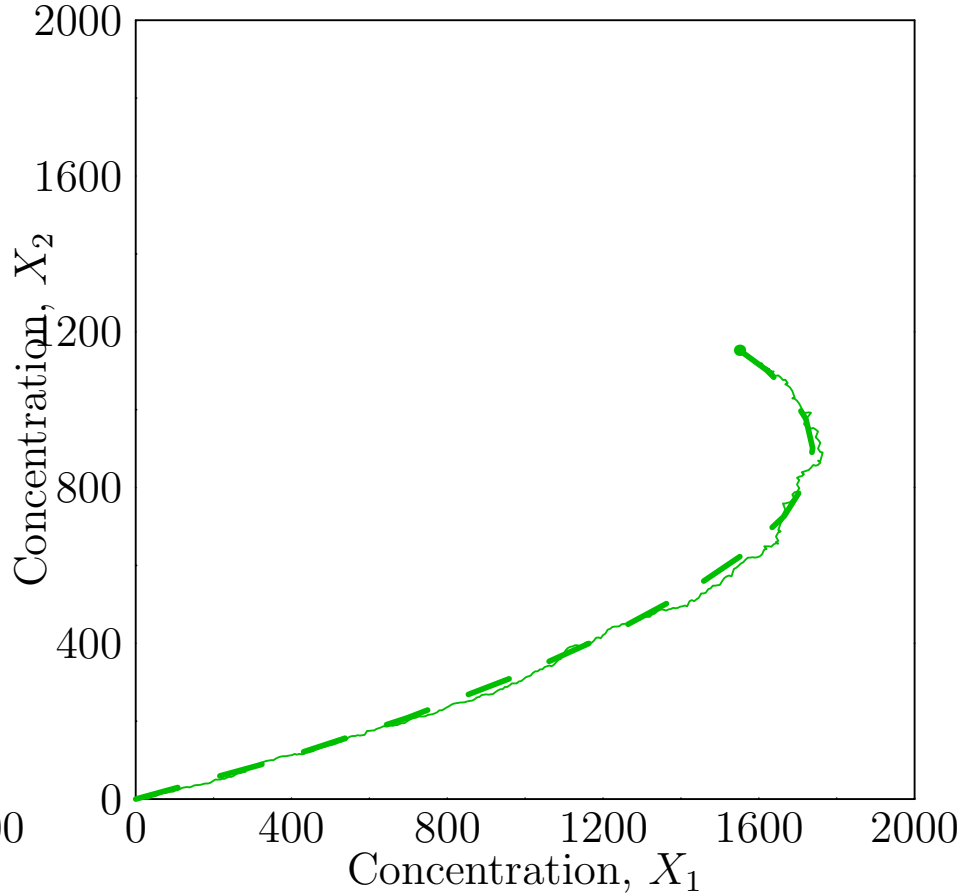

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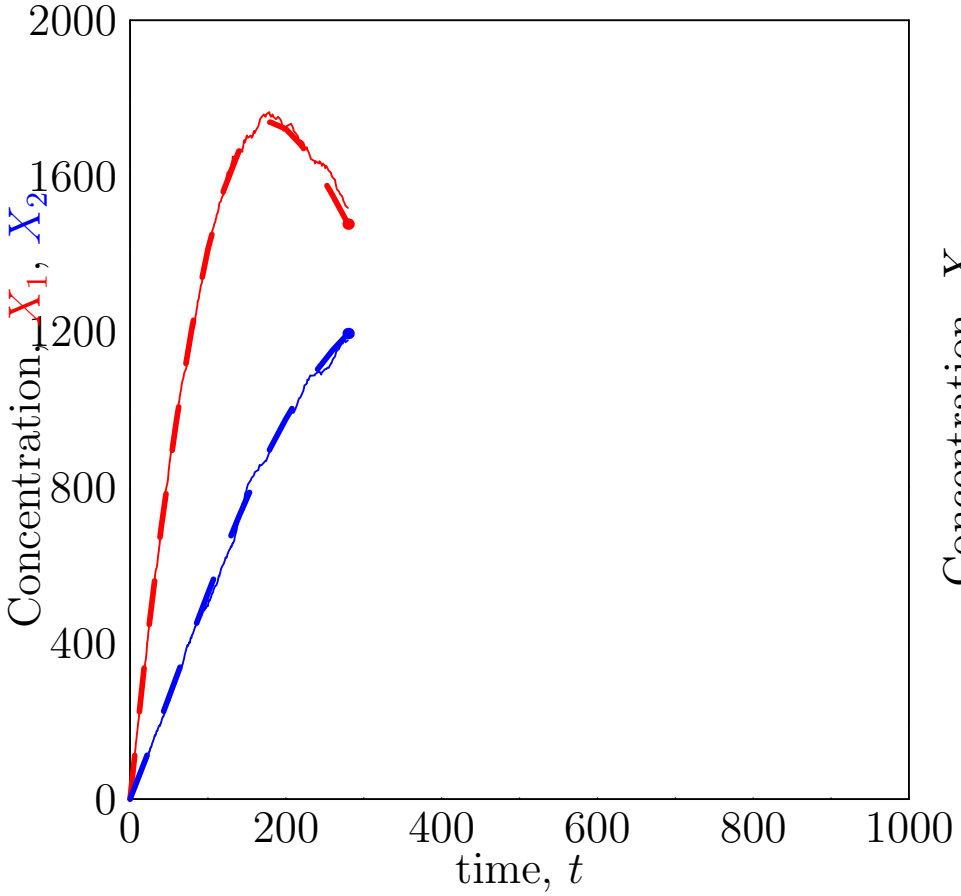

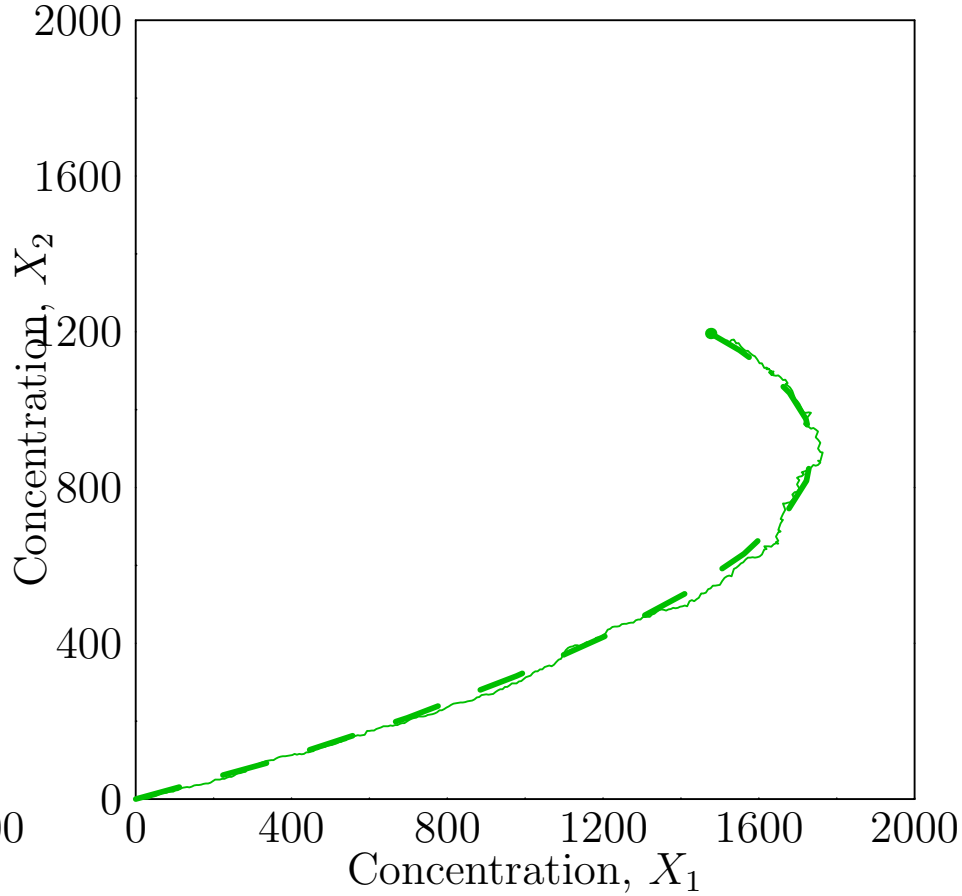

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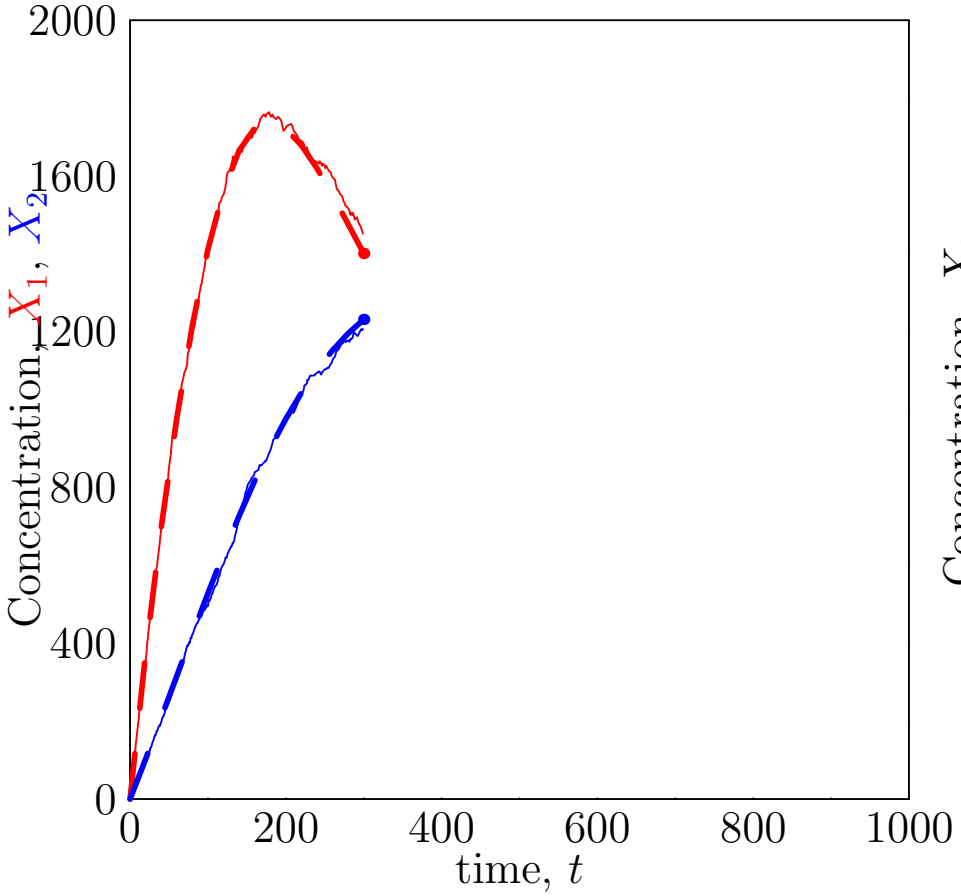

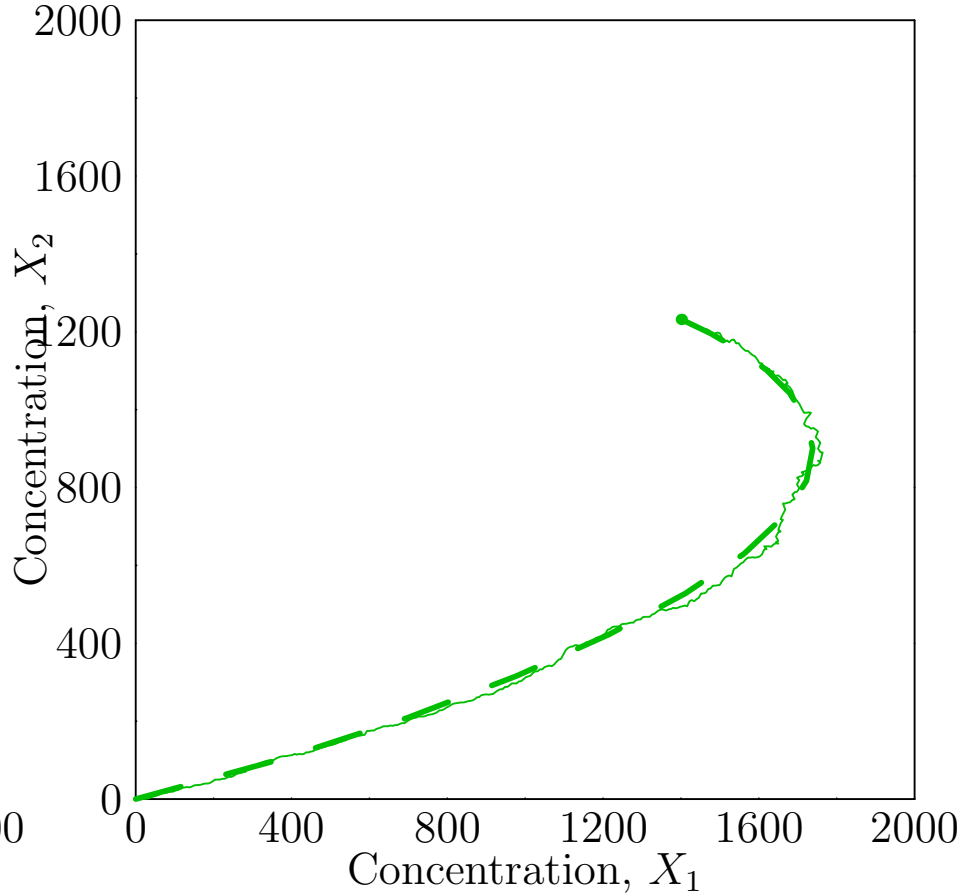

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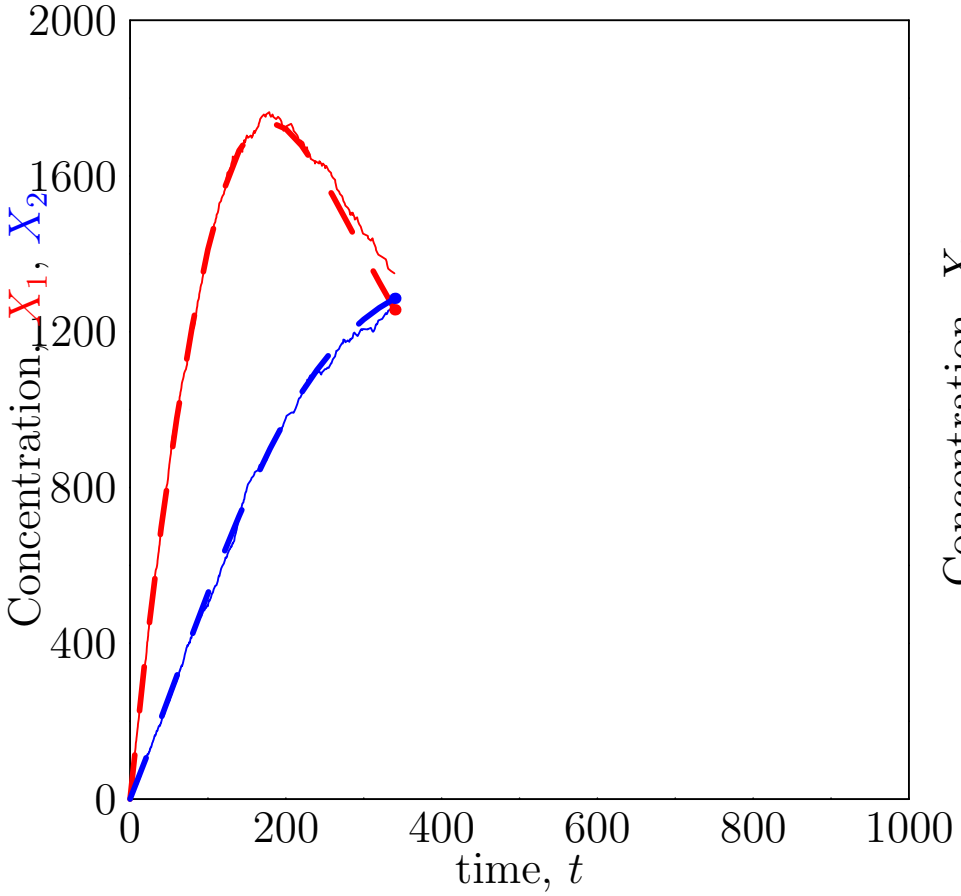

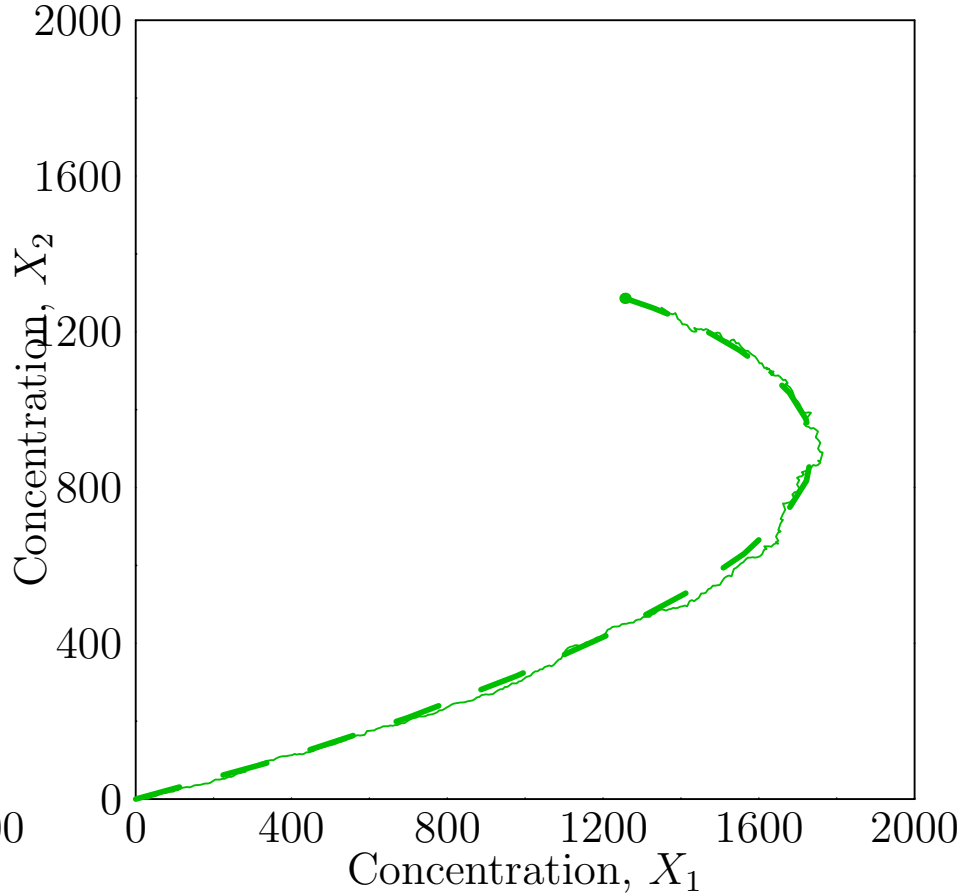

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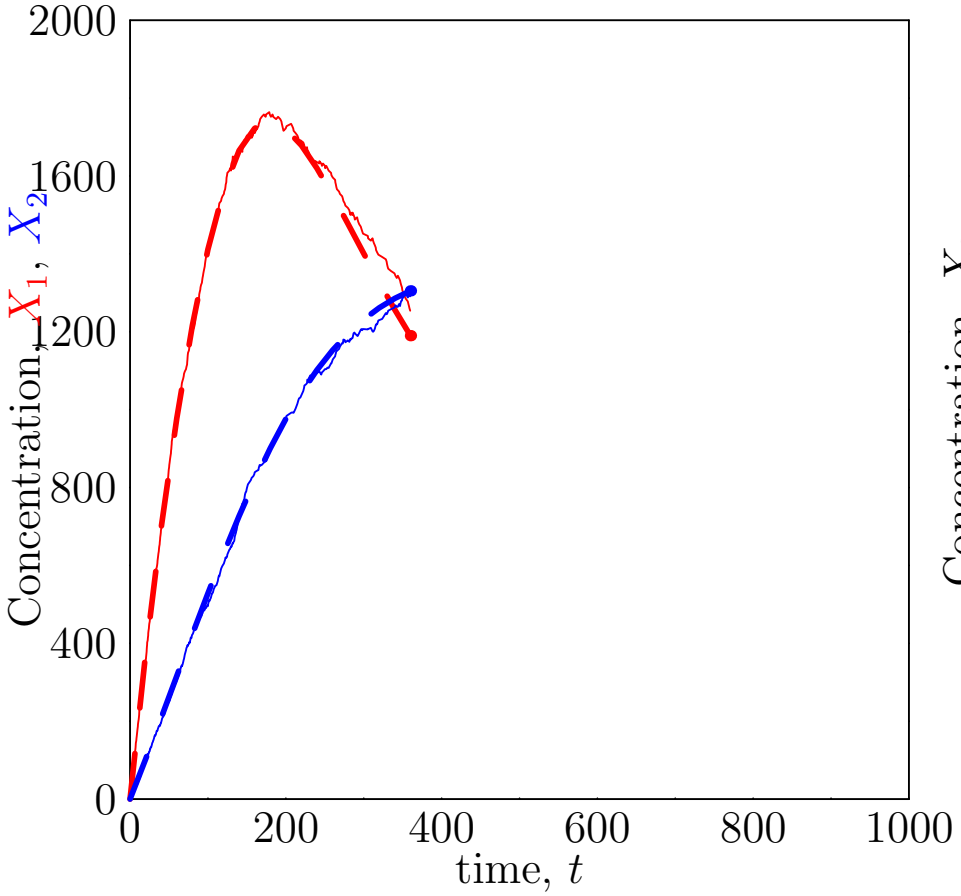

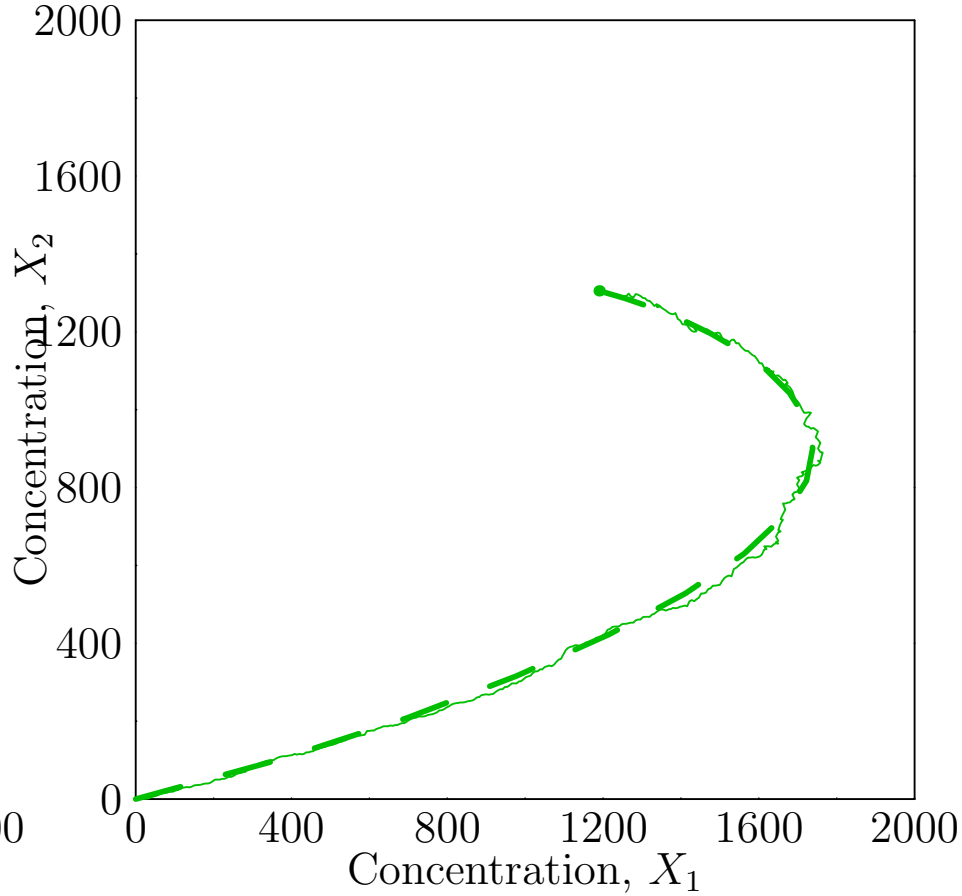

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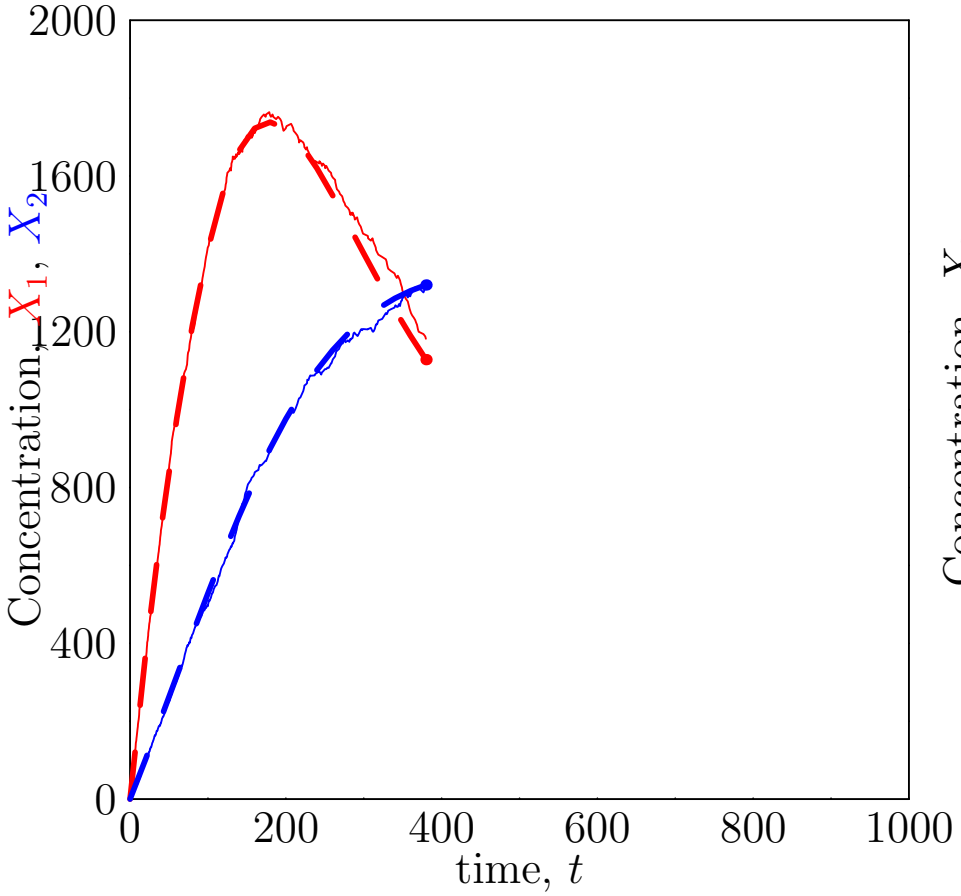

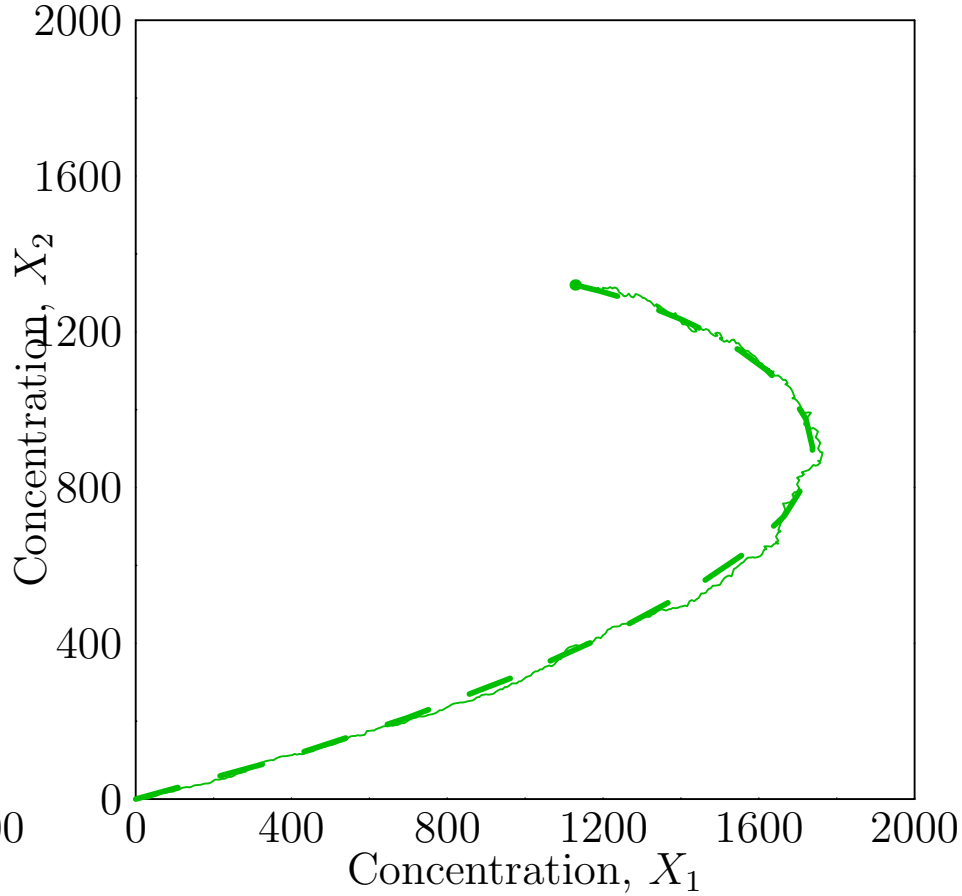

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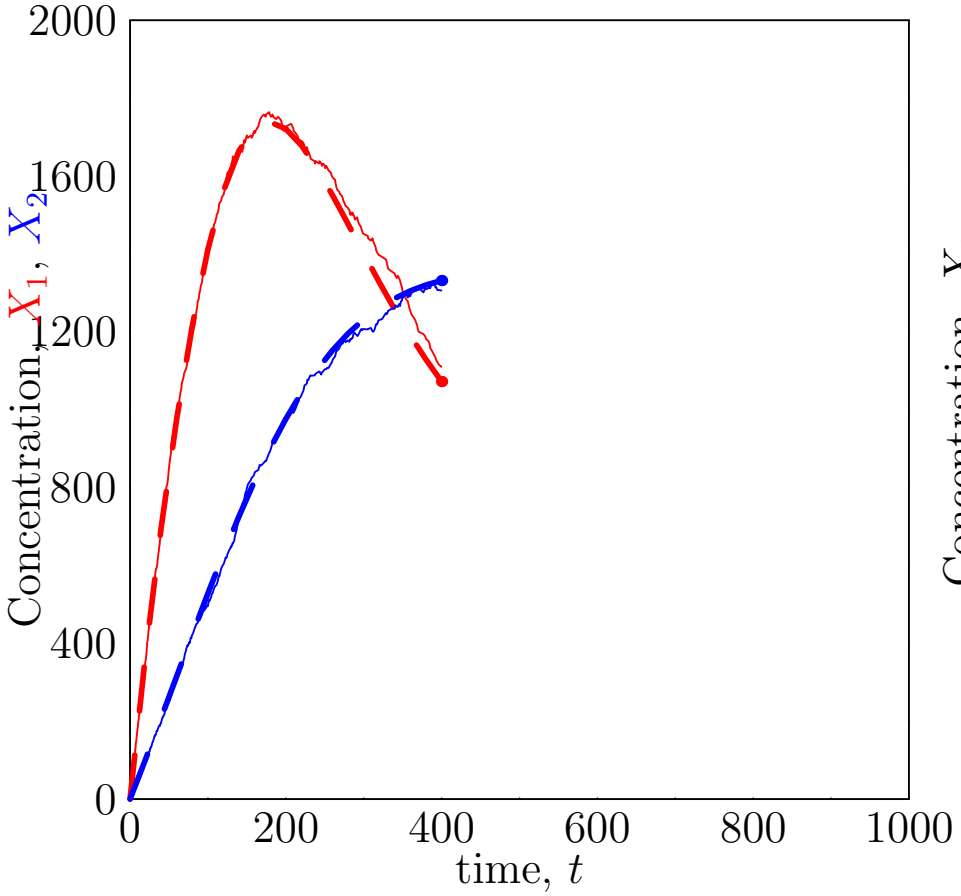

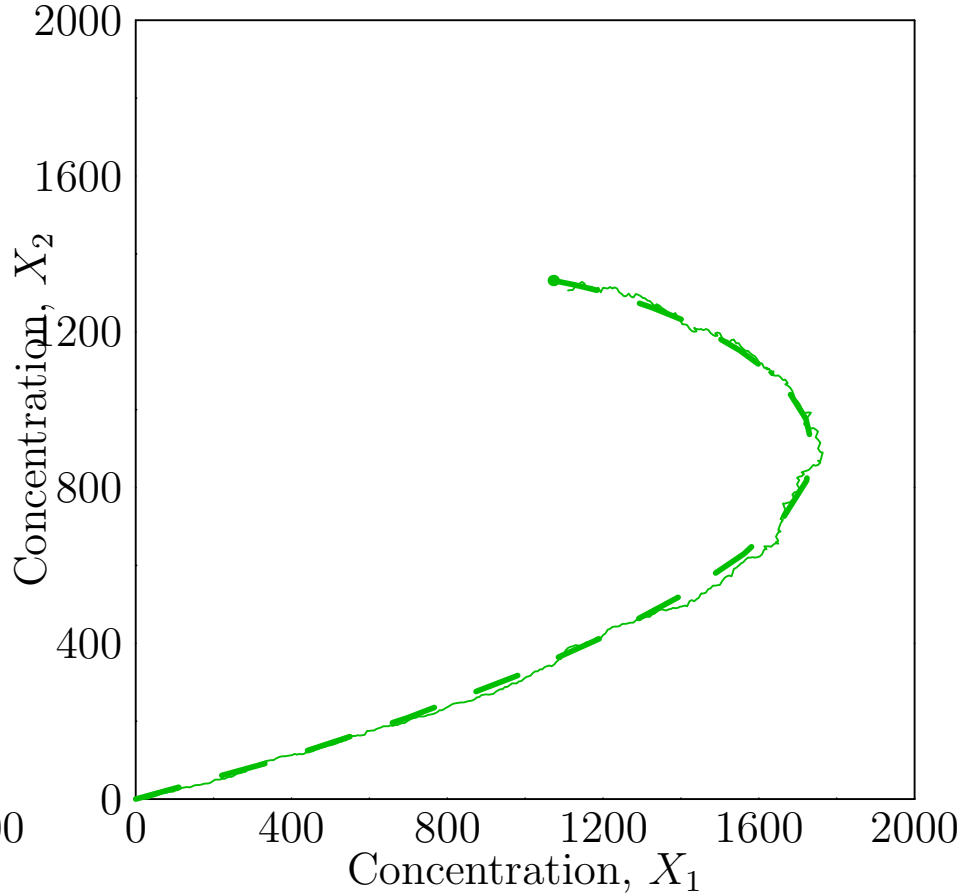

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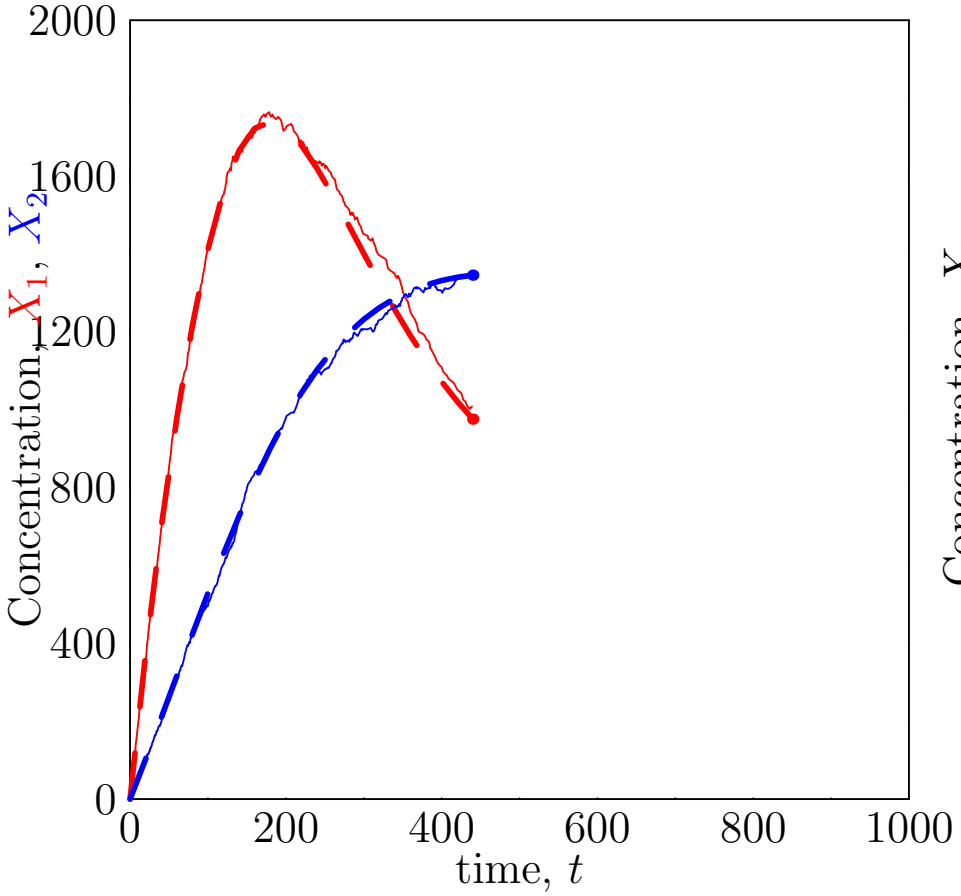

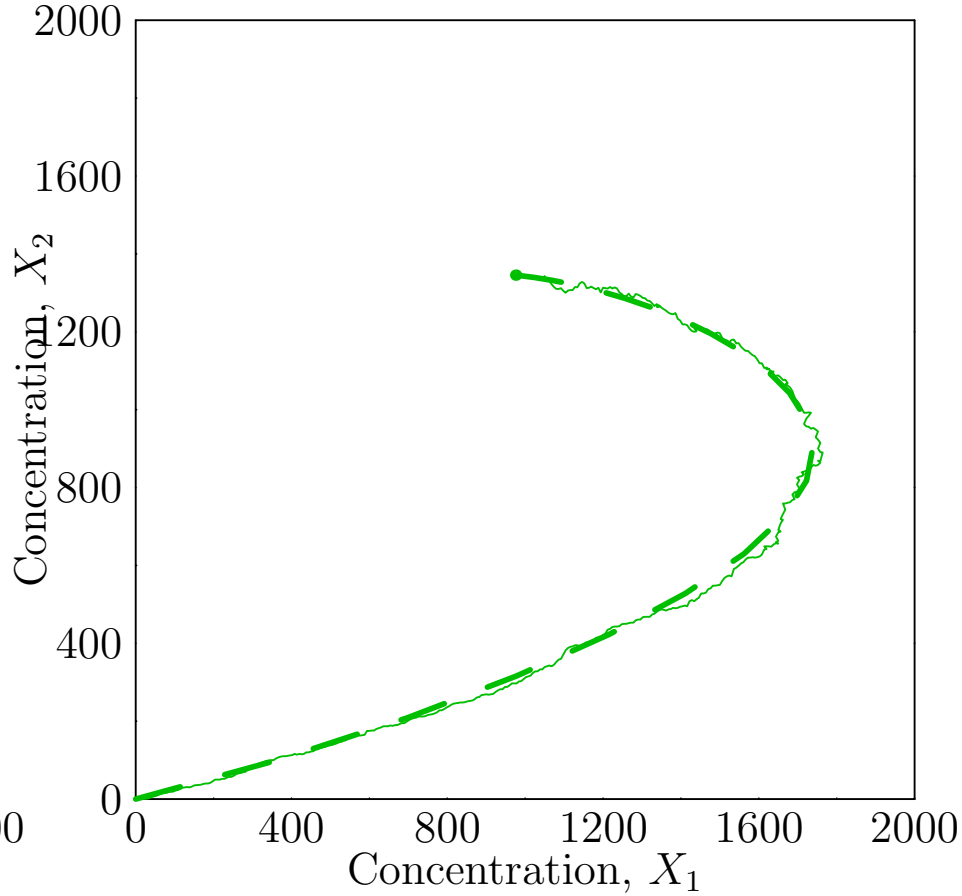

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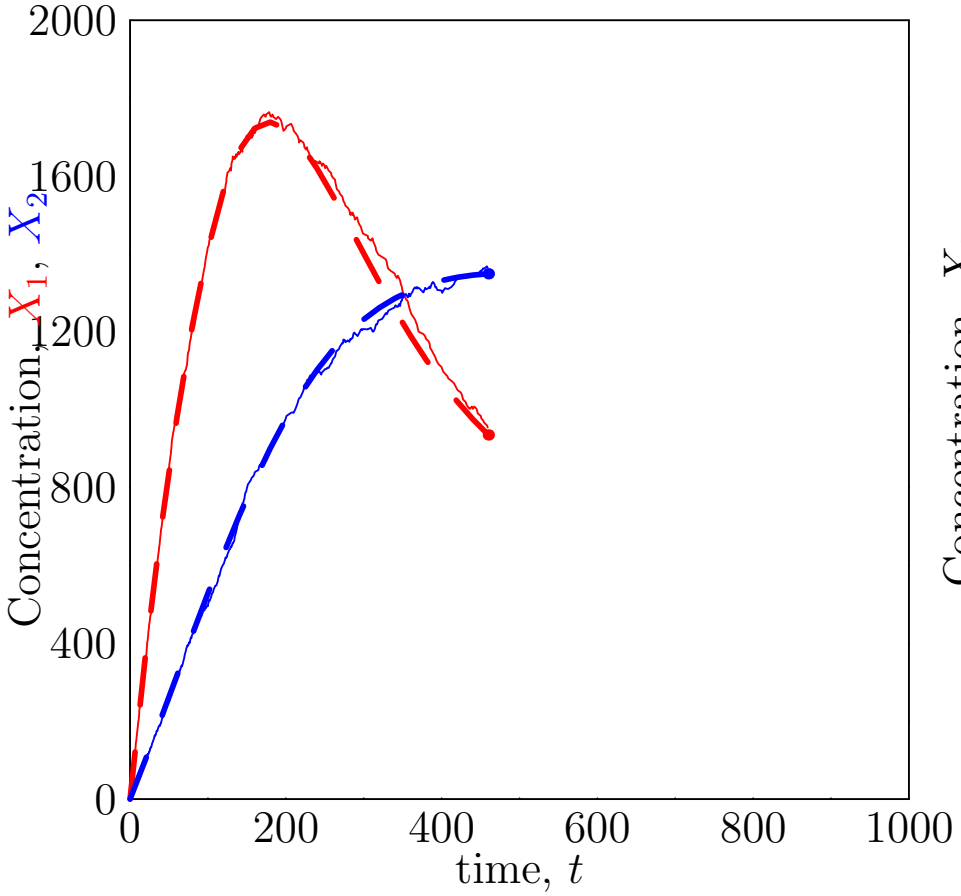

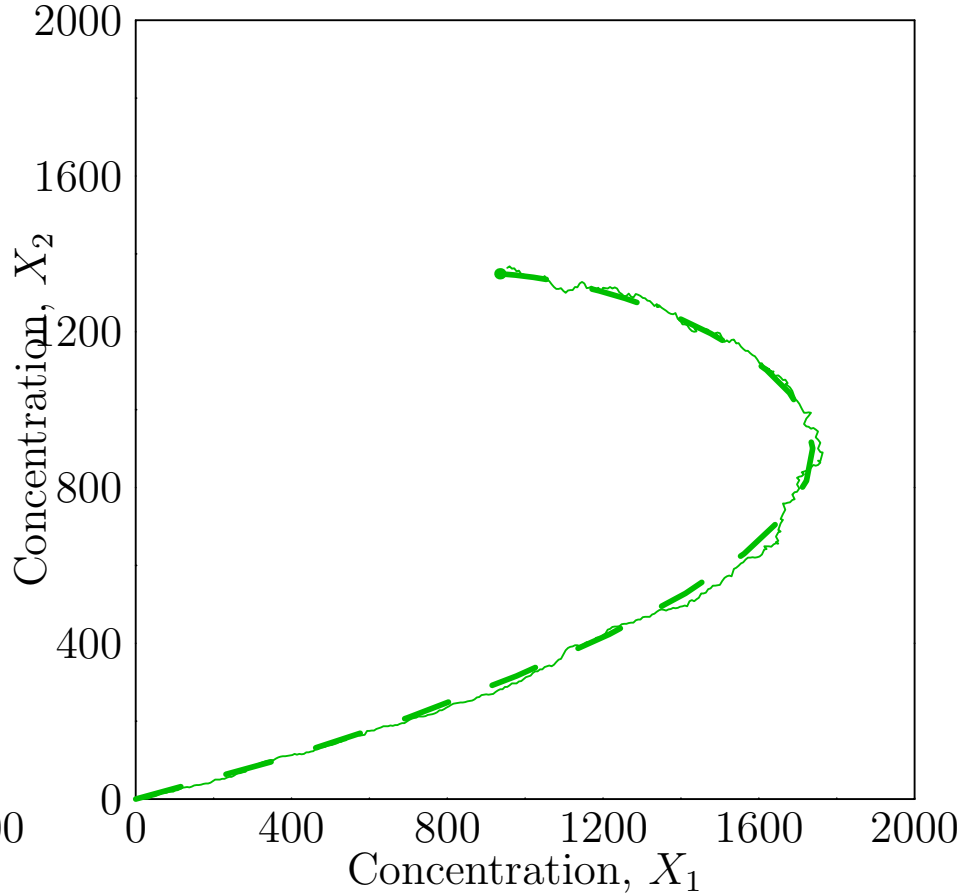

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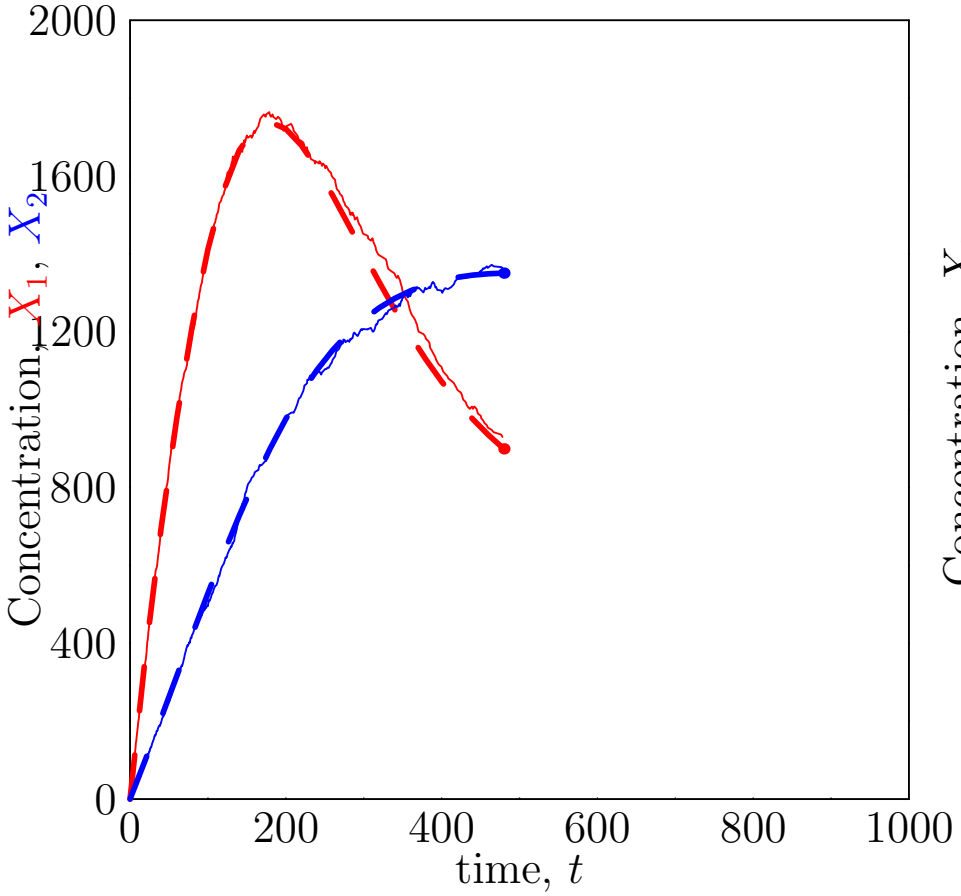

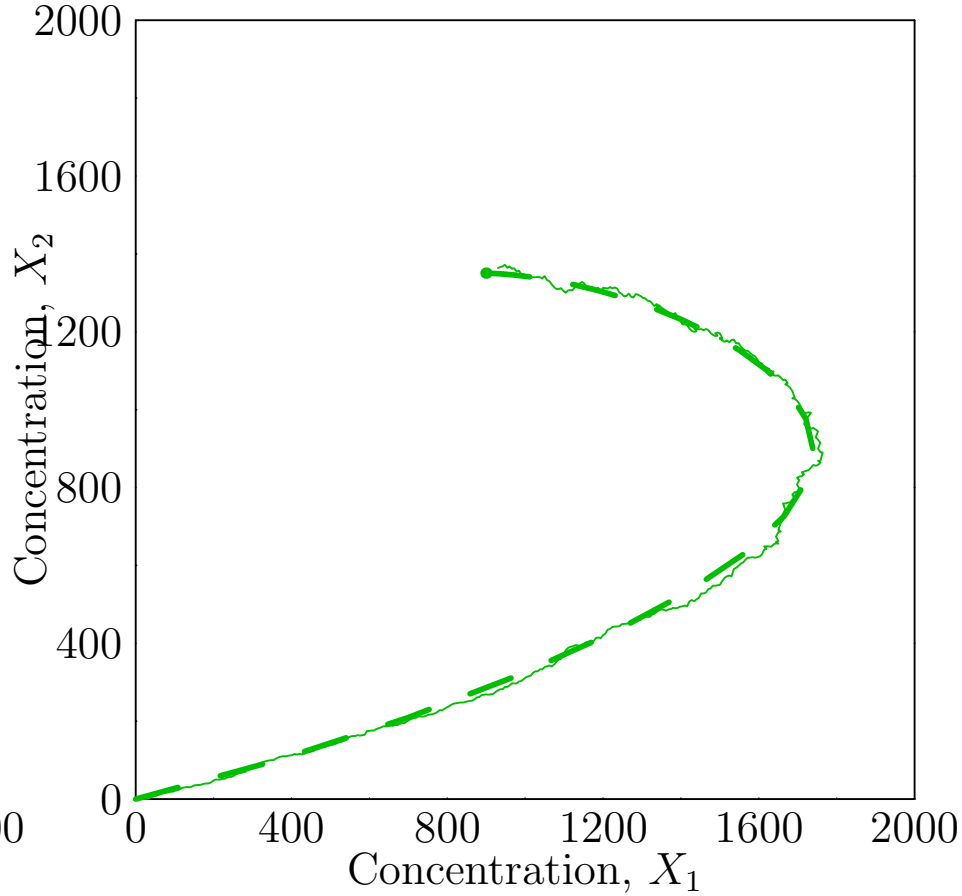

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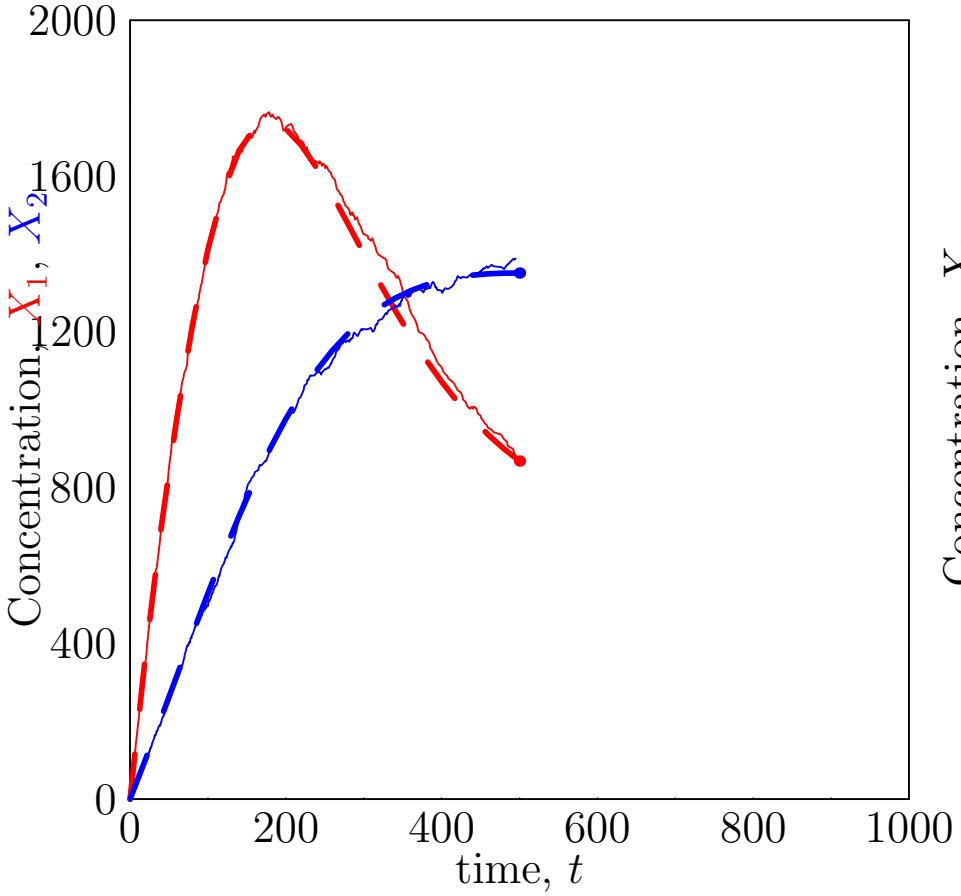

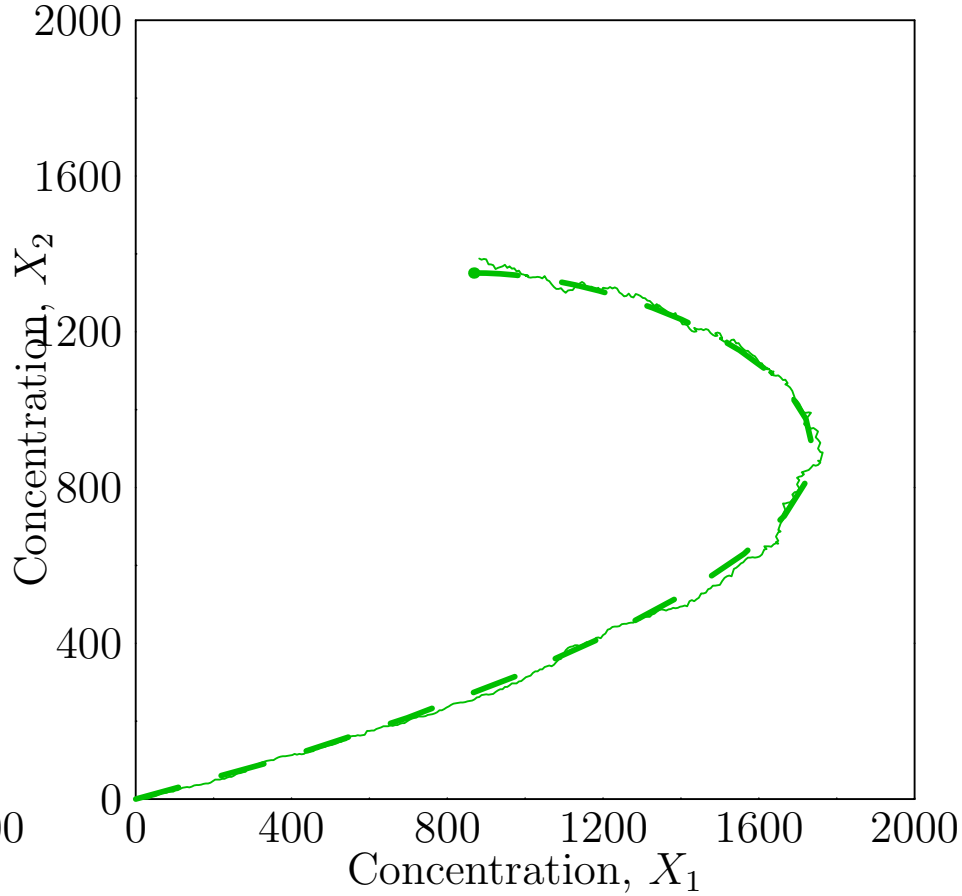

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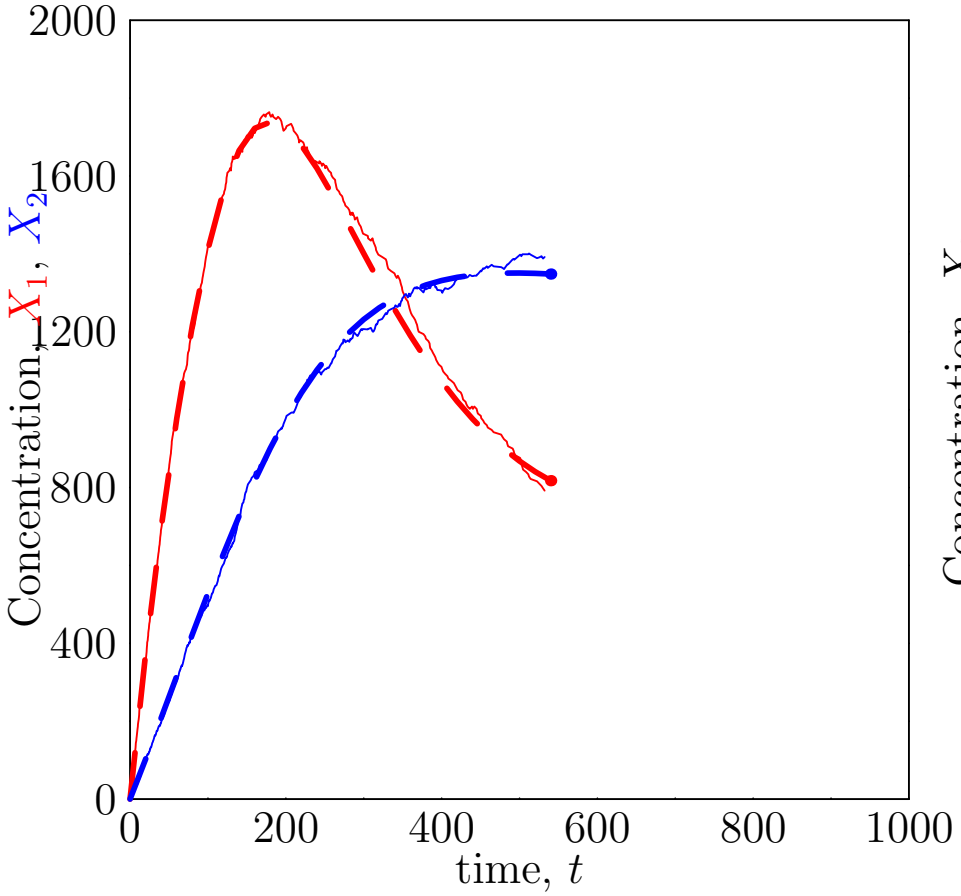

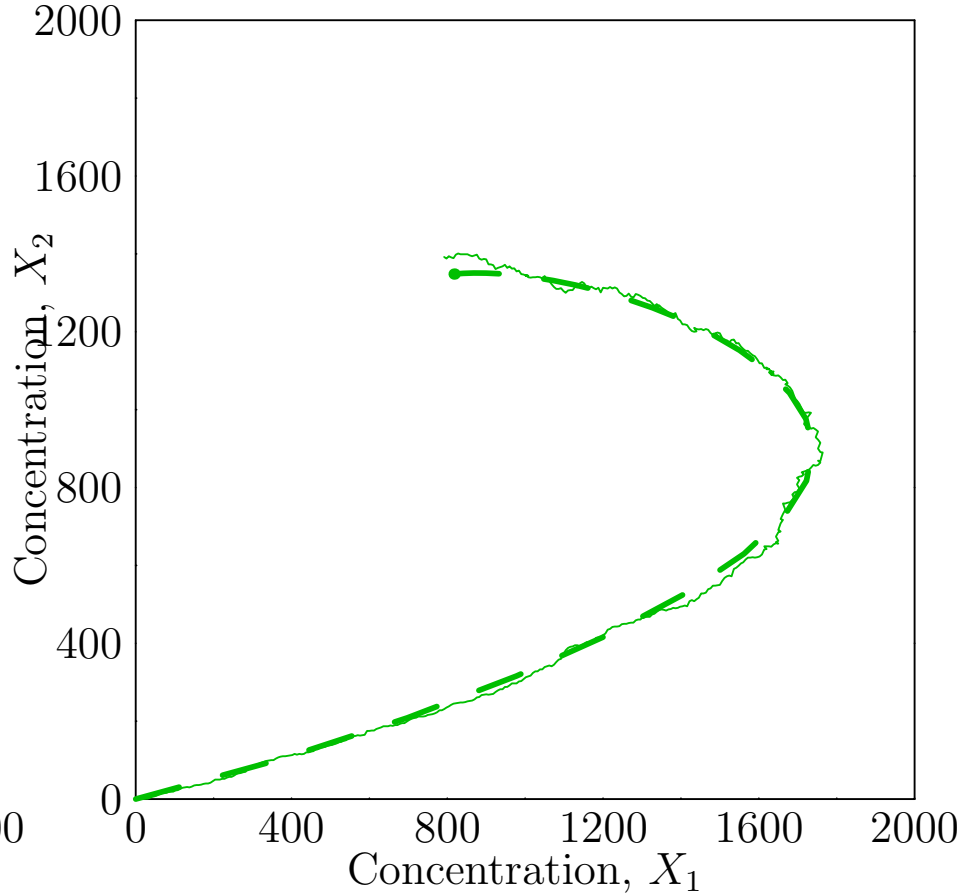

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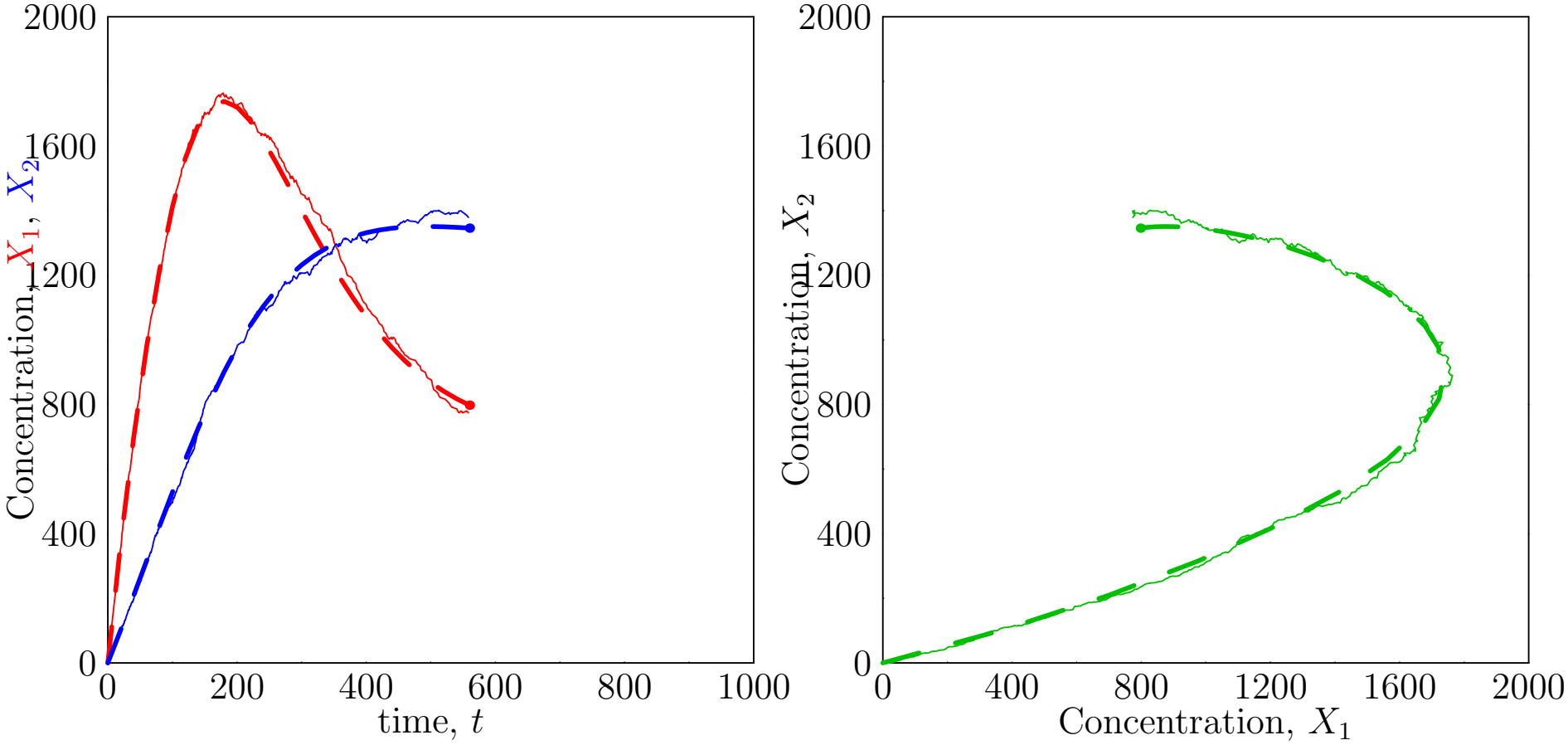

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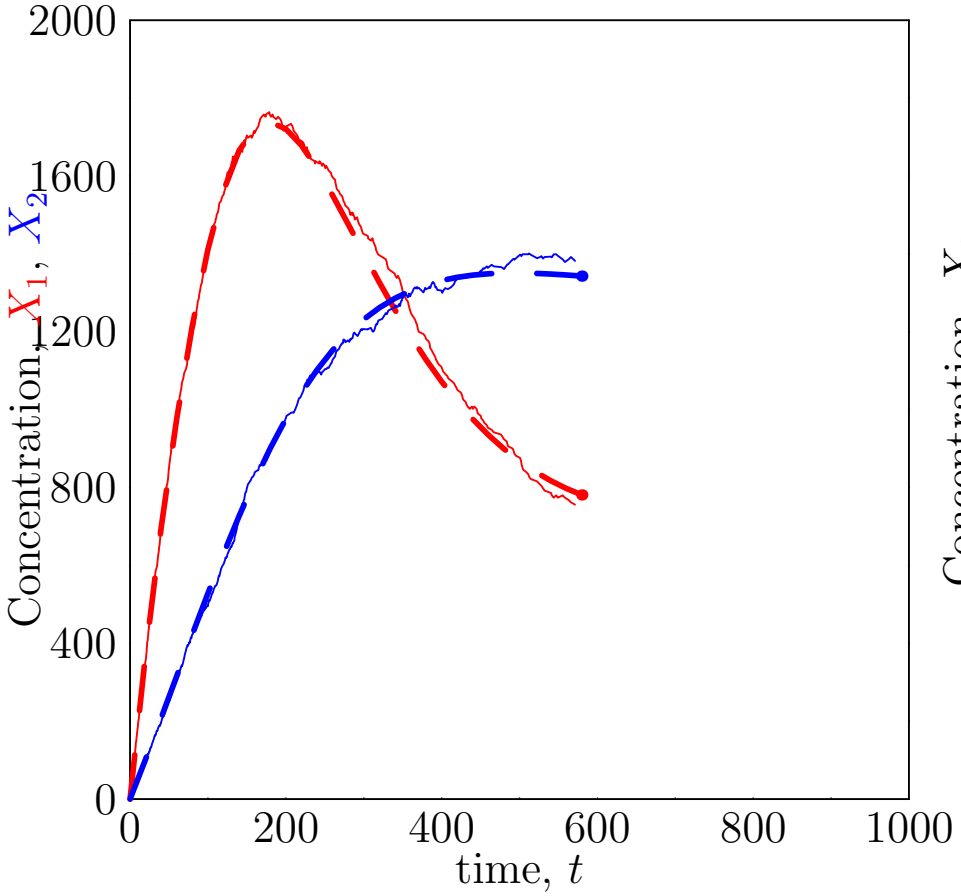

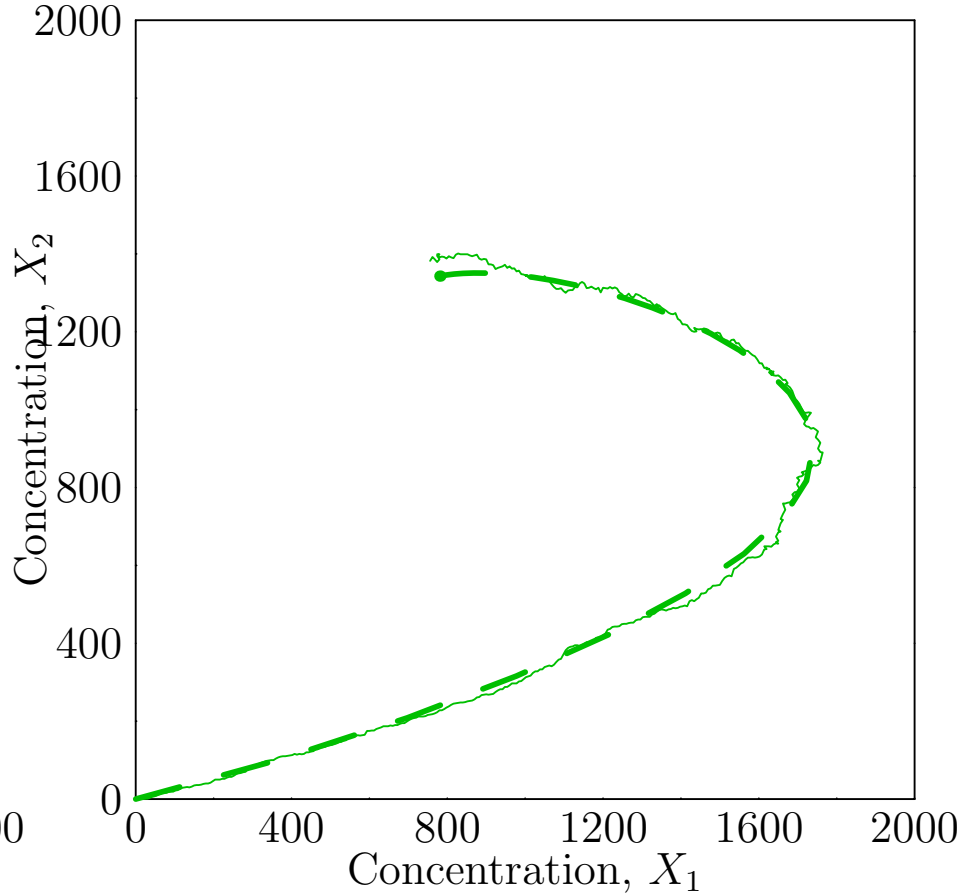

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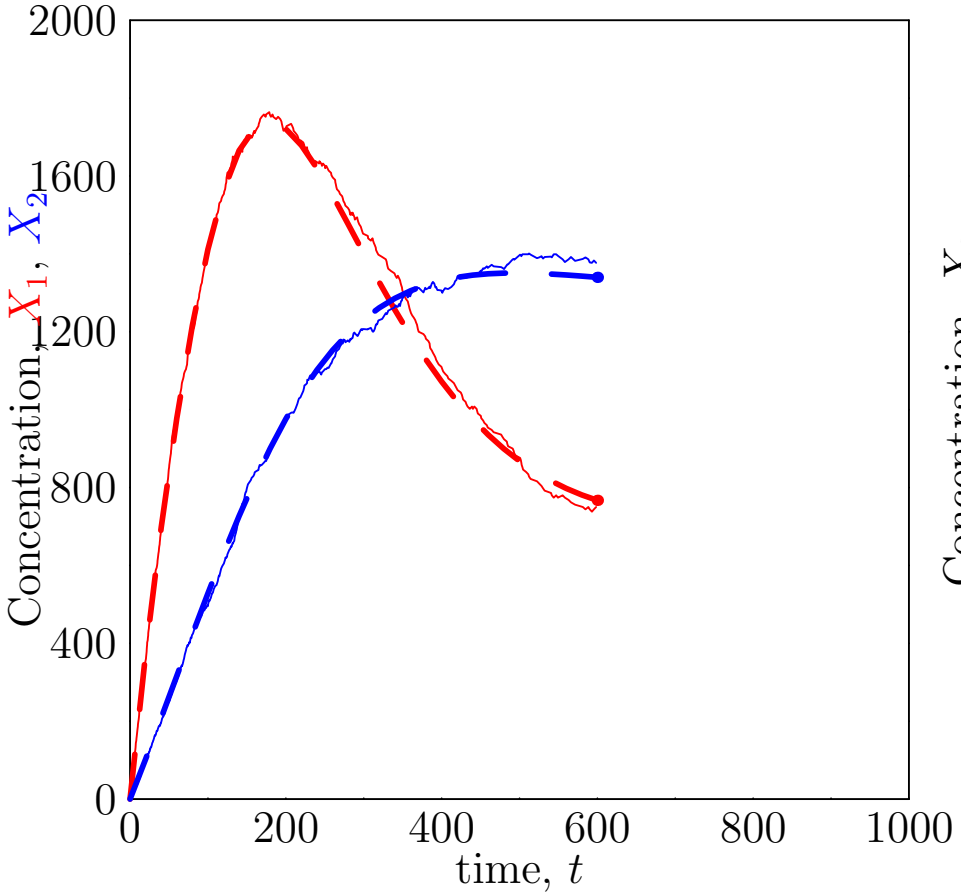

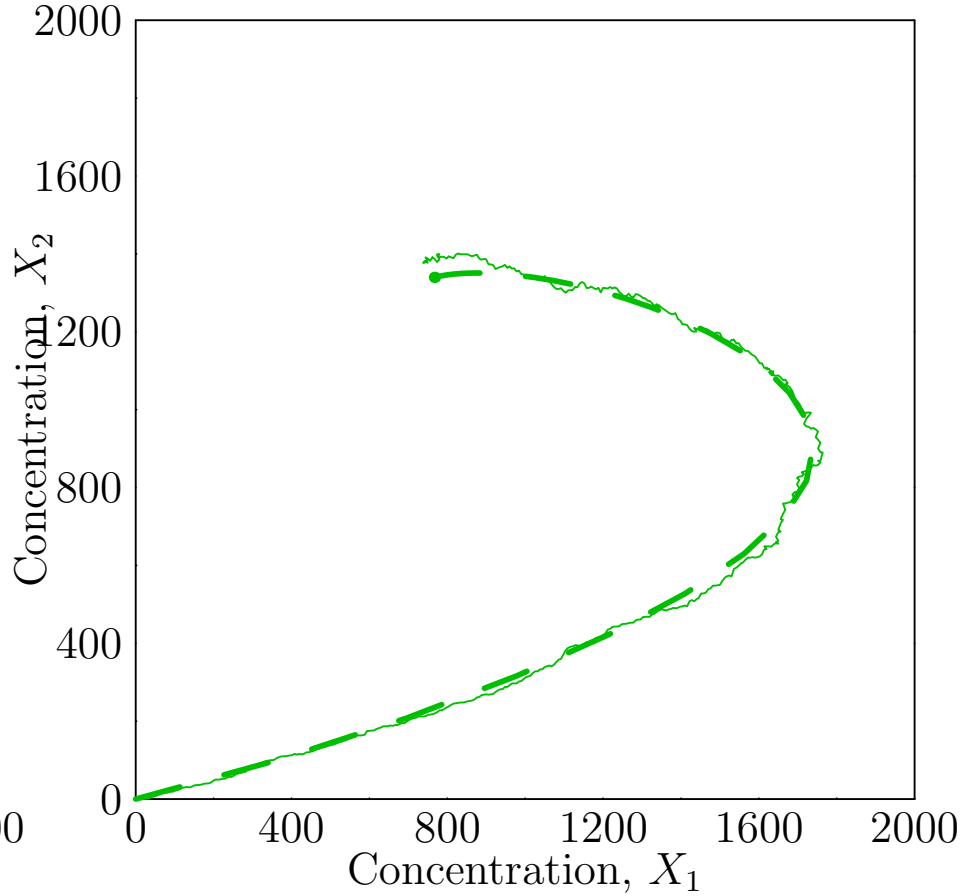

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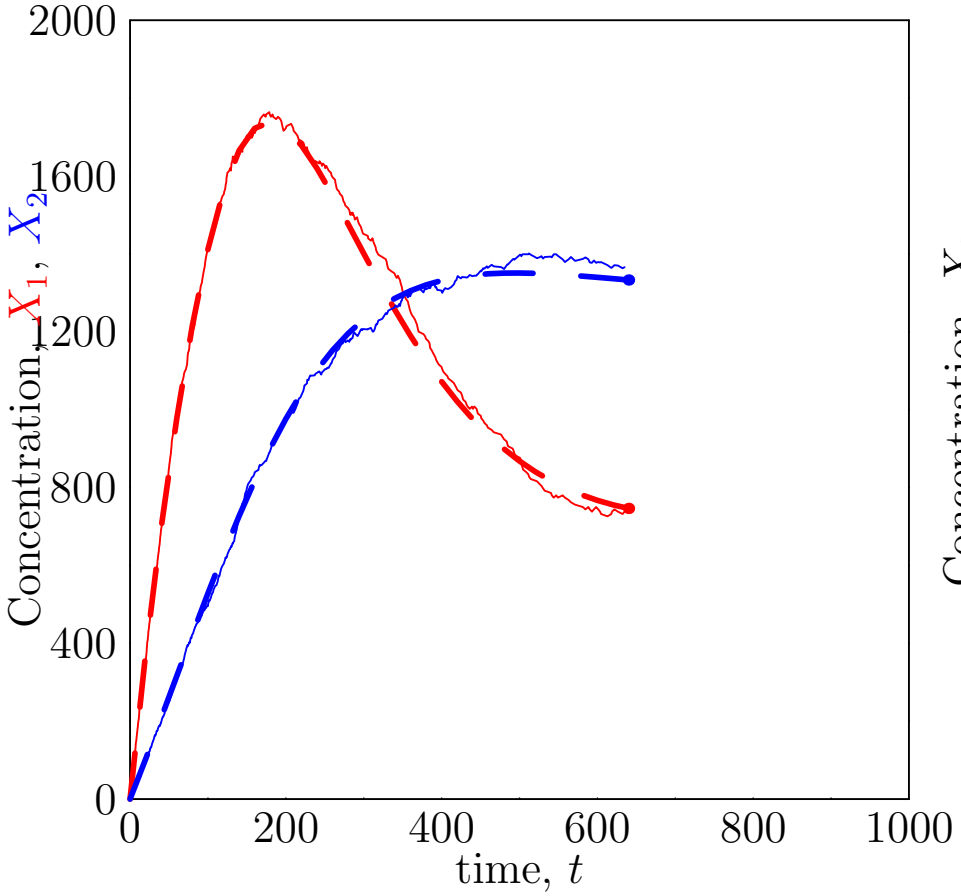

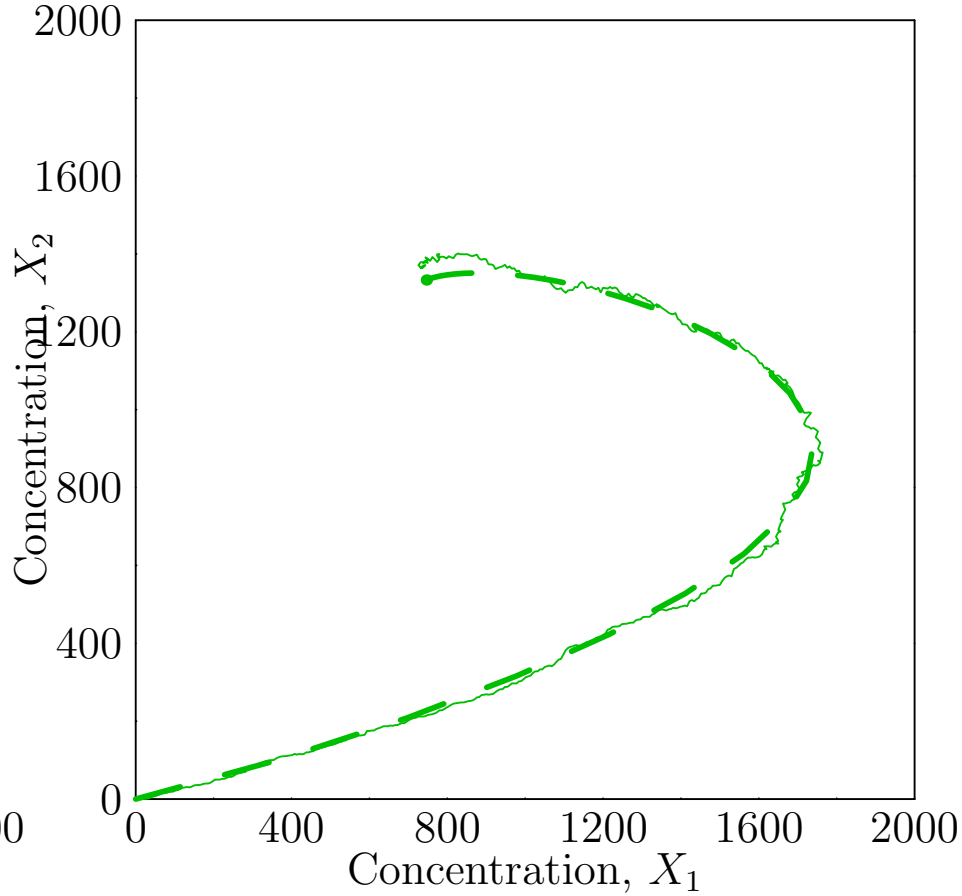

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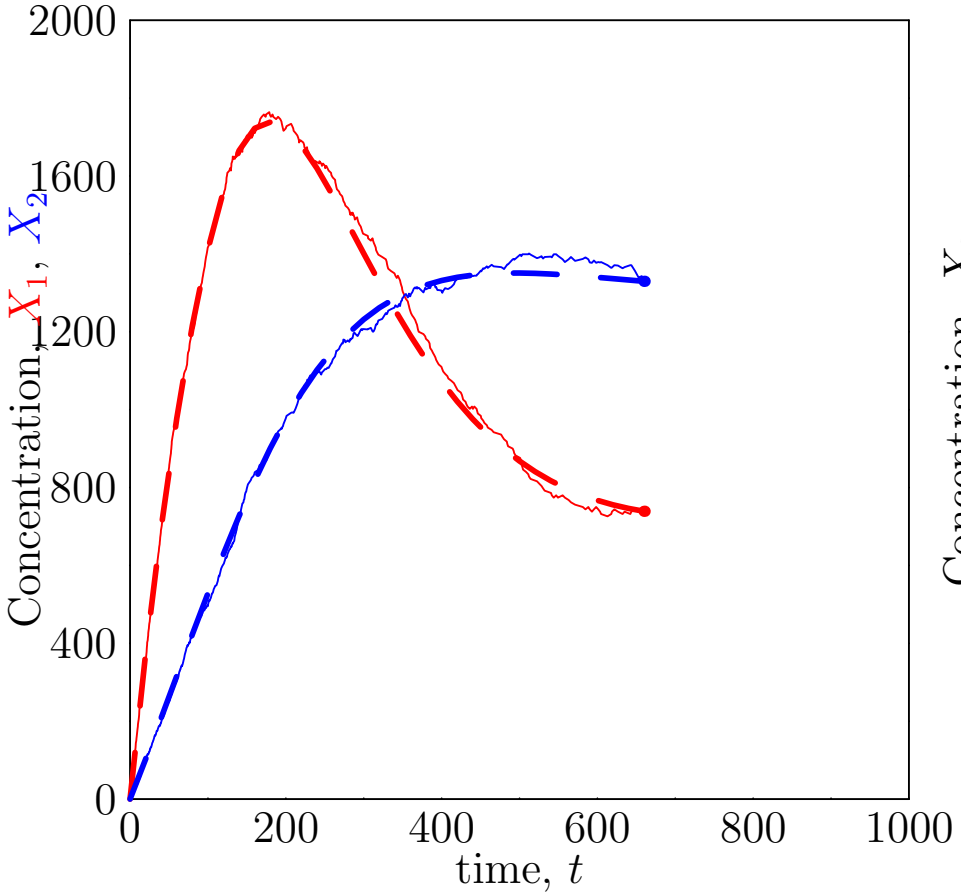

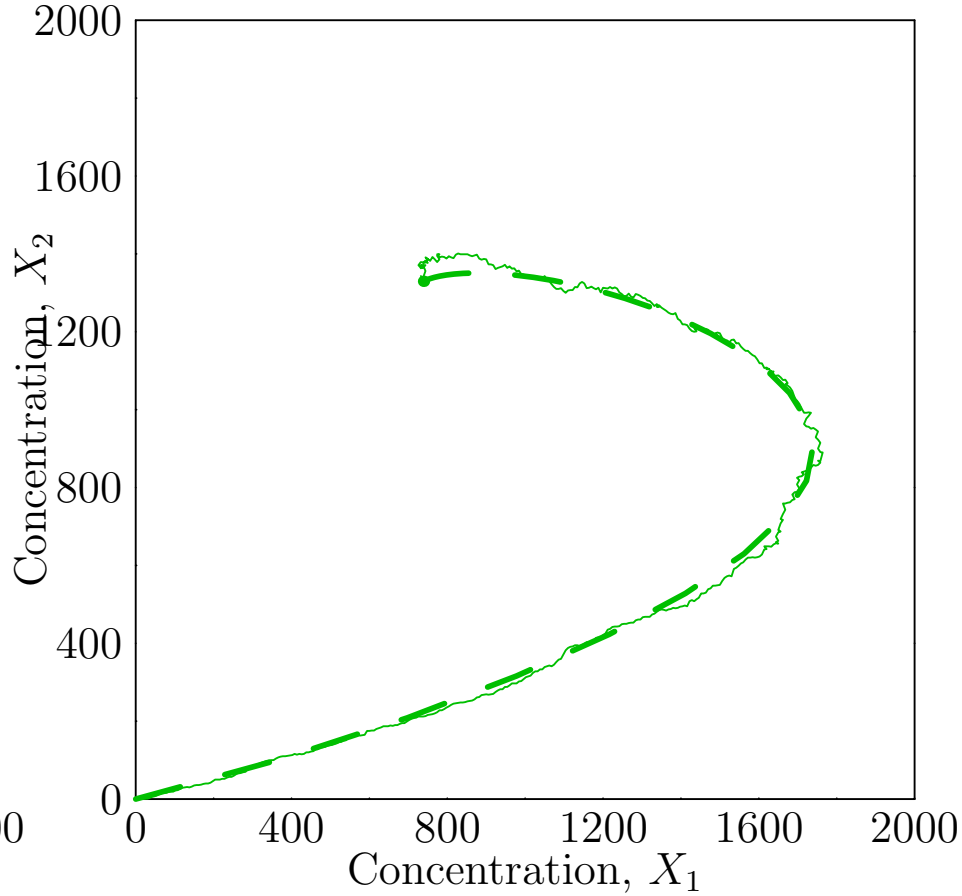

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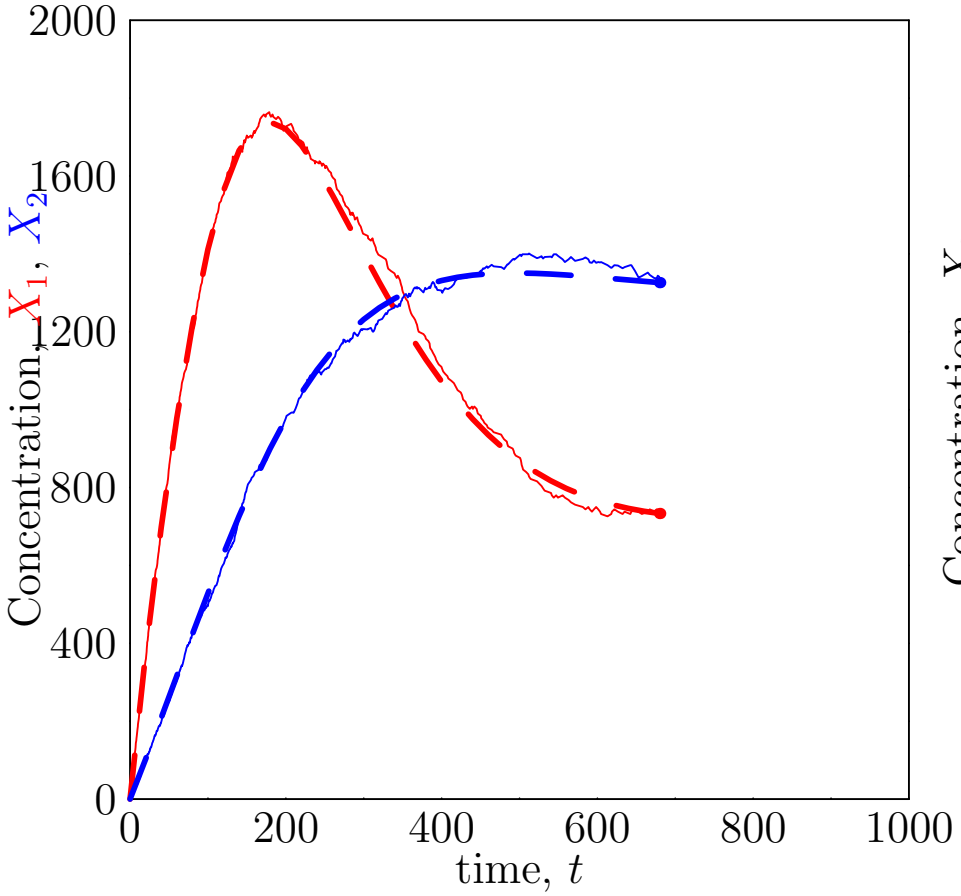

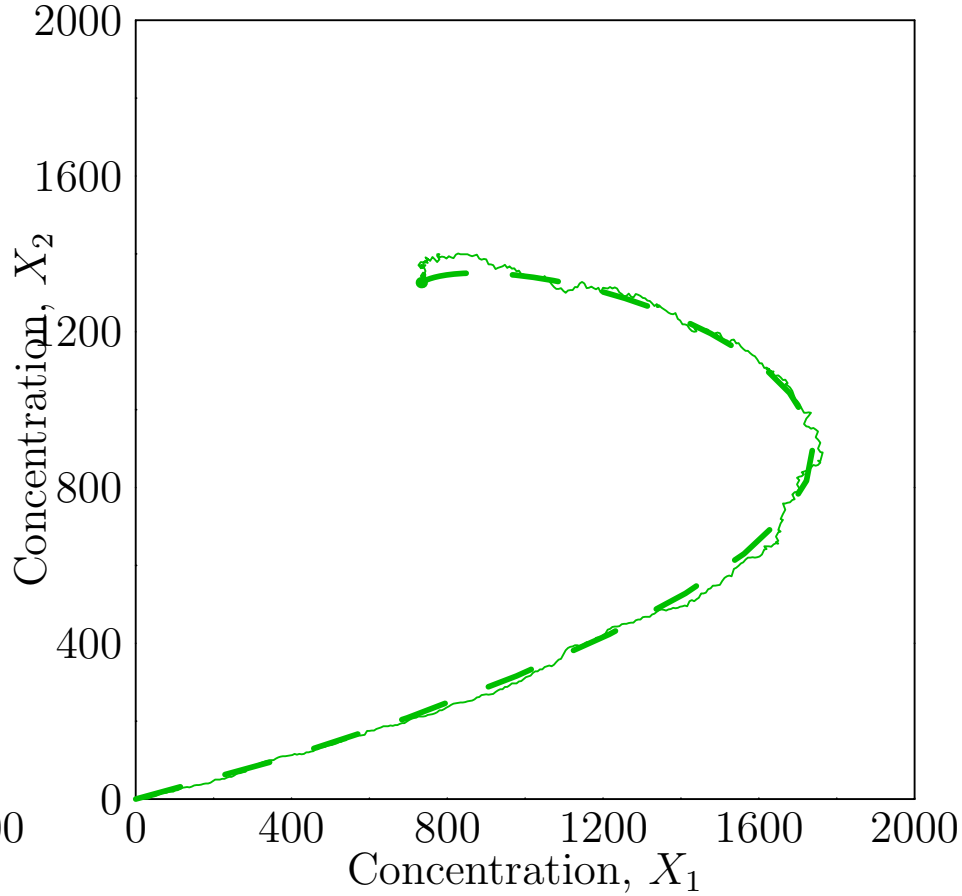

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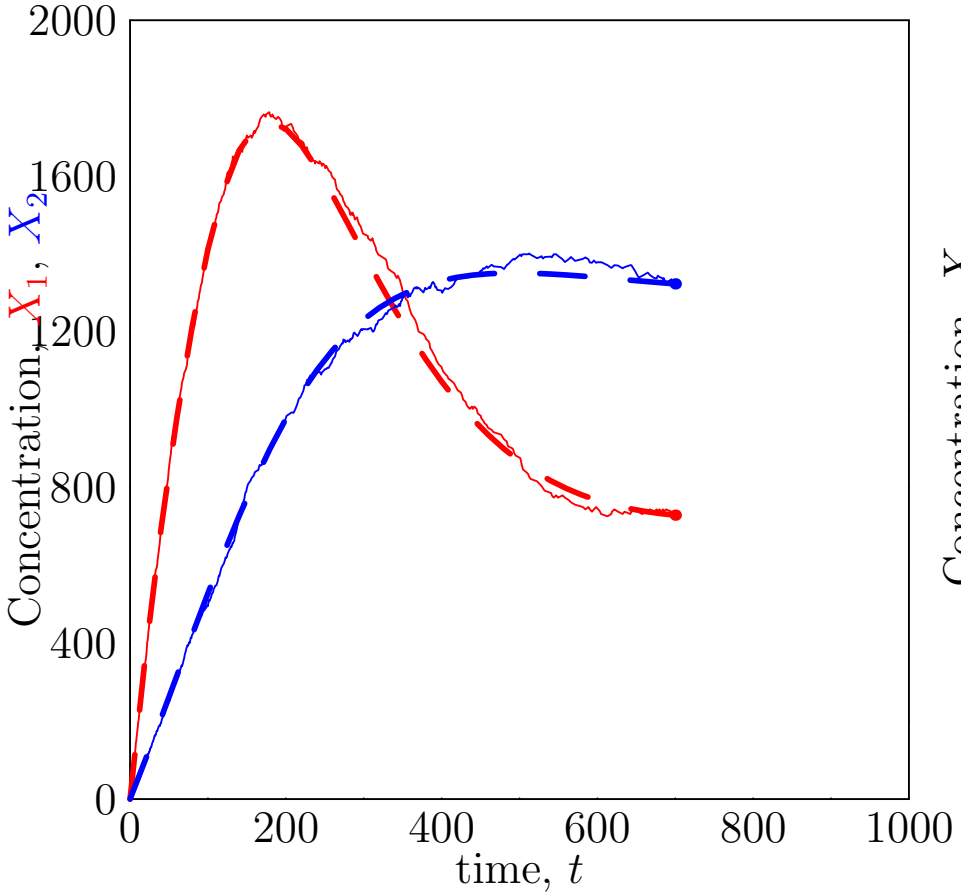

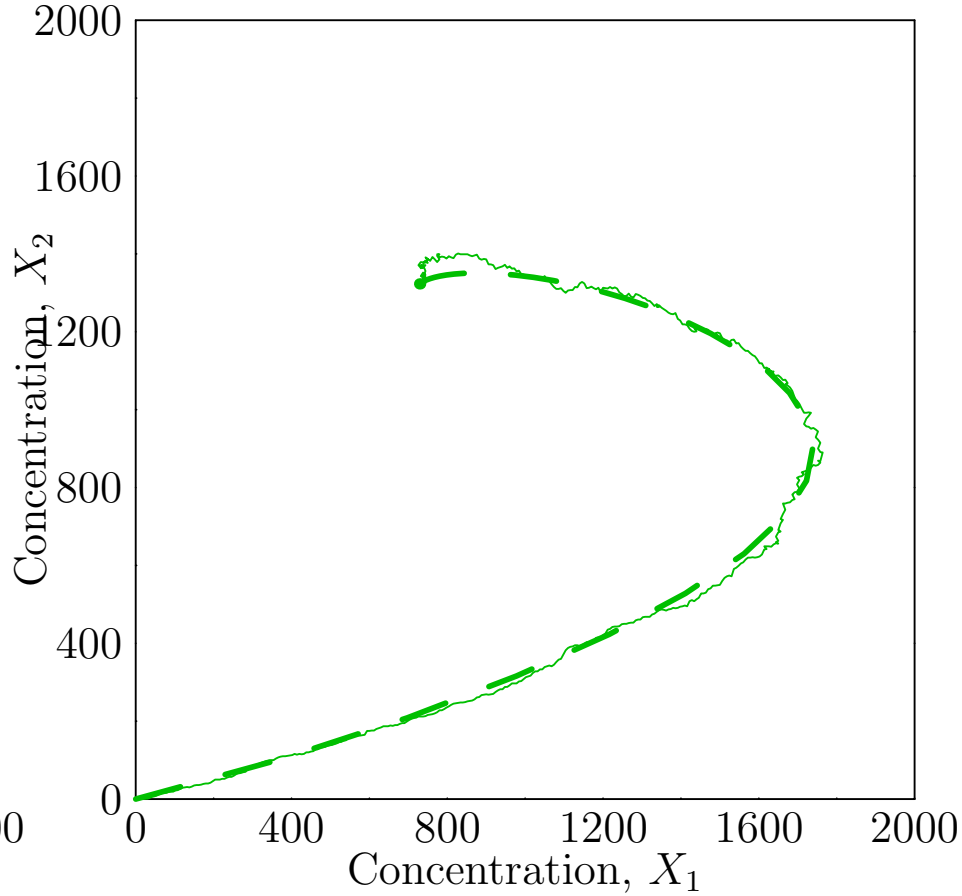

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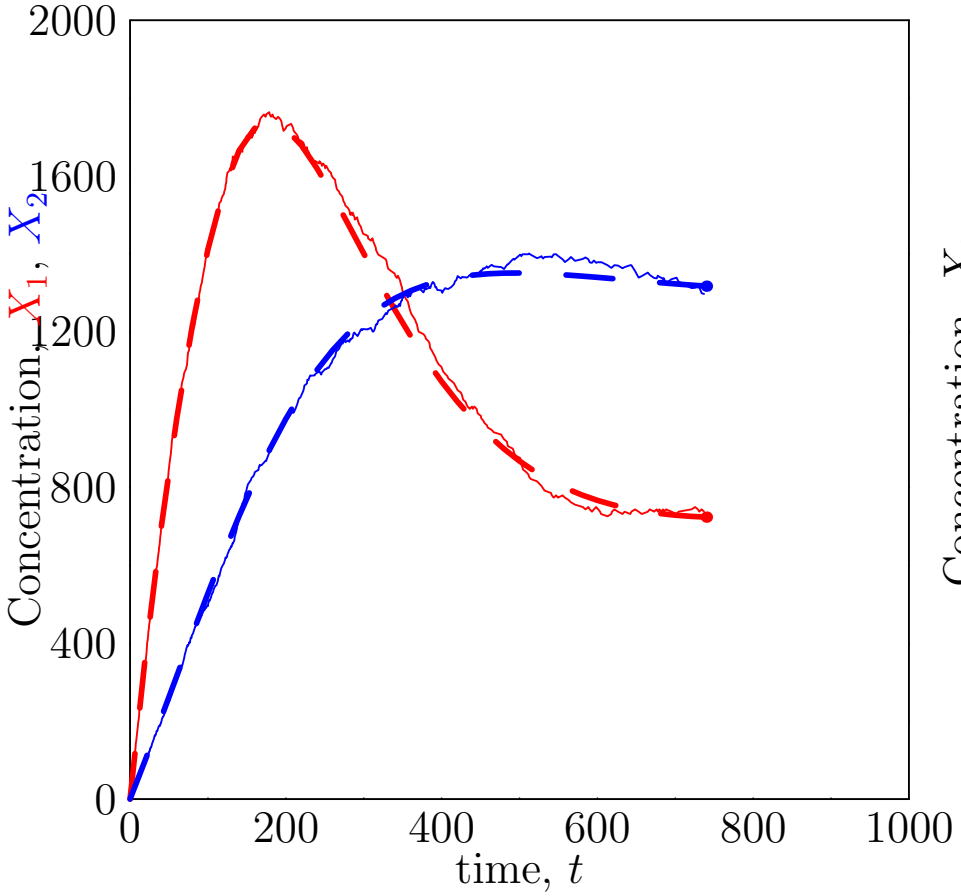

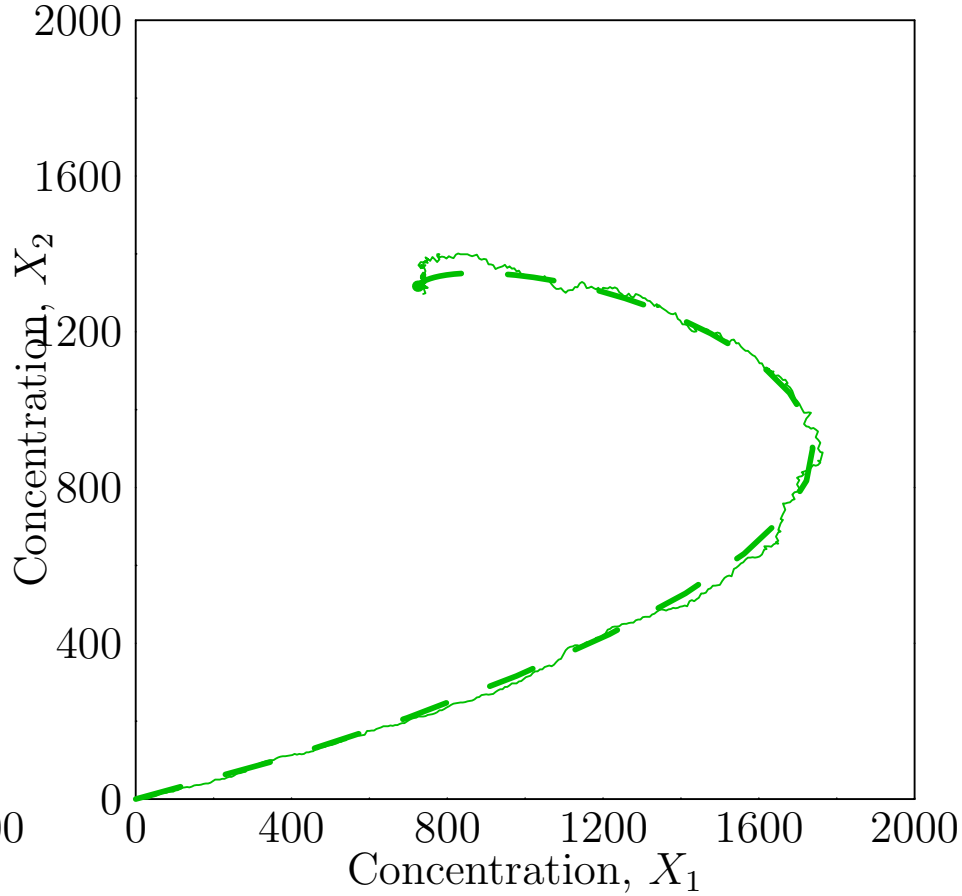

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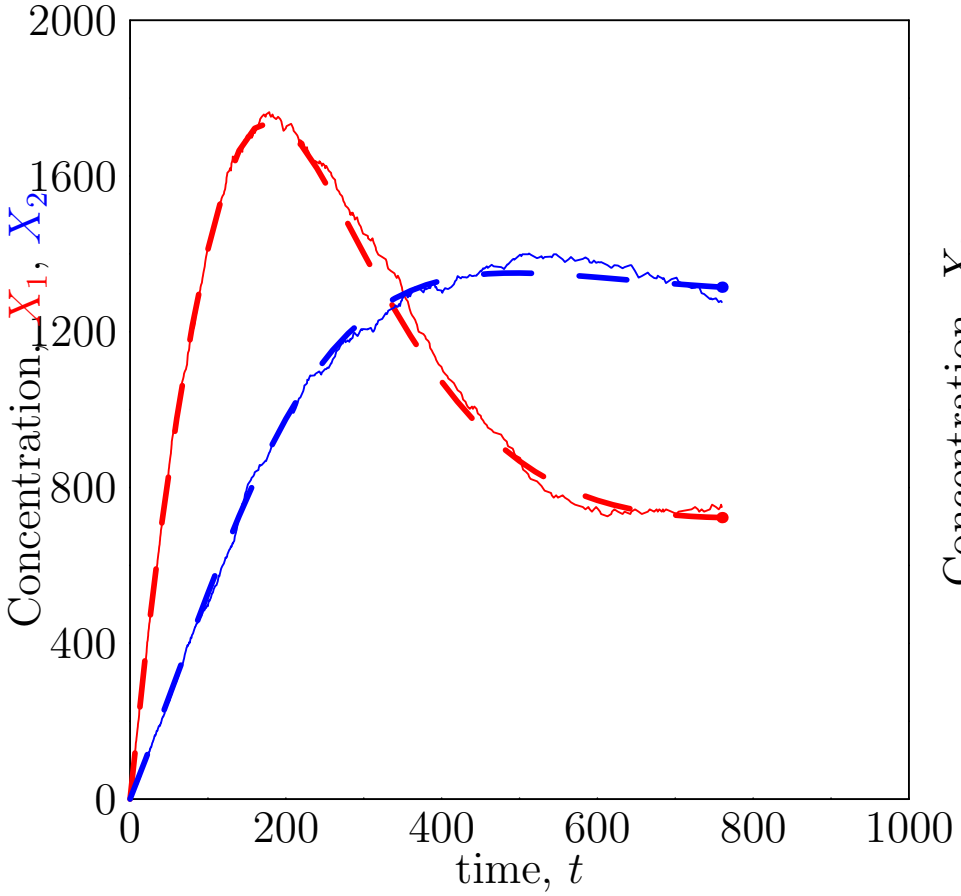

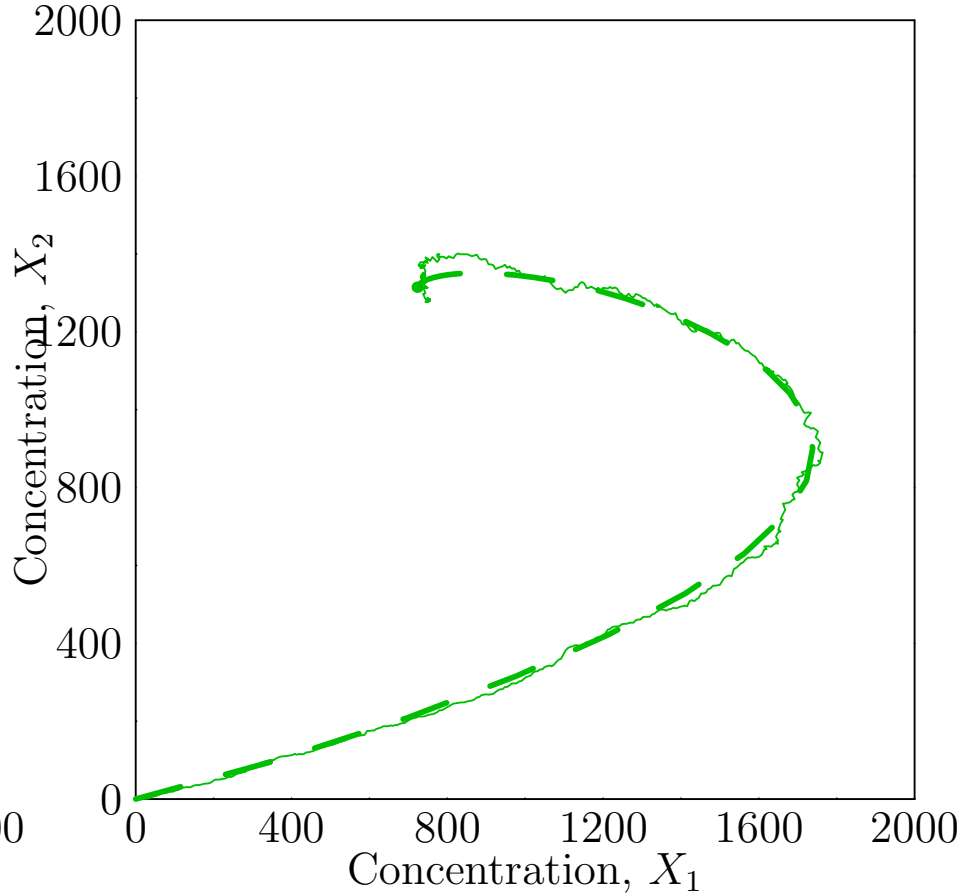

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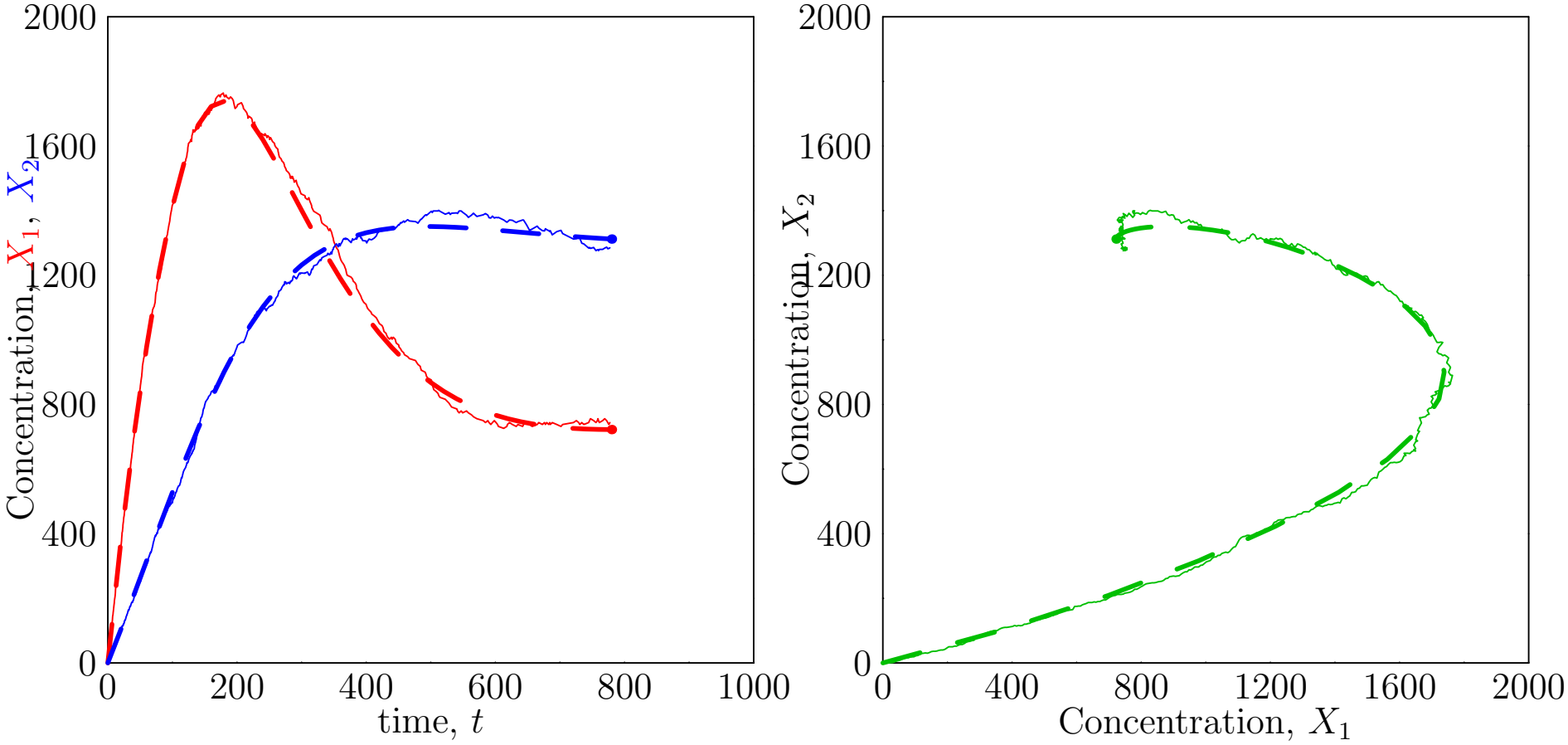

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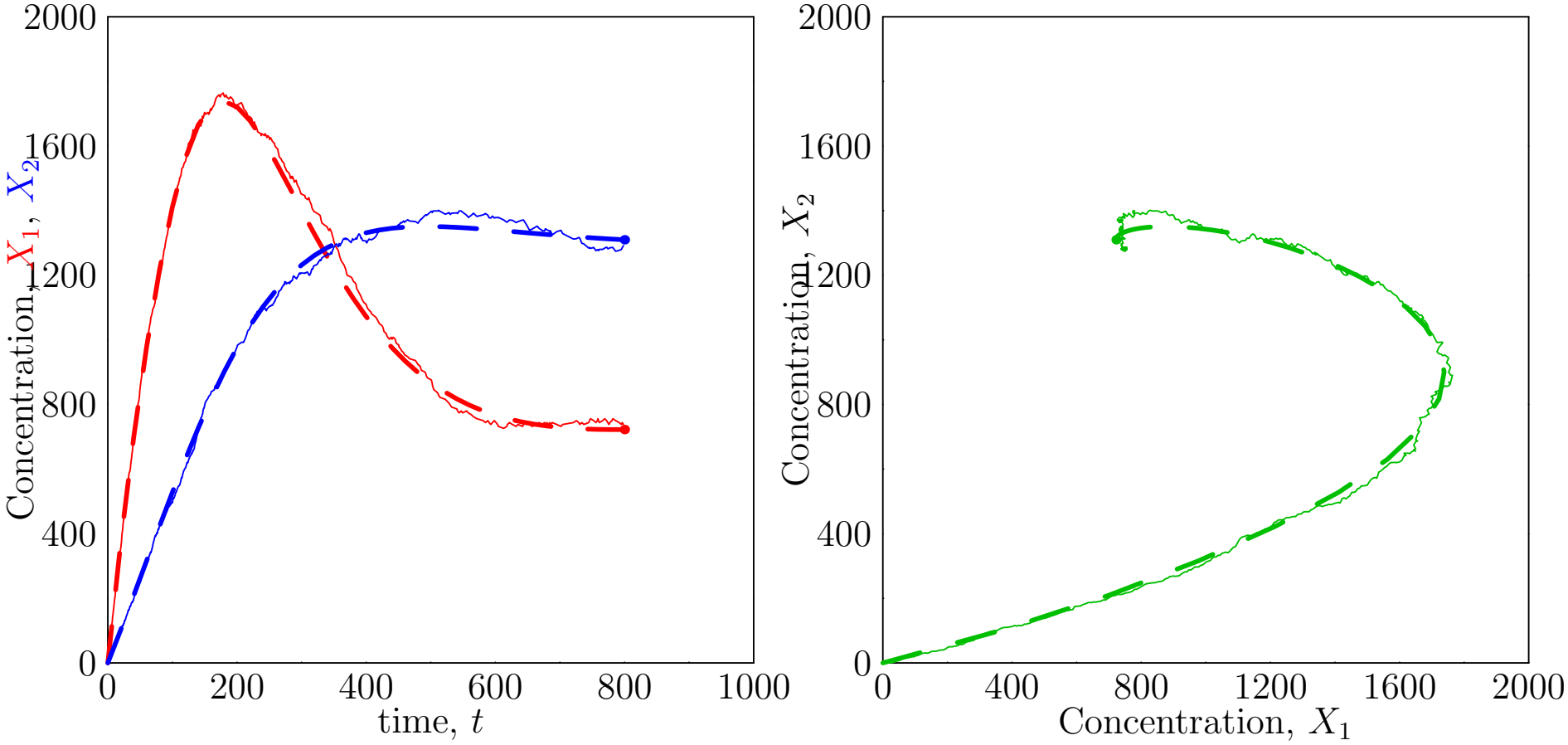

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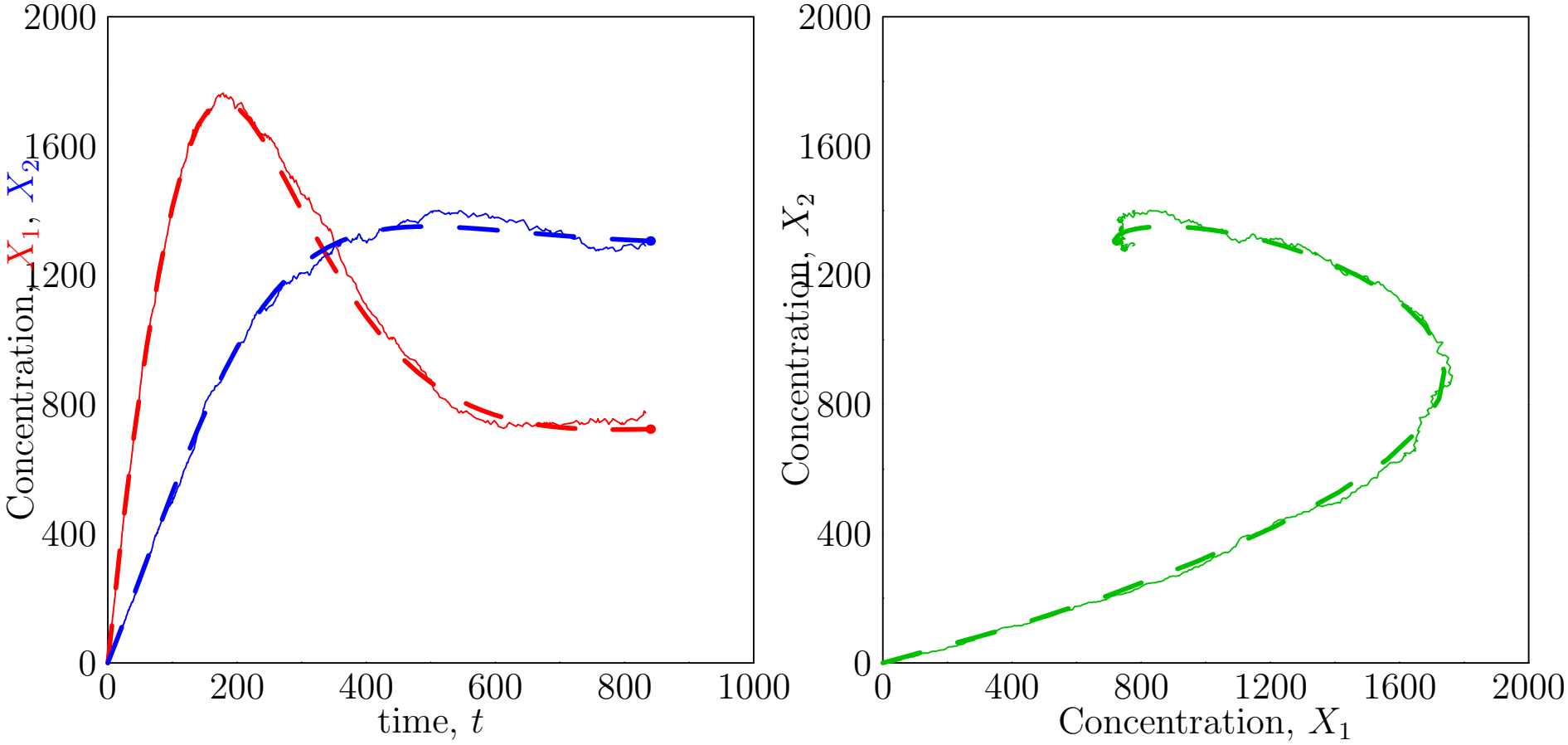

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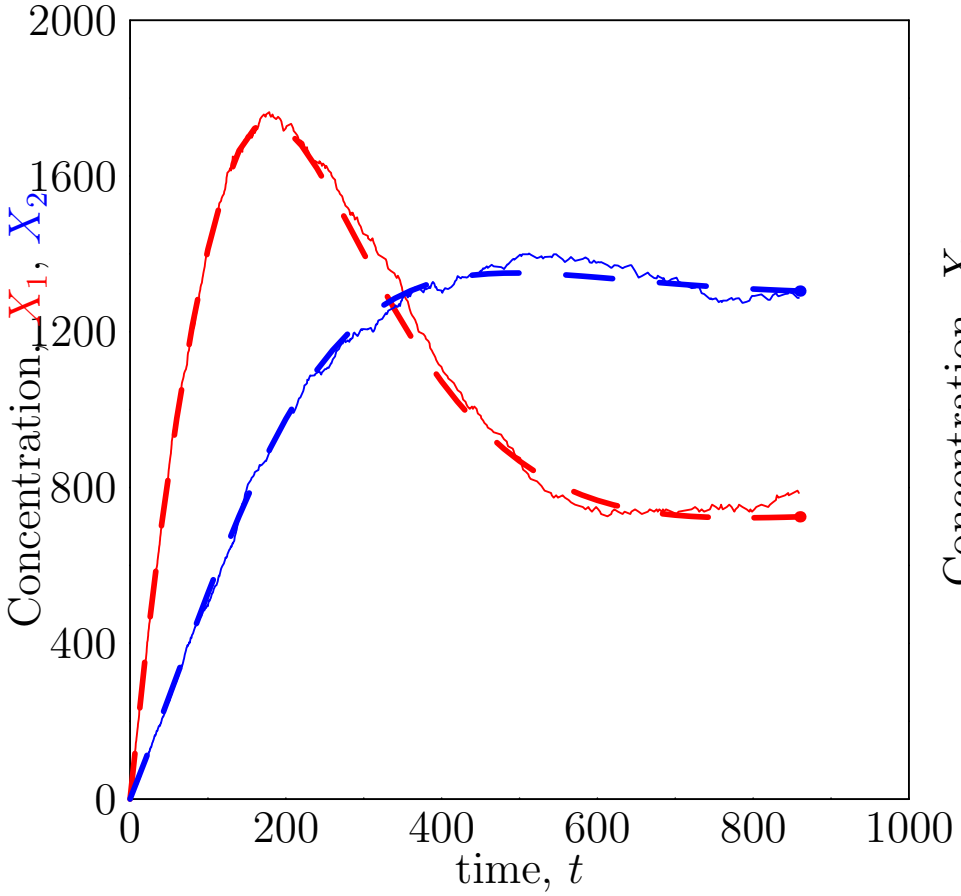

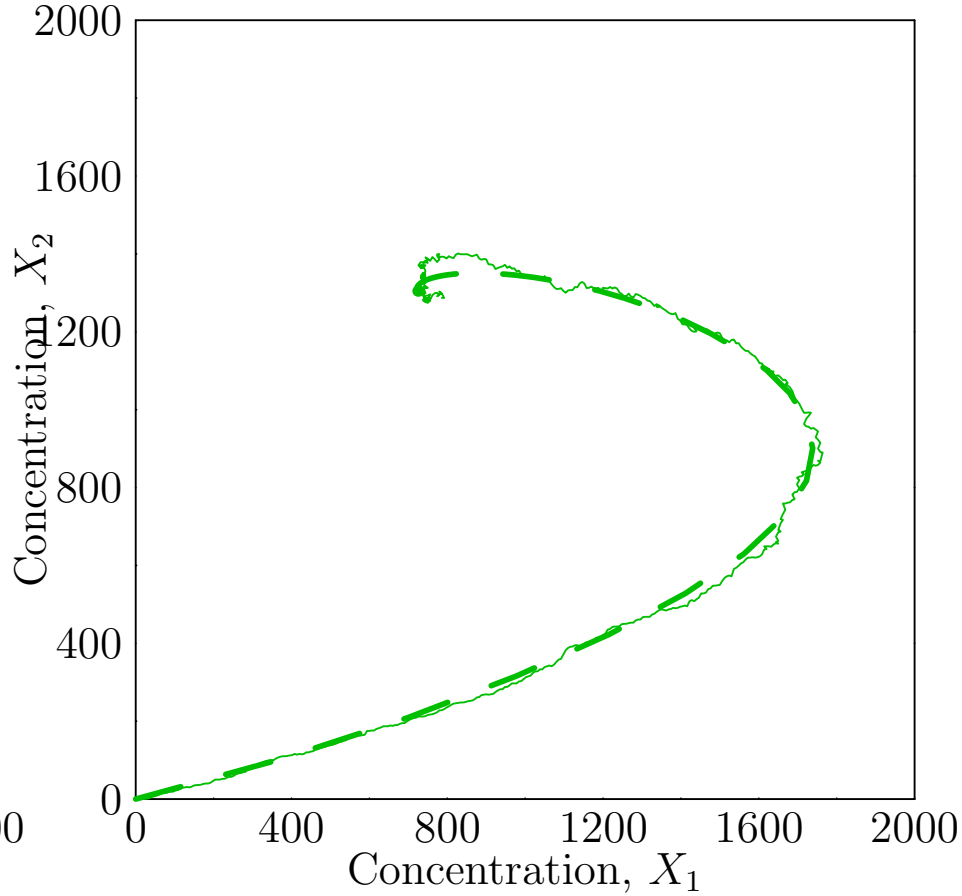

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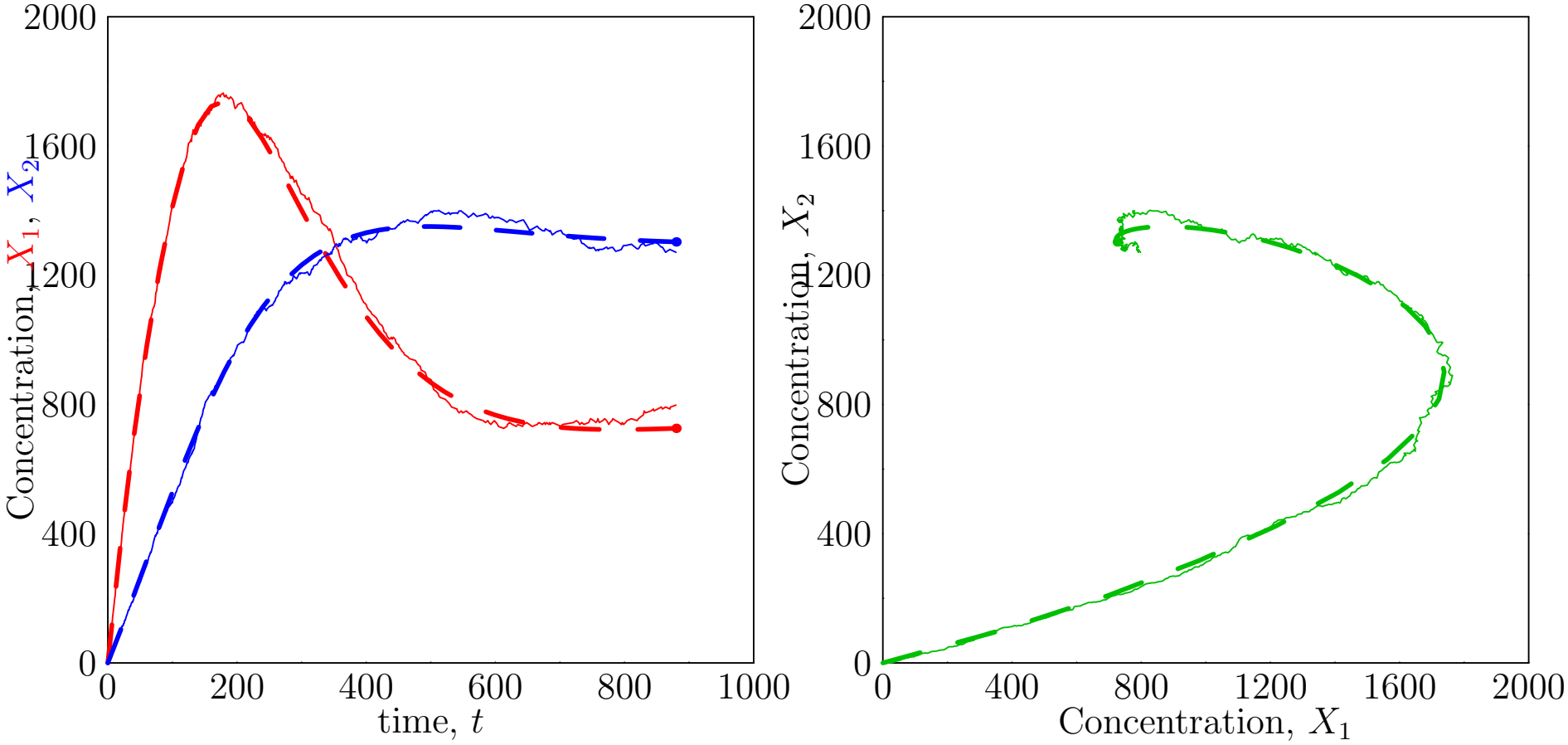

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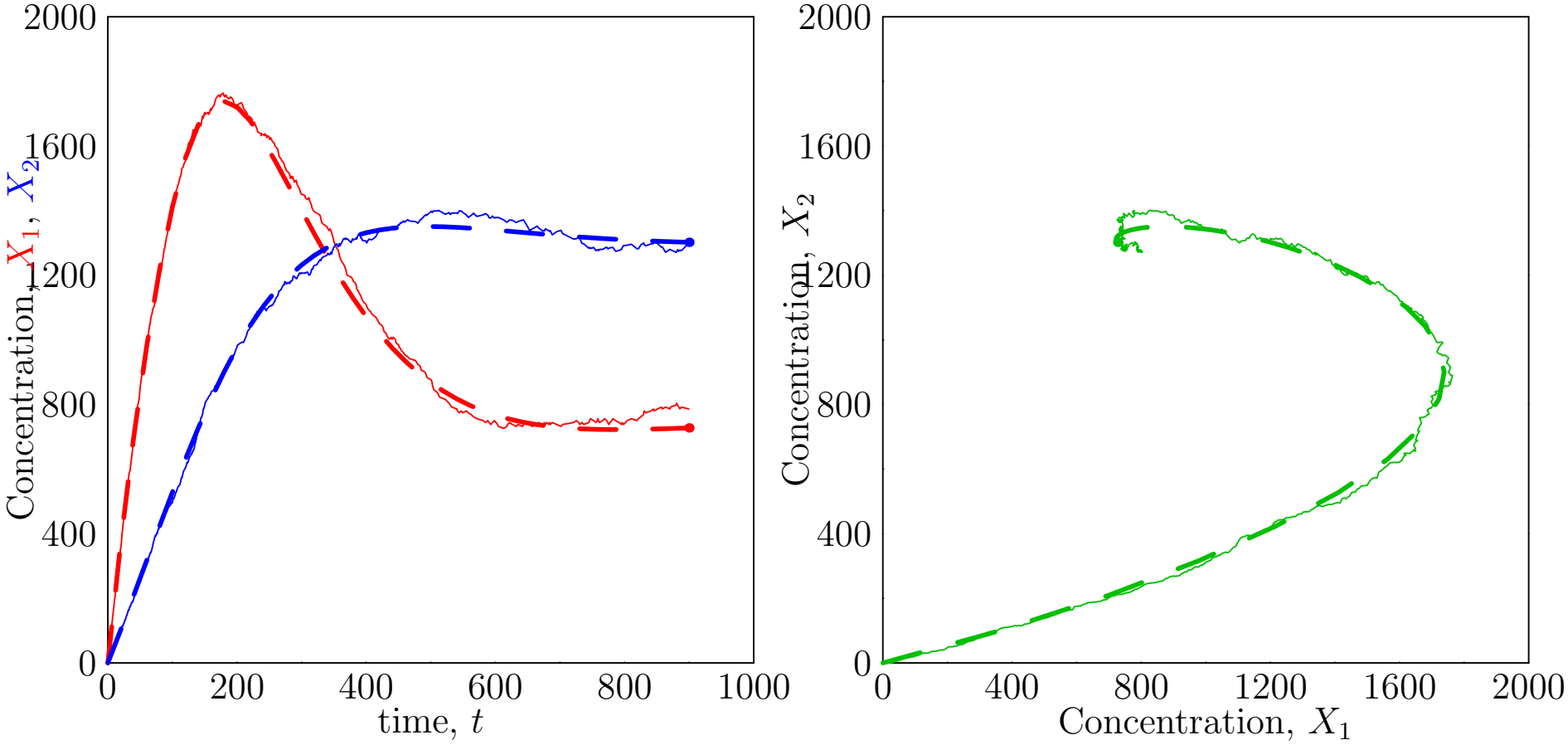

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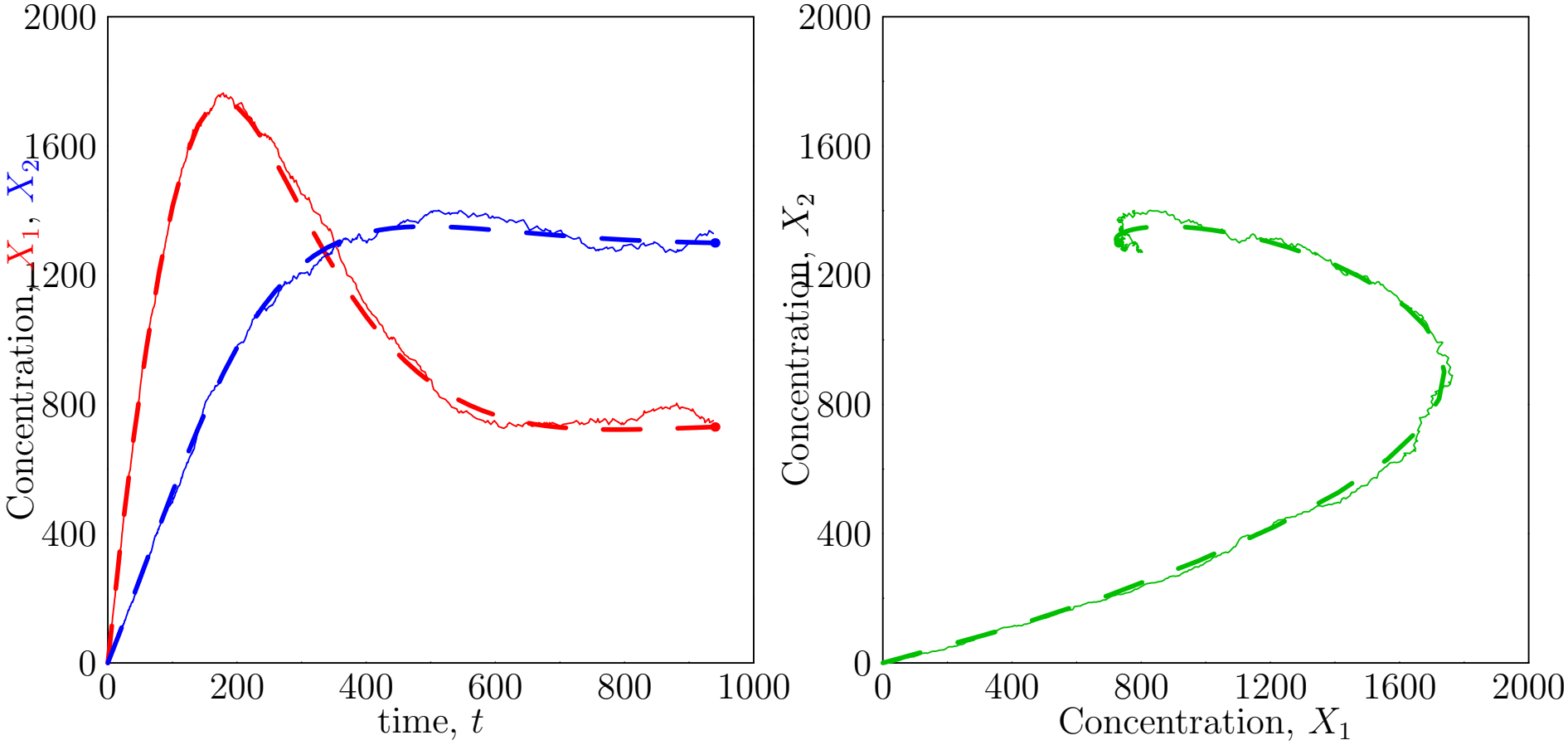

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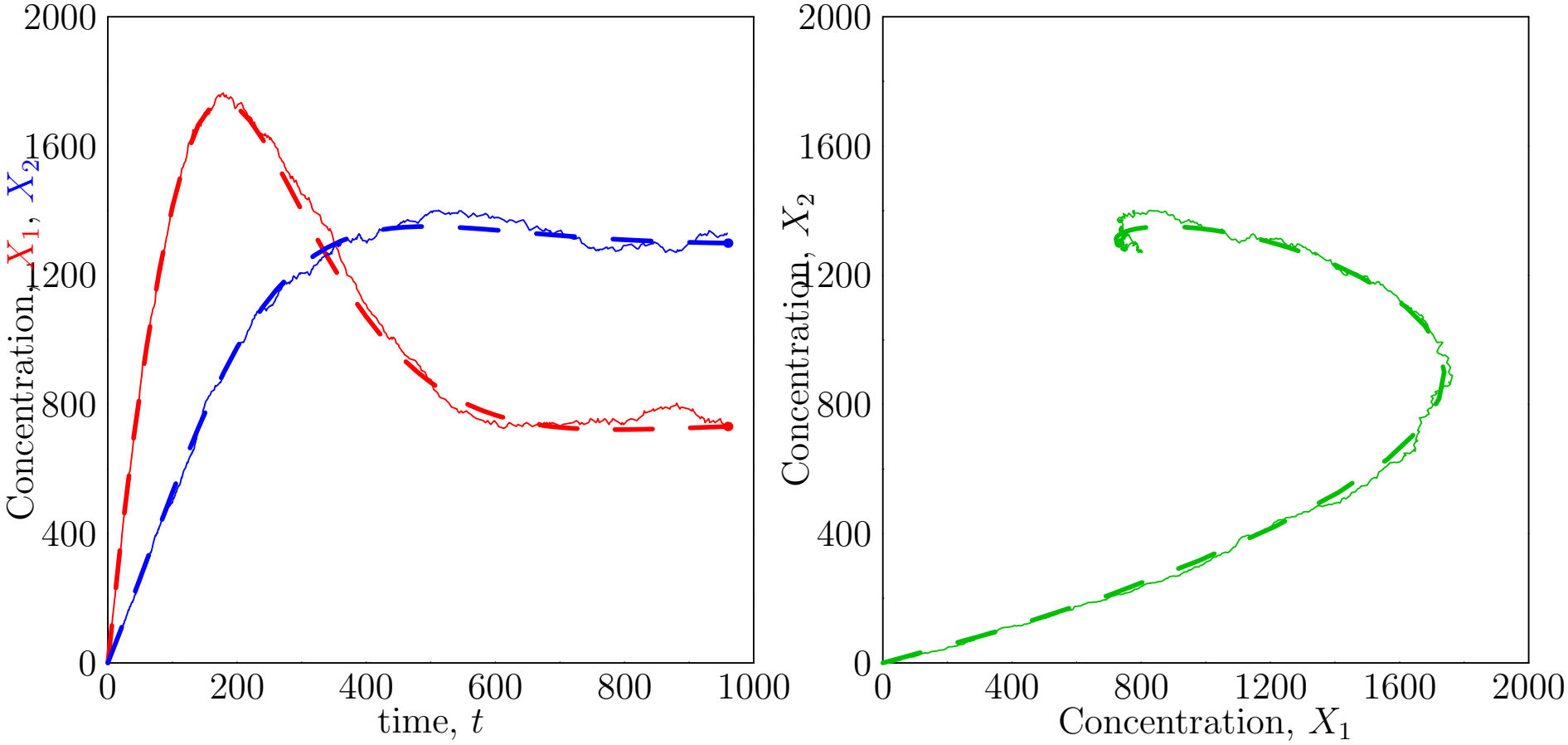

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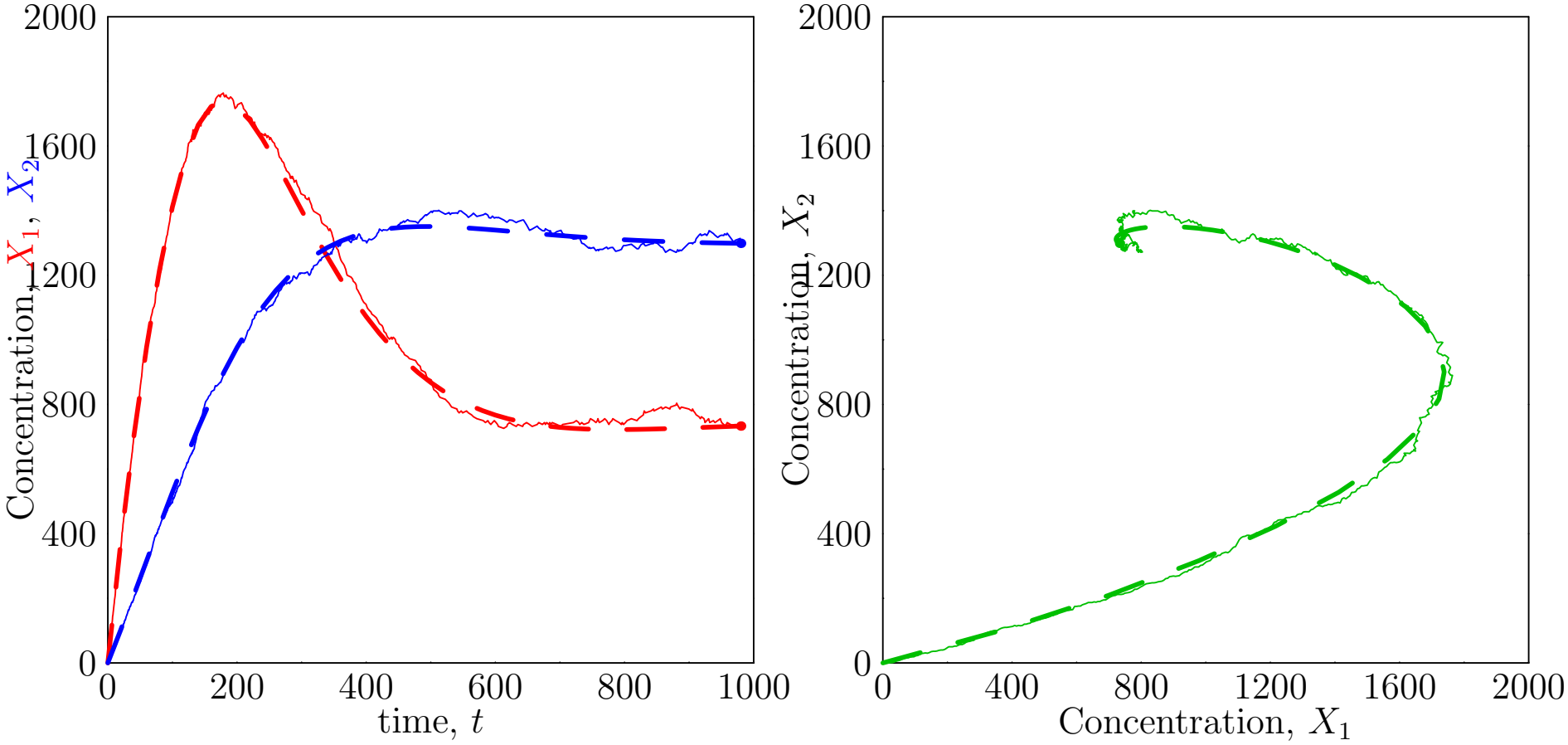

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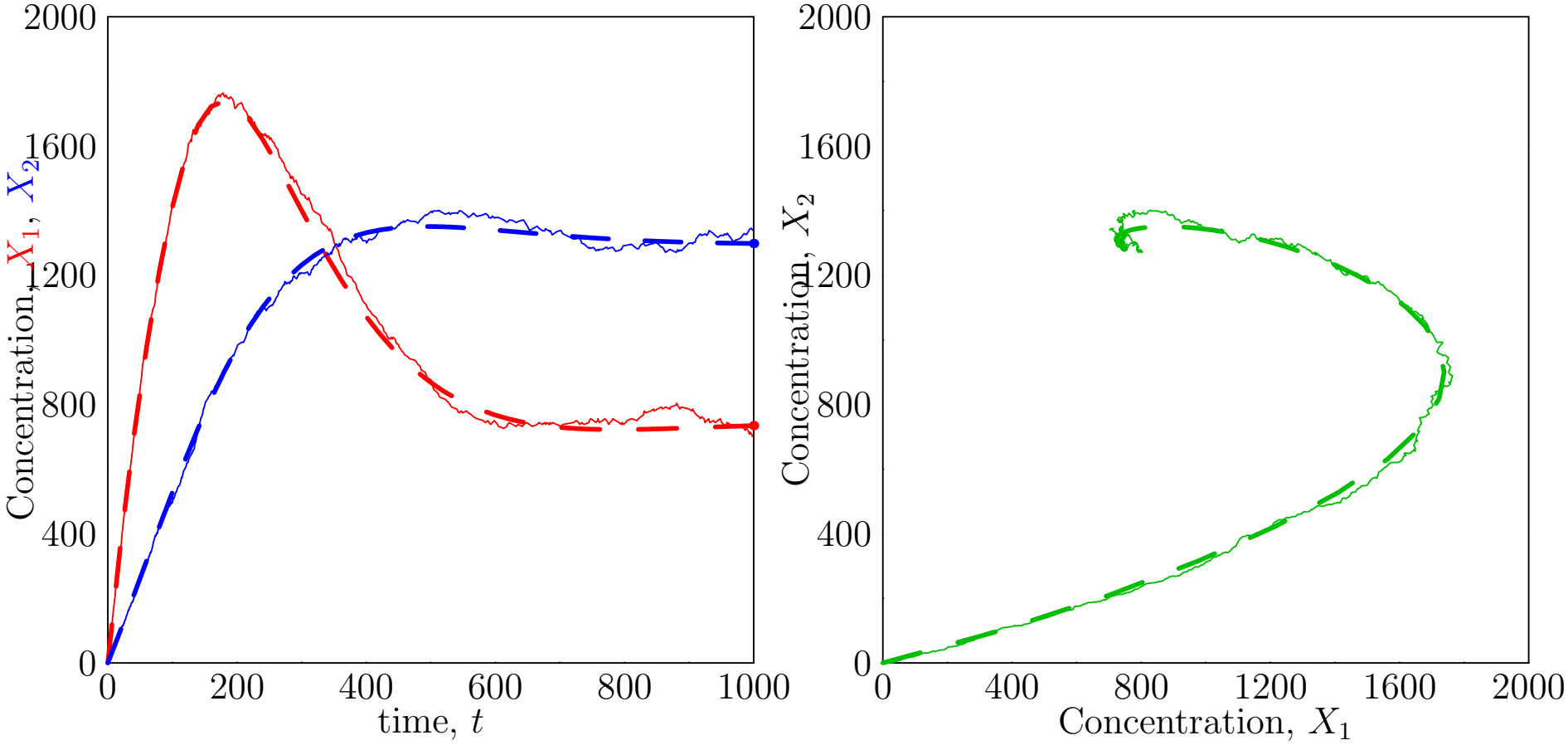

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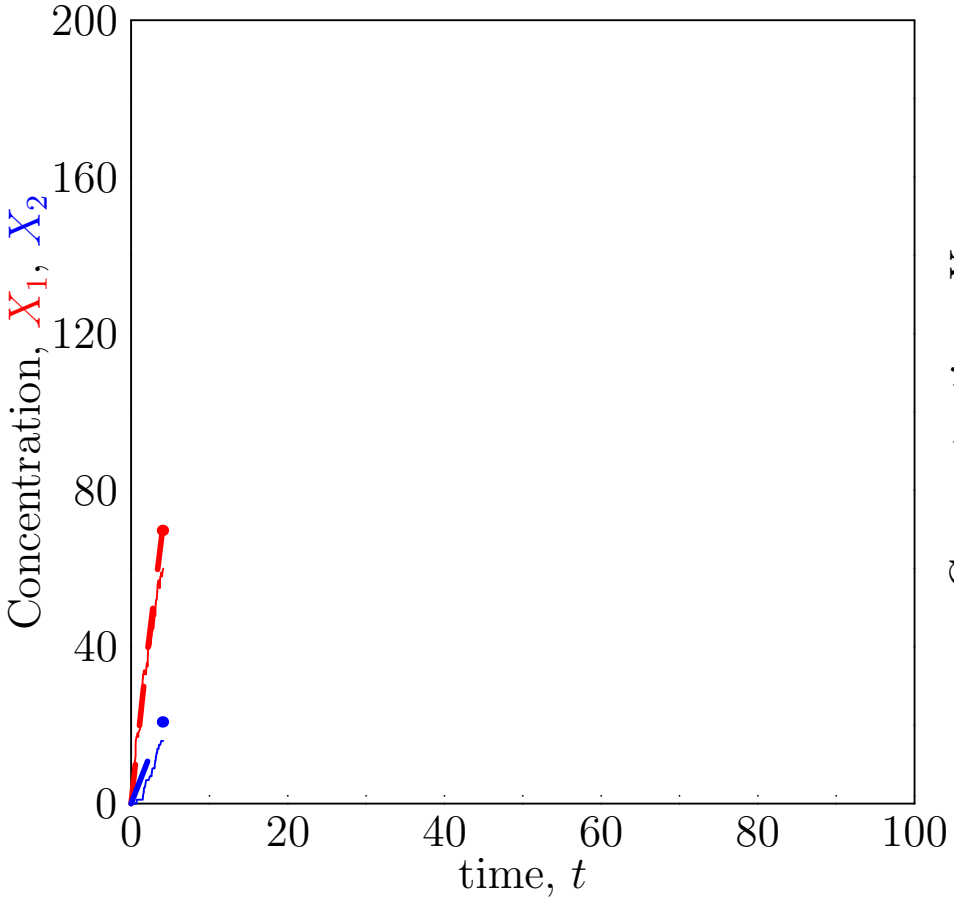

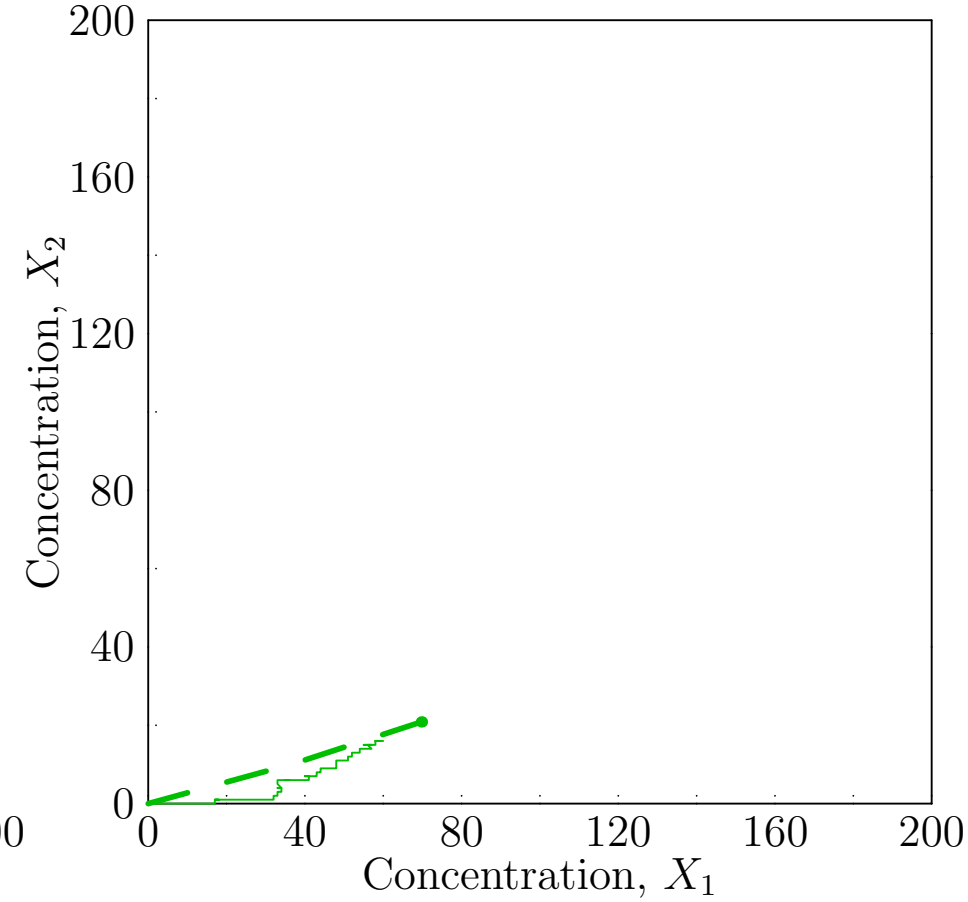

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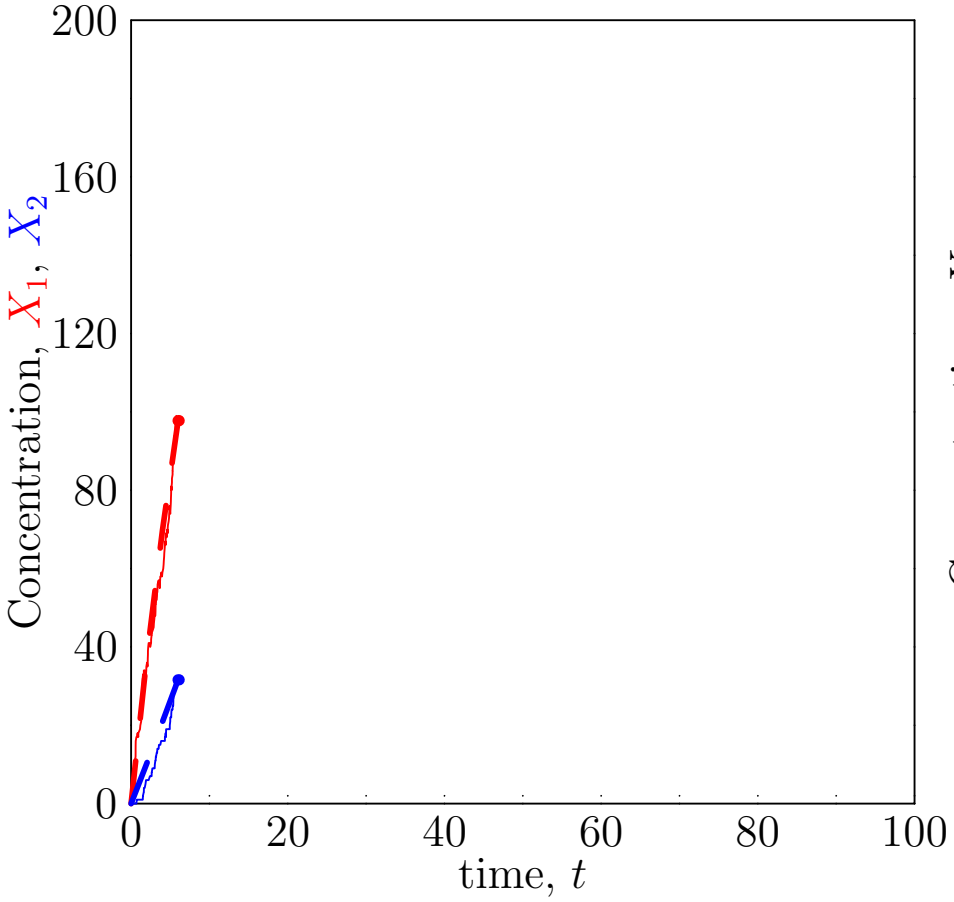

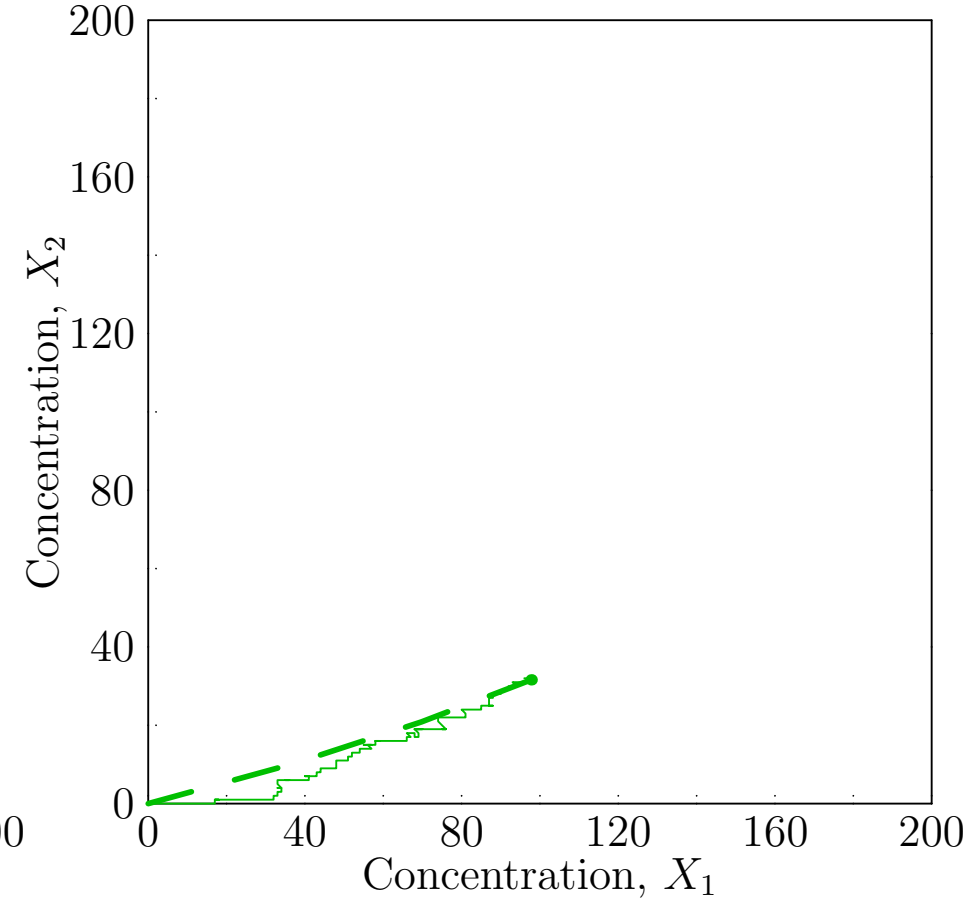

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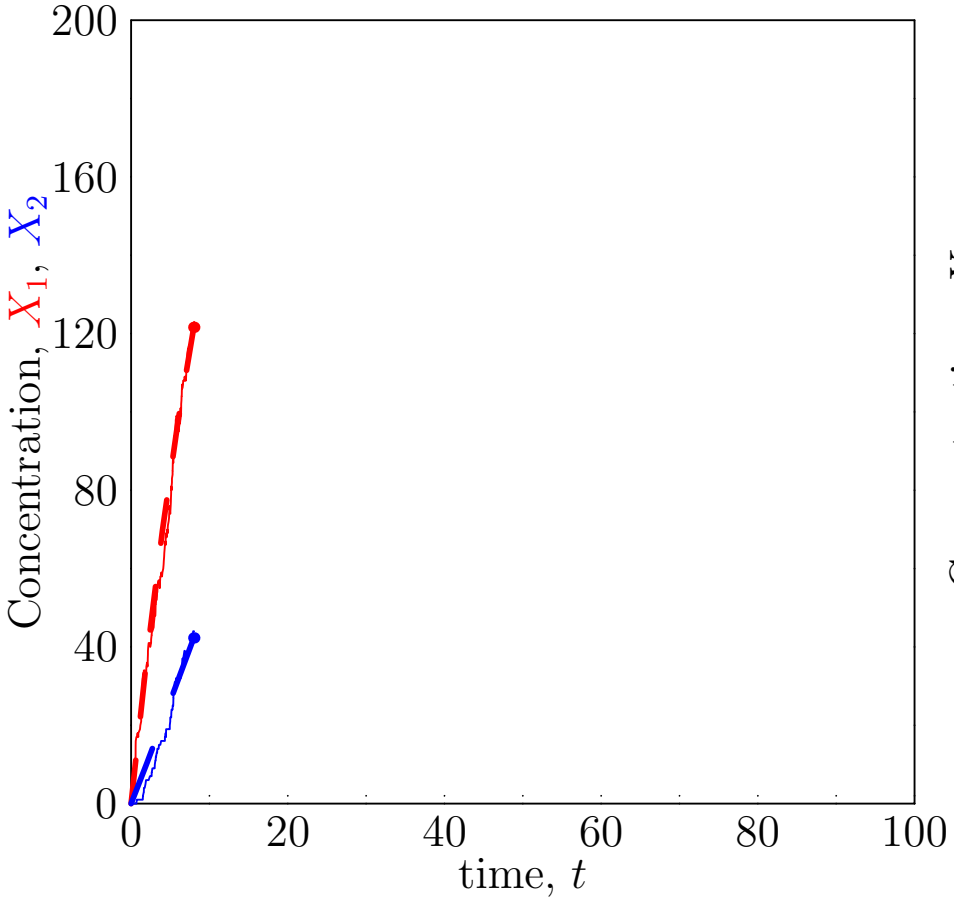

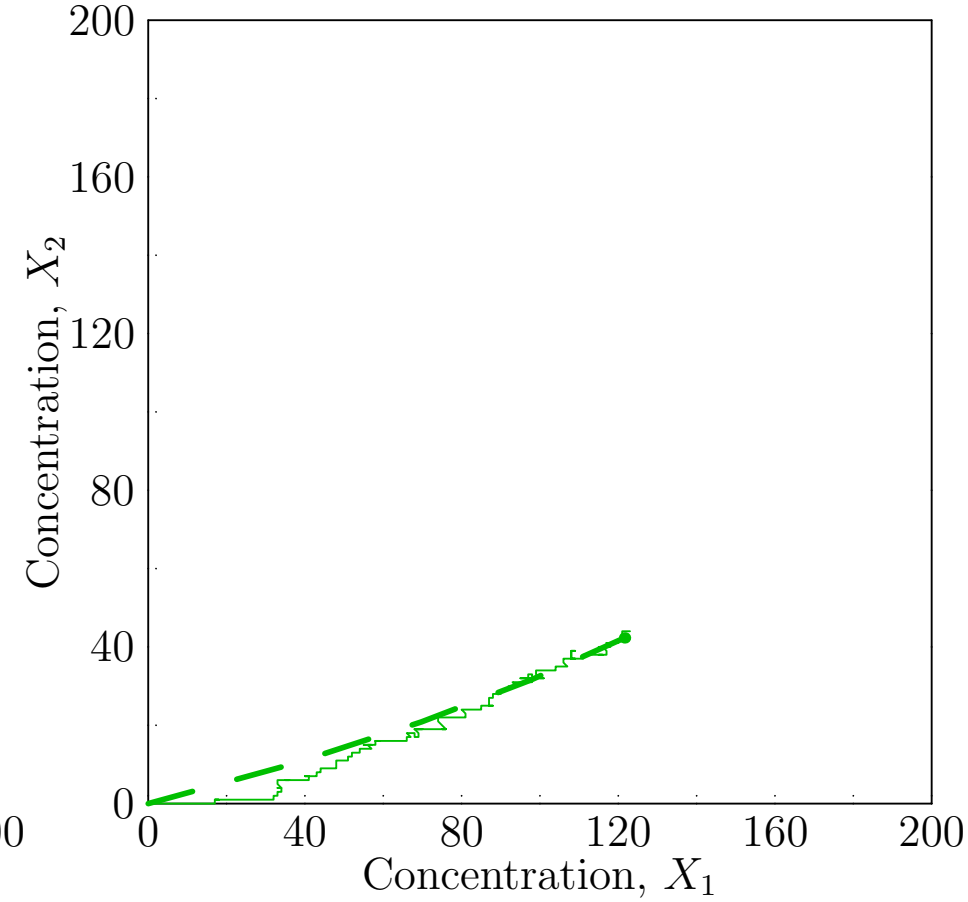

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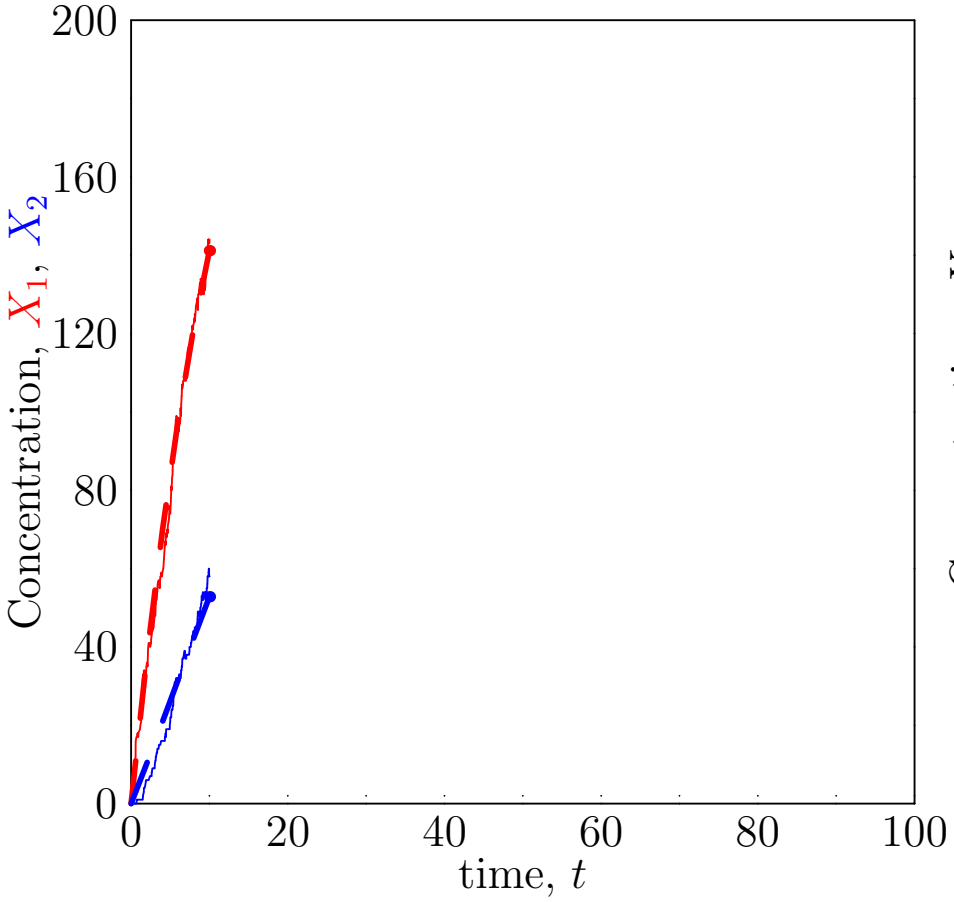

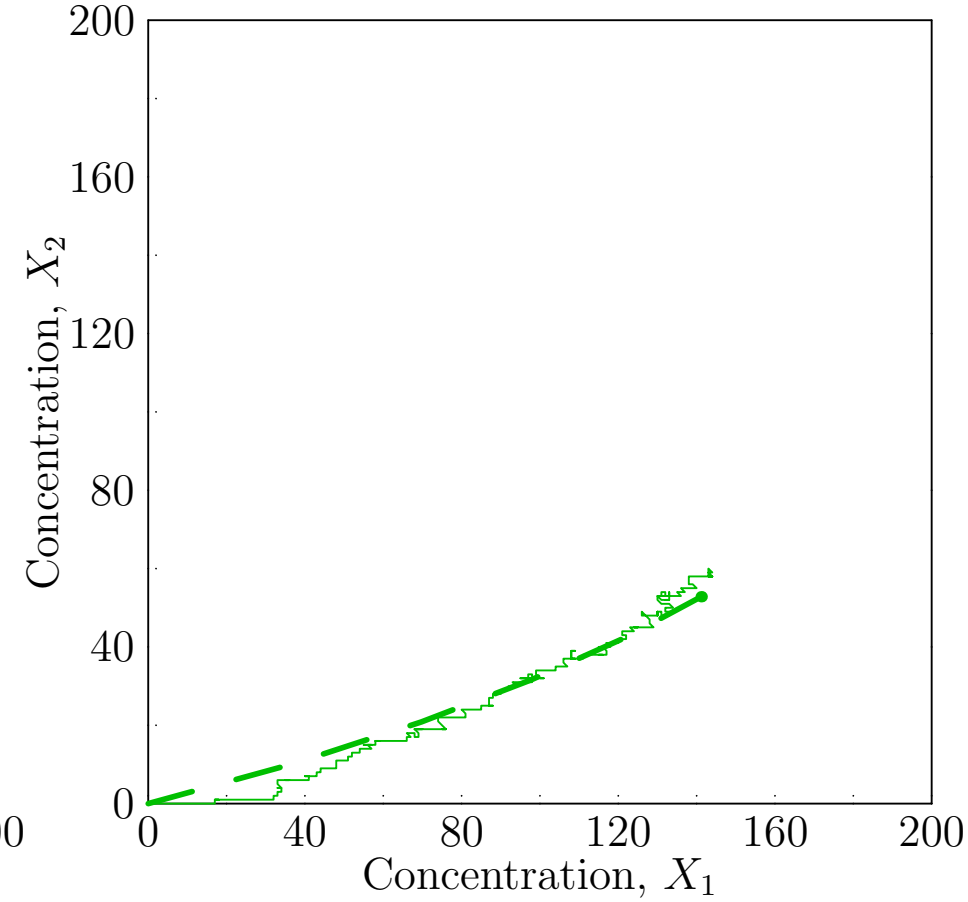

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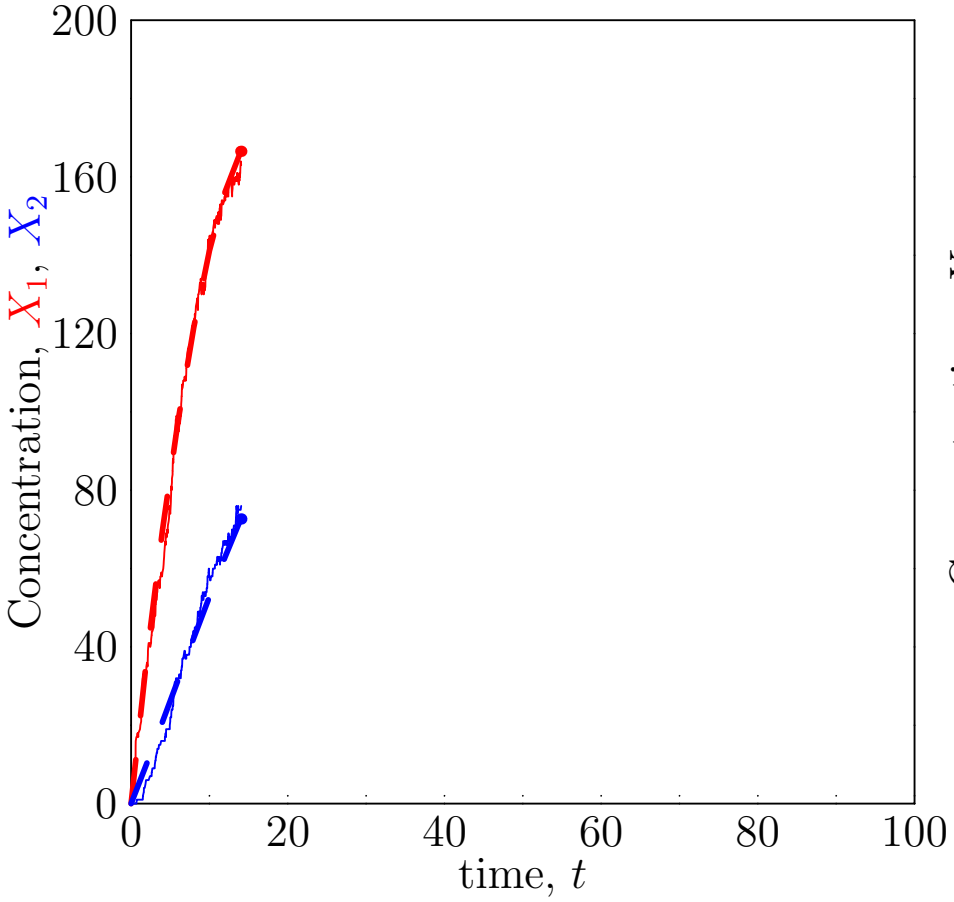

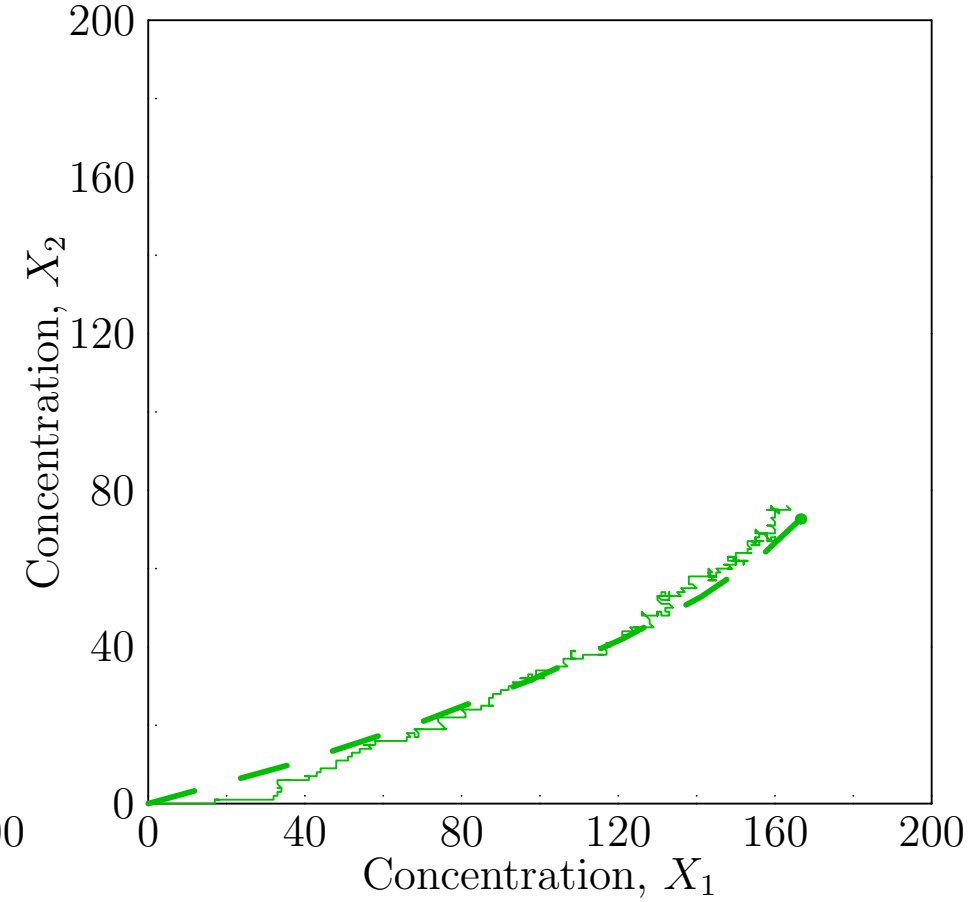

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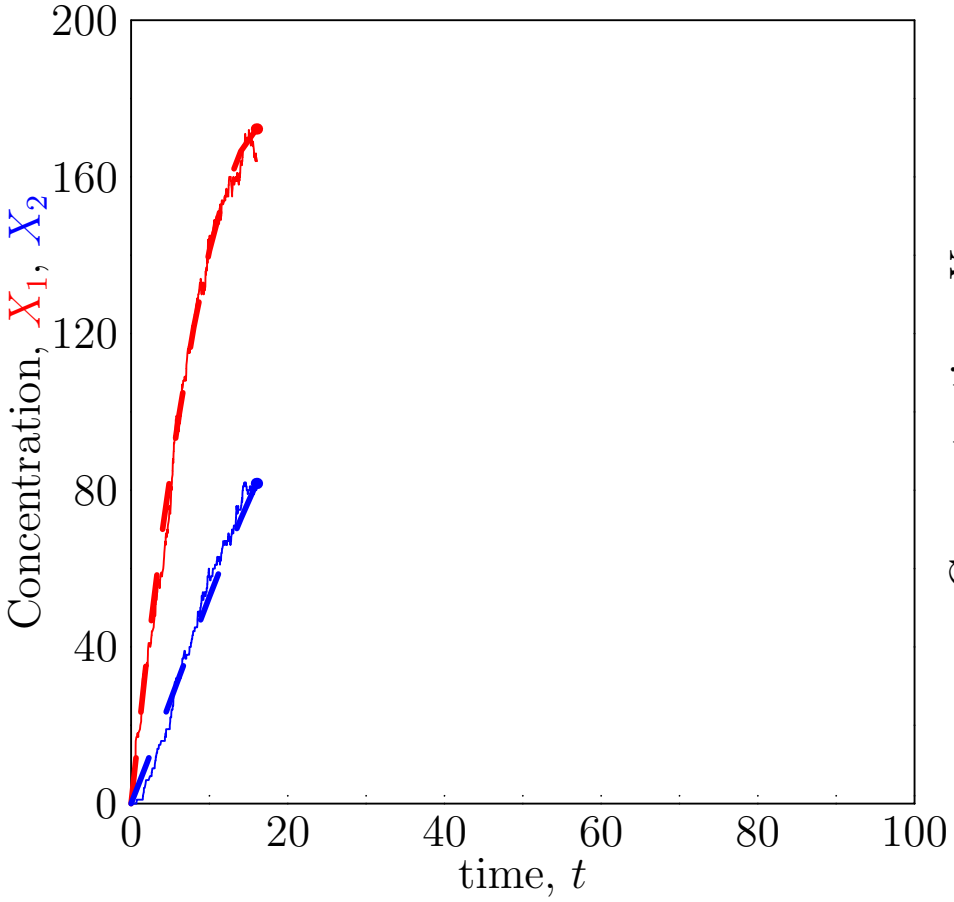

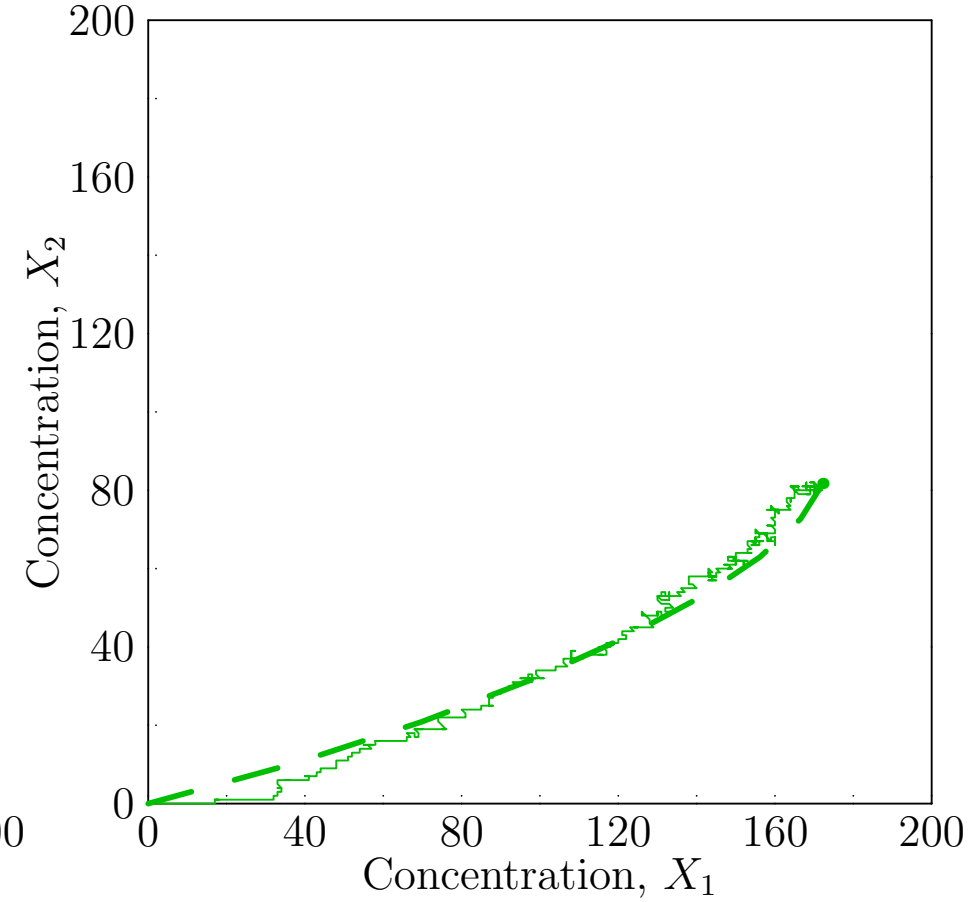

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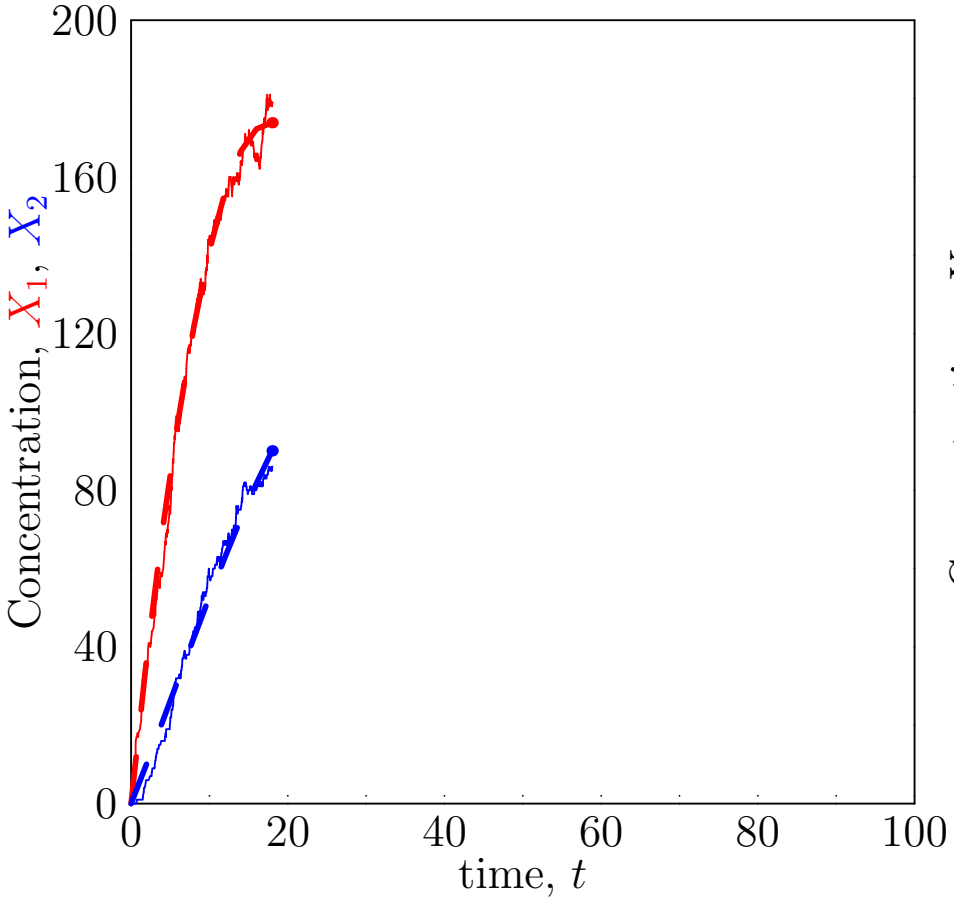

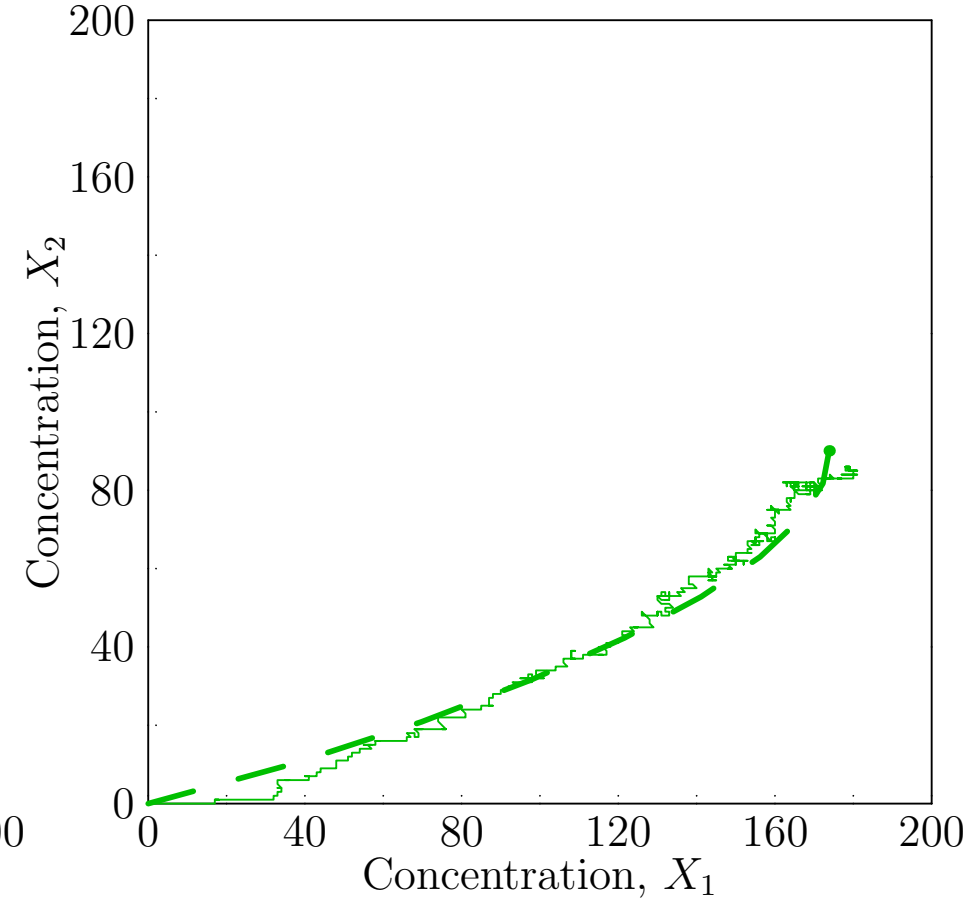

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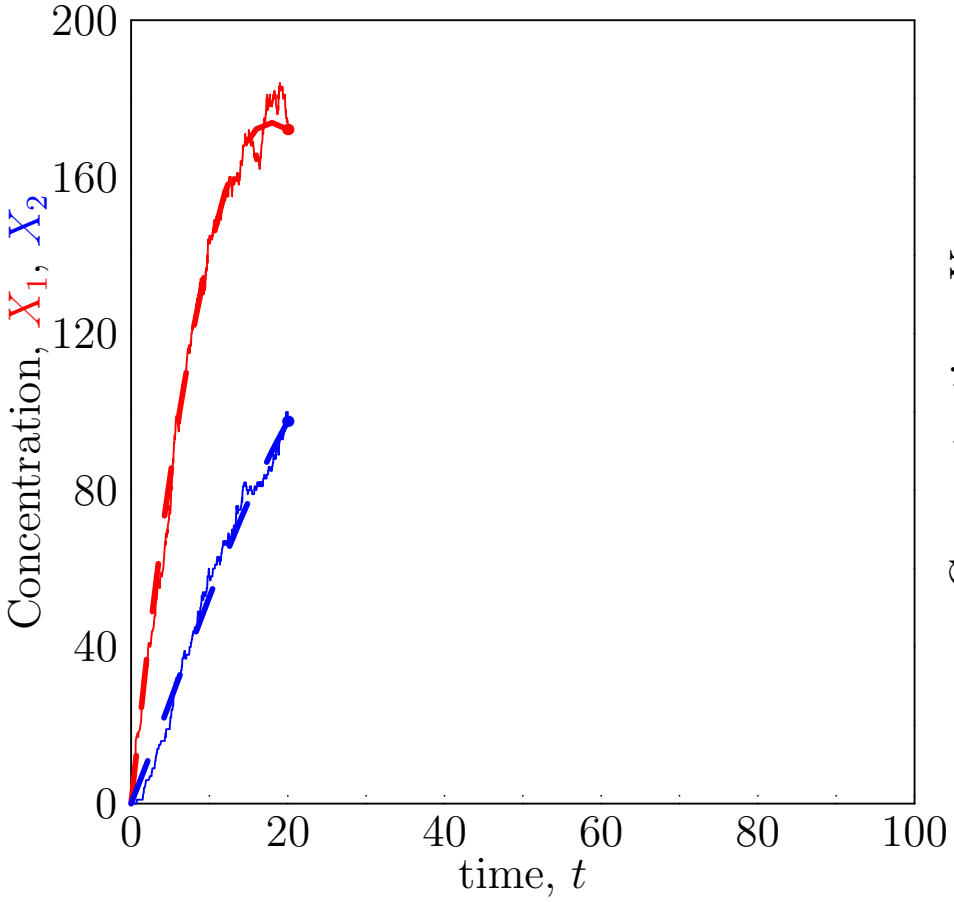

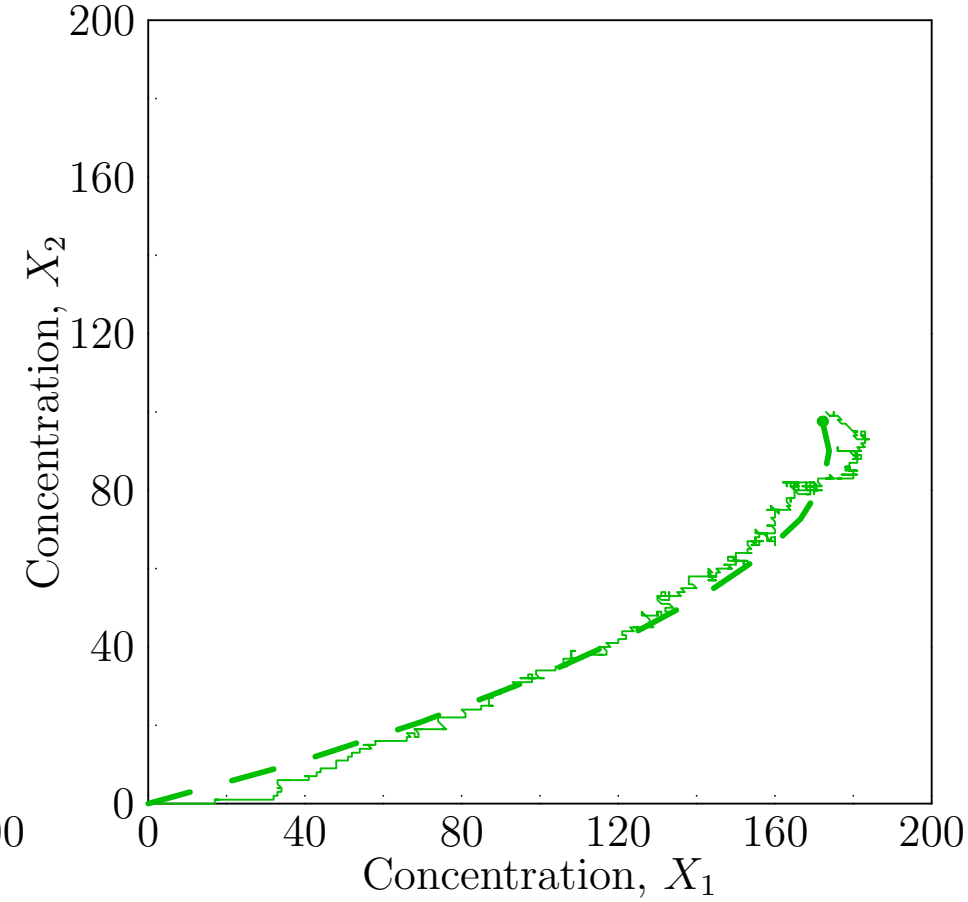

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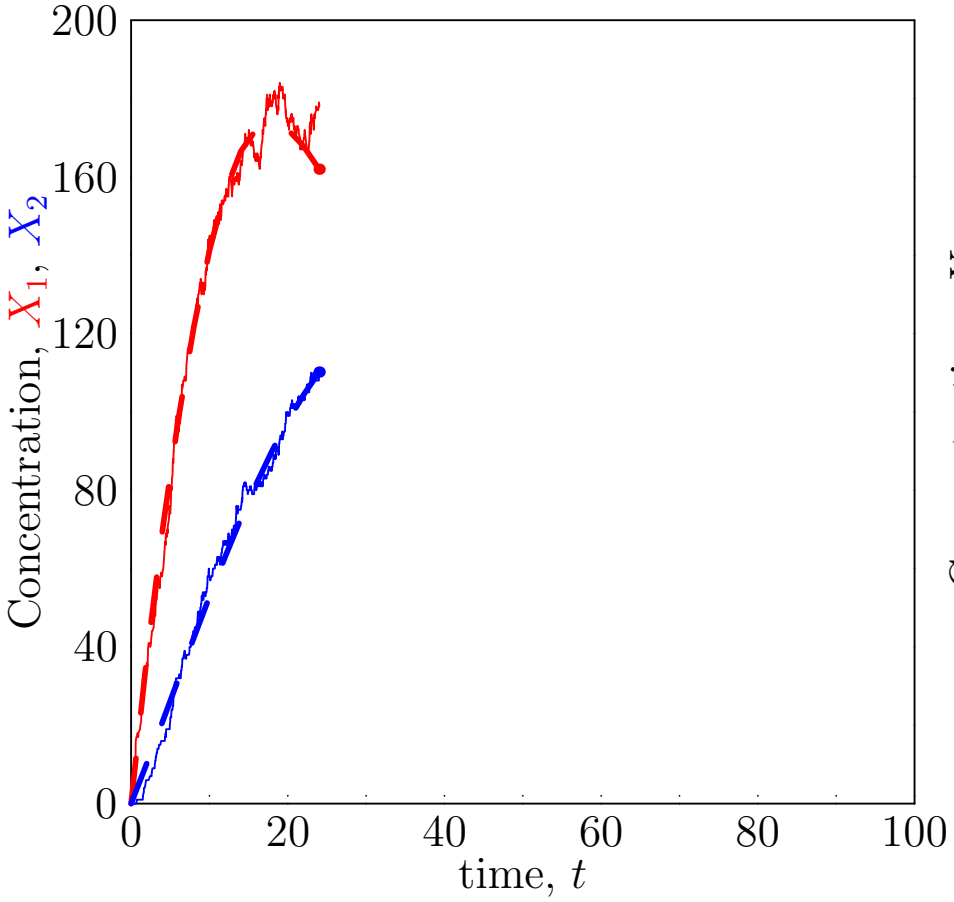

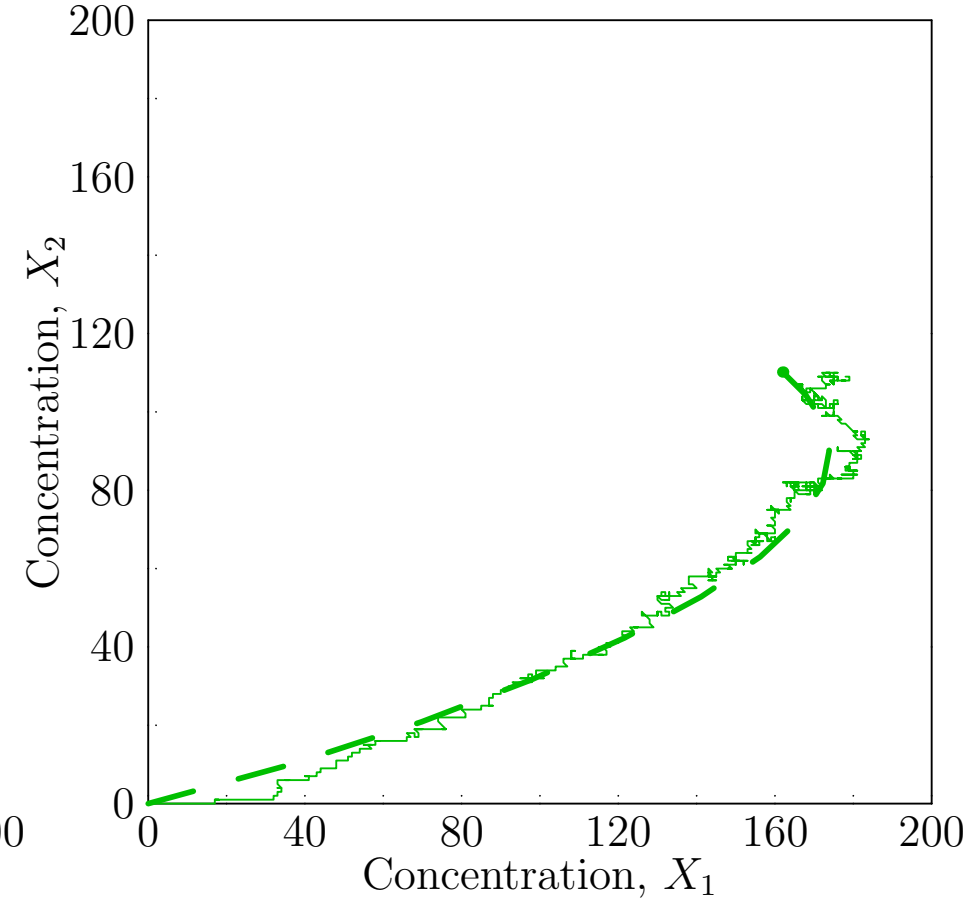

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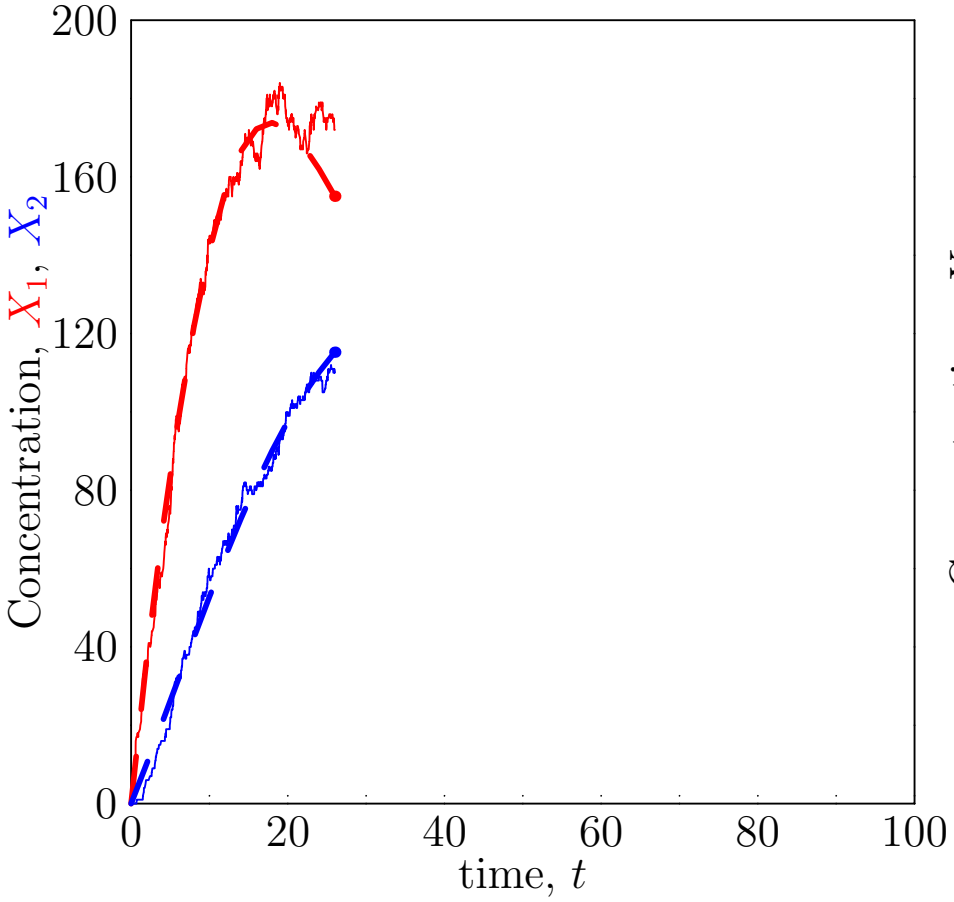

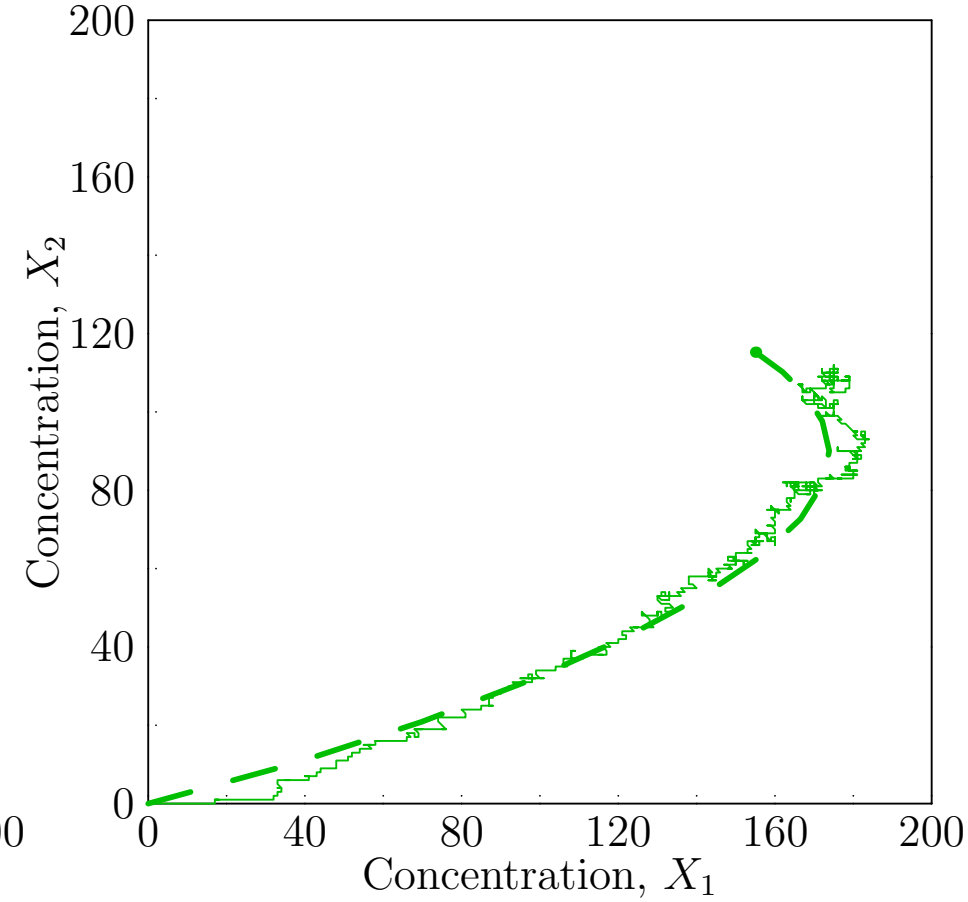

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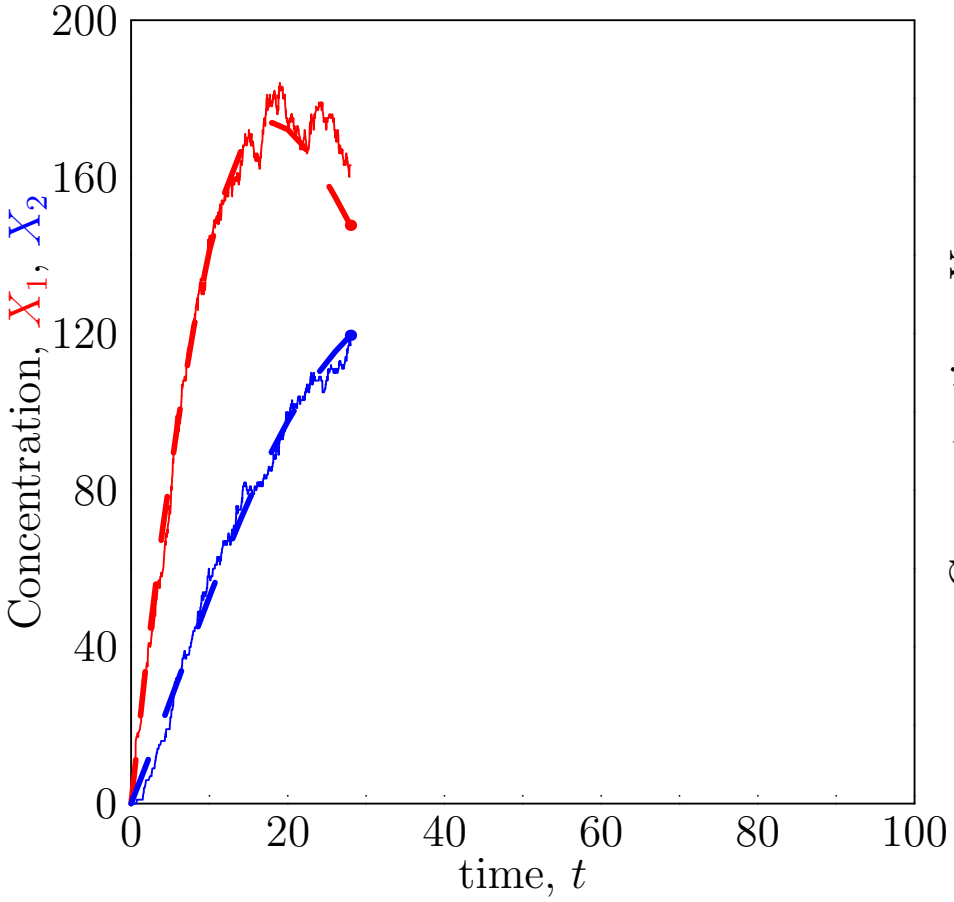

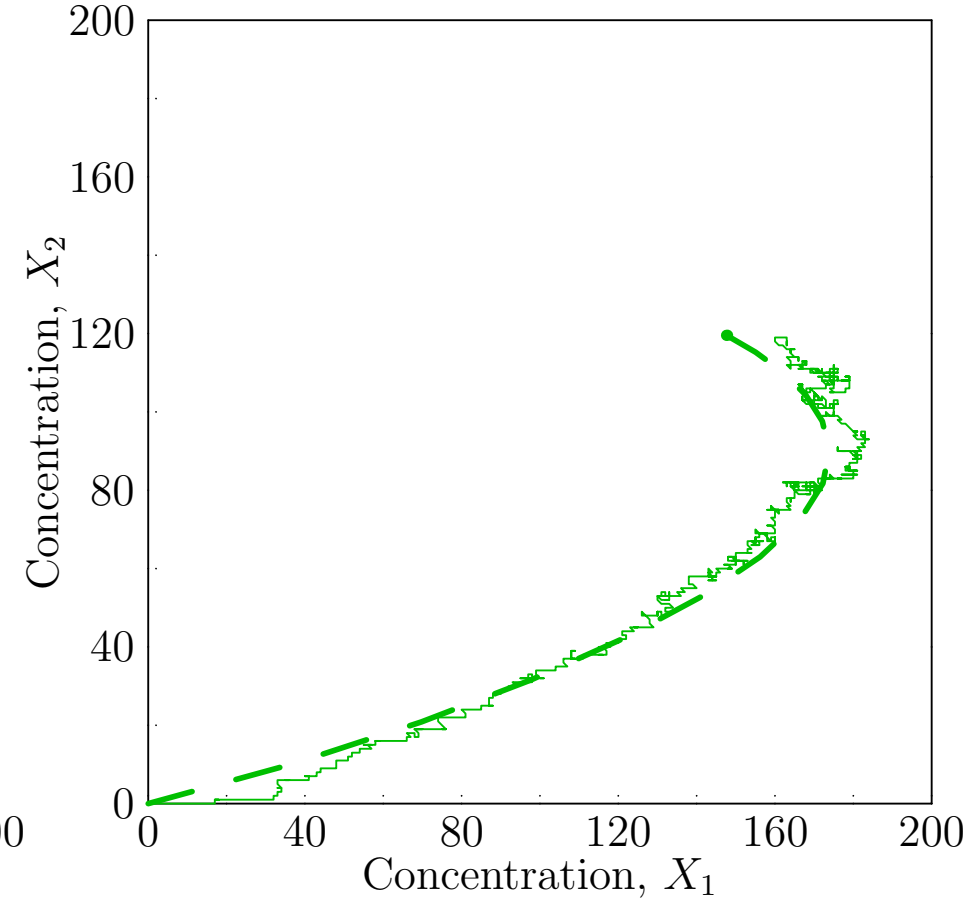

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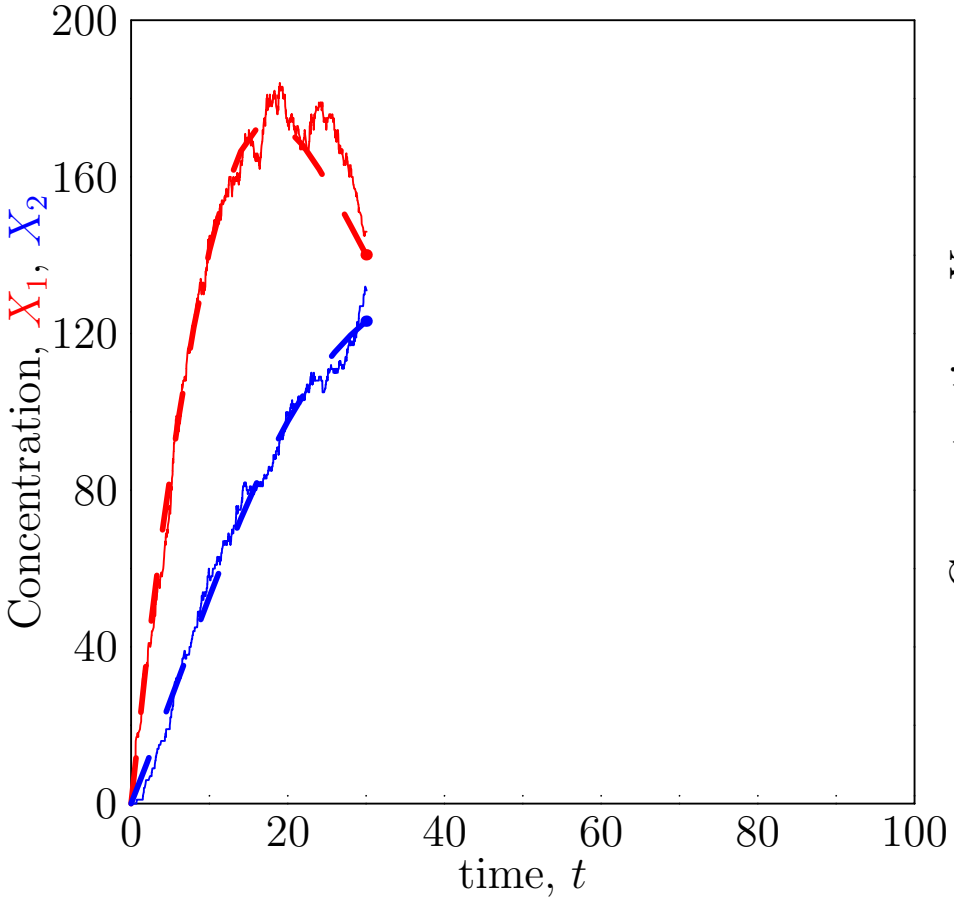

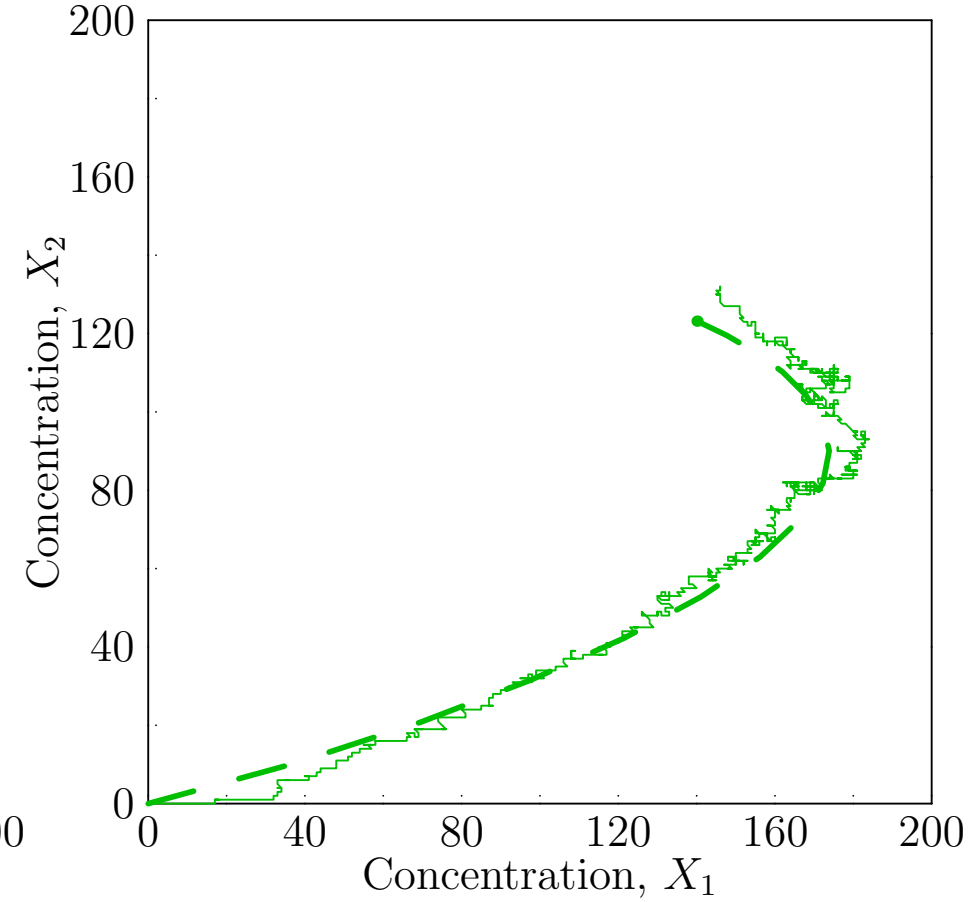

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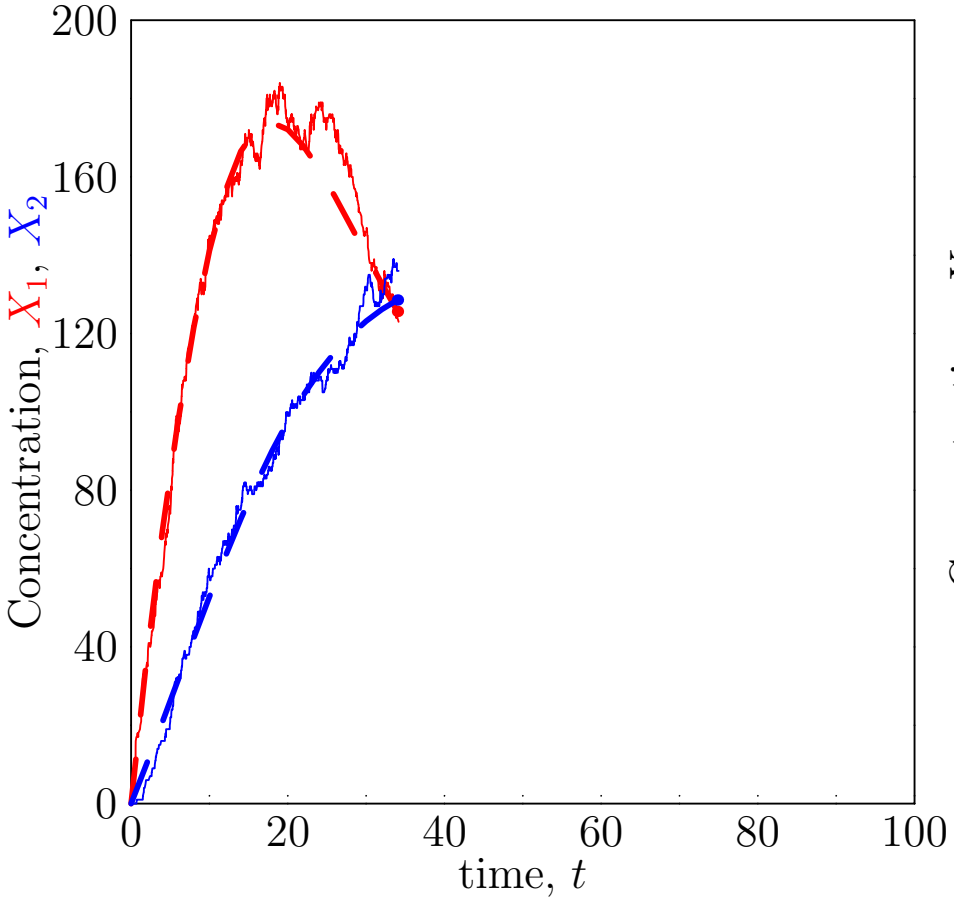

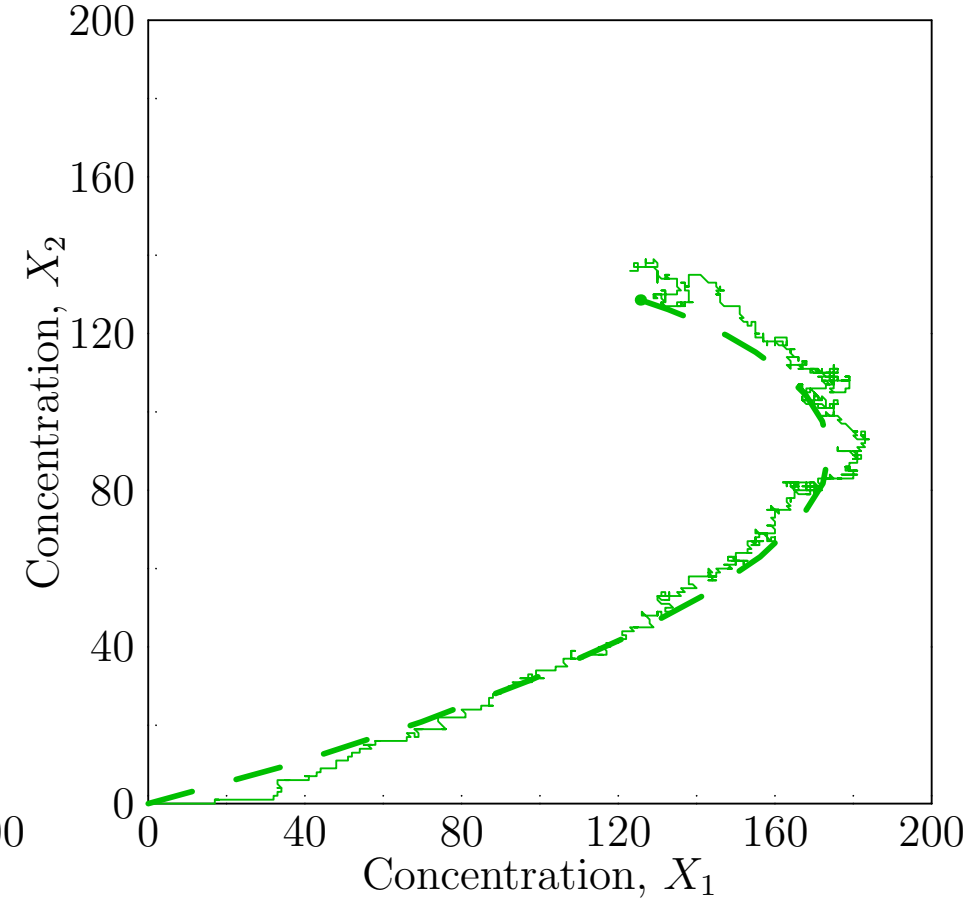

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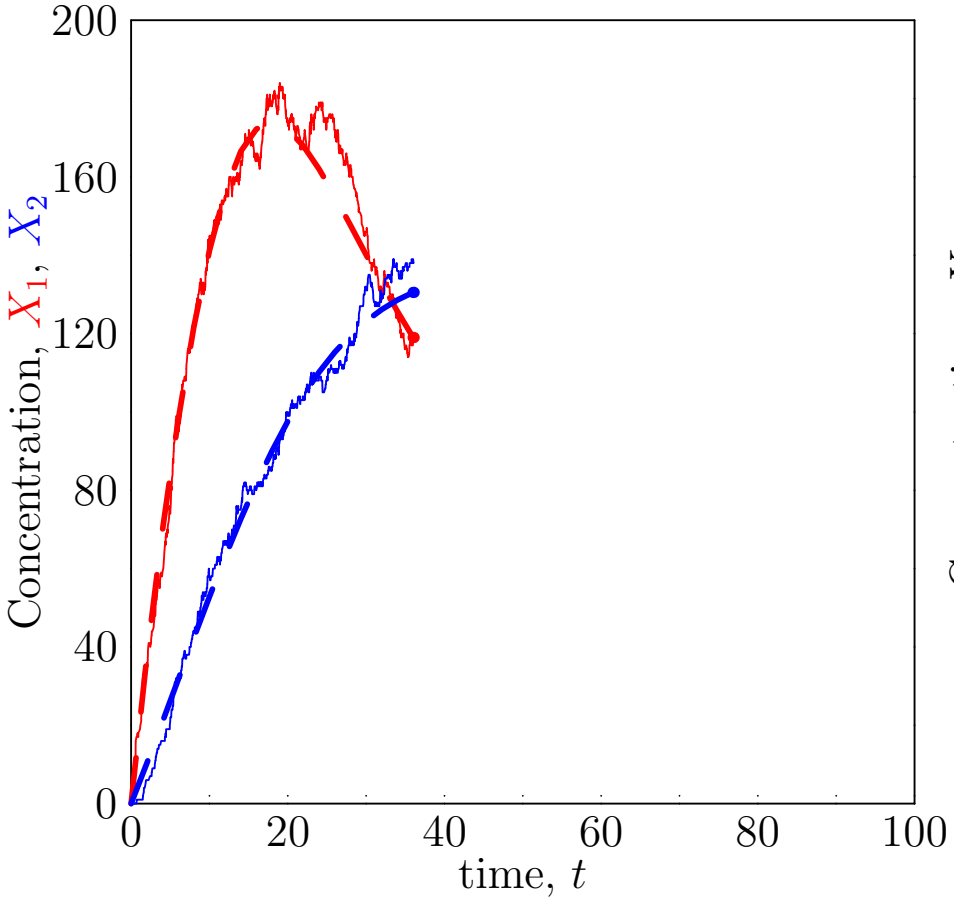

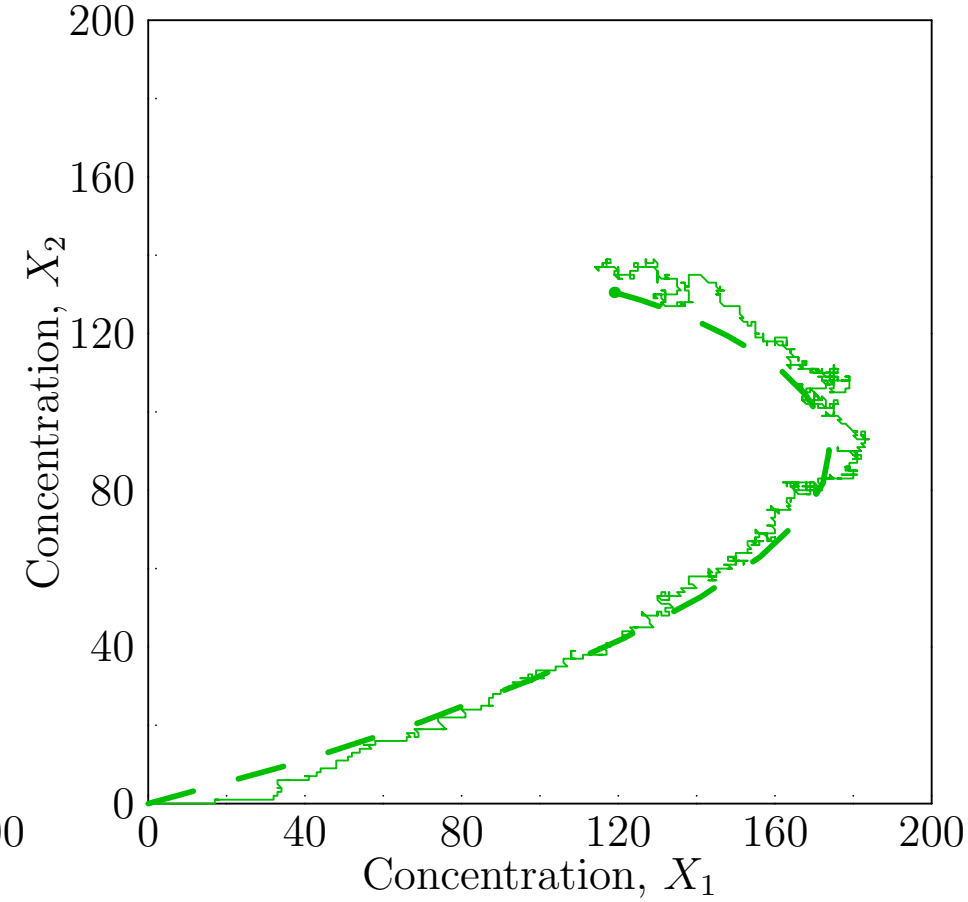

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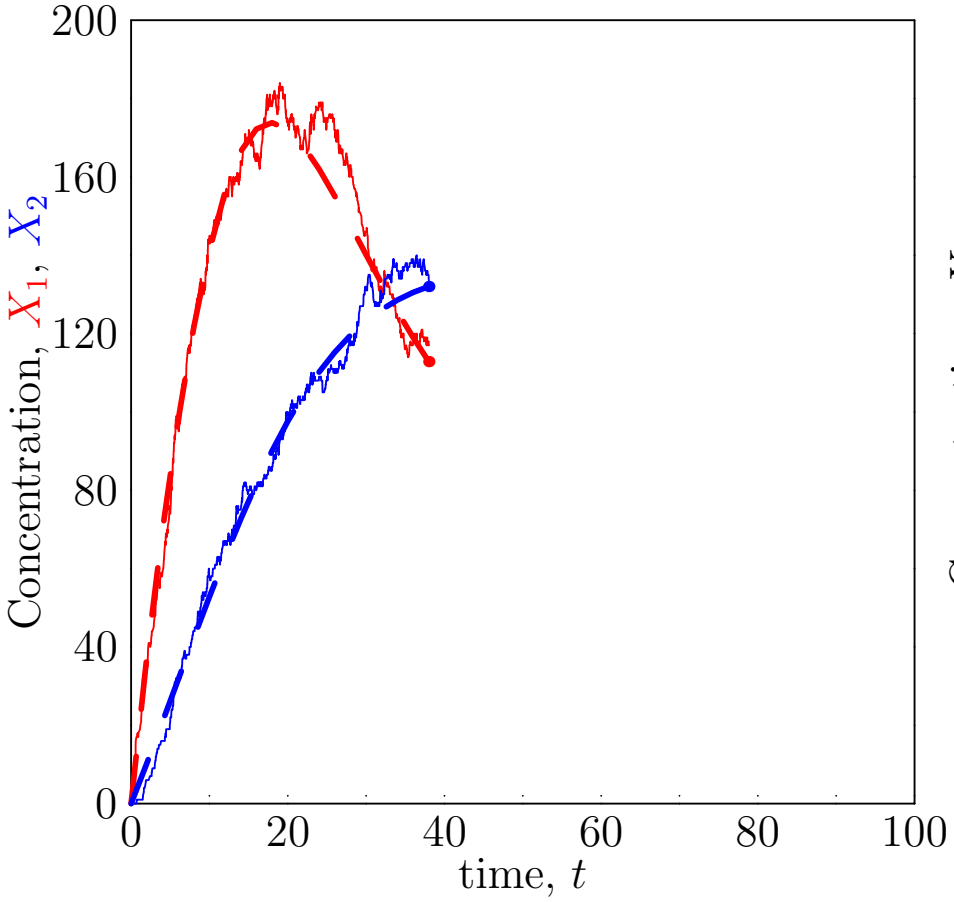

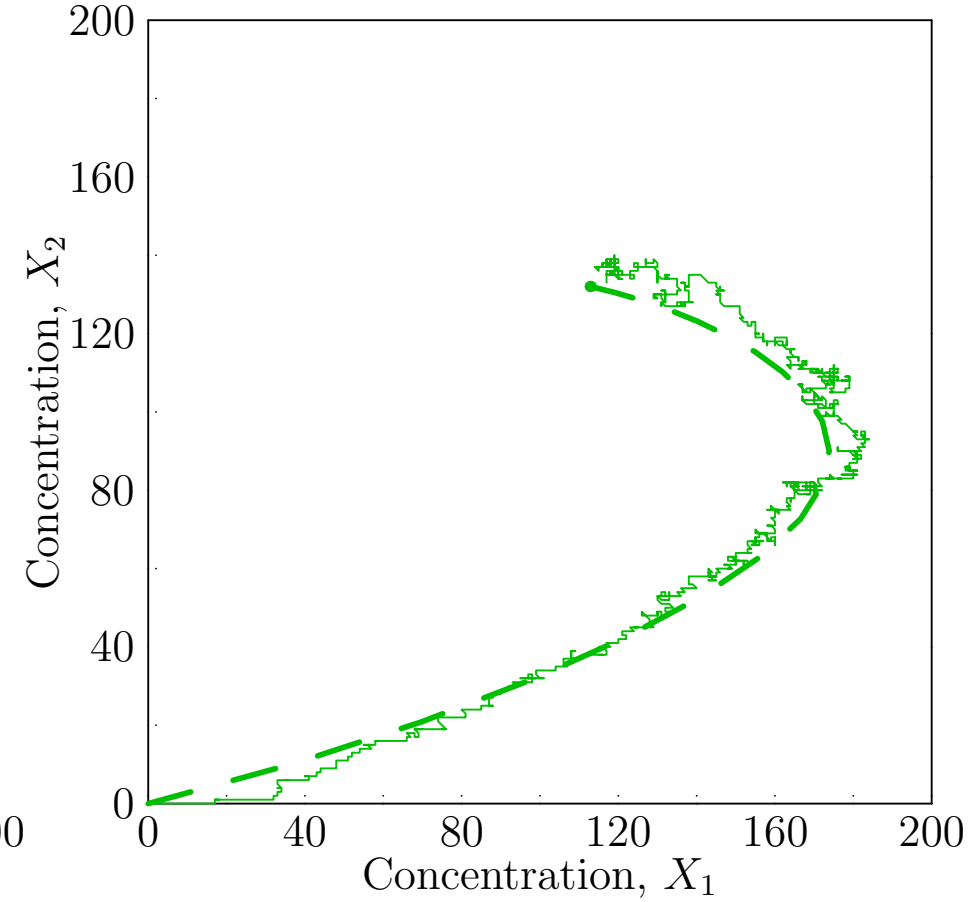

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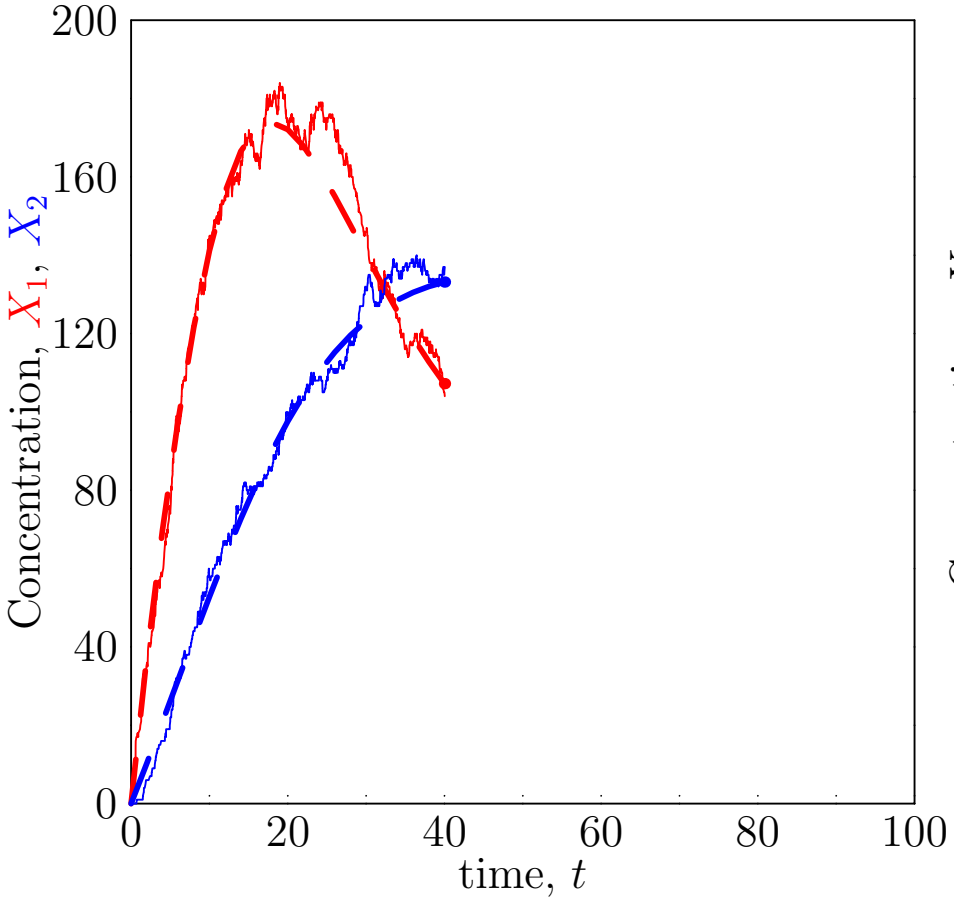

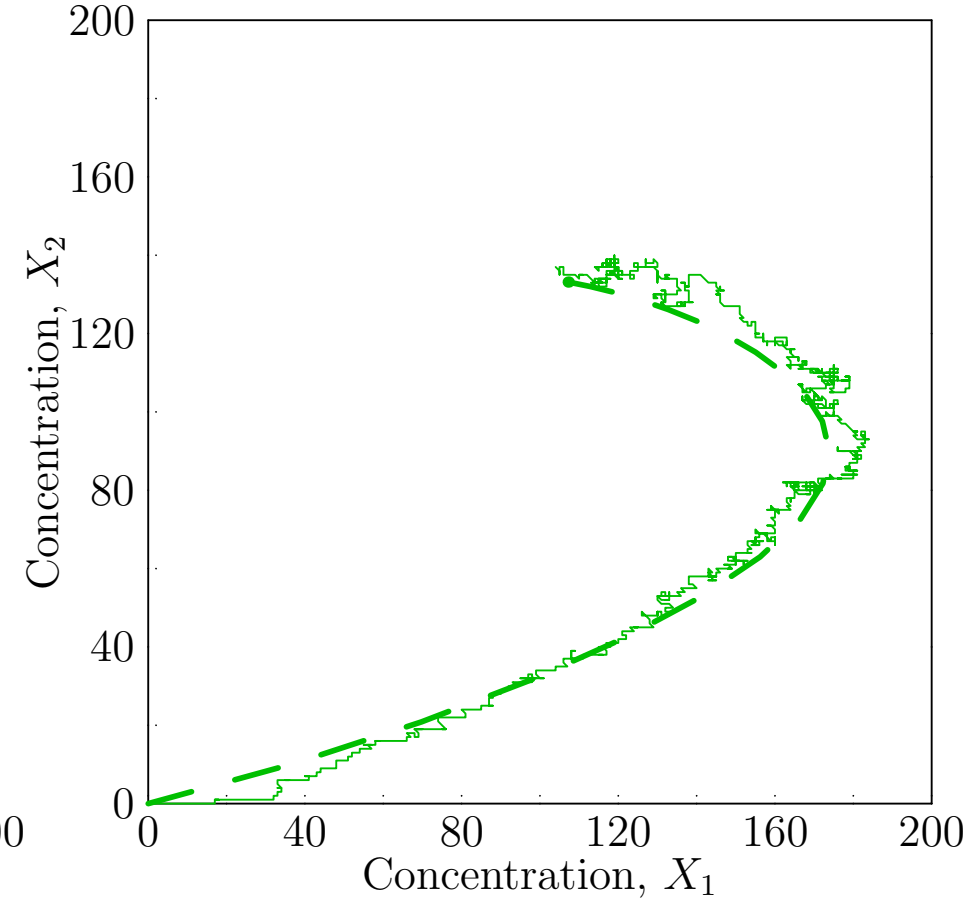

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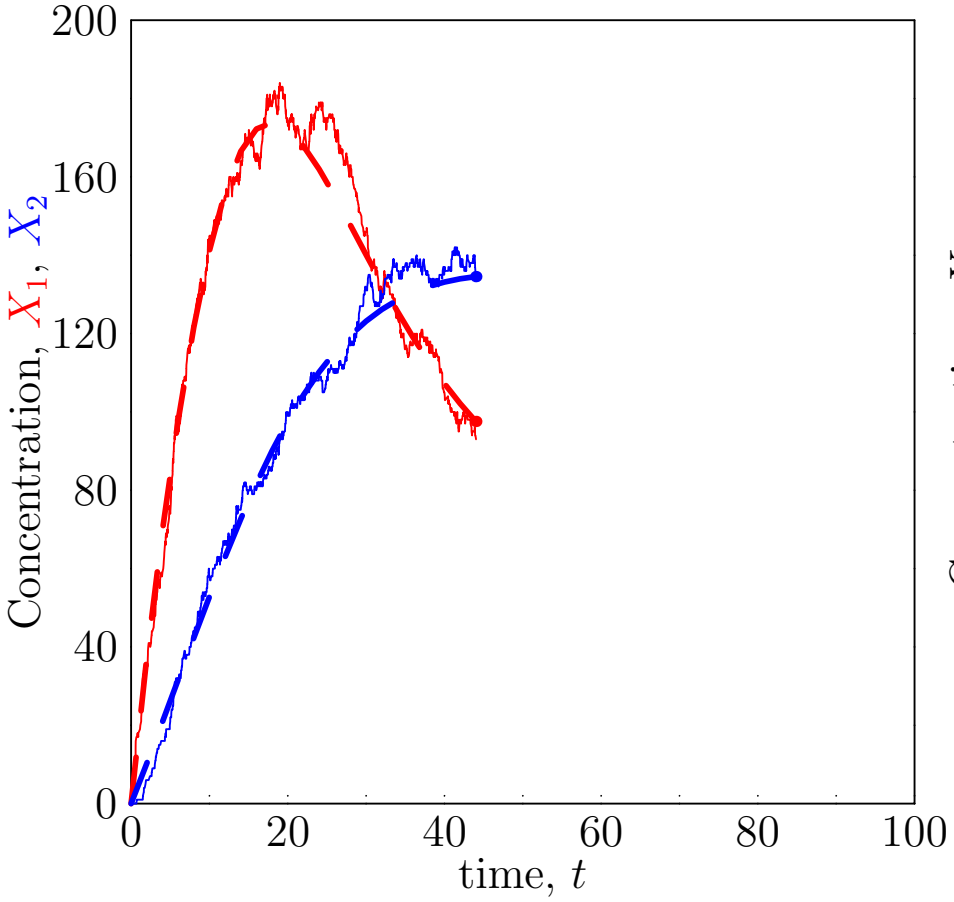

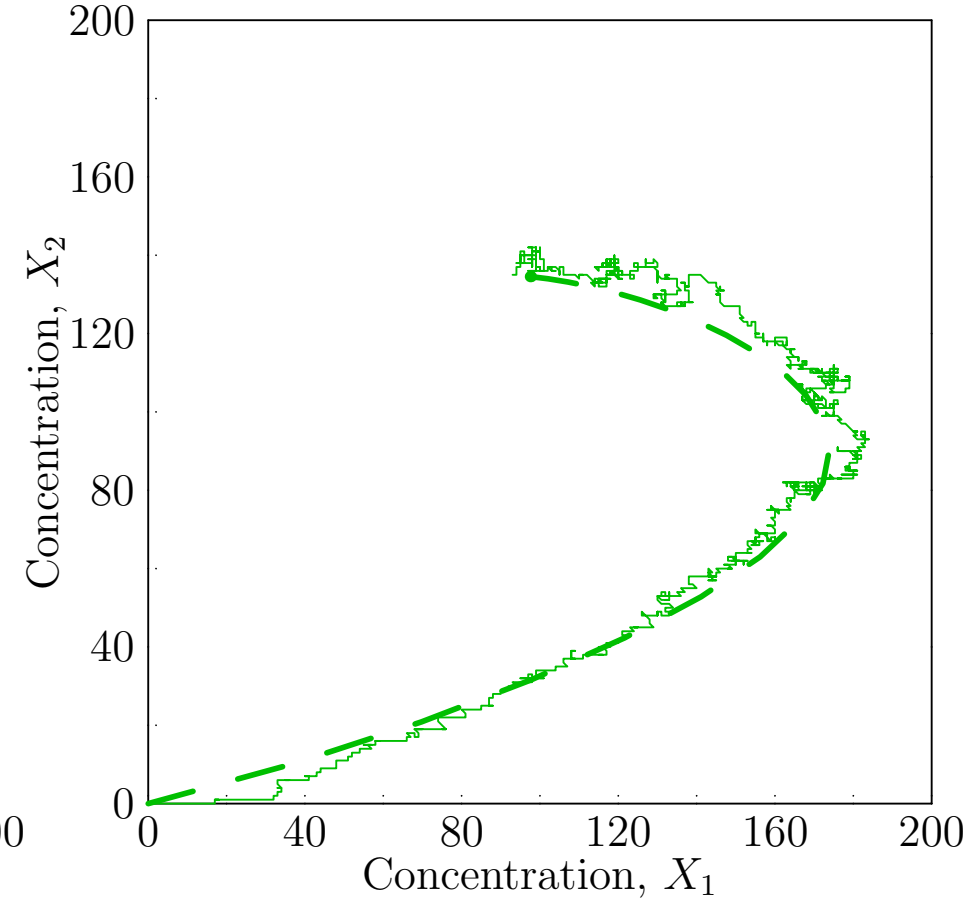

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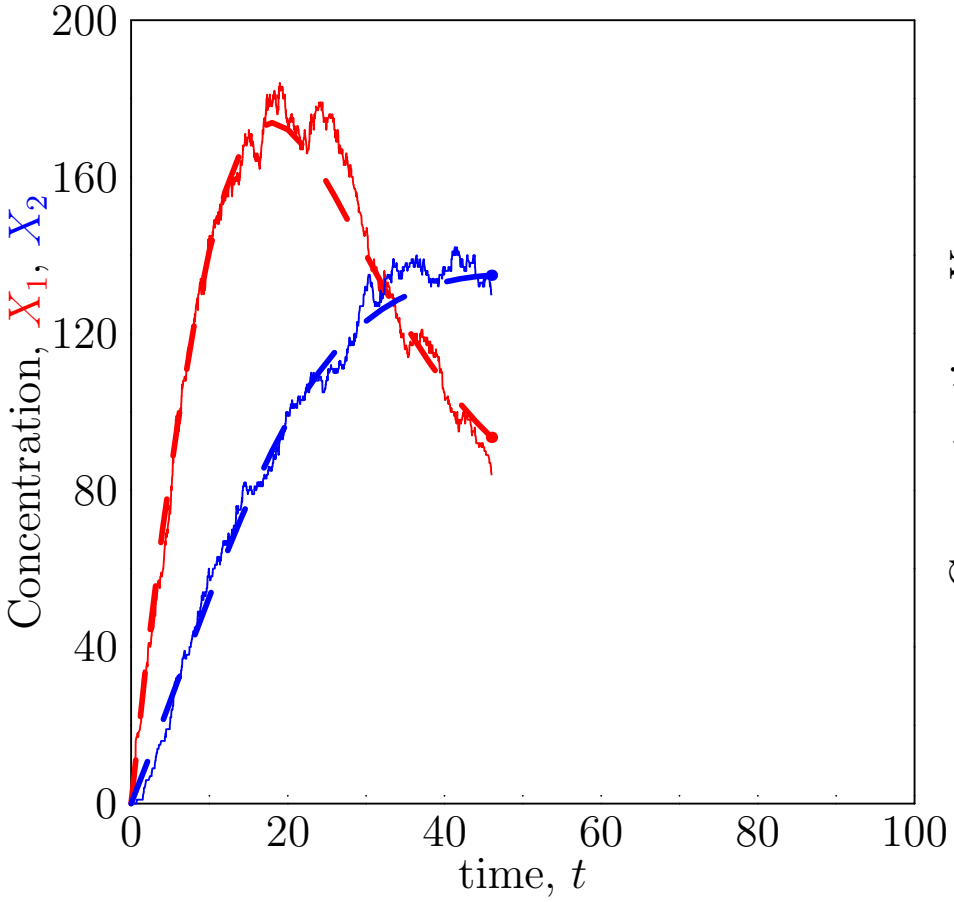

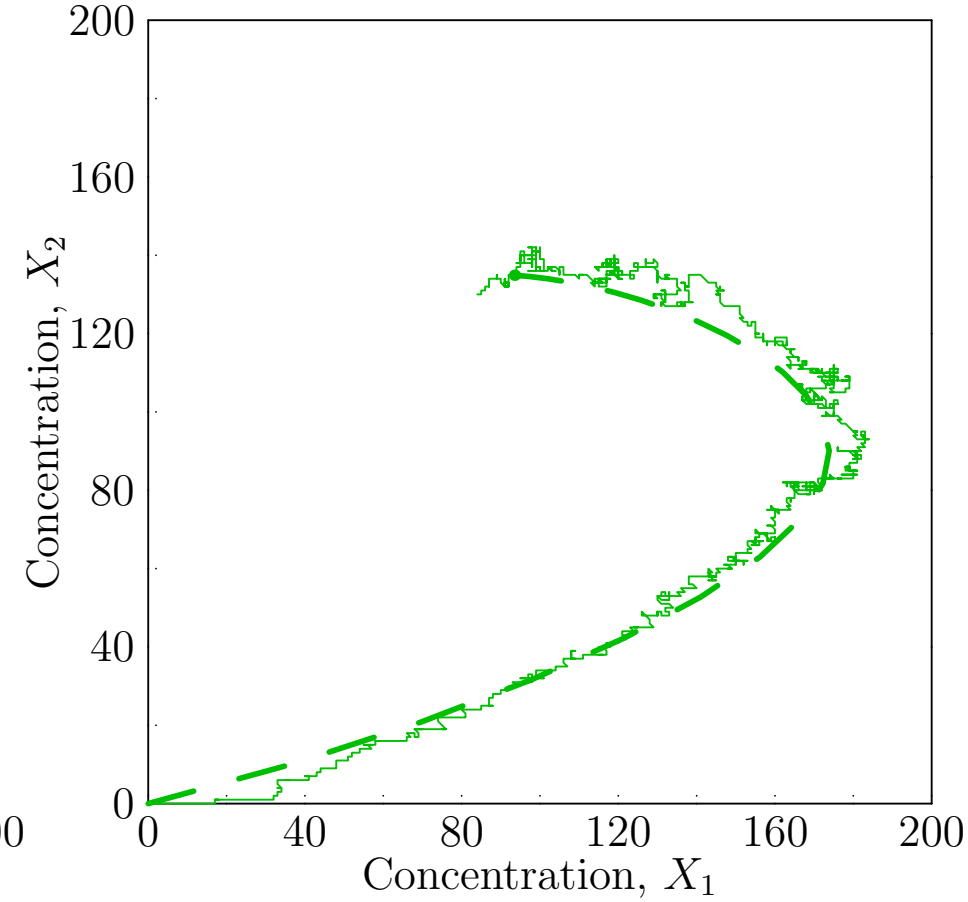

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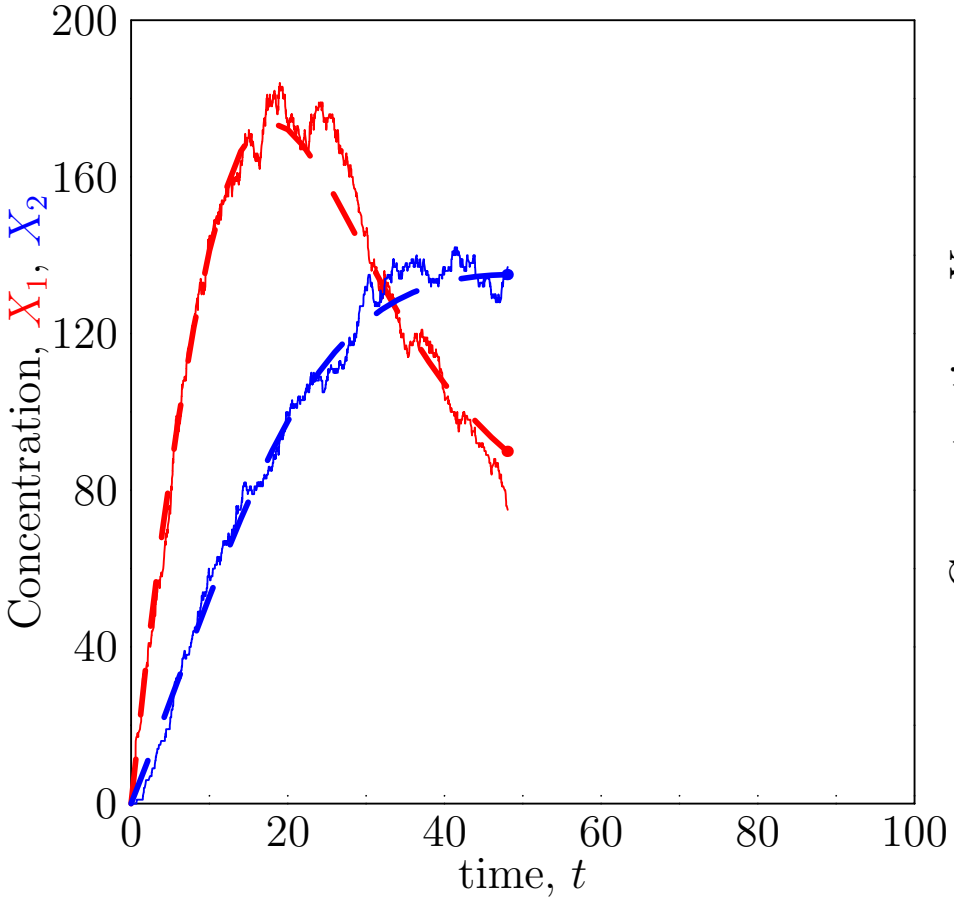

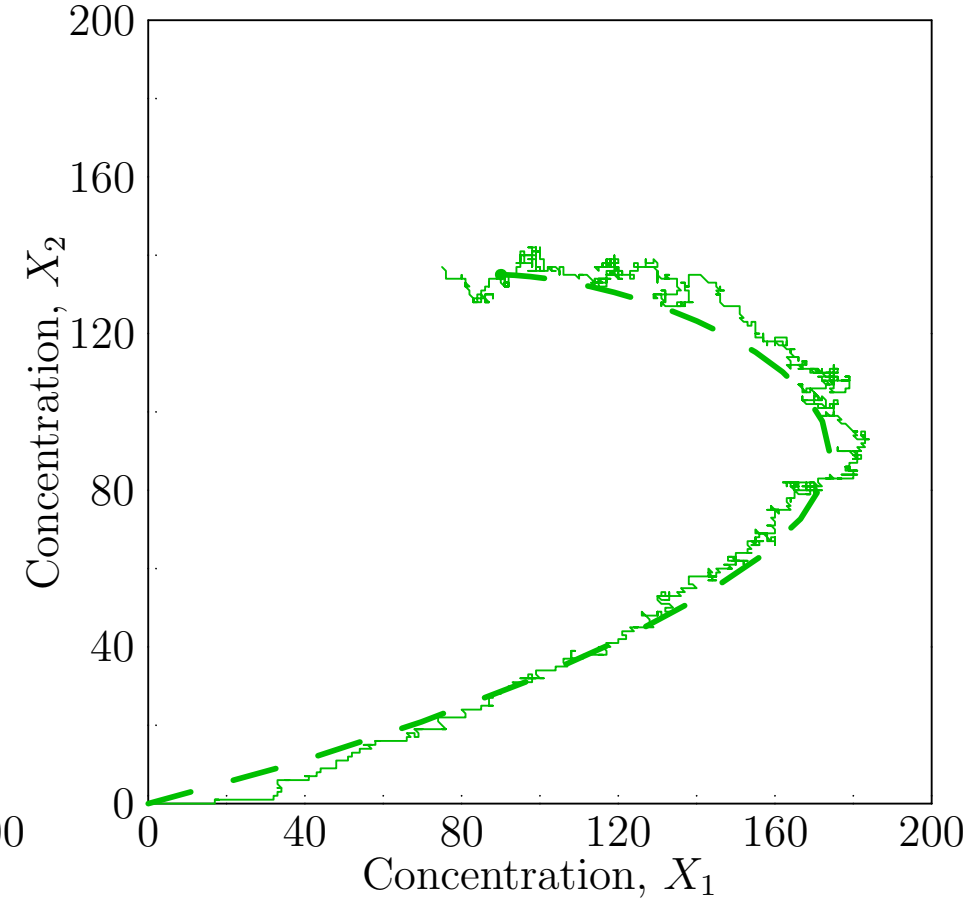

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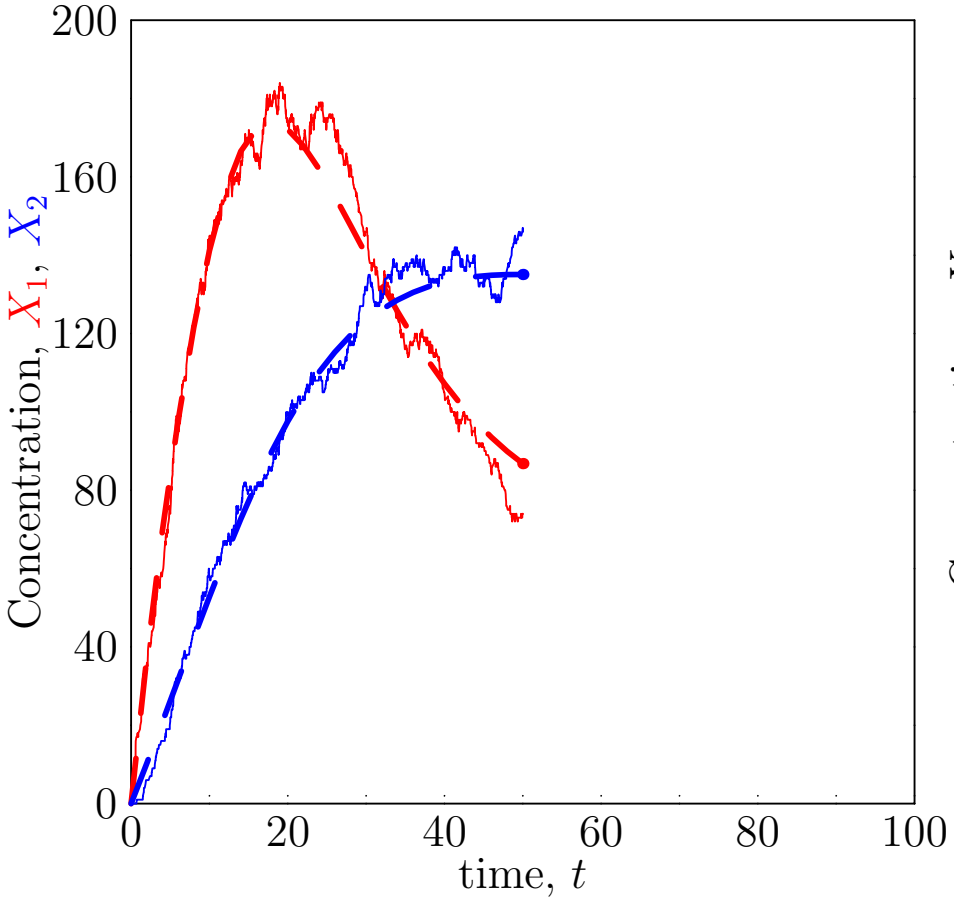

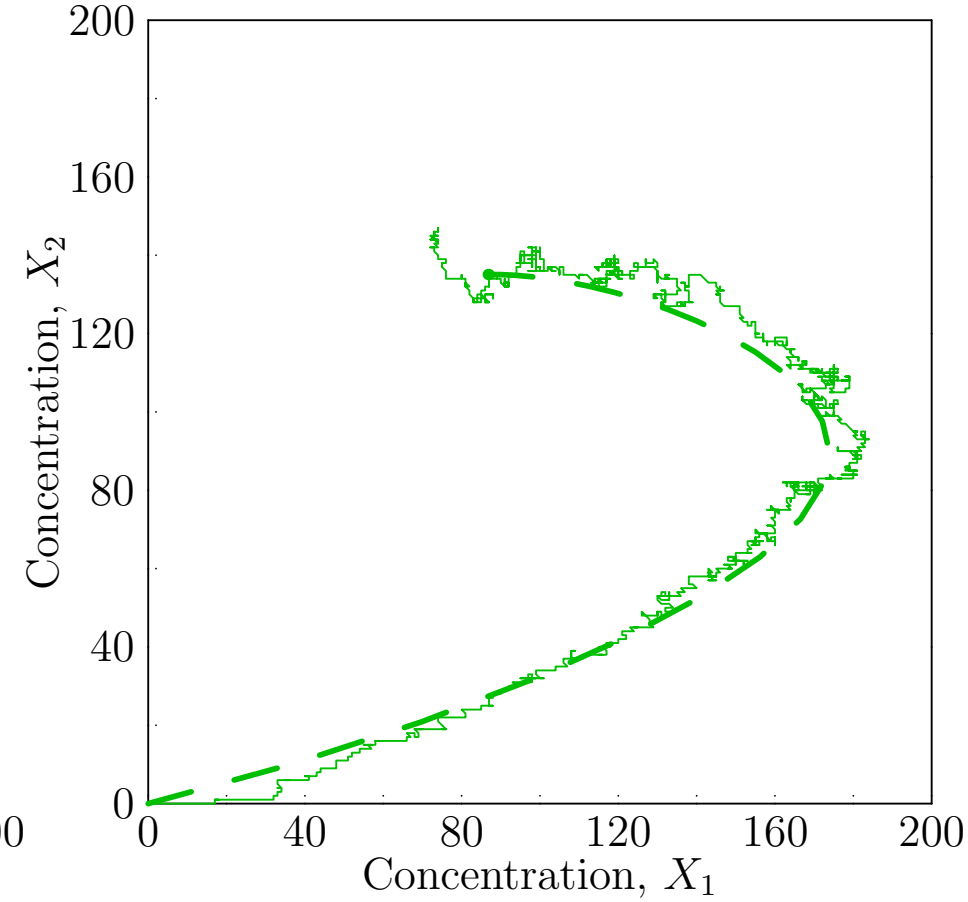

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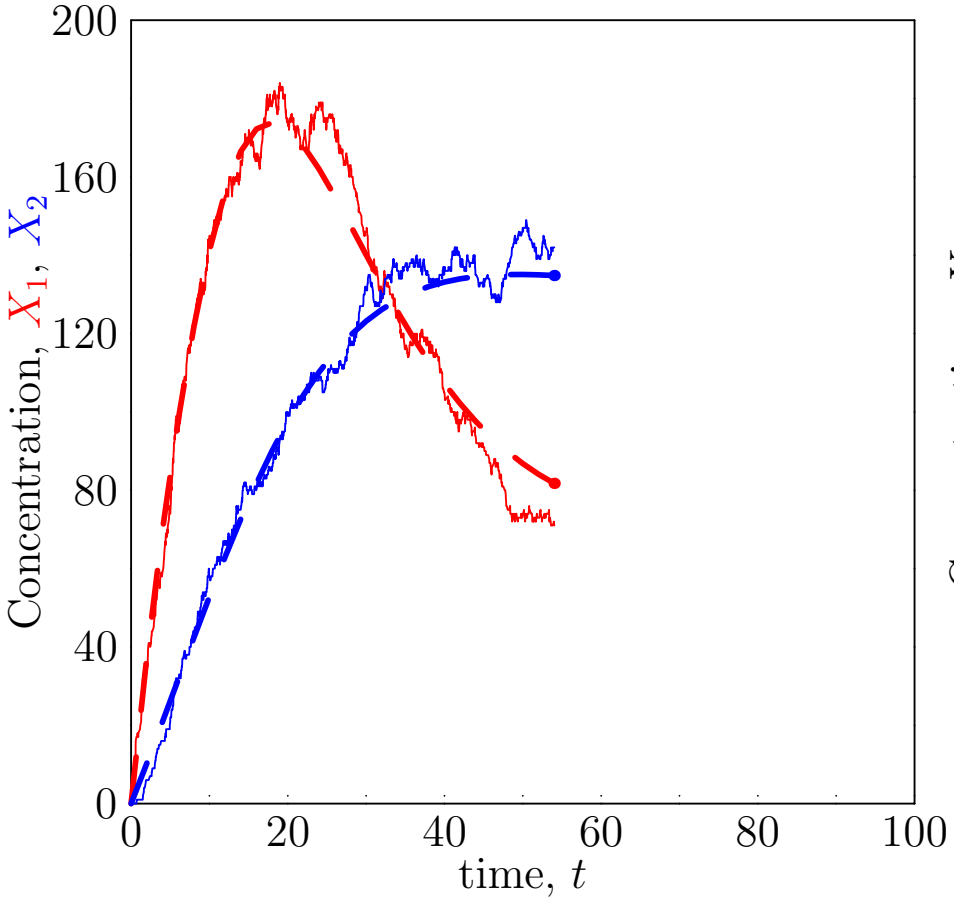

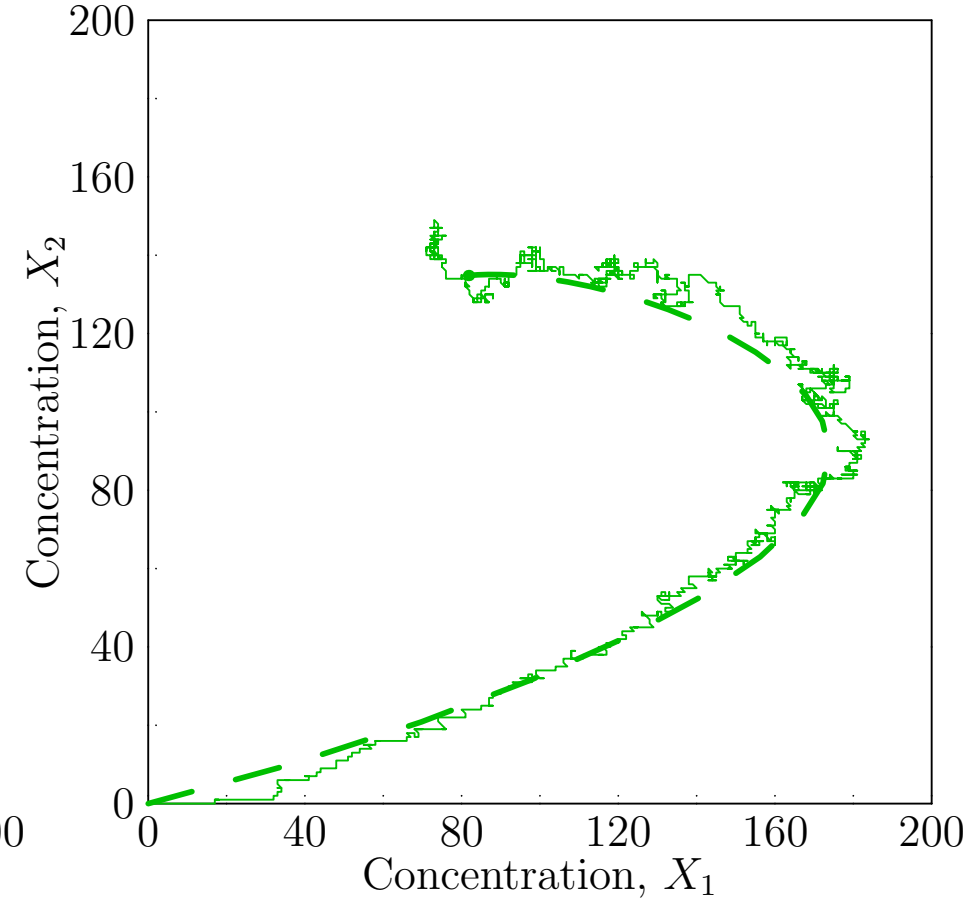

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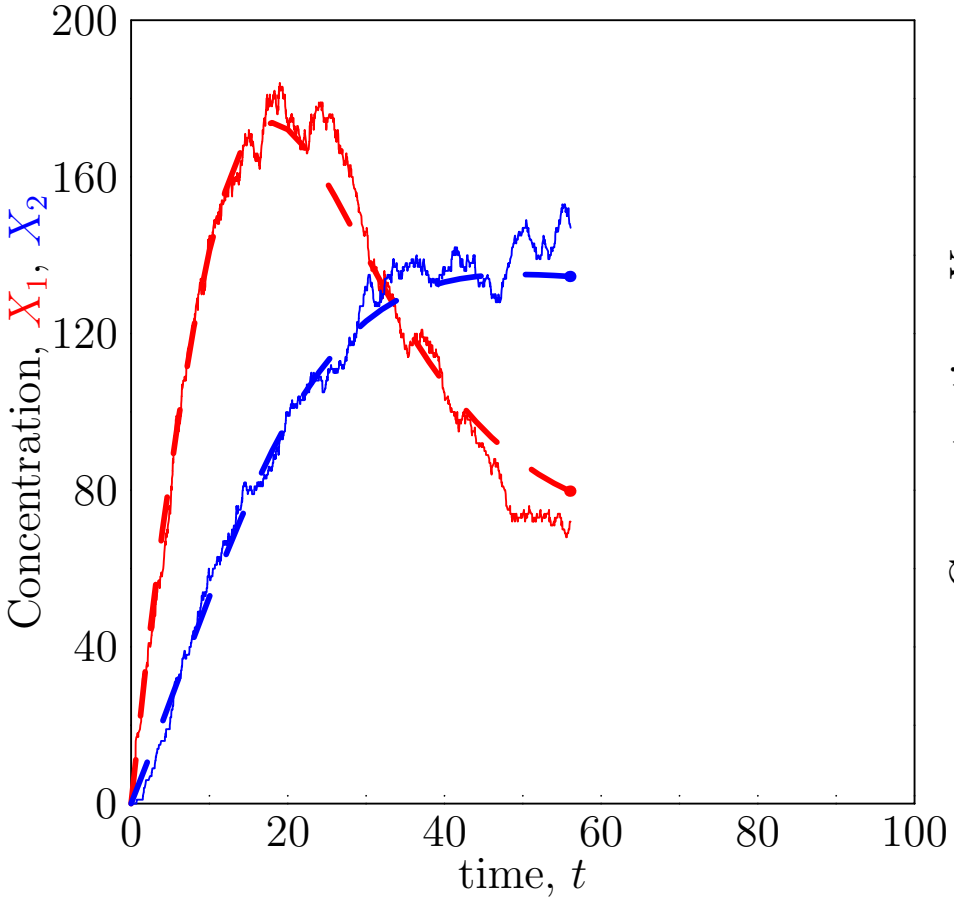

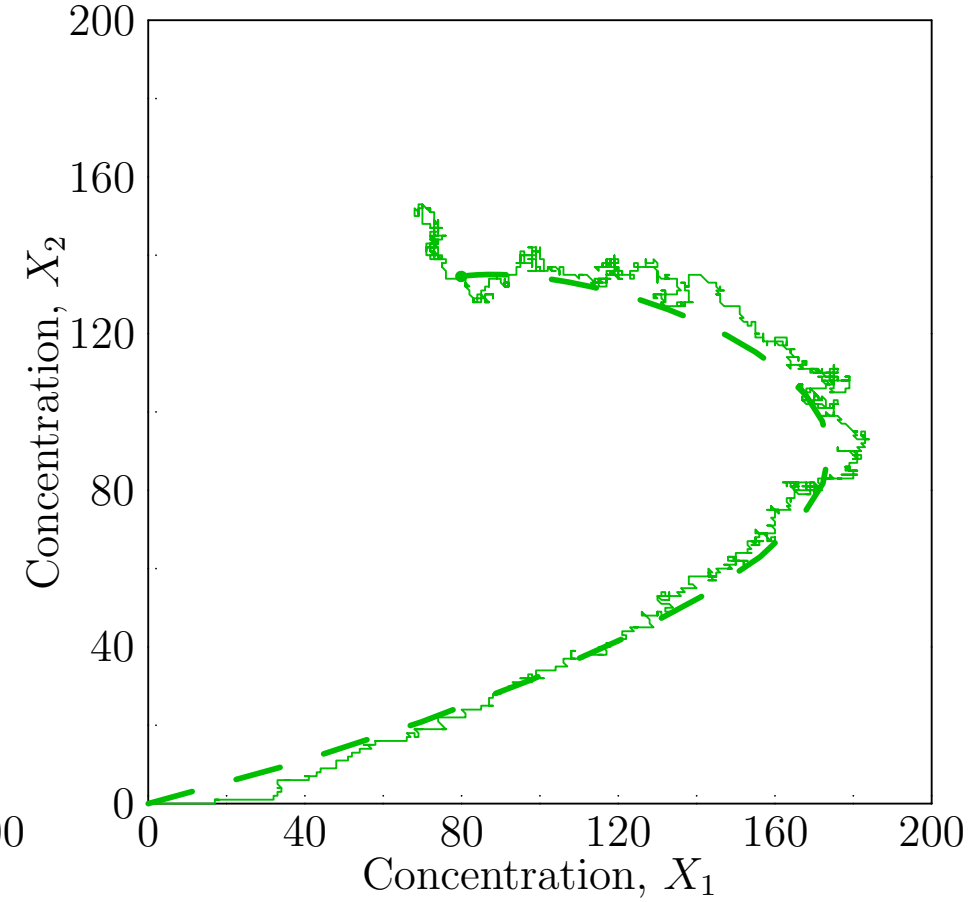

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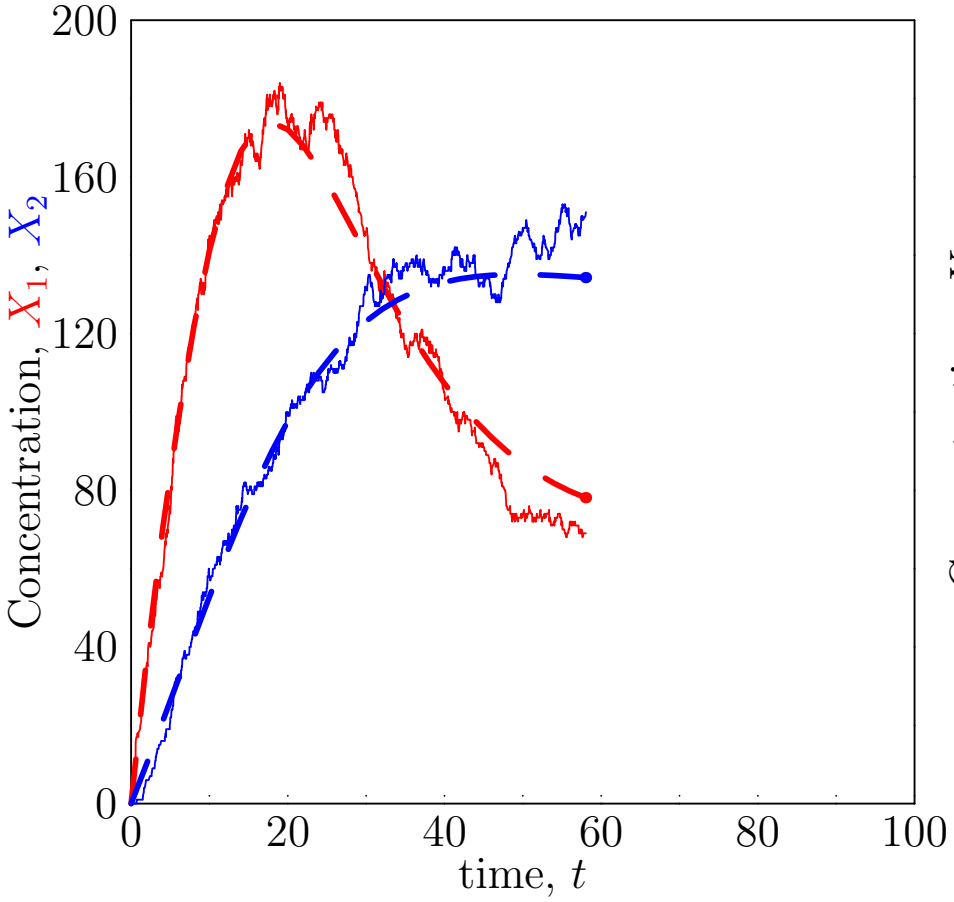

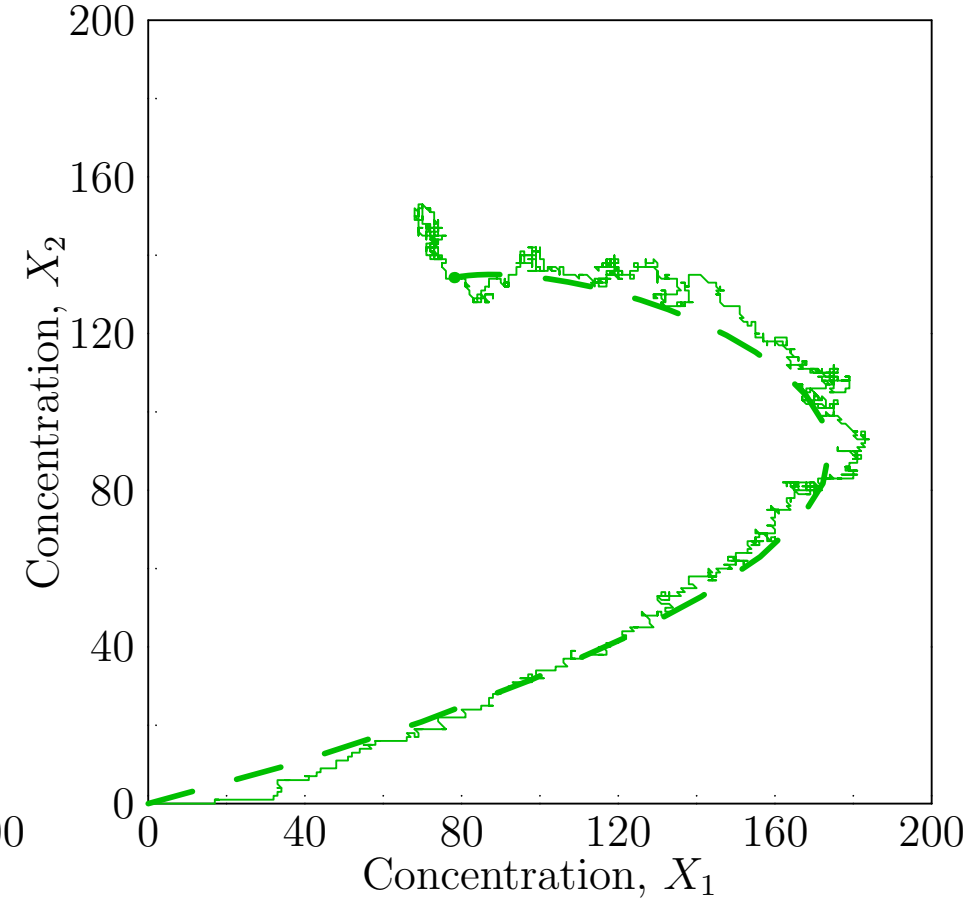

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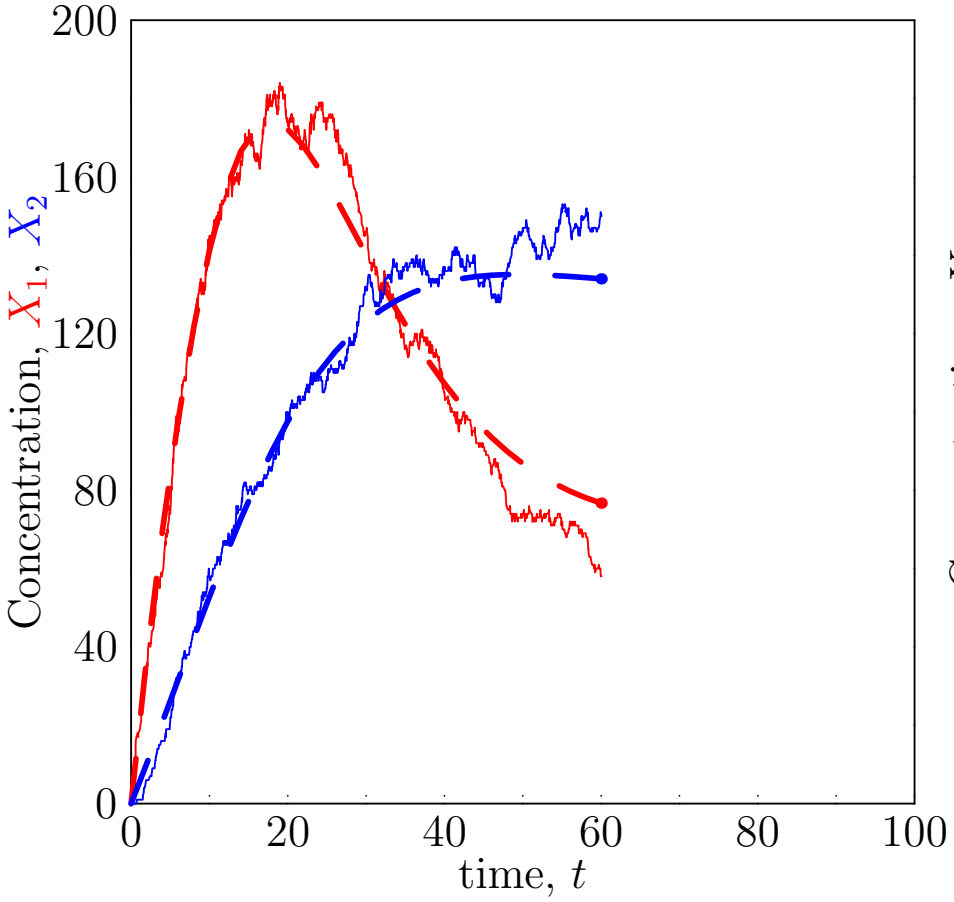

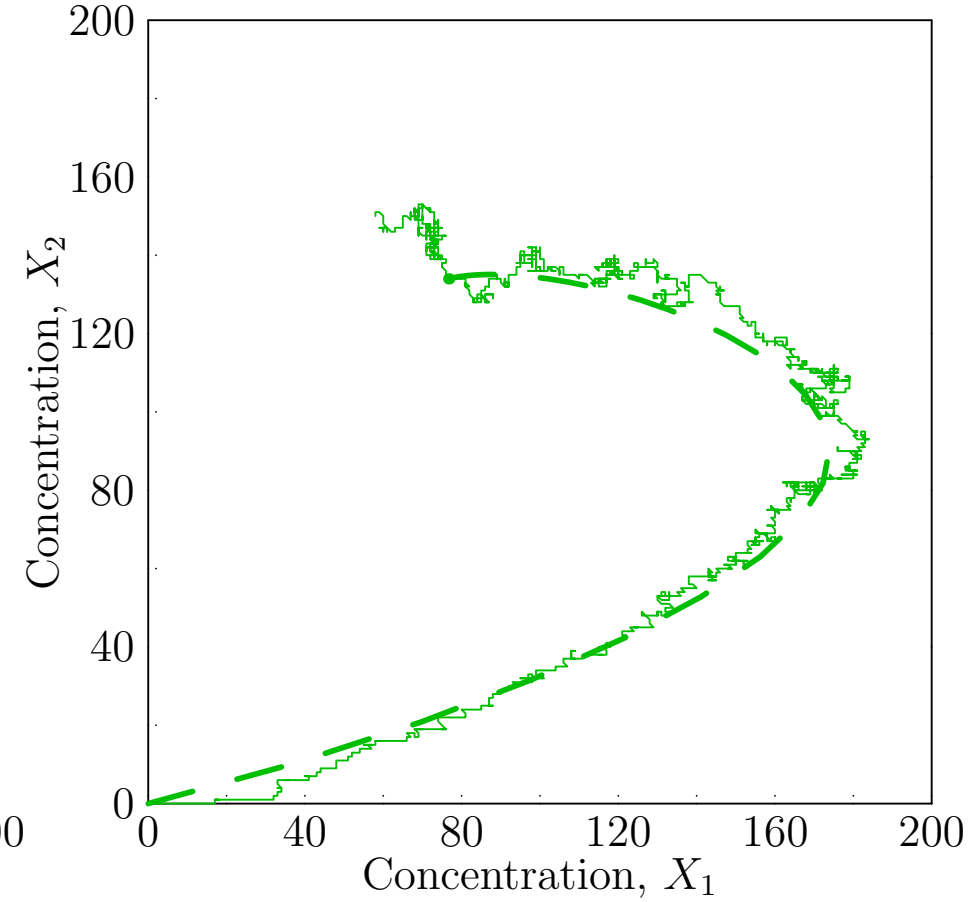

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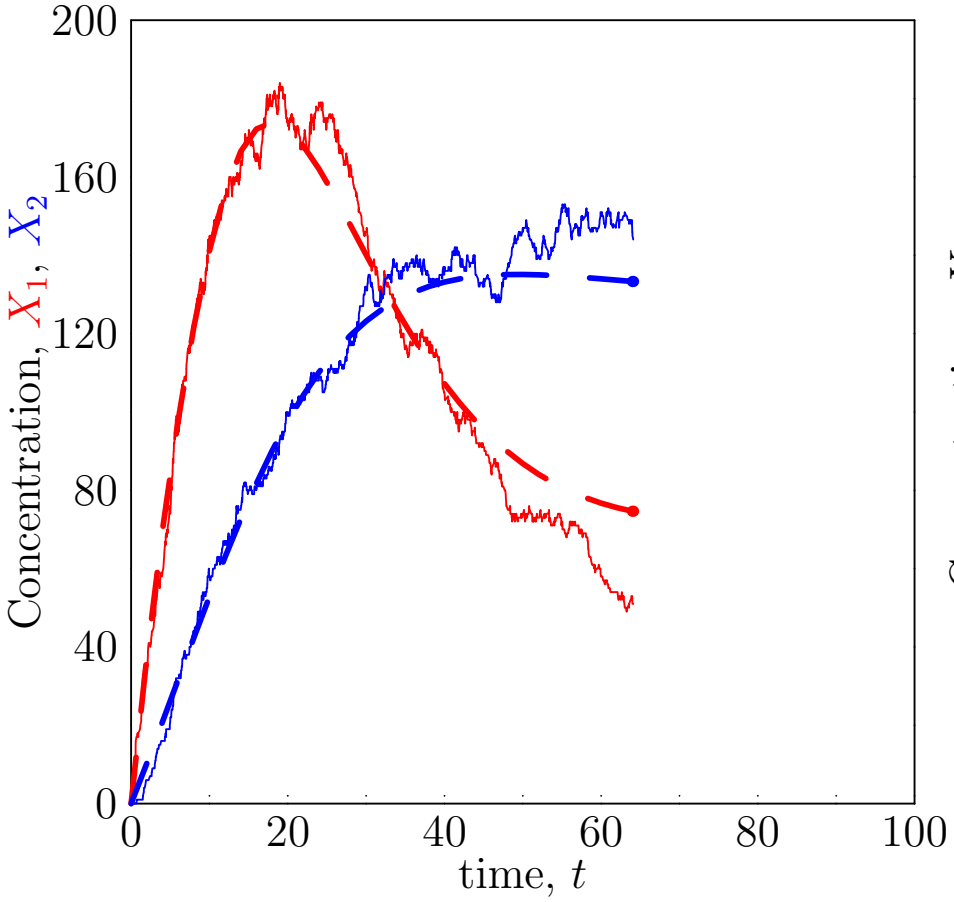

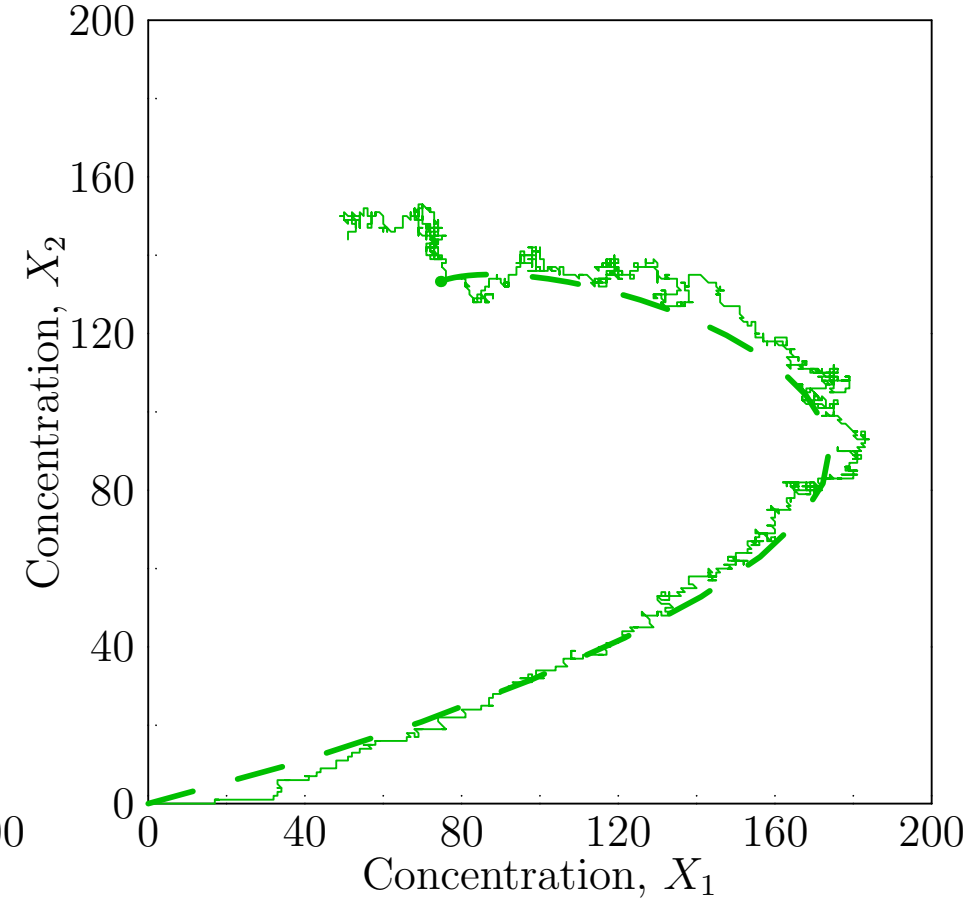

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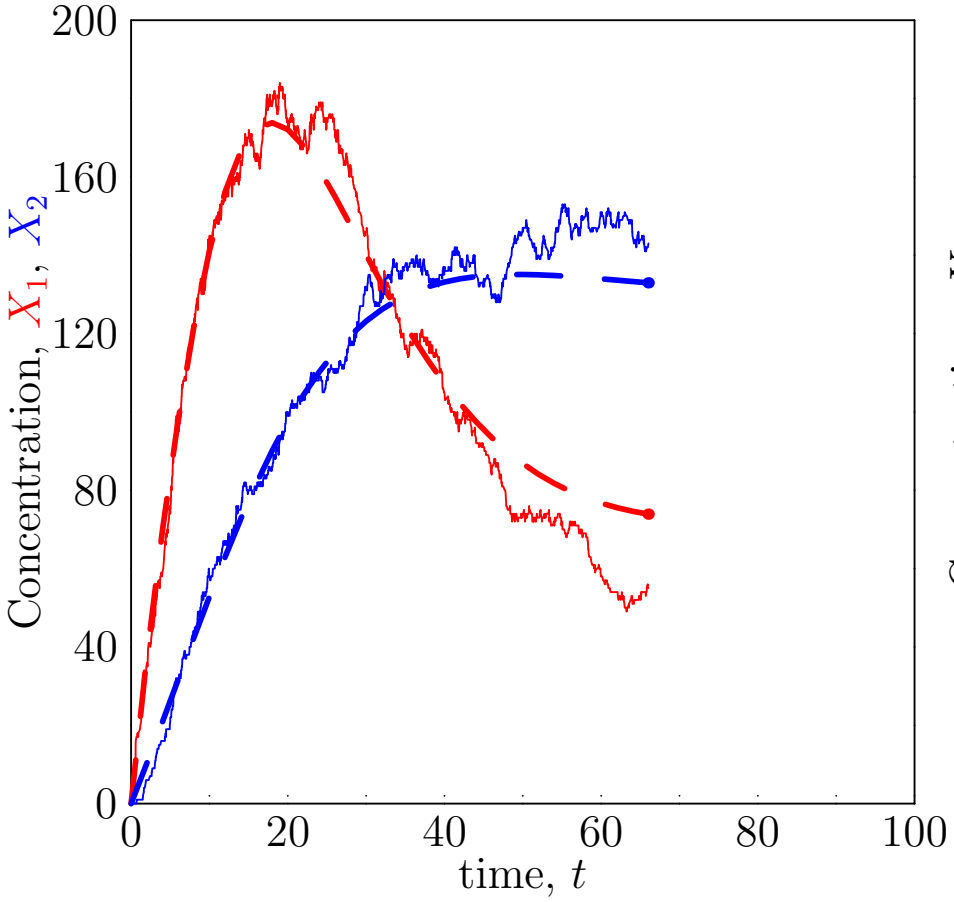

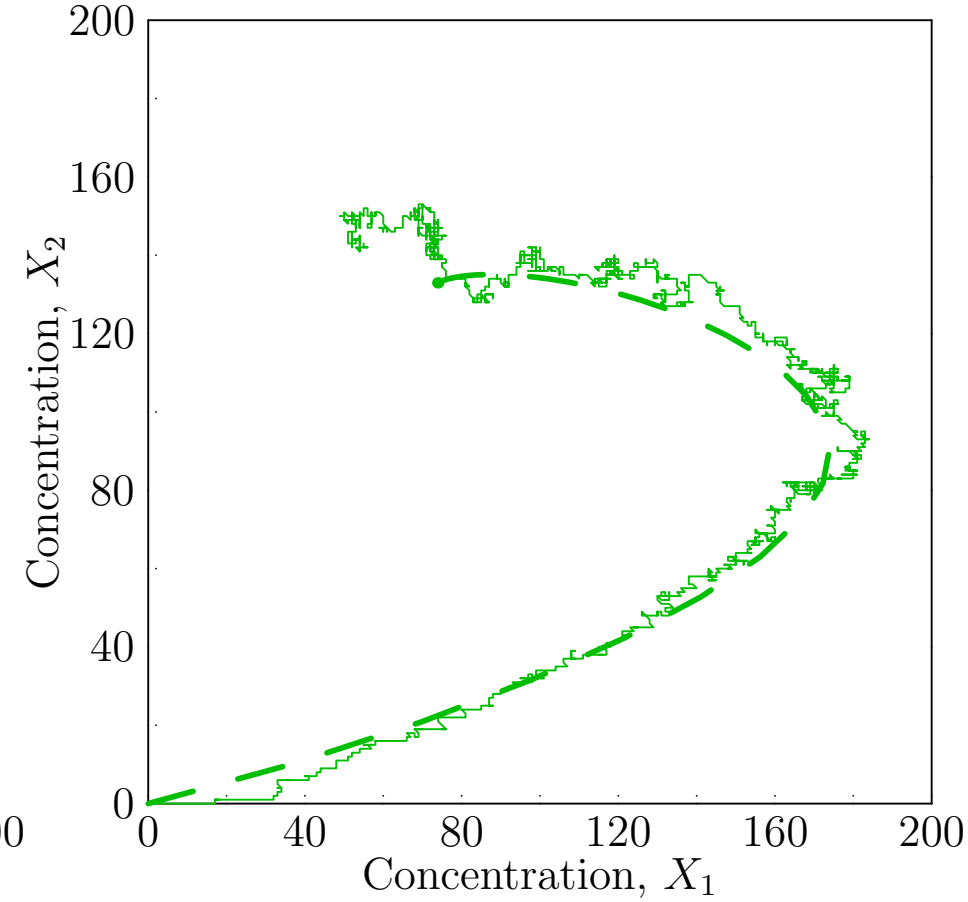

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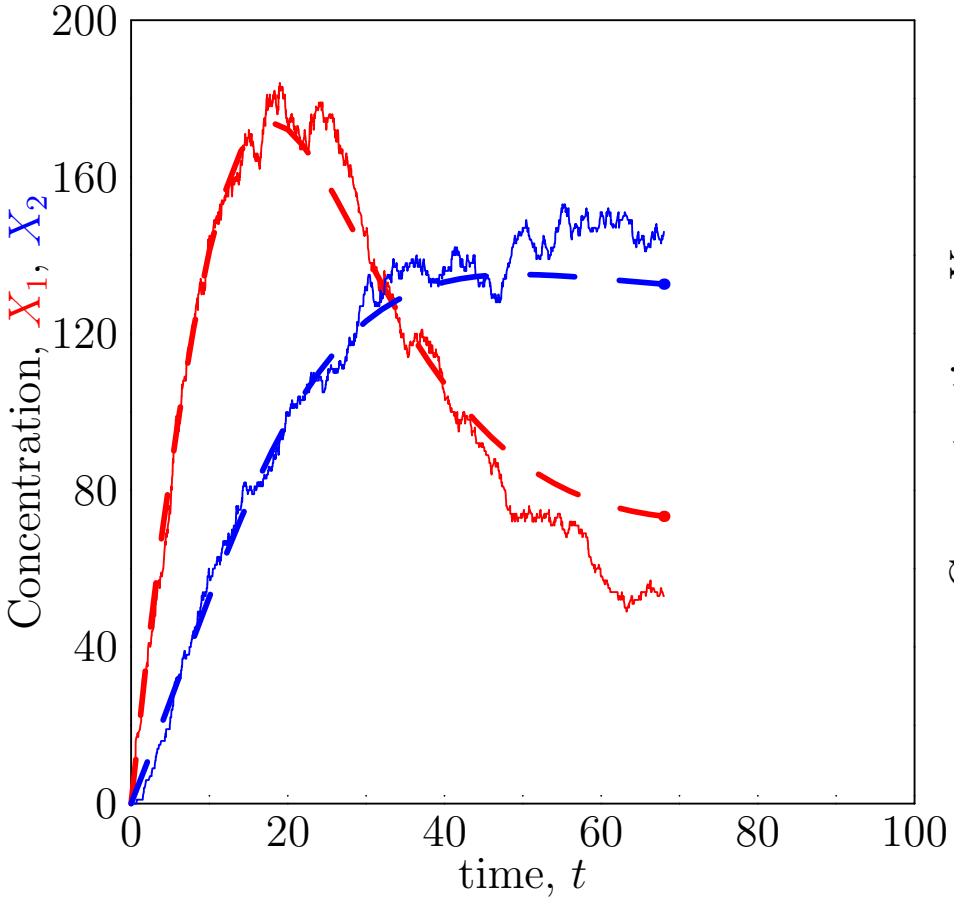

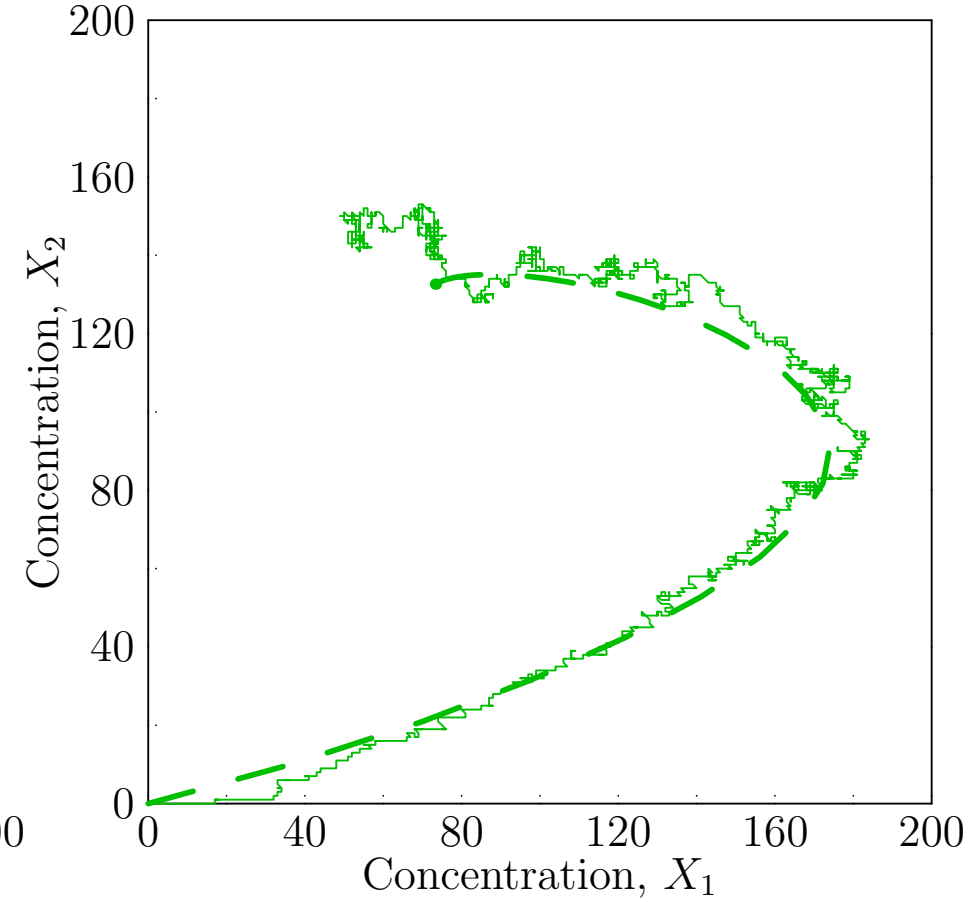

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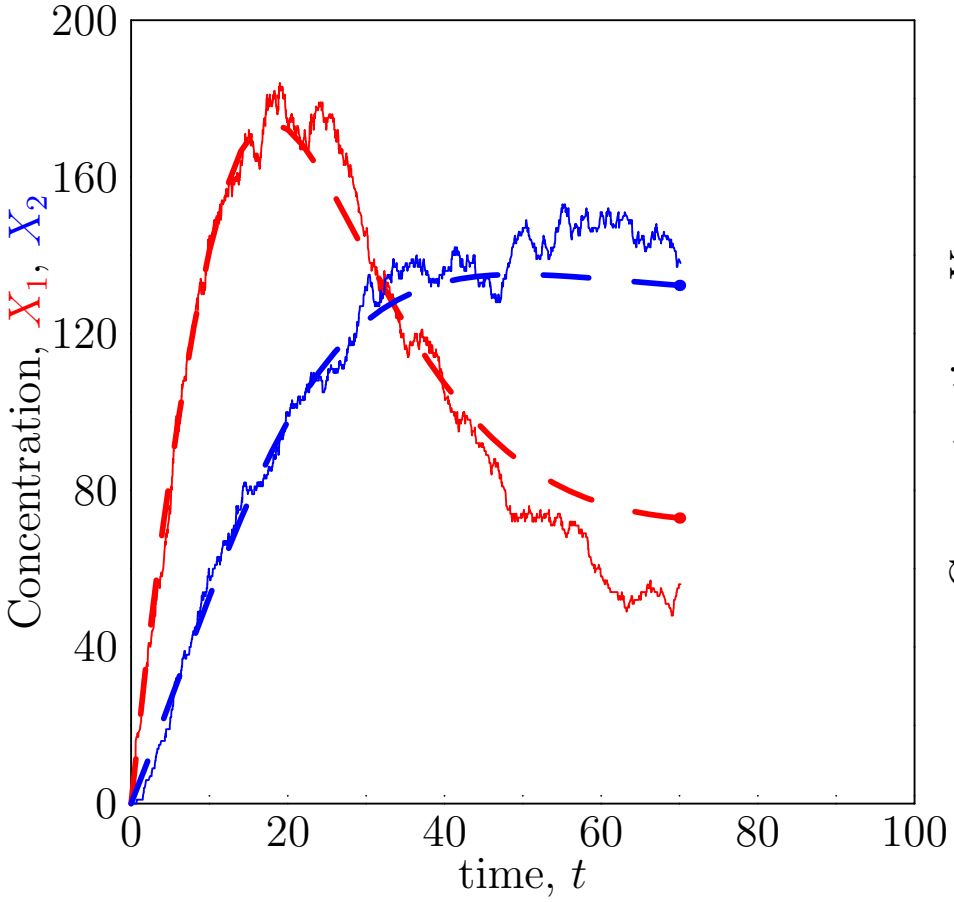

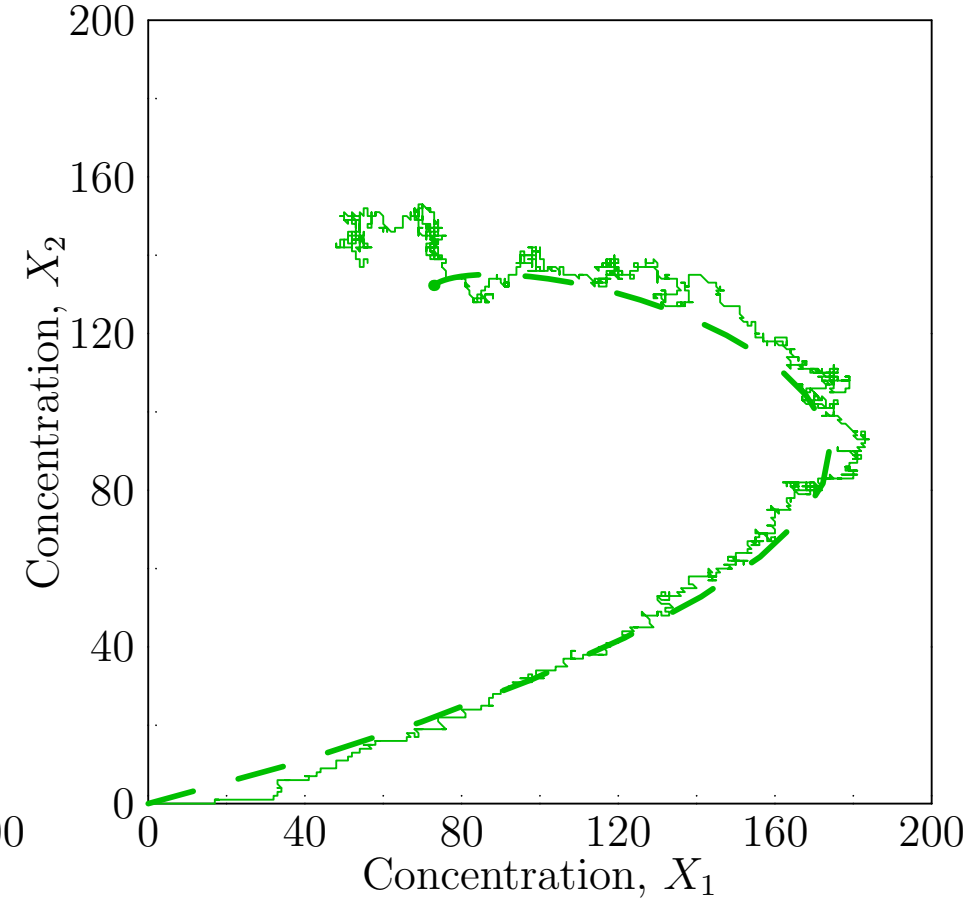

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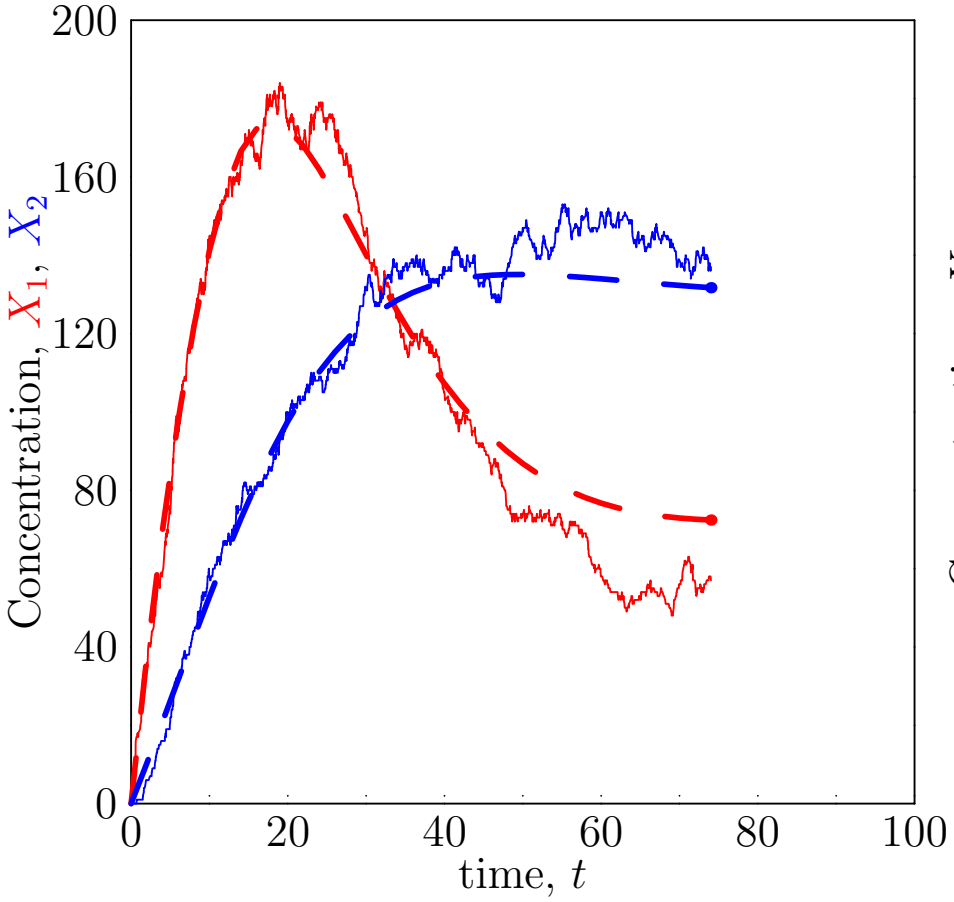

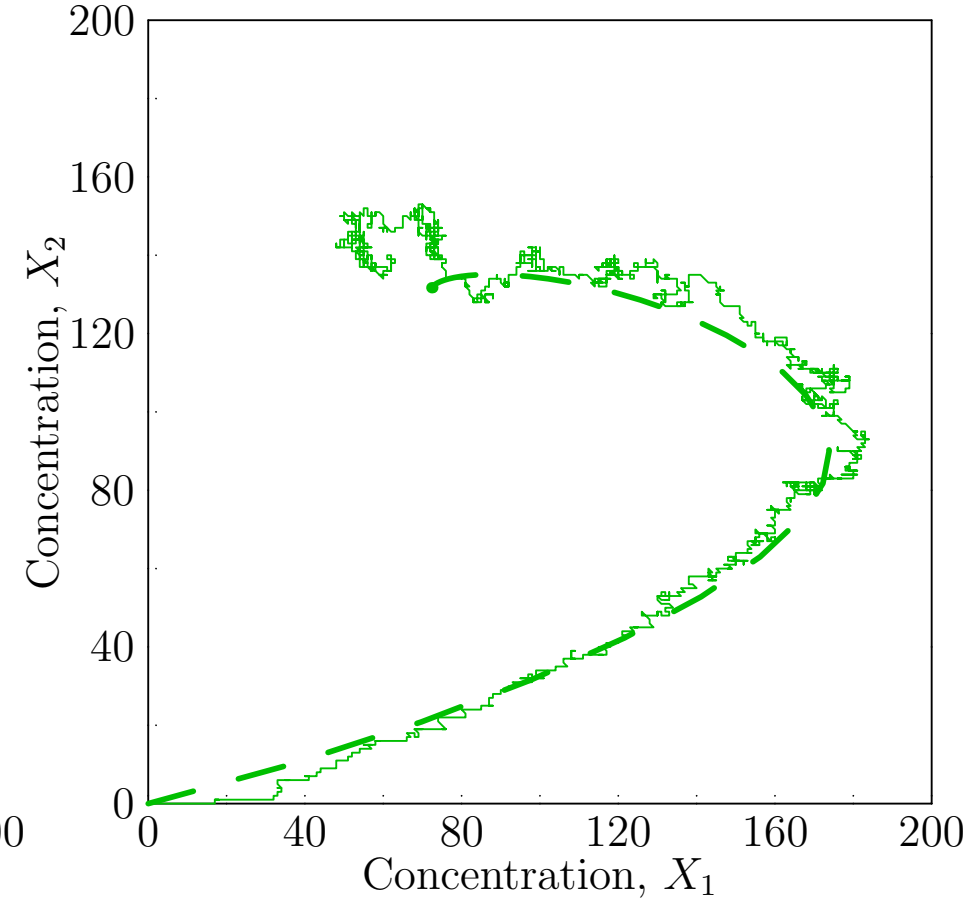

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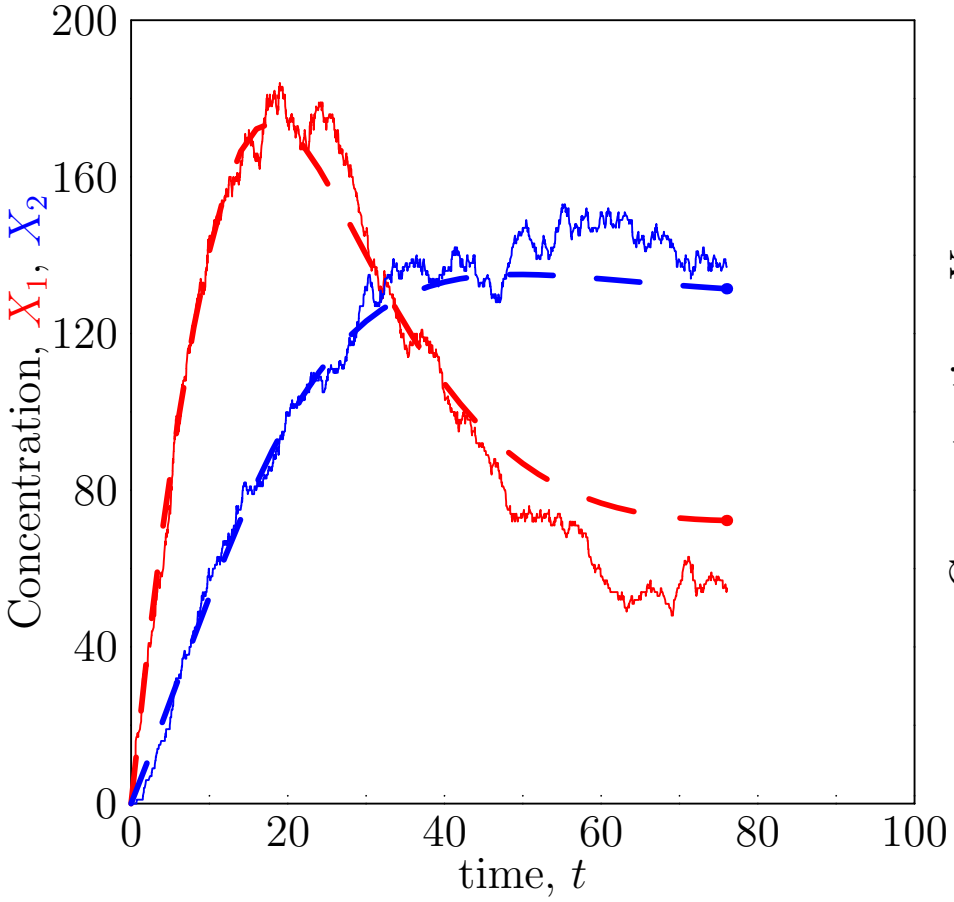

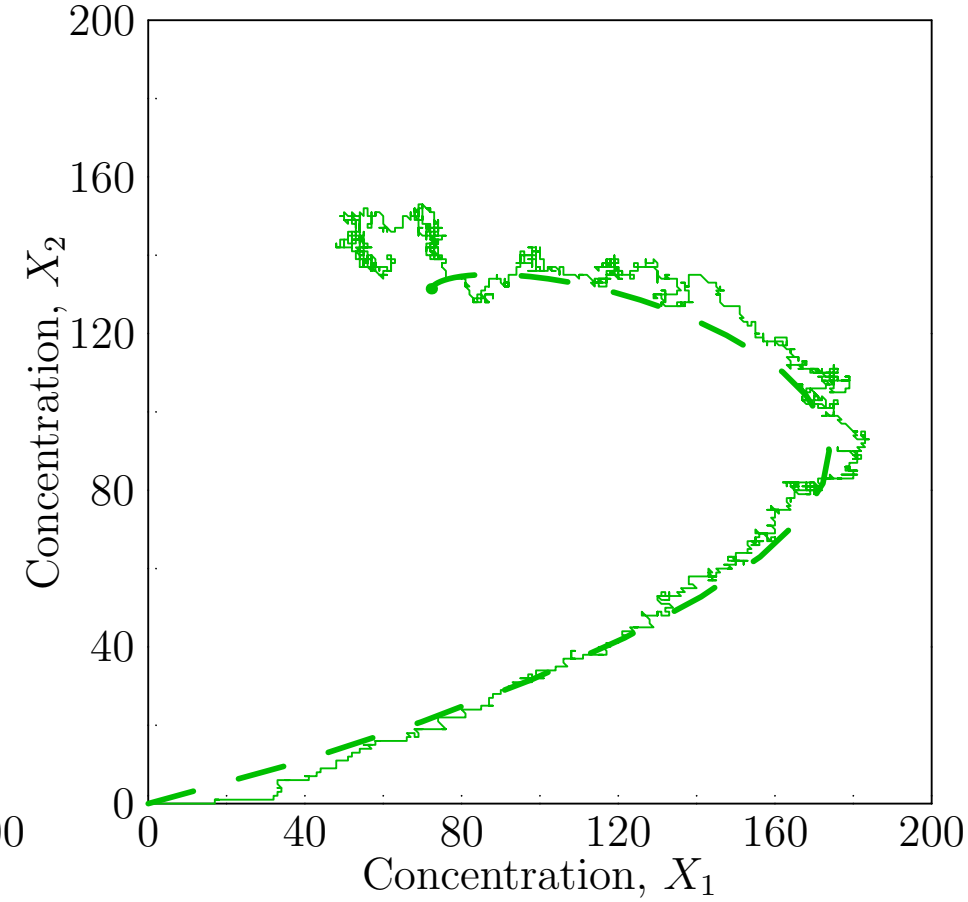

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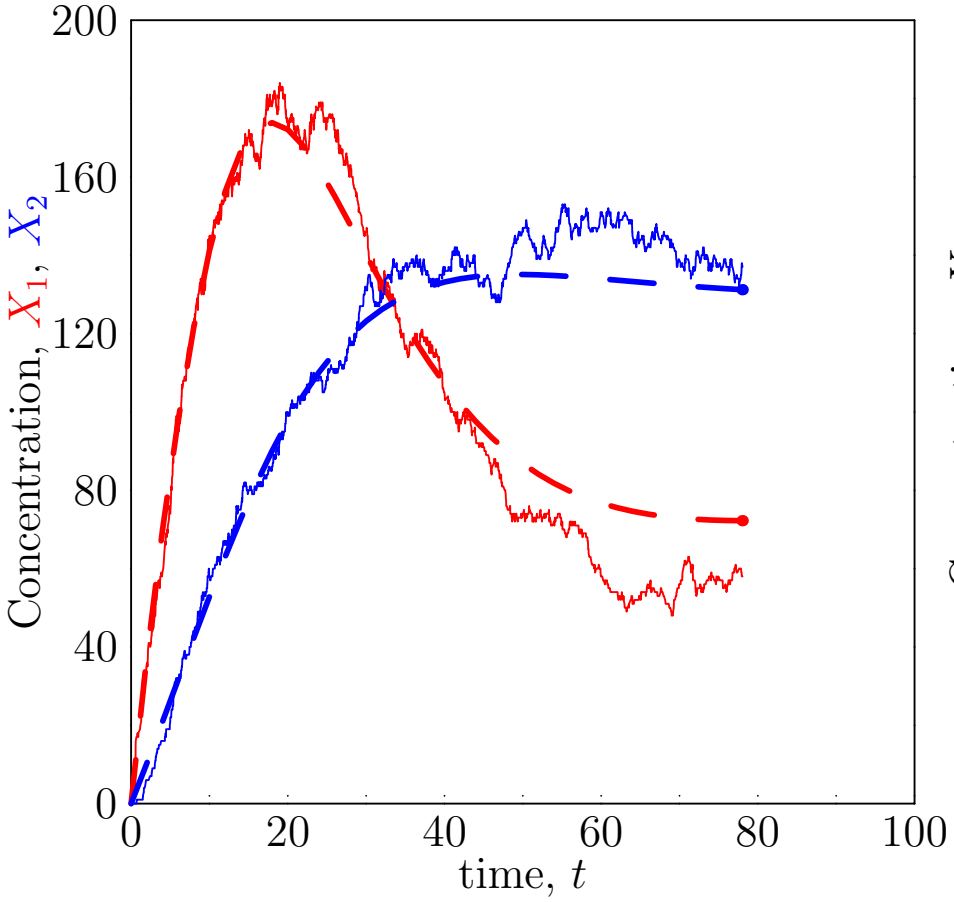

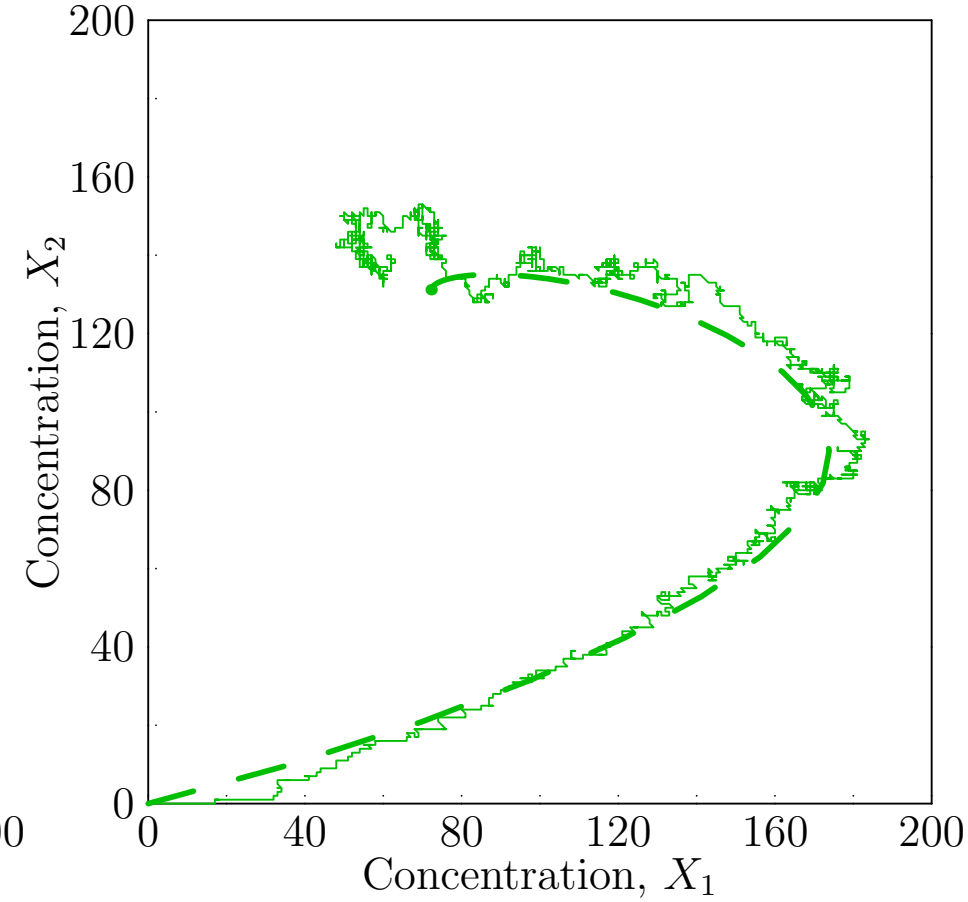

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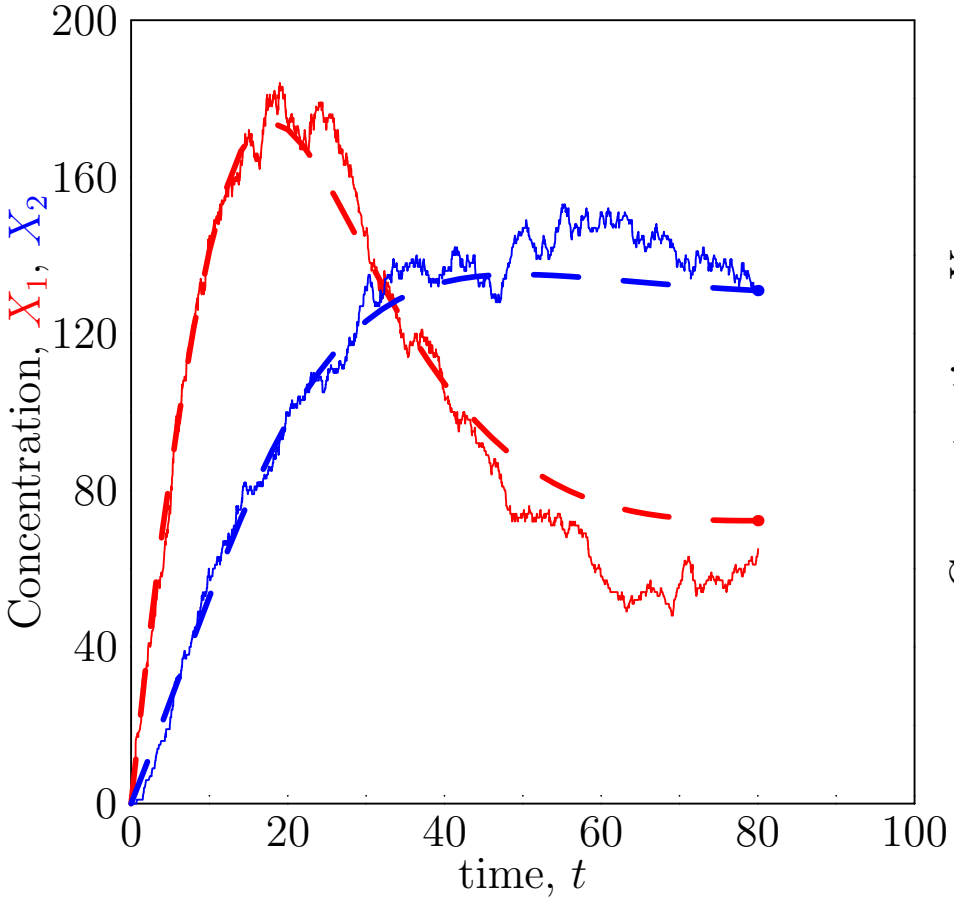

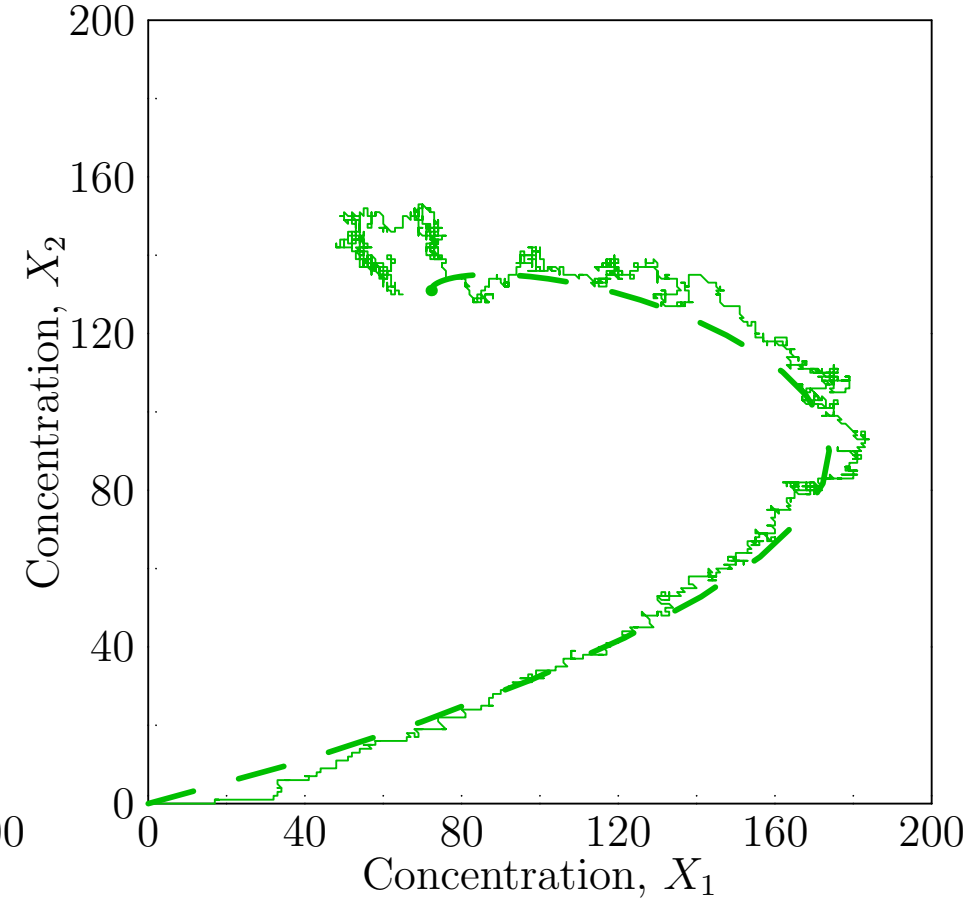

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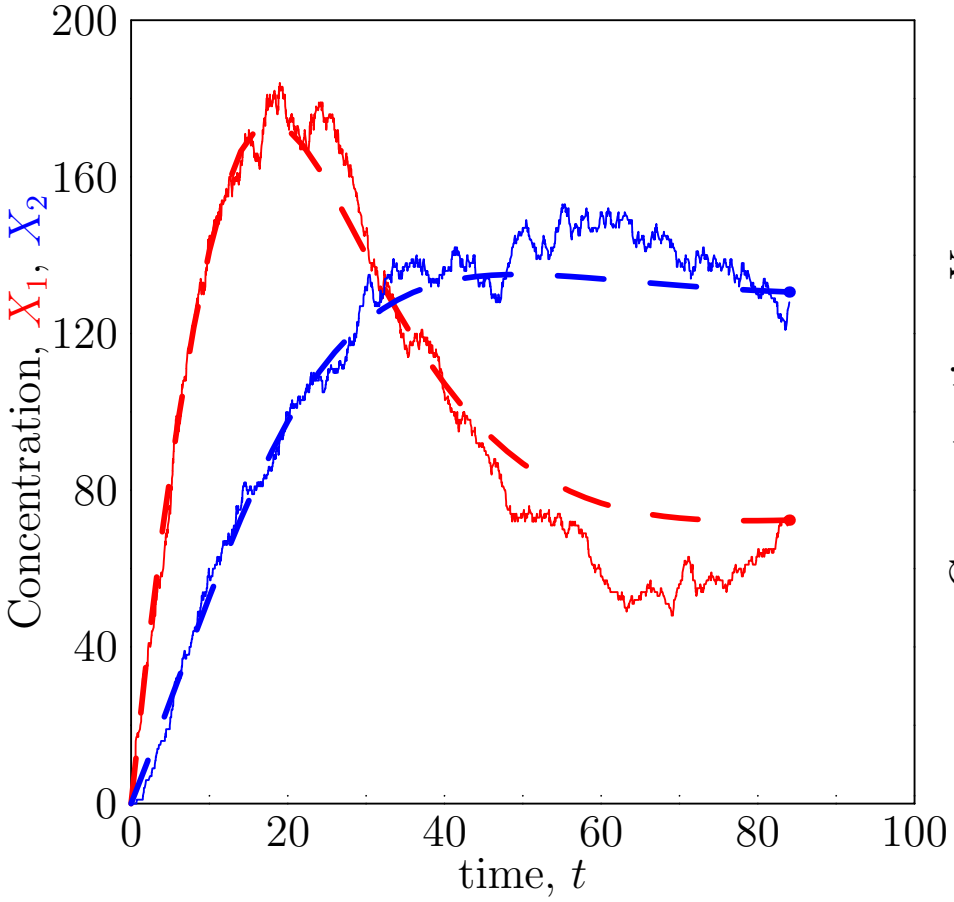

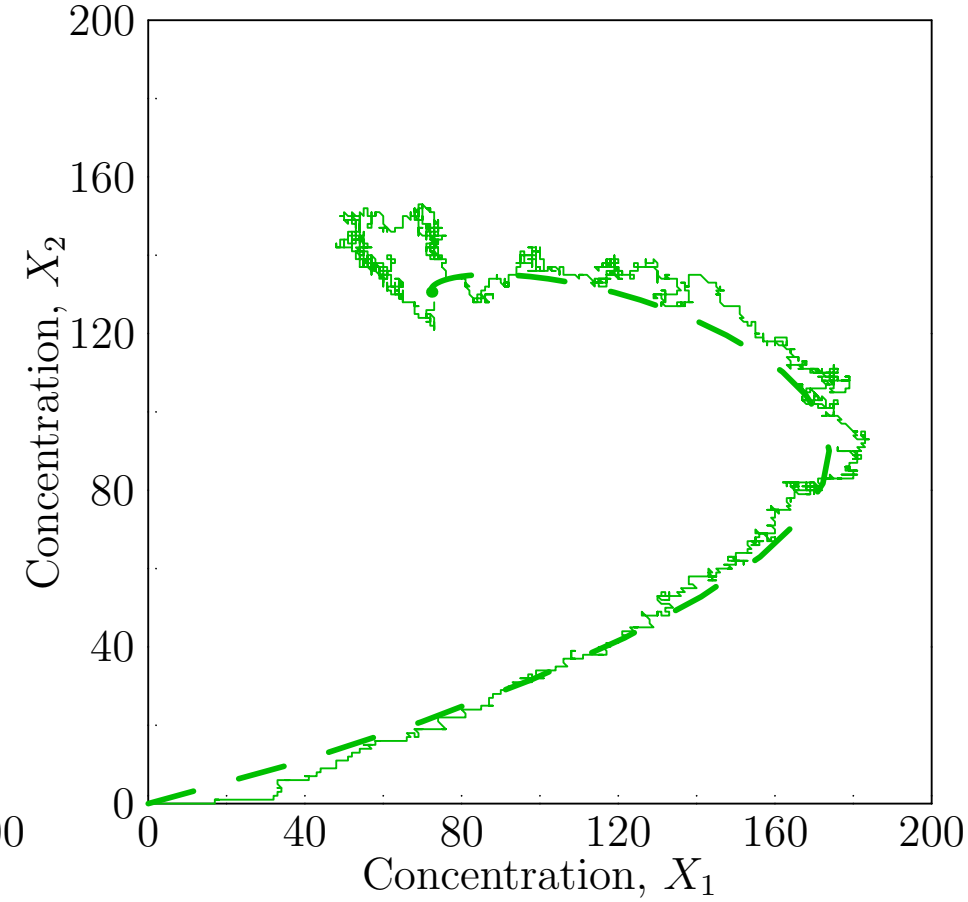

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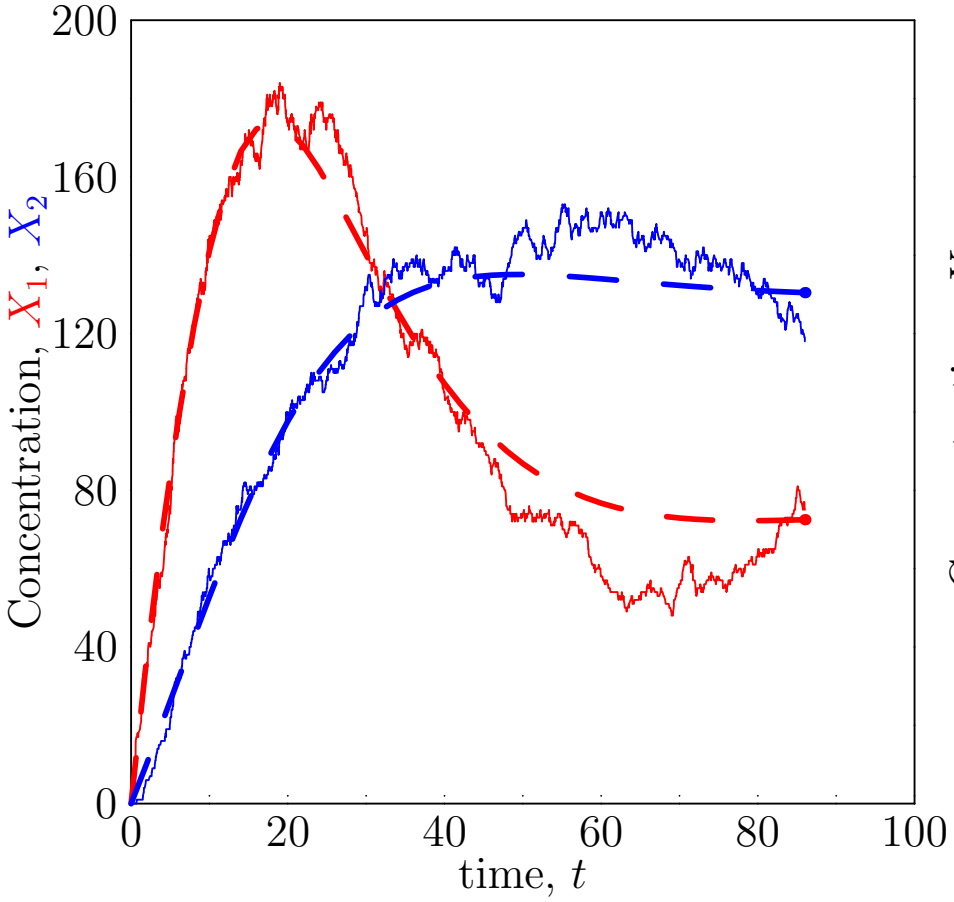

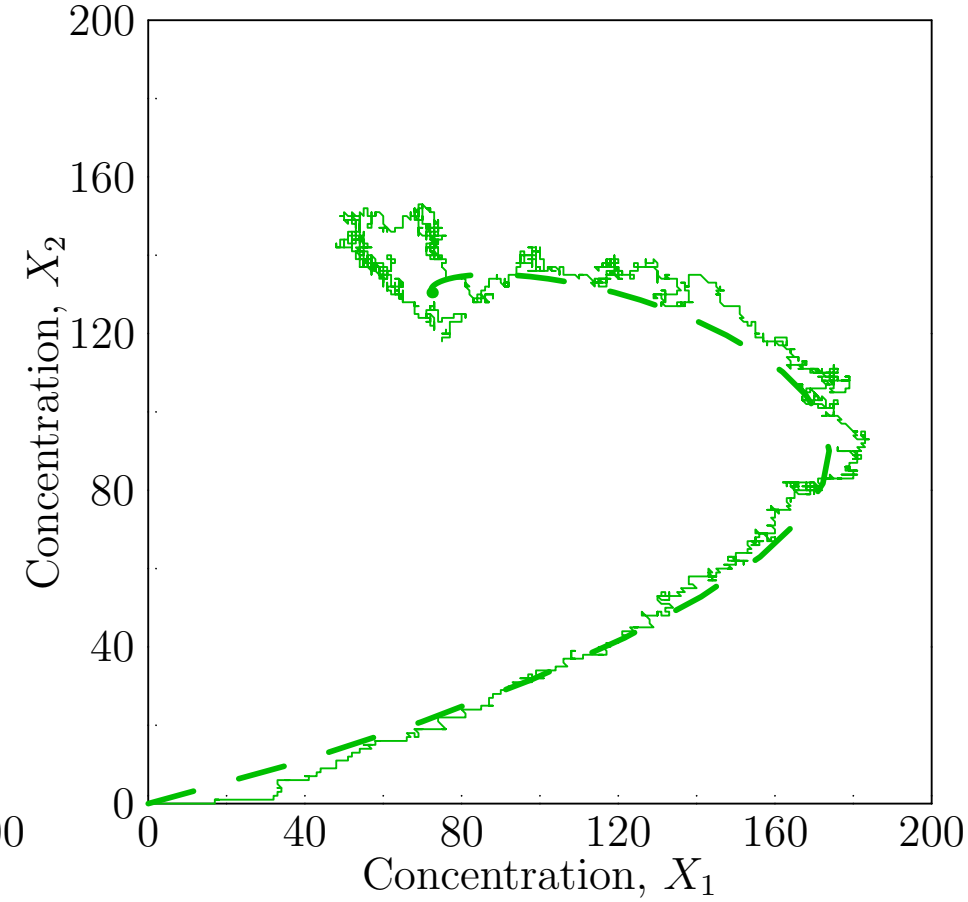

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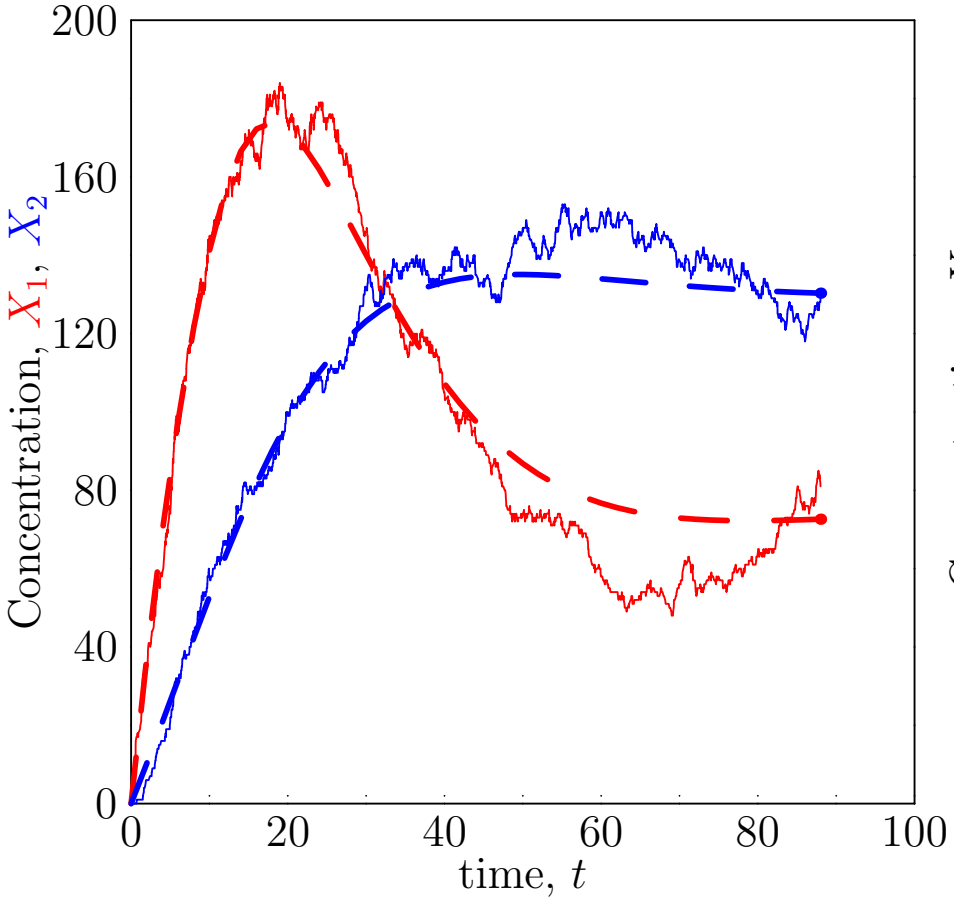

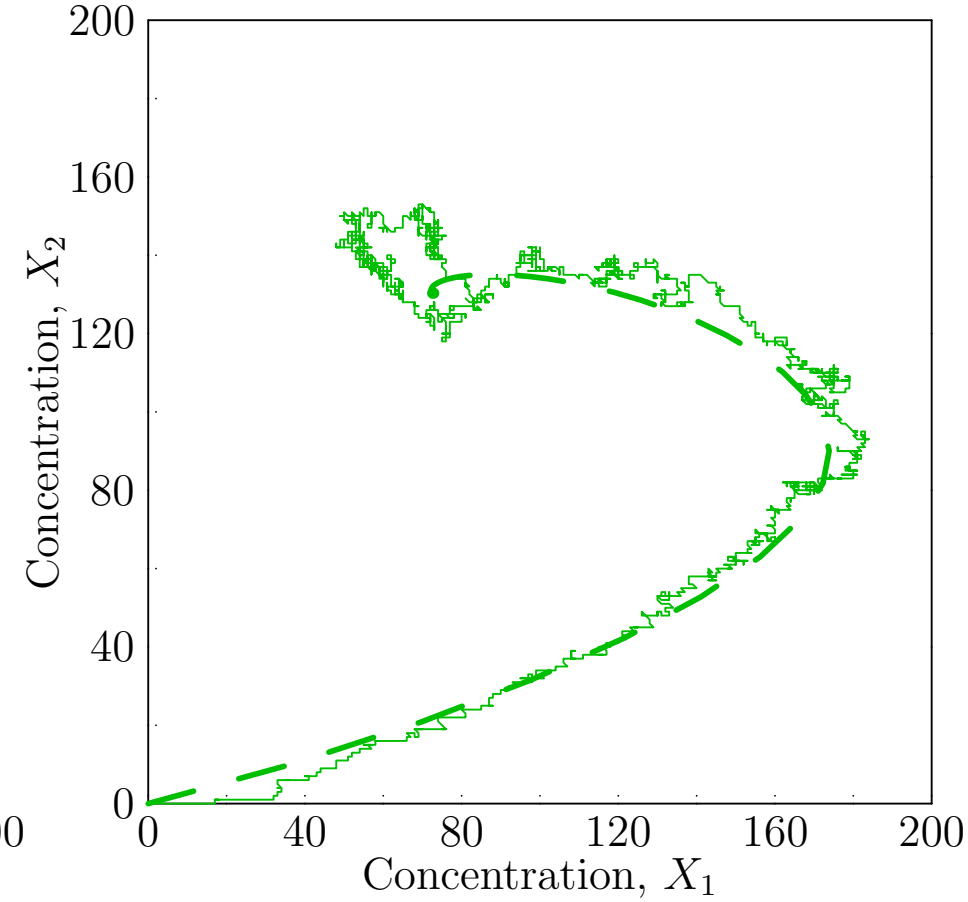

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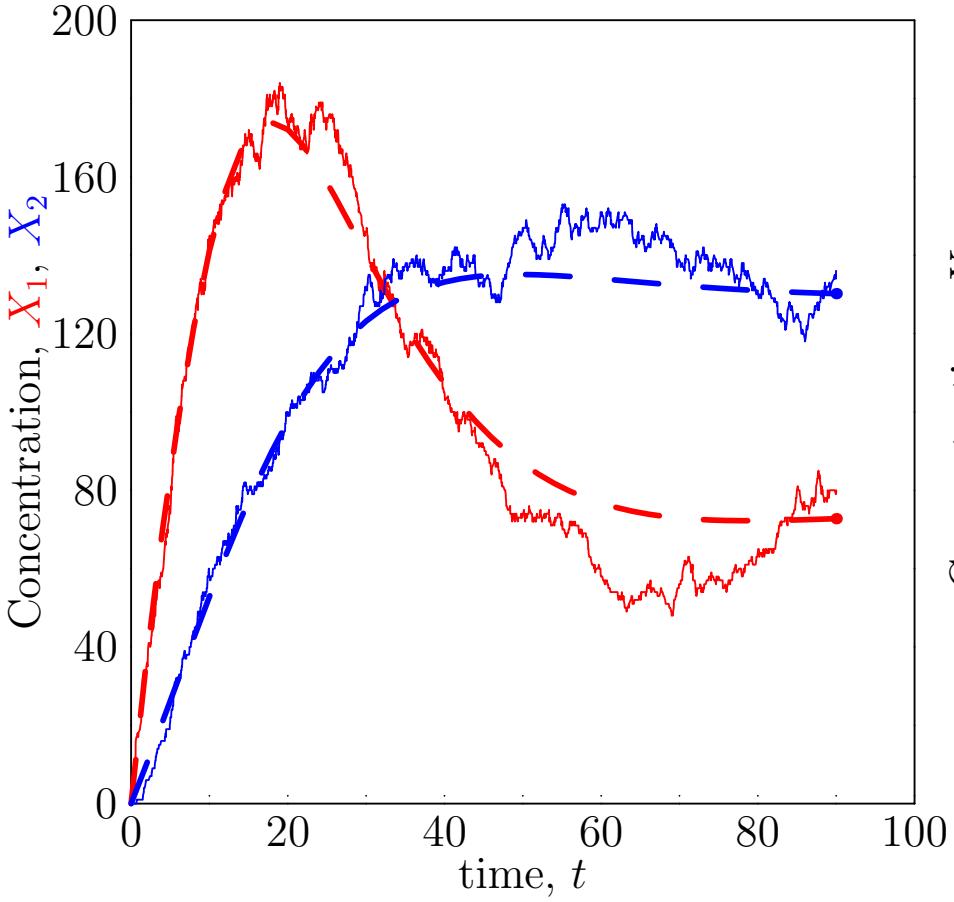

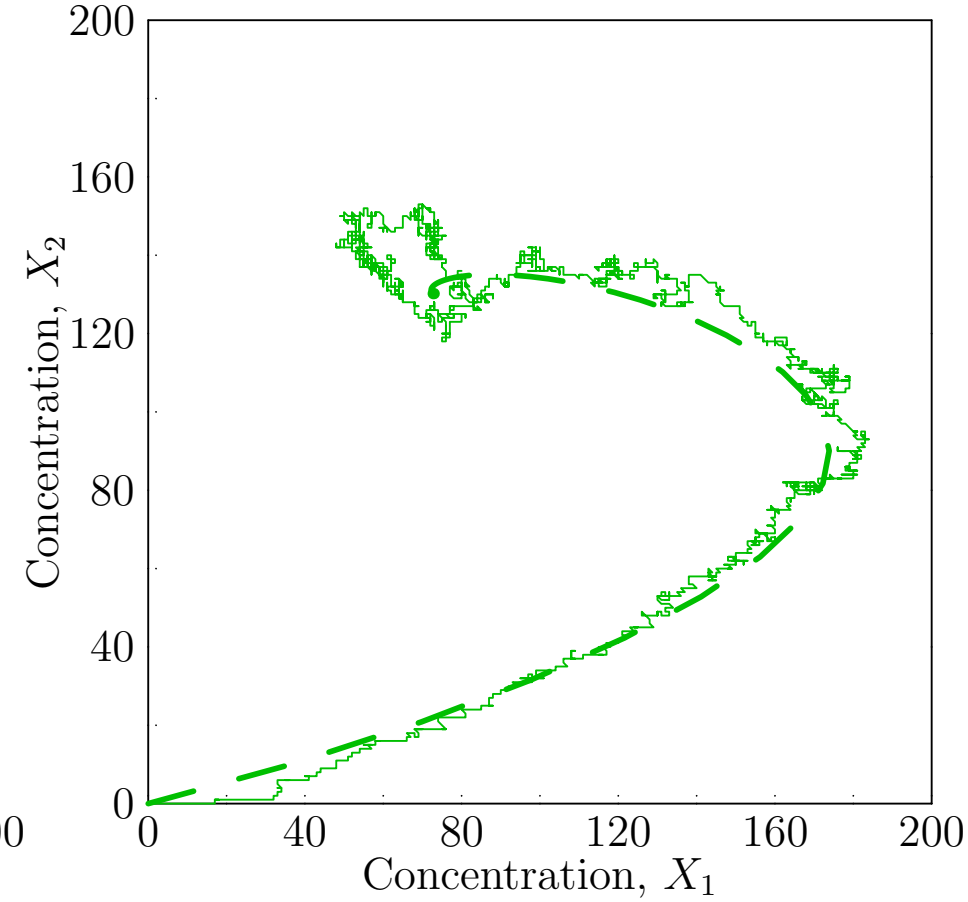

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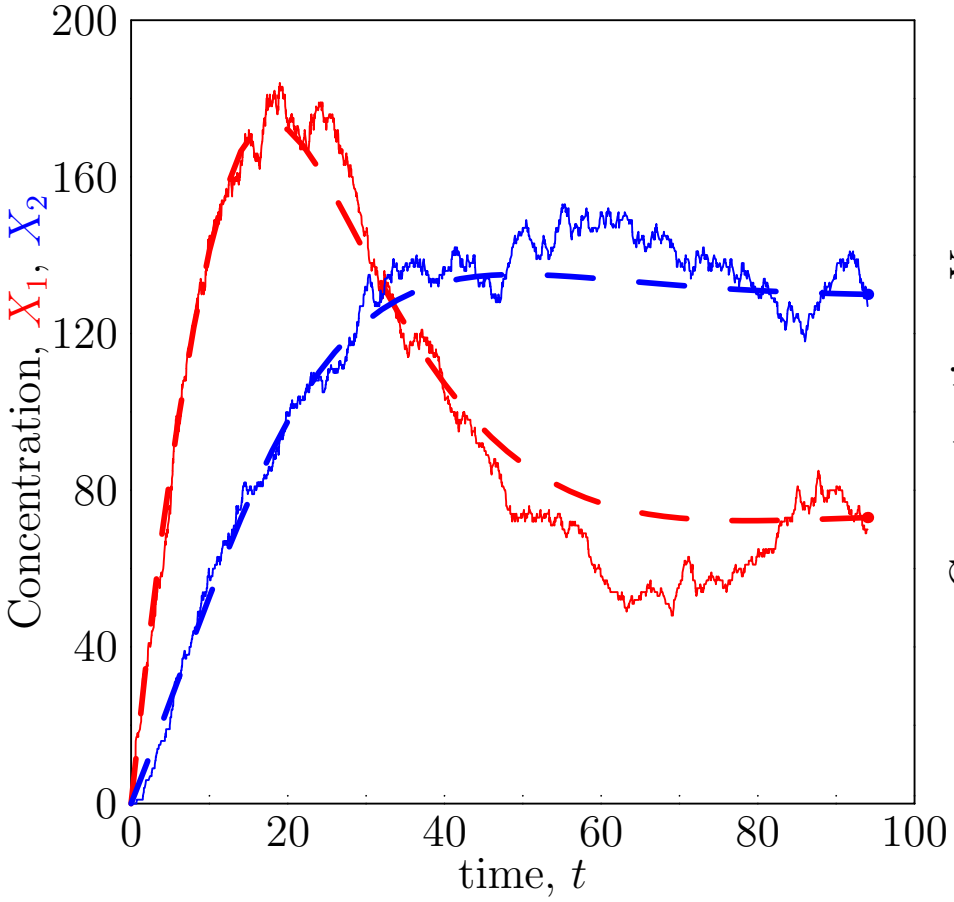

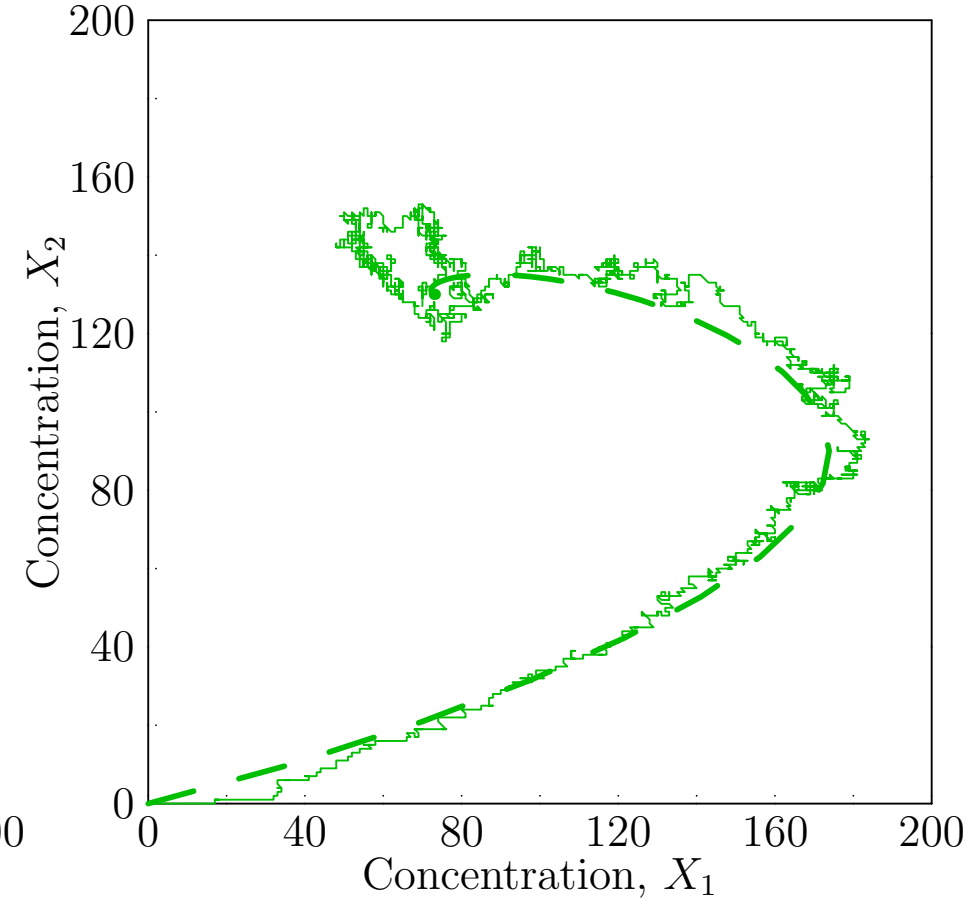

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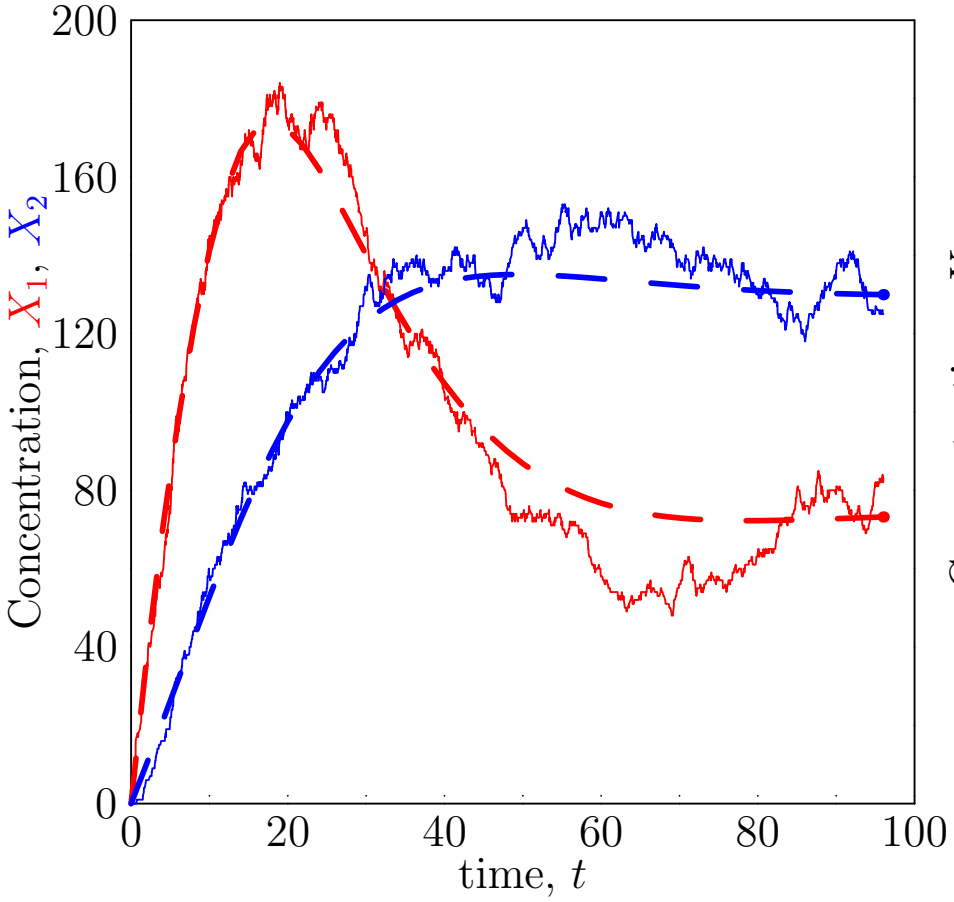

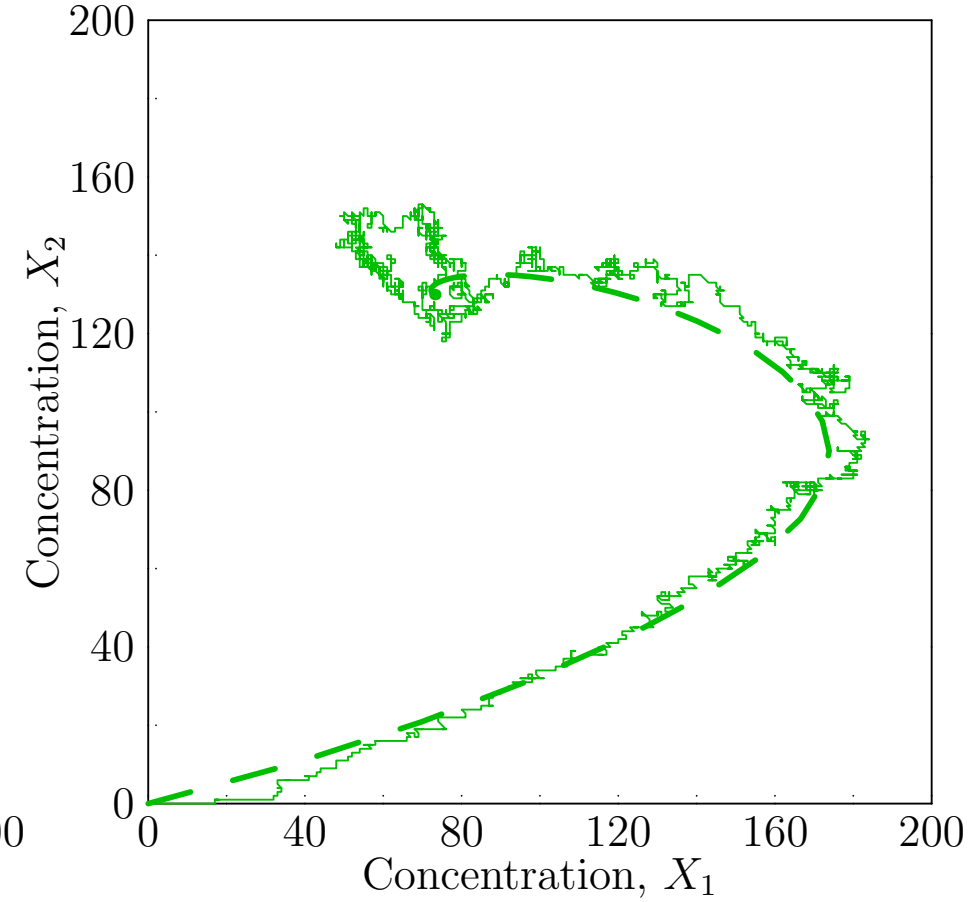

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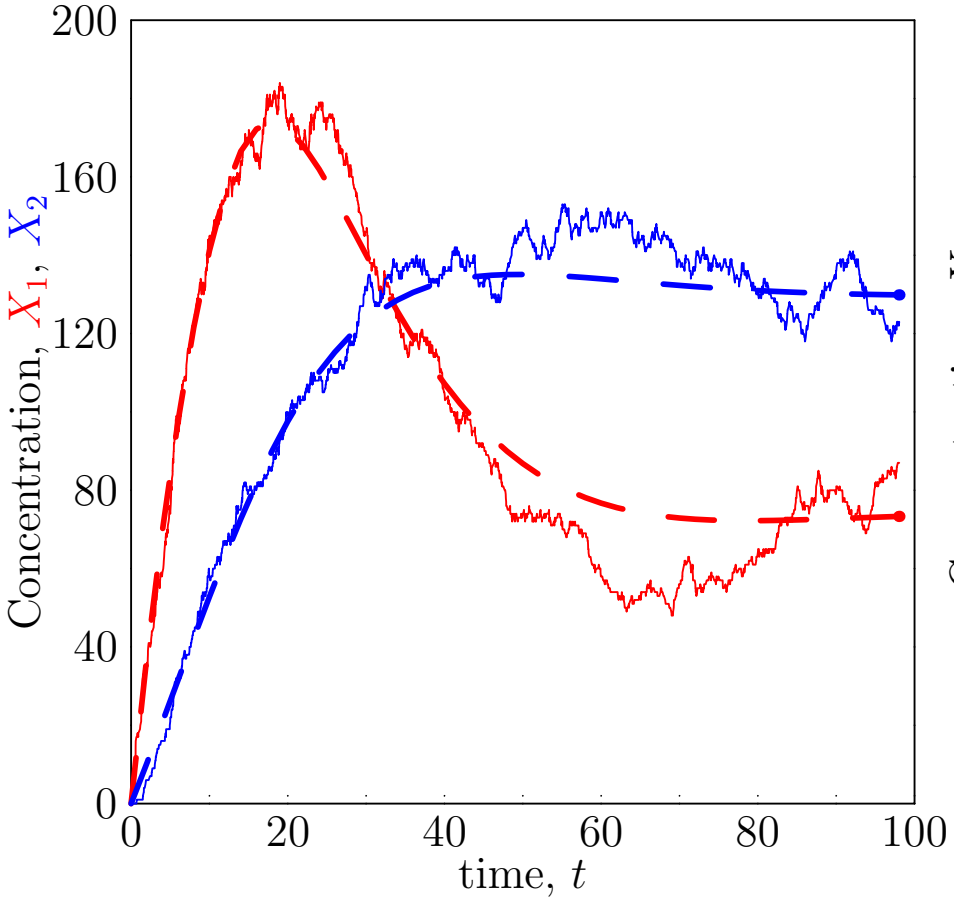

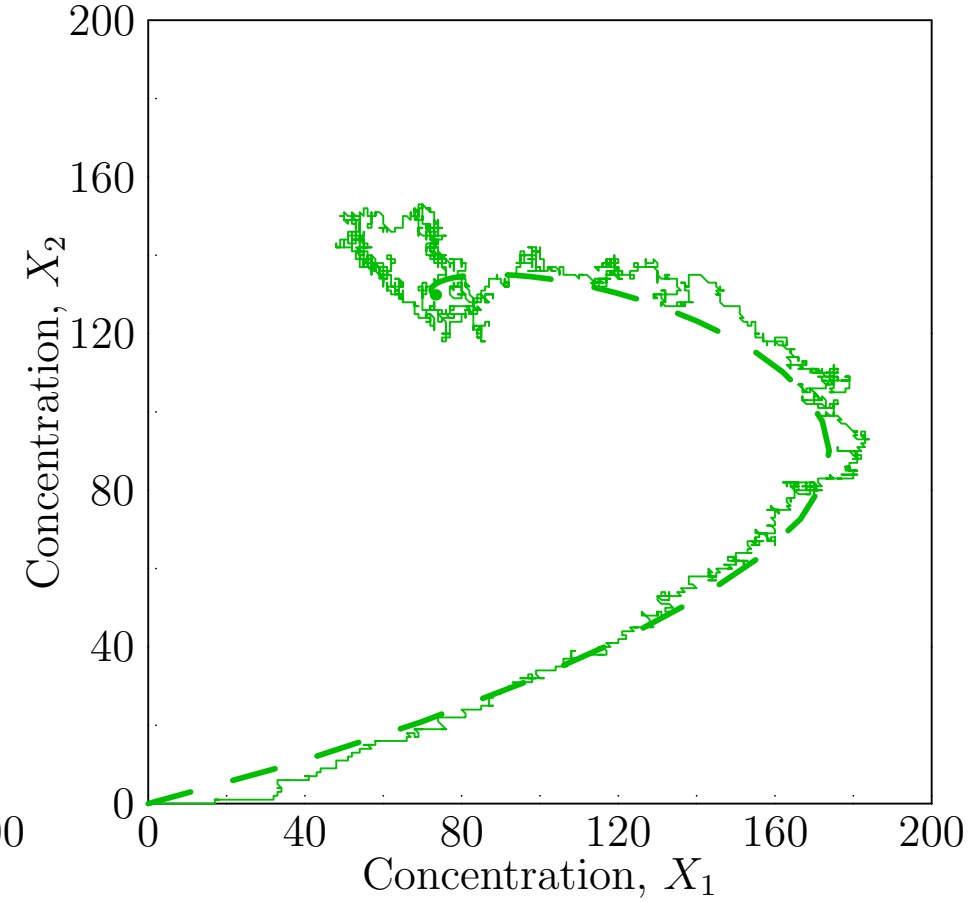

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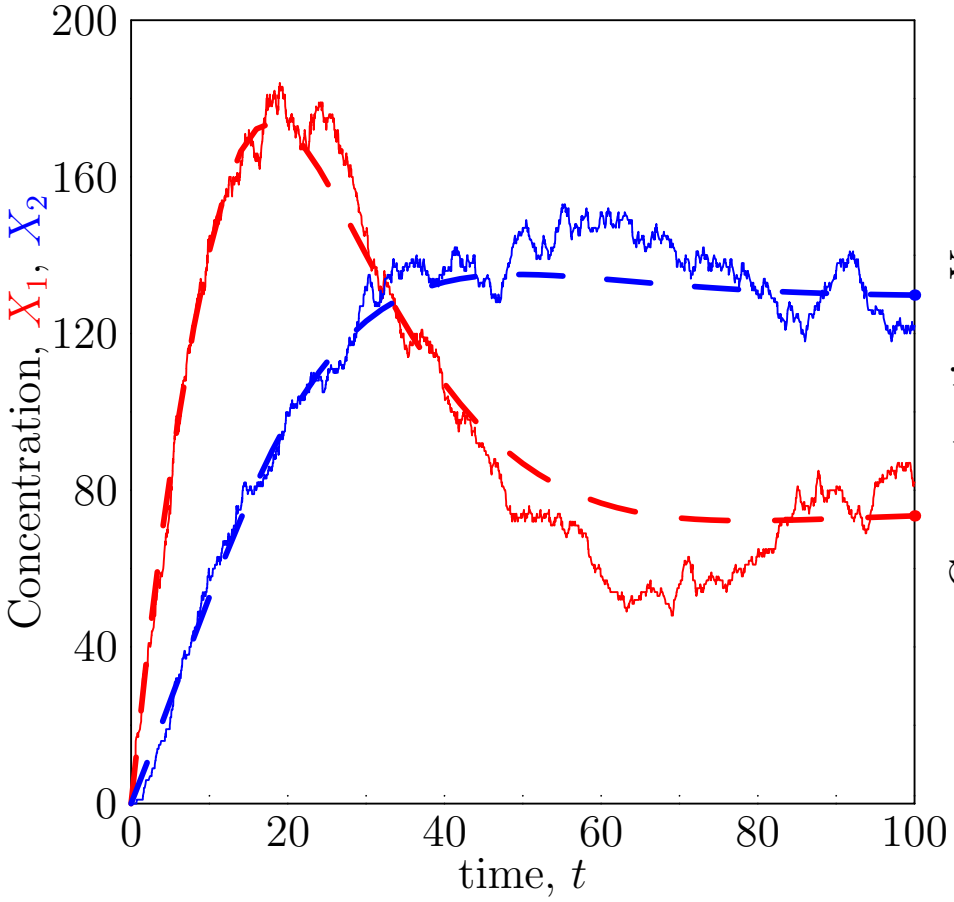

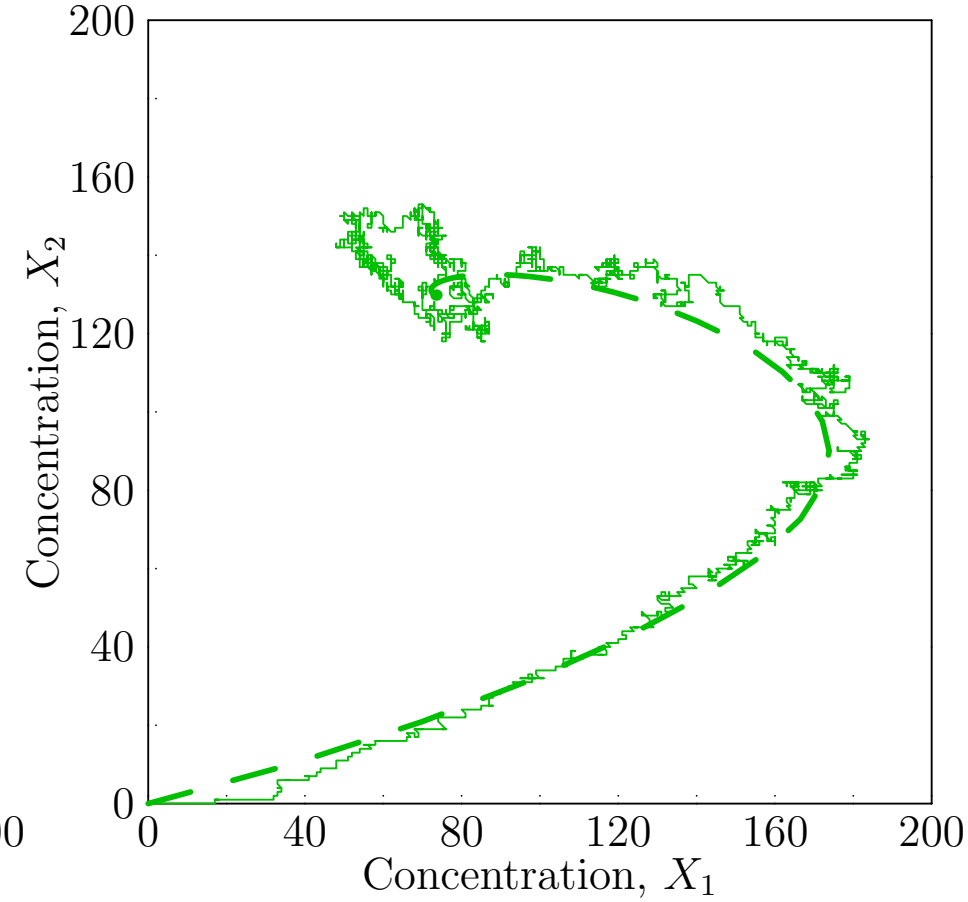

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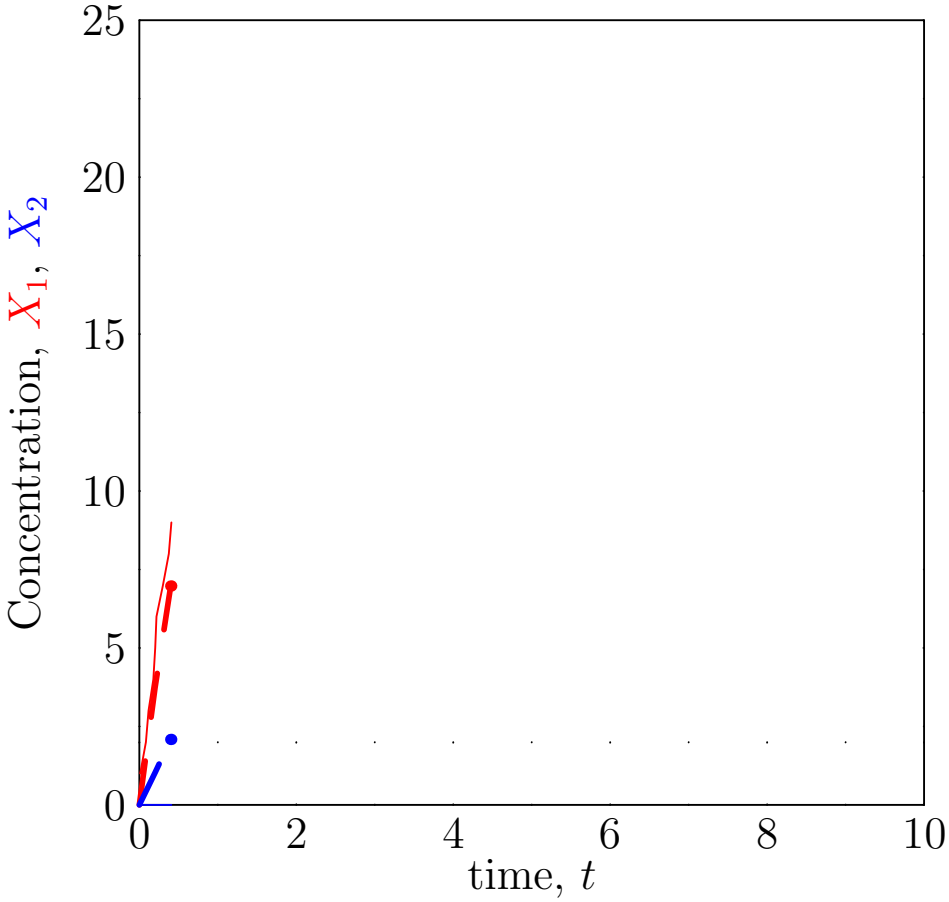

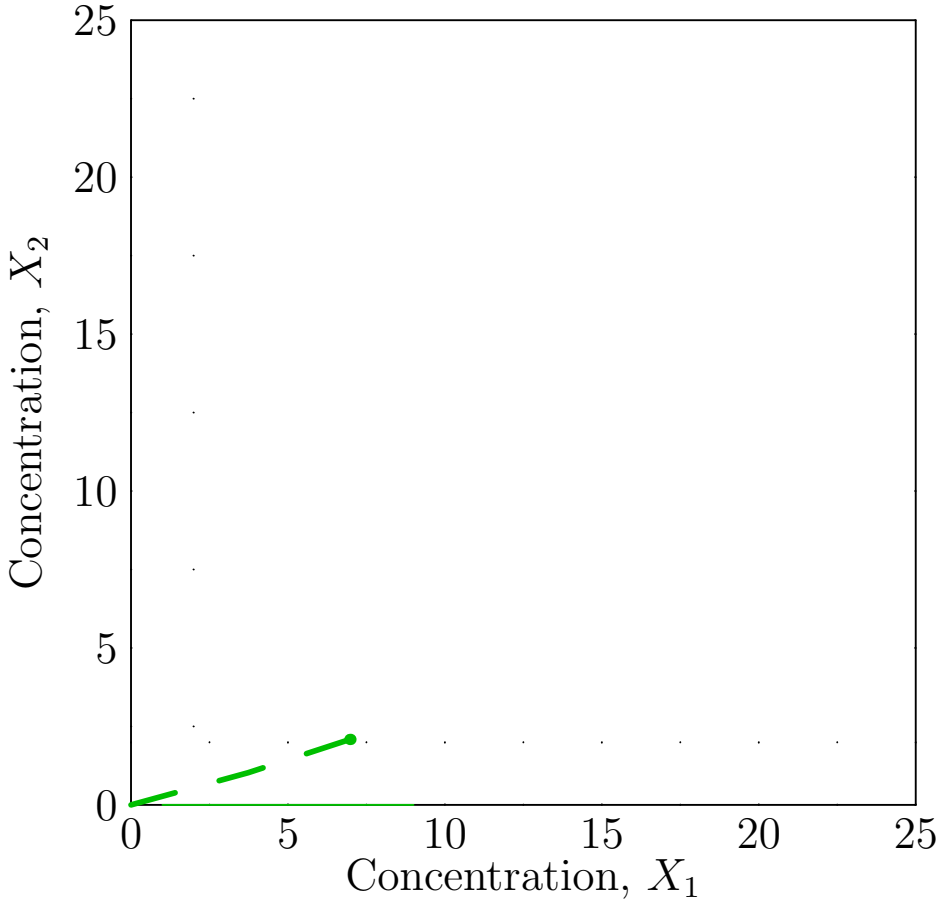

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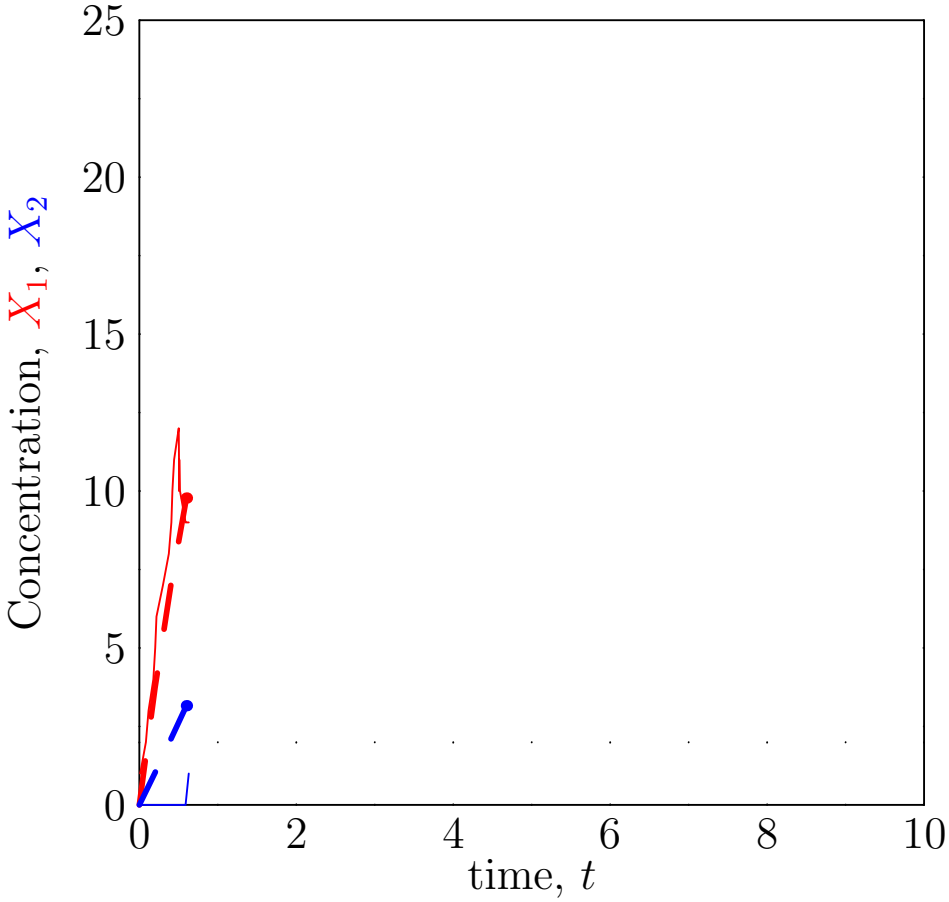

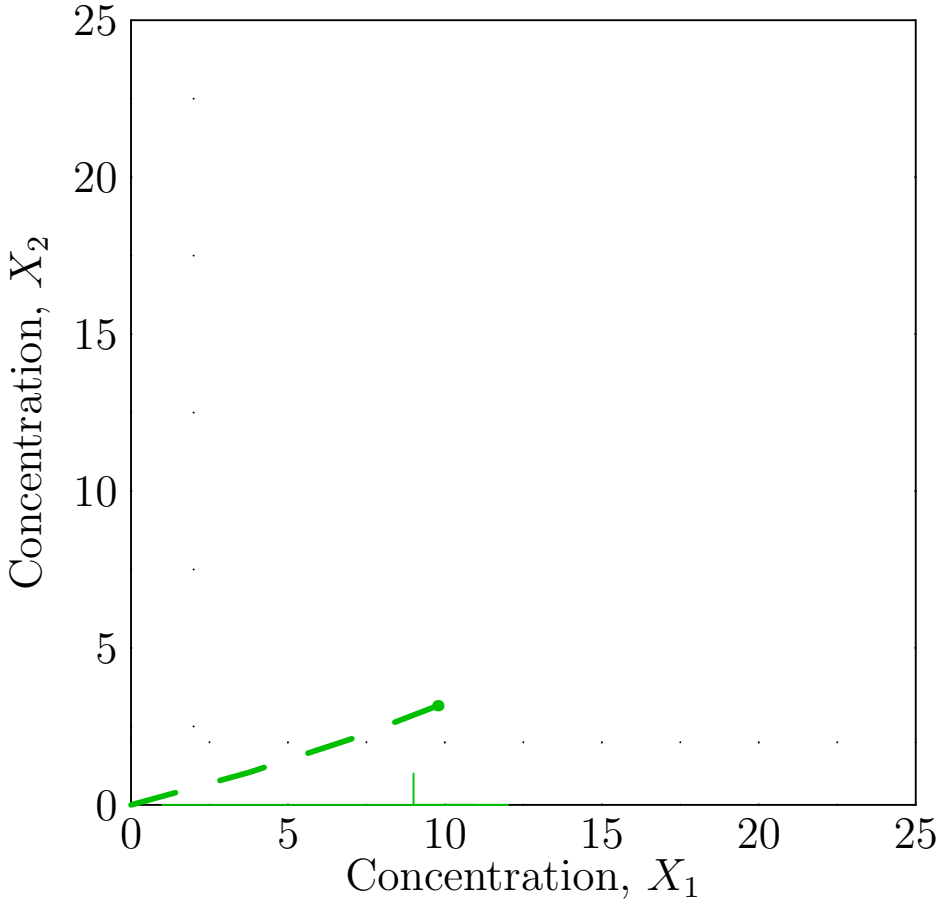

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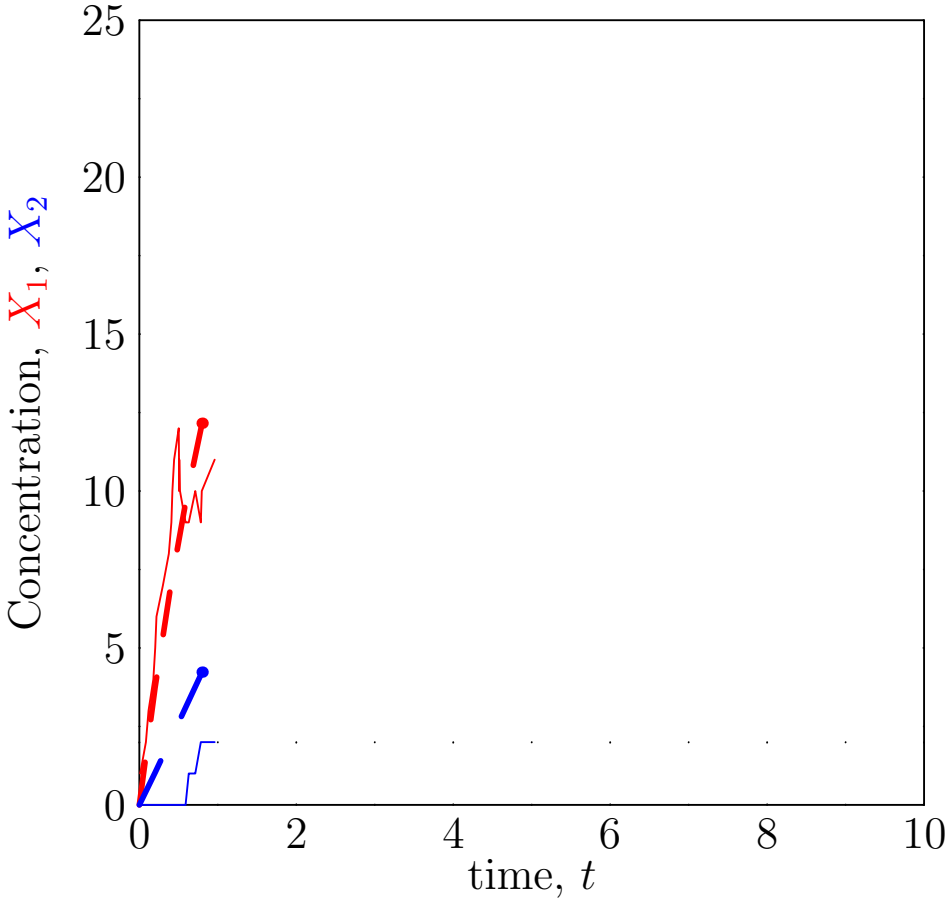

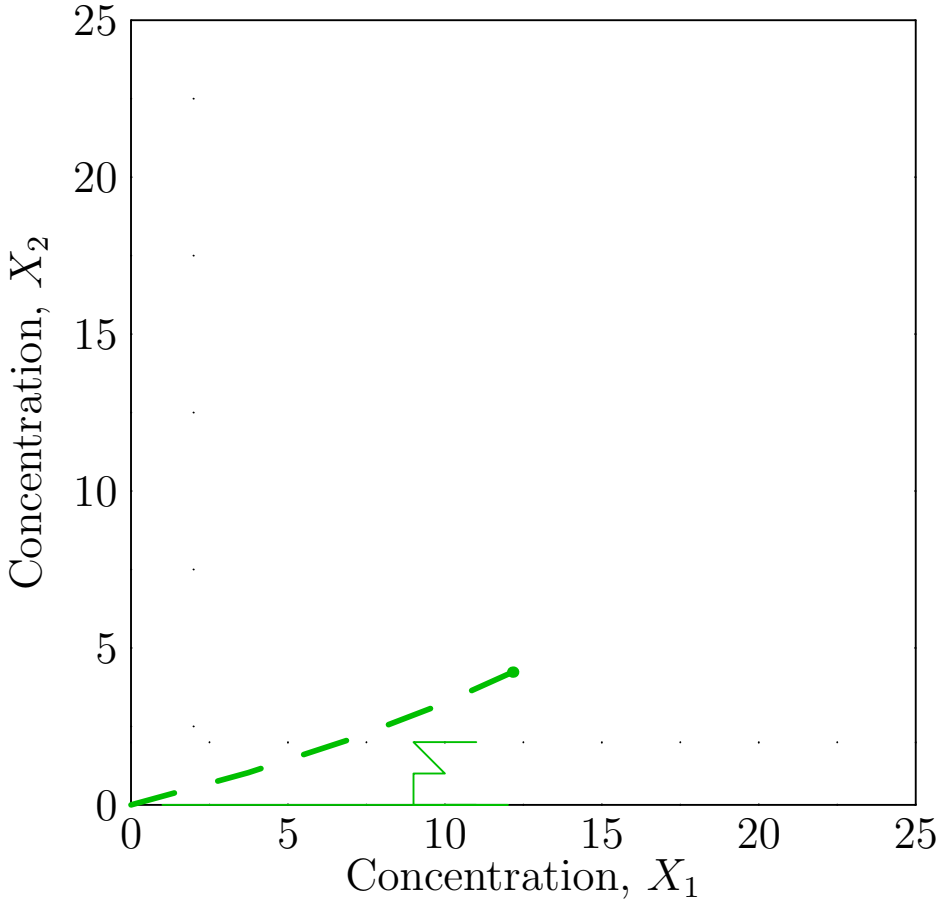

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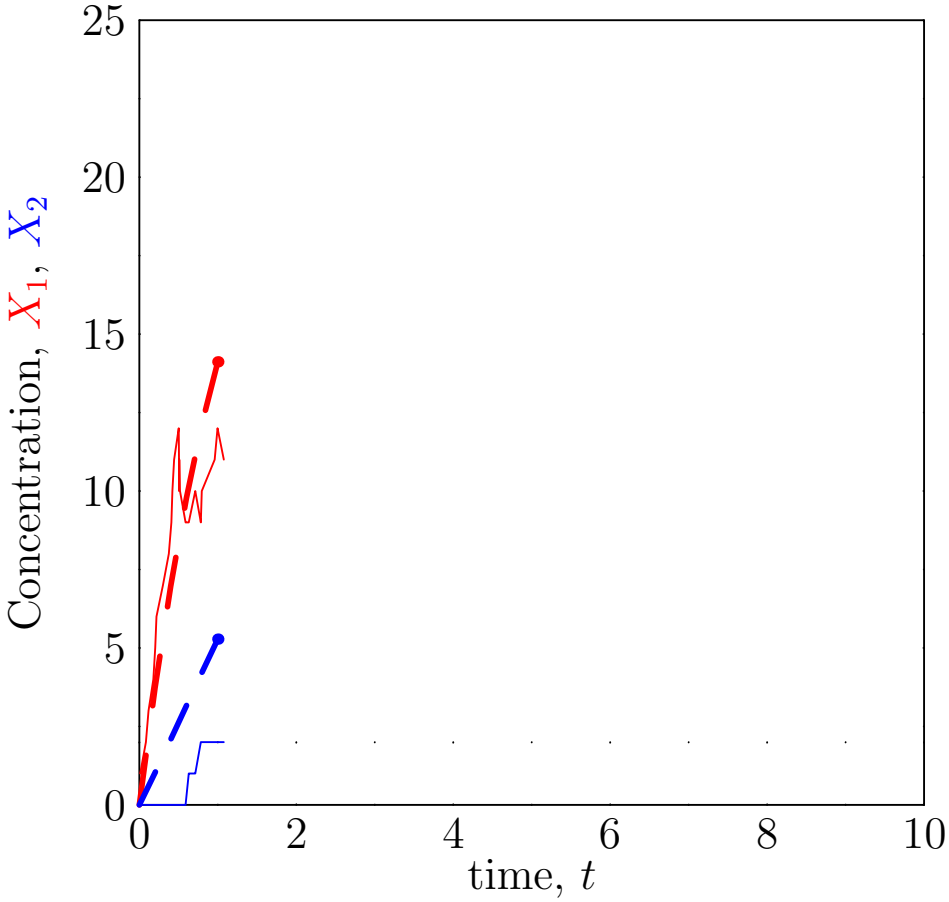

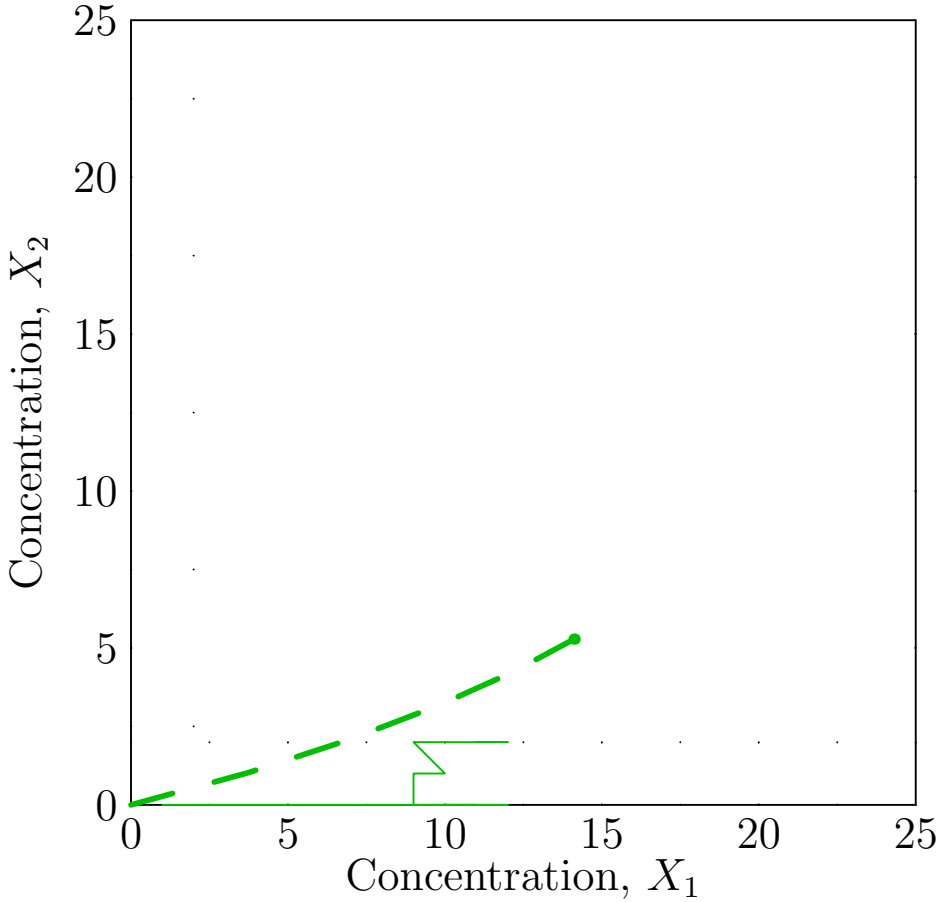

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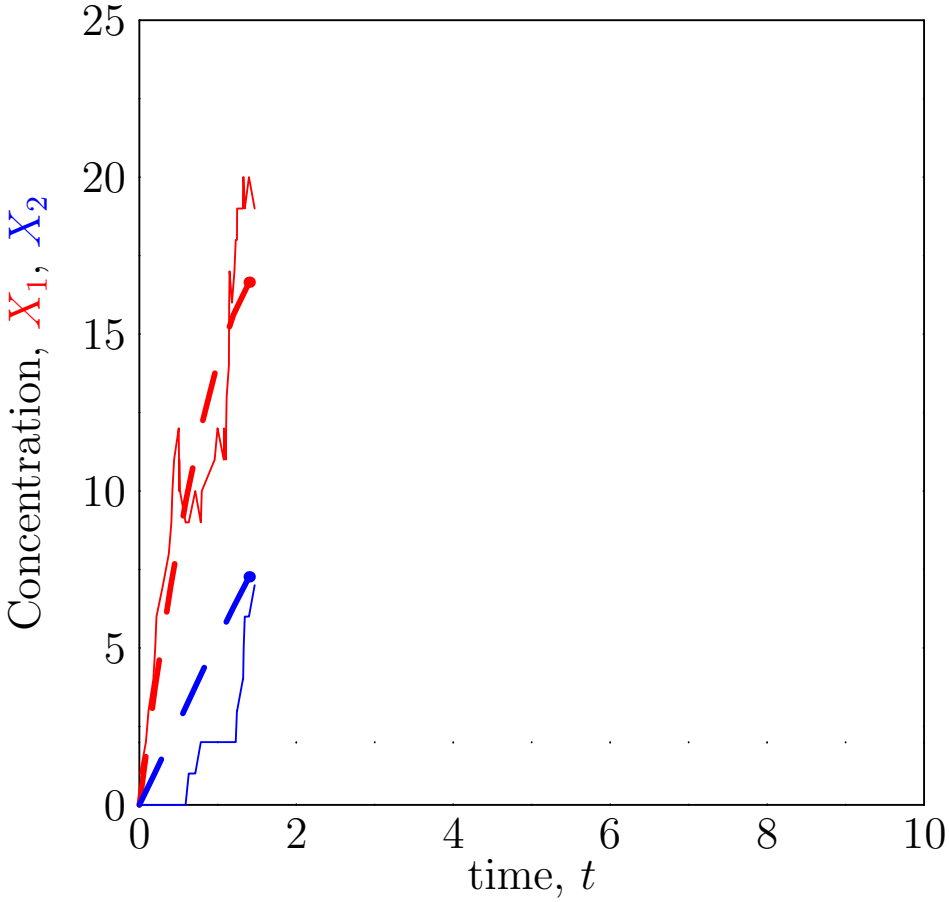

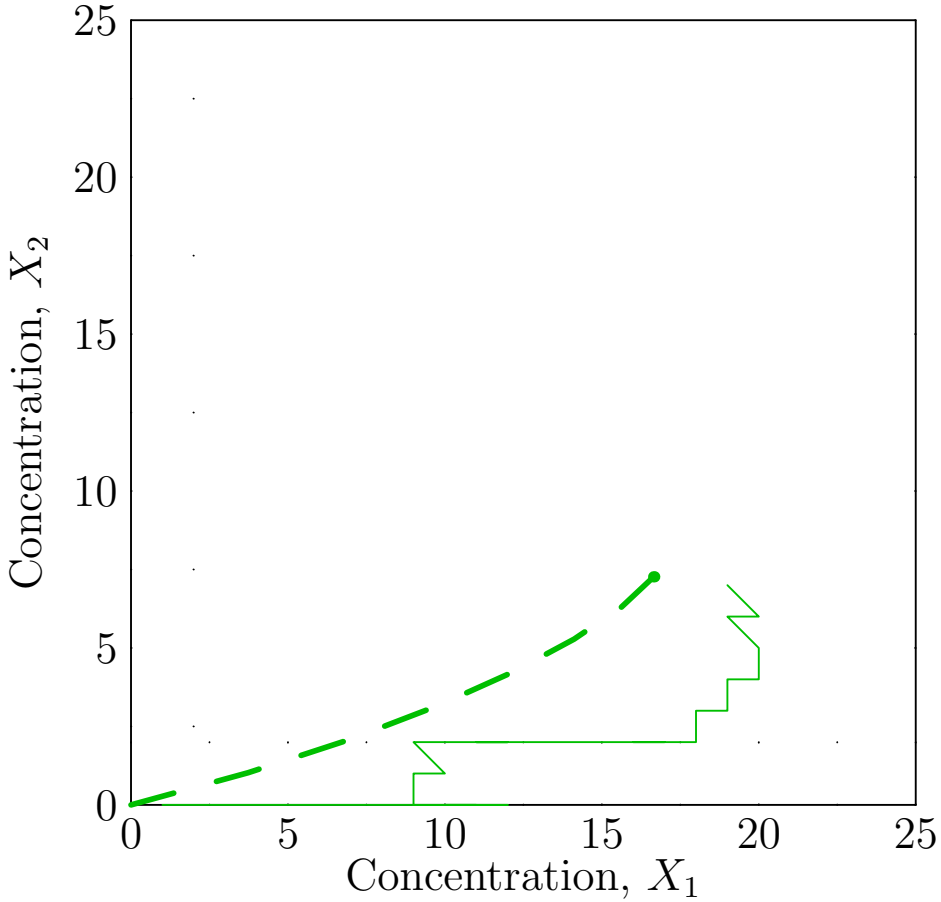

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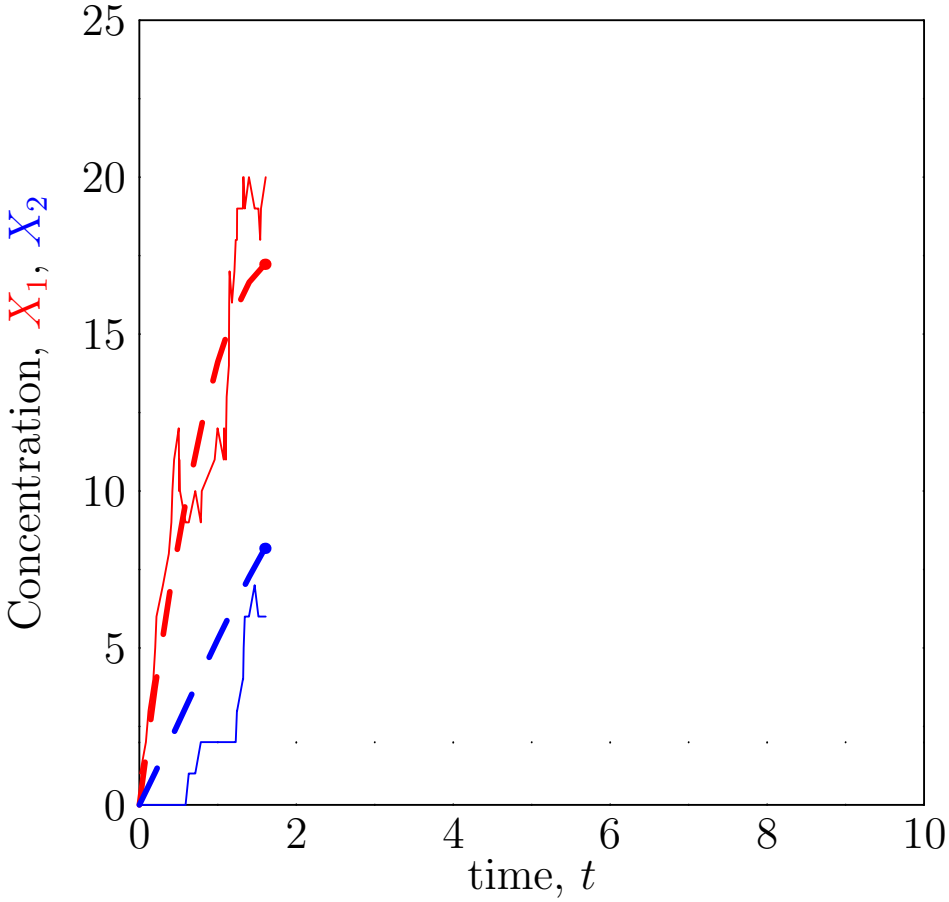

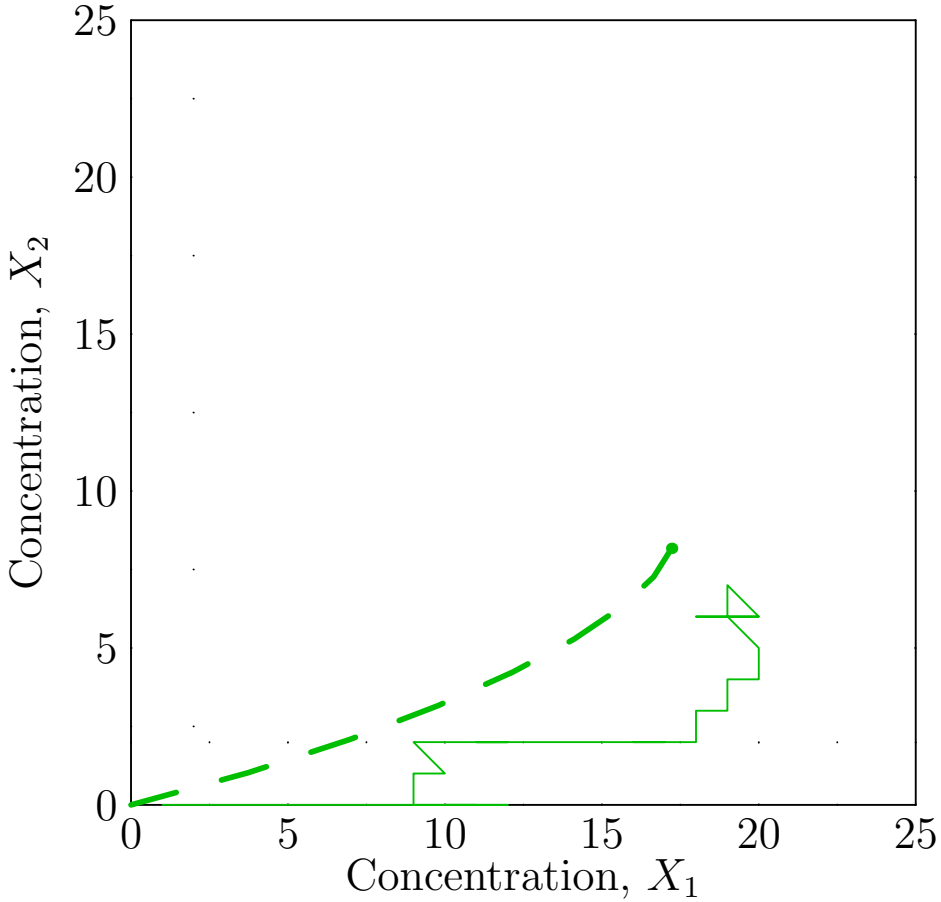

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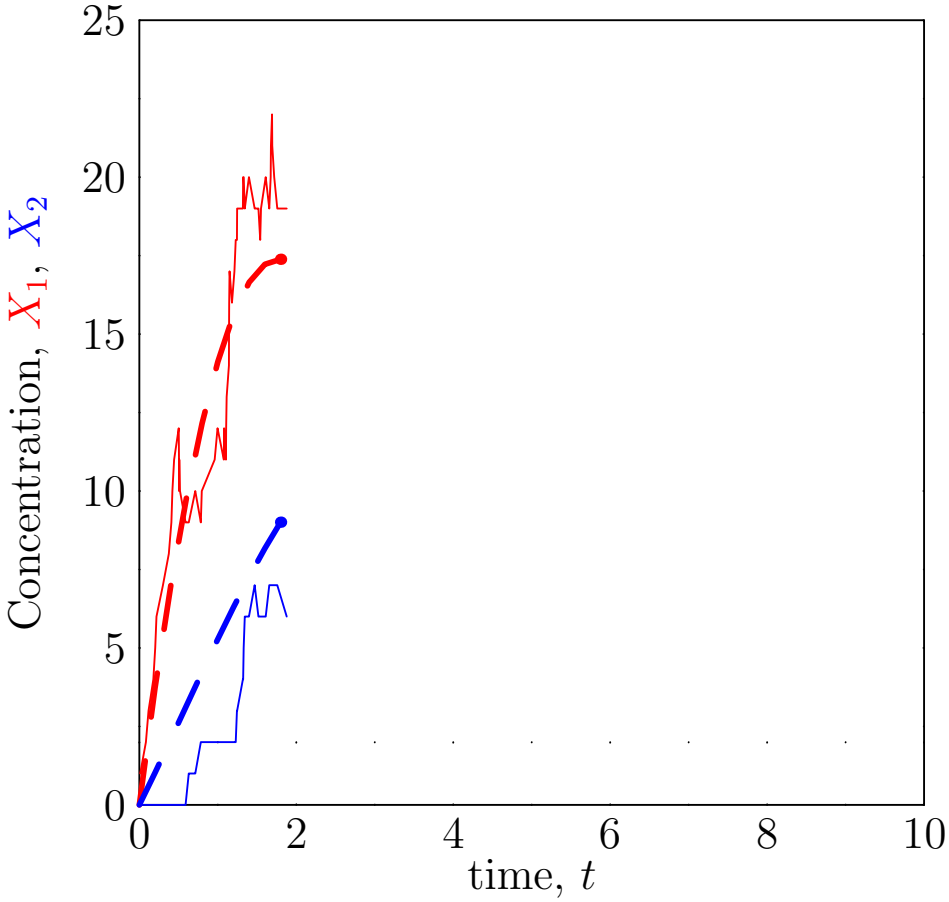

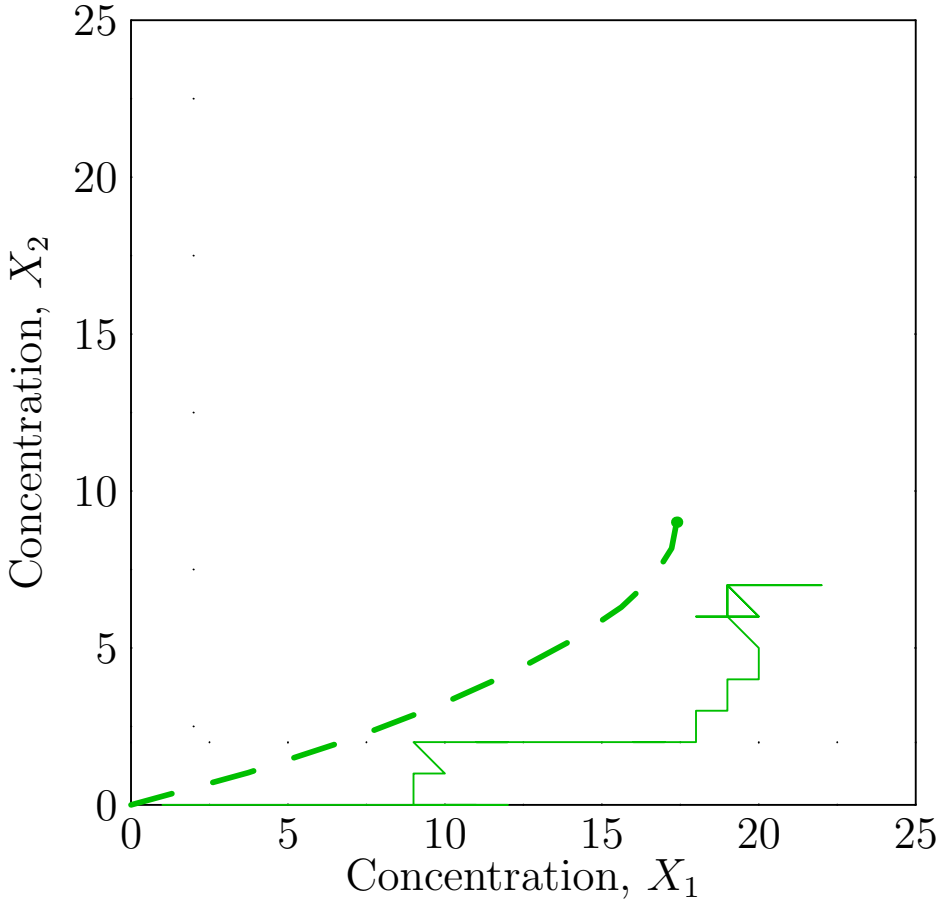

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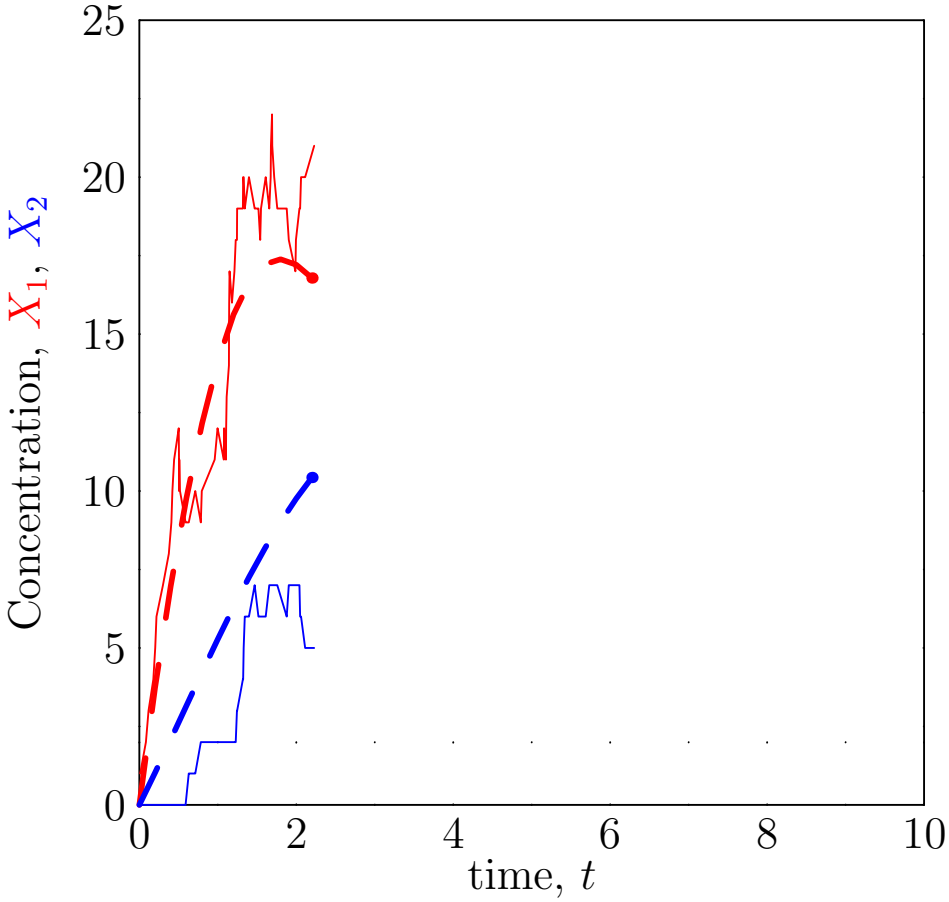

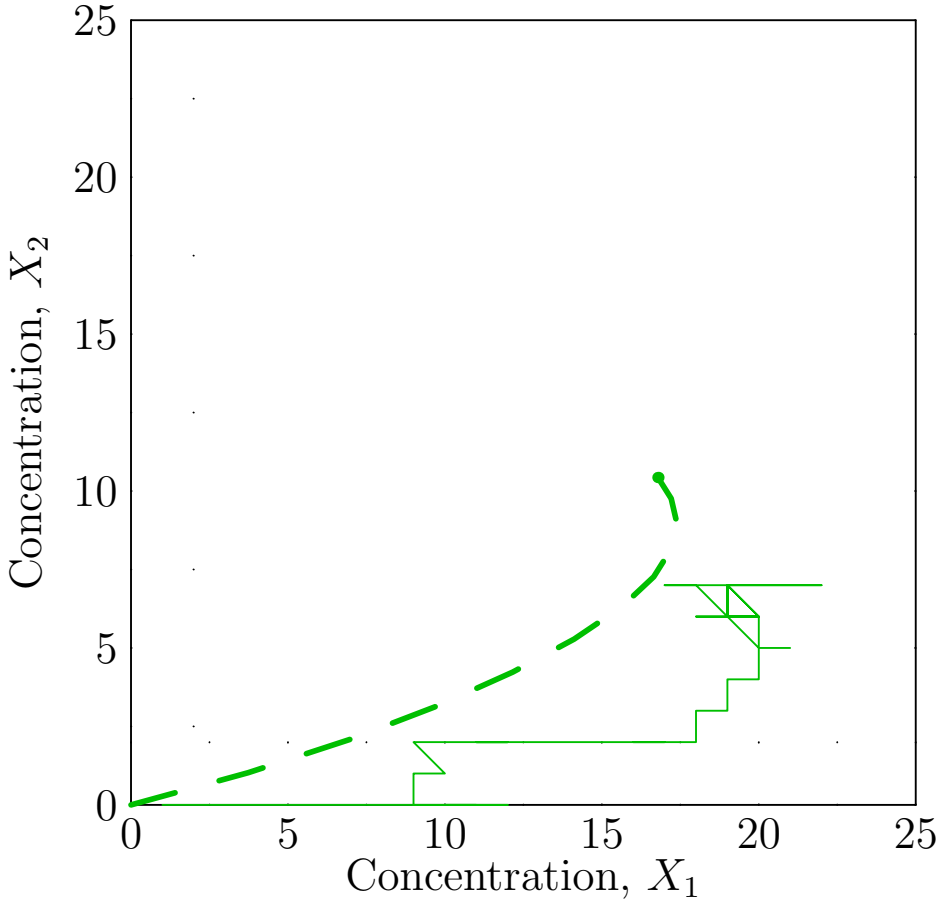

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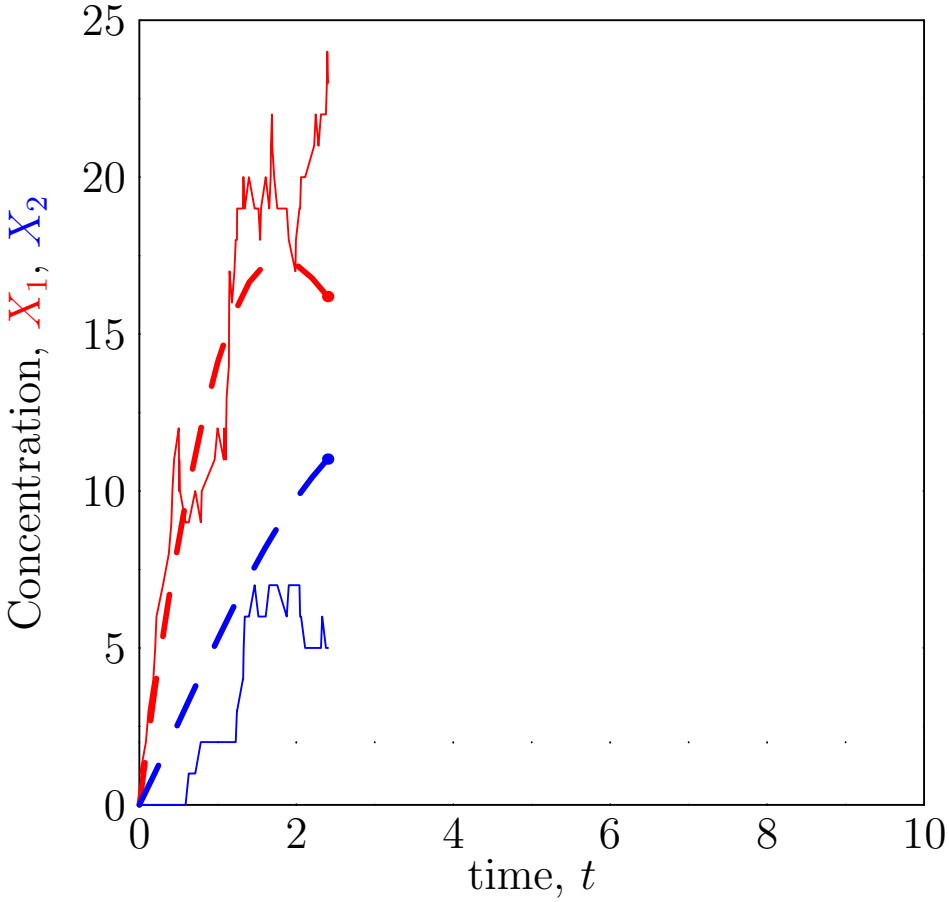

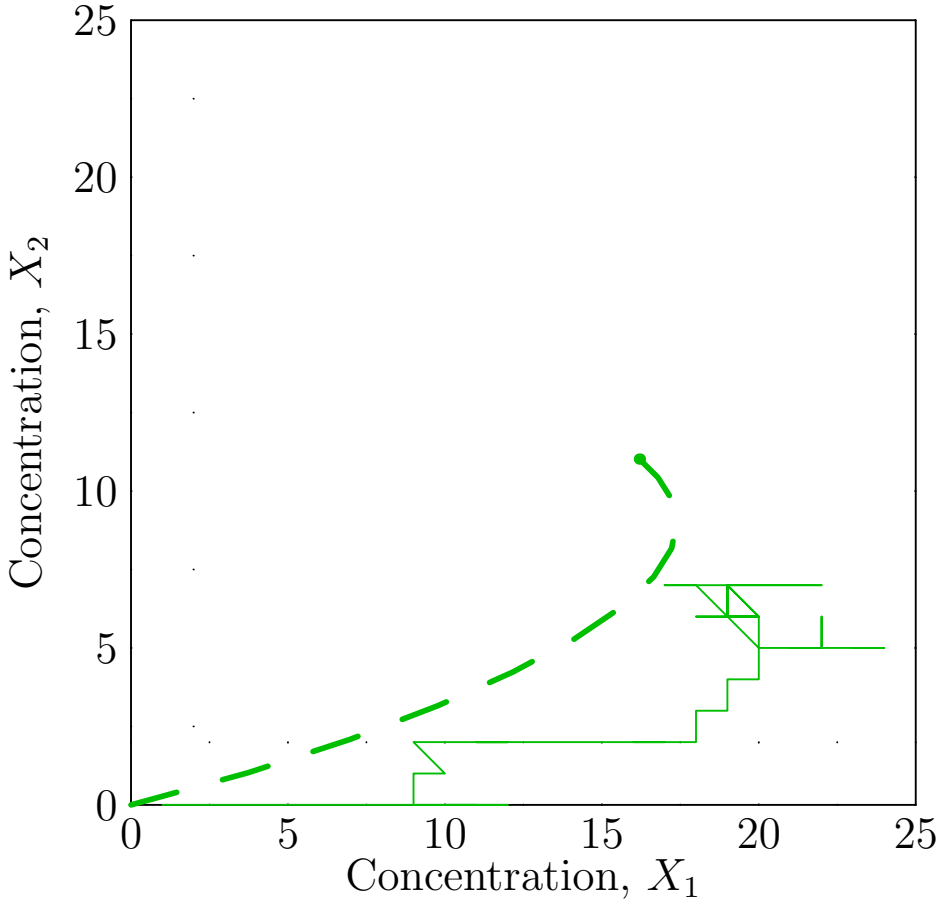

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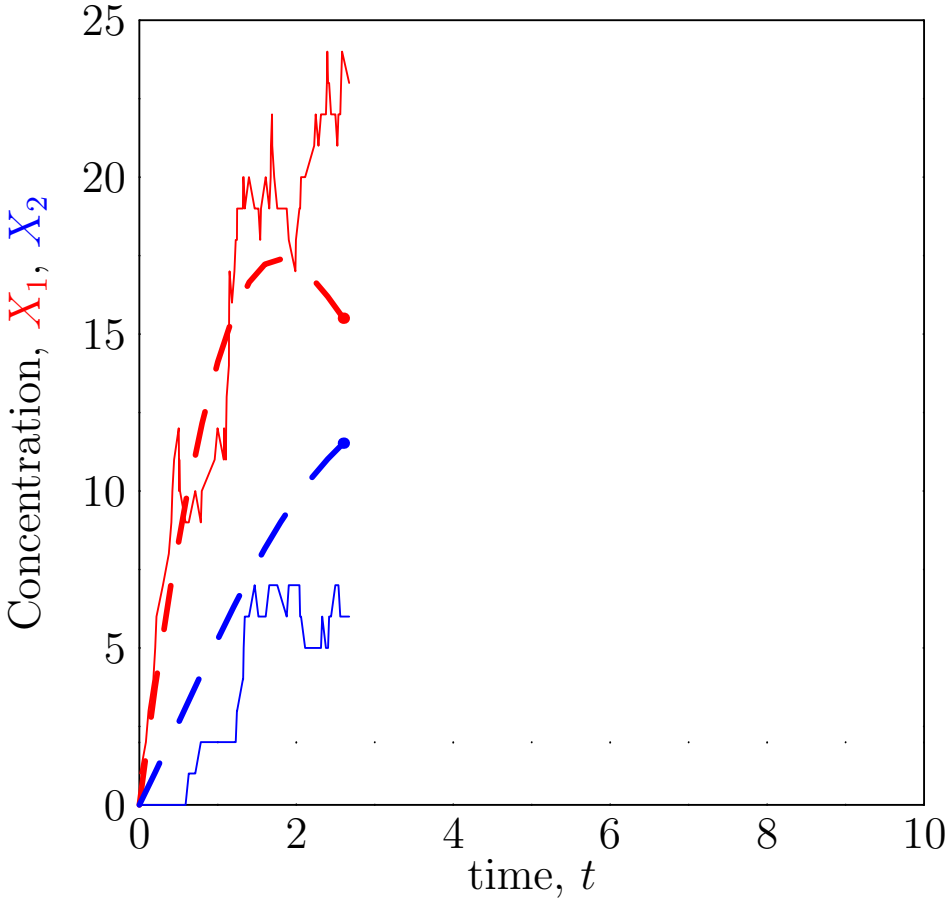

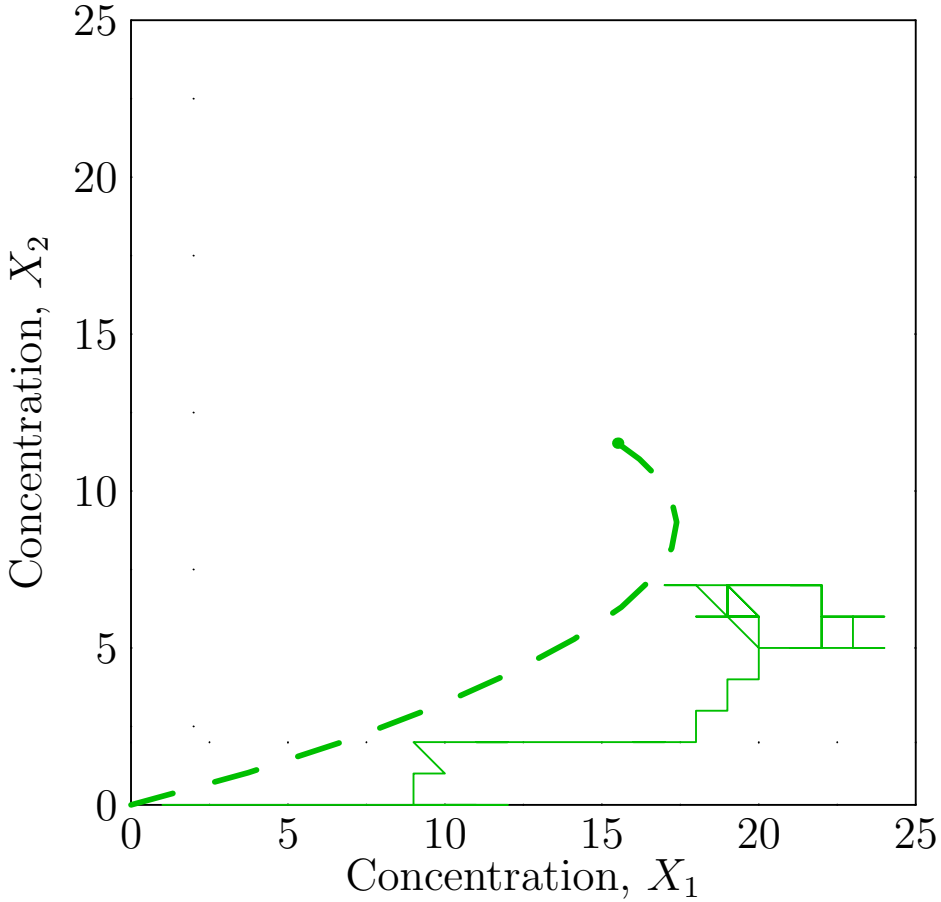

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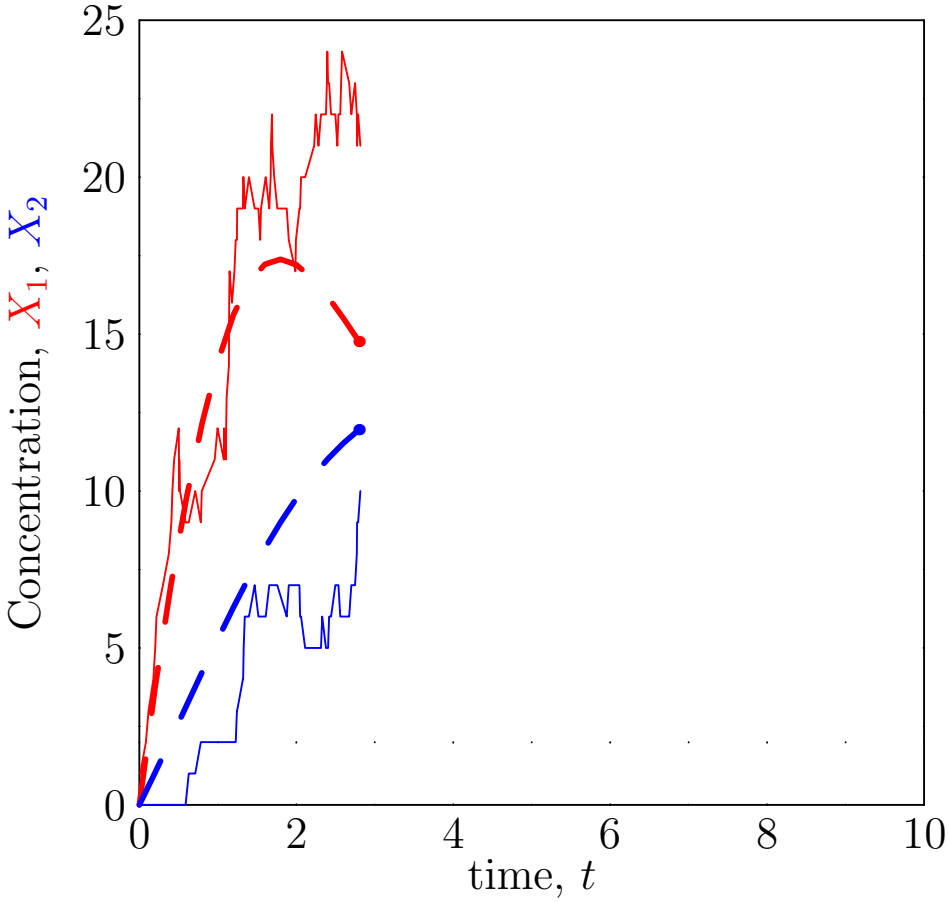

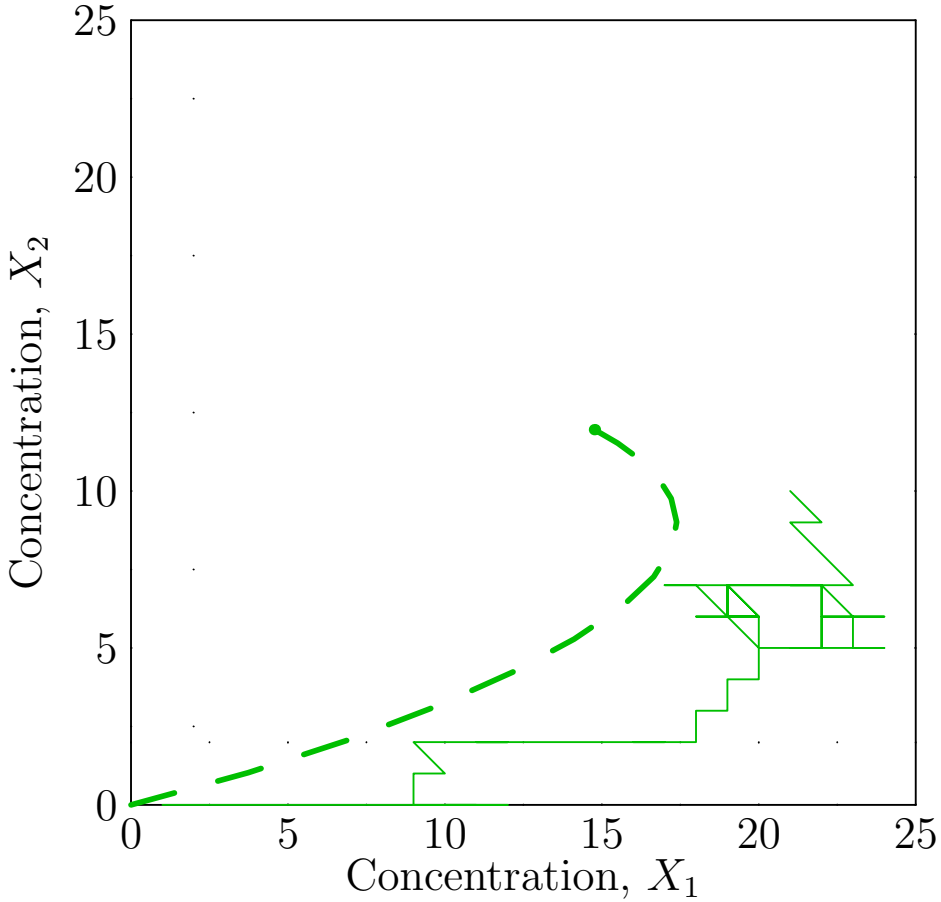

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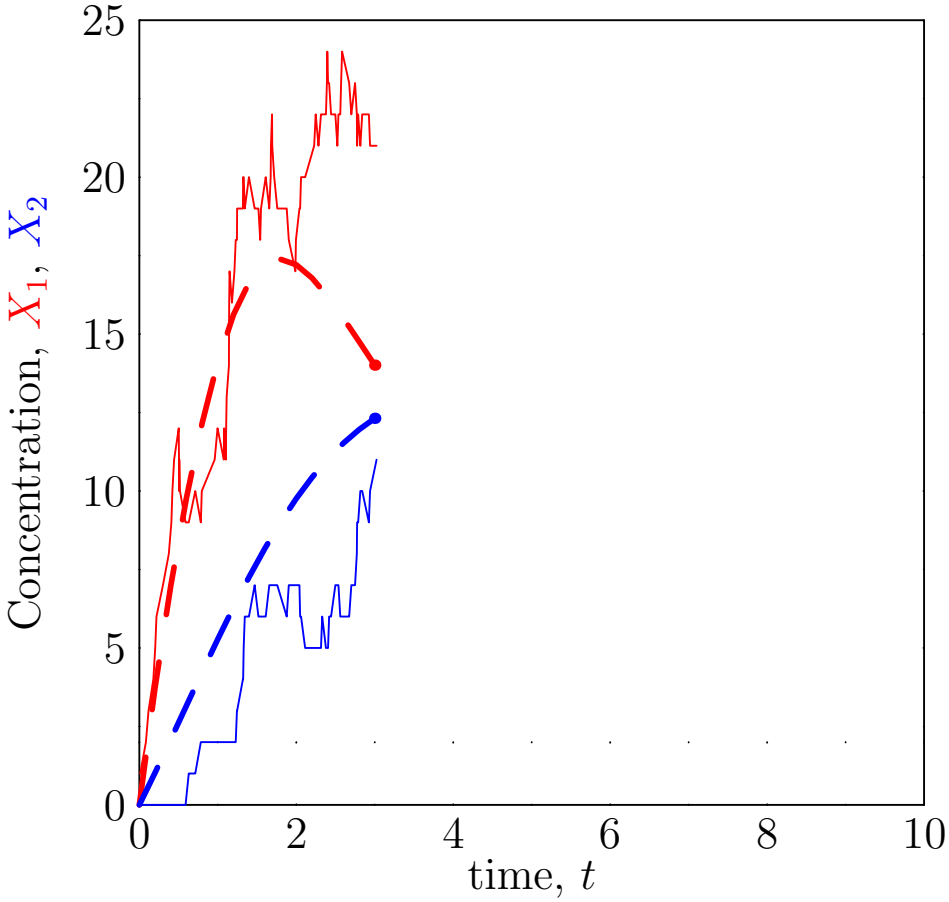

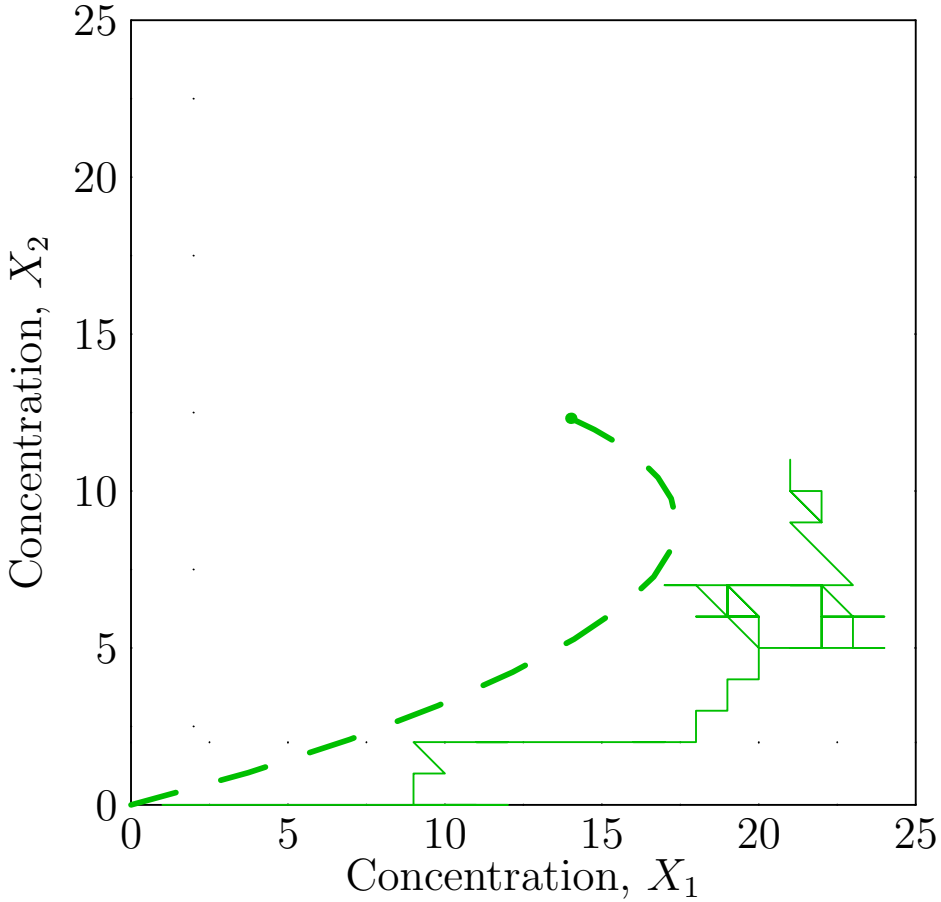

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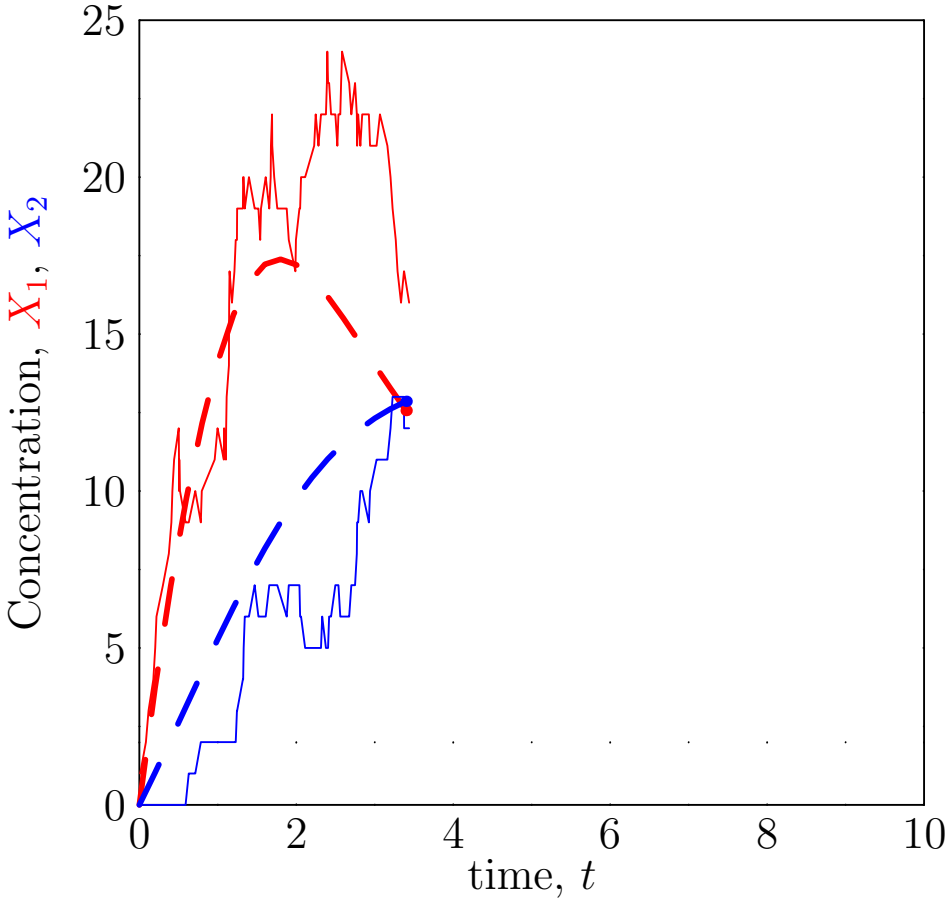

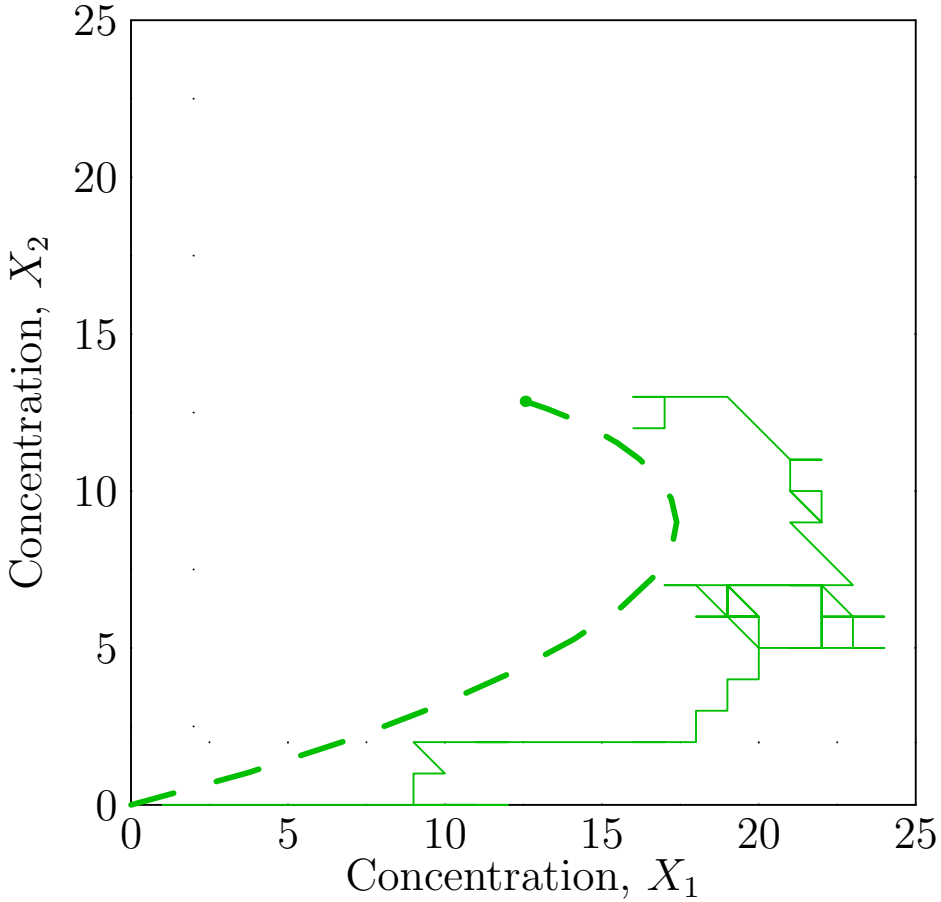

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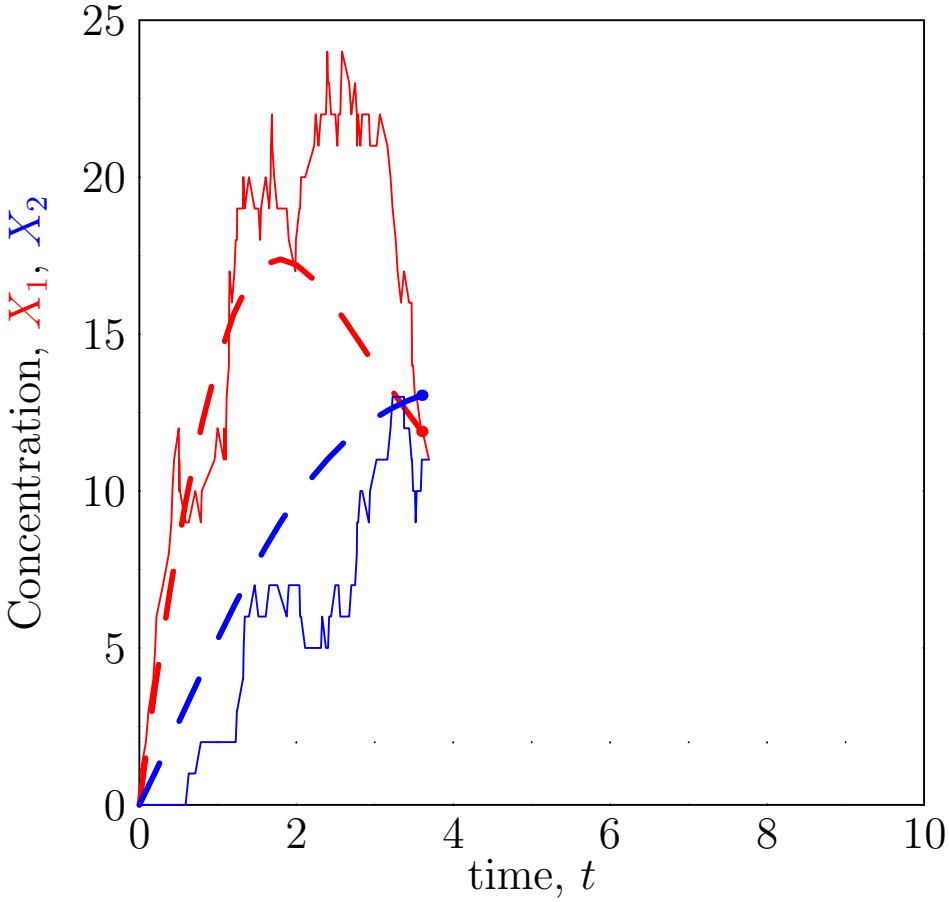

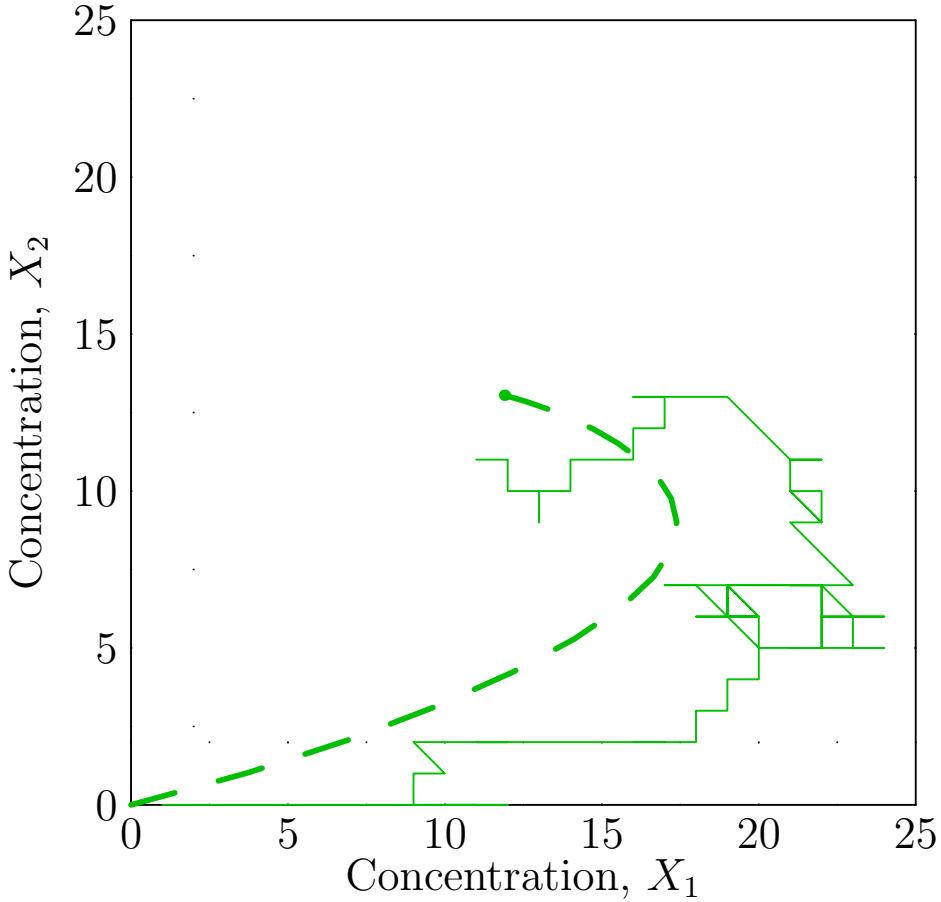

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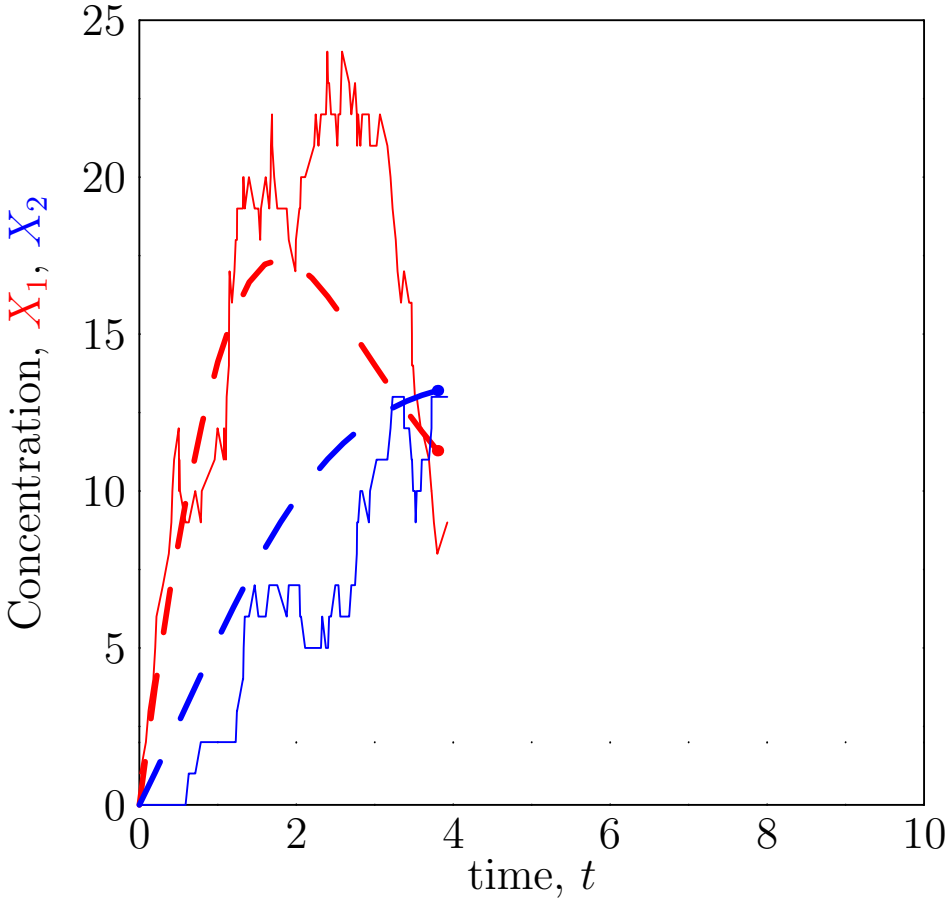

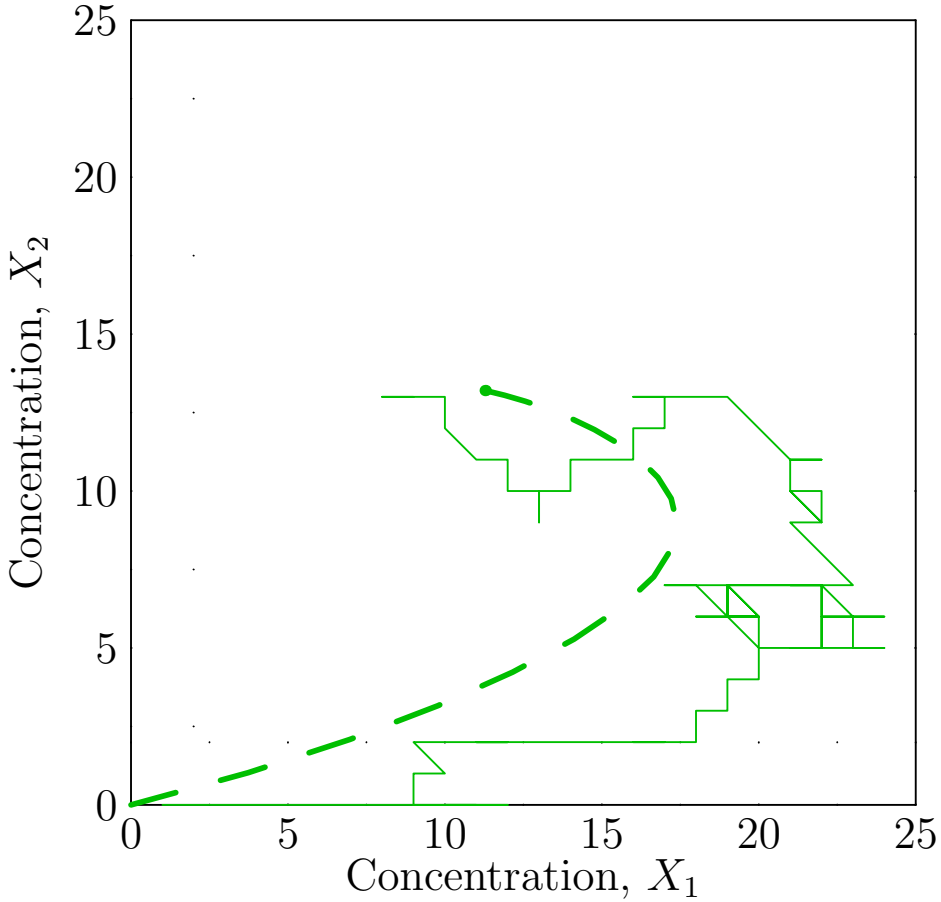

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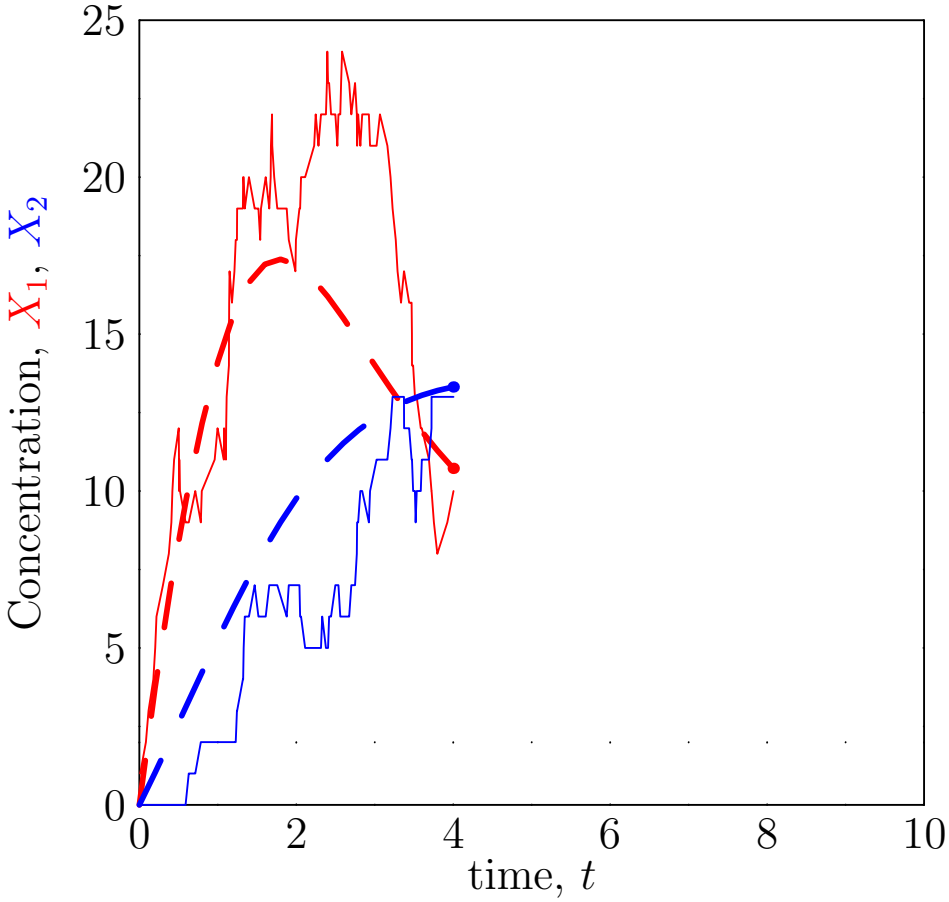

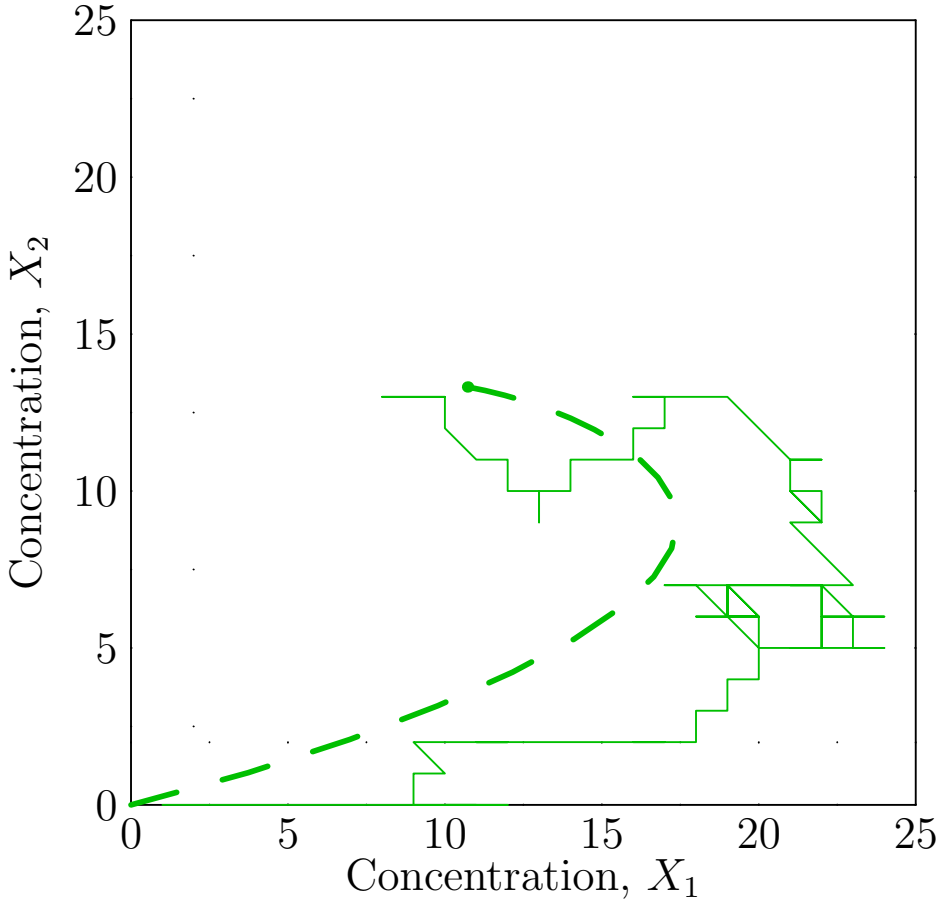

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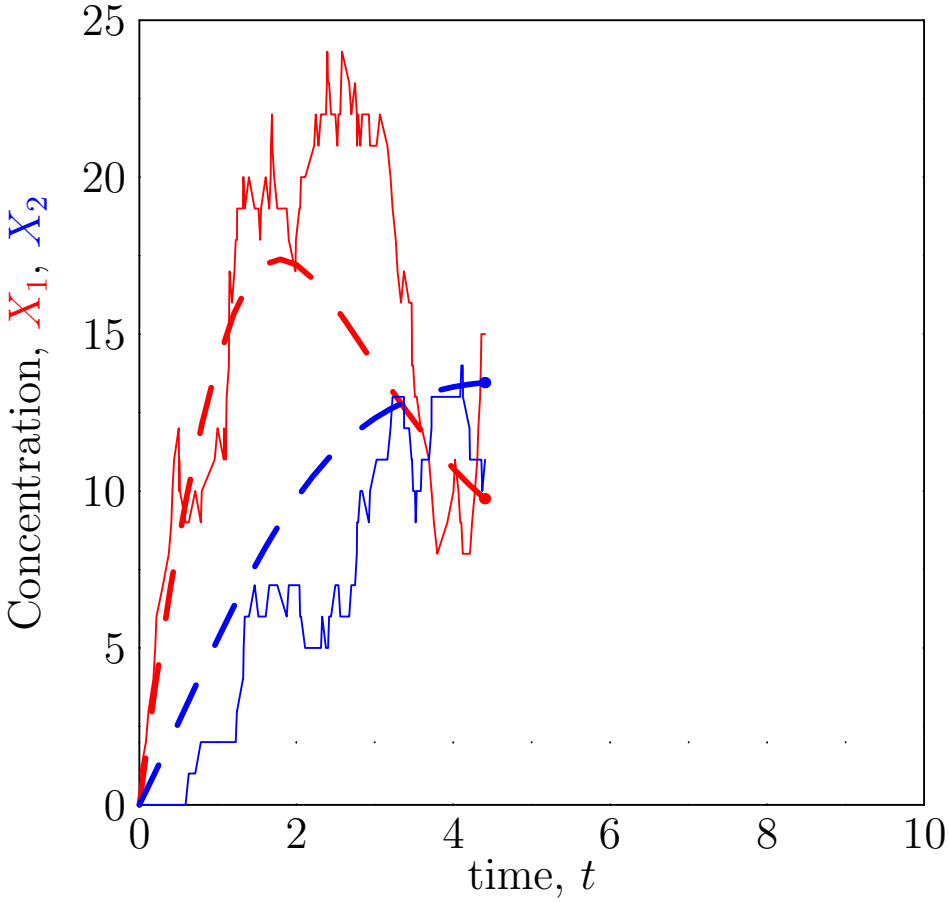

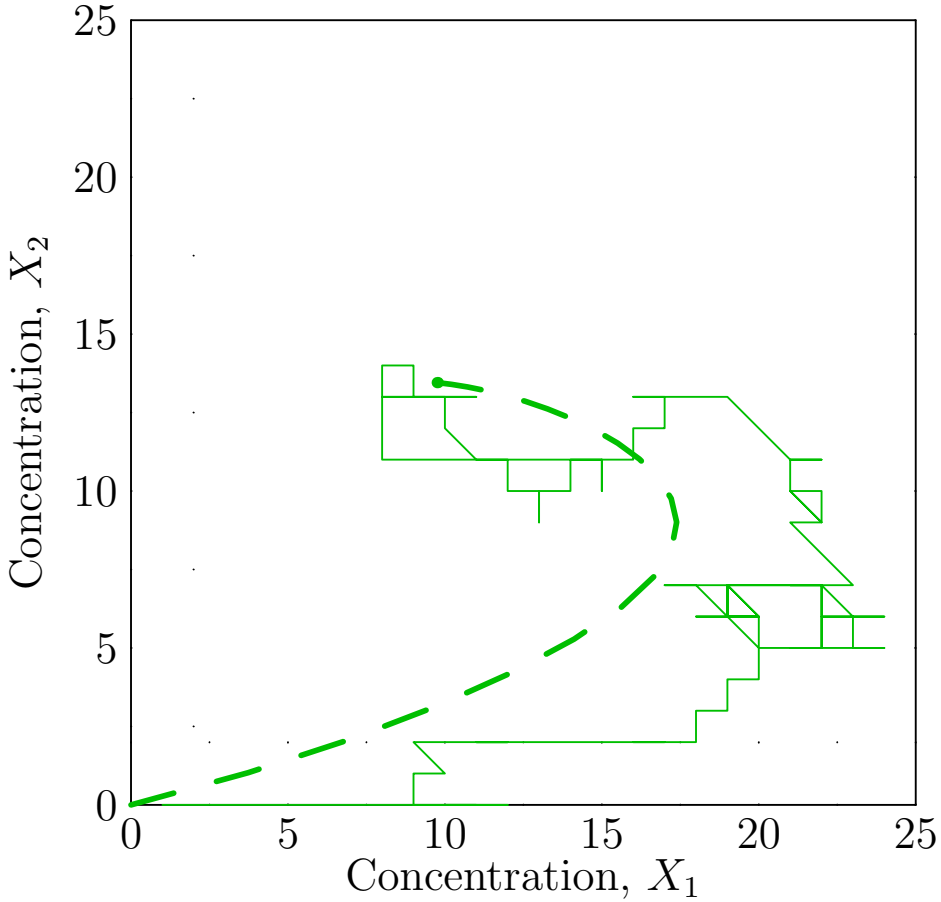

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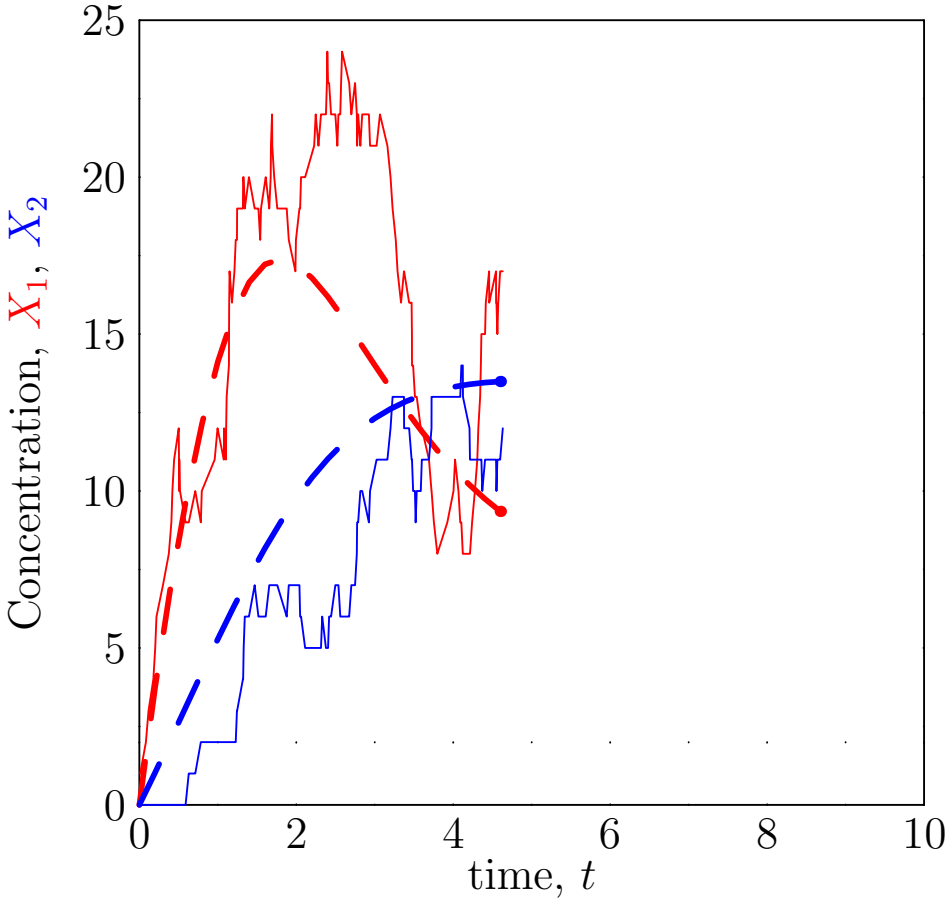

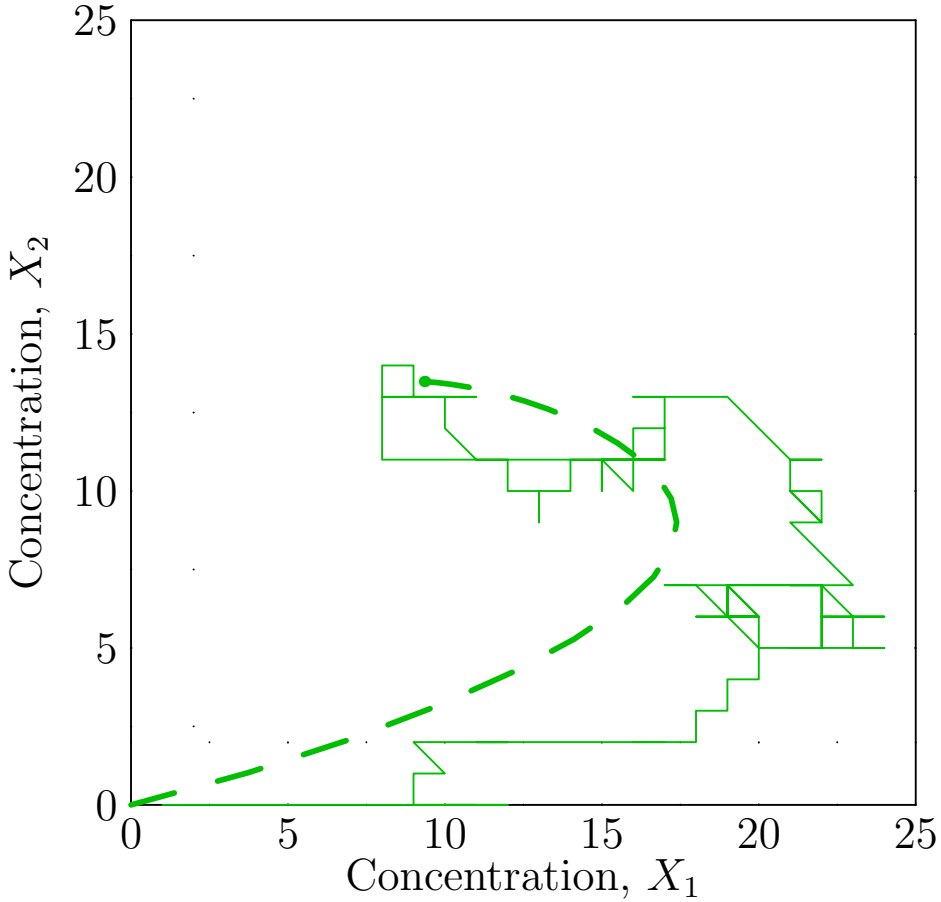

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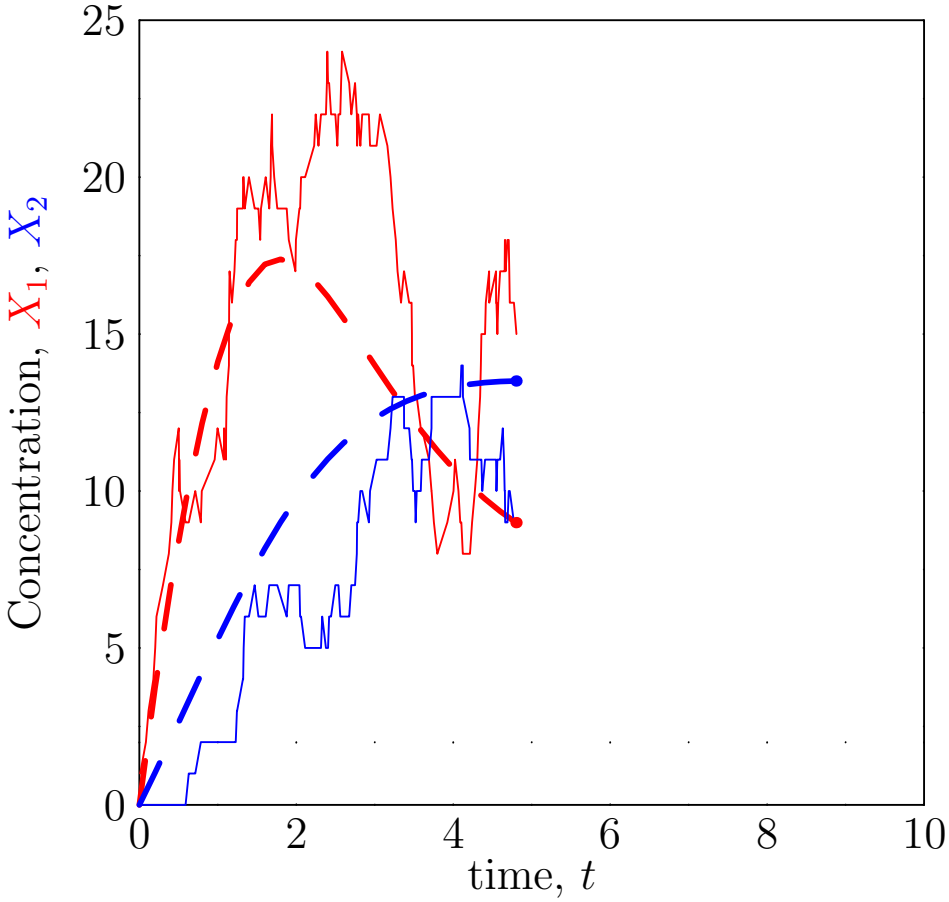

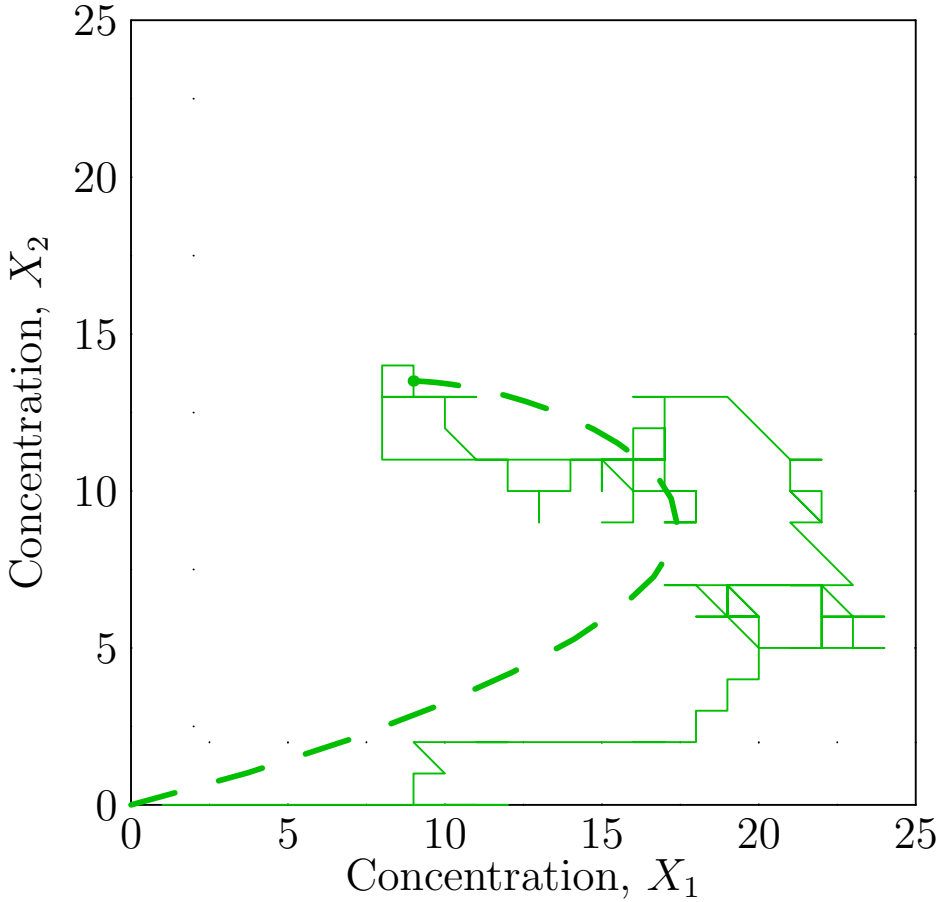

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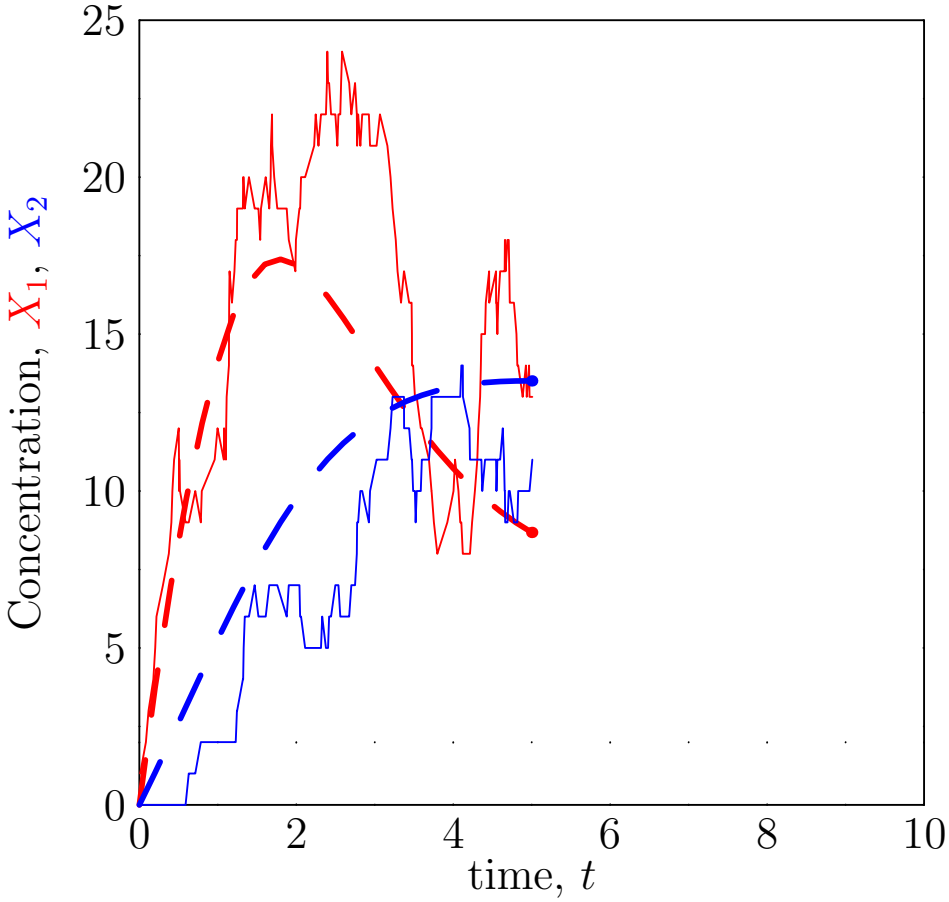

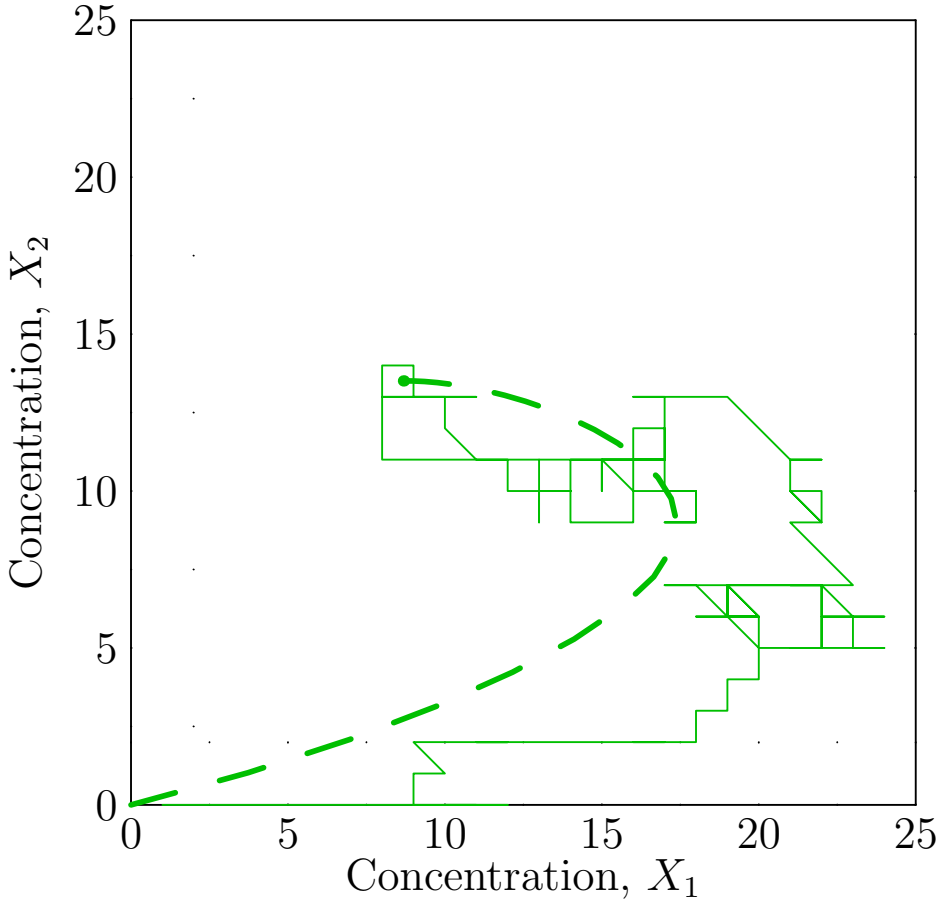

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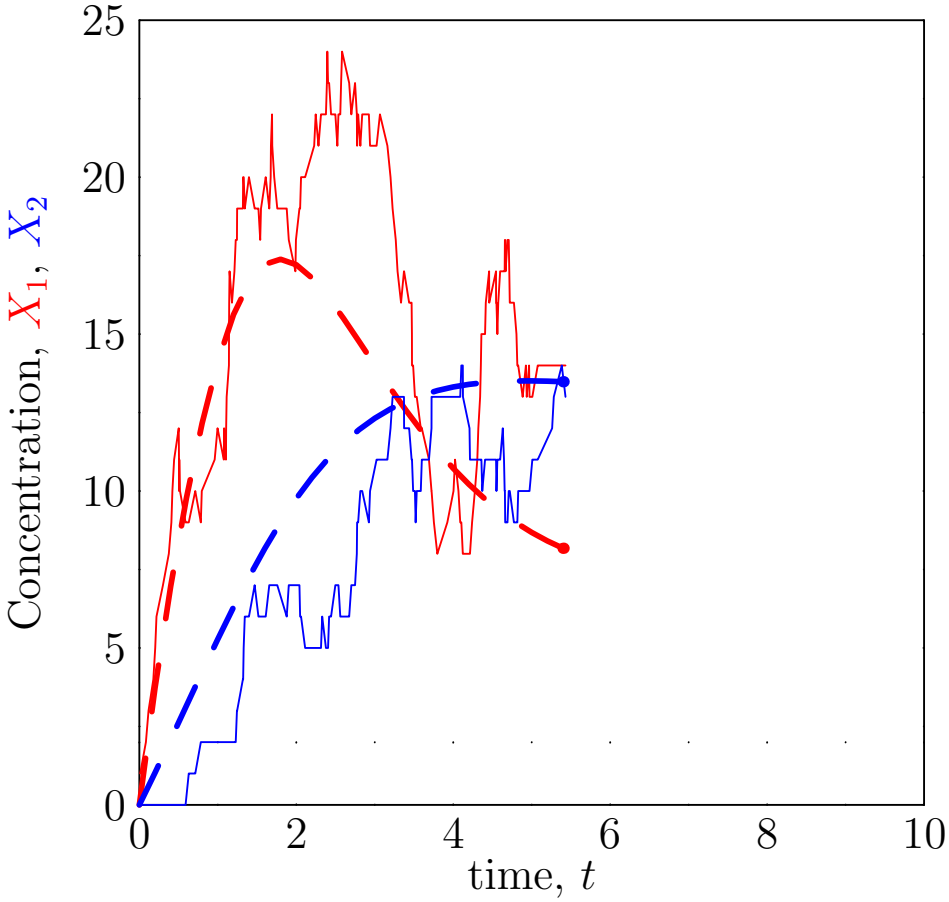

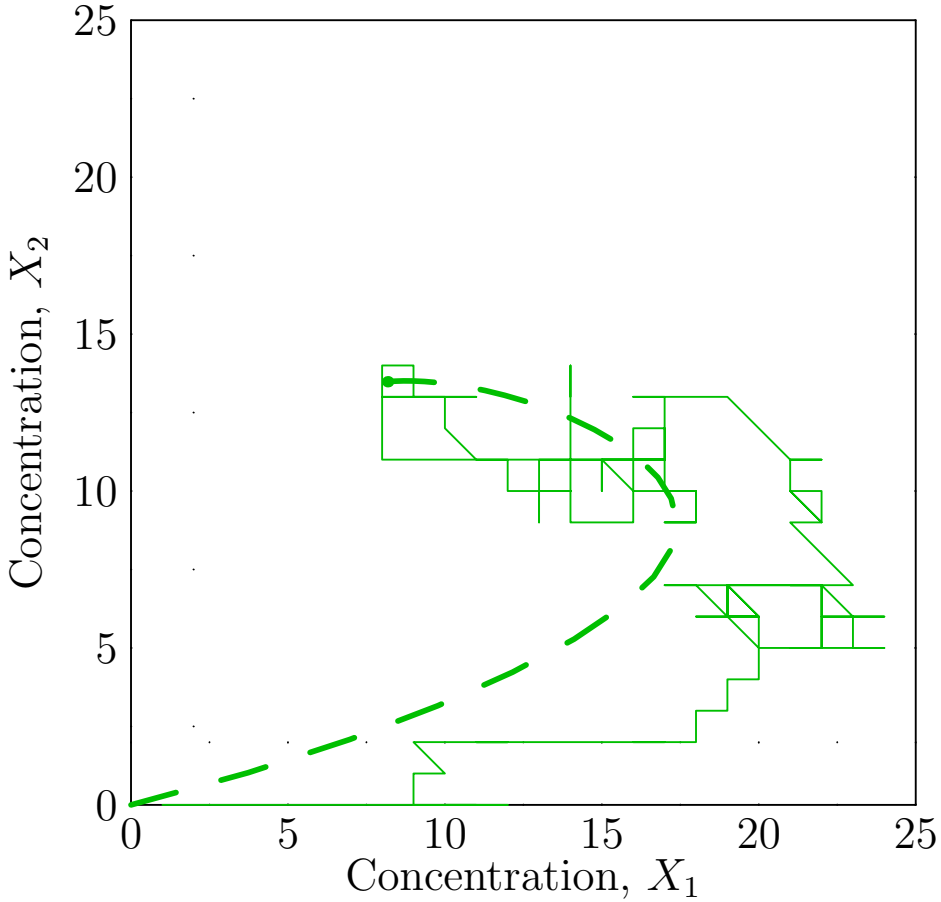

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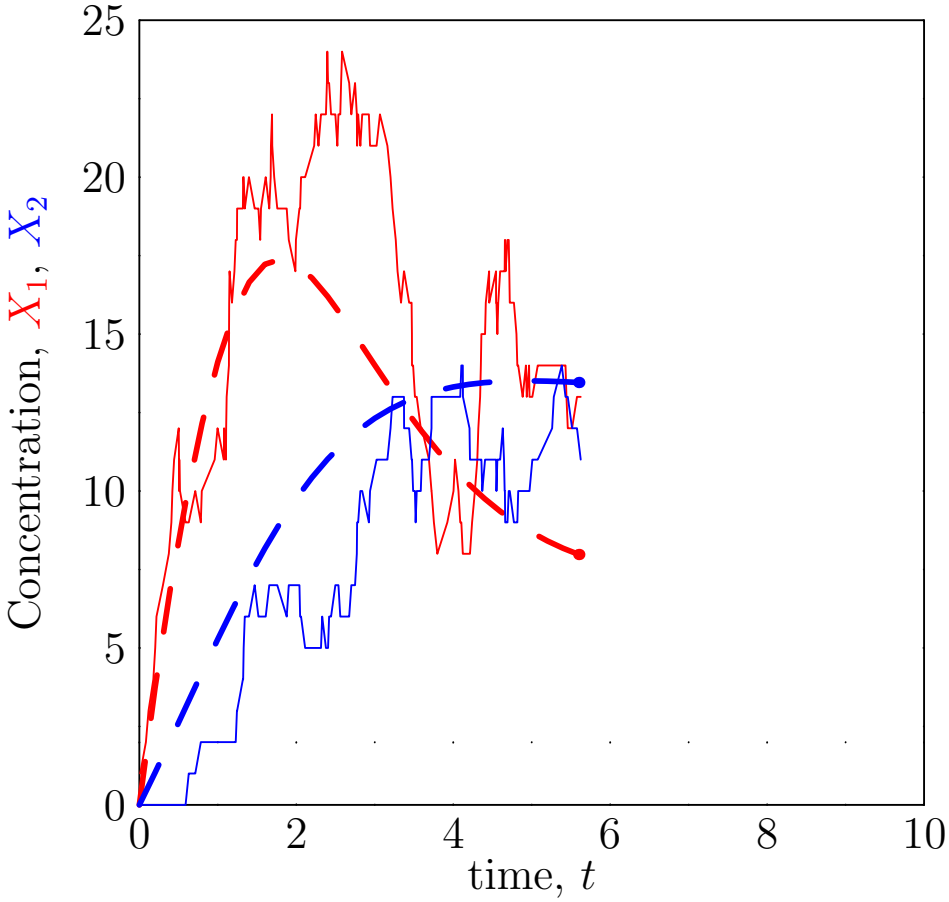

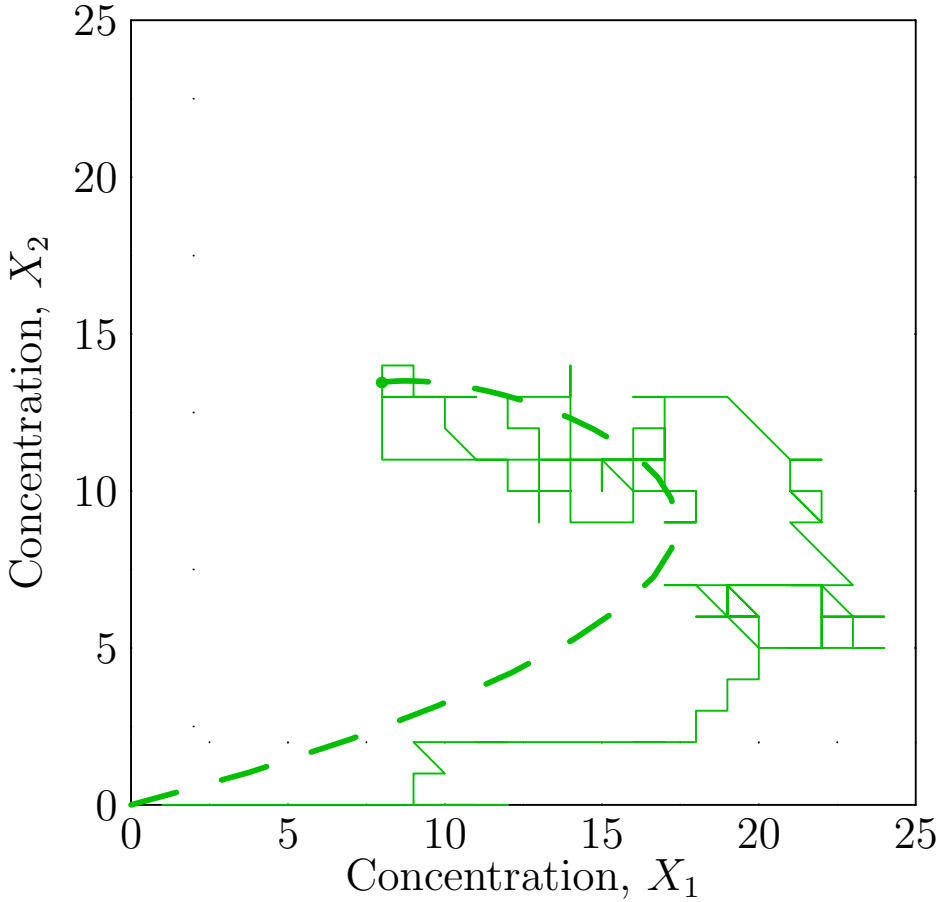

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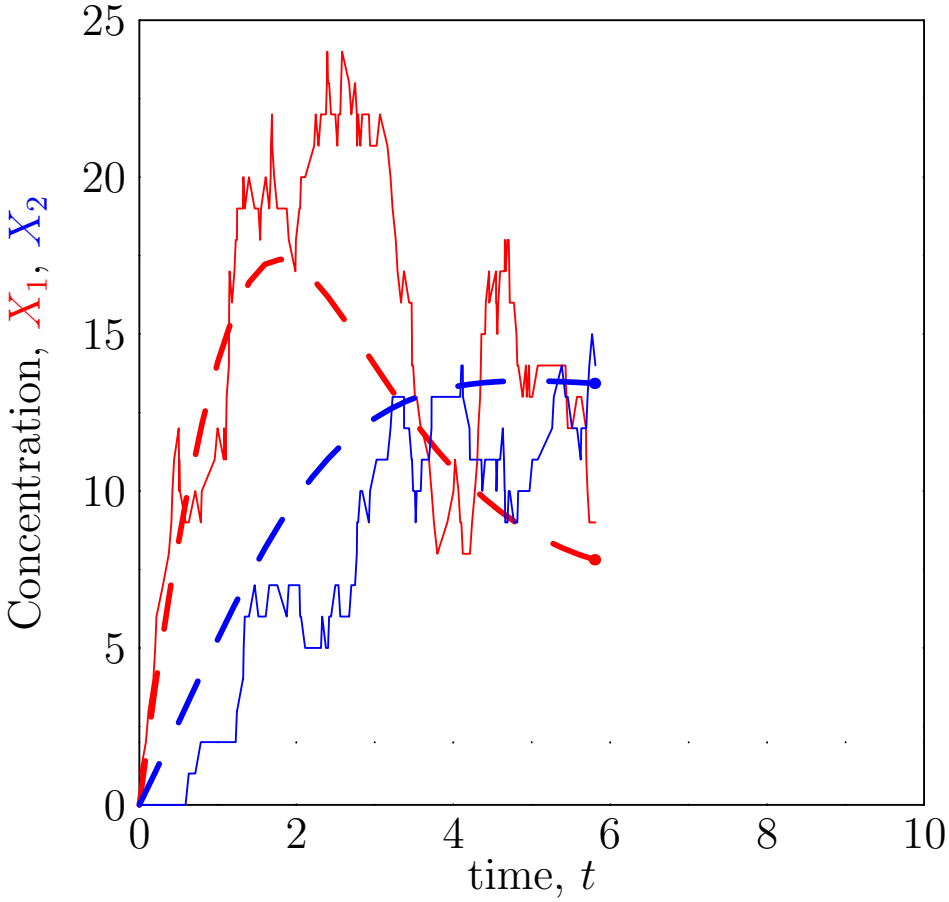

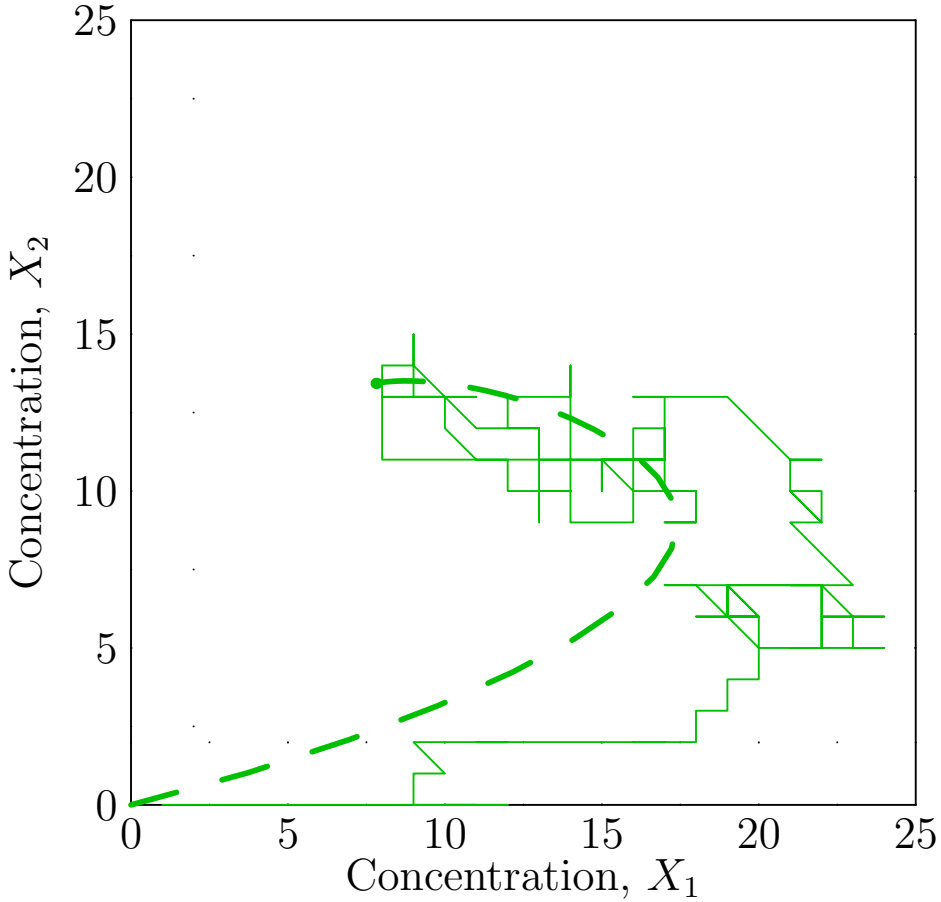

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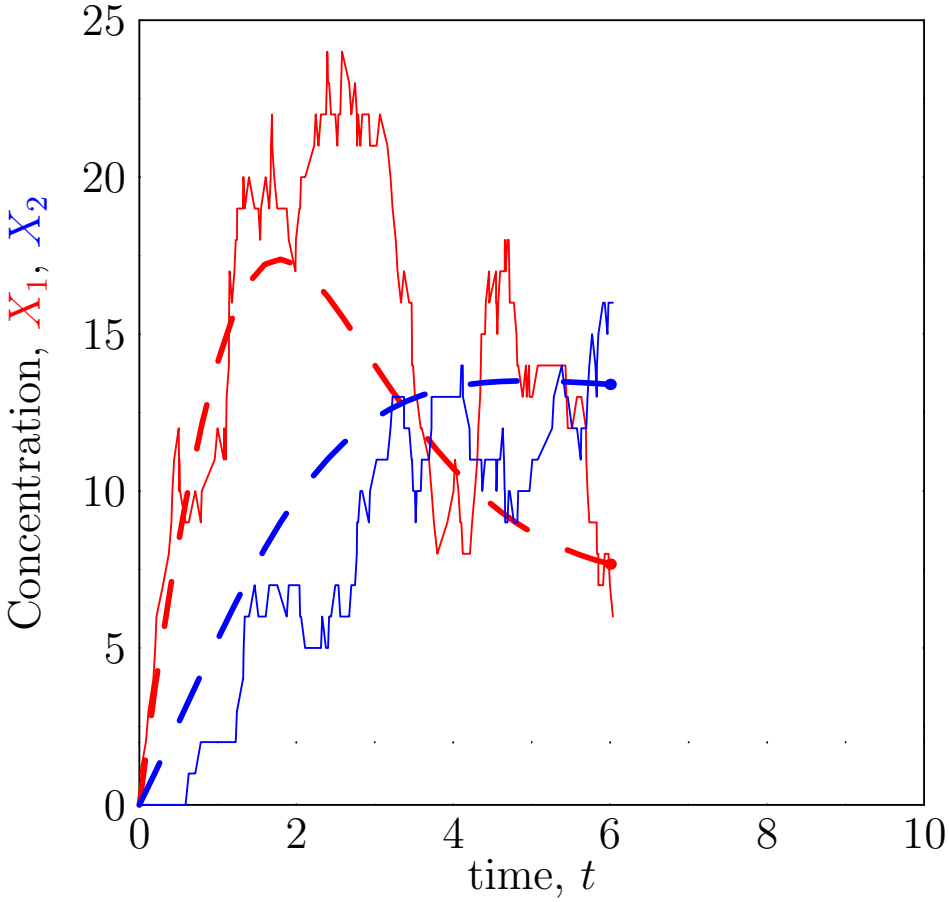

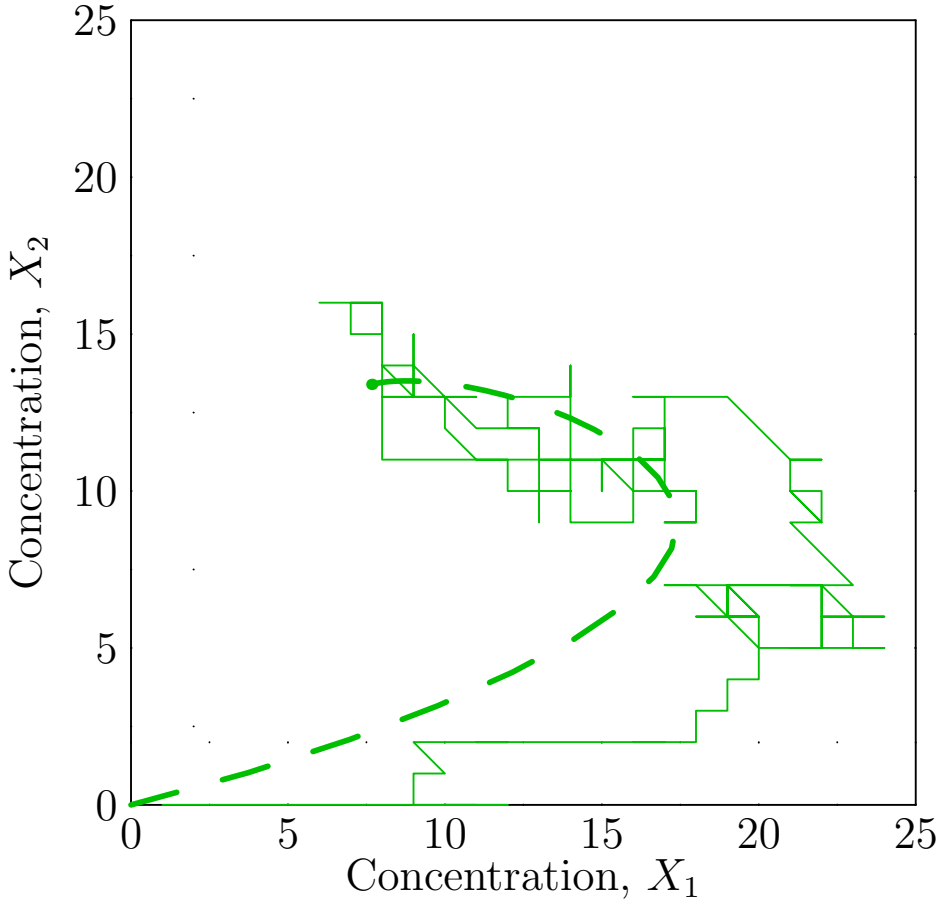

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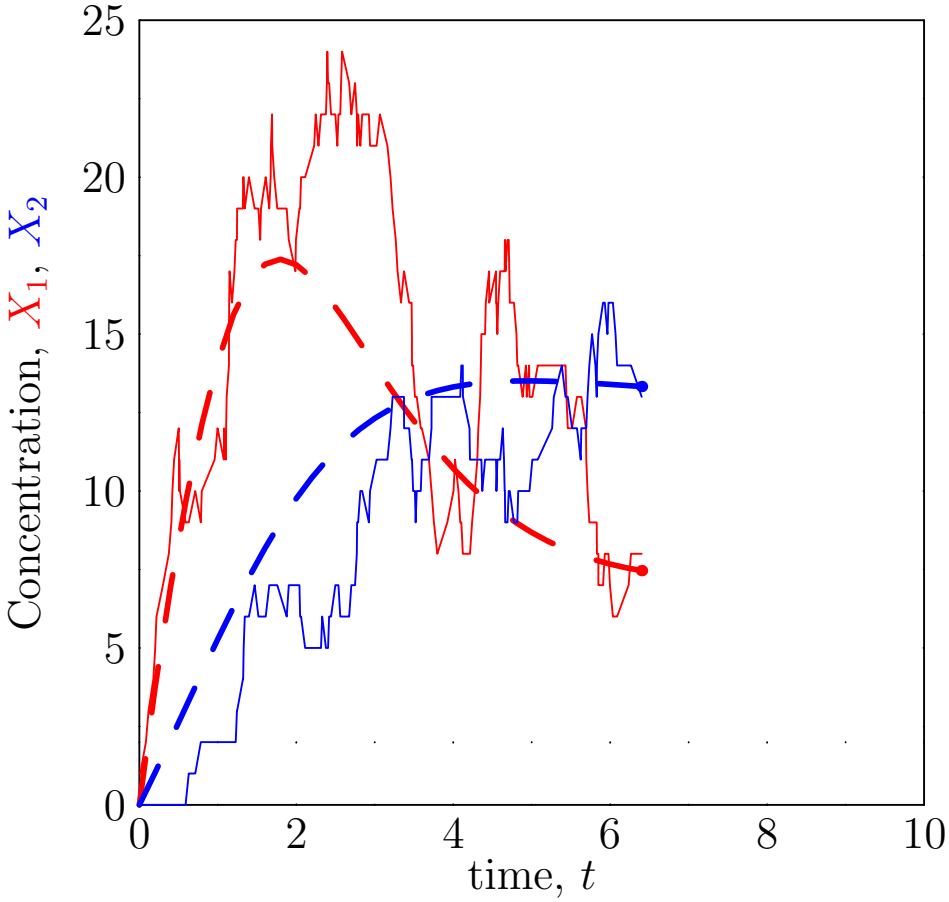

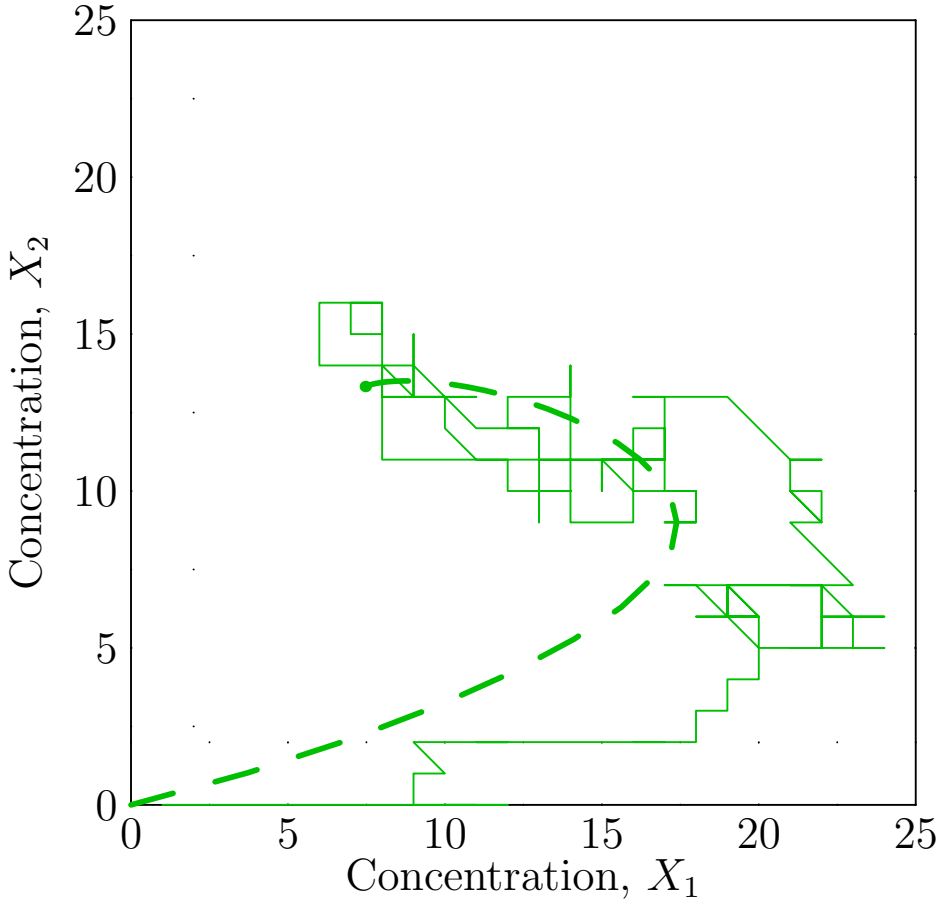

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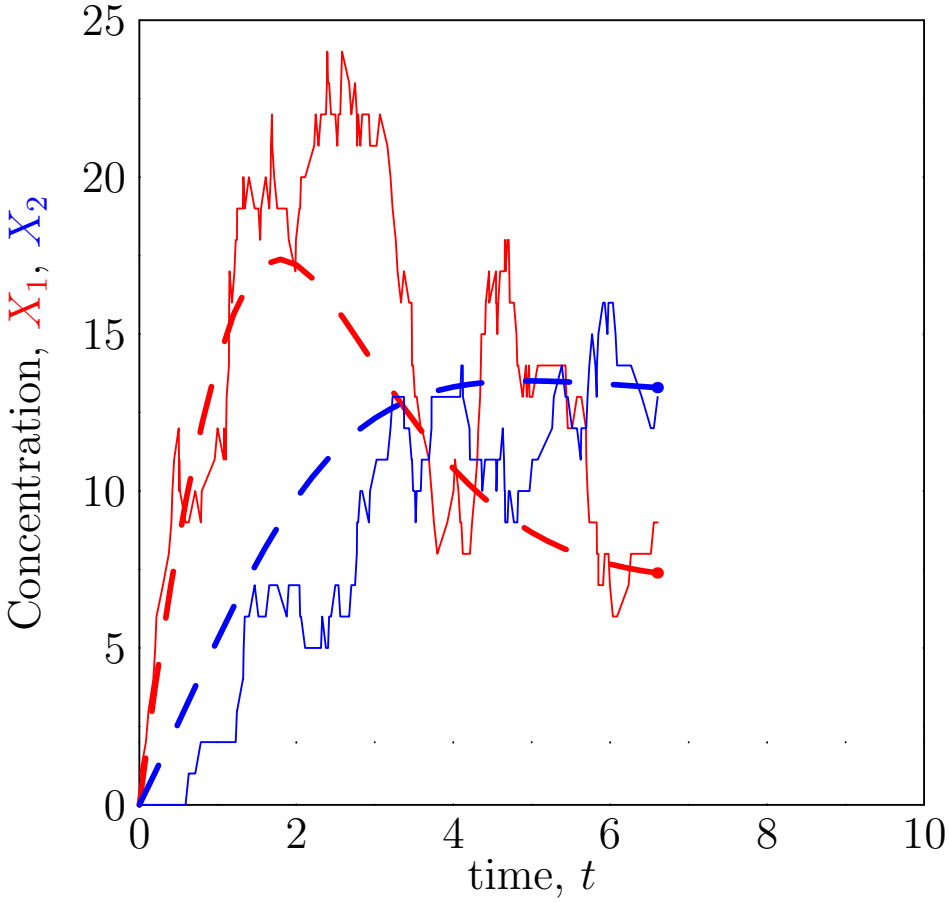

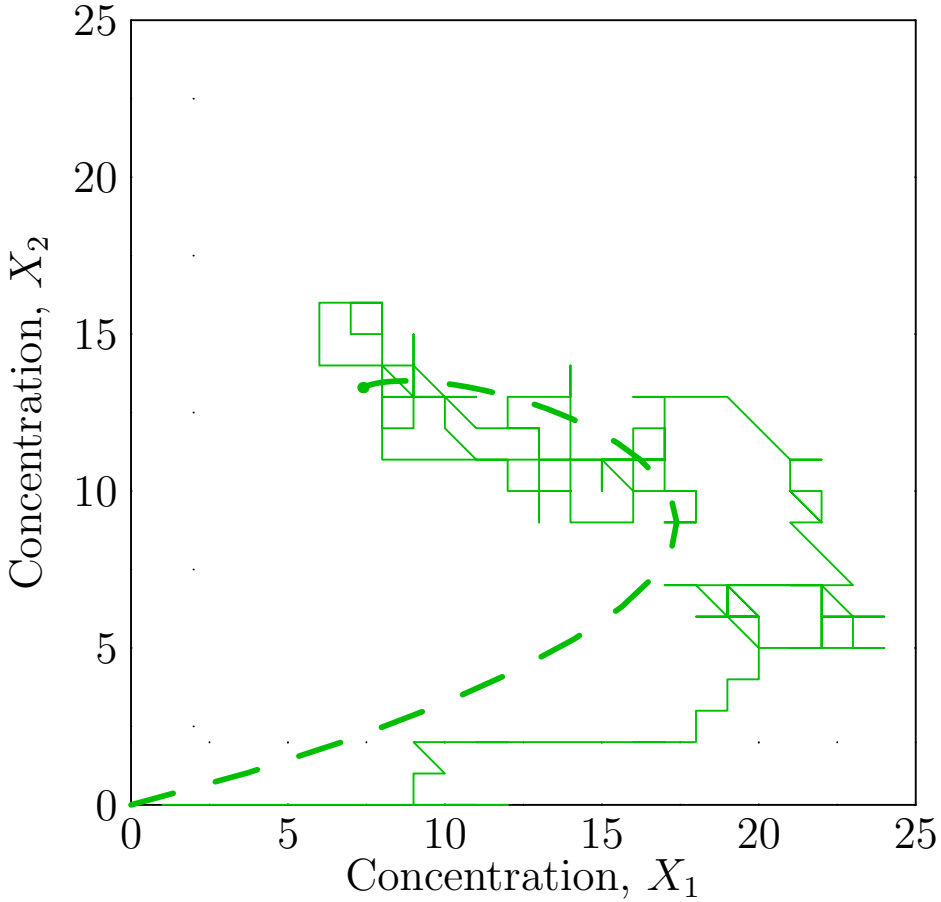

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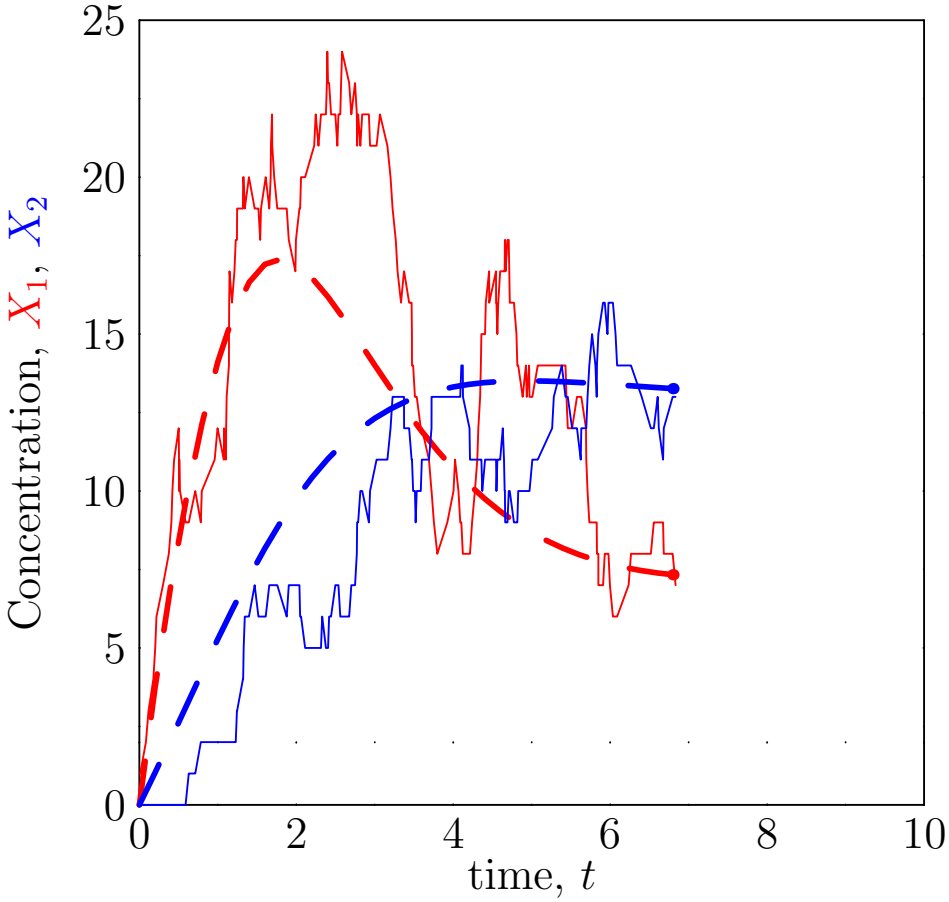

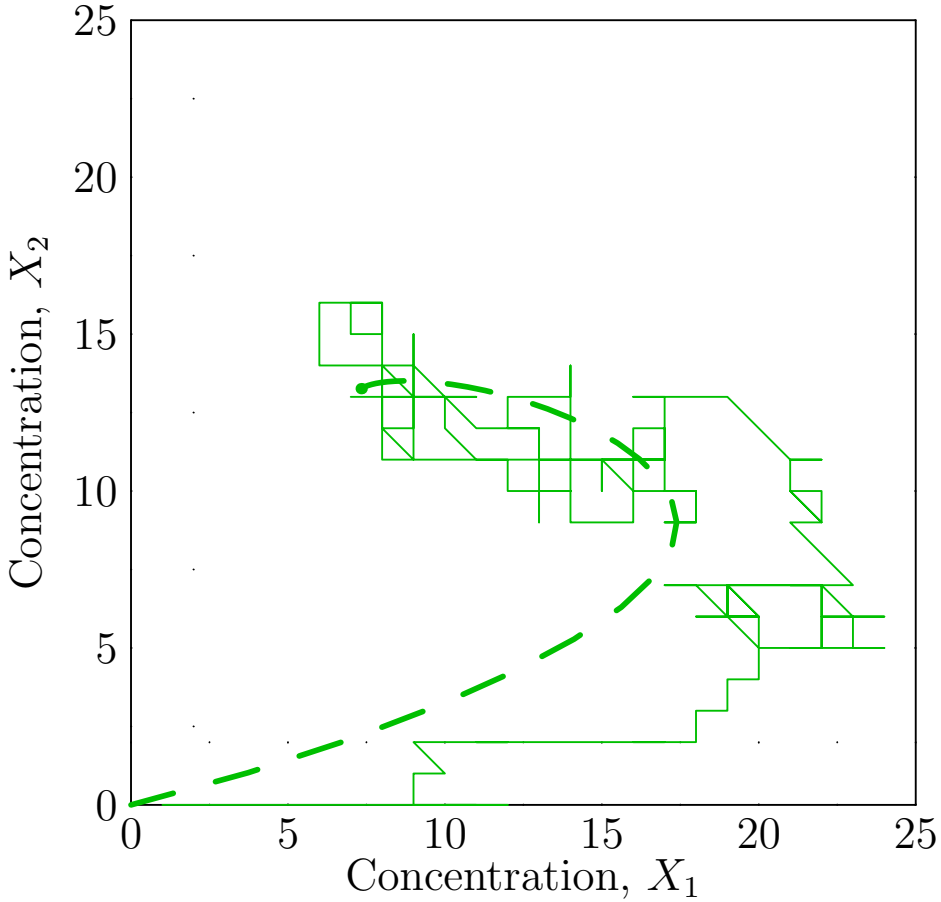

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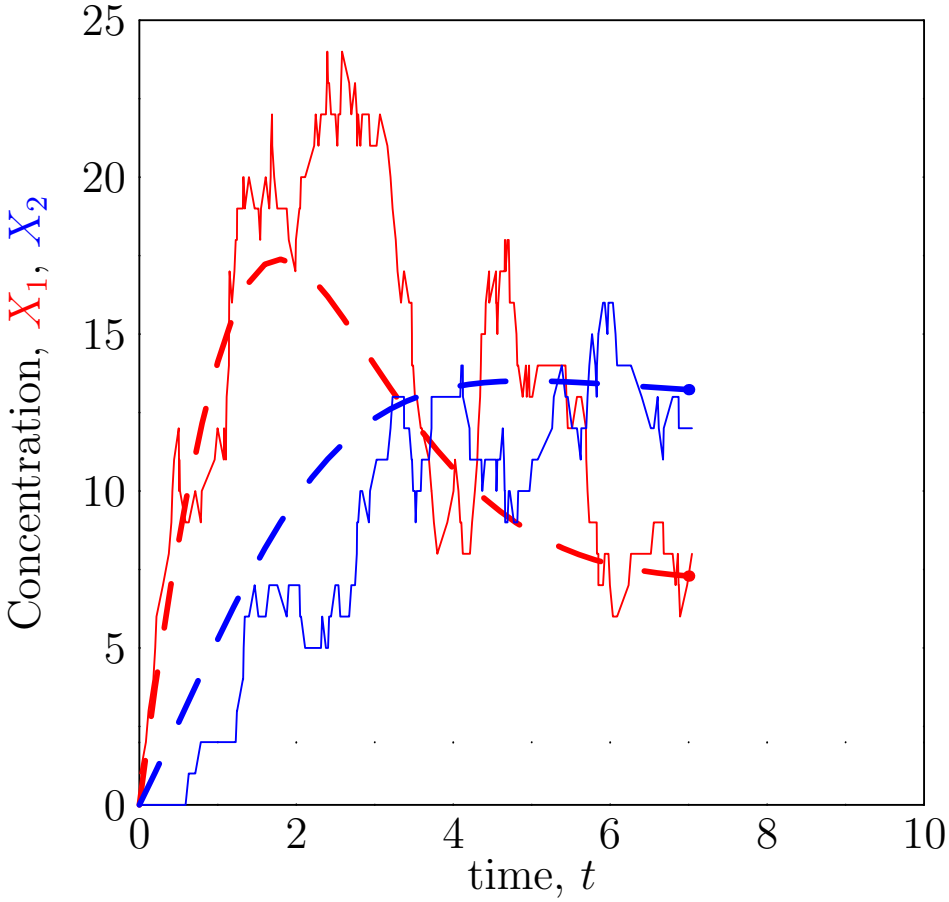

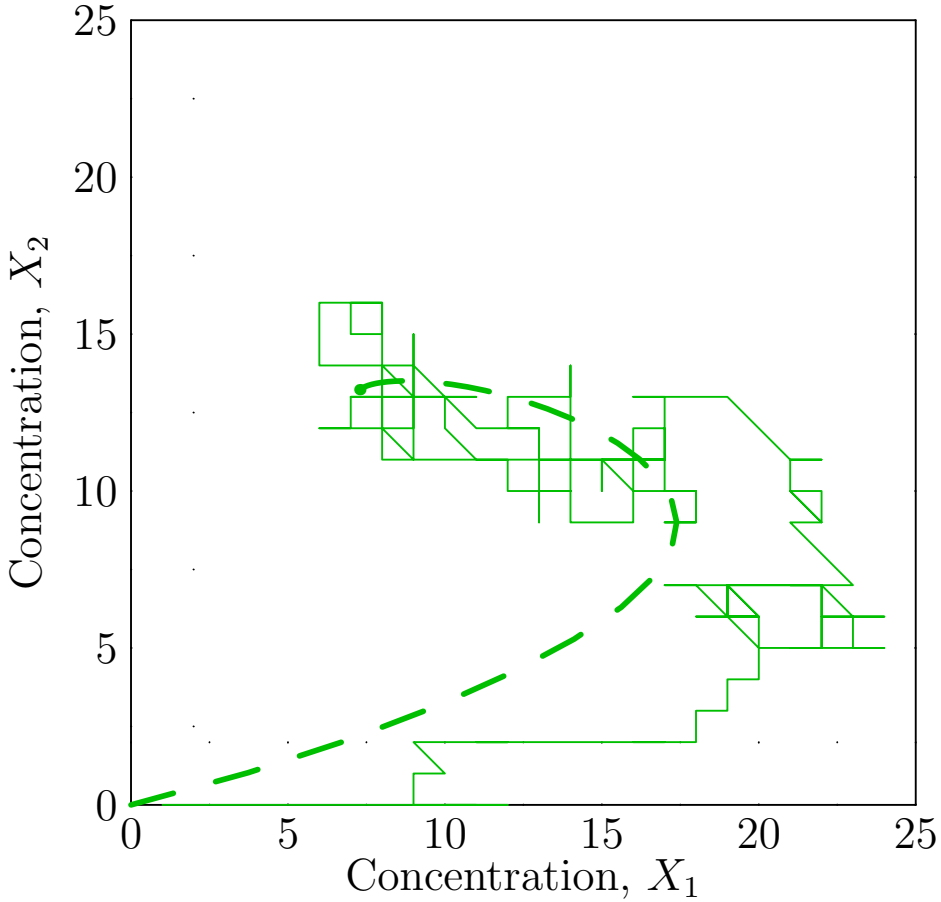

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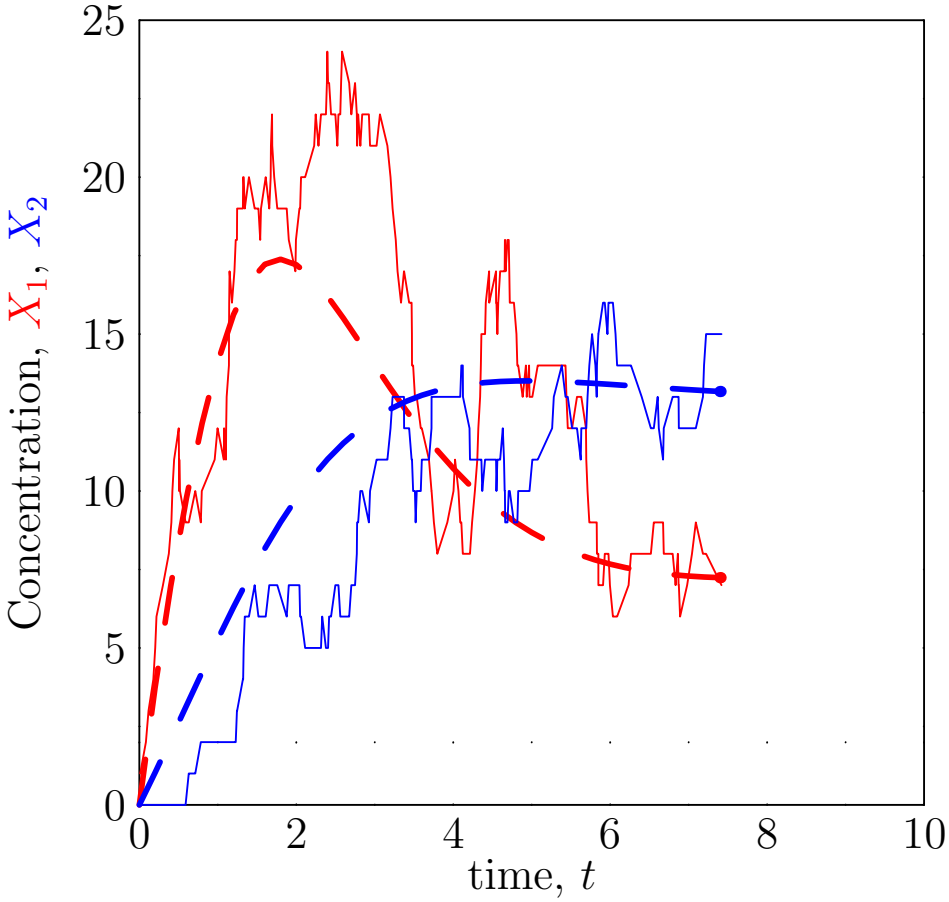

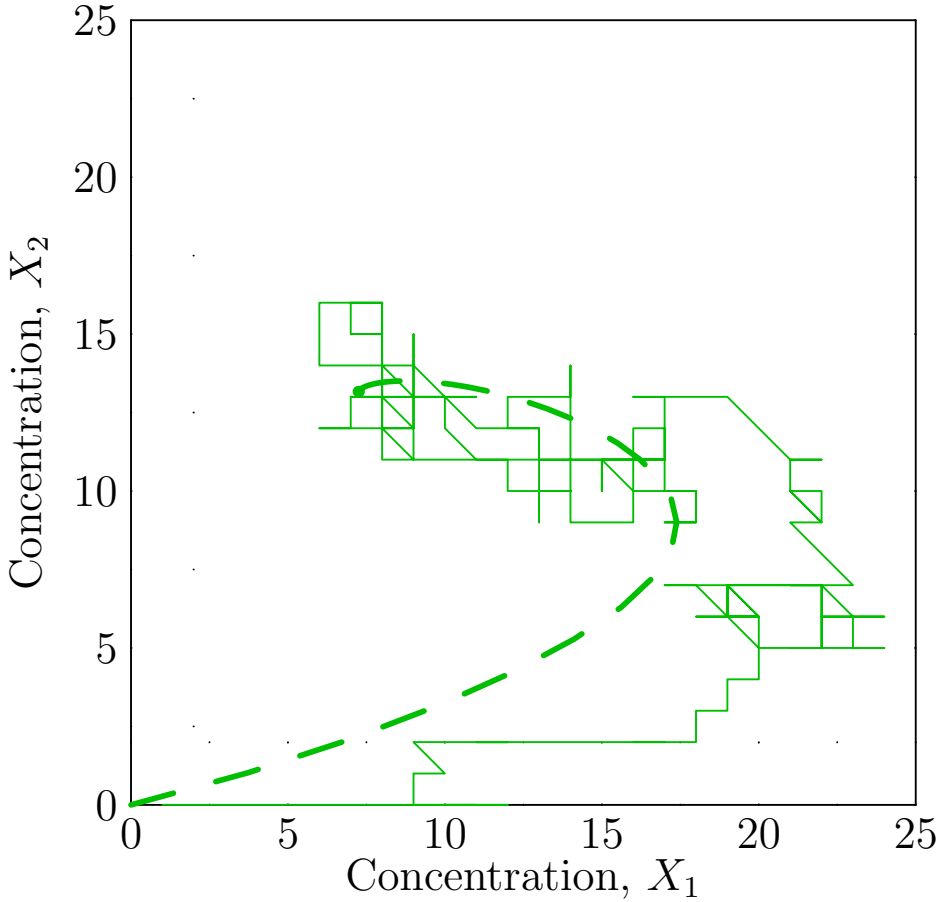

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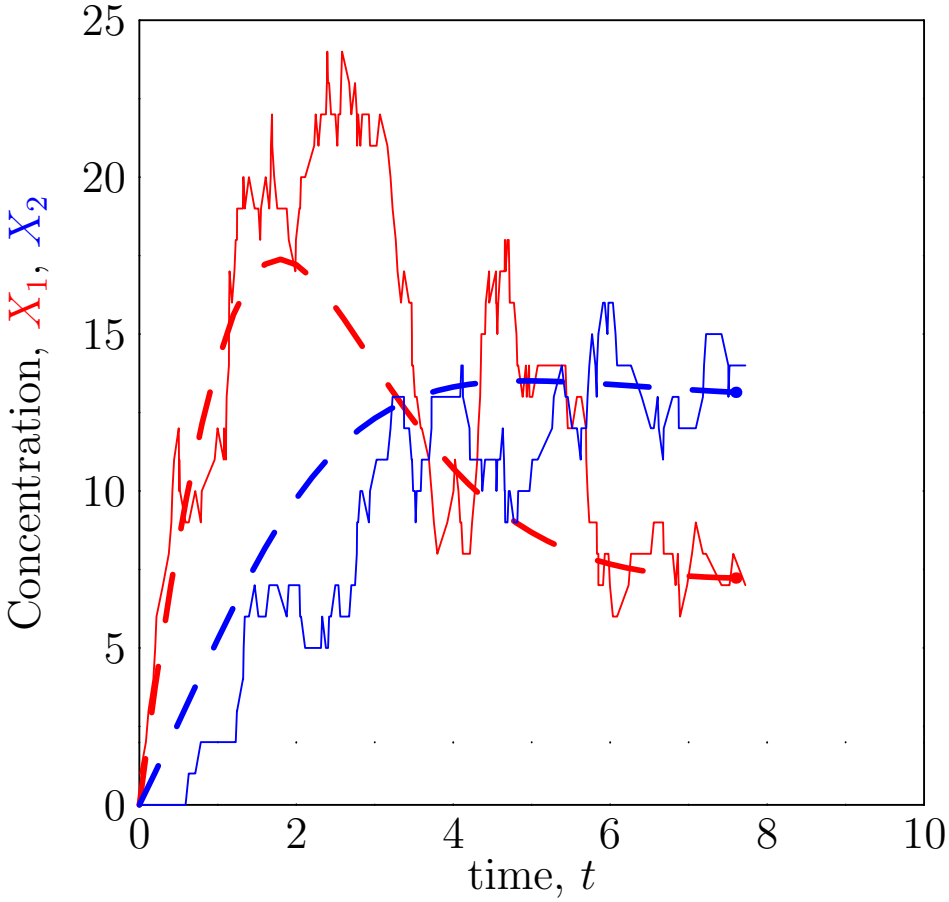

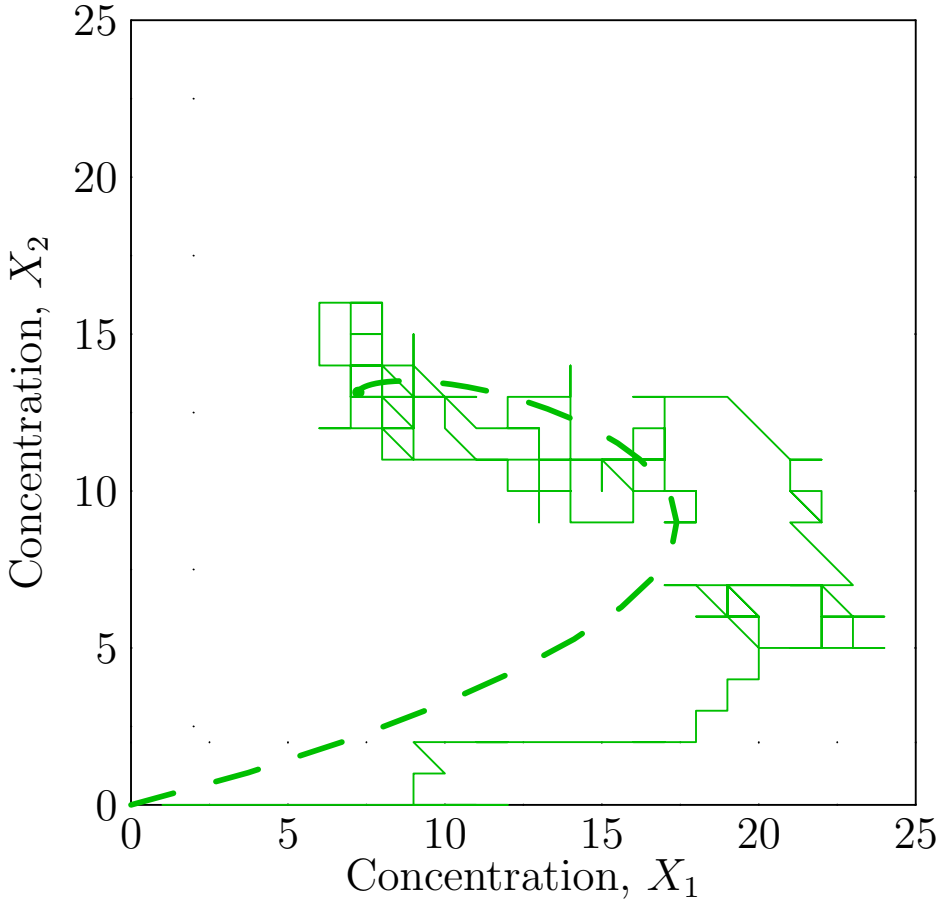

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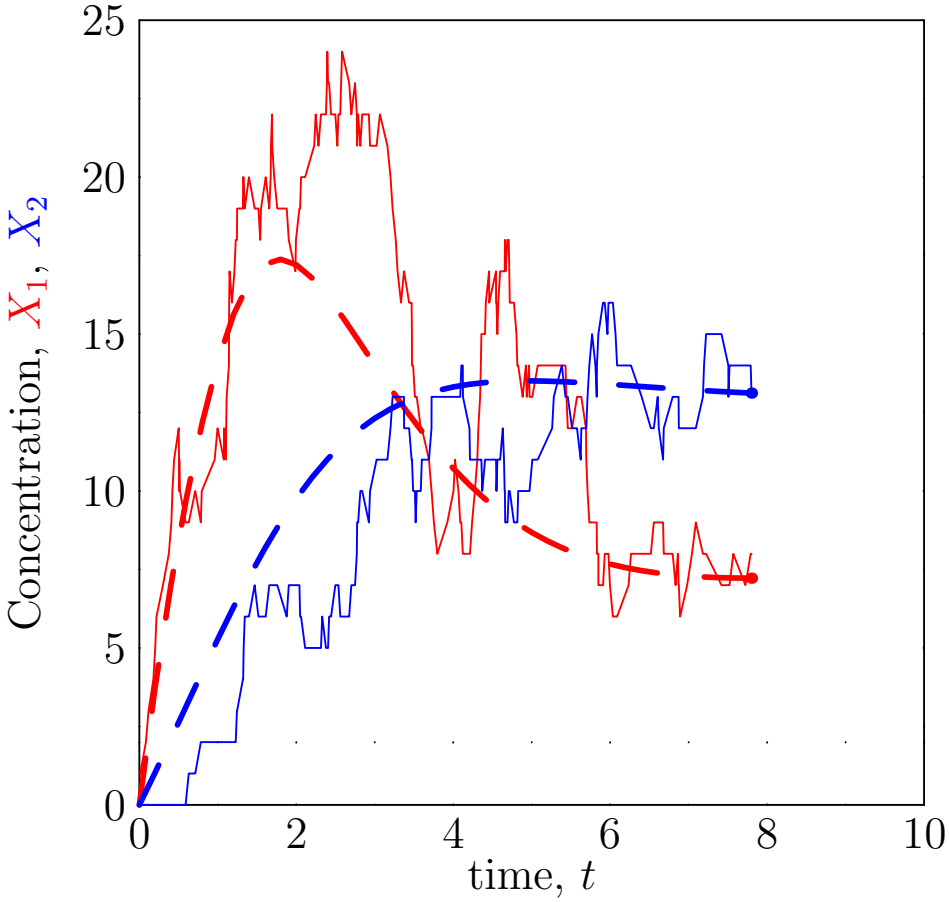

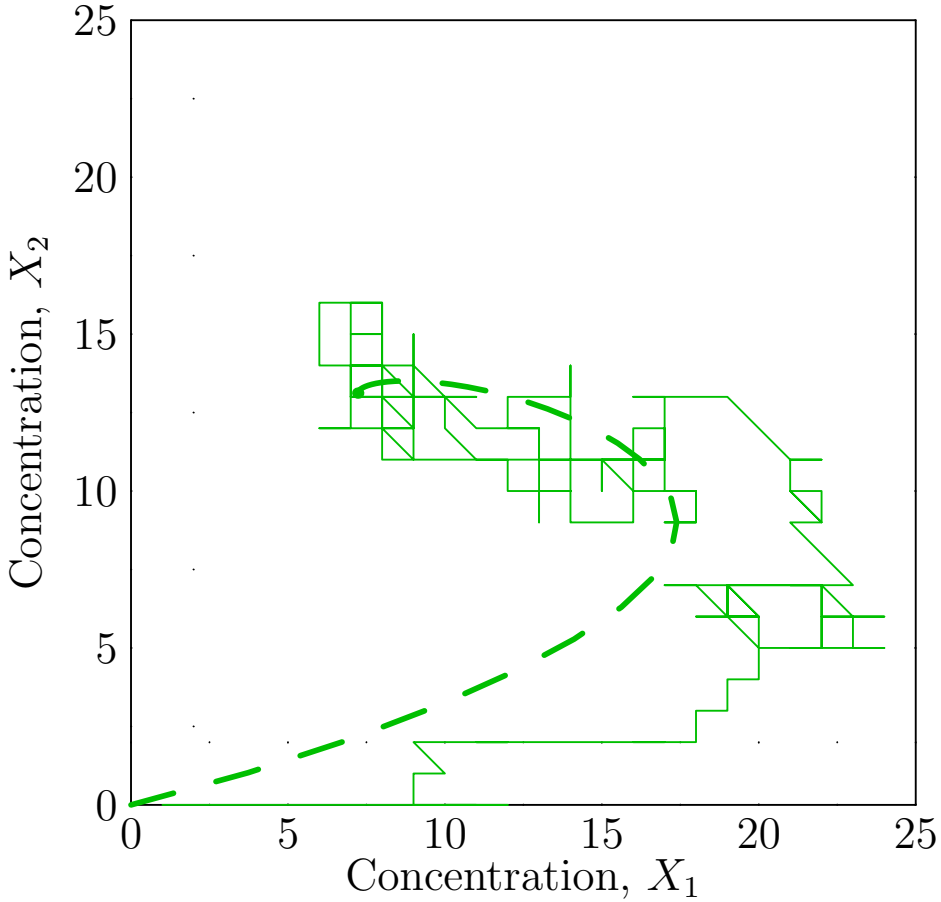

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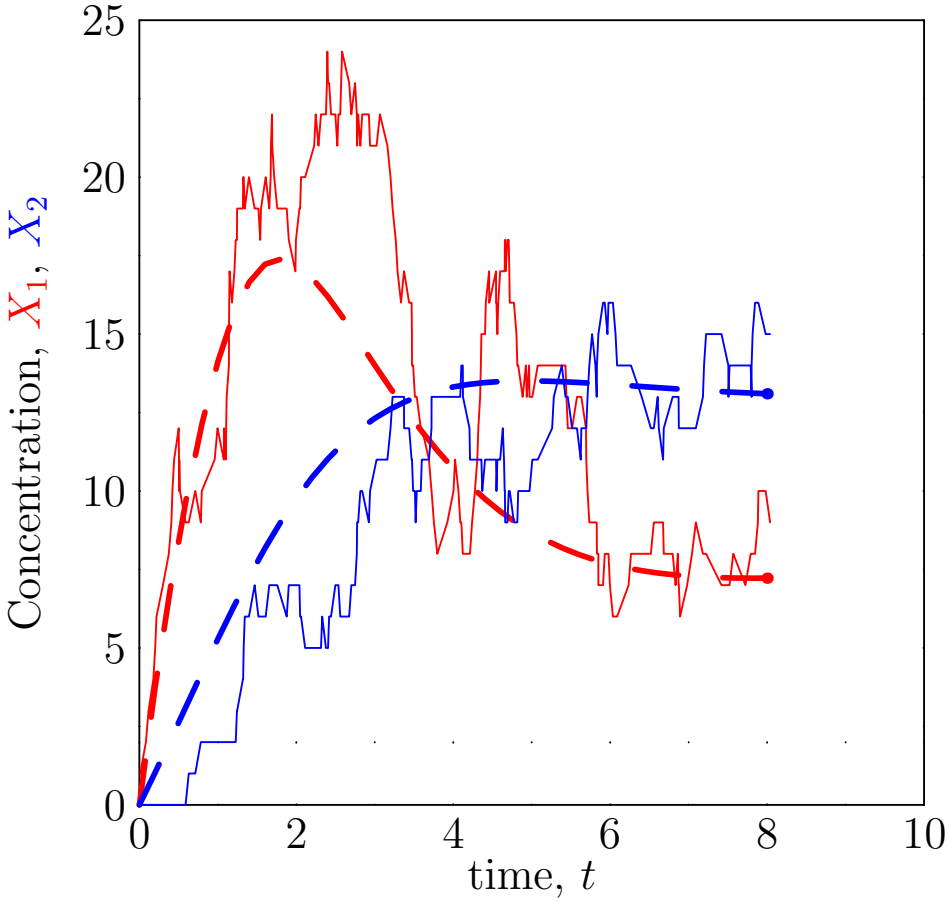

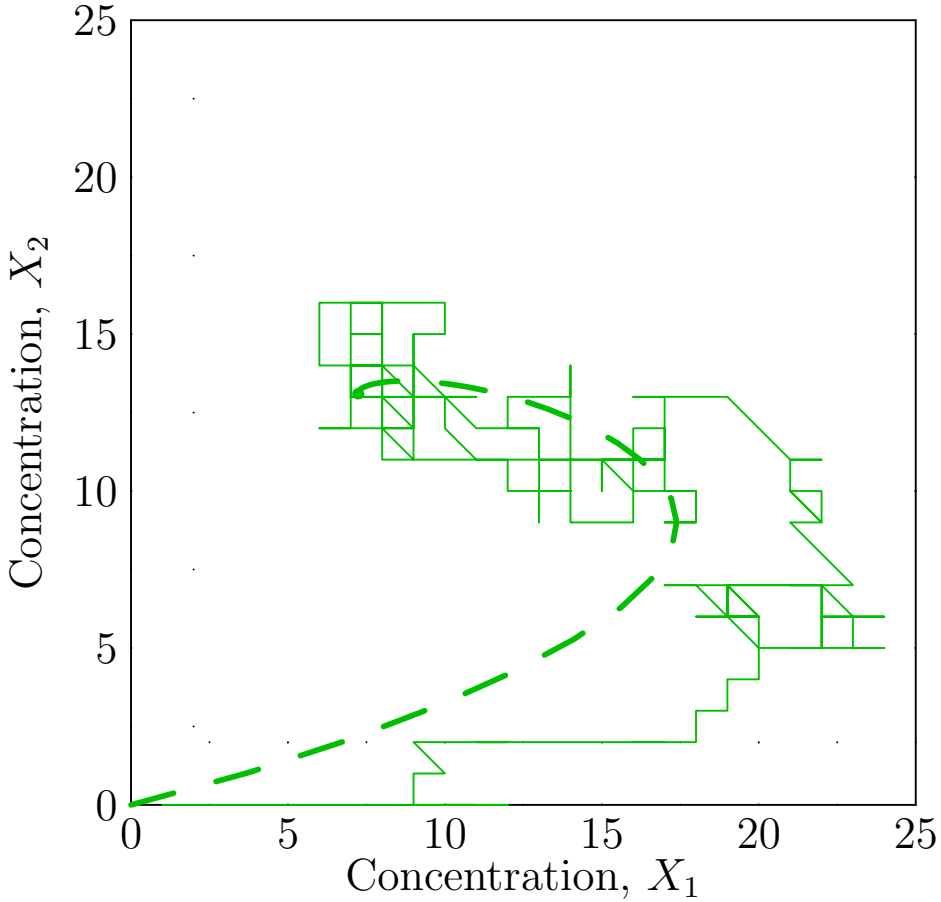

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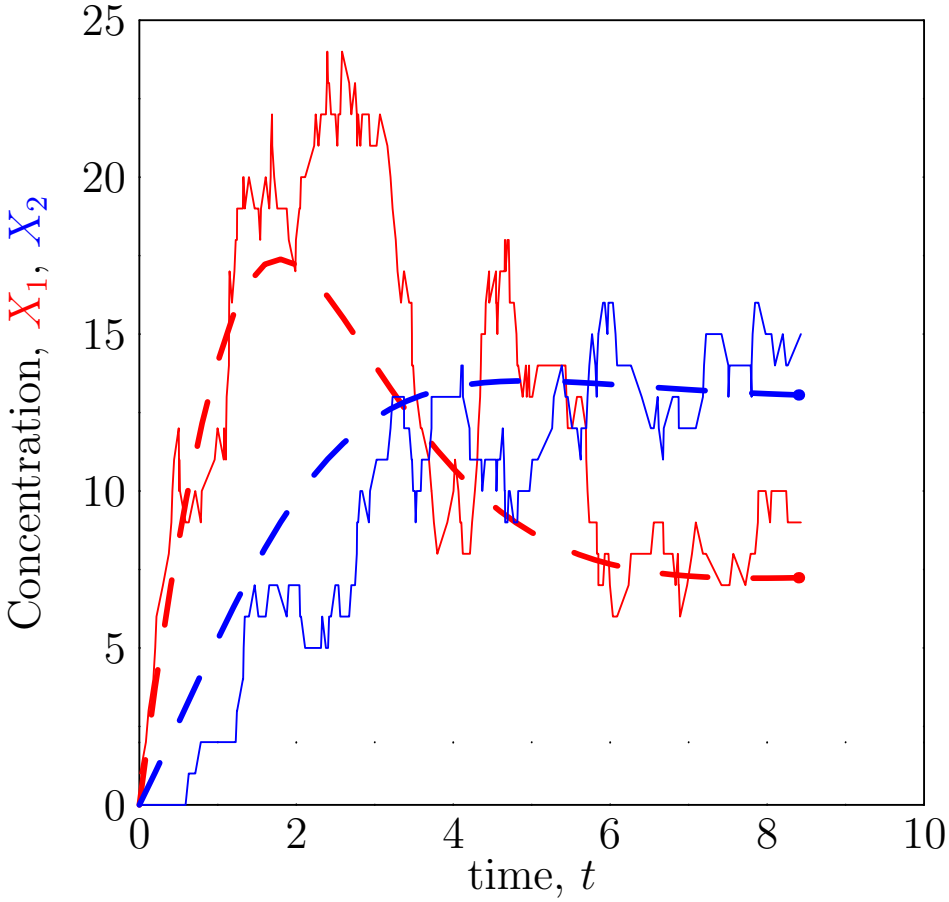

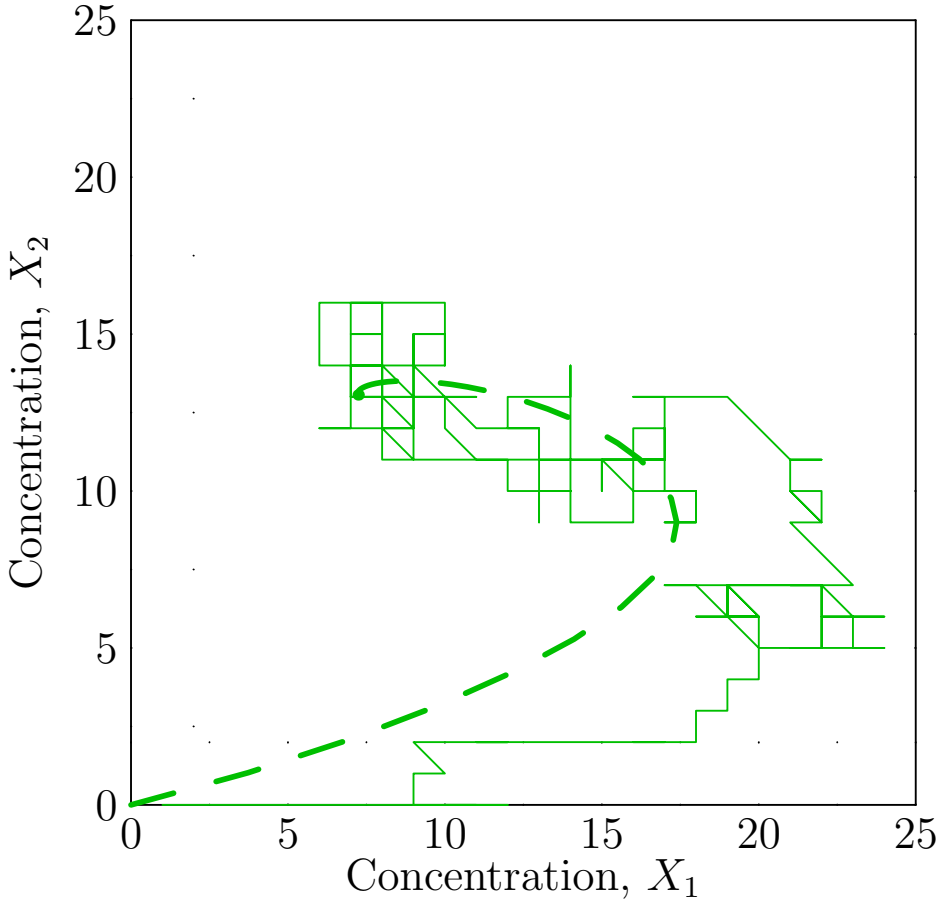

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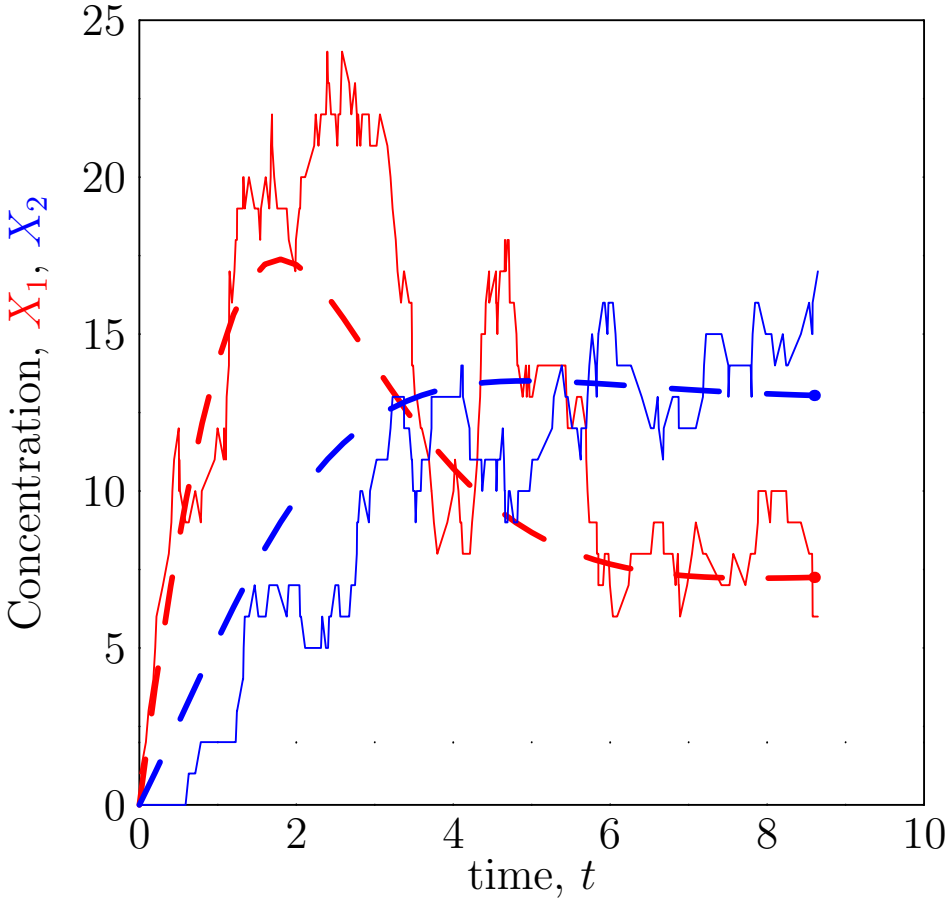

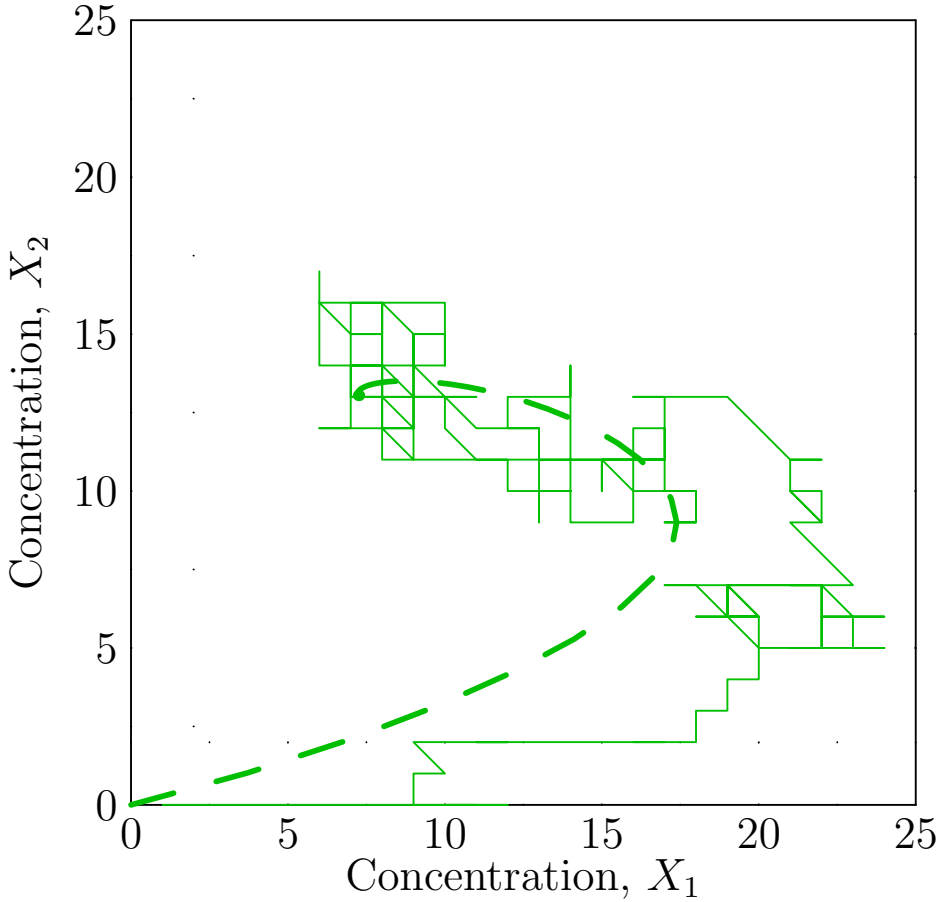

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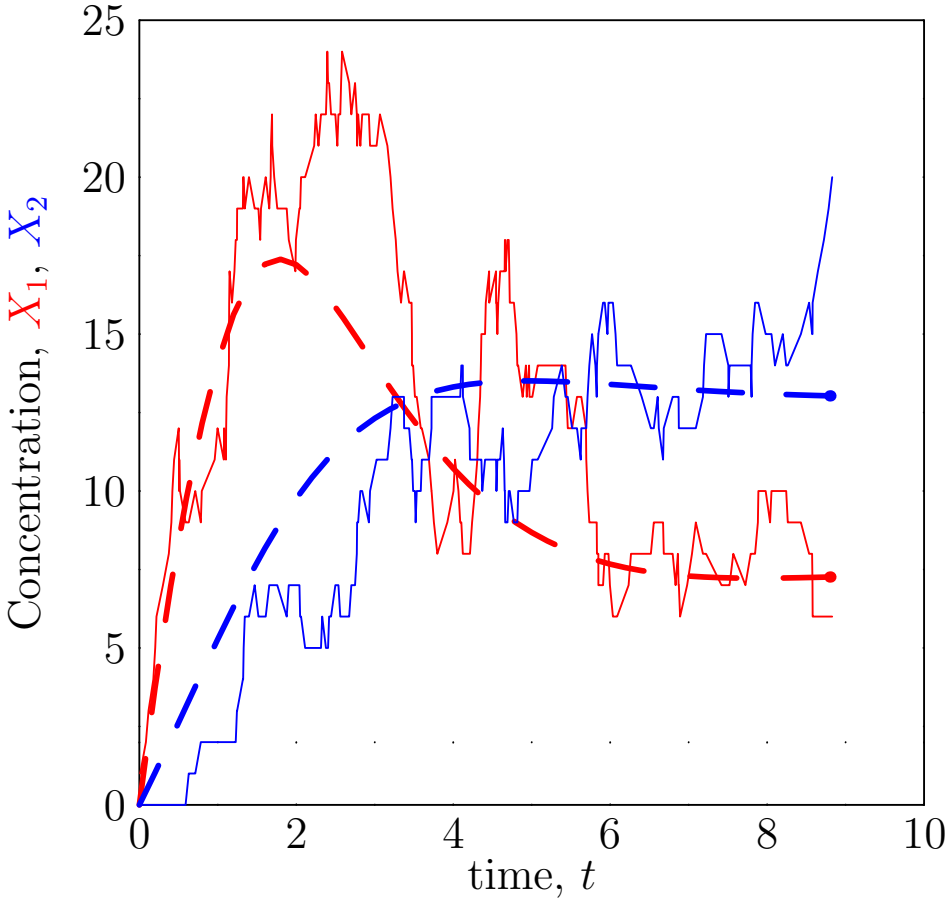

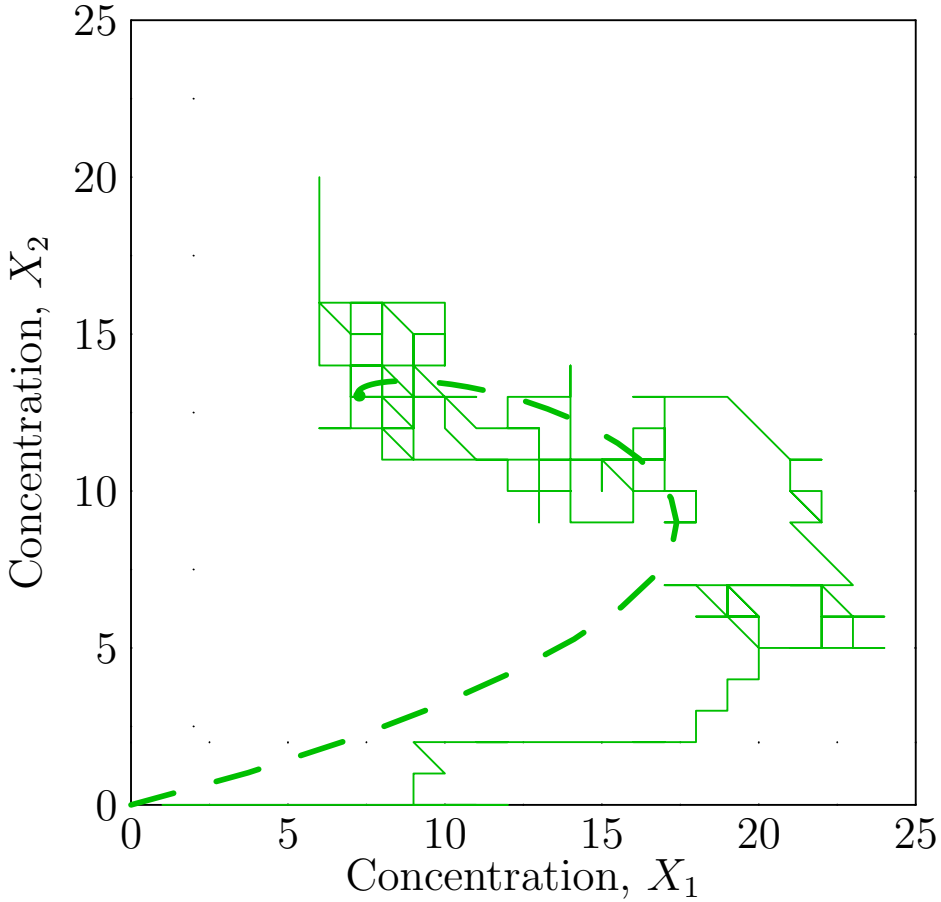

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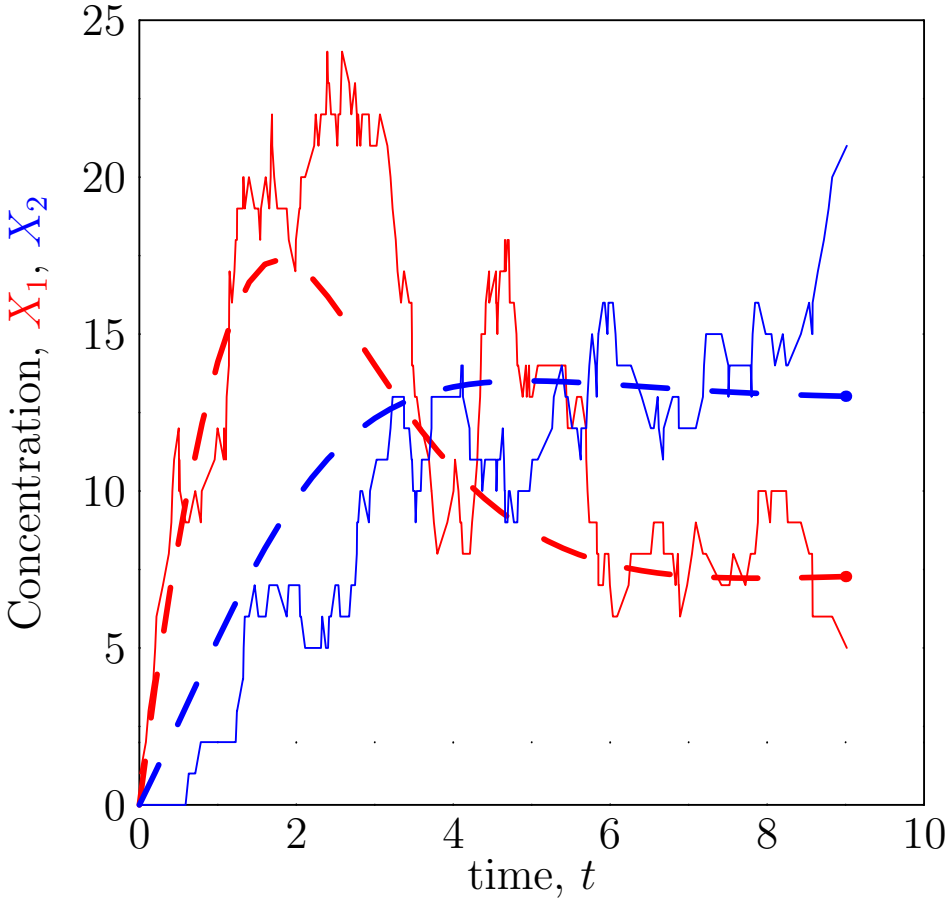

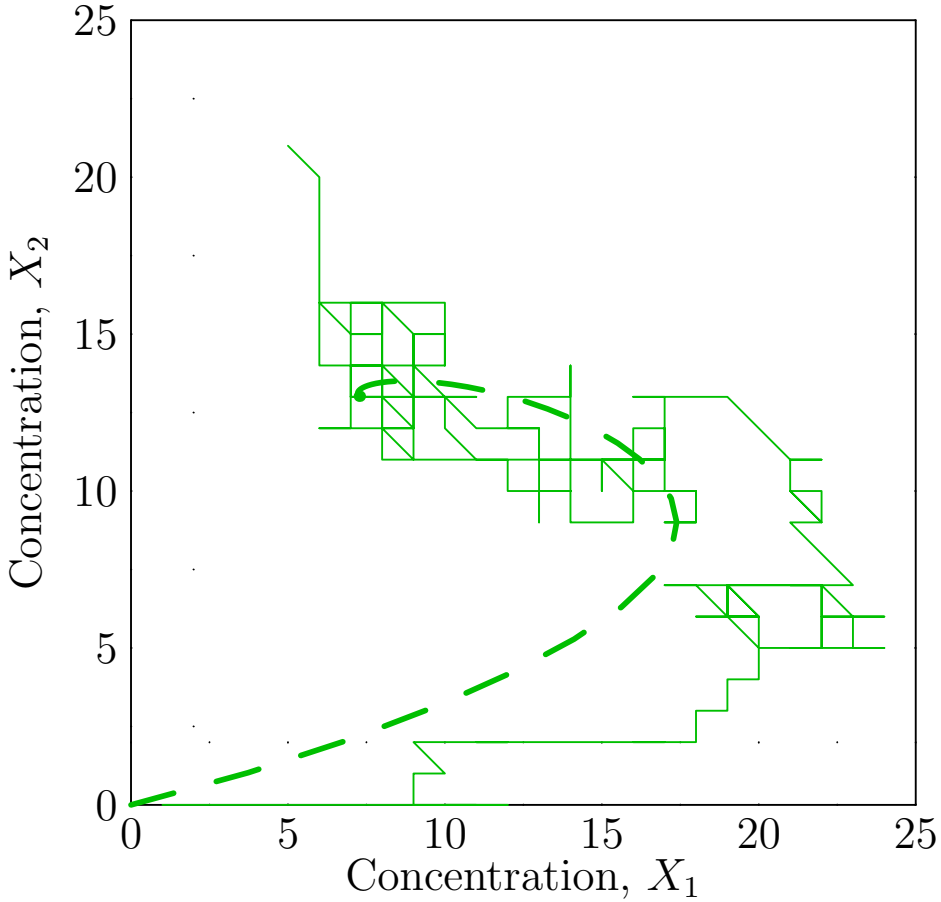

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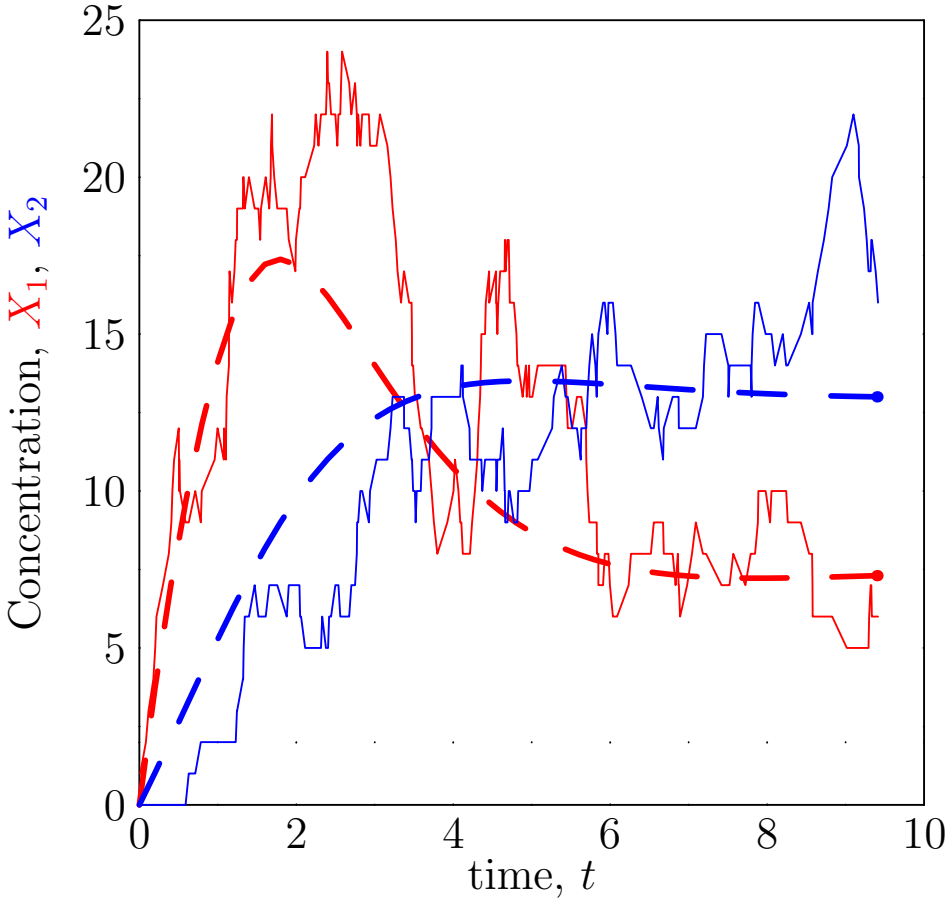

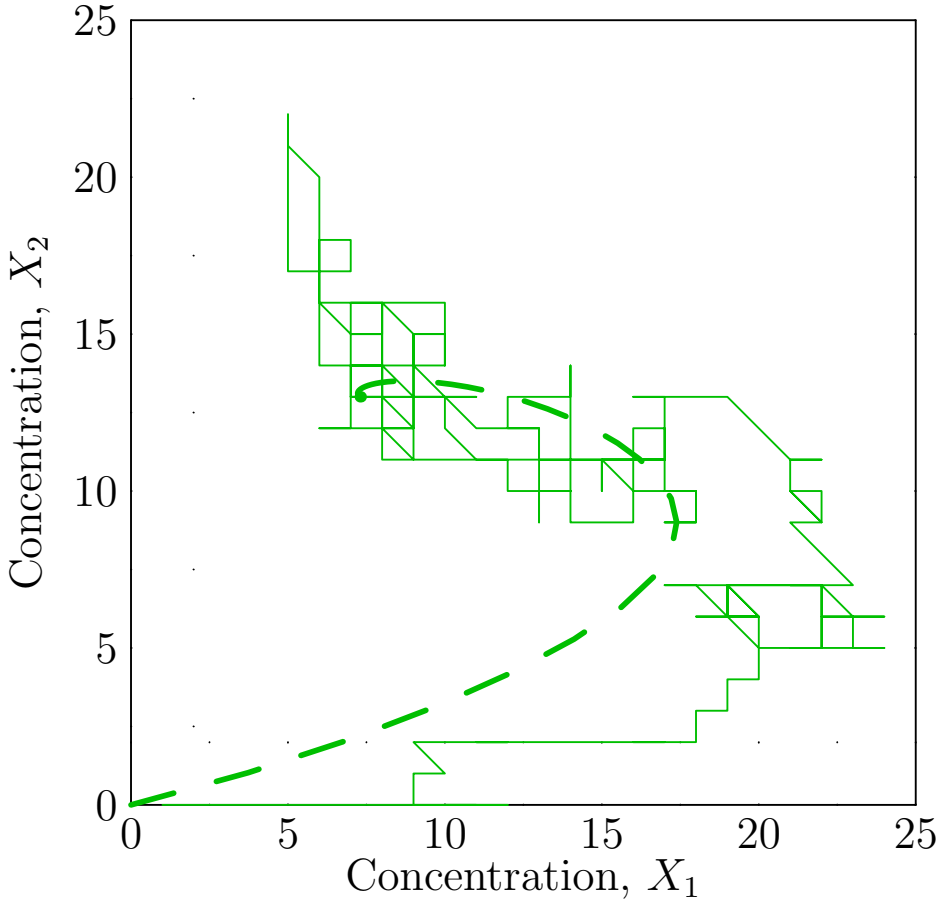

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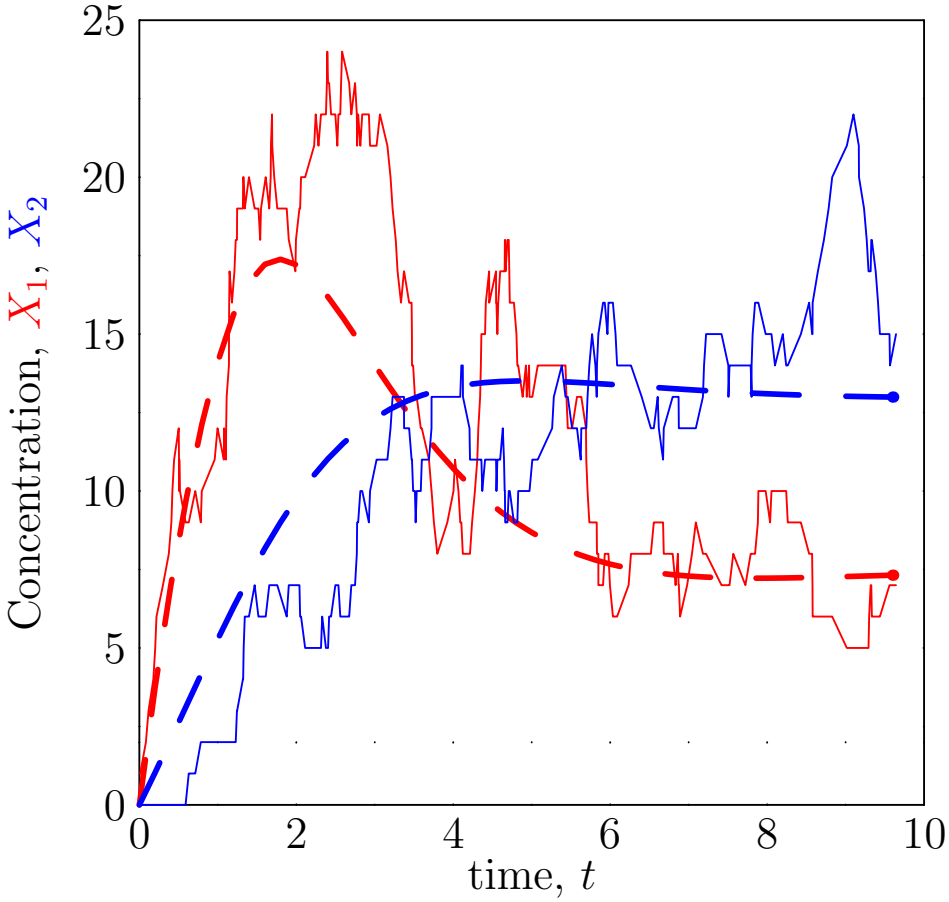

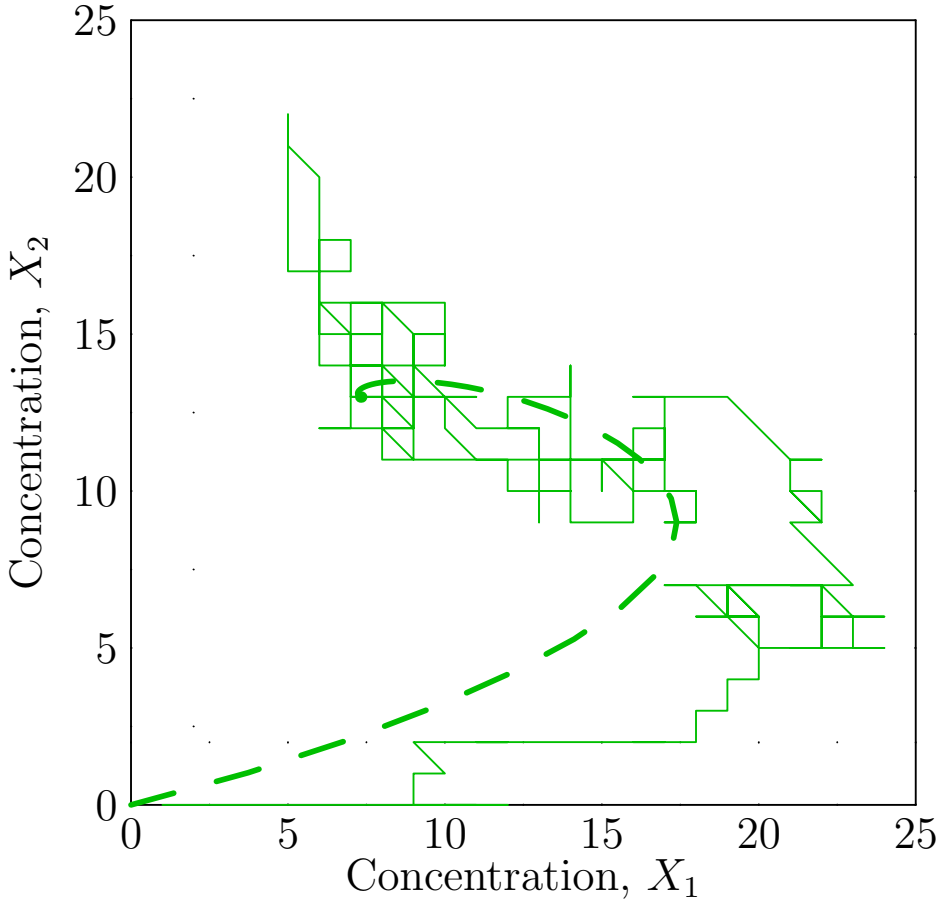

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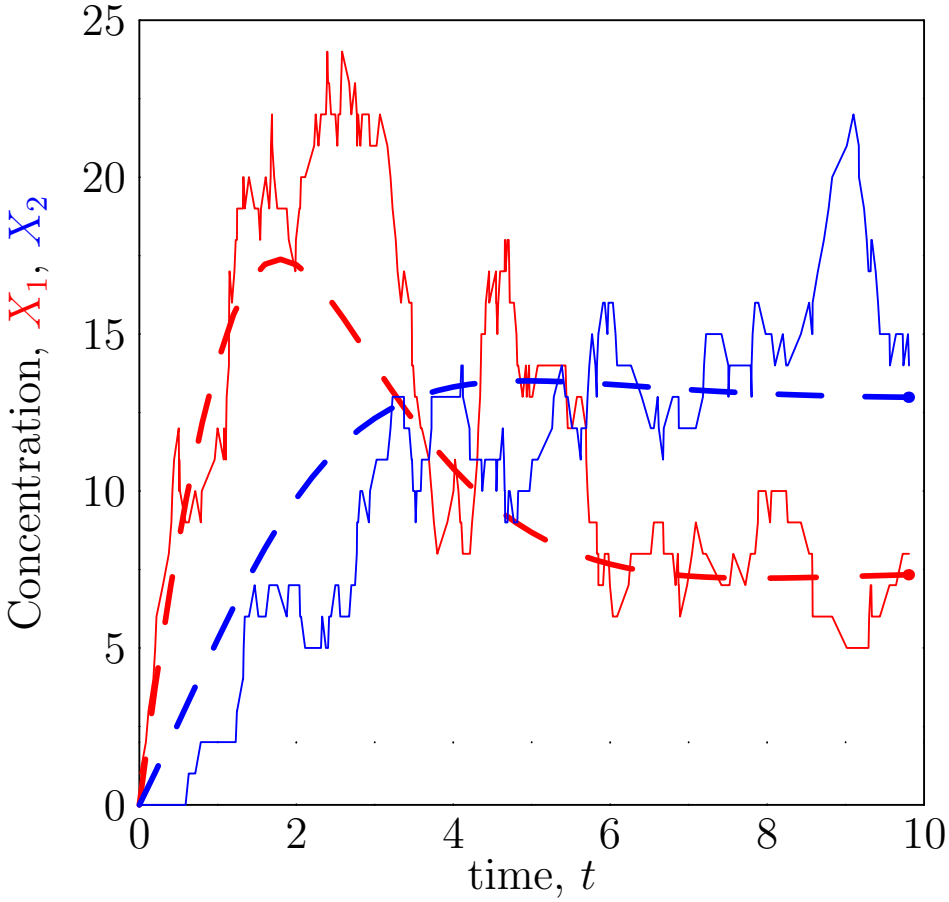
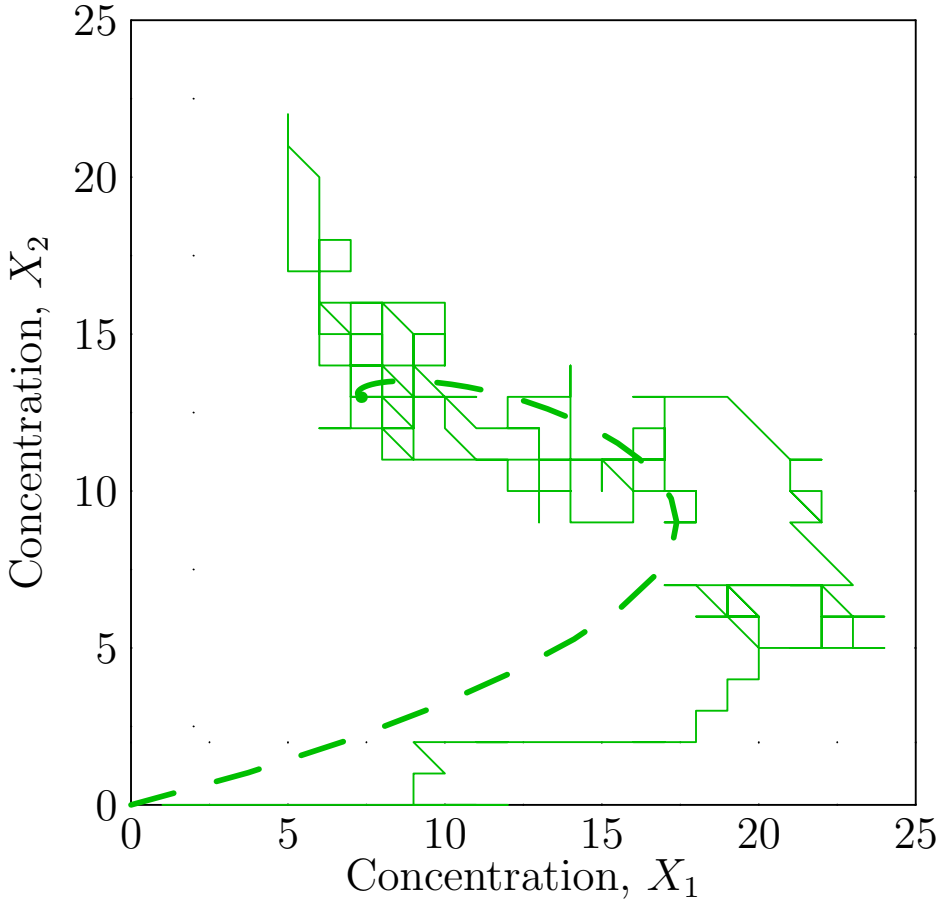
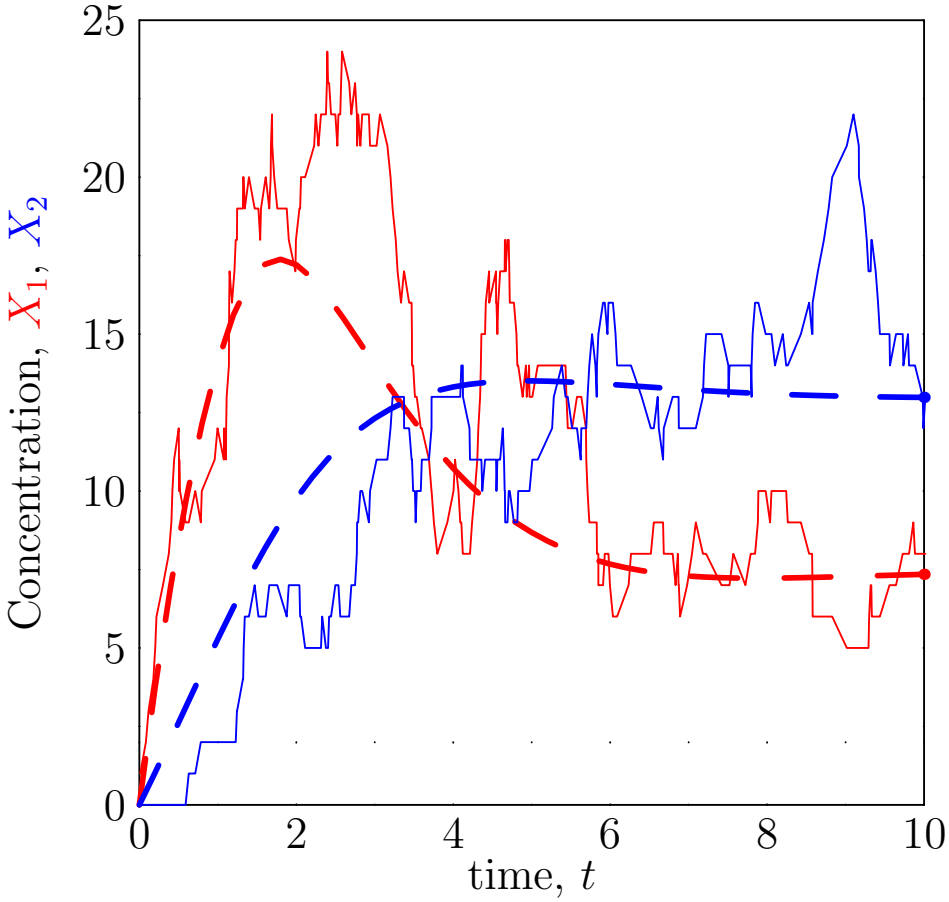
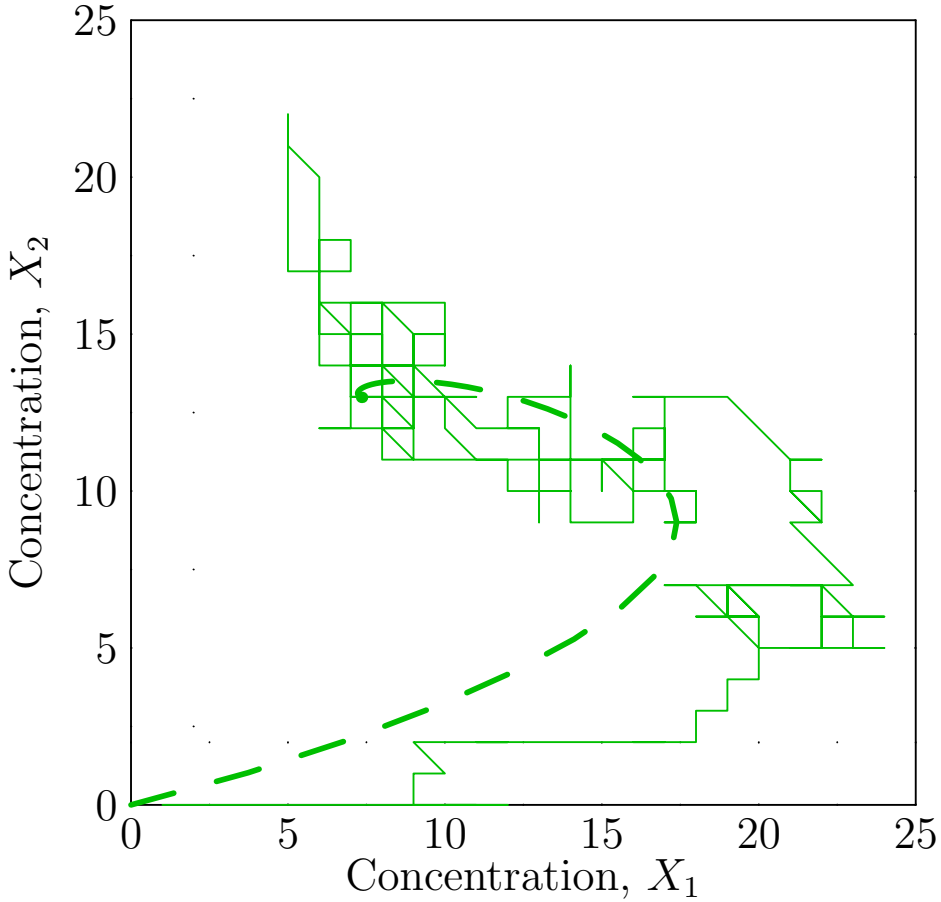




Small Numbers ($r = 10$)





Limitations

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- However, some chemical equations can happen millions of time faster than others
- We need to run long enough to capture the slowest processes
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- Gillespie algorithm provides the best approximation to the true dynamics
- Of course, being entirely numerical it doesn't provide much insight into how changes in parameters change the dynamics
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- There is a well developed system of stochastic differential equations which bridges the gap between Monte Carlo simulation and ODEs, but it is technically challenging (beyond this course) and it is not easy to apply to these systems (it requires putting in the size of fluctuations which can be very difficult to estimate)

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- In small systems with low concentrations this can lead to large fluctuations
- Populations of cell are empirically found to have large fluctuations between individuals
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